KEY PREPARATORY

NATURAL SCIENCE EXAM BOOK

(Grade 11 - 12)

2001 E.C - 2006 E.C EGSECE

With Answers and detailed explanations

2007 Edition

English
Mathematics
Physics
Chemistry
Biology
Civic & Ethical Education

Compiled By:- Anteneh Meheret
G.P

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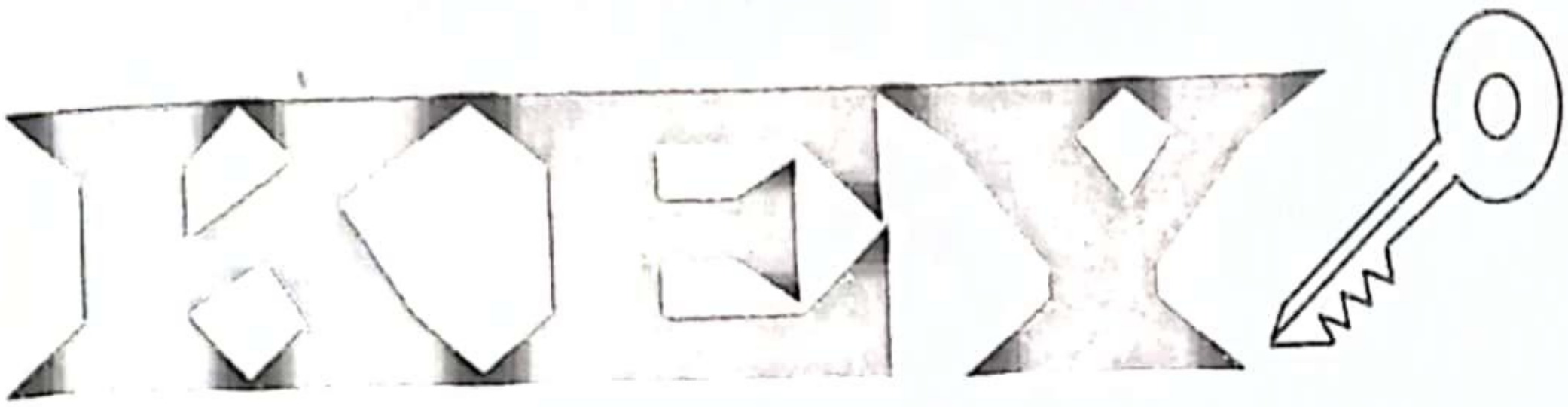

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2001 E.C- 2004 E.C Compiled by – Anteneh Meheret
– Eshetu Biru

2005 E.C - 2006 E.C Compiled by – G.P

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ETHIOPIAN HIGHER EDUCATION ENTRANCE CERTIFICATE ENGLISH EXAMINATION 2001/2009

BOOKLET CODE 12 **SUBJECT CODE 01**

SECTION ONE: Word Order (1-10)

DIRECTION: The following words are not in the proper order. When put in the right order, they give correct English sentences. Choose the correct sentence from the given alternatives and blacked the letter of your choice in the separate answer sheet provided.

1. me they for them to everything do wanted make
   A. They make me wanted everything to do for them.
   B. They wanted to make me do everything for them.
   C. They wanted to do me make everything for them.
   D. They wanted me do to make everything for them.
2. is enjoy too short life it
   A. Too short life; it enjoy.
   B. Enjoy it too short; is life.
   C. Life is too short; enjoy it.
   D. Enjoy too short; it is life.
3. going Awassa week I next am to
   A. I am going to Awassa next week.
   B. I am next week going to Awassa.
   C. I am to going Awassa next week.
   D. Next week, I am to going Awassa.
4. isn't beautiful she is she
   A. Beautiful isn't she, is she?
   B. Is she beautiful, isn't she?
   C. Isn't she beautiful, is she?
   D. She is beautiful, isn't she?
5. work hard to succeed we have to
   A. We have to work hard to succeed.
   B. To succeed hard, we have to work.
   C. To work hard, we have to succeed.
   D. We have to work to succeed hard.
6. he no knows why doesn't one come
   A. No one doesn't knows why he come.
   B. No one knows why he doesn't come.
   C. Why doesn't he come no one knows.
   D. Doesn't one knows why he no come.
7. window wide burglar left the open the
   A. The burglar left the window wide open.
   B. The wide burglar left the window open.
   C. The wide window left the burglar open.
   D. The burglar open the left wide window.
8. develop are weeks materials in the going two we to
   A. The materials are we going to develop in two weeks?
   B. Are we going in two weeks develop to the materials?

C. Are we going to develop the materials in two weeks?
D. Are we to develop the materials going in two weeks?

9. have see you soon to we
   A. We soon you have to see.
   B. We have to see soon you.
   C. Have we to see you soon.
   D. We have to you soon.
10. him tolerate cannot more any I
   A. Tolerate any more I cannot him.
   B. I cannot any more him tolerate.
   C. I cannot tolerate him any more.
   D. Any more I tolerate him cannot.

SECTION TWO: Paragraph coherence (11-20)

DIRECTIONS: when put in the correct order, sentences a-e in each of the questions from 11-20 make up a complete paragraph. From the alternatives given in each case, choose the letter that contains the correct order of the sentences and blacken the corresponding space provided in the separate answer sheet.

11. a. It is very difficult to satisfy everybody.
   b. What we need to make sure is that we are doing the right thing, as our conscience tells us.
   c. The rest is not under our control.
   d. But we expect people to be reasonable.
   e. Surely, there will be somebody to complain about what we do.
   A. a b c d e B. a e b c d C. b c a d e D. a c e b d
12. a. You should therefore have one immediately.
   b. You must then take passport size pictures.
   c. But, first you need to have an ID card from your *kebele*.
   d. You then apply to the immigration department.
   e. You cannot go out of the country without a passport.
   A. e a c b d B. e c b d a C. c b d e a D. e a d b c
13. a. Avoid any distraction once you start your work.
   b. First, find a place that is quiet and comfortable
   c. Take some rest in between to relax your mind and body.
   d. Then, have all the study materials you need ready.
   e. You must be organized when you study.
   A. b e a c d B. e b d a c C. a b c d e D. b d c a b
14. a. We should honour the person, Alexander Fleming, that made this possible.
   b. Before that, people used to die even with simple infections.
   c. Penicillin was discovered in 1928 and used in treating infections from 1948.
   d. He was instrumental in saving countless lives.
   e. There was no way for them to fight diseases.
   A. c b e a d B. b c e a d C. a b c d e D. c e b d a
15. a. My father rises before anyone.
   b. We eat breakfast and jump in the car.
   c. He then wakes up my brother and me.

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   d. Dad drops us off on his way before he gets Mom to work.
   e. Our family has a daily routine to get us to work and school.
   A. a c d b e B. e a b c d C. e a c b d D. d e a c b
16. a. Others put the estimate between 300 and 325 million.
   b. Of these, 145 million live in the USA, about 55 million in the UK and Ireland, and some 30 million in the British dominions.
   c. This in short means that the language is spoken by about one-seventh of the world population.
   d. Exact information on this point is not available, but an estimate of 230 million can not be very wide of the mark.
   e. How many people in the world speak English as a native language?
   A. e d b a c B. a b d c e C. a d b c e D. e a d b c
17. a. Graphology is the study of handwriting.
   b. They also consider the spacing between words.
   c. To do this, they look carefully at the slant of the writing.
   d. For example, a person with a printing-like handwriting is said to be creative and witty.
   e. Graphologists believe they can analyze personalities through their handwriting.
   A. a e d c b B. e b c d a C. e c b d a D. a e c b d
18. a. But we have been together for 15 years.
   b. It is amazing how time flies.
   c. We must be thankful for our lasting friendship.
   d We have lived through bad times and good times.
   e. It looks like it was yesterday when we first knew each other.
   A. b c e a d B. c b e d a C. e d c b a D. b e a d c
19. a. It eats about a ton of food each day.
   b. The lion needs about 15 pounds of food.
   c. Different animals have different food requirements.
   d.The record-holder for the amount of food eaten each day is the blue whale.
   e.The relatively small chimpanzee, for example, eats an average of 4.5 pounds of food each day.
   A. c e b d a B. c b e d a C. b e d a c D. e b d a c
20. a. It saves you from confusing such and other words.
   b. For example, try the words "flammable" and "inflammable".
   c. When you are unsure of the meaning of a word, look it up.
   d. A dictionary is a valuable tool to help you improve your English.
   e. A good dictionary will provide you with the most exact meaning.
   A. c e a b d B. e b a c d C. d e b c a D. d e c b a

SECTION THREE: Reading Comprehension (21-34)

DIRECTIONS: There are two passages in this section. Read each of them carefully and answer the questions according to the information therein. Each

question has four choices, **A-D**. Choose the best alternative and blacken the letter of your choice in the separate answer sheet provided.

Passage I

Global Addiction

1. The 1990s have been designated the Decade Against Drug Abuse by the United Nations. But, with less than three years to go before the end of the decade, governments and health organizations admit that they have made little progress in reducing drug, alcohol and tobacco abuse.
2. Today consumption of all these substances is increasing steadily worldwide. Virtually every country now has problems with illegal drugs. And the world is producing and consuming more alcohol and tobacco than ever. Between 1970 and 1990 beer production alone rose by over 80 per cent. And, if the number of smokers keeps on rising, by the second or third decade of the next century there could be 10 million deaths each year from smoking-related illnesses.
3. Experts like Dr Alan Lopez, Acting Manager of the Substance Abuse Programme at the World Health Organization (WHO), that, taken together, abuse of legal and illegal drugs is a greater health threat than major disease. WHO studies indicate that abuse of tobacco, alcohol, illicit drugs and more mood-altering substances account for between six and seven percent of all disease and injury that occurs in the world each year. "That's a massive amount," he says, "and it's twice as much, for example, as diseases like tuberculosis or malaria."
4. Drugs are also a huge burden on the world economy. In the United States, for example, it's estimated that alcohol and illegal drug use costs the country tens of billions of dollars each year, mainly in health care. When the cost of tobacco-related illnesses is added, that total more that doubles.
5. Drugs are also closely related to crime. Many police forces no longer distinguish between illegal and legal drugs when fighting crime. In Australia, for example, experts estimate that police in some parts of the country spend between 70 and 80 per cent of their time dealing with alcohol-related incidents.
6. International research has revealed that drug use follows different patterns in developing and developed nations. While the number of cigarette smokers declines in countries like the US (down by 0.5 per cent annually) and the UK (down by 2.3 per cent), it is rising by 1.7 per cent a year in the developing world. The Chinese now account for 39 per cent of the world's smokers, and the number of cigarette smoked in the whole Asia region is expected to rise by more than 50 per cent by the end of the century.
7. Aggressive marketing of cigarettes in developing countries is one explanation for the rise, although some countries are starting to limit tobacco advertising. But Dr Alan Lopez of the WHO is still concerned. "There has been a massive increase in cigarette smoking among men in developing countries, and we're now at the point where about half of all men currently smoke cigarettes. Equally, we're concerned about trends in alcohol use, which is rising in developing countries as the multi-nationals begin to sell more and more of their products there."
8. Dr Lopez is also pessimistic about the effect of campaigns to curb illegal drugs like heroin, cocaine and cannabis. In developed countries like the US the numbers of young people taking illegal drugs started to fall in the 1980s. That trend has now been reversed, he says. "We're seeing a steady increase in the prevalence of substances like marijuana and heroin.
9. In fact, so common is drug use in developed countries that illegal drugs have become a part of everyday life for children growing up in big western cities. Recent research in Glasgow, Scotland's second city, reveals how children progress from one drug to another. "Kids of about 10 or 11 start on cigarettes," explains Dr Neil McKegannay, who led the Glasgow research. "A year later they're into alcohol-reporting significant levels of drunkenness – before moving on within a year to cannabis, and then a year or so after that into a broader range of drugs – LSD – and small levels of ecstasy, cocaine and heroin."

Taken from *BBC English*, September 1997

21. The word *Today* (line 1 of paragraph 2) refers to the year(s):
A. 1999 B. 1990s C. 1997 D. 2009
22. In a recent research conducted in Glasgow, children were observed moving from cigarettes, alcohol, cannabis and then LSD. This shows _____.
A. the degree of irresponsibility of parents
B. a significant improvement in drug abuse
C. the decline in the morale of school children
D. the degree of seriousness in the use of alcohol and drugs
23. What per cent of world deaths and injuries would diseases like malaria or tuberculosis make every year?
A. 6 to 7 B. 3 to 4 C. 12 to 13 D. 12 to 14
24. What does the phrase *their products* in Paragraph 7, last line refer to?
A. Tobacco C. Alcoholic drinks
B. Tobacco and alcohol D. All sorts of drugs
25. According to paragraph 2, which one of the following sentences is **not** true?
A. Drug abuse was on the rise.
B. Beer production in the 1990s was high.
C. Towards the end of the 1990s, people smoked more than they did before.
D. By 2030 more than 10 million will have died from smoking-related disease.
26. According to the passage, which of the following is TRUE about the economic impact of drugs in the US?
A. Alcohol and drugs are more expensive than tobacco.
B. Tobacco-related illnesses are cheaper to treat than alcohol and drug-related ones.

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C. Tobacco-related illnesses cost the same money to treat as alcohol and drug-related ones.
D. Tobacco-related illnesses are more expensive to treat than alcohol and drug-related ones.

27. The main idea in Paragraph 1 is that _____.
A. the 1990s were the years against drug abuse
B. anti-drug abuse actions might possibly succeed
C. governments have done little to meet the UN-set target
D. the 1990s have not met the UN-set target against drug abuse

Passage II

Who Does It Better?

1. Who is better at communicating, people or animals? If you think about human inventions such as telephone, the radio, and the Internet, then the answer to this question seems clear. Human beings are "The Great Communicators." However, think about your own personal communication skills. If you compare what you can do with the abilities of certain animals, then the answer is not so simple. Animals can do some things that people cannot.

2. We humans depend on our voices for much of our communication. We use words and sounds to pass information to the people around us. But the sound of our voices cannot travel very far. Even the voice of an opera singer with years of training cannot be heard as well as many animal voices. Think of the elephant, for example. Its voice has great strength because of the elephant's great size, so it can be heard for miles. Elephants can also make very low sounds, sounds that are too deep for any human to hear. These low sounds let elephants communicate over even longer distances. The sounds travel in sound waves through the air and through the ground. How do elephants receive messages like these? No one knows. May be they hear them with their ears, or may be they sense them in some other way. It is possible that these sound waves pass from the ground through the elephants' toenails into their bones and then to their brains.

3. Let's also consider communication through movement. For example, some people dance to share ideas or emotions. When we watch these dancers, we may understand what they are thinking or feeling. But even great dancer's ability to speak through movement can't match the average honeybee's. Bees do a very special task to tell other bees where to find food. The dance tells the other bees which way to go so that they can fly in a straight line to the food. It also tells them exactly how far to go. It gives clear information about both the direction and the distance to a specific place.

4. Many animals communicate through smells. A smell can carry a lot of information. For example, a smell can say, "This is my place – get out!" or a smell can give an invitation. Often a female animal who wants a male to come to her will produce a smell to attract him. It says to the male, "Here I am – come and find me." Many animals receive messages through their noses just as

humans do. For example, your nose might tell you, "There's fresh coffee in the kitchen. It's time to get out of bed." But people don't usually use smells to communicate, and our noses don't receive messages very well. We certainly can't compete with the Great White Shark. A large part of its brain – 14% of it – is just for "reading" smells in the ocean.

5. Our noses are not the best, our voices are not the strongest, and our dancing may not say anything at all. But people are the only ones with words and written languages. So, may be we can still say we are "The Great Communicators."

Taken from *Password 2: A Reading and Vocabulary Text*

28. Which one of the following is true about honeybees?
A. They are poorer at telling distance than the humans.
B. They probably have as good sense of direction as the humans.
C. The humans are probably better at sensing things than honeybees.
D. Honeybees are better at telling both distance and direction than humans.

29. Which one of the following is **not** a function of smell as given in paragraph 4?
A. Ordering B. Inviting C. Informing D. Enquiring

30. Which one of the following animals is likely to have a stronger voice?
A. a lion B. a leopard C. a hyena D. a gazelle

31. The overall idea of the writer is that _____.
A. Humans are poor at dancing
B. Humans are the best communicators
C. Animals are better at smelling than humans
D. Elephants have stronger voice than humans

32. When the author asks "... How do elephants receive messages like these?" (Par. 2), he/she _____.
A. expects us to answer
B. is certain about the answer
C. wants us to debate on the issue
D. provides us with a likely explanation

33. Which one of the following features is possibly peculiar to the elephants?
A. That they have strong voice.
B. That they have massive body.
C. That they can make very low sounds.
D. That they have sounds that travel in waves.

34. Which one of the following best reflects the writer's view in Paragraph 1?
A. Humans are better communicators than animals.
B. Animals are better communicators than humans.
C. Both animals and humans seem to be equally good communicators.
D. It is not clear whether humans or animals have better communication skills.

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SECTION FOUR: Vocabulary (35-62)

A. Completion (35-52)

DIRECTIONS: Questions 35 – 52 are incomplete sentences. There are four alternative words/phrases, A-D, given below each question. Choose the one that best completes the sentence and blacken the letter of your choice in the space provided in the answer sheet.

35. Our rivers are polluted by the _____ materials from the factories.
A. waist B. waste C. wastage D. wasteful

36. Illegal immigrants are usually _____ to their countries.
A. deported B. deporting C. deport D. deportation

37. The _____ in the price of fuel is very welcome.
A. reduction B. reducing C. reduce D. reductive

38. The building is so complex. You can't easily find the entry or the _____.
A. outlet B. departure C. exit D. leaving

39. That he underwent treatment by a traditional doctor has _____ his situation.
A. dropped B. complicated C. explained D. strengthened

40. They found it quite difficult to _____ our terms during the negotiations.
A. expect B. accept C. except D. accord

41. They couldn't _____ with each other, so they had to go their own ways.
A. get up B. get along C. get away D. get round

42. The show was very _____ presented; I enjoyed it very much.
A. artful B. artistic C. artistically D. artistry

43. Don't be _____; stay calm.
A. agitated B. confused C. worried D. relaxed

44. I will not tell your secrets. I promise that my _____
A. mouth is shut C. lips are sealed
B. thoughts are hid D. ears are closed

45. I really trusted him before I found out he was a _____.
A. nail in the mud C. snake in the grass
B. sheep among the wolf D. monk in the monastery

46. So as not to go bankrupt in business, you have to keep your _____.
A. eye on the ball C. ear on the market
B. heart in the field D. mind on the goods

47. The prime minister has _____ his cabinet ministers.
A. elected B. voted C. appointed D. entitled

48. My car is _____. It runs twenty mile a liter.
A. economic B. economical C. with economy D. economically

49. The police used various means to _____ information from the criminal.
A. hide B. investigate C. extract D. explore

50. Read the instructions thoroughly before you activate or _____ your new camera.
A. stop B. turn off C. turn on D. purchase

51. We have to _____ blood in order to save lives.
A. provide B. supply C. donate D. offer

52. You really seem tired. Let me _____.
A. take up B. take off C. take over D. take away

B. Substitution (53 – 62)

DIRECTIONS: Questions 53-62 each has an underlined word or phrase. There are four alternatives A, B, C and D given after each sentence. Choose one word or phrase that best keeps the meaning of the given sentence if substituted for the underlined word or phrase and blacken the letter of your choice in the space provided in the answer sheet.

53. His far-sightedness meant a lot to the crowd.
A. ability to solve problems
B. ability to understand others
C. inability to see from short distance
D. understanding of possible consequences

54. I took your remarks in good part.
A. with no hesitation C. at my departure
B. without being offended D. accepted more than half

55. They ran through the poem.
A. remembered B. studied C. practiced D. skipped

56. Don't you think the present was really priceless?
A. invaluable B. unwanted C. worthless D. not valuable

57. They deliberated upon the issue before they made decisions.
A. carelessly handled C. hurriedly discussed
B. intentionally delayed D. carefully discussed

58. The commandos took a decisive measure to free the hostages.
A. quick B. tentative C. hesitant D. desperate

59. I knew he was being shadowed.
A. secretly followed C. kept in the dark
B. kept in the opposition D. intentionally ignored

60. He never stops talking about her immaculate behavior.
A. rude B. acceptable C. unspoilt D. disappointing

61. It came about in the middle of the night and we could do nothing.
A. bumped into B. happened C. sneaked through D. suddenly attacked

62. She now knows she can longer count on you.
A. notice C. depend
B. keep attracted to D. ask for money

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...7) ...7 are incomplete sentences. There are four ...given below each question. Choose the word that ... and blacken the letter of your choice in the space ...

...join us?
...n't
...No, I don't join you.
C. No, I wouldn't.
D. I wouldn't, please.

64. Leave _____ the printer and the copier out there. They are not functioning.
A. all B. both C. either D. neither

65. Haven't they arrived yet?
A. No, they have.
B. Yes, they haven't
C. No, they haven't.
D. Yes, they have arrived.

66. The sudden death of his boss came as _____ news.
A. surprising B. shocking C. touching D. frightening

67. Why is it that _____ came to your party?
A. anyone B. someone C. no one D. even one

68. Worry not about him! He has never been _____ to me.
A. kind B. happy C. wise D. surprising

69. That was a very stupid mistake; we must all say, _____ again.
A. ever B. never C. none D. no

70. Would you like to come home, darling?
A. No, I don't. B. No, I never do C. No, I never. D. No, I never would.

71. Neither of them _____ interesting to me. Take them away.
A. are B. seem C. is D. has

72. _____ could I read your name. Weren't you there?
A. Nowhere B. Somewhere C. Anywhere D. Everywhere

73. If I got the job, I _____ save some money.
A. will B. shall C. am going to D. would

74. Having been _____ the go all day, I got completely exhausted.
A. in B. through C. on D. upon

75. I have never been dependent on anyone _____ I was ten.
A. since B.until C. while D. when

76. Don't release the prisoner _____.
A. unless he repents
B. unless he has not repented
C. when he repented
D. when he has repent

77. When I was a child, I _____ sing much better.
A. can B. could C. will D. would

78. I found this year's Great Ethiopian Run _____ last year's.
A. as interested as
B. more interesting than
C. most interesting of
D. not interesting than

79. I know you don't want me to, but I _____ it anyway.
A. might have done
B. am going to do
C. could do
D. would have done

80. I have hardly a coin. Can you please _____ me some?
A. lend B.borrow C. take D. steal

81. He had to run for two hours so as _____.
A. he can save his life
B. to save his life
C. he could evade his enemies
D. he should overcome his fears

82. They _____ anything since yesterday.
A. didn't eat
B. had never eaten
C. have never eaten
D. will have never eaten

83. Come on! Let's go for *kitfo*. I _____ my pay.
A. had collected
B. have collected
C. am collecting
D. often collect

84. We have been expecting the book for months, but it has to come out _____.
A. yet B. by now C. still D. up to now

85. By the time he finishes his studies, he _____ spent eight years on it.
A. will have B. has had C. could have D. might have

86. _____ they submitted the proposal, we released the money.
A. Consequently B. In order that C. Because of D. As soon as

87. I _____ a college or a university by October 2009.
A. will join B. should join C. have joined D. will have joined

Section six: ***Communicative Activities (88-103)***

DIRECTIONS: Questions **88 – 98** are presented in a form of dialogue. The part said by one of the speakers is given, and a blank space is left for the other. For each of the blank spaces, four alternative answers, **A-D**, are given. Choose the alternative that is most appropriate to complete the dialogue and blacken the letter of your choice in the space provided in the answer sheet.

88. **Belay:** How can I finish all this in time?
Clerk: _____
A. Why would you work hard?
B. If I were you, I'd ask friends for help.
C. I know, you are so much hardworking.
D. What's the time?

89. **Son:** _____?
Dad: I'll have to talk to your mom.
A. Can I go to the toilet
B. Do chocolates give you illness
C. Dad, what comes after Monday
D. Would you buy me a mobile phone

90. **Bulti:** You shouldn't take any more money from her.
Belay: _____
A. Why shouldn't I?
B. She is not a good friend.

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D. I know. Wealth buys health

C. She did not sound mean.

91. **Director**: Your son stood first in all sections.
Parent: _____
A. How nice to hear that.
B. He wants to study medicine.
C. Your school is one of the top.
D. Wonderful! He has good classmates.

92. **Goitom**: This will get more expensive next week.
Lulu: _____
A. It doesn't look like new.
B. Business men are unkind.
C. Customers have more money.
D. Perhaps it might remain the same.

93. **Hailu**: Shall we have something to eat?
Sileshi: _____
A. Boy, I'm starving.
B. Restaurants are expensive.
C. A hamburger costs two birr.
D. Did you have a good breakfast?

94. **Murad**: Can you tell me the way to the library?
Chaltu: _____
A. It is a kilometer away.
B. It is a famous library.
C. First left, then right.
D. It won't open until 9a.m.

95. **Manager**: John, This is Mr Woyaya, one of my assistants.
John: _____
A. Hi, Mr Woyaya.
B. Hello, Mr Woyaya.
C. How are you, Mr Woyaya?
D. How are you doing, Mr Woyaya?

96. **Student**: _____
Teacher: You had to do it bit by bit.
A. I failed in Mathematics.
B. I get nervous on exam days.
C. I didn't submit my assignment.
D. I couldn't cover all my reading last night.

97. **Husband**: How do you like my new suit?
Wife: _____
A. It is black in color.
B. It is not very expensive.
C. Did you buy it from Merkato?

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D. All your other suits were of higher quality.

98. **Hagos**: Senait is a very intelligent girl.
Tadesse: _____
A. Her boyfriend is an engineer.
B. She lives next door to my brother.
C. Well, I'm not too sure about that.
D. I have also seen her at the supermarket.

Directions: Questions **99 – 103** are presented in the form of a dialogue. The part said by one of the speakers is given, and a blank space is left for the other. Each dialogue is followed by four alternative answers, **A-D**. Choose the answer that is **MOST APPROPRIATE** to complete the dialogue and blacken the letter of your choice in the space provided in the answer sheet.

Conversation between Friends

Hailu: Have you heard the news?
Ayele: No! What?
Hailu: (99) _____
Ayele: Oh, really? Who to?
Hailu: (100) _____
Ayele: Lucky, Asrat. When is the wedding?
Hailu: (101) _____
Ayele: but next week is too soon.
(102) _____
Hailu: At the Hilton, of course.
Ayele: Well, personally I prefer the Sheraton.
(103) _____
Hailu: Yes, you're right. That's their choice.

99. A. Tigist is getting married.
B. Tigist is going to America.
C. Tigist has scored all A's in her exams.
D. Tigist has won the lottery.

100. A. Next week
B. To Asrat, I heard
C. I don't know.
D. What a surprise

101. A. I don't think so.
B. In November.
C. He is thirty five.
D. Next week

102. A. Why don't they do it at the Sheraton?
B. And the wedding party?
C. Where did they meet?
D. Strange, isn't it?

103. A. But, that's their choice.
B. That's a good choice.
C. I wish them a happy marriage.
D. Sheraton is better than Hilton.

SECTION SEVEN: *Writing (104 – 120)*

Directions: The following questions are related to different aspects of writing. Read each of them very carefully and choose the best answer from the four alternative answers, A-D, given. Then blacken the letter of your choice in the space provided in the answer sheet.

104. Which one of the following is correctly spelt?
A. truly B. tention C. testible D. potable

105. Which one of the following sentences is correctly punctuated?
A. while here: Bob was clever.
B. Bob was clever, while here.
C. while here, Bob was clever.
D. Ask Bob how he became clever?

106. What would be an appropriate opening to an application letter for a job?
A. Sirs, B. Dear Sirs, C. Dears, D. My Dear Sir,

107. Which one of the following words is wrongly spelt?
A. sieve B. percieve C. receive D. deceive

108. Which one of the following is most likely a part of the instructions for operating your camera?
A. Always keep the batteries out when not in use.
B. Please, keep the batteries out when not in use.
C. Don't forget to keep the batteries out when not in use.
D. You will need to keep the batteries out when not in use.

109. Which one of the following sentences is correctly punctuated?
A. "When we go to Ambo, said Gari, we will see him."
B. When we go to Ambo" said Gari, "we will see him."
C. "When we go to Ambo," said Gari, "we will see him."
D. "When we go to Ambo," said Gari, "we will see him".

110. "Love is an emotion that binds two individuals in a mutual attention to each other. It manifests itself through devotion to one another and a feeling of loneliness when the two individuals are not together…". This piece of writing is likely to be _____.
A. narrative B. descriptive C. expository D. argumentative

111. Which one of the following is a connecting statement between paragraphs?
A. There have been many attempts to convince them.
B. The issue of weapons smuggling is a sensitive one.
C. Having looked at all these we now propose solutions.
D. The introduction of tax reform dates back to the early twentieth century.

112. "After a two-hour walk, we came to what is suspected to be a stronghold of the bandits. The place was deadly quiet and normal; life seems to have ceased functioning long ago." This is taken from a piece of writing that is most likely:
A. descriptive B. argumentative C. narrative D. expository

113. We kindly request you to leave your room clean. This instruction is most likely found in a _____.
A. laundry B. classroom C. restaurant D. hotel room

114. Which one of the following would you most likely use as an opening to a letter to your close friend, Abebe?
A. Hi, Abebe
B. Dear friend Abebe
C. Hello, Mr. Abebe
D. Dear Mr. Abebe

115. Which one of the following is more appropriate for an academic writing?
A. Earthquakes are sent down on us as punishments for our sins.
B. Earthquakes have killed more people than epidemics in history.
C. Earthquakes can be predicted in order to prevent them from happening.
D. Earthquakes have probably killed more people than epidemics in history.

116. How would you begin a letter to an applicant who has **not** been chosen for employment?
A. We regret to inform you .....
B. We are sad to inform you .....
C. We would like to inform you .....
D. We are pleased to inform you .....

117. "This has always been the weaker version. It is obviously cheaper but you cannot get the same effect on this machine." This is likely taken from a/an _____ piece of writing.
A. descriptive B. expository C. narrative D. argumentative

118. Which one of the following sentences is most likely taken from a descriptive piece of writing?
A. Computers are like TV sets.
B. Computers can store huge amount of information.
C. Computers are widely used in companies.
D. Computers can save information for later use.

119. In which of the following sentences is a hyphen more appropriate?
A. Most important to us was one thing an eraser.
B. We needed the following items Are they available?
C. A pen a pencil and an eraser we needed these items.
D. We needed a pen a pencil and an eraser.

120. Which one of the following sentences is most likely taken from a narrative piece of writing?
A. It was such a delicious meal.
B. Lunch was served at a stop-over
C. I wasn't sure where the lunch came from.
D. This was much better than the lunch we had yesterday.

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ETHIOPIAN HIGHER EDUCATION ENTRANCE CERTIFICATE
ENGLISH EXAMINATION
GINBOT 2002/ JUNE 2010

Booklet code: 11

Directions: The following words are **NOT** in the proper order. When put in the right order, they make correct English sentences. From the given alternatives, choose the one that is correct and **blacken** the letter of your choice in the space provided in the separate answer sheet.

1. Trees rains it when green turn,
   A. Green trees turn, when it rains.
   B. When rains turn, it green trees.
   C. When rains, trees turn it green.
   D. When it rains, trees turn green.
2. Get immediately you have him to go and
   A. Immediately go to have and you get him.
   B. You have to go immediately and get him.
   C. And you have to go get him immediately.
   D. You get him immediately and have to go.
3. Go to why college to need people do?
   A. Do people need why to go to college?
   B. To do why people need go to college?
   C. Why do people need to go to college?
   D. Do why people need to go to college?
4. All in attempt booklet the questions this
   A. Attempt all the questions in the booklet.
   B. In this booklet attempt all the questions.
   C. All this attempt the questions in booklet.
   D. The questions in this booklet all attempt.
5. Mamush child they adult call although he is the an him.
   A. He is an adult although they call him Mamush the child.
   B. Although they call him Mamush the child an adult he is
   C. He is the child an adult although they call him Mamush.
   D. He is Mamush the child although they call an adult him.
6. In your business own mind you that should only bear mind.
   A. Bear in your mind you should mind only that own business.
   B. That you should mind your own business bear in mind only.
   C. Only you should bear in mind that mind your own business.
   D. Bear in mind that you should mind your own business only.

Section two: Paragraph Coherence (7 – 14)

Directions: When put in the correct order, sentences a – e in each of the questions from 7 – 14 make up a complete paragraph. From the alternatives given in each case, choose the letter that contains the correct order of the sentences and **blacken** the corresponding letter provided in the separate answer sheet.

7. a. Peter is a painter.
   b. He went to Art School in 1989.
   c He works in a studio in Addis Ababa.
   d At the moment he is painting my portrait.
   e. His dad bought him for his graduation in 1993.
   A. a c e b d B. a c d e b C. a b c e d D. a d b c e
8. a. First, we saw monkeys.
   b. Then we saw an elephant.
   c. Yesterday we went to the zoo.
   d We saw some other animals, too.
   e. there were three monkeys in the cage
   A. b e a d c
   B. c d a e b
   C. c a e b d
   D. d b a e c
9. a. There are joints at these places.
   b. You need joints to walk and bend
   c. The human body has more than 70 joints.
   d. In many places two bones come together.
   e. Your knees are the joints in the middle of your legs.
   A. c d a b e
   B. d a e b e
   C. d a c e b
   D. b c d a e
10. a. This made the village dull and sleepy.
    b. The weather was hot and extremely scorching.
    c. There were also rumors that the enemy was approaching
    d. Unfriendly as the situation was, not a soul was to be seen anywhere.
    e. Suddenly a too loud and scary explosion broke the monotonous stillness.
    A. b a c d e
    B. b c a d e
    C. b d c a e
    D. b c e a a
11. a. It was, in fact, voraciously striving to make further gains.
    b. Italy was one of the late comers to the Scramble for Africa.
    c. However, at that time, Italy had a problem of sectarianism as well.
    d. Britain, for example, had had her big bite when Italy Joined in the venture.
    e. It was just newly created when others had already had the bulk of their share.
    A. b d c e a
    B. b a e d c
    C. d c a e b
    D. b e d a c
12. a. There are various reasons why gambling is so popular.
    b. The most obvious reason why people gamble is in order to make money.
    c. Even if you know you may not win, there is always a chance that you win
    d. Another reason people gamble is to bring some excitement into their lives.
    e. For may people winning from gambling is the only way they could ever make a lot of money.

    A. a b e d c
    B. c d e b a
    C. d e a c b
    D. a e b d c

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13. a. There is no escaping global warming.
b. Why worry about the changes that are happening today?
c. However, there is a small number of people who think that global warming is a myth.
d. You cannot pick up a newspaper or watch TV without hearing about this major issue.
e. They say that the earth's climate has been changing for as long as anyone can remember.

A. d a c b e
B. a c e d b
C. a d c e b
D. a d e c b

14. a. They can get jobs or provide their parents with free labor.
b. So the answer to the population problem lies in development,
c. This has already happened in many parts of the developed world,
d. If their prosperity can be increased, people begin to have fewer children.
e. The main reason that poor people have lots of children is that children represent money.

A. c e a d b
B. e a d b c
C. c d a b e
D. e a d c b

Section three: Reading Comprehension (15 – 28)

Directions: There are two reading passages in this section. Read each of them carefully and answer the questions that follow. Each question ahs four choices. Choose the best alternative and **blacken** the letter of your choice in the separate answer sheet provided.

Passage One

The most important thing is to understand people. You've got to know what they're thinking. If you can figure that out you can get them to do anything. They come in with an idea about what they want. You get talk about baseball or something like that. If it's a woman, you ask her about fashions. That way they get comfortable with you. You ask them a lot of questions and get them saying 'yes'. Then they just get into the habit of saying 'yes'. In the end, you can put them into anything you want, if you are really good. They need a little car for the city; you send them home with a truck. Of course, I wouldn't really do that. It wouldn't be right. You've got to sell on this job, but you also have to be fair. It's not fair to take advantage of people too much. There are some people in this business who would do anything. But I don't believe in that.

Source: More Reading Power

15. The word **that** in Line 2 refers to _____
A. Knowing what you are doing
B. Getting people to do anything
C. Understanding what people thing
D. Thinking about what people understand

16. The phrase **That way** in Line 6 refers to _____

A. Talking about fashion
B. Talking about baseball
C. Talking about what people like
D. Talking about what the writer likes

17. Which one of the following may **NOT** be needed to make people do what you want?
A. Identifying their interest.
B. Asking them few questions.
C. Understanding their though.
D. Allowing them to talk about themselves

18. You sell on **this job** (Line 11): What job does the speaker mean?
A. Selling cars
B. Designing fashion
C. Buying cars
D. Playing baseball.

Passage Two

1. Biologically, two different types of dreams exist. In the first, we sleep deeply and there is not much activity in the nervous system. Out blood pressure is often low and we don't usually dream. In the second type, our eyes start moving rapidly and our blood pressure rises. This kind of dream is known as REM sleep, (rapid – eye- movement sleep) and it is at this stage that we usually start to dream. Everyone dreams at least four or five times a night, but we don't always remember our dreams. Dreams occur at intervals of 90 minutes and altogether **make up** about 25% of our night's sleep. **Evidence** has shown that a dream period lasts for about 5 – 20 minutes. A new born child on the other hand may dream for up to 50% of his/her sleep. Some recent research found that women could remember more dreams than men and that older people seemed to dream least and have fewer nightmares. Many children thought that their dreams were affected by the books they had been reading.
2. Dream analysts believe that dreams often reflect what is going on in our lives at the time. Dreams express important wishes, fears, concerns and worries of the dreamer. Studies have shown that dreams are more perceptual than conceptual: things are seen and heard rather than subjected to though. In terms of the senses, visual experience is almost always present; auditory experience in 40 – 50%, and touch, taste, smell and pain a relatively smaller percentage. A lot of emotion is felt – usually an emotion such as fear, anger or joy. Most dreams are in the form of interrupted stories made up partly of memory with frequent changes of scene.
3. when we move away from the fact, the questions become more difficult to answer. One question which is often asked is: "Do we dream in color?" The answer to this question is that most dreams are probably in color although people may not be aware of it, either because they have difficulty remembering their dreams or because color is such a natural part of our visual experience. People who are very aware of color while awake, probably notice colors more often in their dreams. Another question which is asked is: "Do animals dream

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too?" They may well do, but we can never be sure. All that can be measured is they have REM (rapid eye movement) which tends to be associated with dreams. People often want to know whether dream which repeats itself over and over again, holds a special meaning. The general answer would appear to be no, and that repetition in dreams is nothing to be concerned about. However, it can be a sign that you have something on your lives, the study and analysis of dreams must be a useful procedure, revealing different aspects of a person's though. Interpreting or analyzing these symbols may possibly lead you to understand yourself.

4. Here are a few examples of what dreams can means. Dreaming about a car, for example, may indicate the way you travel through life. Is the car fast or out of control, or rusty or difficult to start? **Either way**, it might represent something about your life. Successfully flying a plane signified an unusual achievement, but flying in bad weather can mean difficulties in business. Animals are said to stand for aspects of our personality. If you dream about a fierce animal, this might be what you are thinking about yourself or somebody else. If you dream about a friendly animal, then it may point to the opposite – you may be felling confident about yourself or warm towards somebody else. Dreaming about teeth usually suggests power and control. Or of course you could just be having dental problems! Dreams about death imply an end of one thing in our lives and the beginning of another. They don't mean somebody will die. And finally, if you dream about water, it usually signified your emotions, and swimming symbolizes how you will deal with them. Sweet Dream!

Taken from: Improve Your English

19. Which word has a similar meaning as **make** up in Paragraph 1, Line 9?
A. Form B. Create C. Signify D. Constitute
20. Which of the following words can replace the word **evidence** in paragraph 1, Line 10?
A. Fact B. Data C. Research D. Information
21. Which one of the following does repetition in dreams signal?
A. The dreamer's success in life.
B. Things on the dreamer's mind.
C. The way the dreamer travels in life.
D. Difficulty in the dreamer's business.
22. Which of the following is true about people who are aware of colors?
A. They can dream more often than those who are not aware of colors.
B. They can dream in color more often than those who are not aware of colors.
C. They notice color more often in their dreams than those who are not ware of colors.
D. They can remember their dream more often than those who are not aware of colors.

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23. One main reason why dream analysis is said to be important in the passage in that_____.
A. There are good and bad dreams
B. Important dreams repeat themselves
C. Repeated dreams are often of concern to us
D. The result may help us understand ourselves better
24. What does **Either way** refer to in Paragraph 4, Line 4?
A. Traveling by all the types of car mentioned.
B. Traveling by fast cars that go out of control.
C. Traveling by any one of the car types mentioned.
D. Traveling by rusty cars that are usually difficult to start.
25. Which of the following suggests power in dream interpretation?
A. Dreaming about teeth.
B. Dreaming about swimming.
C. Dreaming about fierce animal.
D. Dreaming about dental problem.
26. Which one of the following shows, in dream interpretation, an end of one thing and the beginning of another thing in life?
A. Dreaming about teeth
B. Dreaming about death.
C. Dreaming about swimming.
D. Dreaming about flying a plane
27. Which one of the following is correct according to the information in the passage?
A. Research has shown that animals, like human beings, dream.
B. Research has shown that animal's REM cannot be measured.
C. Flying planes in bad weather would successfully signify achievement.
D. Research has shown that there is likelihood for people to dream in colors.
28. The following four points describe the four sections of the passage. Which alternative shows the correct order of occurrence of the points in the passage?
a. types of dream
b. interpretations of dream
c. dream analysts' belief
d. dream and color
A. c b d a
B. a d c b
C. a b d c
D. a c d b

Section Four: Vocabulary (29 – 54)

A. Completion (29 – 45)

Directions: Questions 29 – 45 are incomplete sentences. There are four alternative words, A –D, given below each question. Choose the word that best completes the sentence and **blacken** the letter of your choice in the space provided in the answer sheet.

29. All over the country, there is a sharp _____ in the cost of living
A. Rice B. Raise C. rise D. arise

30. Who doesn't fancy her? She is so ____ and beautiful
A. Fact B. Slim C. thin D. skinny

31. He is unfriendly; his behavior often _____ me.
A. Repels B. Pulls C. attracts D. contradicts

32. The injection was so _____ that it took him not time to regain consciousness.
A. Efficient B. Effectual C. affective D. effective

33. Look at my teeth! Aren't they as _____?
A. Strong as iron C. white as milk
B. White as show D. wide as a spade

34. Given the extent of the typhoid epidemic in the area, the small amount of money donated is just a drop in the _____.
A. Sea B. Sand C. ocean D. river

35. Take the same _____ as him, and you will end up in Nairobi.
A. Root B. route C. rout D. routine

36. I am _____ that she will arrive this afternoon. She has already left from there.
A. Convincing B. Confidence C. confident D. confidential

37. The film was _____; I did not expect it to be that bad.
A. Interesting C. exciting
B. Discouraging D. disappointing

38. University education requires quite a lot of reading in English. So, good English language ability is _____ for your success at the university.
A. Inaccessible B. Indispensable C. incredible D. inseparable

39. We need time to study for our tomorrow's exam. It is _____ to even think about going to bed at all.
A. Meticulous B. Ambiguous C. ridiculous D. courageous

40. There is a real threat of war everywhere in the world. Ii _____ we will ever see total world peace.
A. Doubt B. Suspect C. feel D. believe

41. She felt very _____ after the death of her husband. She doesn't even like to go out on weekends.
A. Disfavored B. Condemned C. impressed D. depressed

42. I find it difficult to _____ with my boss these days. He really is not approachable.
A. Get on B. Get of C. go on D. get across

43. Don't leave any of your belongings here; we have had a few _____ recently.
A. Theft B. Stealing C. losses D. thieves

44. Dressing smartly for an interview helps you to create a good _____.
A. Thought B. Impression C. response D. expression

45. He completely _____ after a day's hard labour.
A. Wore off C. wore out
B. Wore away D. wore down

B. Substitution (46 – 54)

Directions: Questions 46 – 54 each has an underlined word or phrase. There are four alternatives A – D given after each sentence. Choose one word or phrase that best keeps the meaning of the given sentence if substituted for the underlined word or phrase and **blacken** the letter of your choice in the space provided in the answer sheet.

46. A large number of people hold strong views about cost-sharing in education.
A. Keep B. Suggest C. have D. believe

47. You probably need to do through your plan again
A. Write B. Renew C. check D. implement

48. The Junior officers broke ranks with the leadership. They refused to accept orders.
A. Made a deal C. wanted negotiations
B. Shared authority D. expressed disagreement

49. I don't think this agreement would work. You've been taken for a ride.
A. Really cheated C. given little attention
B. Given only a ride D. considered only temporarily

50. Shouldn't we now give this idea up? To me, it is a lost case.
A. Failure C. cheap idea
B. Bad plan D. misconception

51. She looks off color today. What is the matter?
A. Very sad C. not very healthy
B. Very angry D. shabbily dressed

52. She was put out by the decision to fire her brother form the company.
A. Shattered B. Disturbed C. discouraged D. disappointed

53. Do you know how many were retained from the last year's batch of students?
A. Dismissed B. Accommodated C. promoted D. further kept

54. Would you please turn on the light? It is getting darker in here.
A. Open up B. Push down C. make on D. switch on

Section Five: Grammar (55 – 83)

Direction: Questions 55 – 83 are incomplete sentences. There are four alternative words or phrases, A –D, given below each question. Choose the word that best completes the sentence and **blacken** the letter of your choice in the space provided in the answer sheet.

55. _____ is the effect of large family more apparent than in rural areas of our country.
A. Anywhere B. Everywhere C. Nowhere D. Somewhere

56. The ring leaders could not be traced _____. Perhaps they have left the country.
A. Anywhere B. Somewhere C. nowhere D. everywhere

57. Despite the attractive reward pledged by the government, _____ came forward to give any information.
A. Anyone B. Someone C. no one D. everyone

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58. I have _____ done that kind of job before. Is that interesting?
A. Ever B. Never C. no time D. not never

59. _____ each of the candidate in this hall have a pen and a pencil?
A. Is B. Does C. Do D. Has

60. None of them _____ committed that kind of mistake.
A. Has B. Was C. have D. were

61. Either the customers or the waiter _____ to blame.
A. Is B. Are C. must D. have

62. Do you think this water is safe _____?
A. Drinking B. To drink it C. to drink D. for drinking it

63. She is one of the most _____ people I have ever met; she never talks.
A. Interested B. Interesting C. bored D. boring

64. _____ this exam isn't an easy thing; you really need to read each question carefully.
A. To pass B. To have to pass C. passing D. being pass

65. This question is _____ all the other questions in this booklet.
A. Easier than B.The easiest of C. easy as D. easier as

66. We have little milk left in the fridge. But there is even ____juice.
A. Less B. More C. much D. more little

67. Oh, on! It is already 2 o'clock but I _____ my lunch yet.
A. Do not have C. haven't had
B. Won't have had D. did not have

68. He has improved a lot ever since he _____ her.
A. Married B. Is married C. was married D. has married

69. The deadline is the 30th of June. By 01 July, all applications _ in.
A. Will be B.Will have been C. have been D. would been

70. Look! She is bleeding. I think, she _____ her fingers.
A. Cut B.Was cut C. had cut D. has cut

71. M wallet _____ out of my pocket before I sat down.
A. Has fallen B. Had fallen C. has been falling D. had been falling

72. I am wet; I _____ clothes since morning
A. Washed C. I am washing
B. Was washing D. have been washing

73. I now know you have no money. Don't worry! I ____ lend you some.
A. Will B. Would C. might D. am going to

74. We cannot release him _____ he completes the assignment.
A. If B. When C. unless D. by the time

75. I wouldn't have listened to him if I _____ he were to tell me this.
A. Knew C. had known
B. Did know D. have had known

76. If your brother is in difficult financial circumstances, you ____ help him.
A. Must B. Ought to C. had known D. have got to

77. You had better see a doctor about your chest pain; I think it is getting _____.
A. More worse C. worse and worse
B. Worse and worst D. more and more worse

78. If _____, you wouldn't go swimming in that freezing water.
A. Want not to catch a bad cold
B. Wanted not to catch a bad cold
C. Would want not to catch a bad cold.
D. Would not want to catch a bad cold.

79. You _____ not tell lies if you wanted people to trust you.
A. Have B. ought C. must D. will have

80. You should leave home early _____ to be late for tomorrow's exam.
A. In case not B. So that not C. if not D. in order not

81. My watch _____ be wrong as I set it by the radio clock this morning.
A. Cannot B. May not C. will not D. need not

82. It was _____ boring lecture that every one fell asleep.
A. so B. too C. such D. such as

83. _____ you know what we are getting at, I should tell you everything.
A. Since B. While C. Because of D. As a result of

Section Six: Communicative Activities (84 – 104)

Directions: Questions 84 – 104 are presented in a form of dialogue. The part said by one of the speakers is given, and a blank space is left for other. For each of the blank spaces, four alternative answers, A – D, are given. Choose the alternative that is most appropriate to complete the dialogue and **blacken** the letter of your choice in the space provided in the answer sheet.

84. **Girma:** Would you like some more sandwiches?
Getnet: _____
A. No, thanks. I've had enough.
B. Never at all. I hate sandwiches.
C. No, thanks. I just had my breakfast.
D. No, thanks. Sandwiches are not good.

85. **A:** Isn't boxing a popular sport in this country?
B: _____
A. No, it is! C. Not, never!
B. Yes, it isn't! D. No, it isn't!

86. **Debebe:** May I introduce myself? My name is Debebe.
Kebede: _____
A. Pleased to meet you. My name is Kebede.
B. Allow me to introduce myself. I'm Kebede.
C. I'd like to introduce myself. My name is Kebede.
D. May I introduce myself, too. My name is Kebede

87. **Tourist:** Excuse me, please. How can I get to the museum?
You: _____

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A. The museum is closed today.
B. Do you want to walk or take a taxi?
C. What time do you want to get there?
D. The museum is somewhere around Arat Kilo.

88. **Josie**: _____
Secretary: I am afraid Mr. Abera is not in
A. I want Mr. Abera, please.
B. How are you, Mr. Abera?
C. Is it Mr. Abera's office, please?
D. Can I speak to Mr. Abera, Please?

89. **Teacher**: Which stream would you like to join, Arts or Science?
Student: _____
A. I would rather go for Science.
B. Many students would want Arts.
C. How many teachers teach Art?
D. Excuse me, I would ask my dad.

90. **Selam**: _____?
Hawi: No problem. How much would you like?
A. Your brother told me you were so rich.
B. How annoying! I think I lost my money.
C. Do you have some money in your account?
D. My father is not willing to pay my tuition fee.

91. **Wife**: I am no feeling well these days.
Husband: _____
A. Hospitals are so busy.
B. May be, you'll need a rest.
C. Doctors are not doing their job.
D. What about the new clinic down here?

92. **Peter**: She won a million birr lottery?
Jesie: _____
A. Lotteries are good
B. Does she have a husband?
C. Did she buy the ticket herself?
D. What? That is a big amount of money!

93. **Peter**: Did you know that they sacked your manager?
Player: _____
A. Does he want to be employed?
B. This is ridiculous! Who told you?
C. The national club also did the same?
D. He had a row with the board members?

94. **Tolessa**: The party was great, wasn't it?
wife: _____
A. You left early, didn't you?

B. There are lost of drinks at parties.
C. I didn't know they were getting married.
D. Oh, yeah! The food was particularly delicious.

95. **Customer**: They say your boss is thoroughly rich.
Waiter: _____
A. He has an expensive car.
B. He doesn't live in Ethiopia.
C. People talk about his money.
D. Of course, can't you see all these/

96. **Gemechu**: _____?
Daniel: well, he's short and quite fat, but he has a pleasant face.
A. How is your new English teacher
B. How tall is your new English teacher
C. What does your new English teacher look like
D. What's the face of your new English teacher like

97. **Colleague**: They are closing the unit down
You: _____
A. This is a unit that handles all research work.
B. That is very stupid! It has been efficient all along.
C. Are you kidding? The head has got another appointment.
D. Of course! Don't you understand that most of the staff are competent?

98. **Friend**: _____?
You: I go to the gym two or three times a week.
A. Do you like to play games
B. What is your favorite sport
C. Do you want to lose weight
D. Do you ever do any physical exercise

99. **Girma**: Are you still living in Jimma?
Ujulu: _____
A. This is my 5th year in Jimma
B. To be honest, Jimma is a nice place for living
C. No, I moved to Addis Ababa about six months ago.
D. No, I had been living in Awassa before I went to Jimma.

100. **Girma**: Shall we organize a party to celebrate our tenth wedding anniversary this weekend?
Wife: _____
A. May people enjoy parties.
B. It doesn't sound very practical to me
C. I'm going to Seble's birthday party this evening.
D. Do you remember the party we went to last time?

101. **Almaz**: our French teacher arrived late again this morning
Hawa: _____
A. I'm surprised – he's often late.

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B. I'm not surprised – he's o0ften late.
C. I'm not surprised – he's often on time.
D. I'm surprised – he's often not on time.

102. **Hagos**: How long have you been doing this job?
Kedir: _____
A. It is about 10 years now.
B. I'm usually at home by 2 or 3.
C. I do all the marketing myself.
D. The salary is not good enough

103. **Journalist**: Do you think parents are happy about the amount f their children's TV watching?
Parent: _____
A. No. I don't think so.
B. Yes. Children watch TV too much.
C. Yes. My children watch TV too much
D. No. TV watching does not make children creative.

104. **Father**: what are you going to do with that spade?
Son: _____
A. I am going to have some apple.
B. I want to cut down some apple trees.
C. I am going to plant some apple trees.
D. I want to split the apple tree into two parts.

Section Seven: Writing (105 – 120)
Directions: The following questions are related to different types of writing. Read each of them very carefully and choose the best answer from the four alternative answers, A – D, given. The **blacken** the letter of your choice in the space provided in the answer sheet.

105. Which one of the following words is **NOT** correctly spelt?
A. Cafteria B. Academic C. Extension D. Environment

106. Which one of the following words is wrongly spelt?
A. Believe B. Cumulative C. Deceive D. Accommodate

107. "The building is a small house consisting of two floors. The outside front wall is 9m long and 5m high. There is a door in the middle of the wall at ground level." This piece of writing is likely to be _____
A. Narrative B.Descriptive C. expository D. argumentative

108. "Below are my reasons to claim that the seminar is a better alternative to the lecture method of teaching in higher education." This is likely taken from a/an _____ piece of writing.
A. Narrative B. Descriptive C. expository D. argumentative

109. "We arranged to meet at the foot of the mountain. But we had to wait longer than expected as many did not show up on time. Then at 10 o'clock, after a significant delay, we headed for the tip". This is taken from a piece of writing that is most likely ____.

A. narrative B.expository C. descriptive D. argumentative

110. "The first time I experienced a problem with age was on my fiftieth birthday. Fifty seemed to be too much older than 49. I used to like it when people told me I looked young for my age, but then I realized that they were just telling me how old I was." This piece of writing is likely to be _____.
A. expository B. argumentative C. narrative D. descriptive

111. Which one of the following best illustrates an argumentative writing?
A. Typewriters are not as good as computers.
B. Some of the computers have really huge memory.
C. We have 32 computers distributed among the staff.
D. Computers have the ability to correct wrong spellings.

112. "This brief survey should give you some idea of the great variety of things to be done in Addis Ababa. There is literally something for everyone, at almost any time during the day." This piece of writing is likely taken from the _____ of a written text.
A. Conclusion B. Introduction C. main body D. supporting details

113. "I am writing the letter in response to your advertisement of 12 September 2009 regarding . . ." Which part of a formal letter does the line go in?
A. Salutation B. Introduction C. body D. closing

114. "Dear everyone, we're having a fantastic time here!" this is likely taken from the beginning of a _____.
A. Letter of acceptance C. business letter
B. Letter to close friends D. formal letter

115. "I regret to inform you about the late delivery of your order." This is most likely taken from a letter of _____.
A. apology B. complaint C. enquiry D. rejection

116. The following items (a–e) are components of an application letter arranged in a wrong order. Which alternative shows the correct order?
a) New Thoughts University
b) P.O. Box
c) Ato Bekel Leta
d) Head, Purchasing Department
e) Addis Ababa
A. a c d e b B.c d a b e C) a c b e d D) c d a e b

117. "Please, take a seat in the hall until a nurse calls your name." This is most likely a notice in a _____.
A. nursery B. hospital C. nursing college D. medical school

118. Which one of the following sentences is correctly punctuated?
A. He saw or at least he though, a big snake lying in his bed.
B. He saw or, at least he though, a big snake lying in his bed.
C. He saw or, at least, he thought a big snake lying in his bed.
D. He saw, or at least he though, a big snake lying in his bed.

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119. Which one of the following sentences has a correct punctuation?
A. "But, look here, said Berhan, I can't give you this one"!
B. "But look here," said Berhan, "I can't give you this one!"
C. "But look here," said Berhan, "I can't give you this one"!
D. "But, look here," said Berhan, "I can't give you this one!"

120. Which one of the following sentences is correctly punctuated?
A. If he doesn't obey, then follows punitive measure; he is dismissed for good.
B. If he doesn't obey, then follows a punitive measure: he is dismissed for good.
C. If he doesn't obey, then follows a punitive measure, he is dismissed for good.
D. If he doesn't obey, then follows: a punitive measure – he is dismissed for good .

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DIRECTIONS : The following words are NOT in the proper order. When put in the right order, they make correct English sentences. From the given alternatives, choose the one that is correct and blacken the letter of your choice in the separate answer sheet.

1. And go there is get right.
A. Get it and right go there.
B. Go and get it right there.
C. Go there right and get it.
D. Get there and go it right

2. Clean and fresh breathe must we air.
A. We breathe fresh air and must clean.
B. We clean fresh air and must breathe.
C. We must breathe fresh and clean air.
D. We clean and must breathe fresh air.

3. Can their do me you what looking by uniforms tell as just people?
A. Can you tell me what people do just by looking at their uniforms?
B. What can you tell me people just do by looking at their uniforms?
C. Can you tell me what people just do by looking at their uniforms?
D. What can you tell me just people do by looking at their uniforms?

4. You next do in who class to sit?
A. Who do you sit next to in class?
B. Next who do you sit in to class?
C. To who next do you sit in class?
D. Who next to do you sit in class?

5. It I want heard you.
A. I heard it you want.
B. I want it you heard.
C. I heard you want it.
D. I want you it heard.

Section TWO: Paragraph Coherence (6 – 12)

DIRECTIONS : When put in the correct order, sentences a-e in each of the question from 6- 12 make up a complete paragraph. From the alternatives given in each case, choose the letter that contains the correct order of the sentences and blacken the corresponding letter provided in the separate answer sheet.

6. a. Heat it on a gas stove.
b. Take some water in a kettle.
c. Steam comes out from the kettle.
d. What do you find after some time?
e. When water is heated, it turns into steam
A. d b a c e
B. b d e c a
C. d e b a c
D. e b a d c

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7. a. Noting they can reach is safe.
b. Puppies must be watched constantly.
c. This is because their curiosity is endless.
d. So, if you cannot see or hear them, you had better question.
e they can happily destroy a shoe or a pillow in just a few miuntes.
A. b e d a c
B. b c a e d
C. b c e d a
D. b d c a e

8. a. Like ants they have two body sections.
b. Many people call termites "White ants".
c. Unlike termites, most ants are shiny and dark in color.
d. Termites also differ from ants in matters of colony life.
e. However, their thorax is not clearly separated from the abdomen.
A. b a d c e
B. b d a c e
C. b a e c d
D. b e a d c

9. a. One branch is chemistry.
b. The other branch is physics.
c. physical science has two branches.
d. physics includes the study of energy.
e. Chemistry is concerned with matter in its different forms.
A. e c b d b
B. c b a d e
C. b c d e b
D. c a e b d

10. a That is why we take care of them.
b. Some animals are very useful to us.
c. Domestic animals serve us in many ways.
d. Such animals are called domestic animals.
e. For example, we give them food and shelter.
A. d c e d a B. d c a b e C. b d c a e D. c d a b e

11. a. They use their fellers to find food.
b. Most insects have wings and can fly.
c. Insects are small animals with six legs.
d. Almost all insects have a pair of feelers.
e. Bees and ants are examples of common insects.
A. e c b a d B. c b d a e C. c b a d e D. b a e d c

12. a. Some animals live in the jungle.
b. Such animals are called wild animals.
c. We can see some wild animals in a zoo.
d. They wander in the jungle in search of food.
e. The can also watch wild animals on different TV channels.
A. a b d c e
B. c a e b d
C. d c e d a
D. e d c a b

Section there Reading Comprehension (13 -26)

DIRECTIONS : There are two reading passages in this section. Read each of them carefully and answer the questions that follow. Each question has four choices. Choose the best alternative and blacken the letter of your choice in the separate answer sheet provided

Passage One

1. Every 12 seconds someone is killed by malaria. More often than not the wasted life is that of a child. And the situation is getting worse. Not only is the parasite that causes malarial becoming resistant to treatment. But the mosquitoes that carry the disease are becoming immune to the pesticides that used to kill them. The problem is particularly bad in Africa.
2. Yet, the amount of money spent on research on malaria is tiny compared to other less threatening diseases. According to Tore Godal, Director of the World Health Organization's Tropical Diseases Research Programme, malaria causes " 23% of the world's total burden of disease, but only 0.1 % of the money spent on medical research world wide is spent on malaria"
3. The figures quoted came from a letter published in the world's leading 'science journal, Nature. In it. Scientists, economists and politicians made a powerful plea for international action to fight malaria. They described it as "a disaster looming over Africa", and demanded that the disease be recognized as a major problem, on the scale of the AIDS pandemic.
4. The main reason **this** has not happened is that malaria mainly affects poor countries, which don not have the money to pay for expensive research programmes. Nor are they able to provide profitable markets for new treatments or vaccines. As a result, pharmaceutical companies are not prepared to make long-term investment in research and development.
5. But recently, scientists have become more optimistic that we may develop an effective vaccine. Advances in our understanding of disease- through genetics- have opened up plenty of new possibilities for the treatment and prevention of malaria. Moreover, scientists are confident that genetics offers the best hope of finding a vaccine for the disease.

13. Which of the following is true about malaria treatment?
A. Discovery of vaccines against malaria is a possibility.
B. Scientists are almost at the final stage of developing vaccines.
C. Scientists have recently discovered the best malaria treatment.
D. Genetics has led to an effective treatment and prevention of malaria.

14. Which one of the following best describes the main idea of the passage?
A. Malaria is a life- threatening disease.
B. Malaria has not been given the attention it deserves.
C. Malaria is a disease that affects less developed countries.

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D There are hopes that vaccines against malaria could be found.

15. Which one of the following sentences is true about malarial?
A. Malaria is a diseases that affects Africans.
B. Pesticides are effective in killing mosquitoes.
C. Malaria affects children more than it does others.
D. Malaria causing parasites are responding to treatment.

16. Tore Godal, in paragraph 2, emphasizes that malaria.
A. other diseases are less threatening than malaria.
B. malaria research receives only 0.1% of world money.
C. malaria receives much less money than other diseases.
D. malaria causes nearly a quarter of the world health Problems.

17. The word **this** in paragraph 4, line 1 refers to the fact that
A. malaria mainly affects poor countries.
B. malaria was as deadly a disease as AIDS.
C. malaria failed to win international attention.
D it has not been possible to eradicate malaria.

18. Which of the following is **NOT** explicitly mentioned in the science journal, Nature?
A. Malaria is a disaster looming over Africa.
B. Malaria causes 23% of the world's to total illness.
C. Malaria needs to be taken as seriously as the AIDS pandemic.
D. Scientists, economists and politicians made a powerful plea..

19. Pharmaceutical companies are **NOT** interested in malaria research mainly because:
A. malaria is a disease that affects poor countries.
B. markets for new malaria treatments are not profitable.
C. poor countries may not have the money for expensive research.
D. the companies are not ready for long – term investments in research.

PASSAGE TWO

1. Every society must meet certain basic social needs if it is to survive and provide a satisfying life for its members. For example, children must be raised and cared for; the cultural knowledge of one generation must be passed on to the next; important social values must be shared and upheld; social order must be maintained; goods and services must be produced. Over time, the members of each society create patterns of thought and action that provide and appropriate solution for these recurrent challenges. This pattern of behavior is what sociologists call institutions.
2. An institution is a stable cluster of values, norms, statuses, roles, and groups that develops around a basic social need. Thus, the family institution provides for the care of children. The educational institution transmits cultural knowledge to the young. The religious institution provides a set of shared values and the rituals that reaffirm them. The political institution allocates power and maintains order. The economic institution provides goods and services.
3. One important character of institutions is that they are inherently conservative. Patterns of social behavior become institutionalized, or securely established only when they have been reinforced by custom and tradition to the point where they are accepted almost without question. People tend to resent and resist any attack on the institutions they know. (Imagine, for example, the likely response in North America to a serious attempt to abolish the family, to end compulsory schooling, to replace existing churches with new cults, or to apply communist principles to the economy.)
4. A second character of institutions is that they tend to be closely linked within the social structure. The various aspects of culture tend on the whole to fit together: if they don not, cultural strain and even cultural disintegration result. In the same way, the components of social structure must be fairly well integrated if too much structural strain is to be avoided and the threat of social disintegration averted. For this reason, a society's major institutions tend to uphold similar values and norms, to reflect compatible goals and priorities and to benefit or penalize the same groups and interest. Because all the institutions share common features, any one of them can yield insights into the others and can serve to some extent as a microcosm of the larger society.
5. A third character of institutions is that when they do change, they rarely do so in isolation: significant modifications in any major institution are likely to be accompanied or followed by changes in others. This is, especially, true of changes in the economy for the way people make their living has multiple effects on almost everything else they do. Thus, if a society were to change from a basic subsistence strategy of rural farming to one of urban manufacturing, many other institutional adjustments would be needed to maintain social integration.

20. In paragraph 5, Which institution does the writer use to illustrate his/ her argument?
A. Family B. Political C. Economic D. Educational

21. Which one on the following is **relatively easier** to implement in Ethiopia as implied in paragaph3?
A. promoting inter-ethnic marriage.
B. Making the nomads sedentary farmers.
C. Imposing restrictions against eating raw meat.
D. Limiting the number of children a family can have.

22. If institutions are **NOT** closely linked within the social structure (paragraph5)
A. Culture will remain intact.
B. Cultural strain will take place.

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C. Structural strain will be avoided.
D. Social disintegration will be averted.

23. Which institution is responsible for the safety of our children in school as implied in paragraph 2?
A. Family B. School C. Church D. Kebele

24. Which of the following is true according to paragraph5
A. Institutions are independent of each other.
B. A change in one institution entails a change in another.
C. Our earnings are hardly linked with other things we do in life.
D. Transformation of agriculture into industry may not need adjustments.

25. Which one of the following is **NOT** implied in paragraph 1 as a social need?
A. Parents sending children to school.
B. The young getting married and having children.
C. Elders teaching the young how to defend country.
D. Travelers introducing new culture to other people.

26. This pattern of behavior- (paragraph 1, lines 8-9) means:
A. Encountering recurrent social challenges.
B. Performing those activities in the paragraph.
C. Creating certain patterns of thought and action.
D. Providing and appropriate solution to problems.

SECTION FOUR : Vocabulary (27- 56)

A. Completion(27-46)

Directions: Questions 27-46 are incomplete sentences. There are four alternative words/ phrases, A – D, given below each question. Choose the world that best completes the sentence and **blacken** the letter of your choice in the separate answer sheet.

27. How much would you to ---------- to the children's fund?
A. Donate B. nominate C. denote D. detect

28. It gives me pleasure to ----------you with this clock from us all.
A. Present B. Represent C. offer D. preside

29. This chair has been----------in the family; it was my great grandmother's originally.
A. Handed over B. Given away C. given out D. handed down

30. Last night an armed gang ----------the bank.
A. Stole B. Cheated C. robbed D. snatched

31. How do you -------- a person who boasts a lot?
A. Describe B. Prescribe C. proscribe D. inscribe

32. He eats like a ---------- and drinks like a fish.
A. Lion B. Hyena C. horse D. donkey

33. For no reason, they are intending to ---------- the new building.
A. Destruct B. Demolish C. damage D. demobilize

34. AIDS --------- more rapidly during the 1980s than it did in the latter decades.
A. Spread B. Expanded C. extended D. contracted

35. Oh dear ! I've---------- the milk on the floor.
A. Spoilt B. Soured C. spilt D. splash

36. Sport is ---------- in many schools, but it is compulsory here.
A. Operational B. Professional C. optional D. practical

37. I felt ---------- until I heard the result of my girlfriend's medical test.
A. excited B. enthusiastic C. anxious D. inspired

38. I've been up to my ----------in work all week!
A. Head B. Shoulders C. eyes D. mouth

39. Did you sleep well last night? Yes, thanks. I slept like a -----.
A. Log B. Lied C. cat D. dog

40. He ---------- in a road nearby to escape the shooting spree.
A. Laid B. lied C. lay D. lain

41. He is thick ---------- you can't easily offend him.
A. Skinned B. Headed C. minded D. hearted

42. He is a man who can easily---------- his words. I have seen him betraying many people.
A. take on C. go back on
B. run agains D. go off with

43. You know that I am the ---------- You should take orders.
A. boss B. responsible C. superior D. accountable

44. I felt ---------- that she would pass the exam.
A. Confident C. confidential
B. Confidence D. confidentiality

45. They want to make the college a centre of ----------. The books and other materials they have are exemplary.
A. excelling B. Excellency C. excellent D. excellence

46. If you promise to ---------- this secret, I'll tell you the whole story.
A. Hold B. Retain C. hide D. keep

B . Substitution (47 -56)

DIRECTION : Questions 47- 56 each has an underlined word or phrase. There are four alternatives, A –D, given after each sentence. Choose one word or phrase that keeps the meaning of the given sentence if substituted for the underlined word or phrase and **blacken** the letter of your choice in the separate answer sheet.

47. You were so <u>underdressed</u> for my wedding. Didn't any member of the family make a comment?
A. Sexily dressed C. Cheaply dressed
B. Barely dressed D. Informally dressed

48. If you have something <u>invaluable</u>, let us know. We will come over and see it.
A. precious B. worthless C. cheap D. expensive

49. I wish I could <u>take back</u> what is said to her.
A. recall B. return C. repeat D. withdraw

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50. Why do you always like to pick holes in all that I say?
A. Praise B. criticize C. scrutinize D. supplement

51. His words took my breath away.
A. surprised me C. inspired
B. impressed me D. annoyed me

52. Keep them at arm's length. Things may turn out to be different tomorrow.
A. Get them very close to you.
B. Share part of you secrets with them.
C. Don't get them too friendly with you.
D. Don't allow them to sit too close to you.

53. I was at death's door last week.
A. very ill C. at funeral
B. seriously injured D. mourning

54. To us, the most heartening thing is our customers' satisfaction. Forget the other things.
A. important C. encouraging
B. saddening D. disappointing

55. Switch off the light, please. We are going to bed.
A. Turn off B. Push down C. Disable D. Make off

56. Put in a completely different situation, the child was required to unlearn everything.
A. stop learning C. become illiterate
B. forget old ways D. stop going to school

SECTION FIVE : Grammar (57 – 58)

DIRECTIONS: Questions 57 -85 are incomplete sentences. There are four alternative words or phrases, A – D, given below each question. Choose the word or phrase that **best** completes the sentence and **blacken** the letter of your choice in the separate answer sheet

57. I have -------- received the news that he became the champion.
A. yet B. just C. ever D. since

58. Unless you see the criminal, ---------- !
A. give shout C. don't shout
B. will not shout D. don't have shout

59. I am pleased ---------- that you all have passed this exam.
A. say B.. saying C. to say D. for saying

60. I found this question a bit---------- the rest of the questions.
A. easier than C. easiest of
B. the easiest of D. so easy as

61. I would like to have _____ reliable car. The one I have got keeps on breaking down.
A. more B. the most C. a more D. the more

62. ---------- the electricity bill yet?
A. Have you pain B. Are you paying D. Do you pay

63. If cars ---------- with a restricted speed limit, there would be fewer accidents on the road.
A. produced B. are produced C. produce D. were produced

64. There is nothing ---------- watching a good football match on TV.
A. interesting like C. such as interesting
B. as interesting like D. like interesting as

65. Look at those block cloud! I think, it ----------
A. may rain B. will rain C. is raining D. is going to rain

66. You are mistaken if you think I am being ---------- to you. All I am telling you is that this is a requirement to get service.
A. grateful B. unkind C. forward D. intrusive

67. Isn't it ----------? She would find herself in Japan in two weeks.
A. exciting B. appealing C. interesting D. entertaining

68. What a ---------- day! we hardly had anything to do.
A. tiring B. relaxing C. boring D. frightening

69. This was much ---------- last year. I paid only birr 100 to get it.
A. less expensive C. expensive
B. more expensive D. most expensive

70. The family---------- that car since 1998.
A. had B. had had C. has D. has had

71. I know he---------- to South Africa a couple of times, but not to Kenya.
A. was B. been C. has been D. had been

72. I think it is going to rain. What ---------- we do?
A. shall B. might C. ought D. could

73. I wrote him a month ago, but he ---------- yet.
A. did not reply C. does not reply
B. is not replying D. has not replied

74. Tom Spends--------- of his time playing computer games and listening to music. That is why he scores lwo grades.
A. much B. Some C. little D. enough

75. I looked for it all over the place and it is ---------- . Who could have taken it?
A. nowhere B. somewhere C. any where D. every were

76. Each of us---------- expected to submit a two- page summary by tomorrow morning.
A. is B. are C. will have been D. would have been

77. A: Iam more interested in going to the theater than to the cinema
B: ----------
A. So am I B. So do I C. Neither am I D. Either am I

78. If you are in a hurry, you will have to take a plane.----------

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the train------ the car will arrive on time in the city.
A. Both/ and
B. Not only/ but also
C. Either/ or
D. Neither/ nor

79. What do you see yourself---------- once you have finished your studies?
A. do B. doing C. to do D. will do

80. During weekends, I enjoy ---------- wake up early.
A. not having to
B. not to have to
C. not to
D. not having

81. He didn't report on time; ---------- we penalized him.
A. so B. so that C. because of D. not having

82. ---------- we paid the money on time, they didn't let us go
A. But B. Although C. Even so D. Even then

83. If you wait for me, I ----------
A. may come B. will come C. had come D. would come

84. All the schedules seem to be poorly made, and this one is even -------
A. bad B. very bad C. worse D. worst

85. Neither he nor his father --------- held responsible for what had happened. They were both set free.
A. is B. was C. are D. were

SECTION SIX : Communicative Exchanges (86 – 105)

DIRECTIONS : Questions 86 – 105 are presented in a form of dialogue. The part said by one of the speakers is given, and a blank space is left for the other. For each of the blank spaces, four alternative answers, A-D, are given. Choose the alternative that is most appropriate to complete the dialogue and **blacken** the letter of your choice in the separate answer sheet.

86. Father: What's your new laptop like?
Son: It's the same as my old one,----------
A. Only a lot faster.
B. Only a lot more difficult.
C. Only a bit more friendly
D. Only it's easier to get along with

87. Daughter: I'll ring you when I reach Harar.
Mother: Good ----------
A. When will you return?
B. Have you ever been to Harar?
C. What time will you get there?
D. Harar is an ancient city, isn't it?

88. You : I haven't eaten anything since breakfast.
Colleague:----------
A. You must be very hungry
B. I had bread and tea for breakfast
C. What would you like for break fast?
D. Do you want to join us for coffee?

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89. Shop Assistant: Can I help you?
Customer : ----------
A. Why not?
B. Yes, of course.
C. Do you have a shirt?
D. I'm looking for a shirt

90. Mother: Have we get any more sugar in the kitchen?
Daughter: ----------
A. No, there is none
B. No, there isn't much sugar
C. No, there is little sugar
D. NO, there is nothing

91. Mother : Why is your hair so wet?
Son : ----------
A. I've swum
B. I went to the barber's
C. I've been swimming
D. I'll have it cut this evening

92. Kedir: We've got a new teacher!
Lidiya: ----------
A. What's he like?
B. I think it's time for class
C. When will he give us a test?
D. Our English teacher is coming

93. Your friend: ----------
You: No, I'm single.
A. Are you married?
B. Do you have children?
C. Do you have a girlfriend?
D. Do you want to get married?

94. You: ----------
Colleague: That's a good idea
A. Do you have an appealing idea to go out for dinner
B. How does the idea of going out for dinner appeal to you?
C. Why does the idea of going out for dinner appeal to you?
D. When does the idea of going out for dinner appeal to you?

95. Teacher: These students do not work as hard as the students in the past.
Student : ----------
A. I'm not really sure you're right there.
B. I think many students are not doing well these days.
C. Students are well aware of the importance of hard work.
D. You're right. There are many students in the football field.

96. Patient : My headache is persisting. ----------
Doctor: you better get your blood pressure checked.
A. Can I talk to you?
B. What can I do about it?
C. Is there any good treatment?
D. Do many people complain about this?

97. Peter: This is the cheapest car in town. He shouldn't have gone for it.
John: ----------
A. I can't agree more
B. That car is not new
C. Is he a businessman?
D. Why is that so cheap

98. Brother : Can we have some ice cream in that café?
Sister: ----------

A. Who owns the café?
B. Do they make it there?
C. I'm afraid I have no money.
D. Girls like ice cream very much.

99. Son: ----------?
Dad: I am afraid I need it.
A. Would you borrow my dictionary?
B. Could you please buy me a dictionary?
C. Could you lend me your dictionary, please?
D. How about giving you back your new dictionary?

100. Belile: Mobile phones take too much of my time.
Soma: I know----------
A. They are more handy.
B. I always see you busy with it
C. You've got a beautiful apparatus.
D. The cards are very expensive here.

101. Student: She's going to flunk, I feel. Can we help?
Friend : ----------
A. You know, she is very lazy.
B. I think she has now started working hard.
C. I don't think there is anything to be done
D. I can't understand how she passed her last course.

102. Gutu : she scored an A in mathematics
Didi: ----------
A. She was better at English.
B. I can't believe this is true!
C. Mathematics is an easy course.
D. Which department did she join?

103. Waiter: Coffee or tea, sir?
Customer: ----------
A. How much is coffee?
B. Many people like coffee.
C. I would rather go for coffee.
D. Coffee is a stronger stimulant

104. Lema: ---------
Mulu: 5, 000 Birr.
A. What do you need the money for ?
B. How much do you want to borrow?
C. Do you really need to borrow that much?
D. I wonder if you could lend me some money.

105. Degu: Look at her new hair style. Isn't that fashionable?
Sam :----------
A. She's got curly hair.
C. That's really disgusting!

B. Oh, where's she from?
D. She never bothers about fashion

SECTION SECEN : Writing (106- 120)

DIRECTIONS: The following question are related to different written works. Read each of them very carefully and choose the best answer from the four alternative answers, A – D, given. Then, **blacken** the letter of your choice in the separate answer sheet.

106. " It was great to hear from you" This is likely taken from the beginning of a ---
A. formal letter B. business letter C. letter to a boss D. letter to a friend

107. Which one of the following best illustrates a sentence taken from a descriptive piece of writing?
A. The girl standing by him is confronting the policeman.
B. The girl standing by him is beautiful and wore blue jeans.
C. The girl sanding by him should have really won the contest.
D. The girl standing by him has a father who works in a fish shop.

108. " We had already cruised for one and a half hours when we heard a loud bang. The driver instinctively swerved the vehicle and we came to a safe stop," This is most likely part of a writing that is -------- in nature.
A. narrative B. argumentative C. descriptive D. expository

109. Which one of the following is most likely an instruction in a leaflet packed with a bottle of cough syrup?
A. This is strictly meant for cough.
B. In case of overdose, call your doctor.
C. Children should take only half spoonful.
D. Should this fall to help, report to your physician.

110. " Please find enclosed a photocopy of my credentials. " This is most likely taken from a ----------
A. letter of request
B. letter to close friends
C. letter of acceptance
D. job application letter

111. "I was born in Jimma & moved to Harar with my parents when I was twelve"This statement is likely taken from the beginning of a/ an --------text.
A. narrative B. descriptive C. expository D.argumentative

112. To close a letter to a boyfriend, the most appropriate word is
A. Love
B. Yours faithfully
C. Sincerely
D. Good bye

113. " This bird is one of a group of long- legged wading birds. It has a long beak that curves down. " This statement is likely taken from the beginning of a / an ----------text.
A. narrative B. descriptive C. expository D. argumentative

114. " Let's keep in touch!" Which part of a letter does this line go in?
A. Body B. Closing C. Salutation D. Introduction

115. " The internet has provided great benefits. But at the same time it has also created problems and inequalities. "This piece of writing is likely to be -------
A. narrative B. expository C. description D. argumentative

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116. " I look forward to hearing from you soon," This is most likely taken from:
   a. the body of a business letter.
   b. the introduction of a letter to a close friend.
   c. the closing section of a letter of application for a job
   d. It was not possible to make it even in a four – wheel- drive vehicle.

117. Which one of the following best illustrates a sentence taken from an argumentative piece of writing?
   A. Four-drive venicles are available here
   B. Most four-wheel –drive vehicles consume more petrol
   C. We hired a four-wheel-drive vehicle for the off-road trip
   D. It was not possible to make it even in a four wheel-drive vehicle

118. Which one of the following may NOT be among issues to be explained in a letter of apologies to a customer?
   A. What the mistake was
   B. How the mistake occurred
   C. When the mistake occurred
   D. Who committed the mistake

119. Which one of the following sentences is correctly punctuated?
   A. What do you mean when you say " hyphenation?"
   B. What do you mean when you say " hyphenation" ?
   C. What do you mean when you say, " hyphenation?"
   D. What do you mean when you say, " hyphenation"?

120. Which one of the following is correctly punctuated?
   A. He asked where are you going?
   B. He asked, where are you going?
   C. He asked, " Where are you going?"
   D. He asked, " Where are you going?"

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UINVERSITY ENTERANCE EXAMINATION (UEE)
ENGLISH
Sene 2004/June 2012

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SECTION ONE : Word Order(1-5)

DIRECTION: The following words are **NOT** in the proper order. When put in the right order, they make Correct Sentences, From the given alternatives, Choose the one that is correct and blacken the letter of your choice in the Separate answer Sheet provided.

1. not to training have helped does seem.
   (A) Training does not seem to have helped.
   (B) To seem training does not have helped.
   (C) Training does seem to not have helped.
   (D) Not to have helped training does seem.
2. is on he the by the standing Window chair.
   (A) He is by standing the chair on the Window.
   (B) He is on the Window standing by the chair.
   (C) He is standing on the chair by the window.
   (D) By standing the chair he is on the window.
3. like snore her that hate hearing.
   (A) I hate that snore like her hearing.
   (B) I hate her snore hearing like that.
   (C) I hate hearing her snore like that.
   (D) I hate her hearing like that snore.
4. how are these much T-shirts?
   (A) How much T-shirts are these?
   (B) These are how much T-shirts?
   (C) How much are these T-shirts?
   (D) These are T-shirts how much?
5. I did not anything about her unusual notice at first.
   (A) About her I did not notice anything unusual notice at first.
   (B) I did not notice anything unusual about her at first.
   (C) Anything unusual about her I did not notice at first
   (D) At first anything unusual about her I did not notice.

SECTION TWO: Paragraph Coherence (6-12)

DIRECTION: When put in the correct order, sentences a-e in each of the questions from 6-12 make up a complete paragraph. From the alternatives A-D given in each case, choose the letter that contains the correct order of the sentences and **blacken** the corresponding letter provided in the separate answer sheet

6. a. So, I quietly got out of bed to see what it was.
   b. It was in the middle of the night when we were all fast asleep.
   c. Suddenly, a piercing horrible noise came from downstairs.
   d. I could imagine the chilling fear this

would cause everyone in thehouse

e. But before I walked a step, I saw this huge dark creature holding me by the neck.

(A) b e c a d (C) b c a d e

(B) b c d a e (D) b d c a e

7. a. Right in front of me, there were large plants.
   b. On my left, the ground rose gently to a low hill.
   c. I made my first landing there and began to look around.
   d. A kind of fruit like none I had ever seen before grew on them.
   e. Attracted by their color, I got closer for a thorough examination.

(A) c b e d a (C) c e a d b

(B) c a e d b (D) c a b d e

8. (a) Some eye signals are unconscious.
   (b) We receive 80% of our information about the world through them.
   (c).When we look at something nice, our pupils get bigger.
   (d)Our eyes are the most important of our five senses.
   (e) We also send signals to other people with our eyes.

(A) d c b c e B) d b c a e

(B) c b a e d D)c e b a d

9. (a) Learn as much as you can about them.
   (b) Many zoos breedendangered animals.
   (c) What can you do to help endangered animals?
   (d) Make an effort to support zoos and wildlife groups.
   (e) The more you know about them, the more you can help.

(A) c a e b d C)d a b e b

(B) d b c a d D) b d c e a

10.(a) It lives in man the other stage of its life.
   (b) Bilharzias is a disease found all over the world.
   (c) Three species of Schistosoma cause human disease.
   (d) It is spread by a small fluke or flatworm calledschistosoma.
   (e) Schistosoma lives as a parasite in the water snail during one stage of its

(A) c a b d e (C) c e a d b

(B) c e b d a (D) b d e a c

11. a. On her arm, there is a tattoo.
    b. Juali has an intriguing history.
    c. Physically, she is old and has a wrinkled face.
    d. When she was 10, she became a drug addict
    e. But people know her for the big earrings she wore.

(A) b c a e d (C) b d e c a

(B) c a d b e (D) b d c a e

12. a. In REM sleep, the body is quiet but the mind is active.
    b. These movements can be observed beneath closed eyelids
    c. Others believe that it helps to rid the brain of over stimulation
    d. Some think that REM sleep helps to from permanent memories
    e. During sleep, we experience periods of rapid eye movement(REM).

(A) e a b d c (C) e b a d c

(B) a e d c b (D) d e c a b

SECTION THEREE: Reading Comprehension (13-26)

DIRECTION: There are two passages in this section. Read each of them carefully and answer the questions according to the best alternative and Blacken the letter of your choice in the separate answer sheet provided.

For most people, it is easy to learn to read faster. Our reading rate is often just a matter of habit. But to begin, we may need to try to change some habits and try these tips:

PASSAGE I

1. Nonasserters Pay for not expressing themselves in several ways. The most obvious costs are social ones. Shy people make few new acquaintances and have a hard time building friendships with those people they do meet. They often wind up feeling isolated, even in a crowd, and thus become victims of the unhappy social disease of Loneliness. Even when they do mingle with others ,nonexpressive people are often misunderstood. It's easy to misinterpret the Silence of shyness and discomfort for boredom, hostility or Snobbishness.
2. Besides the Social Consequences, no assertiveness takes a Psychological toll on its Victims. Three attitudes often develop in People who are not able to express the full range of their feelings. Some simply withdraw from any Kind of meaningful contact with others, taking refuge in impersonal activities, watching TV for hours at a time, lavishing all their affection on such intimate objects as cars or motorcycles, becoming preoccupied with earning money, or distracting themselves with liquor or other drugs.
3. Other people deal with their inept communications by becoming cynics, claiming that people aren't worth caring about anyway. Bill was a sad example of this category. He claimed that the best thing about being 18 was his freedom to move away from his parents, whom he shrugged off as"jerks"yet he listened intently in class when other students talked with pride about their families. A third group of nonasserrters react to their condition with despair at themselves and what they see as human condition. They claim that people are disgusting creatures, and that life in this imperfect world is not worth living.
4. Besides social and psychological consequences,nonassertion also can have its psychological costs, often in the form of phychosomatic illnesses. We arenot talking about hypochondria, in which people believe they are ill but aren't, or malingering, in which they pretend to be sick. Psychosomatic disease is real. It differs in no physical way from an organically caused one. The characteristic that distinguishes psychosomatic illness is that while the pain comes from a condition in the person's body, the problem has its origin in some aspect of the person's psychological adaptation to the surrounding.

13. If one is a nonasserter, it is possible that one:

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A. chooses to engage in cooperative activities.
B. aspires to become a business partner to another person.
C. loves driving others around in one's car or other vehicles
D. frequents dealing with own belongings during leisure time.

14. Which of the following is true about Bill's view of other people (paragraph 3)?
A. They are cynical
B. We do not have to care about them.
C. They deal with their inept communications.
D. We need to give them support and attention.

15. Which of the following is true about the physiological problem of nonassertiveness?
A. Malingerers are highly susceptible to the problem.
B. It does not have the same origin as organically caused illnesses.
C. The problem can be diagnosed by testing specimen from a relevant organ.
D. A close look at a sufferer can help to distinguish an organic illness from a psychosomatic one

16. According to Paragraph 2, which of the following is a good example of taking refuge in impersonal activities?
(A) Working hard to make money.
(B) Listening to evening news on the TV.
(C) Taking alcoholic drinks to forget family issues.
(D) Telling others about one 's expensivemobile phone.

17. Which of the Following statements best summarizes the idea of the passage.
(A) The problem of nonassertiveness can be grouped into three.
(B) Nonasserters have real problems which can, in fact, be cured.
(C) Nonassertiveness can impact the life of the individual in many ways
(D) Compared to their counterparts, nonasserters are at some disadvantage.

18. Which one of the following sentences is true about nonasserters?
(A) It is not easy for them to meet new people.
(B) It is difficult for them to make new friends.
(C) It is uncommon for them to be misunderstood.
(D) It is not difficult for them to mingle with others.

19. Which of the following statements reflects the view of a third group nonasserters?
(A) Living in this world is after all useless.
(B) We could make the world a better place to live.
(C) Though disgusting, the human condition can be improved.
(D) There are many good reasons for some to be born into this world.

PASSAGE II

1. A few years ago, a British newspaper published an aid worker's recollection of a shopping trip with Mother Teresa. They were in local supermarket buying supplies for her London mission. When they reached the counter, and were presented with a bill for about £500, Mother Teresa refused to pay, saying, "It is for the poor". She remained unmoved by the demands of the store-keeper and an impasse prevailed until a kindly shopper, at the back of the growing queue, volunteered to pay.
2. This story, and many others, illustrates well Mother Teresa's impressive skills as a manipulator. Some of her supporters might find description insulting, even sacrilegious, but to deny it would be native. She is after all a missionary-a person whose job is to change other people's minds.
3. One of her most famous conversions was the British journalist Malcolm Muggeridge. So taken was he by this little Albanian nun that, in 1969, he made a television film about her called Something *Beautiful for God*. Then he converted to Catholicism. It was a classic missionary's barter: she put him in touch with his soul, he put her in touch with the public, and convinced she was a living saint.
4. But Mother Teresa stopped trying to convert people to Catholicism long ago. Instead, she concentrates looking after society's rejects, and is happy to let them believe what they like.Her first mission, set up in Calcutta in 1950, has fed and cared for an estimated 50,000 homeless, sick and dying people. So it's not surprising that, to Catholics, only the pope is more holy.
5. And she is not only feted by Catholics. Across the world, she is seen as symbol of charity and goodness; a selfless angel in a World where greed and market economics are enriching the few at the expense of the many.
6. But. Mother Teresa is more than just caring soul. Her enormous popularity – helped along by the 1979 Nobel Peace Prize for her efforts to relieve poverty – has given her huge influence among the wealthy and the powerful. And she has not hesitated to use that influence to encourage, cajole and even shame them into giving money to her Missions of Charity, or to improving the lives of the less fortunate.

20. What does the demands of the store-keeper in Paragraph I mean?
(A) That Mother Teresa should move.
(B) That shoppers should Volunteer to pay.
(C) That the prevailing impasse should lax.
(D) That Mother Teresa should settle the bill.

21. Which of the following ,according to the writer, is true about Mother Teresa?
(A) She is native and religious.
(B) She is a missionary and a manipulator.
(C) Normally, she pays for whatever she shops.
(D)Comments about her are sacrilegious toher supporters.

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22. What inspired Malcolm Muggeridge to make the film Something Beautiful for God?
(A) He wanted to publicize Mother Teresa.
(B) He had the motive to impress Mother Teresa.
(C) He was influenced by what Mother Teresa Was doing.
(D) He wanted to use it as a means to convert to Catholicism.

23. Which of the following sentences best describes Mother Teresa as presented in Paragraph 6?
(A) She did a lot more than just caring the poor.
(B) She had enormous recognition that she won the Nobel Peace Prize.
(C) She was influential and popular among the wealthy and the powerful.
(D) She used every means at her disposal to make the wealthy help her charity.

24. Which of the following best reflects the main idea of Paragraph 5 of the Passage?
(A) The existing market economics is favouring the few.
(B) Catholics support and welcome the deeds of Mother Teresa.
(C) Mother Teresa is a selfless angel operating amidst the world of greed.
(D) The world sees Mother Teresa as a symbol of charity world and goodness.

25. Who is accountable for the shopping trip story in Paragraph one of the Passage?
(A) A shopper at the Supermarket.
(B) A Publisher of a British newspaper.
(C) An aid worker who accompanied Mother Teresa.
(D) A journalist who happened to be the supermarket.

26. To Catholics, Mother Teresa was considered the holiest after the Pope because she
(A) helped the society's rejects .
(B) didn't impose religion on people.
(C) set up her first mission in Calcutta.
(D) used to convert people to Catholicism.

SECTION FOUR: Vocabulary (27-57)

DIRECTION: Questions 27—46 are incomplete sentences. There are four alternative words/phrases, A-D given below each question. Choose the one that best completes each sentence and blacken the letter of your choice in the space provided in the answer sheet.

A: Completion (27-46)

27. Apparently, there is a _____ .They were supposed to have been here early morning.
(A) hold in (B)hold up (C) hold out (D) hole down

28. All roads to the city center have been _ because of the ongoing clash.
(A) blocked (B) preven (C) barraged (D) blockaded

29. If food supplies give out, aid workers will have _____for the refugees.
(A) nothing (C) some
(B) Very little (D) abundant

30. If do not believe a word of what you have said. There is often a hugebetween your words and your actions.
A. discrepancy C. duplication
B. disruption D. distortion

31. Before catching the bus, I quickly_____a note to my wife that I would not be home for dinner.
A. dictated C. described
B. sketched D. scribbled

32. A good score in this examination is a__for you to join university next year
A. precedent C. prerequisite
B. precaution D. requirement

33. So far, the parties are _____we don't know who would win.
A. neck in neck C. hand in hand
B. head to head D. neck and neck

34. Watch out! This is a _____ type and you cannot bend it.
A. harsh C. rigid
B. stubborn D. parched

35. House rent is becoming very expensive these days. Considering our income, I feel we pay_____ for it
A. disproportionally C. disproportionately
B. unproportionally D. unproportianately

36. You've got to be_____ whenyou talk to the police. Otherwise, you could end up in a problem.
A. insistent C. constant
B. persistent D. consistent

37. You should be guided by your_____. Don't be pushed around by others.
A. conscious C. conscience
B. consciousness D. conscientious

38. The police used tear gas to _____ the protesters.
A. disperse B. spread C.scour D. diffuse

39. At one time, tigers were hunted almost to _____ for their hides in this country.
A. distinction B. extinction C.exception D.exhaustion

40. I am_____ to you for letting me stay at your house for so long.
A regretful B. grateful C. doubtful D. careful

41. You need to work hard todistractions that interfere with concentration during your reading.
A. regretful B. grateful C. doubtful D. careful

42. I am incredibly pleased with this exam; none of the questions have _____me so far.
A. impressed B. charged C. stretched D. challenged

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A:Completion (27-46)

27. Apparently, there is a _____ .They were supposed to have been here early morning.
(A) hold in B)hold up (C) hold out (D) hole down

28. All roads to the city center have been _ because of the ongoing clash.
(A) blocked B) preven (C) barraged (D) blockaded

29. If food supplies give out, aid workers will have _____for the refugees.
(A) nothing (C) some
(B) Very little (D) abundant

30. If do not believe a word of what you have said. There is often a hugebetween your words and your actions.
A. discrepancy C. duplication
B. disruption D. distortion

31. Before catching the bus, I quickly_____a note to my wife that I would not be home for dinner.
A. dictated C. described
B. sketched D. scribbled

32. A good score in this examination is a__for you to join university next year
A. precedent C. prerequisite
B. precaution D. requirement

33. So far, the parties are_____we don't know who would win.
A. neck in neck C. hand in hand
B. head to head D. neck and neck

34. Watch out! This is a _____ type and you cannot bend it.
A. harsh C. rigid
B. stubborn D. parched

35. House rent is becoming very expensive these days. Considering our income, I feel we pay_____ for it
A. disproportionally C. disproportionately
B. unproportionally D. unproportianately

36. You've got to be_____ whenyou talk to the police. Otherwise, you could end up in a problem.
A. insistent C. constant
B. persistent D. consistent

37. You should be guided by your_____. Don't be pushed around by others.
A. conscious C. conscience
B. consciousness D. conscientious

38. The police used tear gas to _____ the protesters.
A. disperse B. spread C.scour D. diffuse

39. At one time, tigers were hunted almost to _____ for their hides in this country.
A. distinction B. extinction C.exception D.exhaustion

40. I am_____ to you for letting me stay at your house for so long.
A regretful B. grateful C. doubtful D. careful

41. You need to work hard todistractions that interfere with concentration during your reading.
A. regretful B. grateful C. doubtful D. careful

42. I am incredibly pleased with this exam; none of the questions have _____me so far.
A. impressed B. charged C. stretched D. challenged

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43. Don't tell him anything about our plans. He is not_____.
A. dependent C. depending
B. dependable D. independent

44. Every year, a large number of tourists come to Ethiopia, a country rich in cultural_____.
A. heritage C. heredity
B. inheritance D. hereditary

45. Do you understand how much physical labor cans____ the body very quickly?
A. exhaust B. extract C. exert D. extend

46. There is no need to worry. If you apologize, she will_____ you soon.
A. forget B. forbid C. forfeit D. forgive

B. Substitution /47-57/

DIRECTION: Questions 47-57 each has an underlined word or phrase. There are four alternatives, A-D, given after each sentence. Choose one word or phrase that keeps the meaning of the given sentence if substituted for the underlined word or phrase and blacken the letter of your choice in the space provided in the answer sheet.

47. I love antique furniture, but I need your advice before I buy any.
A. old B. modern C. durable D. fashionable

48. The police caught the man who entered thebuilding without the owner's consent.
A. suggestion B. knowledge C. permission D. recommendation

49. Look! Doesn't this girl take after her mother?
A. inherit B. imitate C. resemble D. duplicate

50. I have tried to see if these two can get along. Now I assure you, they are oil and water.
A. really like each other
B. don't seem to agree at all
C. don't have much in common
D. can easily mix with each other

51. If you ask me, all I can tell you is that my memory of the accident is blurred.
A. unclear C. illusionary
B. colorless D. unremarkable

52. Bear with me for a couple of minutes. I'll finish it and leave very soon.
A. Don't disturb C. Give me a hand
B. Listen attentively D. Be patient with me

53. Poverty and crime go hand-in-hand.
A. are mixed C. are merged
B. are separated D.are connected

54. She often gets on my nerves whenever we go on holiday together.
A. irritates me C. invites me
B. pleases me D. imitates me

55. If you want to be someone, keep your nose to the grindstone. That is all I could say .
A. obey your superiors
B. continue working hard
C. work in cooperation with others
D. sniff for what interests the bosses

56. This is a no-poster zone. You ate not allowed to put up any posters around here.
A. fix B. dump C. sell D. read

57. If you are sure it is something you cannot overcome, then why waste the whole day with it? Just forget it.
A. dissolvable C. insurmountable
B. unovercomable D. unsurmountable

SECTION FIVE: Grammar/58-85/

DIRECTIONS: Questions 58-85 are incomplete sentences. There are four alternative words of phrases, A-D given **below** each questions. Choose the word that best completes the sentence and **blacken** the letter of your choice in the space provided in the answer sheet.

58. Haven't you been to the Awash National Park _____? You should really see it.
A. yet C. never
B. ever D. ever since

59. Reta, _____the door please. It is very hot inside.
A. open C. opened
B. has opened D. is opening

60. Only students who do very_____ in this exam can succeed in getting the best field of study available in the country.
A. fast B. good C. hard D. well

61. I have misplaced my new eraser_____ it anywhere?
A. Are you seeing C. Have you seen
B. Do you see D. Have you been seeing

62. She_____ really interested in boys of her own age.
A. was ever C. is never being
B. was never been D. has never been

63. I found_____ who could do the calculation. May be the question was tough.
A. no one C. anyone
B. someone D.everyone

64. Lema____home when I went to his office at half past five.
A. goes C. was going
B. has gone D. had gone

65. Look! Paul_____ to that man over there.
A.speakes C. is speaking

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A.speakes C. is speaking

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D. has spoken

B. speaking

66. If anyone had called, I _____ them she had not been at home.
A. will tell C. will have told
B. would tell D. would have told

67. Sara is_____ a better job because she is not earning enough now.
A. looking at C. looking on
B. looking into D. looking for

68. The burglar in on the run, but_____ in this country could he hide, I believe.
A. anywhere C. nowhere
B. somewhere D. everywhere

69. Inflation is not subsiding at all in this country. Is it?
A. No, it is C. I think, it does
B. Yes, it isn't D. I don't think it is.

70. Everyone of you_____ expected to contribute to contribute to the new enterprise.
A. is B. are C. has D. have

71. She was so_____ to me that she gave me money that would not even buy lunch
A. impolite B. indebted C. rude D. mean

72. The story was somewhat_____ and I could make no head or tail out of it.
A. shocking B. appalling C. dismaying D. perplexing

73. If he killed the man, as the rumour goes, I don't think he_____here
A. would be C. will have been
B. had been D. would have been

74. We fined him_____ he could learn from his mistakes.
A. so as B. because of C. so that D. so as to

75. By the way, do you know how much time_____ to finish the exam?
A. given B. has given C. we are given D. are we given

76. _____students have ever obtained full marks in College Entrance English Examination.
A. Few B. A few C. Many D. Several

77. How long_ this lap-top computer now?
A. did you have C are you having
B. have you had D. were you having

78. ___Lalibella over the last ten years.
A. Only once have I visited B. have to complete
C. Only once had I visited D. Only I have once visited

79. Next year this time, we_ our first year university courses.
A. willcomplet B. have to complete
C. have completed D. will have completed

80. Aspirins _ to help prevent heart disease.
A. know B. knew C. are known D.have known

81. Aren't we so blessed that we have this beautiful country?
A. Yes, we are C. No, we don't
B. Yes, we did D. No, we were not

82. This meal is _____ you. Don't you think we deserve a treat?
A. by B. on C. for D. with

83. It was only a narrow escape. Nohad I left the station than the explosion took place.
A. later B. quicker C.faster D.sooner

84. You should not have asked such an_____ question in public
A. embarrassing C. embarrass
B. embarrassed D. embarrassment

85. _____ can stop them from taking the city. The opposing force has gone in disarray.
A. Anything B)Everything C. Nothing D. Something

SECTION SIX: Communicative Activities /86-105/

DIRECTION: Questions 86-105 are presented in a dialogue form. The part said by one of the speakers is given, and a blank space is left for the other. Four alternative answers, A-D, are given for each of the blank spaces. Choose the alternative that is most appropriate to complete the dialogue, and blacken the letter of your choice in the space provided in the answer sheet.

86. **Stranger:** Excuse me, sir. How can I get to the bus station?
You: _____
A. Sorry, I am a stranger myself.
B. Why do you want to go there?
C. The bus station is far from here.
D. Is it your first time in this city?

87. **Assefa:** How long have you been retired now?
Ayele: _____
A. Just two years.
B. I often keep myself busy.
C. I am looking for a new job
D. Days now seem to be much longer

88. **Man:** Are you married?
Woman: _____.
A. Yes. We do not have a child yet
B. Yes. Actually we met at a wedding.
C. Sure, we both went to the same school
D. Yes, I am. That is my husband over there.

89. **Teacher:** _____
Student: She's a journalist.
A. What is your sister do?
B. What does your sister do?
C. Who is a journalist?
D. Who is your sister

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90. **Daughter:** What time is it?
Mom:_____
A. Breakfast isn't ready.
B. It's just after 7 o'clock
C. Remember to come home for lunch.
D. Do you want to go early this morning?

91. **You:** Why did you leave Jimma, Abdi? You had a good job there.
Abdi: _____
A. Many people live in Jimma.
B. I left Jimma almost a year ago.
C. Yes, but I have a better job here
D. Yes, and Jimma is a nice place for living

92. **Receptionist:**_____
Customer: I want to speak to the manager.
A. What do you want?
B. Whom do you want?
C. What do I help you?
D. What can I help you?

93. **Friend:** I am going to the theatre. Would you like to come along?
You:_____
A. Mmm… I think, I would.
B. Yes please. I don't like cinema.
C. How about going out for dinner?
D. Of course. I should see my boyfriend

94. **Bochera:** The Millennium Dam, I guess, is a noble idea.
Bikila: Certainly _____
A. I think Egyptians would not be happy about it
B. It really is demanding in terms of financial input.
C. Imagine the economic contribution it would make!
D. Did they say it would cost 80 billion Ethiopian Birr?

95. **Brother:** Isn't the day so wet? Even then, I suggest we go for a short walk.
Sister:_____
I can't agree less
A. Do you have a rain coat?
B. It has been like that all morning.
C. If we drive, we can get there faster

96. **Wogari:** Let's go out for dinner.
Kume:_____
A. I don't know what time you have dinner.
B. Are Martha and her boyfriend coming along?
C. I would rather want you to cook us beef steak
D. Did you like the meal we had at Tonta last week?

97. **Berhanu:** Sorry, I used your trainers
You: _____
A. I am sure you know that they are very expensive.
B. These are American-made trainers-so comfortable
C. How disappointing! You know I haven't even used them
D. I think I am going to use them for my competition next week

98. **Taye:** is your ex-girlfriend married?
Kebede:_____
A. She has two children
B. She joined a college
C. I've no idea
D. She's very nice

99. **Stranger:** So, do you live in Addis Ababa?
You:_____
A.

100. **Tadesse:** Do you speak Arabic?
Hailu:_____
A. Very well
B. Only a little
C. I speak English fluently
D. Arabic is spoken worldwide

101. **Senbeta:** This place has enormously changed. Isn't that surprising?
Solomon:_____
A. Yes, I wish you were here
B. They demolished everything
C. Sure. I can't believe it myself
D. All this has changed very fast.

102. **Alem:** Have you heard that Kebede bought a new car?
Chala:_____
A. Do they pay him well?
B. He can now drive us around
C. What? You should be kidding!
D. He is making a lot of money, I guess.

103. **Student:** she is not herself today. What has gone wrong?
Friend:_____
A. She is sometimes angry.
B. Her friend's car has broken down
C. She doesn't like people to talk to her
D. She saw her boyfriend with another girl.

104. **Chane:**_____
Senaet: By bus.
A. Does your Dad walk to work?
B. Why does your Dad go to work
C. How does your Dad go to work?
D. When does your Dad go to work

105. Your Teacher:_____
You: No, she doesn't. She lives in a small town
A. Who lives in a town?
B. Where does your sister live?
C. Does your sister live in a city?

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D. Why does your sister live in a small town?

SECTION SEVEN: Writing/106-120/

DIRECTIONS: The following questions are related to different aspects of writing. Read each of them very carefully and choose the answer from the four alternatives, A-D, given. Then blacken the letter of your choice in the space provided in the answer sheet.

106. "Thenew office building has a contemporary look" This statement is likely taken from the beginning of a/an ____text.
A. narrative B. descriptive C. expository D. argumentative

107. "Please place your purchase here" You would most likely find this notice in a/an____.
A. cafeteria B. exhibition center C. museum D. supermarket

108. "A man and his son were once going with their donkey to a market." This most likely begins a/an_____ text.
A. narrative B. descriptive C. expository D. argumentative

109. "Please see my resume for additional information on my experience." This piece of writing is likely taken from _____
A. business letter C. letter of inquiry
B. Personal letter D. job application letter

110. Which one of the following sentences most likely comes from a descriptive writing?
A. The next stopover was at BishanGari.
B. BishanGari is a little beautiful town in the valley.
C. BishanGari is much better than the other towns around
D. I don't have accurate information about the population of BishanGari.

111. If you take this one, there is a possibility for you to win a new microscope that the company has put on offer. You can also enjoy all benefits that state-of-the-art machines in the category could provide. This is taken from a piece of writing that is most likely:
A. narrative B. descriptive C. expository D. argumentative

112. Right in front of me stood this beautiful lady with her arms round the waist of a young man. The physical makeup of this man is strongly expressive-tells the onlooker that he was definitely revining from some chronic illness that eroded all his stamina. This is taken from a piece of writing that is most likely:
A. narrative B. expository C. descriptive D. argumentative

113. Which one of the following is most likely an instruction on a tube of ointment for skin?
A. For external use only
B. If swallowed, drink a lot of water.
C. In case of overdose, call your doctor
D. To be taken thirty minutes before each meal.

114. Which one of the following is **correctly** spelt?
A. develope B. received C. necessary D. grammer

115. "The claim that zoos educate people about wildlife and the environment is totally unfounded. On the contrary, they encourage people to see animals as a form of entertainment." These statements are most likely taken from the beginning of a/an _____text.
A. narrative B. descriptive C. expository D. argumentative

116. "I thought he was asking me just for a walk and did not hesitate the slightest to give in. He then stated smartly guiding me into a forest where I could see no soul around. Ten minutes into the venture, however, I started realizing that he was trying to use my naivety. He remarked, "You know, I……"This is taken from a piece of writing that is most likely:
A. narrative B. descriptive C. expository D. argumentative

117. Which one of the following words is **correctly** spelt?
A. lisence B. penality C. enterance D. perseverance

118. Which one of the following sentences has a correct punctuation?
A. "The date has been set", said Gurmuand added , "We can't change it now."
B. "The date has been set," said Gurmuand added, "We can't change it now."
C. "The date has been Set," said Gurmu and added, "We can't change it now."
D. "The date has been set," said Gurmu, and added "We can't change it now."

119. "Previous experience as a security guard be an advantage.This statement is likely taken from a_____.
A. Social letter C. job advertisement
B. Letter of application D. business advertisement

120. To write a memo to your close friend, the most appropriate opening is:
A. Hi, C. Dear friend,
B. Hello, D.My dear friend,

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SECTION ONE: Word order (1- 5)

Directions:The following words are not in the proper order. When put in the right order, they make correct English sentences. From the given alternaives, choose the one that is correct.

1. She in do the what you evenings know does.
   A) What she does do you know in the evenings?
   B) In the evenings do you know what she does?
   C) Do you in the evenings know what she does?
   D) Do you know what she does in the evenings?
2. Looking have we over you all been for
   A) We have been looking for you all over.
   B) We all have been looking over for you.
   C) You have been looking
   D) All over you have been looking for we.
3. Him you like son like we and your seeing laugh.
   A) You like your son and we like seeing laugh them.
   B) You laugh like seeing your son and we like him.
   C) We like him laugh and your son like seeing him.
   D) We like seeing him laugh like you your son.
4. get to at tea six up we o'clock drink.
   A) To drink at six o'clock tea we get up.
   B) We get up at six o'clock to drink tea.
   C) We drink up tea to get at six o'clock.
   D) We get up to drink tea at six o'clock.
5. world I the sail to around want.
   A) To sail around the world I watnt.
   B) I want around the world to sail.
   C) I want to sail around the world.
   D) Around the world I want to sail.

SECTION TWO:- PARAGRAPH COHERENCE (6-12)

DIRECTIONS:- when put in the correct order, sentences a-e in each of t questions from 6 – 12 make up a complete paragph. From the alternatives A given in each case, choose the letter that contains the correct order of t sentences.

6. a. It saves time and reduces potential expense.
   b. It also makes the communication easy and very quick.
   c. Effective communication is vital to any business anywhere in the world
   d. For instance, mobile phones help to reduce travel and associated expense
   e. Information technology provides the tools for such communication.
   A) d e a b c C) c d a b e
   B) c e a d b D) c a b e d
7. a. I'm not very patient with people.
   b. I'am also good at mending things.
   c. I don't really want to work with lots of people.
   d. I help my neighbors when something is broken.
   e. I am not good at languages but I'm good at science.
   A) a c d b e C) d b a e c
   B) e b d a c D) b d c a e
8. a. It has only one room.
   b. Konjit's house is a hut.
   c. The room is divided into three sections.
   d. The three sections are bedroom, kitchen and pantry.
   e. A bed and a bigt wooden box are in bedroom section.
   A) e b a d c C) e a c d e
   B) c a d b e D) d e c b a
9. a. The first thing I do is work out.
   b. My morning ritual is the same every day.
   c. After that I put on my make up and leave home .
   d. When I have school, I have to do a lot of things.
   e. When I finish working out, I take a shower, fix my hair and get dressed.
   A) a d c e b C) e a b d c
   B) d e b c a D) b d a e c
10. a. One of my favorite things to do is wash my car.
    b. People say they can tell about a person by their shoes.
    c. Here is the thoughtful process I follow to wash my car .
    d. A clean car says I'm proud and I feel good about myself.
    e. I say that you can tell about a person by how clean his car is.
    A) b e d a c C) e b c d a
    B) c a b d e D) a d b e c
11. a. ***Idir***, in our society, has a long history.
    b. The assistance includes material, financial and labour.
    c. Members contribute certain amount of money every month.
    d. Some Idirs, nowadays, are involved in even more useful activities.
    e. Then they get assistance when the member or one of his family dies.
    A) a d c e b C) d a c e b
    B) a c e b d D) a c d c b
12. a. But unfortunately, it got so furious.
    b. Unlike we thought, it was a wild cat.
    c. I was playing hide and seek with friends.
    d. We were happy thinking we could take it home.
    e. Then my cousin and I discovered an abandoned cat.
    A) e a b d c C) c e a b d
    B) e d a b c D) c e d a b

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SECTION THREE; READING CMPREHENSION (13-26)

Directions: Read the following passage carefully and answer the questions according to the information there in. Each question has four choices, A-D. Choose the best alternative.

1. When I was in junior high school, my parents gave me a typewriter. My friends Amy Neff and Carol Stein often walked with me to my house after school. We closed my bedroom and spent the afternoon giggling-or rolling on the carpet, howling with lunatic laughter-as we tapped out silly limericks and satirical stories on my typewriter. Once we even wrote a play.
2. Our after-school typing was not homework. We rambled wherever our teenage imaginations took us; we explored words widly for the pure fun of it, with no exams or report cards in mind. We practiced literacy risk-takins and problem solving, honed our language skills and learning perseverance, all with no fear of failure- because we were playing. Few children today have time for this kind of after-school play. There is too much homework. Parent's concerns about overload have become a familiar story as schools all over the country pile on homework in a mis-aimed response to criticism that they were teaching too little. Some elementary teachers routinely assign time-consuming work in each of three or four academic subjects every night.
3. Amy Neff, one of the friends who helped me enjoy my typewriter in junior high, learned at a school meeting in May that in fifth grade her 10 year-old daughter would have an hour and a half to two hours of homework seven days a week, plus special projects. " The idea is that, under avalanche of work, children should learn to organize their time efficiently," she wrote me in an e-mail message. "Nobody has mentioned creativity, ideas and content of what is learned. Just doing LOTS, FAST!".
4. The parents and grandparents of today's children had time "playing around" with stamp collectios or model airplanes, writing a fairy-tale, selling lemonade or not realize is that these kinds of interests and hobbies are just as important as academic learning.
5. Many successful adult have a childhood history of freewheel play. Louis Ignarro, a Nobel Prize winning Pharmacologist, had six successively complex chemistry sets and played with model rockets. When Letty cottin Pogerbin, President of the Authors Guid, an association of published writers, was 9, she spent afternoons writing and drawing for a monthly magazine she printed on a primitive press called hectograph. She charged friends and family 3 cents a copy and then mailed each copy with a 3-cent stamp. She recalls now, " I never understood why the magazine did not make a profit." But that is just the point: when kids play, they are free to experiment and learn from experiences, without worrying about how well they are performing.
6. Though homework has a clear benefit in high school, there is no research showing that any amount of it advances the education of elementary school kids. They may be stuffing a great deal of information into their heads, but after an hour or two, children lose any eagerness or joy they have in learning. Theat is important because research has shown that when children enjoy it, their learning is deeper, richer, and longer-lasting.
7. Parents do have some recourse. They can meet with principals and teachers, organize other parents, and protest. Parents in several communities have prevailed upon school districts to enact homework limits. For their children, at least, there may still be time to indulge in the ancient and powerful medium for learning known as play.

13. In paragraph 2, the author maintains that children lean best when:
    A) they are given a lot of homework.
    B) they have time for play after school.
    C) schools do not give them homework.
    D) they do homework with a type writer.
14. As used in paragraph 2, which one can be the best definition of honed?
    A) knew B) learned C) improved D) acquired
15. As implied in paragraph 2, what is the criticism directed to elementary teachers?
    A) They are teaching too little.
    B) They are not giving enough homework.
    C) They teach children a familiar story.
    D) They do not assign interesting work to children.
16. One major issue the author mentioned about the parents and grandparents of today's children is that they:
    A) had no fear or failure at school.
    B) were not given homework to do after school.
    C) had time for out of school interest and hobbies.
    D) did not realize that after school play enhanced learning.
17. Which of the following words did the author use in paragraph 2 to refer to the development of their determined attitude through after school play?
    A) Literacy C) Imagination
    B) Perseverance D) Risk-taking
18. The author her childhood experience to:
    A) show that she had good friends to play with.
    B) describe how she spent her after school hours.
    C) Show how, through play, they were learning.
    D) Tell us that she used a type writer to do her homework.
19. The tone of the author can best be described as:
    A) dismissive B.argumentative C) emphatic D) compassionate
20. The author looks at the issue of homework and after-school play from:
    A) teachers' and parents' perspectives.

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B) her own and her friends' perspectives.
C) school principals' and teachers' perspectives.
D) her friends', teachers', and parents' perspectives.

21. Who sent the author an e-mail message?
A) Amy Neff C) Carol Stein
B) Louis Ignarro D) Letty Cottin Pogrebin

22. Which one of the following is true about homework?
A) It may advance learning in high school.
B) It stuffs kids' head with useful information.
C) In enhances kids' imagination and creativity.
D) It helps elementary school kids organize their time.

23. According to paragraph 7, which one is the recourse parents are reported to have?
A) Enacting homework limits.
B) Meeting with principals and teachers.
C) Protesting against children's overload.
D) Allowing their children to learn through play.

24. Which one of the following is among the major issues raised by the author?
A) Exams and report cards are a threat to childre's learning.
B) Too much play and too little homework advance kids' learning.
C) All successful adults have a childhood history of freewheeling play.
D) Through play, children experiment and learn from their experiences.

25. Elementaty teachers responded to the criticism directed at them by:
A) giving special projects.
B) Telling kids familiar storie.
C) Giving too much homework.
D) Teaching kids to organize their time efficiently.

26. The idea is … (paragraph 3). What does the word idea refer to?
A) Special projects
B) Too much homework
C) Purpose of homework
D) Amy purpose for the e-mail

***SECTION FOUR: VOCABULARY* (27-57)**

A. Completion (27 – 46)

DIRECTIONS: Questions 27 – 46 are incomplete sentences. There are four alternative words/phrase, A-D, given below each question. Choose the one that best completes each sentence.

27. Several students are _____ at the prospect of joining university next year.
A) incited B) relaxed C) surprised D) excited

28. The _____ placed a sticker on each piece of cake she checked for quality.
A) spectator B) supervisor C) inspector D) director

29. The hotel _____ brought our luggage to our rooms.
A) reporter B) porter C) supporter D) importer

30. Linda _____ her request for a vacation to her boss.
A) emitted B) permitted C) submitted D) omitted

31. Lensa, who is always worried about her weight, eats like a/an _____
A) bird B) ant C) pig D) elephant

32. Children are naturally _____ about the world around them.
A) inquisitor B) requisite C) inquisitive D) notorious

33. Many people have a/an _____ about flying and won't travel by plane.
A) hysteria B) nostalgia C) phobia D) accident

34. Laptop computers are _____. They allow work to be done in many different locations.
A) potable B) durable C) portable D) sizable

35. These people seem to be _____. They need additional protein and vitamins.
A) hungry B)malnourished C) starving D) impoverished

36. I appreciate you for the enormous _____. I felt so moved and encouraged.
A) completion B) supplement C) complement D) compliment

37. Children are supposed to _____ their parents.
A) obey B) oblige C) abide D) obligate

38. This time round, we'll need to send out the police in _____ clothes to catch the criminals.
A) simple B) plaine C) casual D) uniform

39. There have been sporadic _____ in all corners of the city almost the whole year.
A) ritots B) stalks C) revellions D) revolutions

40. Olantu was _____ to go to her new school. She did not know any of the students in there.
A) resilient B) reluctant C) restless D) repellent

41. He _____ in bed the whole day. He had a terrible headache.
A) lie B) lay C) lied D) laid

42. We cannot _____ AIDS by eating hot pepper.
A) protect B) spread C) prevent D) project

43. As a child, I sometimes lied to my mother. This led her to _____ me even today.
A) mistreat B) mistake C) mistrust D) mishandle

44. This year's beauty _____ is very stiff. It is difficult to predict the winner.
A) rivalry B) contention C) contest D) competition

45. Ato Lema spent almost three years in prison for a crime he did not _____.
A) commit B) commute C) conduct D) condemn

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46. Have you ever _____ a situation or formed a worong first impression?
A) misjudged B)disrespected C) mislaid D) disrupted

B. Substitution (47 – 57)

Directions: Qusestion 47-57 each has an underlined work or phrase. There are four alternatives A – D given after each sentence. Choose one word or phrase that best keeps the meaning of the given sentence if substituted for the underlined word or phrase.

47. Please, **bear with** me until I finish this. It is very urgent.
A) tolerate B) encourage C) help D) adise

48. This one here is faulty. If you need it badly, we can give you at a **discount**.
A) for less money C) as it is
B) without counting D) for free

49. It was just **a shot in the dark**. No wonder he got it all wrong.
A) a blind money C) a silly mistake
B) a bad proposal D) a deliberate attack

50. She had a hard time **concealing** her emotions.
A) Hiding B) conceibing C) sharing D) correcting

51. Have you seen his handwriting? It was so **illegible** everyone wondered what it meant.
A) ugly B) invisible C) untidy D) unreadable

52. When the thief saw the police officer, his face went **as white as a sheet**.
A) terrified B) excited C) motivated D) annoyed

53. Stop shouting! I am trying to **get a grasp of** what is happening.
A) observe B) absorb C) report D) understand

54. He is leaving. I think he has a bigger fish to fry.
A) something bigger to eat.
B) additional cooking to do.
C) some more money to collect.
D) something more important to do.

55. That was a concert where Teddy Afro **stole the show** but to many it wasn't surprising.
A) sang others' songs.
B) cheated the audience.
C) got attention and praise.
D) took advantage undeservedly.

56. He is **sure of himself** that he can finish it in time.
A) Confidant B) condident C) confidante D) confidential

57. I couldn't still **make up** my mind about what to study at university.
A) limit B) judge C) plan D) decide

SECTION FIVE: GRAMMAR (58 – 85)

DIRECTIONS: Questions 58 – 85 are incomplete sentences. There are four alternative words or phrases,. A – D. given below each question. Choose the word that best completes the sentence .

58. I will see you when I _____ back.
A) get B) will got C) got D) would get

59. Jobs that call for strength and muscle no longer exist in the numbers that they once _____.
A) do B) were C) did D) exist

60. Rains cause soil erosion , _____ away millions of tones of precious top soil every year.
A) carrying C) carried
B) Have carried D) are carrying

61. Talent is like a plant that needs _____, watered, and pruned.
A) Feeding B) to feed C) food D) to be fed

62. The rich aren't _____ the poor.
A) happier than C) as happier as
B) as happier than D) so happier than

63. Last night I _____ sleep well because I had been drinking too much coffee.
A) wouldn't B) couldn't C) can't D) shouldn't

64. If I saw anyone cheating on this exam, I certainly _____ report the case to the invigilator.
A) will B) would C) could D) shall

65. When I heard the news, I _____.
A) delighted C) was delighting
B) was delighted D) have been delighted

66. The suspect told the police officer that he _____ the cashbox.
A) takes B) has taken C) took D) had taken

67. You _____ to do your homework before you went out.
A) are supposing C) supposed
B) were supposed D) were supposing

68. Either of them _____ to blame. They know the only key we have is with them.
A) is B) has C) are D) have

69. Should the machine fail, everybody _____ to evacuate and report to the operator.
A) is B) will C) has D) have

70. Most of the people I met were alert _____the dangers of deforestation. They never allow anyone to cut down a tree.
A) of B) to C) on D) at

71. The movie was so _____ that everybody felt like they were in a different world.
A) thrilling B) horrifying C) confusing D) instigating

72. He fell of his horse a month ago and now he is on a _____ stick.
A) catching B) supporting C) walking D) leaning

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73. It is sad but _____ seems to care about the prisoners. Please make sure they have lunch.
A) nobody B) everybody C) anybody D) somebody

74. Sorry. You can't speak to Mahlet. She _____ a bath.
A) have B) is having C) does D) has

75. I feel that _____ loves me from the group, so I have to join another one.
A) everyone B) someone C) anyone D) no one

76. Whatever the discussion is about, no one _____ pays attention to my suffestion.
A) never B) forever C) ever D) yet

77. Her husband abuses her a lot and now she has decided to run any. She says there cannot be ny _____ life.
A) cheaper B) happier C) poorer D) tougher

78. Shae came _____ just to say hello. She was so kind.
A) by B) on C) in D) for

79. If you see any unattended bag, you _____ immediately report to us.
A) may B) must C) will D) ought to

80. They started early _____ get there before noon.
A) so as to B) because to C) so that D) so that to

81. Applications are accepted only if you submit them no _____ than the end of this monthe. That is the deadline.
A) later B) faster C) after D) sooner

82. Caffeine _____ to cause withdrawal symptoms such as headache and irritability.
A) is known B) has known C) knows D) know

83. I was with her for dinner. She _____ the car which they claim disappeared around 7 pm.
A) won't have stolen
B) would have stolen
C) couldn't steal
D) can probably steal

84. In my life, I _____ had money counted in four digits. Poor me!
A) have never
B) haven't never
C) am never
D) am not have

85. Our departure is only a week away, but we haven't got our tickets _____. Isn't that something to worry about?
A) yet B) still C) now D) still now

***SECTION SIX*: Communicative Activities (86-105)**

DIRECTIONS: Questions 86 – 105 are presented in a form of dialogue. The part said by one of the speakers is given, and a blank space is left for the other. For each of the blank spaces, four alternative answers, A-D, are given. Choose the one that is most appropriate to complete the dialogue .

86. **Tola**: Let's go out for a drink.
Bereket: _____
A) Who else is coming?
B) Why do you like drinks?
C) Drinks are means of socializing.
D) I would rather like going to the theatre.

87. **Secretary**: May I help you?
Visitor: _____
A) Why do you ask?
B) What kind of help?
C) What do you mean?
D) Can I see the boss, please?

88. **Brother**: Life is getting more and more expensive here. Shall we leave for another place?
Sister: _____.
A) How much do you earn now?
B) I guess it is the same everywhere.
C) How much do you need for your food?
D) True! Everybody is complaining these days.

89. **Student** : _____.
Friend: Certainly. Imaginge he wasn't in our group!
A) Do they like Gebre?
B) Isn't Gebre so entertaining?
C) Isn't Gebre always laughing?
D) Is Gebre doing our assignment?

90. **Nana**: Dad is getting us a new TV set. Iam excited.
Chala: _____.
A) I can't believe that!
B) Is your dad rich enough?
C) How much is he going to pay?
D) Many people around do not have a TV set.

91. **Belay**: Everybody should contribute. This is for the good of the Community.
Kiros: _____.
A) I can't agree more.
B) Can everybody affor it.
C) Health centres are important.
D) Each of us is to pay birr 5,000.

92. **Alem**: _____.
Thomas: No way! I'll need it tomorrow.
A) Do you need your money back soon?
B) When should I pay you back the money?
C) Can I pay you back the money next week?
D) Do you need your money back tomorrow?

93. Kuku: _____

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Lili: Delicious! The cook must be great.
A) Was the cook so good? C) What was the dinner like?
B) Did you enjoy the dinner D) What does the dinner like?

94. **Teacher**: What are your favorite things to do at weekends?
Student: _____
A) I like sight – seeing.
B) I don't quite like shopping.
C) I'm going swimming this weekend.
D) I'm not doing anything this weekend.

95. **Teacher**: What was the weather like here yesterday?
Student: _____
A) It rains every day. C) It was sunny and warm.
B) I think it is a lovely day D) I did not out yesterday

96. **Abebe**: Why did you go to Gambella?
Kibret: _____
A) It is very hot there.
B) To vist my birth place.
C) It is a rapidly growing town.
D) The people are very friendly.

97. **Lilu**: Hi Beza. It's me.
Beza: Hello, Lilu! Where _____ all day?
A) were you C) have you been
B) are you D) had you been

98. **Teacher**: How do you like your younger sister?
Student: _____.
A) She's a smart girl.
B) She's my favorite.
C) She's still at school.
D) She's not old enough for marriage.

99. **Bontu**: Did you know she took your camera?
Chaltu: _____.
A) I know she is very bad.
B) Cameras need care to handle.
C) This is not just ordinary camera.
D) How dare she? She didn't ask me for it.

100. **Tamrat**: Well, let's go to the cinema.
Hailu: _____.
A) Do we have time to study?
B) When did we go there last?
C) Ok. Which film do you want to see?
D) Sylvester Stallone is a very good actor.

101. **Abera**: Excuse me. Can you tell me the time, please?
Bekele: _____.
A) It's after breakfast.
B) There is no time around here.
C) I'm sorry. I don't have a watch.
D) Never mind. I don't have a watch.

102. **Passanger** 1: Excuse me, I think those seats facing the front are ours.
Passanger 2: _____.
A) Oh, I'm sorry. We know they were reserved.
B) Oh, what a shame! I like seats facing the front.
C) I'am afraid the seats are not comfortable enough.
D) Oh, I'm sorry. We did not know they were reserved.

103. **Father**: I think you'll pass this examination with very good grades. You have been studying hard over the last months.
Son: _____.
A) I haven't decided yet my field of study.
B) I can't tell you about my excitement, Dad.
C) I think I won't pass the exam. I did not study hard, actually.
D) I don't think I'll pass the exam. I did not study hard enough.

104. **Teacher**: _____.
Student: I get out of bed.
A) What time do you get up?
B) Do you ever get up before six?
C) Who gets up first in your family?
D) What do you do first in the morning?

105. **You**: Good morning. _____.
Guest: No, thanks.
A) Can I help? C) How do you do?
B) What's the matter? D) Can I have coffee please?

SECTION SEVEN: Writing (106-120)

Direction: The following questions are related to different types of writing . Read each of them very carefully and choose the best answer from the four alternative answers, A-D, given.

106. *Once upon a time there was an emperor who loved to spend his money on expensive things.* This is most likely an introductory part of a writing tha is:
A) Narrative B) Descriptive C) Expository D) Argumentative

107. *According to government records, the number of men working or looking for work fell to a record low in the early part of 2012. But I don't think this reflects the relalities of every day life.* This is most likely the introductory part of a writing that is:
A) Narrative B) Argumentative C) Expository D) Description

108. Walk along the road in front of the post office until you come to the football field. Behind the football field there is a street with a vegetable market. There

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is a café opposite the vegetable market. I'm living in the villa located just next to the market. This is most likely taken from a piece of writing that is:
B) Descriptive B) Expository C) Narrative D) Argumentative

109. Love to you all! Which part of a letter does this line go in?
A) Body B) Closing C) Salutation D) Introduction

110. Which one of the following sentences is correctly punctuated?
A) "What is the population of Ethiopia?", the teacher asked.
B) "What is the population of Ethiopia, the teacher asked?"
C) "What is the population of Ethiopia"? the teacher asked.
D) "What is the population of Ethiopia," the teacher asked.

111. One of the reservations that people have about this one is that it has radioactive elements in it. Very important is , however, if it has these elements in the amount that c ould be harmful to the user. Personally, I say this cannot be ....
This is taken from a piece of writing that is most likely:
A) Descriptive B)Argumentative C) Narrative D) Expository

112. Which one of the following sentences has correct punctuations?
A) Mamo said, "Every dog has its day;" however, I didn't understand what it meant.
B) Mamo said, "every dog has its day" ; however, I didn't understand what it meant.
C) Mamo said, "Every dog has its day" ; however, I didn't understand what it meant.
D) Mamo said, "every dog has its day;" however, I didn't understand what it meant.

113. Which one of the following is most likely an instruction on how to operate a camera?
A) Use flashlight only in daylight.
B) Remove battery when not in use.
C) Do not give your camera to anyone.
D) Keep cameras out of reach of children.

114. Not suitable for children under 18! This is most likely a warning:
A) in a movie C) in an X-ray room
B) on a box of whisky D) on a pack of cigarettes.

115. Which one of the following words is **correctly** spelt?
A) restaurant B) extention C) cafeteria D) vollyball

116. Which one of the following words is correctly spelt?
A) scent B) conscent C) omniscien t D) reminiscent

117. We then found bothy lying on the floor soaked in their own blood. There w no question that the shot was from a short distance as something that look like content of the head lay close to what used to be the skull. The po officer stood thunderstruck. ... This is taken from a piece of writing tha most likely:
A) Narrative `B) Argumentative C) Descriptive D) Expository

118. The most appropriate closing to a memo you send to a colleague working in the same office with you would be:
A) Bye, B) Faithfully, C) Good bye, D) Truly yours,

119. We regret having to report delays in dispatching our order of .... This is a typical example of a line taken from a letter of
A) dispatch B) delivery C) collection D) complaint

120. Which one of the following sentences is correctly punctuated?
A) We visit three places: Jimma; Oromia, Shire; Tigray, and Dessie; Amhara.
B) We visit three places: Jimma, Oromia; Shire, Tigray; and Dessie, Amhara.
C) We visit three places, Jimma; Oromia,Shire; Tigray, and Dessie; Amhara.
D) We visit three places: Jimma, Oromia; Shire, Tigray; and, Dessie, Amhara.

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BOOKLET CODE:14
Number of Items:120
SECTIONS ONE: Word Order(1-5)
DIRECTIONS: The following words are NOT in the proper order. When put in the right order, they make correct English sentences. From the given alternatives, choose the one that is correct

1. us him and go up and pick
   A. Let him go and pick up us.
   B. Let him up and go pick us
   C. Let him go up and us pick
   D. Let us go and pick him up
2. Carefully work you did your check?
   A. Did you work your check carefully?
   B. Did carefully you check your work?
   C. Did you carefully check your work?
   D. Your work did you carefully check?
3. People going to many invite party are to your you?
   A. Are you going to invite many people to your party?
   B. Are your many people going to invite you to party?
   C. Are your many party going to you people to invite?
   D. Are you going to many party to invite your people?
4. Many got you how sisters have
   A. How many got sisters you have?
   B. How many you have sisters got?
   C. How many sisters have you got?
   D. How many have you got sisters?
5. for and join come us celebration the?
   A. Come and join for us the celebration.
   B. For us the celebration come and join.
   C. Come and join the celebration for us.
   D. Come and join us for the celebration.

SECTION TWO: Paragraph Coherence (6-10)
DIRECTIONS: When put in the correct order, sentences a-e in each of the questions from 6-10 make up a complete paragraph. From the alternatives given in each case, choose the letter that contains the correct order of the sentences

6. a. This is the planet Venus.
   b. Venus is our nearest neighbor.
   c. However, no plants or animals could exist on Venus.
   d. There is no oxygen in the air and the planet is without water.
   e. In the evening just after sunset we see a bright object in the west.
   A. d e c a b
   B. a e b c d
   C. b c a d e
   D. e a b c d
7. a. Dallol is found in the Afar Region of Ethiopia.
   b. It is one of the lowest and hottest areas of the world.
   c. Temperatures can reach as high as 145°C in the sun.
   d. This makes the place very difficult for wild life to live and vegetation to grow.
   e. There is no rain for most of the year, and the highest annual rainfall is only 200mm
   A. c e a d
   B. b a b c e d
   C. e c a b d
   D. c a b e d
8. a. This may not be practical as it is at the expensive end.
   b. Segregation of waste should also be seriously considered.
   c. Cities must look at their waste economy and invest in re-use.
   d. This helps to clean up relatively less toxic waste to irrigate fields.
   e. For instance, Singapore uses membrane technology to clean up waste water.
   A. c e a b d
   B. c b e d a
   C. c d e b a
   D. c a b e d
9. a. Informative speech is meant to convey knowledge.
   b. She/He should also organize her/his speech coherently.
   c. Its application in other spheres of life is also common.
   d. This type of speech is mostly practiced in the academia.
   e. The user of such a speech should prepare her/himself very well.
   A. a d c e b
   B. a c d e b
   C. a c e b d
   D. a e d b c
10. a. Every year one million children die of malaria.
    b. Malaria is very common in many parts of the world.
    c. However, people must also cooperate to stop its spread.
    d. Governments have special programmes to control malaria.
    e. Many others cannot go to school or work because of malaria.
    A. a b c e d
    B. b a e d c
    C. b d a e c
    D. a e b c d

SECTION THREE: Reading Comprehension) 11-24)
DIRECTIONS: Read the following passage carefully and answer the questions that follow. Each question has four choices. Choose the best alternative

1. If you suffer from shyness, you are not alone, for shyness is a universal phenomenon. According to recent research, "close to 50 percent of the general population report that they currently experience some degree of shyness in their lives. In addition, close to 80 percent of people report having felt shy, at some point in their lives". As shyness is so prevalent in

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the world, it is not surprising that social scientists are learning more about the cause. They have found that shyness in an individual can result from both biological and environmental factors.

2. Recent research reveals that some individuals are genetically predisposed to shyness. In other words, some people are born shy. Researchers say that between 15 and 20 percent of newborn babies show signs of shyness: they are quieter and more **vigilant**. Researchers have identified psychological differences between sociable and shy babies that show up as early as two months. In one study, two-month-olds who were later identified as shy children reacted with signs of stress to stimuli such as moving mobiles and tape recordings of human voices: increased heart rates, jerky movements of arms and legs, and excessive crying. Further evidence of the genetic basis of shyness is the fact that parents and grandparents of non-shy children more often say they were shy as children than parents and grandparents of shy children.

3. However, environment can, at least in some cases, triumph over biology. A shy child may lose much of his or her shyness. On the other hand, many people who were not shy as children become shy as adults, a fact that points to environmental or experiential causes.

4. The environmental cause of shyness may be a child's home and family life. Children who grew up with a difficult relationship with parents or a dominating older sibling are likely to be inhibited in social interactions. Another factor is the fact that today's children are growing up in smaller and fewer relatives living nearby. Growing up in single-parent homes or homes in which both parents work full time, children may not have the socializing experience of frequent visits by neighbors and friends. Because of their lack of social skills, they begin to feel socially inhibited, or shy, when they start school.

5. A second environmental cause of shyness in an individual may be one's culture. In a large study conducted in several nations, 40 percent of participants in the United States rated themselves as shy, compared to 57 percent in Japan and 55 percent in Taiwan. Of the countries participating in the study, the lowest percentage of shyness was found in Israel, where the rate was 31 percent.

6. Researchers Henderson and Zimbardo say, "One explanation of the cultural difference between Japanese and Israelis lies in the way each culture deals with attributing credit for success and blame for failure. In Japan, an individual's performance success is credited externally to parents, grandparents, teachers, and others, while failure is entirely blamed on the person. 'Therefore, Japanese learn not to take risks in public and rely instead on group-shared decisions. " In Israel, the situation is entirely reversed," according to Henderson and Zimbardo. "Failure is extremely attributed to parents, teachers, coaches, friends, and Anti Semitism, and other sources, while all performance success is credited to the individual's enterprise." The consequence is that Isrealis are free to take risk since there is nothing to lose by trying.

7. In addition to family and culture, technology may play a role as well. In the United States, the number of young people who repot being shy has risen from 40 percent to 50 percent in recent years. The rising numbers of shy young people may be,"due in part to the growing dependence on non-human forms of communication, coming about as a result of **our** huge advance in technology. Watching television, playing video games, and surfing the Web have displaced recreational activities that involve social interaction for many young people. Adults, too, are becoming more isolated as a result of technology. Face-to-face interactions with bank tellers, gas station attendants, and store clerks are no longer necessary because people can use machines to do their banking, fill their gas tanks, and order merchandise. College students take online telecourses. Telecommuters work at home, giving up daily contact with coworkers. Everyone texts, e-mails and converses anonymously in online chat rooms. **As a result**, people have less opportunity to socialize in person. Eventually, they increasingly become awkward at it and start avoiding **it** altogether. Source: writing A Lademic English, 2006) In short, they become shy.

11. Which of the following sentences is true according to the passage?
    A. Because of lack of socializing experience, many children grow up in single-parent homes.
    B. Children who get frequent visits from friends or neighbors are less likely to feel socially inhibited
    C. Children who grow up in homes in which both parents work full time have no relatives living nearby
    D. Children who grow up in homes in which both parents work full time have a difficult relationship with parents.

12. Which of the following best summarizes the main causes of shyness?
    A. Culture, home and family
    B. Family relatives and single-parent
    C. Genetics, environment and technology
    D. Television, video games and Web pages

13. As used in paragraph7, who does ***our*** in "...coming about as a result of **our** huge advances in technology " refer to?
    A. Shy adults
    B. The society at large
    C. Readers of the passage
    D. Author of the passage

14. Which of the following does the author mention as evidence of the prevalence of shyness?
    A. Shyness is a universal phenomenon.

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B. Shyness can result from both biological and environmental factors.
C. Research has found that some individuals are genetically predisposed to shyness.
D. Nearly half of the general population asked reported to have experienced some degree of shyness.

15. As used in the text, which of the following does the author mention as evidence of the genetic basis of shyness?
A. Parents of shy children are genetically predisposed to shyness.
B. More parents of shy children admitted they were shy as adults than parents of non-shy children.
C. More parents of shy children admitted they were shy as children than parents of non- shy children.
D. Environment had more impact on the parents of non-shy children than it had on parents of shy children.

16. As used in paragraph 2, which of the following words has nearly the same meaning as **quick sudden movement?**
A. Vigilant B. stimuli C. stress D. jerky

17. In Paragraph 3, the purpose of the author in using the statement " Many people who were not shy as children become shy as adults." is perhaps to show the:
A. genetic basis of shyness.
B. impact of environment over human behavior.
C. relationship between genetics and environment.
D. role of genetics in determining human behavior.

18. As implied in Paragraph 4, which of the following is the benefit of frequent neighbors' or friends' visits to children?
A. It helps children not to become shy.
B. It helps children to grow up quickly.
C. It develops children's socializing skills.
D. It develops a good relationship between children and their parents.

19. According to the text, which one of the following best explains why children feel socially inhibited when they start school?
A. Having few relatives.
B. Growing up in small families.
C. Having inadequate social skill.
D. Growing up in single- parent home.

20. Which of the following statements is true according to the information in the passage?
A. Japanese learn to take risks in public.
B. In Israel, success is credited to parents.
C. In Japan, success is attributed to an individual.
D. In Israel, failure is externally attributed to parents.

21. Which of the following is an example of human form of communication?
A. Surfing the Web in an office.
B. Talking to gas station attendants.
C. Playing video games after school.
D. Watching television in one's room.

22. As used in Paragraph 7, what does it in **Eventually, they increasingly become awkward at it**... refer to ?
A. Technology
B. Face-to- face interaction
C. Anonymous conversation
D. Non-human form of communication

23. Huge advances seen in technology has led to the growing dependence on non-human forms of communication. This, according to the author, has resulted in
A. Watching television too much.
B. an increase in the number of shy people.
C. making face-to – face communication necessary.
D. the difference between Japanese and Israelis cultures.

24. In which of the following counties has culture been reported to have the least effect on an individual's shyness?
A. Japan B. Taiwan C. Israel D. United States of America

A. Completion (25-41)

DIRECTIONS: Questions 25-41 are incomplete sentences. There are four alternative words, A-D, given below each question. Choose the word that best completes the sentence

25. Let us_____ the decision until next week. There could be unexpected changes.
A. Pass B. by pass C. defer D. differ

26. Can you go to the _____ and ask her if they have a pen? I lost mine this morning.
A. station B. stationer C. stationary D. stationery

27. Is it true that your friend was_____ from hospital? That is really good news.
A. relieved B. released C. sent out D. discharged

28. Writing a formal letter in English is not difficult, but you should follow certain_____
A. convictions B. conversations C. conversions D. conventions

29. Newton was a great English scientist. He _____ the law of gravitation.
A. invented B. discovered C. created D. declared

30. Ato Bisrat has been_____ chairman of the new committee.
A. appologized B. appreciated C. applauded D. appointed

31. I_____ them that there had been financial problems earlier in the year.

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A. told B. explained C. revealed D. requested

32. It is a sad sight to see a child_____ with polio.
A. Affected B. afflicted C. affiliated D. affectionate

33. I was _____when I heard the news of her mother's death.
A. ashamed B. impressed C. surprised D. shocked

34. Please_____ yourself from spitting in public places.
A. consign B. restrain C. restore D. contradict

35. You can't go out after mid-night in this town. A state of emergency has been_____ over the last few days.
A. declined B. dropped C. declared D. released

36. Many students in my class were_____for poor attendance.
A. Praised B. legalized C. Penalized D. fined

37. A witness was able to _____ my story that the accident was not my fault.
A. sustain B. suspect C. supplement D. substantiate

38. My parents will be_____ if I get into trouble with the head teacher again.
A. serious B. anxious C. furious D. curious

39. I passed all my exams. This made my mother_____
A. upset B. proud C. annoyed D. anxious

40. He is a very kind person. He gets satisfaction by_____ others' burden.
A. lighting B. slighting C. lightening D. enlightening

41. Do not think we are completely safe. Dangers are still_____.
A. eminent B. imminent C. apparent D. evident

B. Substitution(42-50)

DIRECTIONS: Questions 42-50 each has an underlined word or phra There are four alternatives A, B, C and D given after each sentence. Cho one word or phrase that best keeps the meaning of the given sentence substituted for the underlined word or phrase

42. She **waited on tables** to earn money while she was in college
A. served meals C. painted tables
B. cleaned tables D. sold tables

43. **He turned an excellent essay** in on American Civil War.
A. submitted B. narrated C. read D. edited

44. I don't know how we are going to **cope with** such an increasing cost living this year.
A. explain B. win C. manage D. expect

45. It seems she **turned down** his idea of going out with her. What a shame!
A. hated B. disclosed C. rejected D. postponed

46. I don't **get along** with most o f my classmates.
A. go B. meet C. play D. agree

47. Mary and her sister have been **at odds** with each other for five years. It it is time you intervened.
A. in confusion C. in contrast
B. in unclear terms D. in disagreement

48. I saw the horse **loose** in the field. You should tell me who **did that.**
A. hungry B. free C. lost D. tied

49. In most cultures, the marriage of a man a woman is **accompanied** by some kind of transfer of goods or services.
A. Measured B. known C. followed D. judged

50. It looks as if the whole building could come **tumbling down** at any movement.
A. crack B. demolish C. fall D. move

DERCTIONS: Questions 51-80 are incomplete sentences. There are four alternative words or phrases, A-D, given below each question. Choose the world that best completes the sentence.

51. I turned round to find a man _____a knife at me.
A. Point B. Pointed C. points D. pointing

52. It was agreed that the football match_____
A. be postponed C. Postponed
B. postpones D. was postponed

53. All my friends wanted coffee so I_____ them some.
A. will make B. make C. could make D. made

54. I have searched all over and _____could it be found. Someone must have thrown it out.
A. nowhere B. everywhere C. anywhere D. somewhere

55. I hope you will visit us next summer?_____
A. will you B. won't you C. don't you D. are you

56. It is disheartening that the whole class is away to Lalibela. It means_____is coming for my birthday party.
A. anybody B. everybody C. nobody D. somebody

57. Neither your relatives nor you_____allowed to leave at the moment. Things should clear up.
A. is B. are C. be D. have

58. It has come to my attention that none of them_____ died in the accident.
A. has B. have C. is D. are

59. What they are doing is not relevant_____ the instructions in the book. Better tell them to stop.
A. of B. about C. to D. with

60. My holiday was never_____. I could use every minute for something enjoyable.
A. boring B. worrying C. annoying D. thrilling

61. I haven't seen it, but I could imagine that the Renaissance Dam is _____ than other dams we have in the country.
A. very bigger B. absolutely bigger C. much bigger D. more bigger

62. Has the plane landed_____? People are waiting outside to welcome him.

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A. now B. still C. yet D. till now

63. We have lived in this area since childhood but we_____ seen anything like that before.
A. have never B. haven't never C. have ever D. haven't ever

64. I hope we_____our tenth round by June. We have already done seven now.
A. Have completed C. will complete
B. Should complete D. will have completed

65. I_____ hard for this exam since September and, I think, I did the right thing.
A. had really worked C. really worked
B. have really worked D. will have really worked

66. Unless you carefully_____ the matter, you won't easily find out what is going on.
A. look at B. look into C. look up to D. look upon

67. Rosa dropped_____ to collect her book. She was so cooperative.
A. by B. on C. about D. around

68. We are putting up a notice three weeks in advance_____you have enough time to process your travel.
A. so as B. in order C. so that D. so as that

69. It is now generally accepted that in the future robots will take over many of our tasks, especially the more___ type of jobs.
A. repetitive B. repeating C. repetition D. repeated

70. I think my father_____ in this college since 2000.
A. teaches B. taught C. is teaching D. has taught

71. The telopone_____. Can you answer it, please?
A. can ring B. is ringing C. rang D. has rung

72. My English teacher usually wears jeans, but today he_____ a suit.
A. Wears B. wore C. is wearing D. has been wearing

73. My house_____into last night for the second time.
A. broke C. was broken
B. broken D. has been broken

74. This question is a little_____the next one.
A. less difficult as C. as difficult as
B. difficult than D. less difficult than

75. If I_____ time, I will do some shopping.
A. had B. have C. will have D. am having

76. If all_____well, she should be here before it gets dark.
A. go B. went C. goes D. had gone

77. What _____ it actually like to wake up one day with more money than I can imagine?
A. does B. has C. is D. do

78. Do you find_____difficult to meet new people?
A. them B. it is C. they are D. it

79. She spends hours in her room_____to music.
A. to listen C. listens
B. having to listen D. listening

80. A room _____ for you at the Hilton Hotel.
A. has reserved C. reserves
B. has been reserving D. has been reserved

DIRECTIONS: Questions 81-100 are presented in a form of dialogue. The part said by one of the speakers is given, and a blank space is left for the other. For each of the blank spaces, four alternative answers, A-D, are given. Choose the alternative that is most appropriate to complete the dialogue

81. **Samson**: How much coffee do you drink?
Tewodros:_____
A. At least five cups C. I like coffee
B. Only after lunch D. I don't like strong coffee

82. **You**: How old are you?
Lidet:_____.
A. Twenty-two
B. My sister is 18
C. Old enough for marriage
D. I'm the youngest in the family

83. **Mother:** I' m sick and tired of winter.
Daughter: _____
A. Good idea.
B. Don't worry! It'll be spring soon.
C. Do you want to eat out this evening?
D. Don't you also like going to the cinema?

84. **Secretary:** How can I help you?
Student:_____
A. What is the director doing?
B. Can I use the Director's office?
C. Can I talk to the director, please?
D. By showing me the Director's office.

85. **AKalu:** What was the party like?
Lulu: _____
A. I think, it was OK.
B. It started earlier, didn't it?
C. Do you think people liked it?
D. About a thousand people were invited.

86. **Bachere:**_____
Galgalo: If I were you, I would go home and take rest.

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A. What a bad headache. It is like migraine
B. How about going home? I have a bad headache.
C. I have got a bad headache. I have had it since morning.
D. What a bad headache. I have never had this kind before.

87. **Bacha**:_____
Bachu: Sure, he is packing.
A. What job does your brother do?
B. Is it true he is moving back here?
C. Is it good for him to do a packing job?
D. What is your brother doing these days?

88. **Benti**: This is the Toyota car we bought last week. Isn't it beautiful?
Nani:_____
A. Who bought it? You?
B. What? Is that really your car ?
C. Oh no! I know you can't afford it.
D. What a great car! Really exciting!

89. **Ujulu**:_____
Ariet: Just keep quiet. They say it is his headache again.
A. Getachew has been in bed all day. Anything worng?
B. Getachew didn't greet me today. Did I do anything wrong?
C. Getachew is not his usual self today. He speaks angrily to everybody.
D. Have you seen Getachew today? Something seems to be upsetting him

90. **You**: _____
Yoseph: It is around Arat Kilo, next to Jolly Bar.
A. Is there a post office near here?
B. Where is the nearest post office?
C. Sorry. How can I get to the nearest post office?
D. Excuse me! Does it take long to the nearest post office?

91. **Tolessa**: Would you like to eat out this evening.
Berhe:_____
A. Have you been paid already?
B. Sorry. I would rather revise for my exam
C. Restaurants are very expensive these days.
D. Do you remember the meal we had last time?

92. **Aleltu**:_____?
Thomas: Sorry. I am busy with my assignment.
A. Have you seen the report I compiled?
B. Could you help me organize the report?
C. Have you finished doing your assignment?
D. Would you read the report when you are ready?

93. **Mother**:_____?
You: I put in just a little.
A. Did you put in salt
B. Why did you put in salt
C. When did you put in salt
D. How much salt did you put in

94. **Mother**:_____?
Daughter: Well, I placed the egg carefully in the jar.
A. What did you do next
B. Who placed the egg in the jar
C. Did you place the egg in the jar
D. When did you place the egg in the jar

95. **Stranger**: Excuse me,_____
You: I' m a stranger here myself
A. You tell me where the post office is
B. Tell me where is the post office?
C. Do you tell me where the post office is?
D. Can you tell me where the post office is?

96. **You**: Hello. Where do you live?
Stranger:_____.
A. I' m from Jimma
B. Around Arat Kilo
C. I live with my sister
D. I come by bus

97. **You**: Are you married, Zahara?
Zahara:_____
A. Before three years
B. My husband's name is Jemal
C. I'm 30
D. No, I'm not

98. **Customer**:_____?
Shop Assistant: It's 1500 birr.
A. Whose is this camera
B. It this camera for sale
C. Do you have a camera
D. How much is this camera

99. **Lilu**: Do you speak Italian?
You:_____.
A. Just a little
B. I speak five languages
C. I speak Arabic, too
D. I've never been to Italy

100. **Mulat**: _____?
You: Not enough to pay my bills
A. Do you earn much
B. How much do you pay
C. How much do you earn
D. Is it enough to pay your bills

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DIRECTIONS: The following questions are related to different types of writing. Read each of them very carefully and choose the best answer from the four alternative answers. A-D given.

101. Many people are less prejudiced against foreigners than they used to be in the past.
This is most likely taken from a piece of writing that is
A. descriptive B. argumentative C. expository D. narrative

102. Out in the office behind his house, Lemma is sweating over his newspaper column. His feet are propped up on his desk, and his computer keyboard sits on his lap. This is most likely taken from a piece of writing that is
A. descriptive B. argumentative C. expository D. narrative

103. Coffee was originally used as a food and a medicine by the Arabs, who cultivated it 1,400 years ago in Yemen. Its use as the drink we know today only began to spread after the year 1400. This is most likely taken from a piece of writing that is
A. descriptive B.argumentative C. expository D. narrative

104. Which one of the following sentences is most likely taken from a descriptive writing?
A. It is very tough to get to the tip of Mount Kilimanjaro.
B. At the tip of Mount Kilimanjaro, it is windy and very cold.
C. Many people think Mount Kilimanjaro is in Kenya but it is not.
D. It took us longer than we expected to get to the tip of Mount Kilimanjaro.

105. Which one of the following is a line most likely taken from a narrative piece of writing?
A. The cause of the uproar remains a mystery.
B. Quite a lot of them died suffocated in the crowd.
C. I don't know how many shops were burnt down that day.
D. When the clock struck seven, people started yelling and shouting.

106. Dragged out of their hospital bed in Bangui, the patients were brutally kicked and beaten until they fell unconscious. Crouched in a corner about 30 meters away from the scene, I was watching what was going on. The all of a sudden, a young man of about 23 pulled out a machete... This is taken from a piece of writing that is most likely:
A. narrative C. expository
B. argumentative D. descriptive

107. Why should the poor pay as much as the rich? First, the poor is already poor and nothing is to be drawn out of poverty. Second, there is also moral question of whether it would be just to add another burden to a group already saddled with... This is taken from a piece of writing that most likely:
A. descriptive B. expository C. narrative D. argumentative

108. Everybody was unusually punctual. There was, however, someone very important that was still expected-the driver. While everyone was eagerly waiting to leave the campus, Samuel's phone range. It was the driver. ... This is taken from a piece of writing that is most likely:
A. narrative B .expository C. descriptive D. argumentative

109. Which one of the following is a line most likely taken from a narrative piece of writing?
A. Altogether, he was extraordinarily eloquent.
B. Then the speaker stood up asked everyone for attention.
C. He was so simplistic in his approach and influential in his argument.
D. He had the art of disarming you of your opinion and making you join his bloc.

110. Which one of the following sentences is most likely taken from an argumentative piece of writing?
A. Their price depends on their capacity
B. This is much faster than its predecessor
C. This one transports 500 of them every minute
D. This is a new version of the previous ABB machine

111. Which one of the following words is WRONGLY spelt?
A. arguable B. changable C. dyeing D. cognisance

112. A note of greetings to an old school friend most likely ends in:
A. Sincerely B. Faithfully C. Love D. With regards

113. The phrase I regret to inform you that..... Is a possible introduction used in a letter to someone who:
A. should settle her/his bill.
B. is advised to collect an item.
C. is required to sign her/his contract.
D. did not succeed in a job interview

114. Which of the following sentences is correctly punctuated?
A. Robert Frost never finished college, but he became a celebrated poet
B. Robert Frost never finished college but, he became a celebrated poet.
C. Robert Frost, never finished college, but he became a celebrated poet.
D. Robert Frost, never finished college but he became, a celebrated poet.

115. Take care and thanks again. This is most likely taken from
A. the ending of a formal letter
B. the beginning of a formal letter
C. the ending of a letter to a friend
D. the beginning of a letter to a friend

116. We would love to come. This short extract is taken from a letter which
A. Invites C. accepts invitation
B. gives news D. asks for further information

117. Which of the following sentences is correctly punctuated?

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A. " The frog," states an African proverb" does not jump without a reason."
B. " The frong, "states an African proverb, "does not jump without a reason".
C. "The frog," states an African proverb, " does not jump without a reason."
D. The frog," states an African proverb" does jump without a reason."

118. I hope you are all well. This is most likely taken from
A. the body of a friendly letter
B. the ending of a formal letter
C. the beginning of a formal letter
D. the beginning of a friendly letter

119. Which of the following words is correctly spelt?
A. questionnxaire
B. resemblance
C. extentsion
D. environment

120. Tattoos are put in place by means of a tattoo gun, which has a cluster of needles which jab in and out rapidly. This is most likely taken from a piece of a writing that is
A. descriptive
B. argumentative
C. expository
D. narrative

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Section One: Word Order

1. B
Sub + verbal phrase + obj + verb + complement
They wanted to make me do everything for me

2. C
Sub + link verb + adv(pre-adj.modifier) + adj + imperative
Life is too short; enjoy it.

3. A
Sub + be + v-ing + place adv. + time adv.
I am going to Awasa next weak

4. D
Sub + verb + comp.(adj) + tag question
She is beautiful, isn't she?
- positive statement has negative tag question
- Negative statement has positive tag question

5. A
Sub + v-to have + action verb + adv + action verb (comp)
We have to work hard to succeed

6. B
Sub (Ind. Pro) + perceptive verb + question word + sub. + comp
No one knows why she doesn't come

7. A
Sub + verb + obj. + adv. of manner
The burglur left the window wide open

8. C
Subject – verb inversion to form question with verb to be.

9. D
Sub + v-to have + action verb + obj. + adv. of time
Subject have to see you soon

10. C
Sub + modal verb(neg) + state verb + obj. + adv. of degree)
I cannot tolerate him any more

Section Two: Paragraph Coherence.

11. B
A is the topic sentence. E, B and C are developers they develop the idea under the topic sentence providing argumentative opinion. D concludes the paragraph providing the expectations.

12. A
The topic sentence is E. A, C and B are providing information step-by-step respectively. D winds the steps by providing information about what to do at the end.

13. B
The topic sentence is E. B, D and A provides the procedures respectively. C concludes the paragraph winding the steps.

14. A
The topic sentence is C. B, E and A are developers to the idea under discussion. D concludes the paragraph winding the idea in the preceding sentences.

15. C.
The topic sentence is E. A, C, B and D comes one after the other providing the sequential order of the routine work under discussion.

16. A
E is the topic sentence. D, B and A are developers to the issue under question. They elaborate the issue sequentially. Finally, C comes to wind up the paragraph.

17. D
A is the topic sentence. E, C and B are developers and/or supporters of the idea under the topic sentence successively. Then D, comes to wind the paragraph by providing a concluding remark.

18. D
B is the topic sentence. Then, E, A and D are respectively developing the idea under discussion. Finally, C comes to conclude the paragraph.

19. A
C is the topic sentence E, B, and D supports the topic sentence providing information respectively.

20. D
The topic sentence is D, E, C and B provides supporting information to the idea under discussion respectively. A concludes the paragraph.

Section Three: Reading comprehension

Passage I

21. C
The time when the article was written on the source reads; "Taken from Time English, September 1997".

22. D
The whole paragraph of Nine(9) explains the cause and effect that children growing seriously into alcohol and drug addicts.

23. D
Paragraph three (3) lines 4 – 8, justify this.

24. C
The phrase in paragraph 7 the last line that reads; "…their products", refers to "alcoholic drinks".

25. D
Nothing has been said in paragraph 2 about what will happen in 2030, in the future.

26. A
The cost of treatment for the effect of impact on the economy is not entertained in the passage. Hence, B, C and D are talking about cost of treatment which is non-existent in the passage.

27. D
As indicated in paragraph 1, governments failed to meet the UN-set targets for the year 1990s against drug abuse.

Passage II

28. D
Paragraph three(3) the last part indicates this.

29. C
As indicated in paragraph 4, the contextual meaning of the word "smell" is "informing".

30. C
Most probably the answer to the question is C. Because "Hyena" barks down to the ground. Yet, sound waves travel far and fast on solid maters than on air.

31. B
Paragraph 5, the concluding statement confirms the answer.

32. D
The author intends to provide us with a likely explanations.

33. A
Paragraph 2, lines 6 – 7 that reads; "Its voice has great strength …" justifies the answer.

34. D
Paragraph 1, the whole argument expresses that at the moment the writer was unable to decide whether humans or animals are better communicators.

Section Four: Vocabulary

A. Completion

35. B
A is out of the context and it means joint on the hand. C is noun, but the context requires adjective. D is adjective but it doesn't fit the context.

36. A
The phrase "illegal immigrants" which is used as a noun subject requires past participle adjective to complete the passive expression.

37. A

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The definite article "The" at the beginning of the statement requires noun.

38. C
Entry parallelrily contrasts exit.

39. C.
It is a common understanding that treatment by a traditional doctor usually worsens or complicates health condition.

40. B
The word "negotiations" implies that some one has accepted something.

41. B
The negative expression "couldn't" and "they had to go their own ways", imply that disagreement is created between the two.

42. C
The show that enjoyed someone very much was artistically presented.

43. C
The closing phrase that reads; "stay calm" implies that someone is in a state of worry.

44. C
The phrase "lips are sealed" is a figurative speech that means promised not to utter a word of one's secret.

45. C
The phrase that reads; "... trusted him before I found out ..." implies that the speaker is deceived and disappointed. The word has similar meaning to the figurative speech "snake in the grass" which means disceptive.

46. C
The phrase that reads; "... not to go bankrupt in business, ..." implies the need for information on business.

47. C
The prime minister appoints his cabinet ministers on the mandet he was given by the country's constitution.

48. B
To run twenty miles a litre is too big to someone. This implies that the car consumes too little amount of fuel. So, it is economical.

49. B
To use various means for a police to investigate or elicit information from criminal is natural.

50. C
"Activate" means make ready to use. To make ready for use one has to first turn on the camera.

51. C
Blood is usually donated to users through blood banks to those who are badly need of it.

52. C
"Take over", means to receive the burden, the challenge or responsibility of someone to discharge it according to the provider.

B. substitution

53. A
In this context "far – sighted" means to have the ability to solve problems which is useful or beneficiel to the crowd.

54. A
To take one's remarks in "good part" means to take it with no hesitation and consider as valuable.

55. C
To "run through", a poem here means to practice it with a quick move from the beginning to the end.

56. B
"Priceless", here means extremely important, and it is something that can't be measured with price. It is beyond money can express it.

57. D
"Deliberated up on", contextually means treat or discuss the issue in a careful and serious manner.

58. A
"Decisive measure", contextually means quick and effective measure.

59. D
"Shadowed" here means intentionally and deliberately ignored by someone to give protection or coverage.

60. C
"Immaculate behavior", contextually means perfect, clean and tidy which in turn means unspoiled.

61. B.
The phrase "come about" means something happened un-expectedly.

62. C
The phrase "count on" in this context means depend on or rely on somebody for something.

Section Five: Grammar

63. A
"No, I don't" is short and precise answer to the question and it is complete refusal.

64. B
The connector "and" requires the pronoun both to mean, the two together.

65. C
"yet" in the question implies that the action sought hasn't been completed. So, the response that shows un-attainment of the action is "No, they haven't."

66. B
News of "sudden death" naturally terrifies or shocks the listener because it affects the feeling.

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67. C
The question word "Why" implies suspicion. So, no one appeared to the party.

68. A
The context tells us that there is no need to worry as the action has been usual. So, he has never been kind to the reporter.

69. B
If something has been "stupid mistake", therefore, we need to avoid it altogether.

70. A
Proper way of responding to an invitation provided by one's beloved.

71. C
The phrase "neither of them" requires singular verb. A and B are plural so they don't agree with the noun phrase. Though D is singular, the context in verb to have requires V_3.

72. D
The context in the sentence implies that the speaker is annoyed of calling the name of a person referred to. So, the speaker is accusing the person whose name is repeatedly called for not responding to it.

73. A
In conditional sentence I when the if – clause is simple present, the main clause would be future. In this context, the probability of getting job is less certain. So, "will save" contextually fits the sentence.

74. C
The preposition "on", contextually fits and best completes the sentence.

75. A
The speaker tells us that he has never been dependent since he was ten years.

76. A
B and D are ungrammatical. C doesn't fit the context because 'regret' means regret or apologize or ask forgiveness.

77. B
When + V_2 in the subordinate requires simple past/past conditional in the main clause. Here, past ability is expressed.

78. B
Except B, all other alternatives are ungrammatical.

79. B
Contrasting ideas with present continous imply future actions.

80. B
If someone is completely scarced of money. He has to borrow it from somebody else.

81. B
In this context, the pattern of the sentence "so as" requires full infinitive (to+V_1).

82. C
The phrase "since yesterday" requires present perfect.

83. B
One of the functions of present perfect is to express an action completed at the moment of speech.

84. B
The expectation of the speaker has to be completed by the time of the speech.

85. A
"By + simple present' in the subordinate clause requires future perfect (will +have + V_3) in the main clause.

86. D
The time phrase "As soon as", implies prompt action to follow.

87. D
Future perfect usually has the pattern that is "will + have + V_3 + by + future time adverb".

Section six: Communicative Activities

88. B
Proper way of providing helpful suggestion for someone who is in problem and wondering how to accomplish his task.

89. D
The father's response implies that the son's request is something that needs the mother's agreement.

90. A
Proper way of asking somebody the reason for not doing some action.

91. A
Formal and acceptable way of expressing one's pleasure to a success.

92. D
Proper way of giving argumentative opinion on an issue under discussion.

93. B
Passive or mild way of expressing one's refusal and/or opposing a request.

94. C
Proper way of directing someone to get somewhere on request.

95. D
Proper way of greeting someone who meets for the 1st time and getting introduced oneself to.

96. C
The teacher's suggestion implies that the student would rather had done it bit-by-bit and should have submitted his assignment on time

97. D
The husband's request was to seek some kind of appreciation from his wife. However, the wife's comment is rather disappointing. She belittled the quality by providing that the previous ones are of higher quality.

98. C

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Proper way of providing one's opinion about somebody that someone is not

or certain of it.

Conversation between friends.

99. A
The next word in the conversation implies that it is about marriage.

100. B
The flow in the next conversation again implies the answer.

101. D
The preceding statement justify this

102. B
"Hilton" implies where the wedding party would be.

103. A
Proper way of suggesting the better alternative where the choice being

couples.

Section Seven: writing

104. D
In A, "e" is wrongly used.
In B, the second "t" should have been "s".
In C, "i" is used instead of "a".

105. C
- In A, semi-colon is used wrongly instead of comma.
- In B, the order of the words is wrong and the comma is used wrongly.
- In D, comma should have been used immediate after Bob.

106. B
The formal and appropriate opening of application letter.

107. B
In spelling rules, "e" usually comes after "C" and followed by "i"
some exceptional words like "society" where "i" comes before "e"

108. A
Instruction is neither a forceful order nor a mild request. It is rather
of fact about what one has to do for his own sake.

109. D
- In A, quotation marks after "Ambo" and before "we" are missed.
- In B, quotation marks at the beginning and a comma after "A
  missed.
- In C, the period or full stop is inside the quotation mark.

110. C
Expository writing seeks to inform, explain, clarify, define or instruct
issue under discussion.

111. C
The statement that reads; "Having looked at all these, we
solution", implies that something has already been discussed and
to be forwarded next.

112. A
Descriptive writing vividly portrays a person, place, or thing in such away that the reader can visualize the issue under description.

113. D
- In hotels, people book for room, but usually use the room carelessly leaving trashes in the bed room. In that case, owners usually request people who book rooms in the hotel to leave the room neat and tidy.

114. A
Opening of a letter for a close or intimate friend is usually informal. This is to express one's love friendly and intimacy.

115. B
Academic writing should explain facts based on proof.

116. A
An applicant who has not been choosen for an employment should be informed politely with a formal and in non-offending way that the application is unfortunately not accepted for the reason that would be stated with justifications.

117. D
Argumentative writing usually use comparatives. In this case, "weaker" implies that the piece of writing is argumentative.

118. A
Descriptive writing vividly portrays a thing in such away that the reader can visualize the issue under discussion.

119. A
- In B, we need a full stop, after "items".
- In C, we need comma, after "pen" and colon after "eraser".
- In D, we need comma, after "pen". But in A, we need hypen after "one thing".

120. D Narrative writing sometimes expresses point of views' of speakers.

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Section one: Word order (1-6)

1. **D**. The sentence ahs two parts: subordinate close and main clause. Following is the word formation
when + subject + verb, subject + verb + object (complement)
↓ ↓ ↓ ↓↓
When it rains, trees turn green

2. **B**. The presence of external obligation (have to) needs infinitive. The conjunction 'and' connects two equal forms (verbs "go" and "get"). Besides, the adverb like 'immediately' comes after the verb it modified. Thus, the formation can be
subject + have to/has to + V_1 and adv + and V_1 + obj,
↓ ↓ ↓ ↓ ↓
you have to go immediately and get him

3. **C**. This is a formation of question that seeks information mote than. 'Yes or no' answer such kind of question is formed like below:

When When How Why	+ auxiliary verb	+ subject	+ verb	+ (complement) modifier
↓	↓	↓	↓	↓
Why	do	people	need to go to	college

4. **A**. The word order for imperative is: Action verb + object (complement). The choice under choice 'A' is the best order for imperative sentence.

5. **A**. Normally, the adverbs of concession have two positions in a sentence: At the beginning or at the middle.

Although Though Though Or Sub	+ subject + verb + obj	although though even though	+ sub + Verb + obj, sub + Verb + obj + sub + verb + obj

6. **C**. "mind your own business" meaning that it doesn't concern you and it should be come together. It is possible to use only at first for emphatic purpose or recommendation.

Section Two: Paragraph coherence (7 – 14)

7. **A**. we can use a variety of techniques for coherent paragraph development. One of which is using guiding questions. Now let as sue this technique to close to the best order. Who is the subject whom we age talking for? (peter, the painter). Where does he work? (in a studio). How and when did he get the studio? (His father bought for him as a graduation gift. . .). What is he doing at the moment? (painting a portrait).
8. **C**. The connective words or transitional words (First, then, too) help as to rearrange the best order. The topic sentence is that they went to the zoo. First, they saw monkeys, three in number. Then they saw an elephant and other animals, too.
9. **A**. The topic sentence of the paragraph is the number of joints a human body has (C). In many places (in the human body) two bones come together (d). The demonstrative pronoun "these" (a) refers to the paces of the two bones. The knees (e) are the examples for joints for walk and bend.
10. **A**. After identifying the topic sentence, (b) we should focus on the pronoun (this) and other connective (transitional) words- like, "also" since they pave the road for best coherence. The pronoun "This" refers the antecedent of the extremely hot and scorching weather. The word "also" under choice "C" tells as the presence of additional information that made the village dull and sleepy. The rumors about the approaching enemy gives light what might come next.
11. **D**. The connective (transitional) works like "for example, just newly, in fact, however, at that time "illuminates the way to reconstruct the coherent paragraph. Sentence b is the topic sentence. The pronoun "It" (sentence 'e') scramble for Africa. The word "others" (sentence 'e') is exemplified by Britain (Sentence "d).
12. **A**. The topic sentence is sentence "a" about the various reasons why people gamble sentence "b" is the major reason why people gamble is to make a lot of many sentence "e" is the justification for sentence "b" sentence "d" is also the other reason. Sentence "c" is the clinching sentence.
13. **C**. The issue that is discussed in all sentences is "global warming." The topic sentence is sentence "a", ". . . about this major issue" (sentence 'd') refers to the global warming that stated in sentence "a". The transitional word "However" (sentence 'c') shows the opponent view about global warming. Sentence 'e' reflects the major tenet of the opponent view.
14. **B**. Let us first spot the topic sentence as usual. All except sentence "e" cannot be topic sentences since they (sentence 'a', 'b', c, d) represented by pronouns that refers the antecedent mentioned before. The pronoun "they" (sentence 'a') refers 'children of poor people that is mentioned under sentence e (the topic sentence). Sentence "b" comes after sentence "d" since it is the result of (consequence) the idea mentioned in sentence 'd'.

Section Three: Reading Comprehension (15 – 28)

15. **C**. According to the passage, the word "that" in line 2 refers to understand what people think.

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16. **C**. "That way" in line 6 refers too talking about what people like.
17. **B**. According to the passage, to make people do what we want, the writer advises us to ask them a lot of questions (not few) and get them saying 'yes'.
18. **C**. The phrase "this job" in line 11 refers to getting acceptance by others by approaching their weak side. Hence, selling cars is nearer to this idea than other alternatives.
19. **D**. According to the passage (paragraph 1, line 9), the word "make up" has a similar meaning with 'constitute' which means the amount to be the components of something. For example, twelvemonths constitute (make up) a year.
20. **C**. The word 'evidence' in paragraph 1, line 10 can be replaced by Research without any meaning difference
21. **B**. According to paragraph 3, Line 15, repetition in dreams signals things on the dreamer's mind.
22. **C**. By referring paragraph 3, lines 7 and 8, people who are very aware of colors can remembers more colors in their dreams while awake than who are not aware of colors.
23. **D**. As stated in paragraph 2 and paragraph 4. since dreams often reflect the day-to – day concerns and worries of the dreamer, they help us understand the life of the dreamers including ourselves better.
24. **C**. "The phrase" Either way' in paragraph 4, Line 4, refers to the option (alternative) of traveling by any of the car types mentioned.
25. **A**. According to paragraph 4, lines 11 and 12, Dreaming about teeth signifies the power and control.
26. **A**. As stated clearly in paragraph 4, lines 13 and 14, dreaming about death implies the end of one thing and the beginning of another things in life.
27. **D**. From the alternatives given, choice "D" is stated in line with the idea in paragraph 2. the other choices do not reflect the information stated in the passage properly.
28. **D**. Paragraph 1 deals with the types of dreams (dreams in deep sleeping and REM), paragraph 2 reflects the dream analysts' belief, paragraph 3 is talking about different question about dreams including dream and color. Paragraph 4 about how to interpreter dreams. Therefore, choice 'D' is the correct order points.

Section Four: Vocabulary (29 – 54)

A. Completion (29 – 45)

29. **C**. The adjective 'sharp' needs the noun form of the next word. Choice refers to a kind of cereal (rice) that does not tell us the situation of high living. The noun form of 'raise' refers to the increment in a salary (an increment). The noun form of 'rise' implies increasing of cost of living (price of goods).
30. **B**. All words of the adjectives mentioned shows (indicates) the appearance human beings especially female's appearance. All except 'slim' have negative connotation about the subject they modify it and they cannot be used beautiful. The word 'slim' indicates positive implication that can go together with beautiful.
31. **A**. This reminds us the proverb "birds of a feather flock together." The speaker and the person whom the speaker is talking about have opposite behavior that deters them from intimacy. This situation can best be expressed with repels.
32. **B**. the adjective 'efficient' signifies the ability to perform duties well with respect of time, quality and money. The word is used for persons. Eg. An efficient secretary. The adjective 'effectual' has similar concept like that of efficient but if cannot be used for persons. It particularly refers to bringing about the result required and answering its purpose in respective of time and quality. The words effective signifies the ability to bring about the result intended by focusing on the means and the process. The word affective shows the influence of something on an other.
33. **C**. By looking on teeth, we can tell or compass with seething to show how smart or bad it is. Native speakers of English language compare teeth with milk. However, Amharic Speakers sometimes compass teeth with show.
34. **C**. the idiomatic expression of "a drop in the ocean/ bucket" is used for show how insignificant the a mount of the matter as it is compassed with the required.
35. **B**. The noun form of 'rout' means disorderly retreat. We do not use its verb form since the given sentence requires noun form. The word "route" refers to the way taken/planned from one place to another choice A and D are not appropriate for the context
36. **C**. The adjective "convincing" refers to making somebody feel certain or cause to realize. Whereas, "confident" refers to someone's feeling about the certainty of an event. The noun 'confidence' can not be appropriate for the context since the sentence needs adjective as a complement. The word 'confidential' implies inclination to give confidence to someone or having the confidence of others. The context shows the certainty of the speaker about her arrival on time (confident).
37. **D**. The situation tells us there is a gap of expectation of the speaker before and after watching film. This made the speaker unhappy since he did not get what he hoped for (disappointing). Interesting and exciting can not be sued since they do not go together with the second sentence. The word discouraging refers preventing somebody from doing something. It the speaker was discouraged by the film, he would interrupt his watching.
38. **B**. Since English language is the medium of instruction for university education, the language proficiency of the student is a prerequisite for his/her academic success. This situation can best be expressed by indispensable.
39. **C**. let alone going to bed, thinking about it considered him as foolish and unreasonable. This implies how tensed up he was before the exam. For him, it is ridiculous (deserving to be laughed at). Meticulous means giving or showing much attention to the situation. And 'courageous' also refers to fearless

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condition of the speaker to confront with someone. 'Ambiguous' cannot also go with the situation.

40. **A**. The word doubt indicates the hesitation of the speaker to believe in the occurrence of world peace. He/she has a question about the existence of world peace. The word suspect refers to the idea of feeling the speaker has about the possibility (likelihood) of the world peace. The threatening condition of war and seeing total world place cannot go with choice C and D "Doubt" is the best answer.
41. **D**. The word disfavored refers the state of being out of favor. The word condemned shows blaming somebody for his/her wrong doing. The term impressed indicates the positive feeling a person has on an other thing. We cannot expect such feelings (distaoused, condemned and emplaced) from a woman whose beloved has gone away (dead). The term depressed shows her grief.
42. **A**. The phrasal verb 'get on' in this context refers to working or living with somebody in a sociable way. Whereas, 'go on' refers to pass (of time), behave, take place and the like. The phrasal verb 'get off' (not get of) means to start a journey or doing something. "Get across" also shows the causes of something to be understood. Hence, the phrasal verb 'get on' reflects the situation stated in the first and the second sentence.
43. **C**. The noun theft is uncountable because of this we cannot modifies it by 'a few' (used for quantifying only countable noun). The focal point of the discussion is the belongings (personal property), not the thieves. The term "looses" can be modified by 'a few' since it is a countable noun.
44. **B**. It is widely believed that first impulsion has strong effect on the minds others. Hence, the speaker signifies this.
45. **C**. wore off refers to pass away or cause to pass away.
    Wore away means become impaled, thin and weak as a result of constant use.
    Wore down implies becoming gradually smaller, thinner and weaker.
    Wore out refers to exhaust and tire out or in need of rest. Thus, choice 'C' is the most suitable alternative for the given context.

B. Substitution (46 – 54)

46. **C**. the contextual meaning of the word hold implies to have or keep is one's possession.
47. **C**. The phrasal verb 'go through' refers to examine the details of the plan and can be substituted by 'Check'
48. **D**. Broke ranks in this context refers to refusing to accept the hierarchical order of leadership in military. "Explessed disagreement can substitute the underlined words without any meaning change.
49. **A**. The idiomatic expression "take for a ride" refers to cheating someone to get something (money). Therefore, choice "A" is more appropriate than others.
50. **A**. The expression 'a lost cause (not a lost case as it is mentioned in the question) signifies an ambition, project or aim which seems certain to end in failure because not enough people support it. Therefore, "a lost cause" can be replaced by "failure".
51. **C**. The expression 'off color" shows the unwell of the person in relation to health condition. Hence "not very health" can replace the underlined expression.
52. **B**. phrasal verb "put out" indicates being upset or worried of someone by something.
    - Shattered refers to break down of something suddenly and violently into small pieces.
    - Disturbed signified the state of being upset and worried very much.
    - Disappointed shows someone's being sad by not getting what hopped for
    - Discouraged refers to refrain from doing some thing due to various obstacles.

    Therefore, put out can best be replaced by disturbed.
53. **D**. The word retained refers to keeping or continuing to have or hold the cohort position further kept can substitute the underlined word.
54. **D**. The phrasal verb 'turn on' means to start the flow of electric current by turning a switch. Hence, switch on is a synonym of turn on.

Section Five: Grammar (55– 83)

55. **C**. Certain adverbs, mostly with a restrictive or negative sense. Can for emphasis be placed first in a sentence or clause and are then followed by the inverted form of the verb. The best example for this is the indefinite adverb of nowhere. It can be placed at the beginning of a sentence and is then followed by an inverted verb.

    Example: Nowhere will you find better
    Roses than these.

    Anywhere, everywhere and somewhere cannot be used in such inversion form.
56. **A**. The indefinite adverbs somewhere and anywhere can be used in questions.
    The indefinite adverbs, somewhere, everywhere and nowhere can be used in affirmative sentences.
    The indefinite adverb anywhere can be used in negative sentence.
    Therefore, since the first sentence is negative, we can use 'anywhere'
57. **C**. As we discuses before, the word any or its compounds followed negative sentence. Due to this, we should exclude indefinite pronouns 'someone and everyone. Moreover, the phrase of concession is opposite to the main clause, so should use negative indefinite pronoun no one.
58. **B**. The adverb of frequency 'never' is chiefly used with an affirmative verb. It normally means 'at no time'. The adverb of frequency 'ever' chiefly sued in the interrogative. Its meaning is 'at any time'. It can also be used with negative verb and in affirmative comparison. Hence, never is more appropriate than ever for the given context. Choice 'C' and D are not correct.
59. **B**. Distributive pronouns like everyone, each of, many a, one of, either of, neither of, many a, one, either of, neither of' are always singular and followed by singular verb.

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- Verb to have, be and do have special features since they can be used as auxiliary verb and as a main verb.
- Verb to be is used as a main verb
- Verb to be is used as an auxiliary verb in continuous tense and passive formation
- Verb to have is used as a main verb to show possession or belonging.
- Verb to have is used an auxiliary verb for perfect tense formation.
- Verb to do is sued as a main verb to express an action.
- Verb to do is also sued as an auxiliary verb to negative action verbs and question formation for action verbs.

Won let us the relationship of verb to do and verb to have:

Example – Each of the candidate in this have has a pen and a pencil. (positive sentence)

– each of the candidate in this hall <u>doesn't have</u> a pen and a pencil. (Negative sentence)

– <u>Does</u> each of the candidate in this hall <u>have</u> a pen and a pencil? (question)

These are all because the time, the number and subject influence should be placed on the auxiliary verb (not on the main verb).

60. **C**. We do not sue choice B and D (was or were) since the sentence is not passive.
"None of" as a subject can take either a singular or a plural verb depending on the noun which follows it.
None of + mass noun (uncountable noun) + singular verb
None of + unit noun plural (countable noun) + plural verb
Therefore, the pronoun "them" is plural noun so we need plural verb' have.

61. **A**. When we use Either . . . or and Neither . . . nor the verb must agree with the nearest subject

Neither / Either	+ noun nor	or	singular + noun	+ singular verb
Neither / Either	+ noun Or nor	or	plural + noun	+ plural verb

The nearnest noun of the question is "the waiter" needs singular verb. We c not use 'must' because of the presence of "to". If there were "has" as an optio it would be more appropriate than others.

62. **C**. The infinitive "to drink" can be used after the adjective 'safe as part of complement. Hence, choice 'c' is more preferable then others. We can infinitive after some adjectives.
We can use infinitive after some adjective:

It + be + adj (+ for + object) + infinitive with convenient, safe, unsafe, dangerous, difficult, easily, possible

63. **D**. The adverbs of frequency "Ever" is sued in negative sentences which never is used in positive sentence. Our next step is comparing the two negative participles (bored and boring). The present participle 'boring used as an adjective for active sentence. Its meaning is 'having this effect. The past participle adjective 'bored' is passive and mean affected in this way.
As the context shows, she makes boredom the speaker because she never talks with him. So the sentence is active that needs present participle adjective boring.

64. **C**. Being passed (not being pass) is used for present passive for continuous tense
- The present participial clause (v – ing) usually comes first to explain something or to say why somebody does something.

65. **B**. Comparison has been used in three forms.
- when we want to compare two equal things, we use positive degree/ as + adj/ adv + as)
- when we want to compare two unequal things, we use comparative degree adj / adv + er + than (if th eadj/adv is mono or disyllabic word) more/ less + adj/ adv + then (if the adj/ adv is more than two syllables).
When we compare one thing with more than three things, we use superlative degree.
  - The + adj/adv + est (if the adj/adv is mono or di-syllabic word)
  - The most /least + adj + adv | + (of all)) if the adj/adv is more than two syllables)
- Thus, one question is compared with many other questions need superlative degree. Choice 'B' is an appropriate answer.

66. **C**. when two sentences joined by 'but', the idea of the two sentences is opposite. The adjective of quantity "little" is used in the first sentence, so we use the opposite of this adjective in the second sentence (inch).

67. **C**. "yet means' up to the time of speaking. It is meanly used with the negative sentences. Since the situation (action) started some time before and is going on up to the time of speaking (2 o'clock), it needs present perfect tense.

68. **A**. since means 'from then to the time of speaking. It is used a point or period in time. Since is always sued with a perfect tense and with a clause.
Example: I have worked here since I left phrases and classes with since and every since is also used with perfect tense in end position.
The time of marriage is the starting time and true up to the time of speaking.
The context needs future perfect tense.

69. **B**. The future perfect tense refers to an action which will begin sometime in the definite time in the future. Its form is 'will + have + V_3"

70. **B**. BY now, her finger is bleeding. The cause must have been happened sometime in the past (happened before the time of speaking). The speaker may not know the causes of her finger bleeding due to this, the speaker should focus

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on the action rather than the doer of the action. Therefore, passive formation is appropriate (be + V_3)

71. **B**. If two actions happened one after the other in the past, the first action is expressed by past perfect tense and the second action is expressed by past perfect tense. Past perfect continuous is not appropriate for the context given.

72. **D**. The present perfect continuous tense (has/have + been + V ing) is used in the same way as the present perfect tense but it shows the idea of a continuous or progressive action without any interruption (the of time taken to do the action).

73. **A**. "will" is used to express determination, promise, threat, willingness in the first person.
Example: I will give you a nice present (promise)
I will punish you (threat)
I will many her (determination)
- would is to show (express) willingness or determination.
- am going to indicates what is planned, intended and determined
- might can be used to express possibility
The speaker has understood the problem of the person. Having realized the problem, the speaker promised to help the person by lending some money.

74. **C**. We can use "unless" in place of 'if . . . not unless is followed by positive statement as stated in the question. We do not choose 'if' since it doesn't give sense for the given context.

75. **C**. The statement is conditional type III. The form is:

If clause	main clause
Past perfect	conditional perfect
had + V_3	would + have + V_3

76. **B**. ought to is used to express duty, necessity, advice, moral obligation, the recommendation. The advice or recommendation may relate to duty or practical matters or to what is morally desirable
Example: you ought to respect your elders (duty) you ought to help your parents (moral obligation and recommendation).
The given context is not a strong obligation rather it is morally desirable, so ought to is more appropriate than others.

77. **C**. Gradual increase or decrease is expressed by comparatives joined by and:
Example: The weather is getting colder and colder. Worse is the comparative form of bad

78. **B**. The sentence is constructed by conditional type II. The form of condition type II is

If clause	Main clause
Simple past	would + v,

79. **C**. we do not choose choice A, B and D as an answer since 'have' should be followed by V_3 (past participle) or 'to' 'ought' must be followed by 'to'. The sentence is conditional type II. Then, we use simple past in the if clause and would t V_1 for ordinary sentence but the main clause needs obligation.

80. **D**. we use in order to, to, so as to" to express purpose. So that also used to show purpose but it should be followed by clause. In order not to is usually used to show purpose but it should be followed by clause. In order not to is usually used to show/express negative sentence for purpose.

81. **A**. ":Subject + can" implies – it is possible since the circumstances/ situations permit (this is quite different from the kind possibility expressed by may).
"Need not" can be used in obligation (not in possibility of such kind).
Will used for assumption about the present or future actions.

82. **D**. The sentence is a result clause that needs either "so. . . that or such (a/an) . . . that.
There are one adjectives (boring) and one known between the blank space and the word that.
This situation made us choose such (akn-that structure. Since the noun lecture is countable we can use "such a + adj (boring) + noun (lecture that . . .

83. **A**. choice 'C' (because of) and choice 'D' (as a result of) needs noun phrases instead of clause, so they should be excluded from any preference. Reason clauses can be addressed by words like since, as, and because.

Section Six: Communicative Activities (84 – 104)

84. **A**. Girma invited Genet some more sandwiches. This implies she had eaten enough sandwiches before. So she refused the additional sandwiches politely. "No, thanks. I've had enough is the best answer of all.

85. **D**. The question seeks approval. To approve positively, we say 'yes, it is whereas, to approve negatively, we use, No, it isn't. we use the same auxiliary verb that is used in the question to answer yes/no questions, so yes followed by positively while No followed by negative tags.

86. **D**. When we meet someone (people) for the first time, we introduce ourselves like Debebe does. The reply is also introducing the name. If we use the expression of the first speaker, we usually use "too".
We use the expression "pleased to meet you" (I'm glad/happy to meet you) after introducing ourselves to someone.

87. **B**. This dialogue tells us how to ask the direction. The tourist didn't ask where the museum is, so choice "D" is not appropriate. The expression. How can I get . . .' implies the way to reach the museum. For this reason, our reply might be asking the tourists preference – to go to the museum – walking or taking a taxi.

88. **D**. In most cases, the secretary receives telephone call to the manager to lessen the burden of the manager. The caller uses the expression "can I speak to _____ please?" Josie must use choice 'D' because the secretary's answer follows such expression.

89. **A**. parents may help their children to decide which stream is very good. But the student him/herself can reflect his/her preference. So I would rather go for science is more appropriate.

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90. **D**. In order to win someone's heart and to make him cooperate to our business, we should be more polite and systematic. Hawi's response tells as that selam wants tuition fee from Hawi since her father refuses to pay. Of course, the presence of question mark at the end of selam's expression forces us to focus on choice 'c' However, it doesn't go with Hawi's response.

91. **B**. This dialogue is between a wife and her husband. Form this, some can que that the wife may become pregnant. Thus, the husband may advice her to take rest choice A, C and D can not go with the given context.

92. **D**. Since a million birr from lottery is huge money, it creates great surprise whe we hear at the first time. "What?" shows this great surprise. From the give alternatives, choice 'D' is the best of all for given context.

93. **B**. when a player heard that his manager was sacked (plundered or robbe violently) from the journalist at the first time, he wanted to check the source the information. Choice 'c' shows/reflects the situation in a better way.

94. **D**. The topic of discussion is about the greatness of the party. Tolessa wanted know his wife's feeling about the party. From the context given, we expect the wife either agree or disagree with Tolessa's opinion. Choice 'D' fits context in all respects.

95. C. Having an expensive car may not show the richness of the boss since might get it from someone of relatives as a gift. Living abroad does necessarily show his richness. Refuting the idea is expected from the waiter.

96. **C**. we can deduce Gemechu's question from Daniel's response. Daniel descr the physical appearance of his English teacher. "what does look like" is the expression to describe someone's physical appearance.

97. **B**. Before we proceed the answer, let us have common background colleague is. A colleague is a friend who works together with similar duty to someone. In most cases, the dialogue between colleagues is info style of expression like that of the expression under choice 'B'.

98. **D**. Doing physical exercise by gym is not a game or a favorite sport since is co competition in gym. To choose choice 'c', we need to have inform about the friend's weight even if doing physical exercise may not neces help to reduce weight. Therefore, choice 'D' is more appropriate than the

99. **C**. As we can understand from Girma's questions, ujulu's response is either approval by saying yes or disagree by saying no. choice 'c' is appr for the context.

100. **B**. Going to Seble's birthday party this evening cannot deter from our their party during the weekend. Choice 'D' is also irrelevant respo the context. Choice 'A' also doesn't go with the given context.

101. **B**. If an action happened repeatedly, it might not surprise us. If it occurred at a time unexpectedly, it might cause great surprise. S French teacher is often late to class, Howa may not be surprise action.

102. **A**. Hagos asked Kedir the duration (period) of time. The present perfect continous tense what Hagos used tells us that Kedir is still doing the job. Therefore, choice 'A' is the best answer from the given alternative. It means that Kedir has been doing the job for about 10 year.

103. **A**. The journalist asks the parent what he felt/though/ about the amount of trime children spent for watching TV. The question has two straight forward answers: Yes, I think so or No, I don't think so. Therefore, choice 'A' fits the context.

104. **C**. A space a tool for digging holes to plant trees or to clear the ground. It doesn't help us for cutting down or chopping trees.

Section Seven: Writing (105 – 120)

105. **A**. All except cafeteria are correctly spelt.
The correct spelling is cafeteria (not cafeteria)

106. **B**. From the given alternatives, cumulative is wrongly spelt. The correct one is cumulative.

107. **B**. Description is a writing technique which adds the livelihood and create a visual impression of an object or scene. This is the art of translating perceptions into words. it involves two elements: the object which is seen or heard and the observer he/she sees or hears it. It can be matched with a sort of details we see in vivid and effective photographs. The writer helps as to visualize how the building is/ looks like.

108. **D**. Argumentative writing is another writing technique which intends for convincing the reader to have the opinion the writer has on a subject/ matter). It also involves an attempt/ effort) to persuade the reader to act in a particular manner. For this to happen, the writer must offer the reader particulars of details, whether sensory 9descriptive), quotations, statistical justify his argument about seminar method of teaching.

109. **B**. Expository is a writing technique that expounds and interprets an idea, theory, plan or situation. It answers the question how an why? Similarly, the writer explains the situation how and why they delay to reach the tip of the mountain.

110. **C**. Narration is a writing technique which is used for telling of a story. It normally involves a discussion of events that are true or real events that takes place over a period of time. The writer can connect events in the time span to other events the come before or after. Some of the transitions of time are after words, soon, a day later, then, next, finally, first, second and the like.

111. **A**. Trying to convince using comparison is one way of developing argumentative writing. Comparison and contrast is a method of analyzing likeness or differences between two or more subjects. We may appreciate soccer when we compare it with foot ball; we understand communism when we see it in the light of capitalism. By the same token, the writer can convince us the uses of computer by contrasting it with the typewriter.

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112. **A**. The first sentence intends to summarize the idea of the whole written work. Concluding what the whole text has and what is all about is expressed in brief in the conclusion part.

113. **B**. In the introduction part of the business letter/application letter), the writer should justify what he/she writes for.

114. **B**. this extract must be taken from close friends since the style of words are informal that should not be used formal or business letter.

115. **A**. the writer expresses his/her statement of regret for the inconveniences. For this reason, it must be taken from an apology letter.

116. **B**. This application letter shows the receiver's address. If it were the applicants address, it should have the date and his name should be stated at the end of the letter. Thus, the name must come first, his status (rank) and his organization followed. Next, the P.O. box should be before the city like the order of choice 'B'

117. **B**. This notice must be written in a hospital because the clients (patients) await in the hall until their name are called by the nurse for treatment.

118. **D**. comma has different purpose in writing. Of which, separating extra information from the main idea is one. The idea which clarifies or gives extra information should be enclosed by comma like choice "D"

119. **B**. In direct speech, the introductory verb clause can be used at the beginning, at the middle of the two clauses, and at the end of the sentence as a variety. The exact words of the speaker must be put under inverted comma " " should separate the first clause by comma and any punctuation mark the speaker uses must be in the quotation mark like choice "B" .

120. **C**. From the alternatives given, choice 'C' is more appropriate than other. Since one of the purposes of comma in separating the main clause from a subordinate clause.

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UNIVERSITY ENTRANCE EXAMINATION (UEE)
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Section one : Word order (1-5)

Directions : The following words are Not in the proper order. When put in the right order, they make correct English sentences. From the given alternatives, Choose the one that is correct and blanken the letter of your choice in the separate answer sheet.

1. B and go there it get right

According to its function, a sentence can be said in four Forms.

1. Statement Form: subject + Verb + Object(predicate)
2. Question Form Seeking information
   (why and how question)
3. Seeking approval
   (yes/ no answer question)
4. Imperative Form : Verb + Object (predicate)
   Exclamation form (expressing emotions)
   4.1 What (a) + adj + noun !
   4.2 How + adj
   4.3 Good ! Marvellous ! Splended ! oh !

The above misplaced sentence is an imperative form with double action verbs that are joined by the conjunction " and". The connecter (and) can join two equivalent and grammatically parallel sentences, clauses, phrases or words. Hence, From the given alternatives, Choice 'B' fulfills the above situation. Besides, the two action verbs (Go and get) indicate movement one after the other. Because before you get it right there" , you should "go" to the place (there).

2. Clean and fresh breathe must we air

This misplaced sentence is a statement form
" Subject + verb + Object (predicate) "
The verb comprises the auxiliam (modal) and main verb and the object consist of the noun" air " that modified by two adjectives " fresh and clean"
Let's look at the position of conjunction " and" from the given alternatives

A. Sub + verb + adj + noun and + modal Aux. + adj = wrong
B. Sub + verb + adj) + adj + noun + and modal Aux + verb
   This sentence seems to be grammatically correct, but it is illogical since a sentence should give sensible meaning.
C. Sub + modal Aux + Verb + adj and adj noun. This alternative is grammatically and logically correct
D. Sub + verb (adj) + and modal Aux + verb + adj + noun = wrong

3. Can their do me you what looking by uniforms tell at just people?

Before we rearrange the words of this sentence, let's see the following two situations:

1. A what + noun question form: This form is used for a selection from an indefinite number.
2. Embeded question word (wh – word) questions as noun clause (statement)

Example : Question
What books did you buy?
What time is it?

Embeded Question;	
Can you tell me Could you tell me Could you please tell me	What books you bought? What time it is ?

Form: Question What + noun + aux verb+ sub + verb ?
Embeded Question:
Modal Aux + Sub + verb + obj +
What noun + noun + verb ?
For this reason, alternative "C" is the best word arrangement.

4. A You next do in who class to sit ?
The relative pronoun who in the above sentence is used as an object
Who = as a subject in question
Form : who + (Aux. verb) + main verb + object (predicate) ?
Who = as an object in question
Form : Who/ whom + Aux verb + sub + main verb + object / prepositional phrase.
If the question starts with preposition, whom is used instead of who. Due to this reason, choice " A" is the appropriate answer.

5 C it I want heard you
Some verbs are followed by noun clause as objects of verbs.
Form : subject + Verbs + (that) + sub + verb + object
These verbs include
Hear
Think
Expect
Feel
Hope
Forget
Wish
As can we understand form the form, that is optional
(we can use it or we can omit it). Therefore, choice "C" is the best answer

Section Two : paragraph coherence (6 – 12)

6. D A. Heat it on a gas stove
B. Take some water in a kettle
C. steam comes out from the kettle
D. what do you find after some time?
E. when water is heated, it turns into steam

To put in the correct order, let's find the topic sentence the best core idea of the paragraph is given under "e". To prove the idea that is given in "e" water should be added into a kettle. Then the kettle should be put on a gas stove. After sometime, the steam comes out from the kettle. Choice "D" is the best order

7. C A. Nothing they can reach is safe
B. puppies must be watched constantly
C. This is because their curiosity is endless
D. so, if you cannot see or hear them, you had better question.
E. They can happily destroy a shoe or a pillow in just a few minutes

According to the alternative 20% of the answer is known since order " B" is common in all four alternatives. Then let's ask why that is so as presented in "C". The idea given in "c" is justified in "E" followed by "D" and A hence, choice "C" contains the best order.

8. A. Like ants they have two body sections
B. Many people call termites " white ants"
E. Unlike termites, most ants are shiny and dark in color.
D. Termites also differ from ants in matters of colony life
E. However, their thorax is not clearly separated from the abdomen.

The first sentence is given (that is "b") why are they called white ants" ? Because they have two body sections even though their thorax is not clearly separated from the abdomen. Their difference is mentioned in "C" choice "d" is the additional difference
Alternative "C" has the best order

9. D
A. One branch is chemistry
B. The other branch is physics
C. physical science has two branches
D physics include the study of energy
3. chemistry is concerned with matter in its different forms.

If we locate which one is the topic sentence the details are given with transitional words like " one, the other and the like. Choice "C" is the topic sentence. The next sentence will be choice "a " followed by its definition in "e" the other branch is mentioned in choice "B" with its inclusion in choice "d" . The best order is given in alternative "D"

10. C
A. That is why we take care of them

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B. some animals are very useful to us
C. Domestic animals serve us in many ways
D. such animals are called domestic animals.
E. For example, we give them food and shelter.

The topic sentence is choice "b". The transition words play great role to make the correct order . some of the very useful animals are mentioned in choice "d" as domestic animals". They serve us in many ways , so we take care of them by giving food and shelter. Alternative "C" is the best answer.

11. **B**
A. They use their feelers (feelers) to find food.
B. Most insects have wings and can fly
C. Insects are small animals with six legs.
D. Almost all insects have a pair of feelers
E. Bees and ants are examples of common insects.

The topic sentence can be started by definition like choice "C" followed by choice "b". Having a pair of feelers for finding food is the common feature of the insects. To exemplify this, bees and ants are mentioned as common insects. Alternative "B" is the best order.

12. **A**
A. Some animals live in the jungle
B. Such animals are called wild animals
C. We can see some wild animals in a zoo
D. They wander in the jungle in search of food
E. We can also watch wild animals on different TV channels

To get the best order of arrangement, we can apply the techniques that are mentioned in Question No 11 above. Choice a → b → d → c → e is presented in alternative "A" is the best order

Section Three: Reading Comprehension (13 -26)

Passage one and Two

To answer the comprehension questions successfully, students should develop and apply different skills and techniques. However, scholars in the area recommend the following tips for better comprehending with in the given time.

1. Skimming technique: reading quickly to get the gist or main idea of the whole paragraph

A. Preview the text and the title of the selection before reading it
B. Go through the questions quickly to get a general idea what you are going to do
C. Read quickly the whole of the first paragraph, the first sentence (topic sentence) of the paragraphs and the whole of the last paragraph
D. Read the questions and go to the parts of the passage which enables you to answer it
E. Try to find some thing which makes each multiple choice wrong except the one which is the best answer.
F. write your answer clearly (blanken the circle)

2. Scanning technique: reading quickly to get specific information from the passage

A. Read the question properly and return back to the parts of the passage
B. Read and understand the context of the specific word(s) or phrase (s). In this regard, the neighboring words (before and after the word or phrase (gives us hint that make us reach the best answer.
C. Examine the alternatives what make them wrong except the one which is the best answer.

Remember: Pronouns (this, that , those, it, he, she, you, I, they, we, and the like) are used in place of antecedent or nouns that come before to avoid needless redundancy.

13. **D**
14. **A**
15. **C**
16. **D**
17. **A**
18. **C**
19. **C**
20. **C**
21. **A**
22. **B**
23. **A**
24. **B**
25. **D**
26. **D**

Section Four : Vocabulary (27- 56)

A. Completion (27-46)

As discussed so far, to answer the completion (incomplete sentences) question, understanding the context plays a great role.

27. **A** Two contextual clues are given in this completion sentence. These are " How much and fund"
" How much" is used to ask question for uncountable nous like money
The word" fund" refers to sum of money available for a purpose . Now let's define the given alternatives:
A. Donate (give money for the purpose of charity)
B. nominate (name, appoint or propose for the position)
C. denote (Indicate , be the sign or symbols of or be name of)
D. detect (discover, investigate)
Alternative" A" can go with the context directly

28. **A**

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The speaker feels happy by giving the clock as a gift on behalf of his colleagues

A. Present (give, offer, put forward, submit)
B. Represent (express by a symbol, act in place of submit / give again)
C. Offer (to present for acceptance / refusal express willingness
D. Preside (be chair person, be the head of)

29. **D** The chair has been passed from the great grand mother to the speaker by inheritance. From the given phrasal verbs, handed down " is appropriate to the context

A. Handed over = (delivered a person to authority, transfer,
B. Given away = (allowed somebody else to have, given freely)
C. Given out = (distributed, come to an end, be exhausted
D. Handed down = (passed by inheritance or tradition to children or grand children

30. **C** The phrase " armed gang" mans a group of armed persons working for criminal purpose/ It connotes with commiting crime by force using weapons.

A. Stole = (took some body's property secretary unlawfully/ or illegally)
B. Cheated = (acted in a dishonest way to win an advantage or profit)
C. Robbed = (took some one's property form a place un law fully illegally by force.)
D. Snatched = (put out the hand suddenly and took some one's property and run.)

31. **A** The sentence is a bout explaining or expressing one's physical appearance and emotional personality.

" How .... Describe ..... is used for describing physical / emotional personality

A. Describe = (express what some body looks like, give a picture of in words)
B. Prescribe = (order or advise the use of some thing medicine by doctors)
C. Proscribe (denounce a person or practice as dangerous)
D. Inscribe = (write or mark words, one's name in / on / with something)

32. **C** This question seeks our proverb knowledge. It is common in English as well as in Amharic that is saying "eat like a horse and drink like a fish"

A. Lion : a symbol of strength / bravery. Proverbs associated with lion: Lion's share = larger part . Lion – hearted = brave
B. Hyena : is a flesh – eating wild animal and it's known with its special laughing
C. Horse: proverbs associated with horse
- Eat like horse = eat a lot
- Work like a horse = work hard
- A dark horse = person whose chances of success are not yet known
- a horse of another colour = quite a different matter

- back the wrong horse = support the loser in a contest being treated with proper respect
- straight from the horse's mounth = from a first hand source

D. Donkey : proverbs associated with donkey
- Donkey's years = a very long time
- The donkey work = the hard and boring part of a job
- Talk the hing legs off a donkey = talk for a long time

33. **B** From the given alternatives the verb" demolish" is more appropriate than others

A. Destruct : intentional destruction of a malfunctioning rocket or missile (noun) after its launch
B. Demolish = (pull or tear down, destroy, made an end of old building
C. Damage = (harm or injure that caused loss of value)
D. Demobilize = (release from military service (verb)

34. **A** The statement is about the prevalence of AIDS during the 1980s was greater than the recent time. The words spread is better to describe this situation.

A. Spread = open out, cause to become more widely extended or distributed.
B. expanded = (become larger, not shrink)
C. extended = (reached / stretched) continue in relation to time and space
D. Contracted = made/ became smaller or shorter,) caught an illness

35. **C** The speaker felt guilty due to pouring out the milk on the floor. We are asked the form and the meaning since past participle word is needed after " have" Hence, "splash"

A. spoilt = (make useless / unsatisfactory.)
B. soured= (turn or become sharp tare like vinegar).
C. Spilt = (poured out accidentally / ram over the side of the container). This is the best answer.
D. Splash = (cause a liquid to fly about in drops / make some body or something wet)

36. **C** If two sentences joined by " but" the two sentences should be opposite. For this reason we have to search for the opposite of compulsory that is optional.

A. operational = (ready for use)
B. professional = (engaged in progession)
C. optional = (voluntary)
D. Practical = (useful, not theoretical)

37. **C.** As can we understand from the context, the speaker got a nervous felling in his stomach he heard the result of the medical test. To express such feeling " anxious" is appropriate

A. excited = (full of strong, pleasant feelings)

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B. enthusiastic = (full of strong feeling of admiration or interest
C. anxious = (feeling anxiety; troubled, strongly) wishing
D. inspired = (one secretly suggested by an influencial person who has special information, put unlifting thought / feelings

38 **C** This question seeks our proverb/ idiomatic expression knowledge. When we are deeply involved in some thing like work or problem, we can say we are up to our eyes ears/ eyeballs/ neck/ in work / problem. For this reason, choice "C" is appropriate
Proverbs / idioms/ concerning the following words
A. Head => bang your head against a brick wall (try for a long time to achieve some thing)
- get your head down = (work or study hare, go to bed and sleep)
- get your head round somebody / some thing (understand some thing difficult)

B. Shoulders : a shoulder to cry on = (a person who listens to your troubles and offers sympathy and kindness)
- tell some body from the shoulder = (tell some body honestly and directly)

C. Eyes = be up to your eyes => (have too much of something like work or problems)
D. Mouth = down in the mouth =>) un happy / depressed)
- make some body's mouth water = make some body feel hungry)

39. **A** This question is also a proverb/ idioms) question
A. when we felt tired, we can say " I slept like a log/ top " This means very well/ without waking:
A. Log => (rough length of tree trunk that has fallen or been cut down)
Proverbs : - like a log => (unconscious, immovable)
- Sleep like a log => (sleep soundly and with little or no movement)

B. Dead => (no longer living)
Proverbs:-
Dead men tell no tales => (used as an argument for killing somebody when knowledge of a secret may cause one loss or trouble)
- Wait for a dead man's shoes => wait for some body to die in order to get in to his position)
- C. Cat = (fur – covered domestic animal to catch mice)

Proverbs : - Let the cat out of the bag => (tell a secret with out intending do so)
- Like a cat on hot bricks => (very nervous or jumpy)
- Put / set the cat among the pigeons => (cause alarm and confusion)
- Wait for the cat to jump / see which way the cat jumps => (refuse to give advice, make plans until one sees what other people are thinking or doing)

D. dog : (a domestic animal that gaurds the house)
Proverbs :- a case of dog eat dog => (situation, e.g in business, where ruthless methods are used
- die like a dog / die a dog's death => (die in shame or misery)
- a dog in the manner = (person who prevents others from enjoying some thing that is useless to him.

40. **C** This question demands our knowledge of the confusing words " lie / lie / lay" in their present, past and present participle forms.
The similar sounding verbs lay and lie cause much confusion. Study the following chart.

Present /Future	*Lie*	*lay*	*lie*
Present, past participle Present participle	*lie/ lies. Lied, Lied, lying* } + no object	*lay/ lays laid, laid, laying* } + object	*Lie / lies lay, lain, lying* } + no object
Definition	*To say same thing that Is not true*	*1.to put some or Some one or some thing in a special or the correct position for a purpose. 2. to produce an agg (bird, hen, insect etc.*	*1. to be in a horizontal flat on the ground, bed, etc be at rest 2. to be in a particular position.*
Can appear with these prepositions	*about*	*Down, in next to Out with*	*A head, around, down Behind, In ,On,*

- Let's exclude choice "B" and choice "D" be lied means told un true and " lain" need to have perfect tense formation (has). As shown in the above table " laid" is a transitive verb that need object. For this reason, choice " C" is more appropriate than others.

41. **A** This is a question of figurative speech (proverb) we should find words that collocate with " thick " as a suffix choice "A" and "B" can go with thick.
thick – headed => stupid, dull
There fore, choice "A" (thick skinned) is the best answer

42. **C**. The second sentence gives as hint which phrasal verb is appropriate and the meaning of the phrasal verb should be a synonym to betraying many people. The phrasal verb of go back on is equivalent with betraying and it is the best answer.

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A. Take on = become excited or excited or agitated, become popular .
B. Run against = complete with some body by running in a race.
C. go back on => fail to keep break or withdraw from e.g promise.
D. go off with =>go away with, especially abscond with or steal.

43. **A**. From the context, the speaker seems a person who is higher in rank. A person who gives orders or controls the workers is called "boss" choice "A" is mor appropriate then other.
A. booss = is a person who controls or give orders to workers.
B. responsible = legally or morally liable for carrying out a duty.
C. superior = better than the average a person of higher rank / authority than others.
D. accountable = responsible, expected to give an explanation why something is done/ not done.

44. **A** This questions seeks our knowledge on linking verbs appear, become, feel, get, keep, remain, seem smell, sound, stay, taste that followed by adjectives indicating the subject's emotion or physical or mental conditions.
The linking verb " feel" usually followed by adjectives like angry/ pleased, happy/ sad, hot/ cold, tense/ relaxed, nervous/ confident. For this reason, choice "A" is more appropriate than others.
A. Confident = (certain, full of confidence)
B. Confidence = (belief in one's ability assurance)
C. Confidential = (to be kept secret, have the confidence of another or other)
D. Confidentiality = (contract to keep secrets (not to) reveal information to others)

45. **D** This question seeks our knowledge of the form after the preposition" of " . Noun follows the preposition "of " . Due to this, we should reject choice "A" and "C" Since they are adjectives. Let's compare the two nouns" choice "B" and "D" in terms of meaning "excellence" is appropriate for the context, but Excellency can not go with the context.
A. excelling = doing better than others, being very good
B. Excellency = title of ambassadors, VIP, higer officials
C. excellent = very good, of high quality
D. excellence = the quality of being excellent, great merit

46. **D** Keep a secret" is an expression used for a person wants " not to tell any one else/ to keep the knowledge of others. Choice "D" is the best of all
A. hold = to be able to contain, to keep in one's grasp
B. retain = continue to have or hold, to keep in place
C. hide = conceal, conceal one self
D. Keep = store / continue or maintain in action or conduct

B. Substitution (47- 56)

47. **B** " <u>so underdressed</u> " (passive) => dress inadequately for an occasion or dress less/ fully or formally than an occasion / circumstance demands " Barely dressed" can substitute "so underdressed" without changing the tone and meaning of the sentence

48. **A** ... invaluable in this context implies that something is extremely / very valuable or use ful. It is the opposite of value less / worthless " Precious" can substitute invaluable "without any meaning change

49. **D** "<u>take back</u>" in this context means with draw some thing said or written" the speaker regrets about what he said to her, however it is impossible since it is like " crying over spilt milk.

50. **B** <u>Pick holes</u> is expression(idioms) that is used to criticize and find the weak points in suggestions, plans and arguments"
For this reason, "criticize" can substitute the idiomatic expression of " <u>pick holes</u>" without any meaning change .

51. **A** " <u>took my breath away</u> "This idiomatic expression can be said if some one is surprised/ amazed/ astonished. Therefore, the expression can be substituted with <u>surprised me</u>.

52. **C** " <u>keep them at arm's length</u>" This expression means keep your distance or give a wide berth (place) to someone to keep well him away. Therefore, Don't get them too friendly with you can substitute the expression together with its contextual meaning

53. **A** "...**<u>at death's door</u>** .... " This expression means that the speaker was so ill that he might die due to the illness. Therefore , very ill" can be a synonym for <u>at death's door</u>"

54. **C** heartening This means giving somebody encouragement or hope. For this reason, encouraging can substitute " heartening" without any meaning change

55. **A** <u>Switch off</u> The underlined word <u>switch off</u> means turn off. Its opposite is switch on or turn on. The appropriate answer is choice "A"

56. **B** <u>unlearn everything</u> This expression is used to deliberately forget something that we have learn + as bad or wrong. If so, forget old ways" can replace the given/ underlined words without any meaning change

Section Five : Grammar (57 – 85)

57. **B** Present perfect tense used with " just" for a recently completed action. The adverb " just " should placed between the auxiliary (has or have) and V_3 formof the main verb
A yet = it as an adverb of time, means mainly used with the negative or interrogative " Yet" is normally placed after verb or after verb + object. For this reason, we can't use it for this question
B. just : just as an adverb of time used to say some body did something very recently
C. ever: it means at any time, rather all the time, every time (always). It is chiefly used in interrogative and negative. Thus, it can't be used in this sentence as an answer

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D Since it is used with present / past perfect tense to indicate when an action started and continued up to the time of speaking. It can come after the auxiliary verb or in end position after a negative or question verb.

58. **C** This question demand our knowledge to the form and meaning of conditional type I in negative sentence " If Not " => "unless"

Form	If clause	main clause
1.	If + S + V_1 + O	S + Will + V_1 + Ob
2.	If + S + V_1 + O	S + V_1 + O
3.	If + S V_1 + O	S + has/ have + V_3 + O
4.	If + S + V_1 + O	Imperative V_1 + O

5. If + S + aux verb + not V_1 + O = unless + S+ V_1

A. Give shout = tell to some body (it can't go with the context
B. Will not shout => it lacks subject
C. Don't shout => it fulfills the context / situation as stated in number 5.
D. Don't have shout => it is ungrammatical

59. **C** The speaker felt very happy about announcing / telling the good news of passing the exam. This question is about the form after adjective / participle followed by infinitive with to. Look at the following form.

Form : subject + be + adj/ participle + to + v_1
= sub + be (is, am, are, was, were) + adj / participle (angry, delighted, pleased, relieved, glad, sorry, happy, dismayed) + to +V_1

60. **A** The pre modifier of comparison "a bit" means slightly a little or some what for singular noun. It comes before the comparison to show the difference is very little or small. For example, " it costs a bit more than I wanted to spend" a little bit" can be used as a premodifier of comparison for a small degree.

61. **B** The speaker is comparing his car which is breaking down with another better (reliable) car. For such expression. "a more reliable car" is an appropriate answer.

62. **A** As we discussed so far, in question 57, " yet" is used in negative sentences and questions to talk about some thing that has not happened but that we expect to happen . The speaker want to know whether the person has paid the electric bill up to now (the time of speaking) therefore, present perfect tense is appropriate for this context.

63. **D** The three situations given in this question are
1. conditional type II
2. passive voice
3. past tense

- form for conditional type II If clause main clause ***Simple past would + V_1***
- Passive voice : the speaker wants to emphasize the receiver of the action (the cars) rather than the doer of the action(producer of cars). The form of passive is be (were) + V_3

- The given time is past. For this reason, the alternative A, B, C can not fulfill the above conditions, so the best answer is choice "D"

64. **A** "There is nothing……. Like …….. " is used to say that the speaker enjoys some thing very much. The presence of "noting" doesn't allow us to use choice from B to D except A.

65. **A** From the given context look at those black clouds" we can infer that there may be a possibility of rain but we are not sure. If we are sure and/ or if the speaker doesn't use " I think" choice "B" and "D" can be an answer. Choice "C" shouldn't be answer since the rain have not yet started raining choice "A" is an appropriate answer for the given context.

66. **B** The context needs negative complement. Due to this, "unkind" is more appropriate than others
A. grateful: feeling or showing thanks, pleasant, agreeable or confronting
B. unkind: not showing kindness, lack in
C. forward: directed towards the front or situated in front
D. intrusive Intruding, forcing up on somebody entering a place without invitation

67. **A** From the given emotive adjectives, exciting can be more appropriate than others.
A. exciting: something that is causing somebody to have pleasant fellings.
B. appealing : touching the feelings of sympathy, moving, attractive,
C. interesting : arousing or holding one's attention
D. entertaining : amusing and interesting or pleasing

68. **C.** The second sentence gives us hint that they had done nothing . They spent the whole day idly. Due to this , the day should not be " tiring, relaxing or frightening Rater it should be " boring"

69. **A** The speaker is comparing the price of the material in this year and the previous year. The speaker appreciated price of the previous year since he/ she paid only 100 birr. For this context, choice "A" is more appropriate than others.

70. **D** The adverb " since" is used with perfect tenses that expresses a situation that began in the past (a particular time) and continues to the present. For this reason, we should exclude choice "A" and choice "C" from the possibilities. The past perfect " had" which expresses an activity that was completed before another activity or time in the past. Therefore , present perfect " has had" is appropriate for the context

71. **C** The present perfect also expresses the repetition of an activity before now or un specified time in the past. The exact time of each repetition may not be important.
The phrase " a couple of time " shows the repetition of the avtivity. Choice " C" is more appropriate than other alternatives.

72. **A** This question has two situations/ contexts
1. " is going to " This is used for predication

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The be (is) going to " form can express the speaker's feeling of certainty. The time may not be mentioned, but the action is expected to happen in the near imediate future. In such situations, the following verbs, think, be sure/ afraid, believe can be used.

2. Making suggestions with " shall I / we when shall is used with I or we (first person singular and plural) in a question, the speaker is usually making a suggestion and asking another person if she/ he agrees with this suggestion choice "A" can fulfill the above two conditions.

73. **D** The speaker wrote a letter to the person before one month, however, the person has not replied up to the time of speaking. It is mainly used in negative sentences and questions for all these conditions, present perfect as mentioned under choice "D" should be used.

74. **A** According to the second sentence TOM scored low grades because he used to spend a lot time for playing games and listening music. This implies that he reduced time for studying. Time, not hour, is uncountable. So " much of his time " is appropriate for this context.

NB Expressions of quantity followed by Of + a specific noun

→ All , most, some, any + of + specific plural countable or uncountable noun

Countable : Most of those chairs are confortable

Example uncountable: Most of that furniture is confortable

→ Many (a) few, several, both, two, one + of + specific plural countable noun

→ Example : many of those chairs are confortable

→ Much (a little + of + specific uncountable noun

→ Example : much of that furniture is confortable

75. **A** This is a question of Adverbs of place and direction. Let's examine how and when the alternatives are used:

A nowhere = not any where . Even though the meaning is negative, it is used in positive sentence.

B. somewhere = some particular place
Somewhere can be used in positive or affirmative statements and questions (mostly yes is expected)

C. anywhere = any place (no particular area) Anywere can be used with negative statement like (..... not, hardly, barely, scarcely) and in questions . we can use any where and some where" for question formation without any meaning change, however, if the reply is yes, we should use" somewhere", if the reply is no", we should use " any where

D. Everywhere = in all places (used in a group)

The speaker searched for in all places, but he couldn't find it. Choice "A" can best be matched with the context

76. **C** Future perfect tense: It expresses an activity that will be completed before another time or event in the future. It is normally used with a time expression

beginning with by them, by that time, by the end of next month, by tomorrow morning etc

Its form is " will / shall + have + V_3

This question seeks our knowledge of positive and negative additives. Let's look at the following summary of additives

1. Statements with too and either

Positive	Negative
A. Almaz likes to work	Lemlem doesn't like to work
⇨ Helen does, too	Seleam doesn't either
B. Maskerm is teaching in Addis	Chala is not teaching in Addis
⇨ Hanna is, too	Alazar isn't either.

2. Statements with so/ neither or nor

A. Goitom stayed home.	Tilahun didn't stay home.
⇨ So did Girma	Neither did Eshitu Nor did Eshetu.
B. Bekele can drive a car.	Awoke can't drive a car.
⇨ So can Abera	Neither can Gebremariam nor can wondu.

- For all these reasons, choice "A" is the best answer since verb to be (am) is used in the original sentence.

78. **A** This question is about correlative (paired) conjunctions. Some points about correlatives (paired) conjunctions. Some points about correlatives.

When two subjects connected by "both"

1. .... and, they take a plural verb.

When two subjects are connected by not only .....

2. But also, either.... or, or neither....nor, the subject that is closer to the verb determines whether the verb is singular or plural.

3. The same parallel grammatical form should follow each word of the pair.

Examples

A. Both + noun + and + noun
B. Not only + verb + but also + verb
C. Either + noun + or + noun
D. Neither + adj + nor + adj

- If we substitute all of the alternatives, the grammatical form will be all right nonetheless, the first sentence (conditional type I- likelihood) can go only one of the alternatives. The speaker advises/ gives possibilities) the passanger to take the plane. This implies that the train and the car may not arrive on time. It is not advisable to take either. the train or if he / she in a hurry. For this reason, " neither.... nor" is the best answer

79. **A** This question is about base infinitive and gerund. These are few verbs that followed by

Object + bare infinitive and gerund.

The following basic verbs of sensation followed by:

Object + base infinitive (infinitive with out to) or Gerund

I. Infinitive

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Form : subject + verbs of senses + object + base infinitive

feel
see
hear
mell
watch
notice
listen to

Note * words are sometimes used in this category

Meaning: The Infinitive implies that an action is complete

Example : I saw him enter the shop (I saw the whole action he opened the door, went in, and disappeared)

II. Gerund

Form : subject + verbs of senses + object + Gerund

see
feel
smell
hear
listen to
notice
watch

Meaning : The action in the present participle may either complete or incomplete

Example I saw him entering the shop.

Example I saw him entering the shop

⇨ (I watched the whole action or that I saw only part of it)

80. **A** The question is how gerund (v .ing) is used as a direct object. We can use gerund in affirmative negative or questions.

1. Affirmative

Example : Eshetu finished doing his home work

Bekele enjoyed watching movies

2. Negative

Example : He doesn't like wearing glasses.

They don't mind helping the police

She doesn't miss being with her family.

He likes not wearing glasses.

She regrets not being with her family.

They deny not helping the police

3. Question

DO you like playing the piano?

Did she continue studying English?

Can they begin working on the report?

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Note: Some /partial list of verbs followed by gerund are: enjoy, finish, keep, like, reglet, deny, continue, begin, quit, post pone, resist, stop, tolerate, understand, recommend, remember, imagin, escape, practice.

Due to the above explanation/ characteristic of the verb enjoy, choice is "A" appropriate

81. **A** This question is about cause effect relationship. The first sentence is the cause of the second sentence (second action). For such relationship, we can use the following connectors to join the result with the cause

So, For this reason,
As a result, As a result,
There fore,
Consequently,

Note: "so that " can be grouped with in order to, in order that, so as to, for fear, lest since both of them used to express purpose.

"because of" followed by phrase, not clause is used to express cause / reason and grouped with because, since, as, due to, for owing to, on account of "as a result of" followed by phrase not clause.

82. **B** The first sentence and the second one are contrasted each other. They, therefore, need concession/ contrast conjunction. These include

Although
Even though + S+ V+ O, sentence
Though

Choice " B" can be the best answer.

83. **B** The subordinate clause is conditional type – I since presnt tense preceded by " If", then main clause can be future tense (will + V_1) For this reason, choice "B" can be an appropriate answer.

For better understanding, please refer Q58.

84. **C** The first sentence especially " poorly made" is equivalent with "bad" (positive degree). The second clause contain "even" that is used for as a premodifier of comparative degree.

Example : you know even less about it than I do.

You seem even more smart than usual today

If so, "even" can modify the comparative degree of "Worse"

85. **B** As explained in Q 78 of this examination, when two nouns are connected by "neither…nor, not only… but also and either…. or, the verb should agree with the noun(subject) that nearest to it. Thus, the nearest noun) subject) to the blank space is " his father" which is singular. Besides, the action happened / occurred some times in the past choice "B" fulfills the two situations.

Neither….. nor (none of them) and past tense.

Section Six : communicative exchanges (86 -105)

86. **C** We can grasp something about the new from the son's response. He equates the new labtop with the old one by saying" the same as " with the old one " If so, their difference must be insignificant that deter us not to use ……

a lot" choice "C" especially" ..... a bit" (meaning: a little) can go with the implied meaning.

87. **C** The daughter promised to call her mother when she arrives at Harar. We can understand from her mother's saying "Good" that she wants to know the time her daughter will get Harar. Choice C is the best to fulfill the blank space.

88. **A** The question invtes us to draw logical conclusion from the speaker's expression that he/ she hasn't eaten any thing since break fast up to the time of speaking. The best local conclusion we should draw is she/ he must be very hungry as expressed in choice "A"

89. **D** From the shop assistant's polite request, his/ her customer enters the shop and looking for some thing to buy. Common expressions the shop assistants usually use are
- may I help you?
- What can I help you/ can I help you?
And the customer is expected to tell what he/ she wants to buy/ do . Choice "D" is appropriate

90. **B** From mother's question, we can understand that she wanted some additional sugar from the kitchen. The appropriate reply of the question can be "No there isn't much sugar. This is because the adjective "much meaning a lot/ a great amount can modify negative verb like stated in choice B

91. **A** The mother inquired about why her son's hair so wet because she didn't known the reason. Let's make a reduction of choice "C" since it is present perfect continues that the action does not discontinue at the time of speaking .Choice "D" should also be excluded from the possibility because the action not started at the time of speaking . Choice "B" on the other hand, should be excluded because a barber may not make his hair so wet. Choice "A" can be the best of all since present perfect tense implies that the past activity has strong connection to the time of speaking

92. **A** Kedir's class has got a new teacher and he introduced to Lidiya even though the teacher is not physically there. The possible reply expected from Lidiya might be asking about his physical appearance "How he looks like" Choice "A" is the best response for the given context.

93. **A**. From the second speaker's response " No, I'm single", We can infer that the question is about marriage that happened in the past choice "A" is appropriate for the context.

94. The second idea (response) is about expressing agreement for what was suggested before Choice "A" seeks approval Information by saying "yes or no" choice "C" seeks information about the reason cause of going out for dinner and choice "D" emphasized the time for going out for dinner.

95. **B** The teacher equates the current students with the previous students . He really appreciates the former students and he believes they were better than the current students in achievement. This in return, implies many students are not doing well these days as it expressed under choice "B"

96. **B** According to the patient, his/ her headache is continuing without any change of relief. This may imply that the doctor prescribed some medication before to him/ her. The doctor seemed to be expecting another causes of the headache blood pressure" Choice "A" and choice " D" Can not go with the context . The... good treatment?" should be left for the doctor. What is expected from the patient is telling the doctor about his feeling and sign and symptoms of his sickness. Choice "B" can be appropriate answer.

97. **D** Peter and John seemed to be arguing about the man who bought seemingly the cheapest car in their town. " Peter undermined the car due to its cheapest price and discouraged the man(he) by saying "he shouldn't have gone for it (the expression used to discourage somebody not to try something difficult). From the given alternatives as John's response, He might supposed to raise a question why so cheap that is as mentioned under choice"D"

98. **B.** The given context here is that the brother asked his sister whether/ if they could have some ice cream in the café or not, or the owner of the café, or to what extent girls like ice cream. The best response for the context is choice "B"

99. **C** From the father's (dad) response, we can understand that the father himself needed the dictionary so he didn't want to borrow it to his son. The appropriate question for the given reply is "could you lend me you dictionary, please?"

100. **B** Belile seemed to be spending a lot of his time with the mobile phones. Sonia also understood this situation since she approved by saying" I Know" There fore, I always see you busy with it" is appropriate response for the given context.

101. **C** The student seemed much concerned about her low achievement in the exam and asked his friend whether/ if they could help her. For this context, " I don't think these is any thing to be done is more appropriate than others.

102. **B** Wow how great she Is ! Her scoring an A in math fascinated Gutu and shared his information to Did i This implies she may not be as such student who is expected to score an A in math "the hardest subject" . Choice "B" is more appropriate than others.

103. **C** From the waiter's word," coffee or tea..... we can understand that the customer ordered hot drinks and the waiter wanted to confirm which one the customer preferred to drink. From the given alternatives, Choice "C" can go better with the context.

104. **B** As one can grasp from mulu's response he asked Lema to do him a favour by lending 5000 Birr. Therefore, Lema's question may be some thing like" How much do you want to borrow" as stated in B.

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105. **D** Degu wanted to know whether her new hair style was fashionable or not . Degu's question was not about her hair nature (feature) , not her origin nor attractiveness, but style of the day. Choice "D" is better than others.

Section Seven Writing (106 – 120)

106. **D** From the extract "It was great to hear from you", we can infer that the expression is the introduction of the reply letter to a friend. It is the typical example of letter to a friend.

107. **B** Descriptive writing is the art of drawing pictures by means of words that are very close to our sense organs
(pereception) . Descriptive writing has the following features
1. The description relies on a basic talent that all we have the ability to see, touch, taste, smell or feel about the object in order to recreate people, place, things, emotions and ideas on the reader's mind.
2. It should be developed by choosing only those details most useful in painting a picture for the reader.
3. It can be furthered from top to bottom, left to right, front to back perspectives and then moved carefully from detail to detail like a camera eye.
4. It should create a dominant impression on the subject being described.

Choice "B" is the best as compared with the other alternatives.

108. **A** The piece of writing should be taken from a writing that is narrative in nature. Narrative writing is the technique of a writing that tells a story. It mainly involves a true/ real event which takes place over a period of time. The essence of good narrative writing is organizing the story into beginning, middle, and end or a series of related events. We can understand the theme of the story from the extract that after they had travelled 1 ½ hours, they heard a loud blust (explosion). Intuitively, the driver changed the direction and made a safe stop. This can be a good example of narrative writing.

109. **A** The instruction in a leaflet packed with a bottle of syrup should be clear, brief and easily understandable. Choice A is better than the other instructions

110. **D** This piece of information should be written at the concluding part of "job application letter.

111. **A** From the extract, we can understand the writer narrated his life history in the chronological order – from birth up to the age of twelve. This might be taken from the beginning of a narrative text.

112. **A** A love letter is considered as an informal way of friendly letter. The following closing expression can be used :
Love
Yours affectionfely.
With love
Your loving
With profound love
With enduring love

Yours in love
Love you
With lots of love
Your lover,

113. **C** This explanation concentrates on how a group of birds occurs/ exists. Such kind of expounding or explaining is a typical example of expository writing which focuses on how something is done, how some thing works, or how something occurs/ exists. A variety of techniques is used for the development of the idea. The common method of development techniques are exemplifying, comparing and contrasting, making analogies, reasoning, classifying, defining, describing etc from premise to conclusion

114. **B** " Let's keep in touch! " means inviting other to communicate by means of telephone, letter and the like regularly. This can be written at the end of the letter in the closing part.

115. **D** The writer raised debatable issue about benefits and challenges of internet. According to the writer, internet has created not only problems but also inequalities.
This is a typical example of argumentative or persuasion which emphasis that writers idea is right and reasonable.
The writer is trying to convince others to do something by mentioning major proposition supported with minor proposition details and facts. In addition to this, he may present his own argument logically and convincingly, he/ she must also recognize and deal with the opposing argument called refutation.

116. **C** " I look forward to hearing from you soon can most likely taken from the closing section of a letter. Choice "C" is more appropriate than others.

117. **D** As discussed in Q115 in this examination, choice "D" reflects the writer's debatable issue.

118. **D** For the apology letter, what the mistake was, how and when the mistake occurred should be included

119. **B** The speaker asked for clarification what hyphenation" is. The word" hyphenation is not his word, so he put it in quotation marks. However, the question is his own and should be put outside the quotation mark.

120. **C** The actual words of the speaker should be enclosed with quotation marks. The first letter of the actual speaker should be capital letter choice "C" is properly punctuated.

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UNIVERSITY ENTRANCE EZAMINATION (UEE)
ENGLISH, SENE 2004/ JUNE 2012 ANSWERS

BOOKLET CODE 12

1. **Answer : A**
2. **is on he the by the standing window chair.**
   He is standing on the chair by the window.
   Answer: C
3. **I like snore her that hate hearing. .**
   I hate hearing her snore like that.
   Answer: C
4. **how are these much T-shirts?**
   How much are these T-shirts?
   Answer : C.
5. **I did not anything about her unusual notice at first.**
   I did not notice anything unusual about her at first.
   Answer : B .

SECTION TWO: Paragraph Coherence (6-12)

The Basic Rules of Paragraph Coherence is to Keep One Idea to One Paragraph.. If we begin to transition into a new idea, it belongs in a new paragraph. There are some simple ways to tell if we are on the same topic or a new one. We can have one idea and several bits of supporting evidence within a single paragraph. We can also have several points in a single paragraph as long as they relate to the overall topic of the paragraph. If the single points start to get long, then perhaps elaborating on each of them and placing them in their own paragraphs is the route to go.

Elements of a Paragraph

To be as effective as possible, a paragraph should contain each of the following: **Unity, Coherence, A Topic Sentence,** and **Adequate Development.**

Unity :The entire paragraph should concern itself with a single focus. If it begins with a one focus or major point of discussion, it should not end with another or wander within different ideas. It means that all the sentences refer to the main idea or the topic of the paragraph.

In a unified paragraph, we expect all the sentences to be about the main idea of the paragraph.

Coherence is the trait that makes the paragraph easily understandable to a reader. It cab can created in the paragraphs by creating logical bridges and verbal bridges.The sentences in coherence should be organized in a logical manner and should follow a definite plan of development.

Logical bridges

- The same idea of a topic is carried over from sentence to sentence
- Successive sentences can be constructed in parallel form

Verbal bridges

- Key words can be repeated in several sentences
- Synonymous words can be repeated in several sentences
- Pronouns can refer to nouns in previous sentences
- Transition words can be used to link ideas from different sentences

A topic sentence

A topic sentence is a sentence that indicates in a general way what idea or thesis the paragraph is going to deal with. Although not all paragraphs have clear-cut topic sentences, and despite the fact that topic sentences can occur anywhere in the paragraph (as the first sentence, the last sentence, or somewhere in the middle), In most cases , the topic sentence of a paragraph can be put near the beginning of the paragraph.

Transitions.

For example one paragraph leads on to another-the **main ideas** of each paragraph **seem to connect** but still make sense and that makes us feel that the writing is *constantly flowing*.

Continuity is the key to coherence. Some very well written scholarly works read seamlessly from the first sentence to the last. Coherence is the unifying element in good writing. It refers to the unity created between the ideas, sentences, paragraphs and sections of a piece of writing. Coherence is what gives a piece of writing its flow. It also gives the reader a sense of what to expect and, therefore, makes the reading easier to follow as the ideas appear to be presented in a natural, almost automatic, way.

Common coherence words

We may find this partial list of transitions and conjunctions helpful. Conjunctions are words that link ideas within sentences. (Co-ordinating conjunctions link two equal ideas or components within a sentence e.g., *and, but, yet*. Subordinating conjunctions, subordinate one idea in relationship to another with a sentence e.g., *although, even though, whereas, since, before, during, because*.)

NB :There are many words that have not been included in each category.

Transitions which signal a time relationship	**Transitions which signal an example or additional information**	**Transitions that signal a contrast**	**Transitions that signal a similarity**	**Transitions that signal a cause/effect**
Before	*For example*	*On the contrary*	*Similarly*	*Therefore*
During	*Generally speaking*	*On the other hand*	*Likewise*	*Consequently*
After	*In most cases*	*However*	*Once again*	*Then*
While	*Occasionally*	*Unlike*		*In other words*
Since	*Especially*	*Nonetheless*		*Because*
At the moment		*Nevertheless*		*Since*
By then		*Whereas*		*As a result*
First		*Even though*		*Due to*
				Thus

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Next *Then*	*For instance* *Specifically* *Frequently* *Sometimes* *Moreover*	*Although* *But* *Yet* *Regardless*

6. a. So, I quietly got out of bed to see what it was.
   b. It was in the middle of the night when we were all fast asleep.
   c. Suddenly, a piercing horrible noise came from downstairs.
   d. I could imagine the chilling fear this would cause everyone in the house.
   e. But before I walked a step, I saw this huge dark creature holding me by the neck.

   Answer : B
7. a. Right in front of me, there were large plants.
   b. On my left , the ground rose gently to a low hill.
   c. I made my first landing there and began to look around.
   d. A kind of fruit like none I had ever seenbefore grew on them.
   e. Attracted by their color, I got closer for a thorough examination.**Answer :**D.
8. a. some eye signals are unconscious.
   b. We receive 80% of our information about the world through them.
   c. When we look at something nice, our pupils get bigger.
   d. our eyes are the most important of our five senses.
   e. We also send signals to other people with our eyes.

   Answer : C.
9. a. Learn as much as you can about them.
   b. Many zoos breed endangered animals.
   c. What can you do to help endangered animals?
   d. Make an effort to support zoos and wildlife groups.
   e. The more you know about them, the more you can help.

   Answer: A.
10. a. It lives in man the other stage of its life.
    b. Bilharzias is a disease found all over the world.
    c. Three species of Schistsoma cause human disease.
    d. It is spread by a small fluke or flatworm called schistosoma.
    e. Schistosoma lives as a parasite in the water snail during one stage of its life.

    Answer : D.
11. a. On her arm, there is a tattoo.
    b. Juali has an intriguing history.
    c. Physically, she is old and has a wrinkled face.
    d. when she was 10, she became a drug addict.
    e. But people know her for the big earrings she wore.

    Answer: D.
12. a. In REM sleep, the body is quiet but the mind is active.
    b. These movements can be observed beneathclosed eyelids.
    c. Others believe that it helps to rid the brain of over stimulation.
    d. Some think that REM sleep helps to form permanent memories.
    e. During sleep, we experience periods of rapid eye movement (REM).**Answer: C.**

SECTION THREE: Reading Comprehension (13-26)

Passage I Answers :**13. D ; 14. A; 15.B; 16. A; 17.C ;18. B; 19.A**
Passage II Answers :**20. B; 21.B; 22.C; 23.D; 24.D; 25.C; 26. A**

SECTION FOUR: Vocabulary (27-57)

A. Completion (27-46)

27. Answer: A

A. hold in (trap(v),shut in, capture, hem in keep, imprison, cage in, release (antonym)
B. hold up (n): (delay , hitch, glitch, snag, stoppage , obstruction, difficulty, bottleneck, hindrance; Hold up (v): delay, slow down, slow up , sustain , keep up , Support ; raid= robbery, stickup, assault, mugging
C. hold out (v) : offer, propose, give, present, extend, proffer, withdraw(antonym) ; stand firm(v): stand your ground, persist, endure, withstand, persevere, stand fast, continue, last, give in (antonym)
D. hold down (v) : restrain, limit, restrict, confine, repress, suppress, keep down, let go (antonym); keep(v): retain, , maintain, manage, preserve, lose (antonym)

28. Answer: D

A. blocked (chunk, building block, wedge, mass, obstruct, impede, hinder, jam, prevent, oppose, stop, blockade, bar, check, deter, frustrate, thwart, encourage(antonym); infertile, barren, sterile, fruitless, uncreative)
B. prevented (banned, prohibited, vetoed, not permitted, legitimate (antonym)
C. barraged (n) bombardment, volley), onslaught, hail, torrent, stream, storm, flood, trickle (antonym), dam, dike, bank
D. blockaded (n) barrier, obstruction, barricade, line of defense, cordon, (v) deny access to, defend, block, guard, shut in

29. Answer: A

A. nothing (not anything (pron), not anything, zero, nil, naught, nobody(n) nonentity) and used in positive statement.
B. very little (scanty, insufficient, sparse, scarce, measly, inadequate,not abundant)

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C. some (a number of , a quantity of, a few, a little, not all , certain about, around, roughly)
D. abundant (plentiful, copious, rich, profuse, scarce(antonym)

30. Answer: A
A. discrepancy (inconsistence, difference, incongruity, divergence, disagreement)
B. disruption (disturbance, commotion, trouble, interruption, distraction, interference)
C. duplication (repetition, replication, doubling-up, photocopying ; replica , duplicate,)
D. distortion (buckle, twist, deformation, misrepresentation, falsehood or falsification)

31. Answer : D
A. dedicated (enthusiastic, devoted (wholeheartedly devoted or committed to a goal…)
B. sketched(rough description or explanation, a short written or spoken account that conveys just a general outline or idea, with little detail)
C. described (explain something, or give an account of something by giving details of its characteristics)
D. scribbled (quickly scrawled, written hastily and carelessly, illegible, incomprehensible, scrawled, untidy) . This description can go with the context.

32. Answer: C
A. precedent (example, model, pattern, instance, standard, practice,)
B. Precaution (safety measure, safeguard, provision, protection against possible undesirable event or caution to forestall future trouble)
C. prerequisite (something needed as prior condition)
D. requirement (demand something by law, make something necessary or to have as a necessary precondition)

33.)Answer : D
A. neck in neck. (uncommon)
B. head to head(confronting each other, nose to nose, opposite)
C. hand in hand (in close cooperation, inseparably closely)
D. neck and neck (informal, level in competition and with an equal chance of winning

34. Answer: C
A. harsh (adi, difficult to endure= bleak or inhospitable and therefore difficult to endure, severely critical)
B. stubborn (adi= obstinate, immovable, inflexible, willful; persistent(dogged, tenacious, preserving)
C. rigid (firm and stiff, inflexible, inflexible adhered to , refusing to change)
D. parched (dry= completely lacking in moisture because of hot conditions or lack of rainfall, waterless and unsaturated)

35. Answer: A
A. disproportionally (adv . unequally or out of proportion in quantity, lacking proper compensation, unbalanced, uneven,)
B. unproportionally (uncommon)
C. disproportionately (adj/ adv,asymmetric, excessive, incommensurate, inordinate, irregular,lopsided, nonsymmetrical, out of proportion, overbalanced, superfluous, too much, unequal, uneven, unreasonable,
D. unproportionately (uncommon)

36. Answer: D
A. insistent (adi =persistent in maintaining or demanding, demanding attention,
B. persistent (adi, unrelenting, constant , importunate , relentless , determined, continual, feeble (antonym), lasting; remaining, enduring, permanent, continuing)
C. C, constant (stable, steady, even, unvarying, regular, continuous: endless, relentless, persistent, unremitting, frequent: recurrent, incessant faithful: loyal, trustworthy, devoted,
D. consistent (reliable, steady, constant, unfailing, regular, coherent: in harmony, even, uniform)

37. Answer: C
A. conscious (adj. aware, mindful, cognizant, alert , deliberate,
B. Consciousness (n, awareness, realization, notice, perception)
C. conscience (n, sense of right and wrong, scruples, principles, ethics)
D. conscientious (n, careful, thorough, meticulous, painstaking, reliable, diligent,)

38. Answer: A
A. disperse (scatter, go away, disband, break up, dissolve, separate, diffuse, melt away, disappear)
B. spread (open something to fullest extent, extend widely, extend in time , separate things by stretching, send out in all direction get table ready for meal, extend in range)
C. scour (clean or remove something by rubbing, free something from dirt)
D. diffuse (spread through, scatter or become scattered, make something less intense)

39. Answer: B
A. distinction (difference, high quality , distinguishing feature, mark of high achievement)
B. extinction (fact of becoming extinct , process of becoming inactive, destruction)
C. exception (somebody or something excluded, exclusion, criticism, legal clause,)
D. exhaustion (extremely tired or weak state, act of exhausting something).

40. Answer: B
A. regretful (adj, apologetic, remorseful , repentant, penitent, ashamed)

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B. grateful (adj, feeling or expressing thanks = having or showing the desire or reason to thank somebody, comforting = giving pleasure or comfort)
C. doubtful (adj, uncertain, unsure, hesitant,disbelieving, suspicious, unpromising, insecure, unreliable, questionable, unlikely)
D. careful (painstaking, cautious, watchful, not overspending or being wasteful)

41. **Answer: C**
A. intimidate (persuade or dissuade by frightening, daunt=to create a feeling of fear)
B. dominate (control, rule, dictate, take over , overshadow, tower above , stand out)
C. eliminate (get rid of something, reject somebody or something, defeat somebody in competition, get rid of bodily waste)
D. disseminate (distribute, broadcast, circulate, spread, publish, propagate)

42. **Answer: D**
A. impressed (affect or please somebody greatly, make something clearly understood,)
B. charged (asked money for something, hold somebody financially liable, arrange deferred payment, accused somebody of crime)
C. stretched (extended by force, expand and regain original shape, extend excessively)
D. challenged (stimulating test of abilities, face up to , with particular impairement, lacking in something)

43. **Answer: B**
A. dependent (needing something, not self reliant, contigent)
B. dependable (reliable= able to be trusted , trustworthy, loyal, faithful, responsible,)
C. depending (be contigent, varying according to the circumstances)
D. independent (supporter of self-governing , not controlled by another, able to function by self, self supporting)

44. **Answer : A**
A. heritage (riches of past, something passing from generation to generation)
B. inheritance (inherited wealth or title, ownership or succession by heredity, right to inherit, transmission of genetically controlled characteristics)
C. heredity (ancestry, passing on of genetic factors, set of inherited characteristics)
D. hereditary (transmitted genetically, handed down through generations)

45. **Answer: A**
A. exhaust (tire somebody out, use something up, try out all possibilities, say everyting about something, drain something of its resources, let out waste gases)
B. extract (pull something out, obtain something from source, get something by force)
C. exert (weild, put forth, use, apply, bring to bear, exercise, make use of)

D. extend (open out in to space, increase size of something, occupy distance or space, continue for time , make extra effort, increase amount by adding something)

46. **Answer: D**
A. forget (not remember, leave something behind, neglect somebody or somethin)
B. forbid (order somebody not to do something, not allow something)
C. forfeit (giving something up, penality for wrong doing or breaking law, be deprived of something,)
D. forgive (stop being angry about something, pardon somebody, cancel obligation)

B. Substitution (47-57)

47. I love <u>antique</u> furniture, but I need your advice before I buy any.The word **antique** in this context means the style, traditions, and qualities of ancient times, (especially the art and sculpture of ancient Greece and Rome) or a household object that is valued because of its age. Therefore, **choice C** is more appropriate for the context. **Answer: C**

48. The police caught the man who entered the building without the owner's **<u>consent</u>**.The word**consent**can be substituted by the word **permission** as stated in choice C. **Answer: C**

49. Look! Dosen't this girl **<u>take after</u>** her mother. The phrasal verb **take after** means the girl resembles or look or behave like her mother and this situation seems astonished the speaker. **Answer : C**

50. Ihave tried to see if these two can get along. Now I assure you,they**<u>are oil and water.</u>** The first sentence as a premise gives us hint how the speaker tried to make the friendly with each other,. Unfortunately, he/she seems fed up by them since they are incompatible(unable to go together or cooperate) like **<u>oil and water.</u>**
Answer: B

51. If you ask me, all I can tell you is that my memory of the accident is **<u>blurred</u>**. The introductory expression, ' If you ask me," remind us the speaker is in doubt or uncertain to remember the event since it occurred quickily or not distinctly seen the image. Hence, the word **blurred** means fuzzy or unclear image. **Answer: A**

52. **<u>Bear with me</u>** for a couple of minutes. I'll finish it and leave very soon. The second sentence gives us hint to guess the underlined phrase that the speaker asked the other person to stay with him/her for a few minutes.**Answer:A**

53. Poverty and crime **<u>go hand- in-hand</u>**. From the given context, we can understand the two issues have cause-effect relationship umblically. **Answer: D**

54. She often **<u>gets on my nerves</u>** whenever we go on holiday together. The idiomatic expression of '**gets on my nerves** ' means annoy somebody a lot.
Answer : A

55. If you want to be someone, **<u>keep your nose to the grindstone</u>**. That is all I could say. The underlinened idiomatic expression means continue to work very

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hard like a grindstone is a machine used for grinding , sharpening and polishing knives and tools.

Answer : B

56. That is a no-poster zone. You are not allowed to **put up** any posters around here.
The expression "no-poster zone" implies it is forbidden to attach or stick any poster in the area . The phrasal verb **put up** in this context can be substituted by the verb **fix.**

Answer : A

57. If you are sure it is something **you cannot overcome** ,then why waste the whole day with it ? Just forget it.
A. dissolvable = liquefy or soften, disappear, break up, adjourn , inaugurate
B. unovercomable = not common
C. insurmountable = means impossible or undefeatable , not easy , not fail
D. unsurmountable = not common

Answer : C

SECTION FIVE: Grammar (58-85)

DIRECTIONS: Questions 58-85 are incomplete sentences. There are four alternative words or phrases, A-D, given below each questions. Choose the word that best completes the sentence and blacken the letter of your choice in the space provided in the answer sheet.

58. The question is about the position of time adverbs in the present perfect tense.
A. yet : The time adverb "**yet**" shows that we expect some action to take place in the future. It is used in Yes/No questions and Negative statements at the end of the sentence.
Example : Have you finished **yet**? .(Question)
No, I haven't received the letter **yet**. (Negative)
B. ever : its synonyms are like always, still, forever… and it can be used in a question and at the middle of negative statement.
C. never : for no reason, under no circumstance or not at all and its position is in the middle of the statement , not the last
D. ever since : The adverb " ever since' can be used alone in the end position.
He had a bad summer last year and has been off work **ever since**.(He is off work from the time when). **Answer : A**

59. The context given is about imperative statement. It is a well-known tradition that the presense of comma after the name called in written English like a short pause in spoken English.
Thus, after a comma, the bare verb should be used as stated in choice **Answer : A**

60. Working the exam thoroughly or splendidly can lead to join the best field in the country,
not working fast,good or hard.
A. very fast (adj, like a steak of lightning, , quick as flash or the wind

Remember, then, that the subject is not always the noun closest to the verb. It is the noun (or pronoun) that names what the sentence is about, and it may be separated by several words from the verb.

CASE 2: Reaching Agreement When the Subject Is an Indefinite Pronoun

Remember to add an *-s* to the end of the verb in the present tense if the subject is one of the indefinite pronouns listed below:

- one (anyone, everyone, no one, someone)
- anybody (everybody, somebody, nobody)
- anything (everything, something, nothing)
- each, either, neither. As a general rule, treat these words as third-person singular pronouns (*he, she, it*). In the following sentences, each subject is an indefinite pronoun and each verb ends in *-s*: *Nobody* ***claims*** *to be perfect.*
  Everybody ***plays*** *the field sometimes.*
  Each of the divers has an oxygen tank.
  In that last sentence, note that *has* agrees with the subject *each*, not with *divers* (the object of the preposition).**Answer :A**

71. The context with "so---that" structure given demands our knowledge of the contextual meaning of the words. The alternatives listed A to C cannot show us how **insufficient** the money was to buy lunch let alone the other stuffs and how she stingy or not generous she was. **Answer: D**

72. The idiomatic expression "I could make no head or tail out of it" means no can understand someone or something that someone has said. (Usually with the negative.).Shocking and dismaying are more or less are similar in meaning that how panic or disappointing the story was and cannot describe the notion of the speaker. The term "**appalling**" implies how horrifying or dreadful the story was and cannot go with the context. The word "**perplexing**" signifies how the story was confusing to understand from the very beginning to the end. **Answer: D**

73. This question is about the extension of Conditional Type II. As revealed in the stem of the question , the tense in the **If-Clause** is simple past "**killed**"then the **Main Clause** can be "**would +V1.Answer : A**

74. The question is about adverbs of purpose structure : so that, so as to and so as for.
So that is used with a noun or pronoun and its meaning is with the result or consequence that and its structure is " so that + Clause(SVO). For example, I am going to call her so that she knows I love her. Essentially one action will affect a noun or pronoun. It can be substituted by the phrase "in order that"
So as to is used with a verb. For example, I am going to speak softly so as to not disturb you. Essentially, one action will affect a verb. Its form is " …**so as to + V1**…
So as for is used with a noun or pronoun. For example, So as for leaving tomorrow, the trip is off. Essentially, something is in regard to something else.**Answer: C**

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75. This question demands our knowledge of the embedded question. Sentences containing an embedded question consist of a main clause and dependent (or noun) clause. The embedded question (noun clause) in introduced by wh- where, how, who, whom, whose, what, which, or whether and usually follows the following Phrases :

I don't know…	Can you tell me…
I know…	Do you know…
I wonder…	how can I…

Sometimes the embedded question (noun clause) in the sentence can be changed or reduced to an infinitive or infinitive phrase.

Example Main Clause	Embeded question (Noun Clause)
Can you tell me	how I can wrap this package? where I can find that? what I should bring for the party?
I don't know	how I can wrap this package. where I can find that book. what I should bring for the party.
Can you tell me	how to wrap this package? where to find this book? what to bring for the party?
I don't know	how to wrap this package where to find that book. what to bring for the party.

As can exemplified above, the introductory part of the embedded question is "do you know " and the embedded qution or noun clause starts with " how many time …" followed by SVO (subject + veb+ object). For this reason, choices A, B and D cannot be answers.**Answer : C**

76. To answer this question , we should understand not only the the unique features of quantiers of countable nouns but also the whole sentence. Practically, it is unusual for students to score full mark in the entrance examination. Only some exceptional outstanding students (insignificant number) may earn full mark which can be expressed by " few'(negative meaning) . If this so, the choices B- D have positive meaning that a lot of students have obtainedfull marksin the examination which is contrary to the reality.

Answer : A

77. This question is about the usage of adverbial of time duration. The present **perfect** is used to talk about an action that began in the past and continues up

B. B, very good (adj, great, wonderful, super, tremendous,marvelous, not awful)
C. very hard (very difficult, challenging, troublesome,
D. very well (adv, thoroughly, back to front, splendidly, wonderfully, not badly) **Answer : D**

61. To answer the question with a formal English grammar, we can eliminate the distracters which are expressed under choices A and D since the verb "see" at this context cannot have progressive form "seeing". The present tense question like choice "B" is not appropriate for the context since it cannot be always true or general truth or habitual action that can be expressed with present tense question. The context given needs present perfect question that can express an action which started previously and has strong connection at the time of speaking.**Answer : C**

62. For better understanding of the usage of " ever and never " , let's examine the alternatives.
A. was ever : as stated in question number 58 choice B,it can , at normal condition be used in a question form or in a negative statement , not in a positive statement. Besides, the expression **"her own age "** is not the single incident that occurred in a specific time event as stated in choices A, B, and C. Rather, it can be ranged from some time span in her late childhood to adolescent. Such kind of context can better be expressed with present perfect tense.**Answer : D**

63. The second sentence " May be the question was tough." implies how difficult the question was. If so, choices B, C and D cannot go with the context. **Answer :A**

64. When a short actioninterrupts the previous longer action , the short action can be expressed by **when** and **simple past tense** while the previous ongoing action can be expressed by **past continuous tense (was/were + v.ing) . Answer : C**

65. The expression " **Look!** " can make us visualize that the speaker stretches his/her hand to point to Pual who is talking with the man. The action is going on at the speaking moment which needs present continuous tense as stated under choice "C, is speaking".**Answer: C**

66. The given context helps to infer that the incomplete question is about the Conditional Type III. For this Type, We use **"had + V3"** in the **If Clause** , and **would +have +V3** in the **main clause.** For this reason,choices A, B, and C cannot bethe answer.**Answer: D**

67. This question demands our knowledge cause-effect relationship and prepossions followed by the verb " look –". For better understanding, let's take a look at the synonyms of alternatives.
A. looking at = examination, investigation , searching or studying critically
B. looking into = investigating, exploring , take/have a look at, researching, scruitinizing
C. looking on = be facing, concidering, think about, regarding, perceiving

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D. looking for = searching for, seeking, trying to find, hunting for , rummage around for.

From the aformetioned synonyms listed in the alternatives, "looking for" is the best answer for the context.**Answer : D**

68. The question demands our knowledge of adjective quantifiers of some, any,at, none and their compounds. Let's take a look at turn by turn.

- Any and its compounds may have similar in meaning with some and compounds but used in question and negative statements.
The indefinite prnounAnywhere means anyplace at all can be used question and negative statements.
- Somewhere means in or to some place can be used in a positive question statements.
- Nowhere means not at,to;in any place
- Everywhere means in every place.**Answer : C**

69. It is the question of tags. For positive statements, negative tags is needed vici versa. The response for tags can either be negative or positive statem Choices A and B can not be answer since they are ungrammatical. Choic cannot also be an answer because " **it does"** cannot go with the verb to be as it stated in the premise statement. **Answer : D**

70. This question is about the Tricky Cases of Subject-Verb Agreement. In *present tense, a verb must agree in number with its subject*. That, of cours the basic principle of **subject-verb agreement**. It's a simple enough rule, bu certain occasions even experienced writers can slip up on it. Let's have a lo three of the trickier cases of subject-verb agreement:

1. making subject and verb agree when words come between them;
2. reaching agreement when the subject is an indefinite pronoun and
3. making the verbs *have, do,* and *be* agree with their subjects.

CASE 1: Making Subject and Verb Agree When Words Come Between Them

In determining subject-verb agreement, don't let yourself be confused by words come between the subject and the verb. Let's compare these two sentences:

This box ***belongs*** *in the attic.*

This box of ornaments ***belongs*** *in the attic.*

In both sentences, the verb *belongs* agrees with its subject, *box*. Don't let prepositional phrase in the second sentence fool you into thinking that *ornam* the subject. It is simply the object of the pripositio*nof* and does not affect agreement of subject and verb.

Prepositional phrases (as well as adjective clauses, appositives and participle phrases)) often come between a subject and a verb. So to make sure that a verb agrees with its subject and not with a word in the phrase or clause, mentally cross the interrupting group of words:

One (of my sister's friends) ***is*** *a pilot.*

The people (who survived the explosion) ***are*** *in a shelter.*

A man (chasing unicorns) ***is*** *on the terrace.*

at: My friend is at work.
She is at home.
in: He is in a state of work).
on: He is on duty (scheduled to confusion.
for: I mistook you for someone else.
as: I see her as a good person.
by: They are by themselves (alone).

G. PURPOSE

for: **He bought it for an emergency.**

She went to the city for sightseeing.
He loved her for her thoughtfulness.

The second sentence gives us hint about the context.**Answer: C**

83. This question is about the adverbs like "hardly, scarcely, barely, no sooner". We use them
when telling a story in the past time, a concept which is related to our perception of reality
and part of the existence which is measured in seconds, minutes, hours, days, weeks etc.
Adverbials below are used with partial inversion to refer to the earlier event. In the inversion construction, a verb or an auxiliary precedes the subject. For example:

Here **comes the bus**. *You don't work here,* **do you**?
Hardly **had I arrived** *home when the telephone rang. (I had hardly arrived home when the telephone rang.)*
Scarcely **had she finished** *reading when she fell asleep. (She had scarcely finished reading when she fell asleep.)*
Barely **had they won** *the match when the coach had a heart attack. (They had barely won the match when the coach had a heart attack.)*
No sooner **had the company launched** *its new product than it went bankrupt. (The company had no sooner launched its new product than it went bankrupt.)*
No sooner **did they realize** *that they had made a mistake than the company went bankrupt. (They no sooner realized that they had made a mistake than the company went bankrupt.)*

Note that with *hardly, scarcely* and *barely*,we use *when* to connect the two clauses, while with *no sooner* we use *than* (*sooner* is the comparative form of *soon*).

No sooner ***had I opened*** *one doll than I found another one nested in it.*

If we start the sentence with ***no sooner***, ***had*** is followed by the past participle form of the verb.In this construction we can use either the past perfect or the past simple tense; there will be no difference in meaning:

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No sooner ***had*** *the company* ***launched*** *its new product than it went bankrupt.*
No sooner ***did*** *the company* ***launch*** *its new product than it went bankrupt* **Answer: D**

84. **Such (a) ...that and so... that structure**

A/ I didn't enjoy the book. The story was so stupid.

- We use so + adjective/adverb: so stupid / so quick/ so nice/ so quickly
  I didn't enjoy the book. It was such a stupid story.
- We use such + noun: Such a story / such people
  We use such + adjective + noun:
  such a stupid story / such nice people
- Note that we say such a... (not 'a such...')

B/ So and such make the meaning of an adjective (or adverb) stronger:

- It's a lovely day, isn't it? It's so warm. (= really warm)
- He's difficult to understand because he speaks so quickly.
- Compare so and such in these sentences:
- I like Tom and Ann. They are so nice.
- You can use so...that...:
- The book was so good that I couldn't put it down.
- I was so tired that I fell asleep in the armchair.
- You can leave out that in sentences like this:
- I was so tired (that) I fell asleep.
- We enjoyed our holiday. We had such a good time. (= a really good time)
- I like Tom and Ann. They are such nice people, (not 'so nice people')
- You can use such...that...:
- It was such a good book that I couldn't put it down.
- It was such lovely weather that we spent the whole day on the beach.

Choices C and D cannot go with "such (a(an) + agj/adv + noun ..." structure since they are verb and noun respectively. B cannot also be an answer even though it is an adjective form since it has passive sense , but the context needs an active voice.**Answer : A**

85. Like question numbers 63 and 68 in this exam paper,this question is about indefinite pronoun and their meaning in sentence.

A. Anything: It means whatever thing or no matter which cannot go the second sentence since they opposite each other.
B. The same is true that "everything" means the wholething or all is opposite the second sentence.
C. Nothing which means nobody or zero possibility to stop the opposing force. This can go with the context.
D. Something which means somebody or somewhat cannot go with the second sentence.**Answer : C**

SECTION SIX: Communicative Activities (86-105)

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the present (and will probably continue in the future). This is the important thing to remember about using the present perfect: ***the activity / action / event is still going on.*** It has not finished. This is the difference between the present perfect and the simple past: the simple past is used to talk about an action that started and FINISHED in the past, while the present perfect is used to talk about an action that started in the past and CONTINUES in the present.**How long have you...?** We can use the **present perfect** tense when you want to talk about ***how long*** you have done something, or for what amount of time you have done something. **Answer : B**

78. The question is about the "only" in present perfect tense. It can modify verbs or nouns. It can go anywhere in a sentence. Unfortunately, many people never put it anywhere other than right before the verb, no matter what it is modifying. **Only** comes before the word it qualifies. If we change the position of **only**, the meaning of the sentence can be changed.
The following examples can show this.
Only I spoke to his brother.
This means that you were the only person to speak to his brother.
I only spoke to his brother.
This means that you only talked. You didn't write a letter.
I spoke only to his brother.
You didn't speak to anybody else.
I spoke to his only brother.
He only has one brother and I spoke to him.
When we retun back to the question, the word order of Choices A and C follow the sequence of a question for the statement word order. The action starts from the last ten years and strongly linked to the time of speaking. For such action, present perfect tense is more appropriate than past perfect or other tenses.**Answer : B**

79. The question is about future perfect tense . The **future perfect tense** is formed with **will have + past participle**, as in, "I **will have left** by tomorrow" or "They **will have been** friends for a long time." The form stays the same no matter what subject you use. We use the **future perfect tense** to talk about an action that will be completed sometime in the future. Usually this action will be completed *before* something else happens or by a certain time in the future. For example, we can say, "I **will have graduated** by the time I turn 18." This means that I will turn 18 in the future, and at some point *before then,* I will graduate in the future.**Answer: D**

80. The question is about passive formation. The form of passive is **Object +Verb to Be + V3 + (by + dower of the action).** Passive voice can be used for at least three situation.

1. When we want emphasize the receiver of the action rather than the doer of action,

1. When we do not know the doer of the action,
2. When the doer of action is everybody, many people and the like.

For the above reasons, the "Aspirins…" in the question cannot be the doer of action rather theyare the receiver of the action. All alternatives except choice Care forms of active voice. **Answer: C**

81. It is about the response for tag questions and answers.. As we learnt in our English lessons, positive tags can be used for negative statements and negative tags for positive statements. The auxiliary verb can be used for tag question and answers. As shown in the question the auxiliary verb is the negative form of verb to Be ("Aren't"), If so choices B and C should be eliminated. The past time in Choice D cannot go with the context given.**Answer: A**

82. The question demands our knowledge of prepositions in a sentence. Prepositions relate two events in time or two people or things in space. They form a closed class. They also represent abstract relations between two entities. Usage of Prepositions On, by, for, with Uses of Common Prepositions.They are used to express a number of relationships, including time, location, manner, means, quantity, purpose, and state or condition. The following outline demonstrates the uses of common prepositions.

A. Means or Agent (by, on with)

By : This book was written **by** a famous author.

on: They live on bread and water.

with: He chased the mongoose **with** a stick.

B. Instrument (by, with)

"I heard the news **by** television. (Communication)

"She came **by** bus." (Transportation)

"He opened the door **with** a key." (Instrument or tool)

C. Accompaniment (with)

"I like spaghetti **with** white sauce."

"Mrs. Vajiona went to Thassos Island **with** her husband Georgios."

D. Purpose (for)

"He went to the store **for** milk and bread.

E. MANNER

by: By doing it yourself, you save time.

like: He looks like a hero.

on: I swear it on my word of honor.

with: He ate it with a fork.

in: He left in confusion.

The room was in a turmoil.

You can do it in a day.

F. STATE OR CONDITION

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Narrative Writing

A narrative essay is often written as a story, with a plot, characters, dialogue and climax and is almost always the personal recollection of the writer. The primary purpose of narrative writing is to describe an experience, event, or sequence of events in the form of a story

Goal:	The goal for narrative writing is to tell a story of an experience, event, or sequence of events while holding the reader's interest. Narrative writing is usually characterized by the following: • written in first or third-person • characters, setting, plot • may include some dialogue • events organized in chronological sequence (although flashbacks may be used)
Examples:	Narrative writing appears in and is not limited to poetry, short stories, novels, personal essays, tall tales, and folk tales, to name just a few. It also takes a particular form in scripts and plays.
Characteristics:	The general characteristics of a narrative writing include: **1. revealing something of importance** narratives make a point and that point is usually defined in the first sentence but may appear as the last sentence in the first paragraph **2. depicting characters and setting vividly** bring the story to life by using your senses to depict the events of the story: how does it look, sound, feel, smell. Include the important events that make up the story in enough detail for your readers to understand what happened while avoiding any details that do not relate to the main point **3. showing, not telling:**use vivid and precise verbs when describing your events **4. presenting events in a clear, chronological order:**use a logical progression with good transitions when moving from point-to-point

109. The clinching sentence " Please see my resume for additional information on my experience." can be found at the closing part of the job application letter.

Answer: D

110. From the given alternatives, "… a little beautiful town in the valley" can draw a vivid picture of the town of BishanGari which can be categorized under a descriptive writing.

Answer: B

111. This extract tries to persuade and push indirecltly the readers to buy the goods (commodity) or service. Such kind of piece of writing might be taken from persuasive writing which is slightly related to argumentative writing.

Answer: D

112. Descriptive Writing:It relies on effective use of sensual and visual detail to create a vivid picture of a person, place, object, experience or emotion. The primary purpose of descriptive writing is to describe a person, place, or event so that the topic can be clearly seen in the reader's mind. The writer must use vivid details that paint a picture for the reader.**Answer: C**

Goal:	Descriptive writing vividly describes a person, place, or thing in such a way that the reader can visualize the topic and feel like they are part of the experience. Descriptive writing is characterized by the following: • elaborate use of sensory details • details which go beyond the general • details which enable the reader to picture or relive what the writer is telling
Examples:	Descriptive writing appears almost everywhere and is often included in other genre such as a descriptive introduction of a character in a narrative.
Characteristics:	The general characteristics of a descriptive writing include: **1. vividly descriptive details** elaborate on sensory details including how does it look, sound, feel, smell **2. figurative language may be used** using words, phrases, symbols, and ideas such as simile, hyperbole, metaphor, symbolism and personification in such as way as to evoke mental images and sense impressions **3. showing, not telling** use vivid and precise adjectives, adverbs and verbs when describing the topic

86. The question is about how to ask a direction politely. To reply negatively, the speaker should start " sorry" and then he/she justifies why he/she cannot reply positively. **Answer: A**
87. The **present perfect** is used to talk about an action that began in the past and continues up to the present (and will probably continue in the future). This is the important thing to remember about using the present perfect: ***the activity / action / event is still going on.*** It has not finished. We use ***for*** to say **how long**:
    We have been waiting **for twenty minutes.**
    They lived in Manchester **for fifteen years.Answer: A**
88. The question is about the Yes-no answer with verb to be (Are). If so, the woman can reply with the verb to be (am). **Answer: D**
89. The teacher asked the student what the student's sister does for a living(her job). The best question for the given response can be " What does your sister do?".**Answer: B**
90. The time can be asked in two ways: what is the time? " or " what yime is it?" Without any hesitation," It is just after 7 o'clock." Is the appropriate answer for the context.**Answe : B**
91. Abdi is asked why he left Jimma. He is expected to justify that he has got a better job the current place. **Answer: C**
92. A receptionist is supposed to be polite for a customer to make him/her stay long in the hotel comfortably. The question for the reply is " What can I help you?"**Answer:D**
93. The first speaker invited the second speaker to accompany(go) with him/her to the theater. The best reply to the given context can be " Mmm…I think, I would.**Answer: A**
94. Bochera appreciated the creative idea of the Millennium Grand Dam and Bikila agreed with him by saying " Certainly." The choices A, B, and D cannot go with the given context and the clue.**Answer: C**
95. From the brother's saying , we can deduce something about the weather condition and he wants to have a short walk with his sister despite the wet weather., the sister can reply by saying " Do you have a rain coat?" since it can go with context better than from the other given alternatives. **Answer: B**
96. Wogari took the initiation or reminded Kume to go out for dinner. During the time of speaking, the time of dinner for Wogari can be known , Kume may not say "I don't know what time you have dinner?" Rather, Kume may want to know whether Martha and her friend go with them (join with them) for the dinner.**Answer: B**
97. Berhanu have shown apologetic feeling while he used the trainers (training shoes or sneaker) without permission. The appropriate response can be " How disappointing ! You know I haven't even used them.**Answer: C**

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98. Taye wanted to know whether kebed's former girl friend got married or not. The best answer for such context might be "I've no idea." Since the rest alternatives are far from the context.**Answer: C**

99. From the stranger's part, we can guess that he/she raised such question at the middle of the dialogue. The question posed by the stranger needs "Yes-No "response. If so, " Well, I live near Addis Ababa." Can be best for the context. **Answer: A**

100. Tadesse wants to know whether Hailu speaks Arabic or not. From Hailu's second sentence with "But…" , we can expect negative reply like "Only a little". The rest alternatives can't go with the given context.**Answer: B**

101. The part of the Senbeta's expression tells us that he has returned back to the place after some time and he raised the question " Isn't that surprising?" for comfirmation from Solomon. "Sure. I can't believe it myself." is the best reply to the given context.**Answer: C**

102. From the tone of Alem's question, we feel that she herself is surprised when she knew Kebede bought a new car. Chala can reply " What ? You should be kidding?" is the best answer from the given alternatives.**Answer: C**

103. The speaker student hasn't seen her with such unusual feeling and he/she wanted to know what makes her unsual behavior. From the given alternatives, we can guess that the friend expects that "She saw her boyfriend with another girl." makes her very angry.**Answer: D**

104. From Senaet response, we can guess Chane's question that he might ask her the means of transportation that her father goes to work. The response "By bus" should be a good reply for the question "How does your Dad go to work?" **Answer: C**

105. From the response given, we can understand that the premise question should be a Yes/No response question. The only option that seeks a Yes/No answer question is " Does your sister live in a city? " However, she lives in a small town , not a big city. **Answer: C**

SECTION SEVEN: Writing (106-120)

106. The expression " …a contemporary look" can create a picture of modern building on the readers mind the which is a typical characteristics of descriptive writing. **Answer : B**

107. This piece of notice "*Please place your purchase here.*" Can be found in a supermarket notice since purchasing is taking place in a supermarket than others in the alternatives.**Answer: D**

108. A text which starts like " A man and his son were once going…" can be best taken from a narrative text. Before we proceed to the pertinent question , let's have a short look at about the four writing styles. Generally, there are four types of writing styles: narrative, descriptive, expository and argumentative writing. Literary essays usually use the narrative and descriptive style of writing while academic essays usually employ the expository and argumentative types.

	catalogues, newspaper articles, magazine articles, how-to writing, pamphlets, reports and research papers.
Characteristics:	The general characteristics of an expository writing include: **1. state the main idea** this must be narrow enough to be supported in the writing **2. supporting ideas using quotes, statistics, examples, and/or facts** use any available resource to instruct the reader including published books, television, Internet, etc. **3. information is logically organized** depending on the type of expository that is being written, the information could be organized chronologically, from least important to most important or from most important to least **4. show commitment to the topic** supporting details do not deviate from the topic **5. conclusion:** restate the topic and main supporting ideas

117. Which one the following word is correctly spelt?
   A. lisence (incorrect)= license(correct)
   B. penality (incorrect) = penalty (correct)
   C. enterance (incorrect) = entrance (correct)
   D. perseverance (correct) (meaning=persistence) **Answer : D**

118. This question is about punctuation of the direct speech. In the direct speech , the actual words of the speaker should be enclosed by quotation marks " …". When we want split the actual words or expression for emphatic reason, comma should be placed before the quotation marks. The direct speech will be started with capital letter of the first word inside the quotation marks. **Answer: B**

119. The expression *"Previous experience as a security guard would be an advantage."* can best be extracted from a job advertisement.***Answer: C***

120. A memo for a close friend can start with informal expression like " Hi " rather than formal expressions like "My dear friend or Dear friend and the like.**Answer: A**

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ETHIOPIAN UNIVERSITY ENTERANCE EXAMINATION (EUEE)
ENGLISH ANSWERS GINBOT 2005/JUNE 2013
BOOKLET CODE 12 and SUBJECT CODE 01

SECTION ONE: WORD ORDER

1. D … Subject – Verb agreement in well ordered Questions
2. B….Subject + Verb + Object order of declarative sentence
3. D…. Subject + Verb + Object order of declarative sentence
4. B….. Subject + Verb + Object order of declarative sentence
5. C….. Subject + Verb + Object order of declarative sentence

SECTION TWO: PARAGRAPH COHERENCE

6. C… in arranging jumbled sentence, we have to pay attention to the topic sentences and then the transitions so us to decide what comes next. a, b, d and e can't start the paragraph because the first two begin with the pronoun "it" implying there is a noun mentioned before, choice "d' begins with "For instance" indicating that there is something mentioned and choice "e" consists of "for such communication" implying a sentence with the word "communication" precedes it.
7. A… use the idea in No. 6
8. C… use the idea in No. 6
9. D… use the idea in No. 6
10. A… use the idea in No. 6
11. D… use the idea in No. 6
12. D… use the idea in No. 6

SECTION THREE: READING COMPREHENSION

13. B…he idea of the entire paragraph
14. A… honed in this context means improved, the clue is our language paragraph 3 lines 5 and 6 confirms the answer
15. A…paragraph 2 lines 9 and 10 confirms the answer
16. D…paragraph 2 lines 1-4 confirms the answer
17. C …paragraph 2 lines 1 and 2 confirms the answer
18. C … the entire essay
19. B…paragraph 3 lines 5 and 6 confirms the answer
20. B…paragraph 2 confirms the answer
21. D…paragraph 5 lines 3-7 confirms the answer
22. B…paragraph 6 lines 3 and 4 confirms the answer
23. C…paragraph 7 lines 5 confirms the answer
24. C…paragraph 5 lines 1 and 2 confirms the answer
25. C…paragraph 2 lines 7-9 confirms the answer
26. B …paragraph 3 lines 5 and 6 confirms the answer

SECTION FOUR: VOCABULARY

27. D…. excited means eager and the clue is prospect of joing university
28. C ….the clue her is checked for quality, inspector means checker

113. The instruction on the cover or tube of the medicine should be clear and straight forward. Ointment in the form of cream or lotion isused for external skin, not swallow or drink.**Answer: A**
114. The correct spelling for the given alternatives are :
   A. develope (Incorrect) = develop(correct)
   B. received (correct). Similar spellings are conceive, perceive, receipt, receive, deceive , ceiling and the like).
   C. necessary (incorrect) = necessary (correct)
   D. grimmer(incorrect) = grammar (correct).
   Answer: B
115. Argumentative essays present a thesis statement in which the writer resents the facts and attempts to persuade the reader to agree with the writer'ps viewpoint. This form of persuasive writing has a primary purpose of making a statement that the reader will agree or disagree with, then supporting the statement with specific details that will convince the reader. Argumentative essays are very similar to expository essays because they to require reader to investigate a topic, gather and evaluate evidence, and develop an argument. However, these types of essays involve more research than an expository essay. Again, third person is generally used, so replace "you" with "one."**Answer: D**

Goal:	Persuasive writing gives an opinion in an attempt to convince the reader that this point of view is valid or tries to persuade the reader to take a specific action.
Examples:	A few examples of persuasive writing may be seen in and is not limited to letters to the editor, editorials, advertisements, advice columns, award nominations, pamphlets, petitions and opinion writing.

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Characteristics:	The general characteristics of a persuasive writing include: **1. stating a position** this should be limited in scope and can be readily definable, debatable and meaningful to the writer and reader **2. presenting ideas logically, factually & clearly** possible formats could include a comparison and contrast between the two debatable sides to the topic **3. supporting ideas using quotes, statistics, examples, and/or facts** use any available resource to back your position including published books, television, Internet, etc. Know the difference between facts and truth - "truth" is a commonly held position that cannot be supported with facts. **4. ending with a strong argument, summary, or call for action** clearly redefine the topic and restate your most compelling evidence

116. Expository Writing: It is the best technique for writers ifthey wish to inform their readers some specifics about a certain topic. Expository writing is used when we aim to deposit information to the readers by explaining a concept, principle or a process. It asks the writer toelaborateanalyses, assess, define, discuss, evaluate or illustrate a certain concept.
The primary purpose of expository writing is to provide information such as an explanation or directions. In expository essay writing, the writer has to make sure that he/she is able to answer questions that begin with who, what, when, where, why and how. **Answer: C**

Goal:	The goal for expository writing is to give information such as an explanation or directions. It seeks to inform, explain, clarify, define or instruct.
Examples:	Expository writing appears in and is not limited to letters, newsletters, definitions, instructions, guidebooks,

29. B…the clue is brought our luggage to our hotel rooms, porter means gatekeeper or door keeper
30. C …the clue here is request for a vacation, submitted means presented
31. A …pig is sometimes used as a metaphor for one eating a lot
32. A…children, about things around them are clues to fix the meaning, inquisitive means curious
33. C…won't travel by plane is the clue so phobia means irrational fear
34. C…allow work to be done in different locations help to fix the meaning, portable means transportable
35. B ….need additional pristine and vitamin help to fix the meaning, malnourished means underfed and skinny
36. D …complement means praise
37. Achildren, supposed to are the keys, so obey is the answer that means do as told
38. C …to catch the criminal hinted as to choose casual that means informal
39. A… riot means uprising, unrest
40. B …didn't know any of the stdents there is the clue, reluctant means unwilling
41. B…in bed the whole day and had headache are clues help us decide the answer, lie means recline and the past form is lay
42. A …by eating hot paper is clue, protect means gaurd
43. C lied to my mother is a clue, lie means tell something untrue and lied is the past form
44. C/D…difficult to predict the winner helps to choose the answer, beauty contest means competition
45. A …commit means do something
46. A …first wrong impression is the clue, misjudge means miscalculate
47. C … the clue is it is urgent, helps to fix the meaning
48. A …if you need it badly is the clue to fix the meaning
49. A…no wonder he got it all wrong is the clue
50. A ….hard time, emotions are clues the help to fix the meaning
51. C … everyone wondered what it meant is the clue, unreadable means illegible.
52. A…police, thief, his face are clues to fix the meaning
53. D ….get a grasp of means understand
54. D... a bigger fish to fry means more important
55. C …it wasn't surprise can hint the meaning
56. B …finish in a time is a clue
57. D …make up one's mind means decide or change idea

SECTION FIVE: GRAMMAR

58. A … present parallel construction if there are two clauses in past connected by conjunctions like were we have to use parallel tenses
59. C … simple past tense
60. A… present participle
61. D …. Past construction

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62. A… the comparative and Superlative form of the word happy could be happier/more happy and happiest/most happy
63. B… simple past has the form (…+V-2 +….) and this tense uses time expressions like last + week/year/time/night…
64. B…in conditional type two the if clause is in simple past (… V+ 2…) and the main clause should be in past conditional (…would + V-3…)
65. B… this is past participle for
66. C …. Reporting construction of past tense forms
67. C….simple past construction
68. A…. either of, neither of, each of are usually followed by singular verbs
69. C…..everybody, each, everyone are followed with singular verbs
70. A…. the word alert is coined with the preposition of
71. B… in adverbial clauses of result we use connectors like so…that; such…that with structures (….so + Adjective/adverb + that….) and (….such + a/an + adjective/adverb + noun + that……)
72. C… a compound word…walking –stick
73. A…nobody is negative and means no one.
74. B…present continuous tense (….are/is/am + V-ing…) used to express actions taking place ate the time of speaking
75. D….no one means no person
76. D…we use 'yet' with negative perfect constructions
77. D … comparative form of tough is tougher
78. C… preposition
79. B… in conditional tenses we use simple present in if clause and modals like must, could, may in the main clause.
80. A…in adverbial clauses of result the connectors are like so as to, in order to with the pattern (… so as to/in order to + main verb….)
81. A…later means afterwards
82. A…we say ..something is known to/for…
83. C… simple past construction
84. A …present perfect tense with the pattern of (….has/have + V-3…)
85. A we usually use words like yet, already, just with present perfect construction

SECTION SIX: COMMUNICATIVE ACTIVITIES

86. D…way of expressing inclination to invitation
87. D…when asking capability we usually use "can".
88. B….polite way of expressing agreeing with other's idea
89. D…asking questions
90. A…in expressing
91. A…expressing disagreement
92. C…. in asking permission we use can or may
93. B…asking question
94. A …expressing likes and dislikes
95. C….in talking about whether we use

96. B….giving reasons for something
97. C…present perfect patterns
98. B…expressing ones favorite
99. D …. Asking questions
100. C…we use which for things, who for people and that for both
101. C…expressing politely something you don't know
102. D…expressing politely something you don't know
103. D…expressing ones belief
104. D….asking questions
105. A…in expressing inclination for invitations we use I can't. I have to work.
    - ✓ This evening is no good. I have an appointment.
    - ✓ That's very kind of you, but I've something to do.
    - ✓ Well, I'd love to, but I'm already going out to the restaurant.
    - ✓ I'm really sorry, but I've got something else on.
    - ✓ I really don't think I can.

SECTION SEVEN: WRITING

106. A … .in narratives writing we usually tell stories in a chronological order
107. C….in expository writing we usually focus on giving information, elaborating facts and issues
108. A…in descriptive writing we usually strive to create a vivid experience for the reader. It attempts to create an image to the readers.
109. B….in closing of a letter …we usually use your sincerely, faithfully your, yours in formal letters bye and see you in informal ones.
110. … in using punctuations with quotations it is crucial to comprehend the entire sentence. "Who is in the class?" the teacher asked. Why did the teacher say "stay in the class"?
111. A…in argumentative writing one could attempt to influence others, argue fore or against a certain controversial idea.
112. C… words like however, moreover, therefore, consequently and hence are preceded by semicolon and followed by comma if the connect two segments but if the begin a sentence, the are only followed by comma.
113. A…. in giving instructions we use various expressions like the first thing you do is, to begin with, to start with, first, and, then, nex, after this, following this, finally, lastly etc
114. A… most movies have such declarations
115. A… Restaurant, but extention should be spelt as extension; Cafteria as cafeteria and vollyball as volleyball
116. A…Scent is correctly spelt whereas conscent should be spelt as consent; omniscent as omniscient and remiescent as reminiscent
117. C…in descriptive (see explanation given in No. 108
118. B… in closing a memo written in an office to a colleague we may use informal closings, best, faithfully
119. B… in letter writing, our wrting will depend on our purpose
120. D… the colon is used before lists of things; to indicate series that follow.

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ANSWER EUE ENGLISH EXAMINATION
GINBOT 2006/ MAY 2014

Section one:- word order

1. **D** The two actions go and pick up go one after the other
   Let + object + inf. and inf. + somebody + preposition.
   Let us go and pick him up.
2. **C** Helping verb + subject + adv. of manner + verb + noun phrase
   Did you carefully check your work?
3. **A** Sub – verb inversion to form question with verb to be
   Hv + S + going to + inf. + noun complement.
   Are you going to invite many people to your party?
4. **C** Question word + noun + Hv + pronoun + main verb (v_3).
   How many sisters have you got?
5. **D** In the command the two actions come and join indicate actions one after the other.
   Come and join us for the celebration.

Section two:- Paragraph coherence

6. **D** The right order is e, a, b, c, d. e is the topic sentence. a,b and c are supporting or developing the idea in the topic sentence respectively. d concludes the paragraph.
7. **B** The topic sentence is a b, c and e are describing the place and d concludes by telling as it is difficult for wild life to live and vegetables to grow there.
8. **B** C is the topic sentence .b mentions what should also be taken in consideration. e gives example. d develops the example and a concludes the paragraph.
9. **A** a is the topic sentence. 'This' in d refers to informative speech and followed by c, e and b.
10. **B** The topic sentence is b. The paragraph tells us about malaria. a and e show the effect of malaria. d tells what is done by government and c concludes the paragraph giving suggestion.

Section three:- Reading comprehension

11. **B** Paragraph 4 lines 7 and 8.
12. **C** Paragraph 2 lines 11.
    Paragraph 3 lines 1.
    Paragraph 7 lines 1.
13. **B** Paragraph 7 lines 4.
14. **B** Paragraph 1 lines 8 and 9.
15. **C** Paragraph 2 lines 12 and 13.
16. **D** Paragraph 2 line 9.
17. **B** Paragraph 3 line 4.

18. **C** Paragraph 4 line 7.
19. **C** Paragraph 4 lines 8 – 10.
20. **D** Paragraph 6 lines 9 and 10 for detail information paragraph 6 as whole.
21. **B** Paragraph 7 line 11.
22. **A** Paragraph 7 line 15.
23. **B** Paragraph 7 line 4-6.
24. **C** Paragraph 5 lines 5 and 6.

Section four:- Vocabulary

A. Completion

25. **C** **Defer** means put off to a later time; post pone.
26. **D** **Stationery,** where writing materials are sold
27. **B** **Discharge** means send somebody away; allow to leave a patient from hospital.
28. **D** **Conventions** means consent, agreement.
29. **A** **Invent** means create or design something not existing before.
30. **D** **appointed** means choosen for post.
31. **C** **Reveal** means make known.
32. **A** **Affected** means attacked by a disease.
33. **D** **Shocked** means sudden disturbance of the feeling, terrified.
34. **B** **Restrain from** means hold back, prevent from.
35. **C** **Declared** means announced, make known clearly.
36. **C** **Penalized** means punished.
37. **D** **Substantiate** means give facts to support.
38. **C** **Furious** means violent, angry.
39. **B** **Proud** means having deginity, feel happy.
40. **C** **Lightening** means make or become less heavy, reduce the weight of or burden.
41. **B** **Imminent** means of danger likely to come or happen soon.

B:- Substitution

42. **A** Wait on tables – working or giving service in restaurant.
43. **A** Turn on essay in – give back, submit.
44. **C** Cope with – manage successfully.
45. **C** Turn down – refuse to consider.
46. **D** Get along – agree, be friendly or in harmony.
47. **D** At odds – slight quarrel.
48. **B** Loose – make free.
49. **C** Accompany – go with.
50. **C** Tumble down – likely to collapse.

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Section five:- Grammar

51. **D** The ing- adjective tells what the man is doing.
52. **A** That is subjunctive mood which is the same as should be postponed.
53. **D** Shows something done in the past.
54. **A** Choice A implies the response. I haven't found it any where
55. **B** Question tag- positive statement + negative tag will visit, won't you?
56. **C** There isn't anyone coming or no one appeared to my birthday party.
57. **B** In compound subject joined by neither - - - nor, the verb follows the nearest noun. (Neither your relatives nor **you are** allowed).
58. **B** None of + Pn + pv in present perfect (have died).
59. **C** Relevant is a word followed by 'to' and best completes the sentence.
60. **A** Boring means feeling tired.
61. **B** Extreme adjective (absolutely) + comparative form implies that it is the biggest of all dams in the country.
62. **C** Since people are waiting outside at the moment, adverb of time 'yet' is used.
63. **A** If the verb has two parts adverb of frequency comes within. i.e have never seen
64. **D** Shows something that will take place before another future time/action/ - future perfect.
65. **B** Here the present perfect expresses an action that began at definite time in the past (September) continued up to the present.
66. **B** Look into means investigate, examine.
67. **A** Drop by means pay a casual visit, so preposition 'by' best completes the sentence.
68. **C** So that is a subordinating conjunction introducing purpose clause.
69. **A** Repetitive.
70. **D** Action began in 2000 continued up to now.
71. **B** Shows what is happening at the moment.
72. **C** Action taking place today.
73. **C** Last night shows action completed at a definite time in the past, and the sentence is passive. so was broken is used.
74. **D** The two questions are compared and less difficult than is the appropriate answer.
Precomparative (little) + comparative form + than
75. **B** In conditional I –the if clause uses simple present 'have'.
76. **B** In conditional II when the main clause is in present conditional (should + V_1), the if clause is simple past.
77. **C** Wh. – word + helping verb (is) + pronoun 'it'.
78. **D** Personal pronoun 'it' is used as object (to meet new people).
79. **D** listening means doing something.
80. **D** Action completed in the past at unspecified time and the verb is passive.

Section six:- Communicative activities

81. **A** Asking the amount of coffee he drinks. That is at least five cups.
82. **A** Asking for age. That is twenty two.
83. **B** Suggests as spring is coming soon /approaching/
84. **C** Asking the secretary for permission in order to talk to the director.
85. **A** Asking for opinion about the party, and replied. It was ok. That's giving your opinion.
86. **C** Asking for advice. He needs some one's advice.
87. **B** Asking if someone is sure.
88. **D** Asking for opinion and Nani appreciated the newly bought car.
89. **A** It's his headache again, can be the right reply to the question, Is there anything wrong?
90. **B** Asking for the exact place or location; where the nearest post office is.
91. **B** Sorry, I would rather revise for my exam can be polite reply.
92. **B** The reply sorry, I am busy indicates the proper and polite way of responding to a request for help.
93. **D** Just a little can be reply to "How much salt did you put in?"
94. **A** The answer well, I placed the egg carefully in the jar shows what he did next.
95. **D** The stranger is asking for a place politely.
96. **B** Shows where the stranger lives that is around Arat Kilo.
97. **D** **No, I'm not** can be a reply to the question.
98. **D** The amount given is a reply to the question about price with interrogative adverb- how much.
99. **A** **Just a little** tells if he speaks Arabic or not.
100. **D** 'Not enough' is answer for 'is it enough to pay your bills?'.

Section seven:- Writing

101. **C** It compares and contrasts to explain how people judge against foreigners before and after they have knowledge about them.
102. **A** Descriptive writing is a writing that adds livelihood and created a visual impression of a person.
103. **D** It is taken from a writing which narrates or retells a story.
104. **B** Descriptive writing relies on effective use of sensual and visual detail to create a vivid picture of the mountain. It uses précise modifiers. Adjectives like windy and cold are used to describe Mount Kilimanjaro.
105. **D** Narrative is a writing which is telling a story, what has happened step by step.

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106. D It creates a mental picture to show the event.
107. D Different reasons have been raised to convince and it is argumentative piece of writing.
108. A Narrative story involves a discussion of events that are true or real.
109. B The sequence indicator 'then' shows there are actions happened is sequential order.
110. B Argumentative writing usually uses comparative, In this case,"much faster than" implies that the piece of writing is argumentative.
111. B changeable(x) - changeable
112. D 'With regards' is an appropriate ending of a letter to an old school friend.
113. D The writer is using this expression to inform the person as he hasn't succeeded in a job interview politely with a formal and in non-offending way.
114. A In a compound sentence the two main clauses are joined by ', but'.
115. C It can be a piece of writing in the ending of a letter to a friend.
116. C It is a part of a letter that shows reply to accept invitation.
117. C In direct reporting the actual words are enclosed within quotation marks, separated by a comma from the reporting verb, and if the same actual speech is divided in to two quotation marks are used separately.
118. D It can be beginning of a friendly letter.
119. B a. questionaire (x) - questionnaire
c. extention (x)- extension
d. enviroment (x) - environment
120. C Expository writing explains the process of an issue under discussion

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EUE EXAMINATION MATHEMATICS 2001/2009

Code 24

1. Which of the following is true about the rational function $f(x) = \frac{p(x)}{q(x)}$?
   A. The domain of f is the set of all real numbers where q(x) is well defined and non – zero
   B. If the domain of f is R, then f is a polynomial
   C. The range of f is always R
   D. p(x) is divisible by q(x)
2. Which of the following is true about the function $f(x) = 1 + 3^{-x}$?
   A. f is increasing
   B. The range of f is $\{y: y \geq 1\}$
   C. For $x \leq 0$, $f(x) \in (1, 2]$
   D. The graph of f meets the y – axis at (0, 2)
3. What is the equation of an ellipse with center at (1, 4), vertices at (10, 4) and (1, 2)?
   A. $4(x-1)^2 + 81(y-4)^2 = 324$
   B. $(x-1)^2 + 9(y-4)^2 = 4$
   C. $9(x-1)^2 + 4(y-4)^2 = 1$
   D. $2(x-1)^2 + 9(y-4)^2 = 4$
4. Let f(x) be a polynomial with (x+3) a factor of f(x) and 3 is not a root of f(x) = 0. Then what is the value of f(-3)?
   A. f(3) B. –f(3) C. 0 D. -9
5. Which one of the following sequences is a convergent sequence?
   A. $1, \frac{1}{2}, 1, \frac{1}{3}, 1, \frac{1}{4}, \ldots$
   B. $\{(-1)^n\}_{n=1}^{\infty}$
   C. $\left\{10^{109} - \frac{1}{100}n\right\}_{n=1}^{\infty}$
   D. $\left\{\sin\left(\frac{1}{n}\right)\right\}_{n=1}^{\infty}$
6. What is the coordinate of the point R on the line segment with end points P(-3, 0) and Q(0, -3) such that $\frac{PR}{RQ} = \frac{2}{3}$?
   A. (-1, -1)
   B. (-1, 7/5)
   C. (-9/5, -6/5)
   D. (-7/5, 0)
7. Which one of the following is an arithmetic sequence?
   A. 3, 5, 7, 9, 11, …
   C. -3, 6, -9, 12, -15,…

B. 3, 6, 12, 24, 48,... D. 1,3,6,10,15,21,...

8. Let p: Price is high
q : Items are available
Which of the following is a symbolic form of the argument given below?
Price is high. Items are available is necessary for price not to be high. Therefore, items are not available.
A. $p, q \Rightarrow \neg p \vdash \neg q$ C. $p, \neg p \Rightarrow p \vdash \neg q$
B. $p, \neg p \Rightarrow q \vdash \neg q$ D. $p, \neg q \Rightarrow p \vdash \neg q$

9. The ozone level (in ppb – parts per billion) on a sunny day in a metropolitan area is given by the formula:
$P(t) = 80 + 12t - t^2$, where t is time in hours and t = 0 corresponds to 9 A.M. What is the rate of increase of the ozone level after 3 hours?
A. 6 ppb B. 12ppb C. 107ppb D. 113ppb

10. What is the value of $\int_0^1 \frac{x}{(x^2+1)^2}dx$?
A. $\frac{1}{4}$ B. $\frac{1}{2}$ C. 1 D. $-\frac{1}{4}$

11. The value(s) of x where the graph of $y = \frac{x^2-1}{x^3}$ crosses its horizontal asymptote is (are):
A. x = -2 C. x = -1 and x = 1
B. x = 0 D. $x = -\sqrt{2}$ and $x = 1 + \sqrt{2}$

12. What is the focus of the parabola whose equation is:
$y^2 + 8x + 6y + 25 = 0$?
A. (-2, -3) B. (3, 2) C. (-4, -3) D. (0, -2)

13. Which one of the following is NOT a tautology?
A. $p \vee (q \Rightarrow \neg p)$
B. $p \Rightarrow [(p \Rightarrow q) \vee q]$
C. $[p \vee (q \Rightarrow r)] \Leftrightarrow [\neg p \Rightarrow (q \Rightarrow r)]$
D. $[p \Leftrightarrow (q \wedge \neg r)] \Leftrightarrow [\neg p \Leftrightarrow (\neg q \vee r)]$

14. Which one of the following represents the solution set for the equation $\frac{1}{x-2} = \frac{3}{x+2} - \frac{6x}{x^2-4}$?
A. $\left\{-\frac{1}{2}, \frac{1}{2}\right\}$ B. {-4, 2} C. {-2} D. Ø

15. What is the area under the curve y = 3 sin x over the interval $\left[0, \frac{2\pi}{3}\right]$?
A. 3/2 square units C. 9/2 square units
B. 7/3 square units D. $13\sqrt{2}$ square units

16. What are the values of real numbers u and v that satisfy:
$\frac{u+3i}{4-2i} = \frac{2+vi}{20}$?
A. u = 2, v = 3 C. u = 2, v = 16
B. u = -6, v = 10 D. u = -4, v = 6

17. For a real number s, the function f(x) is defined by
$$f(x) = \begin{cases} \frac{4}{x^2-4} - \frac{1}{x-2}; & for\ x \neq 2 \\ s \quad ; & for\ x = 2 \end{cases}$$
Which value of s make the function f(x) continuous for all x > 0?
A. -∞ B. $-\frac{1}{4}$ C. 0 D. 2

18. What is the solution set of the equation:
$\log_{10} 5x + \log_{10}(x-1) = 2$?
A. {-5, 1} B. {-4, 5} C. {1} D. {5}

19. For $f(x) = x^2\sqrt{2x+12}$, what is the equation of the tangent line to the graph of f at x = 2?
A. y – 4 = 16(x – 2) C. $y = \frac{1}{\sqrt{12}}(x-2) - 16$
B. y = 17(x – 2) + 16 D. y = 8(x – 2) + 32

20. What is the limit of the sequence
$1, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{5}, \frac{1}{5}, -\frac{1}{7}, -\frac{1}{7}, \ldots$?
A. -10 or 1 B. 0 C. e D. does not exist

21. If a line passes through the point (2, 1) and its angle of inclination is 45^0, then which one of the following represents the equation of the line?
A. x = y + 1 C. y = 2x + 1
B. $\sqrt{2}y = 2x + 3$ D. 2y + x = 2

22. Which of the following statements is NOT true?
A. It is possible for a parabola to intersect its directrix.

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B. If the shape of an ellipse is elongated, then its eccentricity is large (close to 1)

C. If the vertex and focus of a parabola are on a horizontal line, then the directrix of the parabola is vertical.

D. In the standard form of the equation of a hyperbola, the larger the ratio of b to a (where a is half of the length of the transverse axis), the larger the eccentricity of the hyperbola.

23. An antiderivative of $f(x) = \frac{1}{1+e^{-x}}$ is:

A. $1 + e^{-x}$
C. $\ln(1 + e^{x})$
B. $\ln(1+e^{-x})$
D. $e^{x} + e^{-x}$

24. The total cost for ordering and storing x units is given by

$C(x) = 3x + \frac{30{,}000}{x}$ birr.

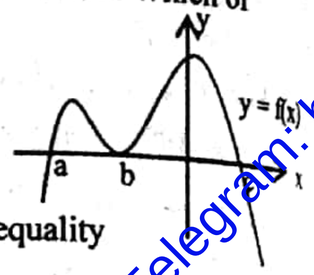
How many units must be ordered to produce a minimum cost?

A. 5
B. 100
C. 300
D. 1000

25. For what value of d is the sphere $x^2 + y^2 + z^2 - 2x + dz = -1$ of radius 7 units?

A. -14
B. $-10\sqrt{2}$
C. $7\sqrt{2}$
D. $10\sqrt{2}$

26. The figure given below is a graph of a polynomial function. Which of the following is NOT true about the function



A. Its degree is at least 4
B. It has three turning points
C. b is a root of odd multiplicity of f(x) = 0
D. the leading coefficient of f is negative

27. Which of the following is the solution set of the inequality

$\frac{1}{1-\frac{1}{x}} \le \frac{x}{x^2-1}$?

A. (-1, 1)
C. $(-1, 0) \cup (0, 1)$
B. $[-1, 0) \cup (0, -1]$
D. $(-\infty, -1) \cup (1, \infty)$

28. $\int 10^{x}\,dx$ is equal to:

A. $10^{x} \ln 10$
C. $10^{x} \ln 10 + C$
B. $10^{x} + C$
D. $10^{x} \log e + C$

29. What is the value of b so that, when $f(x) = 2x^4 + bx^3 - x^2 + 3$ is divided by x +1, the remainder is 7?

A. -3
B. -1
C. 1
D. 3

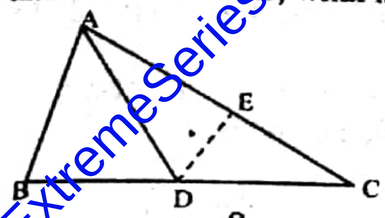
30. For z = x + yi, which of the following represents the circle $(x-3)^2 + (y+1)^2 = 4$?

A. $|z + 3 - i| = 4$
C. $|z + 3 - i| = 2$
B. $|z - 3 + i| = 4$
D. $|z - 3 + i| = 2$

31. A rectangular play ground has length 5 meters more than its width. If 1 meter is added to each side of the rectangle, the area is increased by 154 square meters. What is the length of the original ground?

A. 79m
B. 77m
C. 74m
D. 72m

32. In ΔABC shown below, let $\overline{AD}$ be a median and E be the foot of the perpendicular from D to $\overline{AC}$. If the area of ΔABC is 54 square units and AC = 12 units, what is the length of $\overline{DE}$?



A. 4.5 units
B. 5.4 units
C. 6 units
D. 9 units

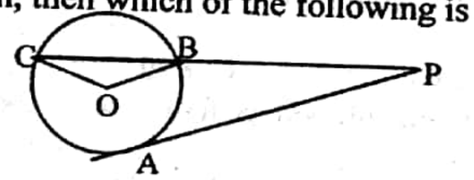
33. Suppose $\frac{x-8}{x^3+4x} = g(x) - \frac{2}{x}$ and $x \neq 0$. Which of the following is equal to g(x)?

A. $\frac{2x-1}{x^2+4}$
B. $\frac{2x+1}{x^2+4}$
C. $\frac{x+2}{x^2+4}$
D. $\frac{x-2}{x^2+4}$

34. If $a_n = \frac{1}{n^2} + \frac{2}{n^2} + \ldots + \frac{n}{n^2}$, then what is the limit of the sequence $\{a_n\}_{n=4}^{\infty}$?

A. 0
B. ½
C. 7/8
D. $+\infty$

35. In the figure given below, the radius of the circle with center 0 is 3cm, $\overleftrightarrow{AP}$ is a tangent line and $\overleftrightarrow{CP}$ is a secant line. If $m(\angle BOC) = 60^0$ and BP = 5cm, then which of the following is the length of $\overline{AP}$?



A. $5\sqrt{2}$ cm
B. $3\sqrt{5}$ cm
C. $2\sqrt{10}$ cm
D. 8cm

36. Which of the following is the inverse of f(x) = 2+ln(x-1)?

A. $g(x) = 2 + e^{x-1}$
C. $g(x) = 2 - e^{x-1}$
B. $g(x) = 1 + e^{x-2}$
D. $g(x) = 1 - e^{x-2}$

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37. If $x_1 = 1 + 3i$ and $x_2 = ½$ are roots of a cubic polynomial f, which expression describes f correctly?
A. $x^3 - 2x^2 + 7x - 6$
B. $x^2 - \frac{5}{2}x^2 + 11x - 5$
C. $x^3 + 3x^2 - 12x + 3$
D. $x^3 + \frac{7}{2}x^2 - 8x + 16$

38. Which of the following is NOT an equation of a hyperbola?
A. $x^2 - y^2 + 4x - 6y = 0$
B. $y^2 - x^2 + 2x - 4y = -3$
C. $x^2 - y^2 = -8$
D. $y = -\frac{4}{x}$

39. A ball is thrown vertically from ground up to a height of 16m. Each time it drops h meters, it rebounds 0.80h m. Noting that the ball travels every height of h twice, what is the total vertical distance traveled by the ball?
A. 40m B. 80m C. 160m D. 320m

40. Which of the following is false about the polynomial $f(x) = x^2(x + 1)^3(x^2 + 1)$?
A. The graph of f(x) necessarily crosses the x –axis at x = -1
B. The graph of f(x) crosses the x-axis at x = 0
C. -1 is a root of f(x) with highest multiplicity
D. The graph of f(x) has at most 6 turning points

41. If f(2) = -3, f'(2) 4, g(1) = -5, g'(1) = 1 and $F(x) = f(2x + 2).g(1 - x^2)$, then what is the value of F'(0)?
A.19 B. 0 C. -20 D. -40

42. The weight w of a person in kilograms and the person's height h in meters are related by the equation:
$\ln(10w) = 4.6 + 3\ln h$
If ln2 = 0.6, ln3 = 1.1 and ln5=1.6, then what is the height of a person whose weight is 60kg?
A. 1m B. $e^{\frac{1}{3}}$ m C. $\sqrt{e}$ m D. 2m

43. What is the sum of the first 20 terms of the sequence 3, 7, 11, 15,…?
A.741 B. 820 C. 899 D. 903

44. On which of the following interval is $f(x) = x^4 - 2x^2 + 10$ strictly decreasing?
A. $(-\infty, 0)$ B. (-1, 0) C. (0, 1) D. $(1, \infty)$

45. Which of the following is equal to $\lim_{x \to 0} \frac{\cos^2 x - 1}{x \sin 2x}$?
A. -1 B.-½ C. 1 D. ½

46. If a geometric progression $G_2 = -10$, $G_4 = -\frac{8}{5}$ and G_5 is positive, then which of the following is a general formula for G_n?
A. $25\left(-\frac{2}{5}\right)^{n-1}$
B. $-25\left(\frac{2}{5}\right)^{n-1}$
C. $\frac{2^{n-1}}{(-5)^{n-2}}$
D. $25\left(-\frac{2}{5}\right)^{n}$

47. If $f(x) = xe^{3x}$, then which of the following is equal to f''(x)?
A. $(1 + 3x)e^{3x}$
B. $(6 + 3x)e^{3x}$
C. $(3+9x)e^{3x}$
D. $(6 + 9x)e^{3x}$

48. Which of the following is not true about $\left\{\frac{e^n}{3^n(n+5)}\right\}$?
A. It converges to 0.
B. It is decreasing
B. It is bounded
D. It diverges

49. Let $f(x) = \begin{cases} 3x - 1, & x \le 2 \\ x + 3, & x > 2 \end{cases}$
Which of the following is true?
A. f'(2) = 1
B. f'(2) =3
C. f'(2) = 5
D. f has no derivative at 2

50. Two regions A and B are separated by a sea. The shores are roughly in a shape of hyperbolic curves with asymptotes y=±3x and a focus at (30, 0) taking a coordinate system with origin at the center of the hyperbola. What is the shortest distance between the regions in kms?
A. $18\sqrt{10}$ B. $9\sqrt{10}$ C. $6\sqrt{10}$ D. $3\sqrt{10}$

51. A translation T(x, y) = (x –h, y – k) takes the line –x + 5y =2 to the line –x + 5y = 10, which of the following relationship is true?
A. h – 5k = 8
B. h – 5k = -8
C. –h + 5k = 10
D. –h + 5k = -10

52. What is the image of point (-2, 3) in reflection about the line x = -1?
A. (0, 3) B. (-1, 2) C. (1, 0) D. (1, 3)

53. Which of the following is a correct identity?
A. $(\sec^2 x)(\tan^2 x) = 1$
C. $\frac{\sin^2 x}{\cos x} + \cos x = \sec x$

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B. $\frac{\sin x}{\csc x} - \frac{\cos x}{\sec x} = 1$ D. $\sin^2 x - \cos^2 x = 1$

54. Let $\vec{u} = (-2, 5)$ and $\vec{v} = (2, 4)$, which of the following is equal to $2\vec{u} - \frac{1}{2}\vec{v} - 3\vec{j}$?

A. (-5, 5) B. (-5, 9) C. (-8, 9) D. (-8, 12)

55. Which of the following is a vector parallel and in opposite direction to $\vec{A} = 3\vec{i} - 4\vec{j}$?

A. $\vec{i} - \frac{4}{3}\vec{j}$ C. $-\frac{3}{2}\vec{i} + 2\vec{j}$

B. $-6\vec{i} - 8\vec{j}$ D. $\frac{3}{5}\vec{i} + \frac{4}{5}\vec{j}$

56. What is the image of the circle $(x-2)^2 + y^2 = 5$ after it is reflected in the line $y = -x$?

A. $x^2 + y^2 + 4x = 1$ C. $x^2 + y^2 - 4y = 1$
B. $x^2 + y^2 - 4x = 1$ D. $x^2 + y^2 + 4y = 1$

57. If $f(x) = 3 - \frac{1}{2}\cos\left(\frac{2\pi}{3}x - \frac{5\pi}{7}\right)$, then what are the amplitude and period of the graph of f?

A. $3, \frac{7}{5}\pi$ B. $\frac{1}{2}, 3$ C. $\frac{1}{2}, \frac{3}{2}\pi$ D. 3; 5

58. If A = (2, -1), B = (3, 2) and D = (-2k, k), then for what value of k is the vector from A to E perpendicular to the vector from A to D?

A. -1 B. 0 C. 3 D. 5

59. Which of the following is equal to $\sec\left(\sin^{-1}\left(\frac{3}{5}\right)\right)$?

A. $\frac{4}{5}$ B. $\frac{5}{4}$ C. $\frac{5}{3}$ D. $\frac{4}{3}$

60. If the standard form of the equation of a line is:
l: $3y - 2x + 3 = 0$, then what is the vector form of l?

A. $r = (2, -3) + t(-2, 1)$ C. $r = (0, 1) + t(2, -3)$
B. $r = (1, -2) + t(0, 3)$ D. $r = (0, -1) + t(3, 2)$

61. If the tail of vector $\vec{u}$ is at the origin and its head is at the point $(2, -2\sqrt{3})$, then what are the magnitude and direction angle, respectively, of $\vec{u}$?

A. -6; 30^0 B. -4; 60^0 C. 12; 60^0 D. 4; 300^0

62. Let $\vec{a}$ and $\vec{b}$ be vectors such that $\|\vec{a}\| = 4$ and $\|2\vec{a} - \vec{b}\| = \|\vec{a} + \vec{b}\|$. Which of the following is equal to $\vec{a} \cdot \vec{b}$?

A. 8 B. 10 C. 12 D. 24

63. At what point on the line $y = 2x$ is the circle with center at (-2, 6) tangent to the line?

A. (0, 0) B. (4, 8) C. (2, 4) D. (3, 6)

64. What is the solution set of the equation $\sin^2 x + 1 = \cos^2 x + \frac{1}{2}$, in the interval $[0, 2\pi]$?

A. $\left\{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11}{6}\pi\right\}$ C. $\left\{\frac{\pi}{6}, \frac{\pi}{2}\right\}$

B. $\left\{0, \frac{3\pi}{2}\right\}$ D. $\varnothing$

65. $\sin^4 A - \cos^4 A$ is equal to:

A. sin2A B. –sin 2A C. cos2A D. –cos2

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ETHIOPIAN HIGHER EDUCATION ENTRANCE CERTIFICATE MATHEMATICS EXAMINATION FOR NATURAL SCIENCE STREAM GINBOT 2002/MAY 2010

BOOKLET CODE:17

Directions: Solve each of the following problems and choose the best answer to the question from the given alternatives. Carefully **Blacken** the letter of your choice on the separate answer sheet provided.

1. Which of the following is equal to $\int_0^{\frac{\pi}{3}} \frac{\tan x}{\cos x} dx$?

   A. $\frac{1}{2}$ B. 1 C. -1 D. $-1+\frac{2\sqrt{3}}{3}$

2. The area of the region bounded between the graph of $f(x) = x^3$ and the x-axis from $x = -1$ to $x = 2$ is

   A. $\frac{15}{4}$ B. $\frac{17}{4}$ C. 4 D. 16

3. Which of the following is true about the polynomial $p(x) = 4 + 5x^2 - (x^2 + 1)^3$? The

   A. Degree is 5.
   B. Leading coefficient is 1.
   C. Constant term is 3.
   D. Coefficient of x^2 is 4.

4. Suppose that the point (2,0) is x-intercept of the graph of the polynomial $f(xi) = x^4 - kw^2 + cx$ and when f(x) is divided by $x - 1$ the remainder is 1. What is the value of k?

   A. -8 B. 8 C. -16 D. 16

5. Which of the following intervals may **NOT** contain a zero of $f(x) = x^4 - 7x^2 + 10$?

   A. (-2, - 1) B. (-1,0) C. (1,2) D. (2,3)

6. For which of the following does its graph lie both above and below the $x-$ axis?

   A. $f(x) = (x + 1)^2 (2 - x)^2$
   B. $f(x) = -(x + 2)^2(x - 1)^2$
   C. $f(x) = (x^2 + 1)(x - 1)^2$
   D. $f(x) = (x + 1)^2 (x^2 - 1)$

7. Which of the following is a rational function?

   A. $f(x) = \frac{\sqrt{2}x^3 - x + \log 2}{x^4 + x^2 + \pi}$

   B. $g(x) = 3x^4 - \sqrt{x^2 + 1} + 2x$

   C. $h(x) = \frac{x^2 - 2^x + x}{x^4 + x + 1}$

   D. $r(x) = \frac{x^2 + \sin x + 2}{x + 1}$

8. If $p(x) = x - 4$, $q(x) = x^2 + 8$ and $r(x) = x^2 - 2x - 8$, then which of the following is the most simplified expression of $f(x) = \frac{x}{p(x)} - \frac{q(x)}{r(x)}$?

   A. $\frac{x}{x-4} - \frac{x^2+8}{x^2-2x-8}$ C. $\frac{2}{x+2}$

   B. $\frac{8x}{x-4}$ D. $\frac{2x+8}{x^2-2x-8}$

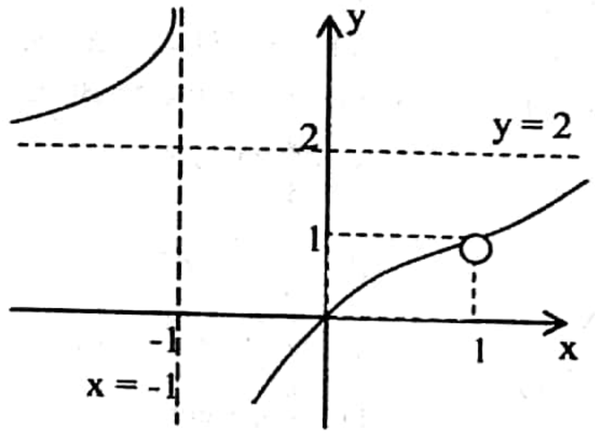
9. What is the solution set of $1 - \frac{1}{x+2} \leq \frac{-1}{x}$?

   A. (-2,0) C. $(-\infty, -2)$
   B. $(0, \infty)$ D. $(-\infty, -2) \cup (0, \infty)$

10. Consider an airplane that flies in straight line from Addis Ababa to city X in a planned time. If it flies at a speed of 1000 km/hr, it will reach at X 30 minutes before the planned time. On the other hand, if it flies at 900 km/hr, it will reach there 20 minutes after the planned time. Which of the following is the distance between Addis Ababa and city X?

    A. 8500km C> 7500km
    B. 8000km D. 7250km

11. Which of the following functions could most likely be sketched as in the figure below?



    A. $f(x) = \frac{2x}{x+1}$

    B. $f(x) = \frac{2x(x-1)}{x^2+1}$

    C. $f(x) = \frac{2}{x+1}$

    D. $f(x) = \frac{2x^2}{x^2-1}$

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Directions: Solve each of the following problems and choose the best answer to the question from the given alternatives. Carefully **Blacken** the letter of your choice on the separate answer sheet provided.

1. Which of the following is equal to $\int_0^{\frac{\pi}{3}} \frac{\tan x}{\cos x} dx$?

   A. $\frac{1}{2}$ B. 1 C. -1 D. $-1+\frac{2\sqrt{3}}{3}$

2. The area of the region bounded between the graph of $f(x) = x^3$ and the x-axis from $x = -1$ to $x = 2$ is

   A. $\frac{15}{4}$ B. $\frac{17}{4}$ C. 4 D. 16

3. Which of the following is true about the polynomial $p(x) = 4 + 5x^2 - (x^2 + 1)^3$? The
   A. Degree is 5.
   B. Leading coefficient is 1.
   C. Constant term is 3.
   D. Coefficient of x^2 is 4.

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   D. $f(x) = (x + 1)^2 (x^2 - 1)$

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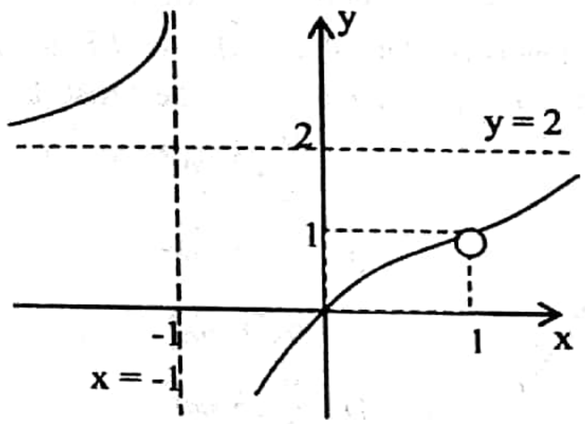

    A. $f(x) = \frac{2x}{x+1}$

    B. $f(x) = \frac{2x(x-1)}{x^2+1}$

    C. $f(x) = \frac{2}{x+1}$

    D. $f(x) = \frac{2x^2}{x^2-1}$

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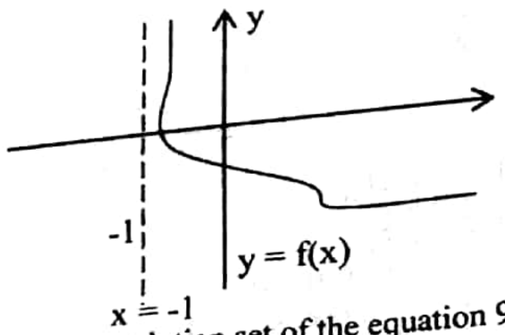
12. The simplified form of $e^{\frac{1-\ln(1/2)}{2}}$ is

A. $\sqrt{2e}$ B. $\frac{1}{2}e^{1/2}$ C. $e^{1/2}in2$ D. $\frac{1}{2}$

13. Which of the following functions can better be represented by the graph shown in the figure?



A. $y = \log_5(x+1)$
B. $y = -\log_5(x+1)$
C. $y = -1+\log_5 x$
D. $y = 1 - \log_5 x$

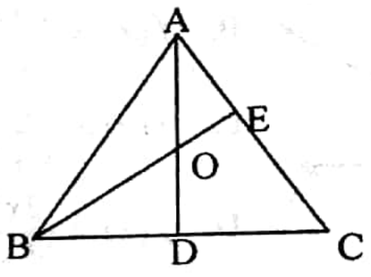
14. The solution set of the equation $9(1-2\log 3^x) = 4$ is:

A. $\left\{\frac{9}{2}\right\}$ B. $\left\{-\frac{9}{2}, \frac{9}{2}\right\}$ C. $\left\{\frac{3}{2}\right\}$ D. $\left\{\frac{\sqrt{3}}{2}\right\}$

15. In a laboratory experiment, suppose that the population p of an organism is increasing exponentially: $p(t) = ae^{kt}$, where a and k are constants and t is the number of days after the start of the experiment. If at the beginning of the experiment the number of the organism was 100 and after 2 days there were 200 organisms, how many of them will be there after 10 days?

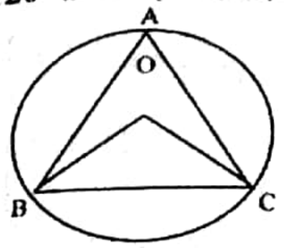
A. 3,200 B. 1,000 C. 800 D. 500

16. The triangle ABC shown below is equilateral and $\overline{AD}$ & $\overline{BE}$ are its medians intersecting at O. if $OD = \sqrt{3}$ cm, the area of the ΔABC is



A. $\frac{9}{4}\sqrt{3}\ cm^2$
B. $\frac{4}{9}\sqrt{3}\ cm^2$
C. $9\sqrt{3}\ cm^2$
D. $\frac{9}{2}\sqrt{3}\ cm^2$

17. In the figure, O is the center of the circle. If $m(\angle A) + m(\angle BOC) = 120^0$ and $m(\angle OBA) = 30^0$, then $(\angle OCA)$ is equal to:



A. 50^0
B. 40^0
C. 30^0
D. 10^0

18. Given three points A(1,0), B(-3,0) and C(1,2), the equation of the line through A and perpendicular to $\overline{BC}$ is

A. $2x+y = 2$
B. $y = -2x + 1$
C. $y = -2x$
D. $x - 2y = 1$

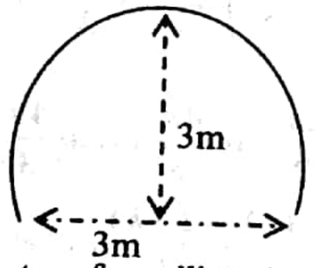
19. Let l be the line containing (1,5) and its angle of inclination be $\theta = 135^0$. Which of the following is the distance from l to (-1, 1)?

A. $2\sqrt{2}$ B. $3\sqrt{2}$ C. $4\sqrt{2}$ D. $6\sqrt{2}$

20. Which of the following is the equation of the circle with center at (2, -1) and tangent to the y-axis?

A. $(x+2)^2 + (y-1)^2 = 1$
B. $(x-2)^2 + (y+1)^2 = 1$
C. $(x-2)^2 + (y+1)^2 = 4$
D. $(x-2)^2 + (y+1)^2 = 4$

21. Consider a gate whose shape is a parabolic arch of height 3m and base width 3m as shown in the figure below. If the origin of the coordinate system is at the vertex, the equation that represents the edge of the gate is



A. $3x^2 + 3y = 0$
B. $4x^2 + 3y = 0$
C. $y = -3x^2 + 1$
D. $3y^2 - 4x = 0$

22. Let the center of an ellipse be at (1, 4) and two of its vertices be at (10,4) and (1,2). What is the equation of the ellipse?

A. $4(x-1)^2 + 81(y-4)^2 = 324$
B. $(x-1)^2 + 9(y-4)^2 = 4$
C. $9(x-1)^2 + 4(y-4)^2 = 4$
D. $2(x-1)^2 + 9(y-4)^2 = 4$

23. What is the length of the minor axis of $x^2 + 9y^2 - 2x = 0$?

A. $\frac{1}{9}$ B. $\frac{2}{3}$ C. 2 D. 6

24. The equation $x^2 - 4y^2 = -1$ represents

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A. A hyperbola with one of its foci at the origin.
B. A parabola with vertex at the origin.
C. A hyperbola with vertices at $\left(-\frac{1}{2},0\right)$ and $\left(\frac{1}{2},0\right)$.
D. A hyperbola with foci at $\left(0,\frac{-\sqrt{5}}{2}\right)$ and $\left(0,\frac{\sqrt{5}}{2}\right)$

25. Which of the following is a point of intersection of the sphere $(x-1)^2+(y-2)^2+(z+1)^2=6$ and the y-axis
A. (0, -2,0) B. (0,2,0) C. (0,-4,0) D. (0,4,0)

26. In the set of complex numbers, the solution set of $x^2-4x+5=0$ is:
A. $\varnothing$ C. $\{2+I, 2-i\}$
B. $\{2+3i, 2-3i\}$ D. $\{1+2i, 1-2i\}$

27. Let z be a complex number such that $\left|\frac{3Z}{-4+3i}\right|=(2-i)\left(\overline{2-i}\right)$. The, the modulus of z is equal to:
A. $\frac{25}{3}$ B. $\frac{10}{3}$ C. $\frac{14}{3}$ D. $\frac{25}{9}$

28. In the set of complex numbers, the solution set of $z^3-8=0$ is
A. {2}
B. $\left\{2,2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right),2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\right\}$
C. $\{2, 2+8i, 2-8i\}$
D. $\left\{2,\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3},\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right\}$

29. If k–th term of a sequence is $a_k=\frac{(-1)^k(2-k)}{2k-1}$, what is its next term a_{k+1} when k is even?
A. $\frac{k-1}{2k+1}$ B. $\frac{3-k}{2k+3}$ C. $\frac{k-3}{2k+1}$ D. $\frac{1-k}{2k+1}$

30. Suppose a geometric sequence $\{G_n\}_{n=1}^{\infty}$ has terms $G_{10}=192$ and G6 = 12. Which of the following is equal to G_2?
A. $\frac{3}{2}$ B. $\frac{3}{4}$ C. $\frac{3}{8}$ D. $\frac{3}{16}$

31. Consider the sequence $\{a_n\}$, $n\geq 1$, where the first term is $a_1=2$, and $a_{n+1}=a_n+4$ for all $n\geq 1$. Then, the sum $a_3+a_4+a_5+\ldots+a_{35}$ is equal to
A. 2460 B. 2458 C. 2442 D. 2450

32. The sum of the series $\sum_{n=1}^{\infty}\frac{(-1)^n}{(3^n)^2}$ is equal to :
A. $\frac{9}{8}$ B. $\frac{9}{10}$ C. $-\frac{9}{10}$ D. $-\frac{1}{10}$

33. Let p(x): x^2+x is positive.
Which of the following is equivalent to $\neg(\exists x)\,P(x)$?
A. $(\exists x)\,\neg(x^2+x>0)$ C. $(\exists x)\,\neg(x^2+x\leq 0)$
B. $(\forall x)(x^2+x<0)$ D. $(\forall x)(x^2+x\leq 0)$

34. If $(P\wedge Q)\Rightarrow(\neg S\vee R)$ is **FALSE**, which of the following is true?
A. $S\Rightarrow R$ C. $(P\wedge\neg Q)\Leftrightarrow R$
B. $\neg P\vee\neg S$ D. $P\wedge R$

35. $\lim_{n\to\infty}\frac{2n^2-n+1}{6n^2+1}$ is equal to
A. 2 B. $\frac{1}{3}$ C. $\frac{2}{3}$ D) ∞

36. Which of the following is a divergent sequence?
A. $\left\{(\sqrt{2})^{-n}\right\}$ B. $\left\{5^{\frac{1}{n}}\right\}$ C. $\left\{\left(\frac{\pi}{3}\right)^n\right\}$ D. $\left\{1+\frac{(-1)^n}{7^n}\right\}$

37. Which of the following is true about $\left\{2+\frac{(-1)^n}{n+1}\right\}_{n=1}^{\infty}$?
A. The least upper bound of the sequence is $\frac{7}{3}$
B. The sequence is monotonic
C. 1.6 is its lower bound
D. $2-\frac{1}{15}$ is a term of the sequence

38. $\lim_{n\to\infty}\frac{x^3+x-n+2}{x^2-x}$ is equal to
A. -2 B. 1 C. 4 D. Does not exist

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39. What is the left hand limit, $\lim_{x\to 0^-} \frac{|x| - xe^x}{x}$?

A. 0 B. -2 C. -1 D. Does not exist

40. Let $f(x) = \begin{cases} a - \cos x, & x \le 0 \\ 3e^x - a, & x > 0 \end{cases}$

For what value of a is f a continuous function?

A. 1 B. $\frac{3}{2}$ C. 2 D. 3

41. If $f(x) = x^3$, the $\lim_{x\to 0^-} \frac{f(2+h) - f(2)}{h}$ is equal to

A. 4 B. -4 C. 12 D. ∞

42. Which of the following is the equation of the line tangent to the graph of

$f(x) = \begin{cases} x - \pi, & x \le 0 \\ 1 - 2\sin x, & x > 0' \text{ at } (\pi, f(\pi))? \end{cases}$

A. $y = 2x - 2\pi + 1$ C. $y = -2x + 2\pi + 1$

B. $y = 2x - 2\pi - 1$ D.

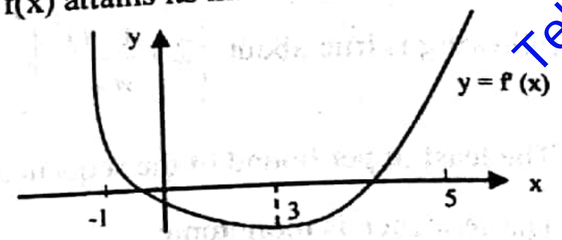
43. Suppose f(x) is differentiable at all $x \in (-\infty, \infty)$ and the graph of $y = f'(x)$ is as shown in the figure below. Which of the following is true?

A. f(x) increases on $[3, \infty)$.

B. f(x) attains its minimum value at $x = 3$.

C. f(x) decrease on $[-1, 5]$.

D. f(x) attains its maximum value at $x = 5$.



44. Let $h(x) = x + 3^x$ and $f(x) = (foh)(x)$ such that $f'(4) = -1$. then, $F'(1)$ is equal to:

A. $1 + 3\ln 3$ B. $-1 - 3\ln 3$ C. $4 + 12\ln 3$ D. $-4\ 12\ln 3$

45. If $f(x) = \ln(2 + \sec x)$, then $f'(x)$ is equal to:

A. $\frac{\tan x}{2 + \sec x}$ B. $\frac{\tan x}{2 + \cos x}$ C. $\frac{\cos x}{1 + 2\cos x}$ D. $\frac{\tan x}{1 + 2\cos x}$

46. Which of the following is the set of all critical numbers of

$f(x) = \begin{cases} x - 1, & x < 1 \\ x^2 - x, & x \ge 1 \end{cases}$

A. $\{1\}$ B. $\left\{\frac{1}{2}\right\}$ C. $\left\{\frac{1}{2}, 1\right\}$ D. $\{\ \}$

47. The rate of certain chemical reaction is given by $r(x) = kx(a - x)$, for $0 \le x \le a$, $a > 0$ and $k > 0$. for what values of x does the rate of reaction increase?

A. $0 \le x \le \frac{a}{2}$ C. $-\infty \le x \le \frac{a}{2}$

B. $\frac{a}{2} \le x \le a$ D. $\frac{a}{2} \le x \le \frac{3a}{4}$

48. Let $f(x) = x^3 - 12x + 1$ be defined on [0,2]. What is the range of f?

A. $(-\infty, \infty)$ C. [-15, 1]

B. $(-\infty, 1]$ D. [-15, 17]

49. Let F(x) and G(x) be anti-derivatives of the function f(x) in the interval [a, b]. Which of the following is necessarily true?

A. F(x) = G(x) for all $x \in [a,b]$

B. F(x) – G(x) = C, where C is a constant, for all $x \in [a,b]$.

C. $\int f(x)dx$ = F(x) + G(x) + C, where C is a constant.

D. $\int_a^b f(x)dx$ = G(b) – F(a)

50. Which of the following is equal to $\int \frac{(e^x + 1)^2}{e^x} dx$?

A. $e^x + e^{-x} + 2x + C$ C. $\frac{1}{3}(e^x + 1)^3 + C$

B. $e^x - e^{-x} + 2x + C$ D. $\frac{1}{3}(1 - e^{-x})^3 + C$

51. If the equation of a line l is given by the dot product: $(x - 1, y + 3).(2, -1) = 0$, which of the following is the equation of the circle with center at (-2, 1) and l is its tangent line?

A. $(x + 2)^2 + (y - 1)^2 = 2\sqrt{5}$

B. $(x + 2)^2 + (y - 1)^2 = 10$

C. $(x + 2)^2 + (y - 1)^2 = 5\sqrt{2}$

D. $(x + 2)^2 + (y - 1)^2 = 20$

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52. Suppose anti-clockwise rotation about the origin through 90^0 is followed by reflection in the x – axis. What is the image of the ellipse given by $\frac{1}{4}(x-1)^2+(y+2)^2=1$ under the two transformations?

A. $4(x-2)^2+(y+1)^2=4$
B. $4(x-2)^2+(y-1)^2=4$
C. $(x-1)^2+4(y+2)^2=4$
D. $(x+1)^2+4(y-2)^2=4$

53. If $\theta = \arccos(½)$, what is the value of $\tan\theta$?

A. $\frac{1}{\sqrt{3}}$ B. $\sqrt{3}$ C. $\frac{\sqrt{3}}{2}$ D. 1

54. If $x = \arcsin\left(\frac{3}{4}\right)$, the value of cos 2x is

A. $\frac{3}{4}$ B. $-\frac{3}{4}$ C. $-\frac{1}{8}$ D. $\frac{1}{8}$

55. Let $f(x) = -3\sin\left(\frac{1}{2}x\right)$. Which of the following is true?

A. The amplitude of the graph of f is – 3
B. The range of f is [-1, 1]
C. f is decreasing on $[-2\pi, 0]$
D. f is increasing on $[\pi, 3\pi]$

56. Which of the following is equal to $\tan\left(x-\frac{\pi}{4}\right)$ if $\cos x \neq 0$?

A. $\frac{\sin x-\cos x}{\sin x+\cos x}$ C. $\frac{\sin x+\cos x}{\sin x-\cos x}$
B. $\frac{\cos x-\sin x}{\cos x+\sin x}$ D. $\frac{\cos x+\sin x}{\cos x-\sin x}$

57. In the interval $[0, 2\pi)$, what is the solution set of the equation $\cos^2 x + \sin x \cos x = 1$?

A. $\left\{0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}\right\}$ C. $\{0, \pi\}$
B. $\left\{0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi\right\}$ D. $\left\{0, \frac{\pi}{4}, \pi\right\}$

58. From the top of a building 50m high, the angle of depression of a parked car is 30^0. How far is the car from the foot of the building measured horizontally?

A. $\frac{50}{\sqrt{3}}m$ B. $55\sqrt{3}$ m C. $50+\sqrt{3}$ m D. $50\sqrt{3}$ m

59. The expression $\frac{\cos A}{1+\sin A}+\frac{\cos A}{1-\sin A}$ is equal to

A. 2 csc A B. 2 sec A C. 2 cos A D. 2 sin A

60. Let ABCD be a parallelogram and $\overline{AC}$ and $\overline{BD}$ be its diagonals. Which of the following is **NOT** true?

A. $\overrightarrow{AB}=\overrightarrow{AC}-\overrightarrow{BC}$ C. $\overrightarrow{AC}=\overrightarrow{BD}-\overrightarrow{2BC}$
B. $\overrightarrow{AB}=\overrightarrow{BC}-\overrightarrow{BD}$ D. $\overrightarrow{CD}=\overrightarrow{BD}-\overrightarrow{AD}$

61. Let A = (0,1), B = (2, 7), and C = (a, b) be points in the coordinate plane. What is (a, b) so that $\overrightarrow{AC}=\overrightarrow{AB}+2\overrightarrow{CA}$?

A. $\left(\frac{1}{3},2\right)$ B. (1, 3) C. (2, 3) D. $\left(\frac{2}{3},3\right)$

62. If $\vec{u}=\overrightarrow{2i}-\vec{j}, \|\bar{v}\|=5$ and $\bar{u}$ and $\bar{v}$ are perpendicular vectors, then $\|\bar{u}+3\bar{v}\|$ is equal to:

A. $\sqrt{55}$ B. $9\sqrt{5}$ C. $\sqrt{230}$ D. 20

63. A translation takes $x^2+(y+1)^2=5$ to $(x-2)^2+y^2=5$. What is the image of (1, 3) under this transformation?

A. (3,4) B. (3, 2) C. (-1, 2) D. (-2,3)

64. Let l be a line given by $x - 2y = 0$. A parametric vector equation of the line that passes through (5, 0) and perpendicular to l is

A. $(x, y) = (5, 0) + t(1,2),\ t \in (-\infty, \infty)$
B. $(x, y) = (5, 0) + t(1, -2), t \geq 0)$
C. $(x, y) = (5 + t, -2t),\ t \in (-\infty, \infty)$
D. $(x, y) = (5 - t, 2t), t \geq 0$

65. Consider a circle that passes through A = (2,3) are centered at C = (1, 1). The equation of the line tangent to the circle at A is

A. $3x + y = 9$ C. $3x - 2y = 0$
B. $X + 2y = 8$ D. $x + y = 5$

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1. Which one of the following is the domain of $g(x) = x - \frac{(x^2+1)}{x^4+x}$

A. $\mathcal{R} \setminus \{0\}$
B. $\mathcal{R} \setminus \{-1\}$
C. $\mathcal{R} \setminus \{0,1\}$
D. $\mathcal{R} \setminus \{-1,0\}$

2. Which of the following is a simplified form of

$\frac{\frac{t-1}{t}}{1+\frac{t}{1-t}} \div \frac{t^2-2t+1}{t}$, $t \neq 1$?

A. t+1 B. t-1 C. 1-t D. –t-1

3. What are the respective values of A, B, and C , so that $\frac{x-1}{x^2+x} = \frac{A}{B} + \frac{BX+C}{x^2+1}$?

A. -1,1,1
B. -1,1,2
C. -1, -2 ,0
D, 3,1,2

4. What is the solution set of $1 - \frac{5}{x^2+4} = \frac{x-1}{x^2-x} - \frac{1}{x}$?

A. {1,-1 } B, {2} C. {-1} D. { 1, 2, -1}

5. Which of the following is a polynomial of degree 3?

A. $x^3 - 3x^{-2} + x$
B. $x^3 - x^2 (1-x) + 1$
C. $x^2 + x^2 (x-1) + \sin(\pi/4)$
D. $x^3 + 3x^2 + \sin(x) + 1$

6. Which of the following is true about $f(X) = x^5(x-1)^4(x^2+3)$

A. 0 is root of $f(x)$ with the lowest multiplicity.
B. The graph of $f(x)$ crosses the x-axis at x=1
C. The graph of $f(x)$ crosses the x-axis only at 0
D. $f(x)$ has three distinct real roots.

7. Let $f(x) = ax^{10} + bx^5 - 1$. If $x + 1$ is a factor of $f(x)$ and when $f(x)$ is divided by $x - 1$ the remainder is 4, then a and b, respectively, equal to .

A. -1, 1 B. 3, 2 C. 1, 5 D, 3,1

8. Which of the following functions touches but never crosses the x-axis?

A. $f(X) = 1 - x^3$
C. $f(x) = (x^2-1)^2$

B. $f(x) = x^4 - 1$
D. $f(x) = x - x^5$

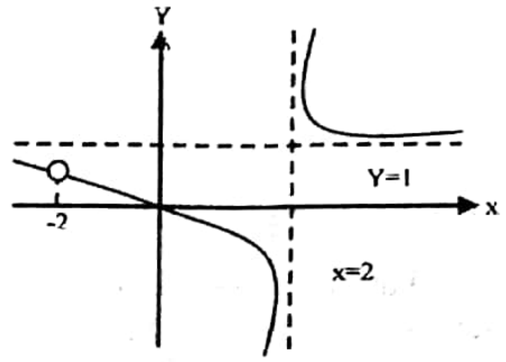
9. What is the solution set of $\log(8x^3) - \log(4x) \geq \log x$

A. $(-\infty, 0) \cup [\frac{1}{2}, \infty)$
B. $(0. \infty)$
C. $(-\infty, 0) \cup (\frac{1}{2}, \infty)$
D. $[\frac{1}{2}, \infty)$

10. Which of the following is NOT true about the graph of $g(x) = \frac{x^2-1}{x^2+1}$?

A. The range of g is $(-\infty, 1)$
B. g is an even function
C. The line $y=1$ is a horizontal asymptote.
D. As $x \to -\infty, g(x) \to 1$

11. Which of the following functions could most likely be drawn as in the figure below?

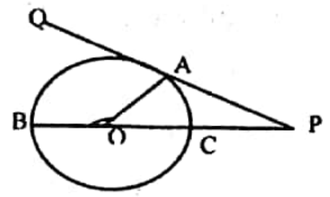


A. $f(x) = \frac{x}{x-2}$
B. $f(X) = \frac{x^2+2x}{x^2-4}$
C. $(fx) = \frac{x-1}{x-2}$
D. $f(x) = \frac{x^3+2x}{x^2-4}$

12. What is the solution set of the equation $(\sqrt{3})^{-x} = \frac{1}{27}(9^x)$

A. $\{\frac{3}{2}\}$ B. $\{\frac{5}{6}\}$ C. $\{\frac{6}{5}\}$ D. $\{\frac{2}{3}\}$

13. In the figure below, 0 is the center of the circle, QP is tangent to the circle at A, the length of BC is $2\sqrt{3}$ cm and the area of the sector AOB is πcm^2 . What is the length of AP?



A. 6cm
B. $\sqrt{3}$
C. $2\sqrt{3}$
D. 3cm

Let (a,b) and (b,a) be point such that $a \neq b$ Which of the following is the equation of the line through (a,b) and perpendicular to the line containing the given points.

A. y= x+ b-a
C. y= -x + b+a

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D. $y = -x - a + b$

B. $y = x - a - b$

14. Let (a. b) and (a, b) be points such that $a \neq b$. Which of the following is the equation of the line through (a, b) and perpendicular to the line containing the given points?

A. $y = x + b - a$ C. $y = -x + b + a$

B. $y = x - a - b$ D. $= -x - a + b$

15. The equation of the circle tangent to the x-axis at (6, 0) whose center is on the line $y = \frac{1}{2}x$ is

A. $(x-6)^2 + y^2 = 9$ C. $(x-6)^2 + y^2 = 36$

B. $(x-6)^2 + (y-3)^2 = 36$ D. $(x-6)^2 = (y-3)^2 = 9$

16. A mirror has an elliptic edge. If the shortest and longest distances between any two points on the edge of the mirror through the geometric center of the mirror are respectively 6 cm and 10 cm, which of the following is as possible equation of the edge of the mirror?

A. $\frac{x^2}{9} + = \frac{(x+1)^2}{5} = 1$

B.. $\frac{x-1)^2}{25} + \frac{y^2}{9} = 1$

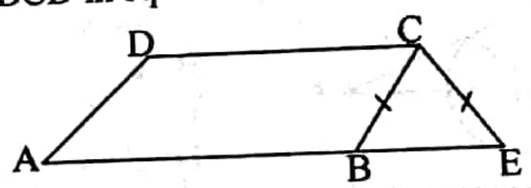
C. $\frac{(x+1)^2}{36} + \frac{x^2}{100} = 1$

D. $\frac{(x+1)^2}{36} + \frac{(y-1)^2}{25} = 1$

17. Suppose that the population P of an organism that increases exponentially is given by $P(t) = 100e^{kt}$, where k is a constant and t is the number of days after the initial count. Beginning from the time of initial count, if the population is 400 after 2 days, what will be the population of the organism after 7 days?

A. 12, 800 B, 1,400 C. 16, 200 D. 6, 400

18. In the following figure ABCD is a parallelogram and $\Delta BCDE$ is an isosceles triangle. If AB= 8units, CE= 6 units and area of ABCD is 40 square units, which of the following is the area of ΔBCD in square units? Assume that A, B and E are on a line



A. $\frac{5}{2}\sqrt{11}$

B. $5\sqrt{11}$

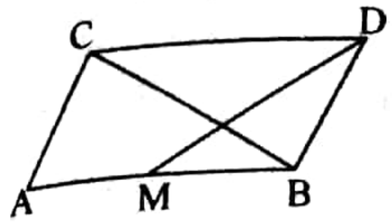
C. 15

D. 27.5

19. If a chord of a circle is at the distance of 4 cm from the center of the circle and the length of the chord is 6cm, how long is the radius of the circle?

A. 10 cm B. 5cm C. 2cm D. 3cm

20. In the figure below, suppose ABCD is a parallelogram and M is the midpoint of AB. Which one of the following is necessarily true ?



A. $AM = \frac{1}{2}$

B. $AM = \frac{1}{2}BC$

C. Area of $\Delta BMD = \frac{1}{2}$ (area of ΔABC)

D. $M(<BMD) = \frac{1}{2}m(<BDC)$

21. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_{n+1} = -na_n$, $a_1 = 1, n = 1,2,3,\ldots$. Which of the following is NOT true about the sequence?

A. a_3 C. $a_n > 0$ if n is odd

B. a_4 D. $a_n < 0$ if n is even

22. Let $S_n = 4n^2$, n =1,2,3,..... be the sum of the first n terms of a sequence $\{a_n\}_{n=1}^{\infty}$. Which of the following is the third term a_3?

A. 8 B. 12 C. 16 D.20

23. Consider the sequence $\{a_n\}_{n \geq 1}$, where the first term is $a_1 = 2$, and $a_n = 3a_{n-1}$ for all $n \geq 2$, Then $\sum_{i=1}^{20}$ a_i is equal to

A. $3^{20} - 1$ C. $1 - \frac{1}{2^{21}}$

B. $1 - \frac{1}{3^{20}}$ D. $3^{-20} - 1$

24. Suppose that every year the amount of atoms of a certain radioactive substance is half of its previous year. If there were 6.4×10^8 atoms of the substance during the initial observation, how many atoms of the substance are present by end of 20 years after the initial observation?

A. 3.2×10^{-20} C. $2\text{-}14 \times 10^9$

B. $2^{-20} \times 10^8$ D. $2^{-14} \times 10^7$

25. A cross-section of a parabolic reflector is shown in the figure below. The opening at the focus, AB, is 12 cm. What is the diameter of the opining, CD, 8cm from the vertex?

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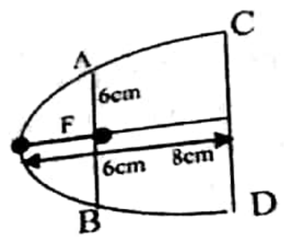
A. $8\sqrt{6}$ cm
B. 12 cm
C. 9cm
D. $6\sqrt{8}$ c



26. The equation $3x^2 - 6x = y^2$ represents
A. an ellipse with center at (1,0)
B. a hyperbola with center at (1,0)
C. an ellipse with center at (-1, 0)
D. a hyperbola with one of its foci at (1, 0)

27. Which one of the following points is outside the sphere $x^2 + y^2 + z^2 - 4z = 0$?
A. (0, -11)
B. (-2,0,2)
C. (1,-1,3)
D. (-1,2,0)

28. Let f(x) $= \frac{\sqrt{x}-2}{\frac{4}{x}+1}$, Then, $\lim_{x\to 4^+} f(x)$ is equal to
A. 0 B. 2 C. 1 D.∞

29. $\lim_{n\to\infty} \frac{6x-12}{x^2-x-2}$ is equal to :
A. 0 B. 2 C. 1 D. ∞

30. Which of the following is equal to $\lim \frac{x \sin 2x}{x-\pi}$?
A. 2π B. π C. 0 D. 1

31. The left hand $\lim_{n\to 0^-} \left(1+\frac{1}{n}\right)^n$ is equal to
A. 0
B. -2
C. -3
D. doesn't exist

32. Which of the following is equal to $\lim_{x\to\infty} \frac{n+\cos 3n}{2-n}$?
A. -1 B. 0 C. 1 D. 3

33. Let f(x) = g($\sqrt{4x+1}$). If g'(3) = $\frac{1}{2}$ then f '(2) is equal to
A. $\frac{2}{3}$ B. $\frac{1}{3}$ C. $\frac{1}{2}\sqrt{5}$ D. $\frac{1}{2}\sqrt{13}$

34. Which of the following is the second derivative of f(x) = ln(x^2+1) ?
A. $1 - \frac{1-2x^2}{x^2+1}$ B. $\frac{-2(x^2-1)}{(x^2+1)^2}$ C. $\frac{-2(x^2-1)}{(x^2+1)^2}$ D. $\frac{2-4x^2}{(x^2+1)}$

35. Let f(x) = x{x}, for all x ∈(-∞, ∞). Which of the following is true?
A. Is not differentiable at x= 0
B. f '(x)= 2/x/at each x ∈(-∞, ∞)
C. f is differentiable only on (0, ∞)
D. f '(x) =2x at each x ∈(-∞, ∞).

36. Let f(x) = $x^2 - 6x^2 + 1$ on [-2,3] Which of the following is a local maximum value of f?
A. f(0) B. f($\sqrt{3}$ C. f(3) D. f(-$\sqrt{3}$)

37. Let f(x)= $\begin{cases} 3a - x, if\ x < 1 \\ b + 1,\ if\ x = 1 \\ ax,\ \ if\ x > 1 \end{cases}$ What is the value of b if f is continuous at x= 1 ?
A. $-\frac{1}{2}$ B. $\frac{1}{2}$ C. -1 D. 3

38. On which of the following intervals is f continuous, where $f(x) = \begin{cases} \frac{x-x^2}{x-1}, & x < 0 \\ 3 - x, & x \geq 0 \end{cases}$?
A. (-∞, 0] B. [-2,1] C. [-1,2] D. [0, ∞)

39. If f(x) = $x^5 + 2x$, then lim $\frac{f(1+h)-f(1)}{h}$ is equal to
A. 2 B. 7 C. 5 D. ∞

40. Let $f(x) = e^{3x-1}$ and $g(x)$ = tan x, If $F(x) = f(x) + f(g(x))$, which of the following is equal to F '(0)?
A. $3e^{-1}$ B. $4xe^{-1}$ C. $\frac{1}{e}(1+\frac{1}{e})$

41. A farmer wants to make three identical rectangular enclosures (same width and same length) along a straight river, as in the diagram shown below. If he has a fencing material of 1,200 meters and the sides along the river need no fence, what should be the dimensions (length/and width w) of each enclosure if the total area of the enclosures is to be maximized?

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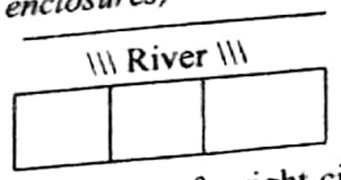
(Take the sides parallel to the river as the length of the enclosures)



A. l =100m, w=225m
B. l = 300m, w= 75m
C. l= 200m, w= 200m
D. l= 200m, w=150m

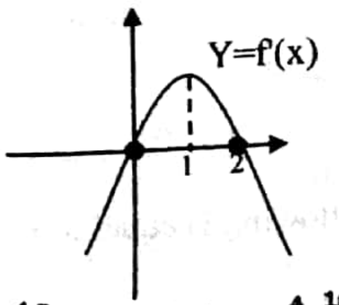
42. The radius of a right circular cylinder of height 10 cm increases at a rate of 3 cm/min. What is the rate of change of the volume of the cylinder when the radius 8 cm?

A. 80 πcm^3 min
B. 320 πcm^3/min
C. 480 πcm^3/min
D. 640 πcm^3/min

43. Which of the following is an antiderivative of $f(x) = \frac{3}{x\sqrt{x}}$?

A. $-\frac{3}{\sqrt{x}} + c$
B. $-6\sqrt{x} + c$
C. $-\frac{6}{\sqrt{x}} + 1$
D. $-\frac{3}{2\sqrt{x}} + 3$

44. Suppose f(x) is differentiable at all $x \in (-\infty, \infty)$ and the graph of y= f'(x) is as shown in the figure below, which of the following is true?



A. f(x) is increasing on ($-\infty$, 1)
B. f(x) attains its maximum
C. f(x) attains its maximum value at x =1
D. f(x) is decreasing on (0, 2)

45. Let $f(x) = x^4 - \frac{16}{3}x^3 + 6x^2$ for all $x \in (-\infty, \infty)$. What is the range of f?

A. $(-\infty, \infty)$
B. [0, ∞)
C. [-9, ∞)
D. $(\frac{5}{3}, -9]$

46. Which of the following is equal to $\int_0^1 e^{-x}\frac{dx}{(1+e^x)}$?

A. $\ell n\,(e-1)$
B. $\ell n\,(1+e) - \ell n 2$
C. $\ell n(2+e)$
D. $\ell n 2$

47. $\int_1^2 x(\frac{1}{x^2} + 2\ \ell n\, x)\, dx$ is equal to

A. $2\ell n\, 2 - 1$
B. $5\ell n\, 2 + \frac{5}{2}$
C. $2\ell n\, 2 - \frac{5}{2}$
D. $5\ell n\, 2 - \frac{3}{2}$

48. The area of the region bounded between the graph of $y = x^2$ and $y = \sqrt{x}$ is

A. 3 sq. unit
B. $\frac{1}{3}$ sq. unit
C. $\frac{2}{3}$ sq. unit
D. 1 sq. unit

49. Which of the following is equal to $\int \frac{x^2-1}{x^3+x}\, dx$?

A. $\ell n|x^2 + 1| + \ell n|x| + c$
B. $\frac{1}{2}\ell n(x^2 + 1) - \ell n|x| + c$
C. $\ell n(x^2 + 1) - \ell n\,|x| + c$
D. $2\ell n(x^2 + 1) - \ell n|x| + c$

50. $\int (2 + \sin^2 x)\cot x\, dx$ is equal to

A. $2\csc x - 2\sec^2 x + c$
B. $2\ln|\sin x| + \frac{1}{2}\sin^2 x + c$
C. $2\ln|\sin x| - \cos^2 x + c$
D. $2\csc 2x + \sin x + c$

51. A 1 m high pole is fixed vertically on the ground at a point 50 $\sqrt{3}$ m far from the foot of a building. If the angle of elevation from the top of the pole to the of the buildings is 30^0, What is the height of the building?

A. 50m B. 51m C. 150m D. 151m

52. If $\vec{A}$ = (3,-3) and $\vec{B}$= (1, -3), What is the unit vector in the direction of $\vec{C} = 3\vec{A} - \vec{B}$?

A. $(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}})$ B. $(\frac{1}{\sqrt{10}}, \frac{-3}{\sqrt{10}})$ C. $(\frac{4}{5}, \frac{-3}{5})$ D. $(\frac{-6}{10}, \frac{8}{10})$

53. If $\vec{u}$ is vector in the direction as vector $\vec{v}$ = (-3, 4) having length 33 times the length of $\vec{v}$, what is the length of the vector $\vec{u} - \frac{3}{2}\vec{v} - 2\vec{j}$, where $\vec{j}$ is te unit vector in the direction of the positive y-axis ?

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A. $\frac{1}{2}\sqrt{118}$ B. $\frac{1}{2}\sqrt{145}$ C. $\frac{21}{2}$ D. $\frac{17}{2}$

54. If θ is the angle between the vectors $\vec{A} = 2\vec{i} - \vec{j}$ and $\vec{B} = \vec{i} + \vec{j}$, then $\cos\theta$ is equal to

A. $\frac{1}{\sqrt{10}}$ B. $\frac{1}{2}$ C. $\frac{\sqrt{3}}{2}$ D. $\frac{2}{\sqrt{5}}$

55. If θ = arc (sin(1/2), then tan(2θ) is equal to

A. $\frac{1}{\sqrt{3}}$ B. 1 C. $\sqrt{3}$ D. $\frac{\sqrt{3}}{2}$

56. What is the amplitude and period, respectively, of the graph of $f(x) = 2 - 3\sin\left(\frac{x+\pi}{4}\right)$?

A. 2, $\frac{\pi}{4}$ B. -3, 8π C. 3, $\frac{\pi}{2}$ D. 3,8π

57. Which one of the following is a simplified form of $(1+\tan^2 x)\sin(x-\frac{3}{2}\pi)$?

A. sec x B. cos x C. –sec x D. –cosx

58. Which of the following is the solution set of $\sqrt{3}\sin 2x = \cos 2x$ on $[0, \pi]$?

A. $\left\{\frac{\pi}{6}, \frac{\pi}{12}\right\}$ B. $\left\{\frac{\pi}{6}, \frac{\pi}{3}\right\}$ C. $\left\{\frac{\pi}{12}, \frac{7}{12}\pi\right\}$ D. $\left\{\frac{\pi}{6}, \frac{2}{3}\pi\right\}$

59. If A(-1,1,0) and B(1,3,2) are endpoints of a diameter of a sphere, then what is the radius of the sphere?

A. 3 B. 6 C. $\sqrt{3}$ D. $\sqrt{12}$

60. Which of the following is true about the point of intersection of the sphere of radius 3 units with center at (-1,3,2) & the z-axis?

A. (0, 0,3) is a point of intersection
B. There are two points of intersection
C. There is no point of intersection
D. There are many points of intersection

61. If $(\rightarrow P \vee Q) \Rightarrow (R \wedge \rightarrow R)$ is true, then which of the following is necessarily true?

A. $R \Rightarrow Q$ C. $P \wedge \rightarrow R$
B. $(Q \Rightarrow P) \wedge R$ D. $(Q \wedge R) \Rightarrow P$

62. Consider the following statement. If x is rational and y is irrational, then x-y is irrational.To show this, start from suppose x-y is rational. Then, x-y = r ∈ Q. Hence x = y + r, and so y = x - r.Since Q is closed under subtraction, this gives y is rational, which is not true.Which of the following is better used in the above procedure?

A. Method of exhaustion C. Indirect proof
B. Proof by contradiction D. Direct proof

63. Which of the following is the reflection of the parabola $y = x^2 + 1$ in the line y=-x?

A. $y = -x^2 - 1$ B. $y = x^2 - 1$ C. $x = y^2 + 2$ D. $x = -y^2 - 1$

64. Consider the circle given by $x^2 + y^2 = 2$ and the line l given by parametric vector equation(x,y)= (2,0)+ t(-1,1). Which of the following is true?

A. l is tangent to the circle at (1/2, $\sqrt{7}$ /2).
B. l intersects with the circle at two distinct points
C. l and the circle have no common point.
D. The distance from the center of the circle to l is $\sqrt{2}$

65. If the midpoint of a line segment is M(-3,3,4) and one of the end points of the segment is at A(0, -1, 3), What is the coordinates of its other endpoints?

A. (-6,7,5) B. (-3,2,7) C. (0,1,3) D. (6,-7,

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1. If $x < 0$, then the simplest form of $f(x) = \frac{x-|x|}{x}$ is equal to:
   A. 2x B. 2 C. -2 D. 0

2. If $f(x) = \frac{\sqrt{x+2}}{x+2}$ and $g(x) = \frac{1}{2} - 2$, then f(g(x)) is equal to :
   A. $\sqrt{x-2}$ B. $\sqrt{x}+2$ C. $\sqrt{x}$ D. $\frac{\sqrt{x}}{x}$

3. If $f(x) = \ell n\left(\frac{x}{x-1} + 2\right)$, for $x > 1$, then which one of the following is the inverse of f ?
   A. $G(x) = \frac{e^x-2}{e^x-3}$ C. $g(x) = \frac{e^x}{e^x+1} - 2$
   B. $G(x) = \frac{e^x-2}{e^x+3}$ D. $g(x) = \frac{x}{e^{x-1}-2}$

4. Which of the following is a simplified form of $\frac{4-(a-2)^2}{a^4+2a^2+1} \div \frac{1-\frac{1}{a^2+1}}{a}$?
   A. $\frac{4-a}{a}$ B.- $\frac{1}{a^2(a^2+1)}$ C. $\frac{a}{a^2+1}$ D. $\frac{4-a}{a^2+1}$

5. If $f(x) = \frac{x-1}{(x-2)^2(x+1}$, which of the following is true about f?
   A. Its graph has an oblique asymptote.
   B. The graph of f does of meet its asymptote.
   C. AS $x \to -1^+$, $f(x) \to -\infty$
   D. As $x \to 2^-$, $f(x) \to -\infty$

6. Line l passes through (0, 5) and (-5, 0). What is the angle between the y – axis and l in radian measure?
   A. $\frac{\pi}{4}$ B. $\frac{\pi}{3}$ C. $\frac{\pi}{2}$ D. $\frac{3}{4}\pi$

7. Consider a circle whose center is on the x – axis. If a line given by y = is tangent to the circle at point (2, 2), what is the equation of the circle?
   A. $x^2 + y^2 = 8$ C. $(x-4)^2 + y^2 = 8$
   B. $(x-2)^2 + y^2 = 4$ D. $(x-1)^2 + y^2 = 5$

8. What is the vertex and the equation of the directix, respectively, of the parabola $x + y^2 + 2y + 1 = 0$?
   A. (0, -1), $x = \frac{-1}{4}$ B. (0, -1), $x = \frac{1}{4}$
   B. (-1, 0), $y = \frac{-1}{4}$ C. (-1, 0) , $y = \frac{1}{4}$

9. The orbit of Mercury around the sum forms an ellipse with eccentricity 0.206, length of the major axis 1.16×10^8 km and the sun at one focus. Which of the following is the best approximation of the maximum distance from Mercury to the sun?
   A. 7.596×10^7 km C. 8.695×10^7 km
   B. 5.695×10^7 km D. 6.995×10^7 km

10. Suppose proposition $p \Rightarrow \neg q$ is false (F), which of the following is true?
   A. $\neg q \wedge (p \Rightarrow q)$ C. $(\neg q \vee p) \Leftrightarrow q$
   B. $p \vee (q \Rightarrow \neg p)$ D. $(p \wedge q) \Leftrightarrow \neg p$

11. Suppose $e(x) \equiv x$ is even ; $p(x) \equiv x$ is prime ; $d(x) \equiv x$ is divisible by 2. Which one of the following has the truth value F on the set of natural numbers?
   A. $(\exists x)[e(x) \wedge p(x)]$ C. $(\exists x)[e(x) \wedge \neg d(x)]$
   B. $(\forall x)[e(x) \Rightarrow d(x)]$ D. $(\forall x)[e(x) \vee \neg d(x)]$

12. Consider the following argument from. Production is high if rain continues. Rain does not continue. Therefore, either production is low or rain continues.
   Let p: production is low
   q: rain continues
   The following table is also given about p and q.

Rows	p	q	¬ p	¬q	¬ p ⇒ q	q ⇒ ¬ p	p V q
1	T	T	F	F	T	F	T
2	T	F	F	T	T	T	T
3	F	T	T	F	T	T	T
4	F	F	T	T	F	T	F

   Which of the following is necessarily true?
   A. The argument form is valid due to row 2.
   B. The argument form is valid due to ropws 2 and 3.
   C. The argument form is invalid due to row 4.
   D. The argument form is invalid due to rows 1 and 3.

13. A school has three classrooms for grade 11, namely 11A, 11B and 11C. the number of students in these classrooms is 28, 20 and 22, recpectively. All the students took an examination and the average score of the students of 11A, 11B and 11C is 60, 70 and 70, respectively. What is the average score in this examination for all grade 11 students?
   A. 66 B. 66.67 C. 65 D. 65.67

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14. You are given a data on the age of students, in a primary school.

Age	8	10	11	12	13
Number of students	5	15	8	10	2

Which of the following is NOT true about the data?
A. The median is 10.5 C. The mean is 10.5
B. The mode is 10 D. The range is 5

15. The following is a table of simple frequency distribution of a data with variable x.

x	1	3	4	5	7
Frequency	2	5	6	5	2

The standard deviation of the data is equal to:
A. $\sqrt{3}$ B. 3 C. $\sqrt{23}$ D. $\sqrt{3.6}$

16. A three – digited library identification card is to be printed from the numbers 0, 1, 2, 3, 4, 5 in such a way that the first is non – zero and no number is to be repeated. How many such cards can be printed?
A. 100 B. 120 C. 150 D. 180

17. A student needs to select 3 books from 3 different mathematics, different physics and 1 history book. What is the probability that one of them is mathematics and the other two are either physics or history books?
A. $\frac{3}{35}$ B. $\frac{9}{35}$ C. $\frac{15}{35}$ D. $\frac{18}{35}$

18. Items produced by certain company are subjected to two kinds of defects D_1 and D_2. Out of the total production, if 5% have defect D_1, 10% have defect D_2 and 2% have both defects, then what is the probability for an item to have defect D_2, given that it has defect D_1.
A. 0.2 B. 0.05 C. 0.1 D. 0.4

19. Let $A = \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -y \\ 0 & 3 \\ 1-x & 2 \end{pmatrix}$ such that $A + 2B^t = 0$.

Then which of the following is the value of y?
A. 0 B. $\frac{13}{2}$ C. -8 D. any real number

20. Let A and B be 3x3 matrices such that

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & -1 & \frac{1}{2} \end{pmatrix} \text{ and } |B| = \frac{1}{10}.$$

Which one of the following is equal to $|AB^T|$?
A. 1 B. 4 C. 100 D. 400

21. What is the solution set of the following system of equations

$$\begin{cases} x + y + 2z = 1 \\ x + 2y + z = 2 \\ -2x - 2y - 4z = -2 \end{cases} ?$$

A. $\{(0, 1, 0)\}$ C. $\{(-3k, k + 1, k) \mid k \in (-\infty, \infty)\}$
B. $(-\infty, \infty)$ D. $\{(3k, k - 1, k) \mid k \in (-\infty, \infty)\}$

22. If $\begin{pmatrix} a & 2 & \beta \\ 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$ and the determinant of the coefficient Matrix is -5, then the value of x is equal to
A. 3 B. $\alpha + \beta$ C. -5α D. 5

23. In the set of complex numbers, the solution set of $x^2 - 2x + 5 = 0$ is
A. $\emptyset$ C. $\{1 + 2i, 1 - 2i\}$
B. $\{2 + I, 2 - i\}$ D. $\{2 + 4i, 2 - 4i\}$

24. Which one of the following is the simplest form of $\frac{4-3i}{3+4i} + \overline{1 - 2i}$?
A. 1 + i C. 2 - i
B. 1 + 3i D. 1 – 3i

25. Let z be a complex number. Which of the following is the solution set of $z^3 - iz = 0$?
A. $\left\{\pm\frac{1}{\sqrt{2}}(1 + i)\right\}$ C. $\{\pm\sqrt{2}(1 + i)\}$
B. $\left\{0, \pm\frac{1}{\sqrt{2}}(1 - i)\right\}$ D. $\left\{0, \pm\frac{1}{\sqrt{2}}(1 + i)\right\}$

26. Let $a_n = n + \cos n\pi, n = 1,2,3,\ldots$ be the n^{th} term of a sequence. Then which of the following is true?
A. The fifth term is 6 C. $a_7 + a_8 = 15$
B. The 10^{th} term is 9 D. $a_7 + a_6 = 1$

27. If $\{A_n\}$ is an arithmetic sequence with first term $A_1 = 5$ and fifth term $A_5 = 21$, then the partial. Sum $\sum_{n=1}^{\infty} A_n$ is equal to
A. 760 B. 780 C. 860 D. 870

28. What is the sum of the series $\sum_{n=0}^{\infty} 2^n 3^{2-n}$?

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A. 6 B. 18 C. 27 D. ∞

29. A business association gets a net profit of Birr 3,000 at the end of each month. Just after the fifth year, its amount was Birr 230,000. What was the starting amount of the business?
A. Birr 47, 000 C. Birr 53, 000
B. Birr 50,000 D. Birr 56, 000

30. To which of the following is $\lim_{n\to\infty} \frac{\sqrt[3]{n}-6n+5}{4n+1}$?
A. $\frac{-3}{2}$ B. 0 C $\frac{3}{4}$ D. $\frac{5}{4}$

31. To which of the following is $\lim_{x\to 1^-} \frac{x^2-1}{|x-1|}$? equals to ?
A 4 B. 0 C. - D. -4

32. Let $f(x) = \begin{cases} \frac{a}{x^2+3}, if\ x \le 1 \\ \frac{x-1}{\sqrt{x}-1}, if\ x > 1 \end{cases}$
What is the value of a if f is continuous at x =1?
A. 0 B. 2 C. 4 D. 8

33. $\lim_{x\to 0} \frac{x\csc 3x}{x+1}$ is equal to:
A. 2π B. π^2 C. 1 D. 0

34. $\lim_{x\to\infty} \left(1+\frac{1}{x}\right)^{2x+4}$ is equal
A. e B. e^2 C. e^4 D. 6

35. If $f(x) = \pi^2$, then f '(1) is equal to
A. 2π B. π^2 C. 1 D. 0

36. Let $f(x) = \frac{6x}{x+a}$. For what value of a is f ' (a) = 1?
A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $\frac{3}{2}$ D. 3

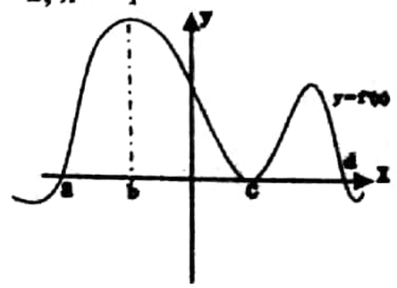
37. Which one of the following is necessarily true about a function f(x)?
A. If f is continuous at x+a, then f is differentiable at x=a.
B. If f is not differentiable at x=a, then $\lim_{x\to a^-} f(x) \neq \lim_{x\to a^+} f(x)$.
C. If f is differentiable at x+a, then $\lim_{x\to a^-} f(x) = f(z) = \lim_{x\to a^+} f(x)$.
D. If the derivative f '(a) = 0, then f attains its maximum value at x=a

38. If $g(x) = \frac{f(x)}{x+1} + (f(x))^2, f(1) = 8\ and\ f'(1) = 2, then$ f ' (1) is equal to
A. 36 B. 31 C. 25 D. 16

39. If $f(x) = xe^{3x} - \cos(2x), then\ f''(0)$ is equal to
A. 0 B. 2 C. 6 D. 10

40. At what value (s) of x does $f(x) = \frac{1}{3}x^3 - \frac{1}{2}x^2 - 2x + 1$ have a local maximum?
A. x= 2 C. x=5
B. x = -1 D. x = 2, x = -1

41. Look at the following graph of f '(x).
Which of the following is true about f?
A. f is increasing on [c,d]
B. f if decreasing of [b,c]
C. f has a local minimum at d.
D. f has a local extreme value at c.



42. A rectangular field of length l and width w meters for w < l has perimeter 400 meters. If a circular region of area w^2 is to be reserved for office purpose, what should be the length of the field so that the area of the remaining region is maximum?
A. 50 meters C. 120 meters
B. 100 meters D. 150 meters

43. On which of the following intervals is the graph of $f(x) = \ell n(x^2 + 1)$ concave upward?
A. $(-\infty, -1] \cup [1, \infty)$ C. $(-\infty, 0]$
B. $[0, \infty)$ D. [-1, 1]

44. Water is poured into a cylindrical tanker of radius 5 meters at a rate of 10 meter cube/min. what is the rate of change of the height of the level of water when it rises to 3 meters?
A. $\frac{3}{25\pi}$ m/min C. $\frac{1}{2\pi}$ m/min
B. $\frac{2}{5\pi}$ m/min D. $\frac{3}{5\pi}$ m/min

45. Which of the following is an antiderivative of f(x) = tan x?
A. $\frac{1}{2}\tan^2 x$ C. ℓn (cos x)
B. $\sec^2 x$ D. ℓn(sec x)

46. Which of the following is equal to $\int \frac{x}{x^2+2x+1} dx$?

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A. $\ell n\ (x^2+2x+1)+c$

B. $\ell n\ |x+1| - \frac{1}{x+1} + c$

C. $\ell n\ (x^2+2x+1-\frac{1}{x+1} - c$

D. $\ell n\ (x^3+2x+1)+\frac{1}{x+1}+c$

47. $\int \frac{x^{2x}-4x}{xe^{2x}}$ dx is equal to

A. $\ell n\ |x| + 2e^{-2x} + c$

B. $\ell n\ |x| - 4e^{2x} + c$

C. $\ell n\ |x| + 4e^{2x} + c$

D. $-\frac{1}{x^2}2e^{-2x} + c$

48. Which of the following is equal to $\int_0^1 \frac{(x-1)^2}{x+1}$ dx ?

A. $-\frac{3}{2} + \ell n16$

B. $-\frac{5}{2} + \ell n15$

C. $-\frac{5}{2} + 4\ell n2$

D. $\frac{5}{2} + \ell n16$

49. If $f(x) = \begin{cases} \sin x, & x \leq 0 \\ 3x\sqrt{x^2+1}\ x & > 0 \end{cases}$, which of the following is equal to

$\int_{\frac{-\pi}{2}}^{\sqrt{8}} f(x)dx$?

A. 17 B. 25 C. 27 D. $3\sqrt{8}$

50. The volume of the solid which is generated when the region bounded by $y=\sqrt{x+1}$ and the x-axis from x = 0 to x = 2 is rotated about the x-axis is equal to

A. 4π B. 3π C. $\frac{4}{3}\pi$ D. $\frac{3}{4}\pi$

51. Which of the following is a vector that lies on the line through (0,0) and (2,4)?

A. $\vec{u} = (2,1)$

B. $\vec{u} = (-1,-2)$

C. $\vec{u} = (\frac{1}{2},-2)$

D. $\vec{u} = (-2,-6)$

52. Let i and j be the standard unit vectors in the directions of positive x-axis and positive y-axis, respectively, and $\overline{AB}$ be a vector from the point A(-1,0) to the point B(2,2). If $\overline{V} = 3\ \overline{AB}\ -3\bar{i} + 2\bar{j}$, then the unit vector in the direction of $\overline{V}$ is equal to

A. $\left(\frac{3}{5},\frac{4}{5}\right)$

B. $\left(\frac{-3}{5},\frac{-4}{5}\right)$

C. $\left(\frac{-3}{5},\frac{4}{5}\right)$

D. $\left(\frac{3}{5},\frac{-4}{5}\right)$

53. Which of the following is a vector equation of the line tangent ot the circle $x^2 + y^2 + 2x - 7 = 0$ at (1,2) ?

A. $(x,y) = (0,3) + \lambda(-1,2)$

B. $(x,y) = (1,2) + \lambda(2,-1)$

C. $(x,y) = (0,3) + \lambda(1,-1)$

D. $(x,y) = (1,2) + \lambda(-1,2)$

54. Let ℓ be the line whose equation is $2x - y = 10$. Which one of the following is the equation of the following is the equation of the image of after a reflection in the line y=2x-5 followed by a rotation through the angle of 90° about the origin?

A. x + 2y=0

B. 2x+y=0

C. x +2y = 5

D. x-2y=5

55. Which one of the following is a simplified form of $\csc(\frac{\pi}{2} - x)\cos x - \sin^2 x$?

A. 2cos x

B. $\cos^2 x$

C. 2 sin x

D. sin 2x

56. Cos (tan (0,5)) is equal to:

A. $\frac{1}{\sqrt{5}}$ B. $\frac{1}{\sqrt{3}}$ C. $\frac{2}{\sqrt{5}}$ D. $\frac{2}{\sqrt{3}}$

57. A patrol boat on a sea sailed from its station 10 km to the south and then changed its course and sailed $5+\sqrt{6}$ km in the direction N 60° E. then, what is the distance the boat should travel in order to return to its station by the shortest route?

A. $4\sqrt{6}$ km

B. 9 km

C. $3\sqrt{6}$ km

D. 7 km

58. If a sphere with center C(0, 1, 1) interests the z-axis at P(0, 0,3), then the radius of the sphere is equal to

A. 5 B. 3 C. $\sqrt{3}$ D. $\sqrt{5}$

59. Suppose A and B are the end points of a diameter of the sphere whose equation is $x^2 + y^2 + (z+2)^2 = 1$. If A = (1,0,-2), then B is equal to

A. (0, 1, -2)

B. (0, 0, -1)

C. (-1, 0, -2)

D. (0, -1, -2)

60. Let ℓ_1 and ℓ_2 be two lines in space intersecting at the origin, (0,0,0). If ℓ_1 and ℓ_2 pass thorugh point A(1, 1, 0) and V(0, 1, 1) respectively, then the angle between ℓ_1 and ℓ_2 is equal to

A. 30° B. 45° C. 60° D. 90°

61. Let $\bar{a} = \overline{-i} + 3\bar{k}$ and $\bar{b} = \overline{-i} + \bar{j}$ be vectors in the space. Which one of the following is the cosine of the angle between $\bar{a}$ and $\bar{a} - \bar{b}$?

A. $\frac{9}{10}$ B. $\frac{3}{3}$ C. $\frac{3}{\sqrt{10}}$ D. $-\frac{9}{10}$

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62. Let $\bar{V} = 3\,\bar{i} + 4\bar{k}$, when $\bar{i}$ and $\bar{k}$ are the standard unit vectors in the directions of positive x-axis and positive z-axis, respectively, and $\overrightarrow{AB}$ be a vector from the point A(0,1,2) to a pint B in space iv $\overrightarrow{AB}$ is parallel to $\bar{V}$ and $\|\overrightarrow{AB}\| = 10$, then point B is at

A. (-6, -1, 10)
B. (6, -1, -10)
C. (6, 1, -6)
D. (-6, -1, 6)

63. If x and y are non-negative integers, which of the following is NOT true?

C. $(\forall x)(\exists y)(y > x^2 - 1)$
D. $(\exists x)(\forall y)(y > x^2 - 1)$
C. $(\exists y)(\forall x)(y \leq x^2 - 1)$
D. $(\exists y)(\exists x)(y \leq x^2 - 1)$

64. Consider the following statement: $\frac{x^2+2}{x} \neq 0$ for every real number x

To show this, a person constructed the following proof.

"Proof: Take x = 1 Then, $\frac{x^2+2}{x} = \frac{1^2+2}{1} = 3 \neq 0$

In the same way, if we take x= n for any real number n we get

$\frac{x^2+2}{x} = \frac{n^2+2}{n} \neq 0$

It follows that $\frac{x^2+2}{x} \neq 0$, for every number x,"

Which one of the following is true about this proof?

A. The proof is correct by the principle of induction.
B. The proof is correct by the method of exhaustion.
C. The proof is correct and it uses the method of direct proof.
D. It is not a valid proof since its argument cannot lead to the conclusion.

65. Let $a_n = n_2 - n$, when nЄN (set of natural numbers) Which one of the following is true, when k is an arbitrarily chosen natural number and a_k is integer?

A. a_n is not a multiple of 2 for some nЄN because $a_1 = 0$.
B. a_n is a multiple of 2 for all nЄN because $a_1 = 0$ and if $a_k = 2m$, then $a_{k+1} = 2(m+1)$
C. a_n is a multiple of 2 for all nЄN because $a_1 = 0$ and if $a_k = 2m$, then $a_{k+1} = 2(m+k)$.
D. a_n is a multiple of 3 for all nЄN because $a_1 = 0$ and if $a_k = 3m$, then $a_{k+1} = 3(m+k)$

ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (UEE)
MATHEMATICS FOR NATURAL SCIENCE STREAM
GINBOT 2005/ JUNE 2013

BOOLET CODE:10

1. If $\{a_n\}$ is a sequence such that $a_1 = 2, and\ a_{n+1} = a_n + 4$ for all $n \geq 1$, then $\sum_{n=1}^{35} a_n$ is equal to:
   A) 2460 B) 2458 C) 2450 D) 2442
2. If $f(x) = x^2 + 2\ell nx$, then what is $\lim_{h\to 0} \frac{f(2+h)-f(2)}{h}$?
   A) 5 B) 4 C) 2 D) 0
3. If $f(x) = e^{2x} + x - 3\cos(x)$, then what is $f''(x)$?
   A) $e^{2x} + 1 - 3\sin(x)$ B) $e^{2x} + 1 + 3\sin(x)$ C) $4e^{2x} - 3\cos(x)$ D) $4e^{2x} + 3\cos(x)$
4. which one of the following intervals does $f(x) = x^4 + 4x$ increase?
   A) $(-\infty, -1]$ B) $(-\infty, 0]$ C) $[-1, \infty]$ D) $(-\infty, \infty]$
5. Which one of the following is the simplest form of $|3 + 4i| - \frac{25i}{3+4i}$
   A) $5 - 5i$ B) $5 + 5i$ C) $1 + 3i$ D) $1 - 3i$
6. If $Z = \cos\left(\frac{\pi}{10}\right) + \sin\left(\frac{\pi}{10}\right)$, then what is the value of Z^5?
   A) $\frac{\pi}{2} + \frac{\pi}{2}i$ B) $\frac{1}{2} + \frac{1}{2}i$ C) i D) $1 + i$
7. What is the value of $|x| + 2x$ if $x < 0$?
   A) $-3x$ B) $3x$ C) $-x$ D) x
8. Suppose $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ If X is a 2×2 matrix such that $AX\text{-}A^T = 2A$, then what is the value of x?
   A) $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ B) $\begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$ C) $\begin{pmatrix} 3 & 6 \\ 6 & 9 \end{pmatrix}$ D) $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$
9. Suppose that A and B are 3×3 matrices, I is the identity matrix of order 3 such that AB=2I. If $\det B = |B| = 6$. What is $\det(A^T)$?
   A) $\frac{1}{3}$ B) $\frac{4}{3}$ C) 12 D) 48
10. What is the 50^{th} term of the sequence 3,10,17,24,31…?
    A) 310 B) 346 C) 510 D) 531
11. For real numbers x and y, which one of the following statements is true?
    A) $(\forall x)(\exists y)(x^2 + y + 1 = 0)$ B) $(\exists x)(\forall y)(x^2 + y + 1 = 0)$ C) $(\exists y)(\forall x)(x^2 + y + 1 = 0)$ D) $(\forall y)(\exists x)(x^2 + y + 1 = 0)$
12. Which one of the following functions has **NO** vertical asymptote?

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A) $f(x) = \ell n(x+1)$ C) $f(x) = \frac{x^2-9}{x-3}$

B) $f(x) = \frac{x^2+1}{x^3+8}$ D) $f(x) = \frac{x-1}{x^2-x}$

13. Let p, q and r be propositions such that $p \Rightarrow (r \vee \neg q)$ is false. Then, which one of the following propositions is true?
A) $p \Rightarrow r$ B) $\neg r \Rightarrow q$ C) $\neg p \Leftrightarrow q$ D) $q \Leftrightarrow r$

14. If $f(x) = \frac{1}{e^x+1}$, then which of the following is equal to $f^{-1}(x)$?
A) $\ell n(1-x) - \ell n(x)$ C) $\ell n\left(\frac{1}{x+1}\right)$
B) $e^{-x} + 1$ D) $\frac{1}{e^{-x}+1}$

15. If $x^2 - 6x + y^2 + k = 0$ is equation of a circle with radius 2, then what is the value of k?
A) 13 B) 5 C) 4 D) -4

16. If a line with angle of inclination of $\frac{3}{4}\pi$ passes through $(0,1)$, which one of the following is the equation of the line?
A) $y = -x + 1$ C) $y = -x - 1$
B) $y = x + 1$ D) $y = x - 1$

17. If $g(x) = xf(x) - \sqrt{f(x)}$ and $f(2) = f'(2) = 4$, then which of the following is equal to $g'(2)$?
A) 11 B) 8 C) 2 D) 0

18. What is the sum of the series $\sum_{n=1}^{\infty} \frac{2^{2n+1}}{5^{n-1}}$?
A) 40 B) 20 C) 10 D) 8

19. What is the value of $\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^{-\frac{x}{2}}$?
A) $\frac{1}{\sqrt{e}}$ B) $\sqrt{e}$ C) e^{-2} D) ∞

20. Among students who took a quiz, 15 students scored 6,20 students scored 7,10 students scored 8 and 5 students scored 10. What is the average score of the students?
A) 7.8 B) 7.5 C) 7.2 D) 7.0

21. A parabola with focus (3,-1) has directrix y=3. Which one of the following is the equation of the parabola?
A) $(x-3)^2 = -4(y+1)$ C) $(x-3)^2 = 4(y+1)$
B) $(x-3)^2 = -8(y-1)$ D) $(x-3)^2 = 8(y-1)$

22. How many four digit even numbers can be formed from 1,2,3,4 and 5 the numbers start with 3?
A) 40 B) 50 C) 100 D) 120

23. A satellite moves along a hyperbolic curve whose horizontal transverse axis is 24 km and an asymptote $y = \frac{5}{12}x + 2$. Then, what is the eccentricity of the hyperbola?
A) $\frac{5}{13}$ B) $\frac{12}{13}$ C) $\frac{13}{12}$ D) $\frac{15}{3}$

24. What is an antiderivative of $f(x) = \frac{2}{4x^2+4x+1}$?
A) $\frac{1}{2x+1}$ B) $\frac{-2}{2x+1}$ C) $-\frac{1}{2x+1}$ D) $\ell n(4x^2+4x+1)$

25. At which vale(s) of x does $f(x) = \frac{1}{4}x^4 - 2x^2$ have a local maximum?
A) $x = 4$ C) $x = -2$ *and* $x = 2$
B) $x = 0$ D) $x = 0$ *and* $x = 2$

26. In the set of complex numbers, which one of the following is the solution set of $z^3 - iz^2 + 2z = 0$?
A) $\{0\}$ B) $\{0, -i\}$ C) $\{0, -i, 2i\}$ D) $\{0, i. -2i\}$

27. The population of certain country is currently 80 million with growth rate of 2% per year.

Given: $(0.02)^9 = 5.12 \times 10^{-16}$, $(1.02)^9 = 1.19$
$(0.02)^{10} = 1.024 \times 10^{-17}$, $(1.02)^{10} = 1.22$

Which one of the following is the best approximation of the population after 10 years?
A) 81.9 million C) 95.2 million
B) 86.8 million D) 97.6 million

28. The volume V of a melting ice cube after t seconds is $V = 2000 - 40t + 0.2t^2$ $(in\ cm^3)$. How fast is the volume changing when t=40 seconds?
A) 24 cm^3/sec C) -15 cm^3/sec
B) 15cm^3/sec D) -24cm^3/sec

29. Which one of the following is equal to $\lim_{n \to \infty} \frac{1-n-3n^2}{6n^2+1}$?
A) $\frac{1}{6}$ B) $-\frac{1}{2}$ C) $-\frac{1}{6}$ D) $-\infty$

30. What is the solution set of $\frac{2}{x} - \frac{x-2}{x^2-2x} = 1 - \frac{2x-2}{3x-2}$?
A) $\{1, -2\}$ B) $\{1,2\}$ C) $\{-1\}$ D) $\{1\}$

31. Let $f(x) = \begin{cases} 3^x + k, & if\ x \leq 0 \\ 3\frac{\sin(2x)}{x}, & \text{if } x > 0 \end{cases}$ if f is continuous at $x = 0$, then what is the value of k?
A) 6 B) 5 C) 2 D) 0

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A) $f(x) = \ell n(x+1)$

B) $f(x) = \frac{x^2+1}{x^3+8}$

C) $f(x) = \frac{x^2-9}{x-3}$

D) $f(x) = \frac{x-1}{x^2-x}$

13. Let p, q and r be propositions such that $p \Rightarrow (r \vee \neg q)$ is false. Then, which one of the following propositions is true?

A) $p \Rightarrow r$ B) $\neg r \Rightarrow q$ C) $\neg p \Leftrightarrow q$ D) $q \Leftrightarrow r$

14. If $f(x) = \frac{1}{e^x+1}$, then which of the following is equal to $f^{-1}(x)$?

A) $\ell n(1-x) - \ell n(x)$

B) $e^{-x} + 1$

C) $\ell n\left(\frac{1}{x+1}\right)$

D) $\frac{1}{e^{-x}+1}$

15. If $x^2 - 6x + y^2 + k = 0$ is equation of a circle with radius 2, then what is the value of k?

A) 13 B) 5 C) 4 D) -4

16. If a line with angle of inclination of $\frac{3}{4}\pi$ passes through (0,1), which one of the following is the equation of the line?

A) $y = -x + 1$

B) $y = x + 1$

C) $y = -x - 1$

D) $y = x - 1$

17. If $g(x) = xf(x) - \sqrt{f(x)}$ and $f(2) = f'(2) = 4$, then which of the following is equal to $g'(2)$?

A) 11 B) 8 C) 2 D) 0

18. What is the sum of the series $\sum_{n=1}^{\infty} \frac{2^{2n+1}}{5^{n-1}}$?

A) 40 B) 20 C) 10 D) 8

19. What is the value of $\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^{-\frac{x}{2}}$?

A) $\frac{1}{\sqrt{e}}$ B) $\sqrt{e}$ C) e^{-2} D) ∞

20. Among students who took a quiz, 15 students scored 6,20 students scored 7,10 students scored 8 and 5 students scored 10. What is the average score of the students?

A) 7.8 B) 7.5 C) 7.2 D) 7.0

21. A parabola with focus (3,-1) has directrix y=3. Which one of the following is the equation of the parabola?

A) $(x-3)^2 = -4(y+1)$

B) $(x-3)^2 = -8(y-1)$

C) $(x-3)^2 = 4(y+1)$

D) $(x-3)^2 = 8(y-1)$

22. How many four digit even numbers can be formed from 1,2,3,4 and 5 the numbers start with 3?

A) 40 B) 50 C) 100 D) 120

23. A satellite moves along a hyperbolic curve whose horizontal transverse axis is 24 km and an asymptote $y = \frac{5}{12}x + 2$. Then, what is the eccentricity of the hyperbola?

A) $\frac{5}{13}$ B) $\frac{12}{13}$ C) $\frac{13}{12}$ D) $\frac{15}{3}$

24. What is an antiderivative of $f(x) = \frac{2}{4x^2+4x+1}$?

A) $\frac{1}{2x+1}$ B) $\frac{-2}{2x+1}$ C) $-\frac{1}{2x+1}$ D) $\ell n(4x^2+4x+1)$

25. At which vale(s) of x does $f(x) = \frac{1}{4}x^4 - 2x^2$ have a local maximum?

A) $x = 4$

B) $x = 0$

C) $x = -2$ *and* $x = 2$

D) $x = 0$ *and* $x = 2$

26. In the set of complex numbers, which one of the following is the solution set of $z^3 - iz^2 + 2z = 0$?

A) $\{0\}$ B) $\{0, -i\}$ C) $\{0, -i, 2i\}$ D) $\{0, i. -2i\}$

27. The population of certain country is currently 80 million with growth rate of 2% per year.

Given: $(0.02)^9 = 5.12 \times 10^{-16}$,	$(1.02)^9 = 1.19$
$(0.02)^{10} = 1.024 \times 10^{-17}$,	$(1.02)^{10} = 1.22$

Which one of the following is the best approximation of the population after 10 years?

A) 81.9 million

B) 86.8 million

C) 95.2 million

D) 97.6 million

28. The volume V of a melting ice cube after t seconds is $V = 2000 - 40t + 0.2t^2$ $(in\ cm^3)$. How fast is the volume changing when $t=40$ seconds?

A) 24 cm^3/sec

B) 15cm^3/sec

C) -15 cm^3/sec

D) -24cm^3/sec

29. Which one of the following is equal to $\lim_{n \to \infty} \frac{1-n-3n^2}{6n^2+1}$?

A) $\frac{1}{6}$ B) $-\frac{1}{2}$ C) $-\frac{1}{6}$ D) $-\infty$

30. What is the solution set of $\frac{2}{x} - \frac{x-2}{x^2-2x} = 1 - \frac{2x-2}{3x-2}$?

A) $\{1, -2\}$ B) $\{1,2\}$ C) $\{-1\}$ D) $\{1\}$

31. Let $f(x) = \begin{cases} 3^x + k, & if\ x \leq 0 \\ 3\ \frac{\sin(2x)}{x}, & \text{if } x > 0 \end{cases}$ if f is continuous at $x = 0$, then what is the value of k?

A) 6 B) 5 C) 2 D) 0

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32. Which one of the following is equal to $\int(1+x)3^x dx$?
A) $(1+x)3^x - 3^x + c$
B) $(1+x)3^x + (log_3 e)3^x + c$
C) $(1+x)3^x log_3 e - (log_3 e)^2 3^x + c$
D) $(1+x)3^x log_3 e - 3^x(log_3 e) + c$

33. A committee consisting of 3 students is to be selected from 10 candidates among which 4 are girls. What is the probability that at least one girl is selected?
A) $\frac{5}{6}$ B) $\frac{2}{3}$ C) $\frac{1}{3}$ D) $\frac{1}{6}$

34. A group of six students take their seats at random in a round table for a discussion. What is the probability that two specific students do NOT sit together?
A) $\frac{3}{5}$ B) $\frac{2}{3}$ C) $\frac{2}{5}$ D) $\frac{1}{3}$

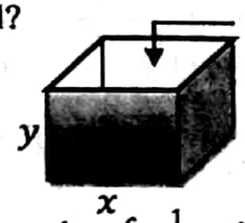
35. Given $f(x) = \ln(x-1)$ *and* $g(x) = \sqrt{1-2x}$, which one of the following is the domain of fog?
A) $\{x \in \Re : x > 1\}$
B) $\left\{x \in \Re : x \leq \frac{1}{2}\right\}$
C) $\{x \in \Re : x < 0\}$
D) $\left\{x \in \Re : x > \frac{1}{2}\right\}$

36. The mark that students scored in an examination is grouped in class intervals as shown in the following table.

Class Interval (Mark)	Number of Students
55 – 64	8
65 – 74	12
75 – 84	20
85 – 94	6
95 – 100	4

What is the median of the mark?
A) 25.0 B) 75.5 C) 77.0 D) 79.5

37. A box seen below is to have a square base, an open top and volume of 32 cubic units. If x is the length of each side of its base and y is its height, how many units should x and y be in order to make the box with the smallest amount of material?



A) $x = 4$, $y = 2$
B) $x = 2$, $y = 8$
C) $x = \sqrt{8}$, $y = 4$
D) $x = \sqrt{2}$, $y = 16$

38. Which one of the following is equal to $\int \frac{1}{x^2+x}\, dx$?
A) $\ell n|x^2 + x| + c$
B) $2\ell n|x+1| + \ell n|x| + c$
C) $\ell n|x| - \ell n|x+1| + c$
D) $\ell n|x| + \ell n|x+1| + c$

39. Consider the following argument:
"If he does not love her, she will not marry him.
He loves her. Therefore, she will marry him"
If '$p \equiv$ *He loves her*' *and* '$q \equiv$ *She will marry him*', which one of the following is the correct representation of the argument and its validity?
A) $\neg p \Rightarrow \neg q, p \vdash q$; Valid argument
B) $\neg p \Rightarrow \neg q, p \vdash q$; Invalid argument
C) $p \Rightarrow q, p \vdash q$; Valid argument
D) $p \Rightarrow q, p \vdash q$; Valid argument

40. What is the value of $\int_0^{\frac{\pi}{2}} 2x cosx\, dx$?
A) $\pi - 2$ B) $\frac{\pi}{2} + 1$ C) $\pi + 2$ D) $\frac{\pi}{2} - 1$

41. A box contains 5 white, 6 red and 4 black balls of identical size. If 3 balls are randomly taken out of the box one after the other, what is the probability that the first ball is white and both the second and third balls are red?
A) $\frac{2}{15}$ B) $\frac{3}{15}$ C) $\frac{4}{75}$ D) $\frac{5}{91}$

42. Which one of the following is the equation of the line tangent to the graph of $f(x) = \frac{1}{x+1} + \cos x$ *at* $(0, f(0))$?
A) $x + y = 1$ C) $x + y = 2$
B) $x - y = -2$ D) $x + 4y = 2$

43. Which one of the following is equal to $\lim_{x \to 1} \frac{\sqrt{x}-1}{x^2-1}$?
A) ∞ B) 0 C) $\frac{1}{4}$ D) 0

44. Which one of the following is equal to $\lim_{x \to 1} \frac{1-x}{1-\frac{1}{x^2}}$?
A) 1 B) 0 C) $-\frac{1}{2}$ D) Doesn't exist

45. Which one of the following is equal to $\int \frac{\ell n(xe^x)}{x} dx$?
A) $\ell n|x| + \frac{1}{2}e^x + c$
B) $\frac{1}{2}(\ell nx)^2 + x + c$
C) $\ell n|x| + e^{2x} + c$
D) $-\frac{1}{x^2} + (\ell nx)^2 + c$

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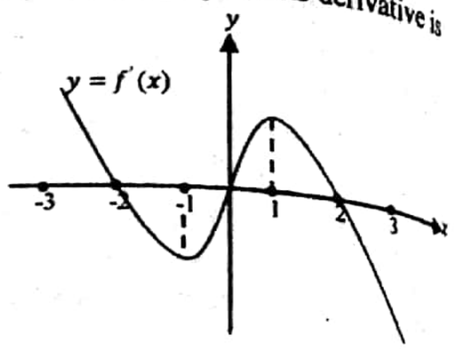
46. Consider the system $\begin{cases} ax + y + z = 1 \\ x + 2y + 4z = 0 \\ 5x - y + z = 0 \end{cases}$ if the determinant of the coefficient matrix is 2, then what is the solution of the system of equations?

A) $\left(3a, \frac{19a}{2}, \frac{-11a}{2}\right)$ C) $\left(\frac{3}{a}, \frac{-19a}{2}, \frac{11}{2}\right)$

B) $\left(3, \frac{19}{2}, \frac{-11a}{2}\right)$ D) $\left(\frac{3}{2}, \frac{19}{2}, \frac{9}{2}\right)$

47. Suppose f is differentiable on $(-\infty, \infty)$ and the graph of its derivative is as shown below.



Which one of the following is true about f?

A) f is decreasing on $(-\infty, -1] \cup [1, \infty)$

B) f has a local minimum at $x = -2$

C) f is concave down on $[0, \infty)$

D) f is concave up on $(-1,1)$

48. Which one of the following is true about the derivative of $f(x) = x|x|$?

A) f is not differentiable at $x = 0$

B) $f'(x) = 2|x|$, for every $x \in (-\infty, \infty)$

C) $f'(x) = 2x$, for every $x \in (-\infty, \infty)$

D) $f'(x) = |x| + x$, for every $x \in (-\infty, \infty)$

49. What is the area of the region between the graphs of $y = x^2$ and $y = -x + 2$, where $0 \le x \le 2$?

A) 3 B) 2 C) $\frac{3}{2}$ D) $\frac{2}{3}$

50. For what value of b does the parabola $p(x) = ax^2 + x + b$ pass through the points $(-1,5)$ and $(2,-1)$?

A) 9 B) 3 C) -3 D) -15

51. If $f(x) = 2 - \frac{1}{2}\sin\left(\frac{\pi}{2}x\right)$, then which one of the following is the amplitude and period of f, respectively?

A) $\frac{1}{2}$ and 4 B) $\frac{-1}{2}$ and 4 C) 2 and π D) $\frac{1}{2}$ and π

52. Which one of the following is equal to $\sec\left(\frac{\pi}{2} - x\right) sin^3x + \cos 2x$?

A) $2cosx$ B) $2sinx$ C) cos^2x D) sin^2x

53. Suppose $P(1,2,1)$ and $Q(1,0,2)$ are points in space and $\vec{A} = \overrightarrow{PQ}$. If $\vec{B}$ is parallel to $\overrightarrow{PQ}$ and $\vec{A}.\vec{B} = -10$, then which one of the following is true?

A) $\vec{A}$ and $\vec{B}$ has the same direction

B) $\|\vec{B}\| = 10\|\vec{A}\|$ C) $\|\vec{B}\| = \frac{1}{10}\|\vec{A}\|$ D) $\|\vec{B}\| = 2\|\vec{A}\|$

54. What is the solution of $cos^2x + \frac{1}{2}\, sin2x = 1$ in the interval $[0,2\pi)$?

A) $\left\{0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}\right\}$ C)$\{0, \pi\}$

B) $\left\{0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi\right\}$ D) $\left\{0, \frac{\pi}{4}, \pi\right\}$

55. If $\vec{u} = (-3, x)$ and $\vec{v} = (x, y - 2)$ are vectors, what is the value of y so that $\vec{u} + \vec{v} = 3\vec{u} - \frac{1}{2}\vec{v}$?

A) $\frac{2}{3}$ B) $-\frac{10}{3}$ C) -4 D) $-\frac{22}{3}$

56. Which one of the following points is closer to the sphere $x^2 + y^2 + z^2 - 2x + 6z + 9 = 0$?

A) (1,0,0) B) (0,0,0) C) (0,−1,0) D) (0,0,−1)

57. Which one of the following describes the principle of Mathematical Induction on a set of natural numbers?

A) If an assertion is true for a natural number n, then it is true for $n + 1$.

B) If an assertion is true for 1 and it is true for $n = 1$, then it is true for some n.

C) If an assertion holds for $n = 20$ and for an $n \ge 20$, then it is true for n implies true for $n + 1$.

D) If an assertion is true for $n = 1$ and is true for $n = k$, whenever is true for $n = k + 1$.

58. What is $cot(arcsinx)$ if $0 < x < 1$?

A) $\frac{x}{\sqrt{1-x^2}}$ B) $\frac{1}{x}\sqrt{1-x^2}$ C) $\sqrt{1-x^2}$ D) $\frac{1}{\sqrt{1-x^2}}$

59. Suppose $\vec{A} = 2\vec{j} - \vec{k}$ and $\vec{B} = 5\vec{i} + 15\vec{k}$, where $\vec{i}, \vec{j}$ and $\vec{k}$ are the standard unit vectors in the directions of positive x,y and z axis, respectively. Which one of the following is the unit vector in the direction of $\vec{A} + \frac{1}{5}\vec{B}$?

A) $\frac{3}{5}\vec{i} + \frac{4}{5}\vec{k}$ C) $\frac{4}{5}j - \frac{3}{5}k$

B) $\frac{1}{3}\vec{i} + \frac{2}{3}\vec{j} + \frac{2}{3}\vec{k}$ D) $\frac{2}{3}i - \frac{1}{3}j + \frac{2}{3}k$

60. A line given by a vector equation r(t)=(0,3)+t(1,1) is tangent to a circle at point (0,3). If the radius of the circle is $\sqrt{2}$ which one of the following is the center of the circle?

A) (1,4) B) (1,-4) C) (-1,2) D) (1,2)

61. Suppose the following statements are the premises of an argument.

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" He was lazy or he did not like the classroom.
If he was lazy, he could not pass the exam.
He passed the exam."
Which one of the following can be a conclusion that makes the argument valid?

A) He did like the classroom.
B) He did not like the classroom.
C) If he was not lazy, he did like the classroom.
D) He was not lazy and he did like the classroom.

62. What is the image of the ellipse whose equation is $2(x+2)^2+(y-1)^2=2$ under a translation that takes (2,1) to (4, followed by a rotation of 90^0?

A) $x^2+2y^2=2$ C) $2(x-4)^2+y^2=2$
B) $2x^2+y^2=2$ D) $(x-4)^2+2y^2=2$

63. Let $\vec{a}=2\vec{\imath}+(x-1)\vec{\jmath}+\vec{k}$ and $\vec{c}=\vec{\imath}-\vec{\jmath}+y\vec{k}$ be vectors. If $\vec{a}.\vec{c}=$ and $\|\vec{a}\|=3$, which one of the following is a possible value of y?

A) -4 B) -1 C) 3 D) 4

64. If $\vec{A}$ is perpendicular to $\vec{B}$, what is the cosine of the angle bet $\vec{A}$ and $\vec{A}-\vec{B}$?

A) $\frac{\|\vec{A}-\vec{B}\|}{\|\vec{A}\|}$ B) $\frac{\|\vec{A}\|}{\|\vec{A}-\vec{B}\|}$ C) $\frac{\|\vec{A}-\vec{B}\|}{\|\vec{B}\|}$ D) $\frac{\|\vec{B}\|}{\|\vec{A}-\vec{B}\|}$

65. Which one of the following is necessarily true?

A) If $\|\vec{A}\|=\|\vec{B}\|$, then $\vec{A}=\vec{B}$.
B) $\|k\vec{A}\|=k\|\vec{A}\|$, for any real number k.
C) If $\vec{A}$ is parallel to $\vec{B}$, *then* $\vec{A}.\vec{B}=0$.
D) If $\vec{u}$ is a unit vector in the direction of $\vec{A}$, then $\vec{A}.\vec{u}=\|\vec{A}$

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ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (EUEE)
MATHEMATICS FOR NATURAL SCIENCE STREAM
GINBOT 2006/JUNE 2014

BOOKLET CODE: 17

DIRECTIONS: for each of the following problems, choose the best answer from the given alternatives and carefully. Blacken the letter of your choice on the separate answer sheet provide.

1. A water tank is a circular cylinder with base radius 2m and height 3m. If the tank is empty and water is pumped into it at rate of $2m^3/min$, how long does it take for the tank to be full?

A) 1.5 min B) $\frac{3}{2}\pi min$ C) 6π D) 12 min

2. If $A=\begin{pmatrix}2&0&-1\\1&2&0\\0&0&-1\end{pmatrix}$ and $(2A+B)^T=A^TA$, then which one of the Following is equal to B?

A) $\begin{pmatrix}1&0&-2\\2&0&0\\0&0&4\end{pmatrix}$ C) $\begin{pmatrix}8&0&-4\\4&8&0\\0&0&-4\end{pmatrix}$
B) $\begin{pmatrix}0&0&0\\0&0&0\\0&0&0\end{pmatrix}$ D) $\begin{pmatrix}1&2&0\\0&0&0\\-2&0&4\end{pmatrix}$

3. Which one of the following represents a geometric sequence?

A) $3,1,\frac{1}{3},\frac{1}{9},\frac{1}{27},\ldots.$ C) 1,3,6,10,15,...
B) $\frac{1}{2},-\frac{1}{3},\frac{1}{4},-\frac{1}{5},\frac{1}{6},\ldots$ D) -3,6,-9,12,-15,...

4. The sequence $\left\{\frac{(n-1)(2n+1)}{1-n^2}\right\}_{n=1}^{\infty}$ converges to:

A) $-\infty$ B) -2 C) 0 D) 1

5. If F(x) is an ant derivative of $f(x)=1-\frac{2}{x^2}$ *and* $F(1)=0$, then F(2) is equal to :

A) 0 B) $\frac{1}{2}$ C) $-\frac{1}{2}$ D) 3

6. Which of the following expression is a polynomial expression?

A) $x^2-3x+sinx$ C) $\frac{2+\pi}{1+\pi^2}$
B) $\frac{4x^3+12x^2-x}{\pi x^2}$ D) $2\text{-}3x^{\frac{2}{3}}+7x^{5/2}+3x^{-1}$

7. What is the distance from the origin to the line that passes through (1,0) and (0,1)?

A) $\frac{1}{2}\sqrt{2}$ B) 1 C) $\frac{1}{2}$ D) $\sqrt{2}$

8. If the list of a measurement is 10, α, 5, α, 5,10,20,15,20,5 with mean $\bar{x}$?
   A) $10\bar{x} - 90$
   B) $9\bar{x} - 90$
   C) $5\bar{x} - 90$
   D) $5\bar{x} - 45$
9. The total cost (in Birr) of producing x iron sheets per day is C(x) $=1{,}000+100x-0.5x^2$, $0 \le x \le 100$. What is the marginal (rate of change of) cost at a production level of 80 iron sheets?
   A) 8.5 B) 20 C) 1,800 D) 5,800
10. If z=x+yi is a complex number, then $|z|^2 + \frac{1}{2}(z-\bar{z})^2 = 1$ is equivalent to which one of the following equations?
    A) $x^2 - y^2 = 1$
    B) $x^2 - 3iy^2 = 1$
    C) $x^2 - y^2 = 2$
    D) $2x^2 - y^2 = 2$
11. The following is the frequency distribution of a grouped data.

Class intervals	Frequency (f)
3-7	2
8-12	2
13-17	10
18-22	6

What is the mean and standard deviation of the distribution respectively?
   A) $15, 2\sqrt{5}$
   B) $15, \sqrt{7.5}$
   C) $12.5, 5\sqrt{2}$
   D) $12.5, \sqrt{15}$
12. If $(p \vee q) \Rightarrow (\neg r \wedge r)$ is true, then which one of the following is necessarily true?
    A) $(p \vee r) \Rightarrow q$
    B) $\neg q \wedge r$
    C) $\neg p \Leftrightarrow r$
    D) $\neg p \vee r$
13. What is the actual value of the sum $\sum_{n=1}^{\infty}(\frac{2^n+5^n}{10^n})$?
    A) 0.325 B) 1 C) $\frac{5}{4}$ D) $\frac{37}{9}$
14. If $f(x) = \frac{x+1}{x-1}$ *and* $f(a) = 5$ *then* $f(2a)$ is equal to:
    A) 2 B) 4 C) 6 D) 8
15. If $f(x) = \sqrt[3]{1+e^{-x}}$, which of the following is equal to $f^{-1}(x)$?
    A) $\ell n\left(\frac{1}{x^3-1}\right)$
    B) $\frac{1}{\ell n(x^3-1)}$
    C) $\ell n(1-x^3)$
    D) $(1+e^{-x})^3$
16. What is the equation of the directrix for the parabola whose equation is $y^2 + 8x + 6y + 25 = 0$?
    A) Y=3 B) x=2 C) x=0 D) x=4
17. Given that $\lim_{x\to 3} f(x) = 5$ *and* $\lim_{x\to 3} g(x) = 11$, what is the value of $\lim_{x\to 3}\left(\frac{(f(x)-g(x))(g(x))}{g(x)^2-f(x)^2}\right)^n$?
    A) $-\frac{66}{96}$ B) $-\frac{1}{16}$ C) 0 D) does not exist
18. If $f(x) = \frac{x^2}{1+xg(x)}$, $g(2) = 1$ *and* $g'(2) = 10$, *then which* one of the following is equal to $f'(2)$?
    A) -8 B) $-\frac{8}{9}$ C) $\frac{4}{3}$ D) $\frac{8}{9}$
19. If w= $\frac{16i}{1+i} + (1-3i)^2$ *and* $z = |w| + \overline{w}$, which one of the following is the simplest form of z?
    A) $\sqrt{2} + 2i$
    B) 2+8i
    C) 4-2i
    D) 2-2i
20. If M= $\begin{pmatrix} 0 & 1 & 2 \\ 3 & -1 & 0 \\ 5 & 2 & 4 \end{pmatrix}$ and $A^T M = 2I$, *where A is a* 3 × 3 matrix and I is the identity matrix of order 3, then what is det(A)?
    A) 0.2 B) $\frac{4}{17}$ C) 0.8 D) $\frac{1}{17}$
21. If distinct codes (words0 of eight letters are formed by rearranging the letters in the word'ABBEBAYE', how many of the codes begin with B or Y?
    A) 840 B) 630 C) 1680 D) 420
22. What should be the value of k so that the system of equation
$$\begin{cases} x - y + z = 1 \\ -x + 5y - 4z = 1 \\ 2x + 2y - z = k \end{cases}$$ has a solution?
    A) 0 B) 1 C)-4 D) 4
23. Let $f(x) = \begin{cases} a\frac{\sin x}{x-|x|}, & if\ x < 0 \\ e^{-x} + \cos x, & if\ x \ge 0 \end{cases}$
    If f is continuous at x=0, then what is the value of a?
    A) 4 B) 2 C) $\frac{1}{2}$ D) -4
24) What is the area of the region between the graphs of $y=-x^2+2$ and $y=|x|$, where $-1 \le x \le 2$?
    A) $\frac{11}{6}$ B) $\frac{25}{6}$ C) $\frac{7}{3}$ D) $\frac{11}{3}$
25) What is the derivative of $f(x) = \int_0^{(x^2+\pi} \frac{dt}{sint+1}$?

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A) $\frac{\cos x}{\sin(x^2+\pi)+1}$

B) $\frac{2x}{\sin(x^2+\pi)+1}$

C) $\frac{2x\cos x}{\sin(x^2+\pi)+1}$

D) $\int_0^{2x} \frac{dt}{\sin t+1}$

26) If two lines y=x and y=x-4 are tangent to a circle at (2,2) and (4,0) respectively, then what is the equation of the circle?

A) $(x-2)^2+y^2=4$

B) $(x-4)^2+(y-2)^2=4$

C) $(x-3)^2+(y-1)^2=2$

D) $(x-1)^2+(y+1)^2=10$

27) Which one of the following is the set of all critical numbers of $f(x)=\frac{1}{3}x^3-|4x-1|$?

A) $\left\{\frac{1}{4},2\right\}$

B) $\left\{-2,\frac{1}{4},2\right\}$

C) $\{-2,2\}$

D) $\left\{\frac{1}{4}\right\}$

28) Which one of the following is equal to $\int_0^{\pi/2} \frac{x-\sin x}{\sec x}dx$?

A) $\frac{\pi-3}{2}$ B) $\frac{\pi-1}{2}$ C) $\frac{3-\pi}{2}$ D) $\frac{\pi+3}{2}$

29) Which one of the following is true?

A) A polynomial can have infinitely many vertical asymptotes

B) The graph of a rational function can never cross its horizontal asymptote.

C) The graph of $f(x)=\frac{3x-1}{x-1}$ has no horizontal asymptote.

D) The graph of $f(x)=\frac{x^3-x}{x^2-x}$ has no vertical asymptote

30) The simplified form of the derivative of $f(x)=\frac{1+\sin x}{\cos x}$ is

A) Sec x + tan x B) $\frac{1+\sin x}{\cos^2 x}$ C) $\frac{1}{1+\tan x}$ D) $\frac{\cos x}{\sin^2 x}$

31) If Qi, Di and Pi are respectively the i^{th} – quartile, decile and percentile of a data arranged in an increasing order, then which one of the following is necessarily true?

A) $Q_2 \frac{Q_1+Q_3}{2}$

B) $D_3 > P_{25}$

C) $P_{25} > Q_1$

D) $Q_2 = mean of the data$

32) If $f(x)=x^2\sqrt{2x+12}$, what is the slope of the tangent line to the graph of f at x=2?

A) -4 B) 2 C) 18 D) 17

33) If $f(x)=e^{2x}\sin x$, then $f''(x)$ is equal to

A) $3e^{2x}\sin x-4e^{2x}\cos x$

B) $4e^{2x}\sin x+2e^{2x}\cos x$

C) $e^{2x}(3\sin x+4\cos x)$

D) $e^{2x}(4\sin x-3\sin x)$

34) If the equation $(x-2)^2-(y-2)^2=1$ represents a hyperbola, which one of the following represents equation of an asymptote to the hyperbola?

A) y=4-x

B) x+y=1

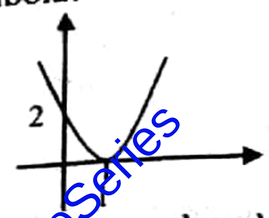
C) x=2-y

D) x+2y=3

35) What is the sum of the series $\sum_{n=1}^{\infty}(-1)^n 3^{-2n}$?

A) $\frac{-1}{8}$ B) -0.13 C) -0.1 D) $\frac{1}{8}$

36) Which of the equations below is represented by the following parabola?



A) $y=x^2+2$

B) $y=(2x-1)^2$

C) $y=2(x-1)^2$

D) $y=(2x+1)^2$

37) A company produced 25,000 bulbs and randomly tested 2% of the product. Among the tested bulbs, if 40 have defect of type D_1, 60 have defect of type D_2 and 25 have both types of defects, what is the probability that a bulb produced by the company has none of the defects?

A) 0.95 B) 0.80 C) 0.85 D) 0.20

38) A semi-elliptical arc over a tunnel for a road through a mountain has a major axis of length 80 meters and a height of 30 meters at the center. What is the equation of the semi- elliptical arc over the tunnel, if the center is considered as the origin?

A) $\frac{x^2}{6400}+\frac{y^2}{900}=1$

B) $\frac{x^2}{1600}+\frac{y^2}{900}=1$

C) $\frac{x^2}{900}+\frac{y^2}{6400}=1$

D) $\frac{x^2}{8100}+\frac{y^2}{6400}=1$

39) Suppose AX=b, where A is a 3×3 martix, $b=b_1,b_2,b_3)^T$ and X= $(x,y,z)^T$. What one of the following is necessarily true about this system of linear equations?

A) The system has a solution only when det(A)≠ 0.

B) The Cramer's rule is suitable to solve the system if two rows of A are identical

C) If det(A) ≠ 0 and the second column of A is a multiple of b, then x=0

D) If b=0, then $X=(0,0,0)^T$ is the only solution of the system.

40) If $y=\sin(3x^2)$, *then the simplified form of* $\frac{d^2y}{dx^2}$ *is*:

A) $-6\sin(3x^2)$

C) $6\cos(3x^2)-36x^2\sin(3x^2)$

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B) $\cos(6x) - 6\sin(3x^2)$ D) $x^2\cos(3x^2) + 6\sin(3x^2)$

41) If S is a set with 10 elements and A⊆S, what is the probability that A has 3 or more elements?

A) $\frac{7}{10}$ B) $\frac{8}{11}$ C) $\frac{121}{128}$ D) $\frac{7}{128}$

42) If $a_n = \left(\frac{n+3}{n+1}\right)^n$, then the limit of the sequence $\{a_n\}_{n=1}^{\infty}$ is equal to:

A) 1 B) $\frac{1}{2}e$ C) e^2 D) $+\infty$

43) Which one of the following is equal to $\lim_{n\to\infty}\left(\frac{x}{x+2}\right)^{-3x}$?

A) e^6 B) e^{-3} C) $e^{-3/2}$ D) e^{-6}

44) If a box with square base open top is made from 1,200cm2 material what is the largest volume of the box in cm³?

A) 4000 B) 8000 C) 15,000 D) 3000

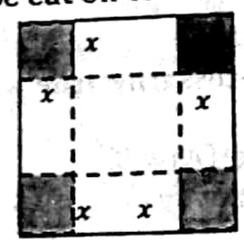
45) Which one of the following is true about the graph of $f(x) = \frac{x^2-1}{x-2}$?

A) x=0 and x=1 are its vertical asymptotes.
B) y=1 is its horizontal asymptote
C) y=x-1 is its oblique asymptote
D) It is almost the same as the horizontal line y=-1 as $x \to \pm\infty$

46) Which one of the following is a valid logical argument?

A) $p \Rightarrow q, q \backslash -q$ C) $\neg P \wedge q, q \Rightarrow r \backslash -r$
B) $p \Leftrightarrow q, p \Rightarrow q \backslash -q$ D) $\neg p, p \vee q, r \Rightarrow q \backslash -r$

47) Suppose that equal squares are cut from each of the four corners of a square cardboard whose sides are 72 centimeters long. [See the figure below.] The resulting flaps are then folded up to form a box without a top. How long should be each of the four squares that have to be cut off to maximize the volume of the box?



A) 6cm
B) 12cm
C) 15cm
D) 24 cm

48) What is the area of the region between the graph of $f(x) = -x^2 + 4x - 3$ and the x-axis from x=0 to x=3?

A) $-\frac{2}{3}$ B) $\frac{2}{3}$ C) $\frac{4}{3}$ D) $\frac{8}{3}$

49) Which one of the following is equal to $\int \frac{x+\ell n(x+1)}{(x+1)^2}dx$?

A) $\ell n(x+1) + \frac{x}{x+1} + c$ C) $(x+1)^2 - \frac{1}{x+1} + c$
B) $(x+1)^2 + \frac{1}{x+1} + c$ D) $\frac{x\ell n(x+1)}{x+1} + c$

50) If $F(x)=f(2x+2).g(1-x^2), with\ f(2) = -3, f'(2) = 4,$ g(1)=-5, and $g'(1) = 1$, then what is the actual value of $F'(0)$?

A) -40 B)-20 C) 0 D) 19

51) Suppose that and airplane is descending at a speed of 50 miles per hour at an angle of 30^0 below the horizontal line. What is the x-and y- components, respectively, of the velocity of the plane?

A) $50\sqrt{3}, 25$ C) $25, -25\sqrt{3}$
B) $-25, 50\sqrt{3}$ D) $-25\sqrt{3}, -25$

52) If a point (2,5 is reflected under a line to the point(-3,1), what is the line of reflection?

A) $2x + 3y = 7$ C) $8y + 10x = 19$
B) X+3y=7 D) 2x+3y+5=0

53) Let the equation $x^2 + 2x + y^2 = 8$ represents a circle. Then which one of the following lines cut the circle at exactly two points?

A) $4x + 3y + 19 = 0$ C) $2y = 5x + 43$
B) $x + 4y + 14 = 0$ D) $2x = y - 50$

54) Suppose $\vec{A} = 2\vec{\imath} - \vec{\jmath} + 2\vec{k}$ and $\vec{B}$ is a vector in space such that $|\vec{B}| = \vec{A}.\vec{B}.$ *if* $\vec{u}$ is the unit vector in the direction of $\vec{B}$, then $|\vec{A} + \vec{u}|^2$ is equal to

A) 16 B) 12 C) 10 D) 14

55) If A- (-2,3), B=(3,1) and C is any other point on the plane, then which one of the following is the coordinate form of $\overrightarrow{AC} - \overrightarrow{BC}$?

A) (-5,2) B) (5,-2) C) (1,4) D) (-1,-4)

56) What is the equation of a line that passes through the point (-1,2) and parallel to the vector (1,-1)?

A) $2x - y = 1$ C) $x - 2y = 3$
B) $x + y - 1 = 0$ D) $y - 2x + 1 = 0$

57) Suppose P and Q are points in space such that the midpoint of $\overline{PQ}$ is on the negative z-axis and the distance between P and Q is 6. If P=(2, −1, 0), then what is the coordinate of Q?

A) (−2, 1, 4 C) (2, −1, −6)
B) (2, −1, 6 D) (−2, 1, −4)

58) Which one of the following is equivalent to $\neg[(\forall x)(P(x) \Rightarrow Q(x))]$?

A) $(\forall x)(\neg P(x) \Rightarrow \neg Q(x))$ C) $(\exists x)(\neg P(x) \wedge Q(x))$
B) $(\exists x)(\neg P(x) \Rightarrow \neg Q(x))$ D) $(\exists x)(P(x) \wedge \neg Q(x))$

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59) An observer on level ground is at a distance $10\sqrt{3}$ m from a building. The angles of elevation to the bottom of the windows on the second and third floors are 30^0 and 60^0, respectively. What is the distance h between the bottoms of the windows?

[*you may use the values*: $sin30^0 = cos60^0 = \frac{1}{2}$ *and* $sin60^0 = cos30^0 = \frac{\sqrt{3}}{2}$]

A) 15m B) 20m C) $15\sqrt{3}$m D) 32m

60) The following is an assertion of a person and his proof. "For any natural numbers n, $n! < 10^n$.

Proof:

Step 1 let n=1 Since $1! = 1$ and $10^1 = 10$, *it it true that* $1! < 10^1$

Step2 Let n=2. Since $2! = 2$ and $10^2 = 100$, it is true that $2! < 10^2$

Step3Letn=3. Since $3! = 6$ and $10^3 = 1000$, it is true that $2! < 10^2$

Step4 Continuing in this manner, we can see that whenever $k! < 10^k$ is true, then $(k+1)! < 10^{k+1}$ is also true.

Therefore, by induction, $n! < 10^n$ for all natural numbers."

Which one of the following is true about the proof?

A) The proof is correct by the principle of mathematical induction though Step 2 and Step 3 can be omitted.

B) The proof is correct by the principle of mathematical induction though Step 2 and Step 3 are necessary since they provide additional information.

C) The proof is invalid because Step 4 did not justify the desired induction step.

D) The proof follows the technique of a proof by exhaustion.

61) What is the image of the line given by $(x, y) = (-1,0) + t(3,6), t \in$ under the translation that takes (1,0) to (0,1) following by the reflection about the line y= 2x?

A) $y = 2x + 3$ C) $y = 2x + 6$

B) $y = 2x - 3$ D) $y = 2x - 5$

62) If $\theta = 2\arctan\left(\frac{1}{2}\right)$, then which one of the following is equal to $\sec(\theta)$?

A) $\frac{25}{3}$ B) $\frac{4}{5}$ C) $\frac{5}{3}$ D) $\frac{5}{4}$

63) If a translation T takes the circle $x^2 + y^2 - 2x + 6y + 3 = 0$ into the circle whose equation is $(x + 2)^2 + (y - 4)^2 = 7$, then what is the image of the origin under T?

A) (-3,7) B) (1,2) C) (1,-3) D) (-2,4)

64) If ℓ is the line passing through(0,2) and parallel to $V = \vec{i} + 3\vec{j}$, which one of the following is true about ℓ and the circle $(x - 2)^2 + (y - 1)^2 = 5$?

A) ℓ is tangent to the circle at (0,2)

B) ℓ *is tangent to* the circle at some point P, where P≠ (0,2).

C) ℓ intersects the circle at two distinct points.

D) The distance between ℓ and the center of the circle is greater than $\sqrt{5}$

65) Suppose$\vec{A} = 3\vec{i} - 4\vec{j}$ and $\vec{B}$ is a vector in the xy-plane such that the angle between $\vec{A}$ and $\vec{B}$ is $\frac{\pi}{3}$. if $\vec{u}$ is the unit vector in the direction of $\vec{B}$, then $\vec{A}.(\vec{A} - 2\vec{u})$ is equal to:

A) 20 B) 25 C) 15 D) 30

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1. A: $f(x) = \frac{p(x)}{q(x)}$

- Domain of f(x) is $\{x \in R: q(x) \neq 0\}$
- B) is false for example $f(x) = \frac{x^2 - 1}{x^2 + 1}$ is a rational function with domain R but f is not a polynomial.

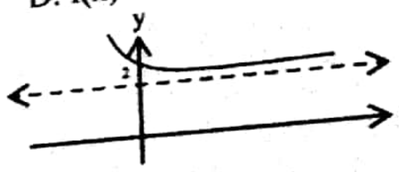
C) is false because the range depends on the nature of f. For example $f(x) = \frac{x^2 - 1}{2}$ is a rational function whose range is $\{y: y \geq -½\}$ and not R.

D) p(x) may not be divisible by q(x).

2. D: $f(x) = 1 + 3^{-x}$ has the following graph

$f(0) = 1+3^{-0} = 1+1 = 2$

$\therefore$ The y-intercept of f(x) is (0, 2)

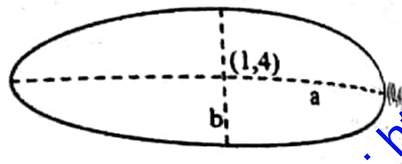


3. A: Given:

Ellipse with center (1, 4)

Vertices V_1 (10, 4)

V_2(1, 2)



Required: Equation of the Ellipse

a = 10 – 1 = 9

b = 4 – 2 = 2

The equation of ellipse with center (h, k), major axis 2a. minor axis 2b is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

Substituting h, k, a and b; we get

$$\frac{(x-h)^2}{9^2} + \frac{(y-4)^2}{2^2} = 1$$

$$\Rightarrow \frac{(x-1)^2}{81} + \frac{(y-4)^2}{4} = 1$$

$$\Rightarrow 324\left[\frac{(x-1)^2}{81} + \frac{(y-4)^2}{4}\right] = 1 \times 324$$

$$\Rightarrow 4(x-1)^2 + 81(y-4)^2 = 324$$

4. C: If f(x) is a polynomial with (x+3) as a factor then f(-3) = 0

5. D: $\lim_{n\to\infty} \sin^{\left(\frac{1}{n}\right)} = 0 \quad \therefore \left\{\sin\left(\frac{1}{n}\right)\right\}_{n=1}^{\infty}$ is convergent

6. C:

Given: $\frac{PR}{RQ} = \frac{2}{3} = \frac{r_1}{r_2}$

Required: Co-ordinates of R

$$R = \left(\frac{r_2x_2 + r_1x_2}{r_1 + r_2}, \frac{r_2y_1 + r_1y_2}{r_1 + r_2}\right)$$

$$= \left(\frac{3(-3) + 2(0)}{2+3}, \frac{3(0) + 2(-3)}{2+3}\right)$$

$$= \left(\frac{-9}{5}, \frac{-6}{5}\right)$$

7. A: 3, 5, 7, 9, 11, . . . is an arithmetic sequence with $A_1 = 3$ and d = 2

8. B: Given: p = price is high

q = Items are available

price is high. Items are available is necessary for price not to be high. Therefore, items are not available.

$p, \neg p \Rightarrow q \vdash \neg q$

9. A: Given: $p(t) = 80 + 12t - t^2$

$\Delta p(t) = 12 - 2t$

$\Delta p(3) = 12 - 2(3)$

$= 12 - 6$

$= 6$

10. A: $\int_0^1 \frac{x}{(x^2+1)^2} dx = ?$ Let $u = x^2 + 1$

$du = 2x\,dx$

$\frac{du}{2} = x dx$

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$$\int_0^1 \frac{x}{(x^2+1)^2}dx = \int_0^1 \frac{1}{2u^2}du = \frac{1}{2}\int_0^1 \frac{1}{u^2}du = \frac{1}{2}\left(\frac{-1}{u}\right)\Big|_0^1$$

Substituting $u = x^2 + 1$

$$= \frac{1}{2}\left(\frac{-1}{x^2+1}\right)\Big|_0^1$$

$$= \frac{-1}{2(1+1)} - \frac{-1}{2(0+1)} = \frac{-1}{4} + \frac{1}{2} = \frac{1}{4}$$

11. C: $y = \dfrac{x^2-1}{x^3}$

Required: intersection point(s) of the graph and horizontal asymptote

Horizontal Asymptote is $y = 0$

Intersection $\Rightarrow \dfrac{x^2-1}{x^3} = 0 \Rightarrow x^2 - 1 = 0$

$\Rightarrow x = \pm 1$

So, the intersection points are $x = -1$ and $x = 1$

12. C: Given: Equation of parabola $y^2 + 8x + 6y + 25 = 0$

Required: focus

$(y-k)^2 = 4p(x-h)$ general form of parabola with focus $(h+p, k)$

$y^2 + 8x + 6y + 25 \Rightarrow y^2 + 6y + 9 = -8x - 25 + 9$

$\Rightarrow (y+3)^2 = -8x - 16$

$\Rightarrow (y+3)^2 = 4(-2)(x+2)$

$h = -2$, $p = -2$ and $k = -3$

focus $(h+p, k) = (-2-2, -3)$

$= (-4, -3)$

13. A: Let $p \vee (q \Rightarrow \neg p)$ is false. That is p is false and $q \Rightarrow \neg p$ is also false

But, since p is false $\neg p$ is true

This means, $q \Rightarrow \neg p$ is never false when p is false.

Thus; $p \vee (q \Rightarrow \neg p)$ is never false.

14. D: Required: Solution set of $\dfrac{1}{x-2} = \dfrac{3}{x+2} - \dfrac{6x}{x^2-4}$?

$$\frac{3}{x+2} - \frac{6x}{(x-2)(x+2)} - \frac{1}{x-2} = 0 \qquad x \neq -2, 2$$

$$\frac{3(x-2) - 6x - (x+2)}{(x-2)(x+2)} = 0$$

$$\frac{3x - 6 - 6x - x - 2}{(x-2)(x+2)} = 0$$

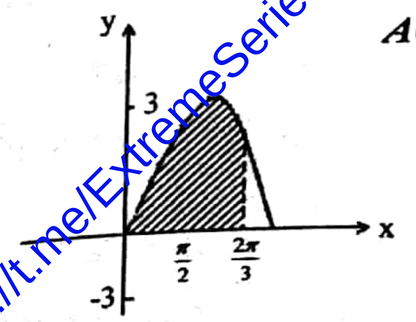
$-4x - 8 = 0$

$-4x = 8$

$x = -2$

$\therefore$ solution set $= \varnothing$

15. Required: Area under $y = 3\sin x$. on $\left[0, \dfrac{2\pi}{3}\right]$



$$A(x) = \int_0^{2\frac{\pi}{3}} 3\sin x dx$$

$$= -3\cos x\Big|_0^{\frac{2\pi}{3}}$$

$$= -3\left(\cos\frac{2\pi}{3} - \cos 0\right)$$

$$= -3\left(\tfrac{-1}{2} - 1\right)$$

$$= -3\left(\frac{-3}{2}\right) = \frac{9}{2}$$

16. C: Given: $\dfrac{u+3i}{4-2i} = \dfrac{2+vi}{20}$?

Required: $u = ?$ $v = ?$

$20u + 60i = (4 - 2i)(2 + vi)$

$20u + 60i = 8 + 4vi - 4i + 2v$

$20u + 60i = (8 + 2v) + (4v - 4)i$

$\Rightarrow 20u = 8 + 2v$ and

$60 = 4v - 4$

$\Rightarrow 64 = 4v \Rightarrow v = 16$

From, $20u = 8 + 2v$, substituting $v = 16$, we get

$20u = 8 + 32$

$20u = 40 \Rightarrow u = 2$

So, $u = 2$ and $v = 16$

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17. B: Given: $f(x)=\begin{cases}\dfrac{4}{x^2-4}-\dfrac{1}{x-2}; & x\neq 2\\ s \qquad\qquad ; & x=2\end{cases}$

f is continuous for all $x>0$

Required: value of s

$$\lim_{x\to 2} f(x)=\lim_{x\to 2}\left(\frac{4}{x^2-4}-\frac{1}{x-2}\right)$$

$$=\lim_{x\to 2}\left(\frac{4-(x+2)}{x^2-4}\right)=\lim_{x\to 2}\left(\frac{4-x-2}{x^2-4}\right)$$

$$=\lim_{x\to 2}\left(\frac{-x+2}{x^2-4}\right)$$

$$=\lim_{x\to 2}\left(\frac{-(x-2)}{(x-2)(x+2}\right)$$

$$=\lim_{x\to 2}\frac{-1}{x+2}$$

$$=\frac{-1}{4}$$

f is continuous at $x=2$, thus $f(-2)=-¼=5$

18. D: Required: Solution set of $\log_{10}^{5x}+\log_{10}^{x-1}=2$

$$\Rightarrow \log_{10}^{5x(x-1)}=2$$

$$\Rightarrow 10^2=5x(x-1)$$

$$\Rightarrow 100=5x^2-5x$$

$$\Rightarrow 5x^2-5x-100=0$$

$$\Rightarrow x^2-x-20=0$$

$$\Rightarrow (x-5)(x+4)=0$$

$$\Rightarrow x=5 \qquad x=-4$$

Solution set = {5}

19. B: Given: $f(x)=x^2\sqrt{2x+12}$

Required: Equation of tangent line to f at $x=2$

$$f'(x)=2x\sqrt{2x+12}+\frac{x^2}{\sqrt{2x+12}}$$

$$f'(2)=16+\frac{4}{\sqrt{16}}$$

$$=16+1$$

$$=17$$

The equation is $y-f(2)=f'(2)(x-2)$

$$y-16=17(x-2)$$

$$y-16=17(x-2)+16$$

20. C: Required: lim it of $1-\frac{1}{3},\frac{-1}{3},\frac{1}{5},\frac{1}{5},\frac{-1}{7},\frac{-1}{7},\ldots$

The limit of $1,\frac{-2}{3},\frac{2}{5},\frac{-2}{7},\frac{2}{9},\frac{-2}{11},\ldots$ is 0

21. A: Given: * (2, 1) on l

* inclination of l is 45^0

Required: Equation of l

Slope of $l=\tan 45^0=1$

The equation is $\frac{y-1}{x-2}=1 \qquad \Rightarrow y-1=x-2$

$$\Rightarrow y=x-1$$

$$\Rightarrow x=y+1$$

22. A: because the vertex of a parabola lies at the mid point between the focus and the directrix.

23. C: Given: $f(x)=\dfrac{1}{1+e^{-x}}$

Required: Antiderivatine of f(x)

To find Antiderivative of $f(x)=\int\frac{1}{1+e^{-x}}dx$

let $1+e^{-x}=u$

$du=-e^{-x}dx$

$du=-(u-1)\,dx$

$$\Rightarrow dx=\frac{du}{-(u-1)}$$

$1+e^{-x}=u$

$\Rightarrow \quad e^{-x}=u-1$

$\Rightarrow -e^{-x}=-(u-1)$

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$$\int \frac{1}{1+e^{-x}}dx = \int \frac{1}{u}\frac{du}{-(u-1)} = -\int \frac{du}{u(u-1)} = -\int\left(\frac{-1}{u}+\frac{1}{u-1}\right)du$$

$$= \int \frac{du}{u} - \int \frac{du}{u-1} = \ln u - \ln(u-1)$$

Substituting $u = 1 + e^{-x}$

$$= \ln(1+e^{-x}) - \ln(1+e^{-x}-1)$$
$$= \ln(1+e^{-x}) - \ln(e^{-x})$$
$$= \ln\left(\frac{1+e^{-x}}{e^{-x}}\right) = \ln\left(\frac{1}{e^{-x}}+1\right) = \ln(e^{x}+1)$$

24. B: Given: $C(x) = 3x + \frac{30{,}000}{x^2} = 0$

Required: value of x for which C(x) is minimum

$C'(x) = 0$

$$\Rightarrow 3 + \frac{-30{,}000}{x^2} = 0$$

$$\Rightarrow \frac{-30{,}000}{x^2} = -3$$

$$\Rightarrow 10{,}000 = x^2$$

$$\Rightarrow x = 100$$

25. A: Given: a sphere $x^2 + y^2 + z^2 - 2x + dz = -1$ with radius = 7

Required: Value of d

$$x^2 - 2x + y^2 + z^2 + dz = -1$$

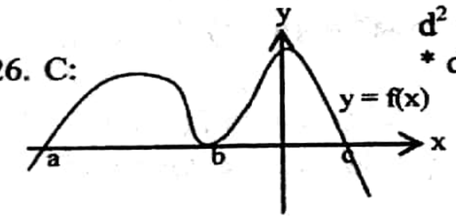
$$x^2 - 2x + 1 + y^2 + z^2 + dz + \frac{d^2}{4} = -1+1+\frac{d^2}{4}$$

$$(x-1)^2 + (y-0)^2 + \left(z+\frac{d}{4}\right)^2 = \frac{d^2}{4}$$

$$\frac{d^2}{4} = 7^2 \Rightarrow d^2 = 4 \times 7^2$$

$$d^2 = 196 = \pm 14$$

26. C:



* degree = at least 4
- No of turning p + s = 3
- b is a root of multiplicity = even
- leading coefficient = negative

27. C: Required: Solution set of $\frac{1}{1-\frac{1}{x}} \le \frac{x}{x^2-1}$

For $x \neq 0 \Rightarrow \frac{1}{1-1} \le \frac{x}{x^2-1}$

$$\Rightarrow \frac{1}{1-1} - \frac{x}{x^2-1} \le 0$$

$$\Rightarrow \frac{1}{1-1} - \frac{x}{x^2-1} \le 0$$

Using sign chart

		-1		0		1	
x^2	+	+	+	0	+	+	+
x + 1	-	0	+	+	+	+	+
x - 1	-	-	-	-	-	0	+
(x+1)(x-1)	+	0	-	-	-	0	+
$\frac{x^2}{(x+1)(x-1)}$	+	∄	-	0	-	∄	+

Solution set = (-1, 0) ∪ (0, 1)

28. D: $\int 10^x dx = ?$

$$\frac{10^x}{\ln 10} + C = \frac{10^x}{\frac{\log_{10}^{10}}{\log e}} + C = 10^x \log e + C$$

29. A: Given: remainder = 7

When $f(x) = 2x^4 + bx^3 - x^2 + 3$ is divided by x + 1

Required: value of b

$f(-1) - 7 \Rightarrow 2(-1)^4 + b(-1) - (-1)^2 + 3 = 7$

$$\Rightarrow 2 - b - 1 + 3 = 7$$
$$-b = 7 - 4$$
$$b = -3$$

30. D: Given: Z = x + yi and circle $(x-3)^2 + (y+1)^2 = 4$

Required: vector form of the circle

Center = (3, -1)

Equation is $|(x-3) + (y+1)i| = 2$

$$\Rightarrow |x - 3 + yi + i| = 2$$

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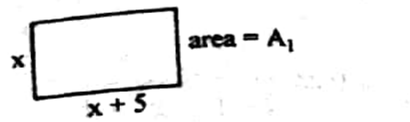
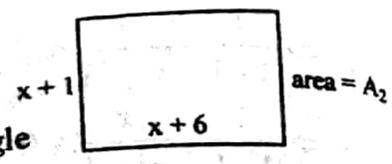
$\Rightarrow |x + yi - 3 + i| = 2$
$\Rightarrow |z - 3 + i| = 2$

31. A: Given:
Rectangle





$A_2 = A_1 + 154$
Required: length of original rectangle
i.e: x + 5
$A_2 - A_1 = 154$
$(x + 1)(x + 6) - x(x + 5) = 154$
$x^2 + 7x + 6 - x^2 - 5x = 154$
$2x = 154$
$2x = 148$
$x = 74$
$x + 5 = 74 + 5$
$= 79m$

32. A: Given: $\overline{AD}$ is median
DE ⊥ AC
area (ΔABC) = 54 sq units
AC = 12 units
Required: DE
$a(\Delta ABC) = 54 \Rightarrow a(\Delta ADC) = \frac{1}{2}(54) = 27$
$\Rightarrow 27 = \frac{1}{2}\left(\overline{AC}\right)\left(\overline{DE}\right)$
$27 = \frac{1}{2} \times 12 \times \overline{DE} \Rightarrow 27 = 6\overline{DE}$
$DE = \frac{27}{6} = 4.5$ units

33. B: Given: $\frac{x-8}{x^3+4x} = g(x) - \frac{2}{x}$
Required: g(x)

$$g(x) = \frac{x-8}{x^3+4x} + \frac{2}{x} \qquad x \neq 0$$

$$= \frac{x-8}{x(x^2+4)} + \frac{2}{x}$$

$$= \frac{x-8+2(x^2+4)}{x(x^2+4)}$$

$$= \frac{x-8+2x^2+8}{x(x^2+4)}$$

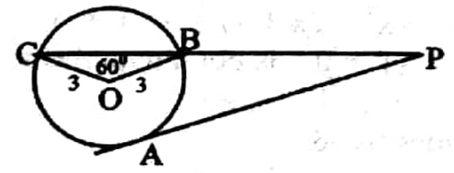
$$= \frac{x+2x^2}{x(x^2+4)} = \frac{x(1+2x)}{x(x^2+4)} = \frac{1+2x}{x^2+4}$$

34. Given: $a_n = \frac{1}{n^2} + \frac{2}{n^2} + \ldots + \frac{n}{n^2}$
Required: $\lim_{n\to\infty}\{a_n\}$

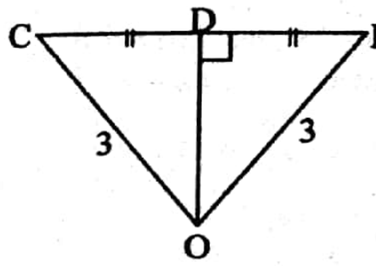
$$a_n = \frac{1+2+3+\ldots+n}{n^2} = \frac{\frac{n(n+1)}{2}}{n^2} = \frac{n^2+n}{2n^2} = \frac{1+\frac{1}{n}}{2}$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{1+\frac{1}{n}}{2} = \frac{1}{2}$$

35. C: Given



Required: AP
ΔBOC is equilateral triangle
BD = 1.5cm
BC = 3cm



$\Rightarrow (AP)^2 = PB \times PC$
$\left(\overline{AP}\right)^2 = 5 \times (5 + 3)$
$= 5 \times 8$
$= 40$
$\overline{AP} = 2\sqrt{10}$

36. B: Given: $f(x) = 2 + \ln(x - 1)$
Required: $f^{-1}(x)$
$y = 2 + \ln(x - 1)$

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inverse $x = 2 + \ln(y - 1)$

$\Rightarrow x - 2 = \ln(y - 1)$

$\Rightarrow e^{x-2} = y - 1$

$\Rightarrow y = 1 + e^{x-2}$

37. B: Given: $x_1 = 1 + 3i$ $\quad x_3 = 1 - 3i$

$x_2 = ½$ roots of f(x) with deg 3.

$f(x) = (x - x_1)(x - x_2)(x - x_3)$

$f(x) = (x - ½)(x - (1 + 3i)(x - (1 - 3i)$

$= (x - ½)(x^2 - 2x + 10)$

$= x^3 - \frac{5}{2}x^2 + 11x - 5$

38. B, D: Required: Equation of a hyperbola

A) $x^2 - y^2 + 4x - 6y = 0$

$(x^2 + 4x + 4) - (y^2 + 6y + 9) = 4 - 9$

$(x + 2)^2 - (y + 3)^2 = -5$

Dividing both sides by -5, we get

$$\frac{(y+3)^2}{5} - \frac{(x+2)^2}{5} = 1$$

$$\frac{(y+3)^2}{(\sqrt{5})^2} - \frac{(x+2)^2}{(\sqrt{5})^2} = 1 \quad \textit{hyperbola}$$

B) $y^2 - x^2 + 2x - 4y = -3$

$(y^2 - 4y + 4) - (x^2 + 2x + 1) = -3 + 4 - 1$

$(y - 2)^2 - (x - 1)^2 = 0$ is not a hyperbola

C) $x^2 - y^2 = -8$

Dividing both sides by -8

$$\frac{y^2}{8} - \frac{x^2}{8} = 1$$

$$\frac{y^2}{(\sqrt{8})^2} - \frac{x^2}{(\sqrt{8})^2} = 1$$

D) $y = \frac{-4}{x}$ is not a hyperbola

39. C: $S = 2(16 + 0.8 \times 16 + (0.8)^2 \times 16 + \ldots)$

$G_1 = 16$

$r = 0.8$ $\quad S = 2 \times \frac{G_1}{1-r} = 2 \times \frac{16}{1-0.8} = 2 \times \frac{16}{0.2} = 160$

40. B: $f(x) = x^2(x + 1)^3(x^2 + 1)$

$x = 0$ is a root of f(x) with multiplicity 2. Therefore, the graph of f(x) turns at $x = 0$ and does not cross.

41. D: Given: $f(2) = -3$, $f'(2) = 4$

$g(1) = -5$, $g'(1) = 1$

$F(x) = f(2x + 2).g(1 - x^2)$

Required: F'(0)

$F'(x) = f'(2x+2).(2x+2).g(1-x^2)+g'(1-x^2).(1-x^2)'.f(2x+2)$

$= f'(2x+2).(2).g(1-x^2)+g'(1-x^2).(-2x).f(2x+2)$

$F'(0) = f'(2).(2).g(1) + g'(1).(0).f(2)$

$= 4 \times 2 \times (-5) + 1 \times 0 \times (-3)$

$= -40 + 0 = -40$

42. C: Given: $\ln(10^w) = 4.6 + 3\ln h$

$\ln 2 = 0.6$

$\ln 3 = 1.1$

$\ln 5 = 1.6$ $\quad w = 60$

Required: h = ?

$\ln(10 \times 60) = 4.6 + 3\ln h$

$\ln 600 = 4.6 + 3\ln h$

$\ln 2 \times 3 \times 10^2 = 4.6 + 3\ln h$

$\ln^2 + \ln 3 + 2[\ln 5 + \ln 2] = 4.6 + 3\ln h$

$0.6 + 1.1 + 2(1.6 + 0.6) = 4.6 + 3\ln h$

$6.1 - 4.6 = 3\ln h$

$1.5 = 3\ln h$

$$\ln h = \frac{1.5}{3} = 0.5 = \frac{1}{2}$$

$e^{½} = h$

$\sqrt{e} = h$

43. B: Given: 3, 7, 11, 15, ...

Required: sum of 20 terms

The sequence is arithmetic with $a_1 = 3$ and $d = 4$

$$S_n = \frac{n}{2}(2a_1 + (n-1)d)$$

$$S_{20} = \frac{20}{2}(2(3) + (20-1) \times 4)$$

$$= 10(6 + (19 \times 4)$$

$$= 10 \times 82 = 820$$

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44. C: Given: $f(x) = x^4 - 2x^2 + 10$
Required: interval where f(x) is strictly decreasing
f(x) is strictly decreasing on
$(-\infty, -1) \cup (0, 1)$

45. B: Required: $\lim_{x\to 0} \frac{\cos^2 x - 1}{x \sin 2x}$

$$= \lim_{x\to 0} \frac{-(1-\cos^2 x)}{x \sin 2x}$$

$$= -\lim_{x\to} \frac{\sin^2 x}{x(2\sin x \cos x)}$$

$$= \frac{-1}{2} \lim_{x\to 0} \frac{\sin x \sin x}{x \sin x \cos x}$$

$$= \frac{-1}{2} \lim_{x\to 0} \frac{\sin x}{x \cos x}$$

$$= \frac{-1}{2} \lim_{x\to 0} \left(\frac{\sin x}{x} \cdot \frac{1}{\cos x} \right)$$

$$= \frac{-1}{2} \lim_{x\to 0} \frac{\sin x}{x} \cdot \lim_{x\to 0} \frac{1}{\cos x}$$

$$= \frac{-1}{2} \times 1 \times 1$$

$$= \frac{-1}{2}$$

46. A: Given: $G_2 = -10$

$$G_4 = \frac{-8}{5}$$

$G_5 > 0$

Required: $G_n = ?$

$G_2 = G_1 r$

$$G_4 = G_1 r^3 \Rightarrow \frac{G_1 r^3}{G_1 r} = \frac{\frac{-8}{5}}{-10}$$

$$\Rightarrow r^2 = \frac{8}{50}$$

$$\Rightarrow r^2 = \frac{4}{25}$$

$$r = \pm \frac{2}{5}$$

Since $G_5 > 0$, $r < 0$ and $G_1 > 0$

i.e. $r = \frac{-2}{5}$

$$G_1 = \frac{G_2}{r} = \frac{-10}{\frac{-2}{5}} = 25$$

$G_n = G_1 r^{n-1}$

$$= 25\left(\frac{-2}{5}\right)^{n-1}$$

47. D: Given: $f(x) = xe^{3x}$
Required: f''(x)

$$f'(x) = (xe^{3x})' = (x)'e^{3x} + (e^{3x})'.x$$
$$= e^{3x} + 3e^{3x}.x$$
$$= e^{3x} + 3xe^{3x}$$
$$f''(x) = (e^{3x} + 3xe^{3x})' = (e^{3x})' + (3xe^{3x})'$$
$$= 3e^{3x} + 3e^{3x} + 3xe^{3x}.3$$
$$= 3e^{3x} + 3e^{3x} + 9xe^{3x}$$
$$= 6e^{3x} + 9xe^{3x}$$
$$= (6 + 9x)e^{3x}$$

48. D: $\left\{\frac{e^n}{3^n(n+5)}\right\}$

Since $\lim_{n\to\infty}\left[\frac{e^n}{3^n(n+5)}\right] = 0$, the sequence ionverges to 0

49. D: Given: $f(x) = \begin{cases} 3x-1, & x \le 2 \\ x+3, & x > 2 \end{cases}$

Required: f(2)

$$f'(2) = \lim_{x\to 2} \frac{f(x) - f(2)}{x-2}$$

But, $\lim_{x\to 2^-} \frac{x+3-5}{x-2} = \lim_{x\to 2^-} \frac{x-2}{x-2} = \lim_{x\to 2^+} 1 = 1$

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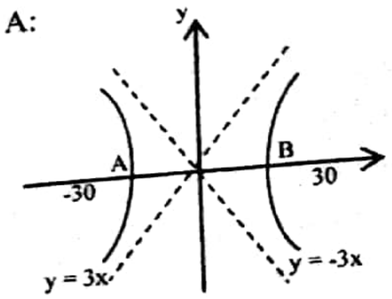
and $\lim_{x\to 2^-}\frac{3x-1-5}{x-2}=\lim_{x\to 2^-}\frac{3x-6}{x-2}=\lim_{x\to 2^-}\frac{3(x-2)}{x-2}=3$

$\Rightarrow \lim_{x\to 2}\frac{f(x)-f(2)}{x-2}$ does not exist

Therefore, f(x) does not have a derivative at x = 2.

50. A:



Asymptote $y=\frac{a}{b}x$

$\Rightarrow \frac{a}{b}=3$

$a = 3b, c = 30$

$b^2 = c^2 - a^2$

$b^2 = 30^2 - a^2$

$b^2 = 30^2 - (3b)^2$

$b^2 = 900 - 9b^2$

$10b^2 = 900$

$b^2 = 90$

$b = 3\sqrt{10}$ and $a = 3b$

$a = 3 \times 3\sqrt{10}$

$a = 9\sqrt{10}$

$\therefore d = 2a = 2\times 9\sqrt{10}$

$= 18\sqrt{10}$

51. A: $T(x, y) = (x - h, y - k)$

l: $-x + 5y = 2 \longrightarrow$ l': $-x + 5y = 10$

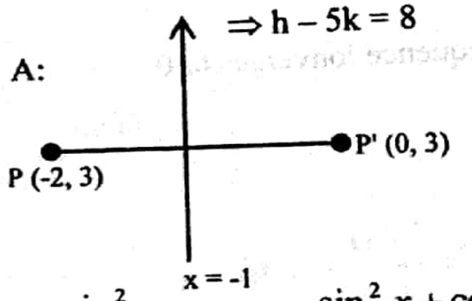
Let: (-2, 0) is on l; (-2 –h, -k) is on l'

$\Rightarrow -(-2 - h) + 5(-k) = 10$

$\Rightarrow 2 + h - 5k = 10$

$\Rightarrow h - 5k = 8$

52. A:



53. C: $\frac{\sin^2 x}{\cos x}+\cos x=\frac{\sin^2 x+\cos^2 x}{\cos x}=\frac{1}{\cos x}=\sec x$

54. A: Given: $\vec{u} = (-2, 5)$

$\vec{v} = (2, 4)$

Required: $2\vec{u} - \frac{1}{2}\vec{v} - 3j$

$\vec{u} = -2i + 5j$

$v = 2i + 4j$

$2u - \frac{1}{2}v - 3j = 2(-2i + 5j) - \frac{1}{2}(2i + 4j) - 3j$

$= -4i + 10j - i - 2j - j = -5i + 5j = (-5, 5)$

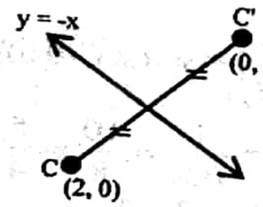
55. C: $\vec{A} = 3i - 4j$

Slope of $\vec{A} = \frac{-4}{3}$ and $\vec{A}$ is 4th quadrant vector

$\frac{-3}{2}i + 2j$ has a slope $\frac{2}{\frac{-3}{2}}=\frac{-4}{3}$ same as $\vec{A}$ but opposite to $\vec{A}$ in the 2nd quadrant

56. D: Given: a circle: $(x - 2)^2 + y^2 = 5$

Required: Image circle after reflection y = -x



$(x - 2)^2 + y^2 = 5$

Center = (2, 0)

radius = $\sqrt{5}$

C'(0, -2)

y = -x

Equation of image circle is $x^2 + (y + 2)^2 = 5$

$\Rightarrow x^2 + y^2 + 4y + 4 = 5$

$\Rightarrow x^2 + y^2 + 4y = 1$

57. B: Given: $f(x)\ 3 - \frac{1}{2}\cos\left(\frac{2\pi}{3}x-\frac{5\pi}{7}\right)$

Required: Amplitude, Period

For $f(x) = C + A\cos B(x - D)$, Amplitude = |A| and

$\text{period} = \frac{2\pi}{B}$

$f(x) = 3 - \frac{1}{2}\cos\frac{2\pi}{B}\left(x-\frac{15}{14}\right)$

$\Rightarrow$ Amplitude = |A| = |-½| = ½

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and Period $= \frac{2\pi}{B} = \frac{2\pi}{\frac{2\pi}{3}} = 3$

58. D: Given: A = (2, -1)
B = (3, 2)
D = (-2k, k)
Required: value of k for which AB ⊥ AD

Slope of AB $= \frac{3}{1}$, slope of AD $= \frac{k+1}{-2k-2}$

AB ⊥ AD $\Rightarrow 3\left(\frac{k+1}{-2k-2}\right) = -1$

$\Rightarrow 3k + 3 = 2k + 2$

$\Rightarrow k = -1$

59. B: Required: $\sec\left(\sin^{-1}\left(\frac{3}{5}\right)\right)$

$\sec\left(\sin^{-1}\left(\frac{3}{5}\right)\right) = \sec(\sin^{-1}(0.6)) = \sec 37^0 = \frac{1}{\cos 37^0}$

$= \frac{1}{0.8} = \frac{10}{8} = \frac{5}{4}$

60. D: Given: l:3y -2x + 3 = 0
Required: vector form of l.
Let $p_1(0, -1)$ and $p_2(3, 1)$ be points on l
V(3 -0, 1-(-1)) = v(3, 2) is a vector parallel to l
Thus, r = (0, -1) + t(3, 2) or r = (3, 1) + t(3, 2) are
Vector forms of the equation of l

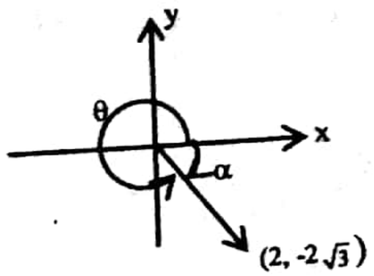
61. D: Given: vector $u = (2, -2\sqrt{3})$
Required: Magnitude and direction angle

$|u| = \sqrt{x^2 + y^2} = \sqrt{2^2 + (-2\sqrt{3})^2}$

$= \sqrt{4 + 12}$

$= \sqrt{16}$

$= 4$



$\tan\alpha = \frac{-2\sqrt{3}}{2}$

$= -\sqrt{3}$

$\Rightarrow \alpha = -60^0$

$\Rightarrow \theta = 300^0$

62. A: Given: |a| = 4
$|2a - b| = |a + b|$
Required: a.b
Let $a = a_x i + a_y j$ and
$b = b_x i + b_y j$

Then $|a| = 4 \Rightarrow \sqrt{a_x^2 + a_y^2} = 4$

$\Rightarrow a_x^2 + a_y^2 = 16$.....(1)

$|2a - b| = |2a_x i + 2a_y j - (b_x i + b_y j)$

$= |(2a_x - b_x)i + (2a_y - b_y)j|$

$= \sqrt{(2a_x - b_x)^2 + (2a_y - b_y)^2}$.....(2)

$|a + b| = |a_x i + a_y j + b_x i + b_y j|$

$= |(a_x + b_x)i + (a_y + b_y)j|$

$= \sqrt{(a_x + b_x)^2 + (a_y + b_y)^2}$.....(3)

$|2a - b| = |a + b| \Rightarrow$ equating (2) and (3)

$\Rightarrow \sqrt{(2a_x - b_x)^2 + (2a_y - b_y)^2} = \sqrt{(a_x + b_x)^2 + (a_y + b_y)^2}$

$\Rightarrow 4a_x^2 - 4a_x b_x + b_x^2 + 4a_y^2 - 4a_y b_y + b_y^2 = a_x^2 + 2a_x b_x + b_x^2 + a_y^2 + 2a_y b_y + b_y^2$

$\Rightarrow 3a_x^2 + 3a_x = 6a_x^2 b_x + a_y b_y$

$\Rightarrow 3(a_x^2 + 3a_y^2 = 6(a_x b_x + a_y b_y)$

$\Rightarrow 3 \times (16) = 6(a.b)$

$\Rightarrow 48 = 6(a.b)$

$\Rightarrow a.b = \frac{48}{6} = 8$

63. C: Given: l: y = 2x
Circle with center (-2, 6)
Required: point of tangency

Slope of l = 2

$\Rightarrow$ slope of OP is –½

Thus, $\frac{b-6}{a+2} = \frac{-1}{2}$

$\Rightarrow 2b - 12 = -a - 2$

$\Rightarrow 2b = -a + 10$.....(1)

Since P(a, b) lies on l, b = 2a .....(2)

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Substituting b on eq(1), we get

$2(2a) = -a + 10$

$4a = -a + 10$

$5a = 10$

$a = 2$

Substituting a = 2 on eq (2), b = 4

So, P(2, 4)

64. A: Given: $\sin^2 x + 1 = \cos^2 x + ½$

Required: solution set on $[0, 2\pi]$

$\sin^2 x + 1 = \cos^2 x + ½$

$1 - ½ = \cos^2 x - \sin^2 x$

$½ = \cos 2x$

$$\Rightarrow 2x = \frac{\pi}{3} + 2\pi n,\ n \in N,$$

$$\Rightarrow 2x = \frac{\pi}{3} + 2\pi n,\ 2x = \frac{5\pi}{3},\ 2x = \frac{7\pi}{3},\ 2x = \frac{11}{3}\pi$$

$$x = \frac{\pi}{6} \qquad x = \frac{5\pi}{6} \qquad x = \frac{7\pi}{3} \qquad x = \frac{11}{3}\pi$$

solution set: $\left\{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{3}\right\}$

65. D: Required: $\sin^4 A - \cos^4 A = ?$

$\sin^4 A - \cos^4 A = (\sin^2 A)^2 - (\cos^2 A)^2$

$= (\sin^2 A - \cos^2 A)(\sin^2 A + \cos^2 A)$

$= \sin^2 A - \cos^2 A$

$= -(\cos^2 A - \sin^2 A)$

$= -\cos 2A$

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1. B Required: equivalence of $\int_0^{\pi/3} \frac{\tan x}{\cos x} dx$

$$\int_0^{\pi/3} \frac{\tan x}{\cos x} dx = \int_0^{\pi/3} \frac{\sin x}{\cos^2 x} dx$$

Let $\cos x = u \Rightarrow du = -\sin x\, dx$

$$= \int_0^{\pi/3} \frac{-du}{u^2} = -\int_0^{\pi/3} u^{-2} du = -\frac{u^{-2+1}}{-2+1}\Big|_0^{\pi/3}$$

$$= u\Big|_0^{\pi/3} = \frac{1}{\cos x}\Big|_0^{\pi/3} = \frac{1}{\cos \pi/3} - \frac{1}{\cos 0}$$

$$= \frac{1}{1/2} - \frac{1}{1}$$

$$= 2 - 1$$

$$= 1$$

2. B. **Given:** $f(x) = x^3$

X axis (y = 0)

Required: Area bounded by f and y = 0 from -1 to 2

$$A(x) = \int_0^2 x^3 dx - \int_{-1}^0 x^3 dx$$

$$= \frac{x^4}{4}\Big|_0^2 - \frac{x^4}{4}\Big|_{-1}^0$$

$$= \left(\frac{16}{4} - \frac{0}{4}\right) - \left(\frac{0}{4} - \frac{1}{4}\right) = \frac{16}{4} + \frac{1}{4} = \frac{17}{4}$$

3. C. **Given:** $f(x) = 4 + 5x^2 - (x^2 + 1)^3$

Required: true statement about P(x)

$P(x) = 4 + 5x^2 - (x^2 + 1)^3 = 4 + 5x^2 - (x^6 + 3x^4 - 13x^2 + 1)$

$= 4 + 5x^2 - x^6 - 3x^4 - 3x^2 - 1$

$= -x^6 - 3x^4 + 2x^2 + 3$

- degree of p(x)n is 6
- leading coefficient is – 1
- constant term is 3
- coefficient of x^2 is 2

So, only C is true

4. B. **Given** $f(x) = x^4 - kx^2 + cx$

$(2,0)$ is the x – intercept of $f(x)$

When $f(x)$ is divided by $x - 1$, remainder $= 1$

Required: Value of K

$(2,0)$ is x – intercept $\Rightarrow 2^4 - k(2)^2 + c(2) = 0$

$\Rightarrow 16 - 4k + 2c = 0$

$\Rightarrow -4k + 2c = -16 \ldots (1)$

Remainder $= 1 \Rightarrow f(1) = 1$

$\Rightarrow 14 - k(1)^2 + C(1) = 1$

$\Rightarrow 1 - k + C = 0$

$\Rightarrow k + C = 0$

$\Rightarrow C = k \ldots\ldots (2)$

Placing (2) into (1), we get: $-4k + 2k = -16$

$\Rightarrow -2k = -16$

$\Rightarrow k = 8$

5. B. **Given** $f(x) = x^4 - 7x^2 + 10$

Required: Interval that doesn't contain zero of $f(x)$

$x^4 - 7x^2 + 10 = 0 \Rightarrow (x^2)^2 - 7x^2 + 10 = 0$

Let $x^2 = a$

$\Rightarrow a^2 - 7a + 10 = 0$

$\Rightarrow (a - 2)(a - 5) = 0$

$\Rightarrow a = 2$ or $a = 5$

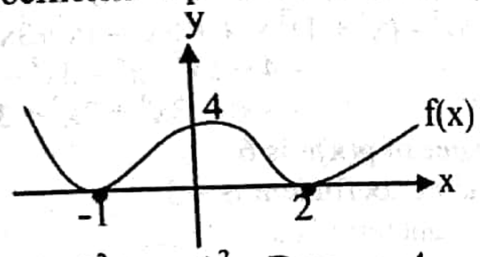
When $a = 2$, $x^2 = 2 \Rightarrow x = \pm\sqrt{2}$

When $a = 5$, $x^2 = 5 \Rightarrow x = \pm\sqrt{5}$

$(-1,0)$ doesn't contain any of these four zeros.

6. D. **Required:** function whose graph lies both above & below x-axis

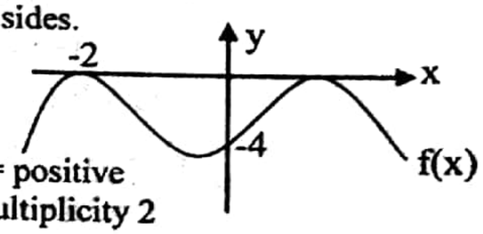
A. $f(x) = (x + 1)^2(2 - x)^2$ has x – intercepts at $x = -1$ and $x =$ both with multiplicity 2. its y – intercept is $(0, 4)$ and the leading coefficient is positive. Hence, it turns up on both sides,



B. $f(x) = -(x + 2)^2 (x - 1)^2$ – Degree $= 4$

- leading coefficient = negative
- zeros at $x = -2$, $x = 1$ with multiplicity 2.
- Y intercept $(0, -4)$

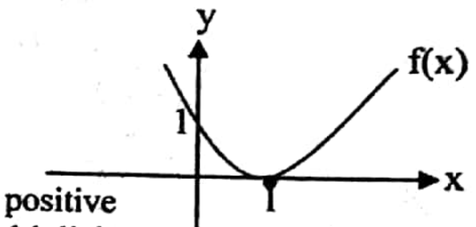
Thus, $f(x)$ turns down on both sides.



C. $f(x) = (x^2 + 1)(x - 1)^2$

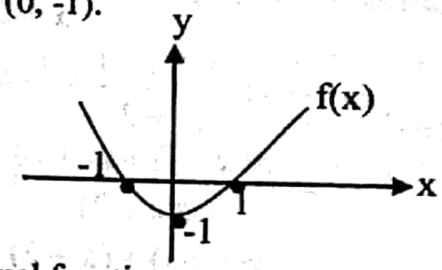
- Degree 4
- Leading coefficient = positive
- Cero at $x = 1$ with multiplicity 2
- Y intercept $(0, 1)$.

Thus, $f(x)$ turns at $(1, 0)$ up on both side



D. $F(x) = (x + 1)^2 (x^2 - 1)$

- Degree 4
- Leading coefficient = positive
- Zero at $x = -1$ with multiplicity 3 and at $x = 1$
- Y intercept at $(0, -1)$.



7. A. **Required:** identifying a rational function

$f(x) = \dfrac{\sqrt{2}x^3 - x + \log 2}{x^4 + x^2 + \pi}$ is rational function $\left(\dfrac{polynomial}{polynomial}\right)$

The inclusive of $\sqrt{x^2 + 1}$, in B; 2^x in C) and sinx in D) can't make the functions rational.

8. C. **Given:** $P(x) = x - 4$, $g(x) = x^2 + 8$, $r(x) = x^2 - 2x - 8$

Required: The most simplified form of $\dfrac{x}{p(x)} - \dfrac{q(x)}{r(x)}$

$$\frac{x}{p(x)} - \frac{q(x)}{r(x)} = \frac{x}{x-4} - \frac{x^2+8}{x^2-2x-8} \quad x \neq -2,4$$

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$$= \frac{4}{x-4} - \frac{x^2+8}{(x+2)(x-4)}$$
$$= \frac{x(x+2)-(x^2+8)}{(x+2)(x-4)}$$
$$= \frac{x^2+2x-x^2-8}{(x+2)(x-4)}$$
$$= \frac{2x-8}{(x+2)(x-4)}$$
$$= \frac{2(x-4)}{(x+2)(x-4)}$$
$$= \frac{2}{x+2}$$

9. A. **Required** Solution set of $1-\frac{1}{x+2} \le \frac{-1}{x}$

$$1-\frac{1}{x+2} \le \frac{-1}{x} \Rightarrow \frac{x+2-1}{x+2}+\frac{1}{x} \le 0$$
$$\Rightarrow \frac{x(x+1)+(x+2)}{2(x+2)} \le 0$$
$$\Rightarrow \frac{x(x+1)+(x+2)}{2(x+2)} \le 0$$
$$\Rightarrow \frac{x^2+2x+2}{x(x+2)} \le 0$$

Science $x^2 + 2x + 2 > 0$ for any x (check the discriminate) we f
solution for $x(x+2) < 0$
$\Rightarrow x > 0$ and $x + 2 < 0$ or $x < 0$ and $x + 2 > 0$
$\Rightarrow x > 0$ & $x < -2$ or $x < 0$ and $x > -2$
$\Rightarrow \varnothing$ or $(-2, 0)$

10. C **Given:** at 1000 km/hr, time = $(t - ½)$ hrs
at 900 km/hr, time = $(t - ½)$ hrs
Required: distance covered
The time taken to cover a distance d at 1000km/hr is

$$t-\tfrac{1}{2} = \frac{d}{1000} \quad \ldots\ldots (1)$$

The time taken to cover a distance d at 900 km/hr is

$$t+\tfrac{1}{3} = \frac{d}{900} \quad \ldots\ldots (2)$$

Solving (1) and (2) simultaneously, we get

$$\frac{-1}{2}-\frac{1}{3} = \frac{d}{1000}-\frac{d}{900}$$
$$\Rightarrow \frac{-5}{6} = \frac{-d}{9000}$$
$$\Rightarrow d = \frac{5}{6} \times 9000 = 7{,}500\text{km}$$

11. B **Required:** The function that could represent the graph.
1. The graph has a hole at $x = 1 \Rightarrow$ There is a factor $(x - 1)$ on both the numerator and the dominator of f(x).
2. $x = -1$ is a vertical asymptote $\Rightarrow x + 1$ is a factor at the denominator
3. $y = 2$ is a horizontal asymptote $\Rightarrow$ quotients of the leading coefficients of the numerator & denominator = 2.
4. The x – intercept is at $x = 0 \Rightarrow$ There is a factor x on the numerator

The properties mentioned (1) to (4) above suggest that f(x) shall be $\frac{2x(x-1)}{x^2-1}$

12. A. **Required:** Simplified form of $e^{\frac{1-\ell n\left(\frac{1}{2}\right)}{2}}$

$$e^{\frac{1-\ell n\,1/2}{2}} = e^{\frac{1}{2}(\ell ne - \ell n\,1/2)}$$
$$= e^{\frac{1}{2}(\ell n\, e/1/2)}$$
$$= e^{\frac{1}{2}(\ell n 2e)}$$
$$= (2e)^{\frac{1}{2}}$$
$$= \sqrt{2e}$$

13. B. **Required:** Function that better represent the graph.
- Vertical asymptote $x = -1 \Rightarrow$ Domain of f(x) is $\{x : x > 1\}$

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- As x gets larger f(x) is smaller and is below x – axis

⇒ The coefficient onust be negative.

Those properties suit $f(x) = -\log_5(x+1)$

14. No Answer **Required**: solution set of $9^{(1-2\log 3x)} = 4$

$9^{(1-2\log 3x)} = 4 \Rightarrow 9^{\left(1-2\log_{3^2}^{x^2}\right)} = 4$

$\Rightarrow [9]^{\log_3^{(y_{x2})}} = 4$

$\Rightarrow [3^2]^{\log_3^{(y_{x2})}} = 4$

$\Rightarrow 3^{2\log_3^{(y_{x2})}} = 4$

$\Rightarrow 3^{\log_3^{(y_{x2})^2}} = 4$

$\Rightarrow \left(\frac{3}{x^2}\right)^2 = 4$

$\Rightarrow \frac{3}{x^2} = \pm 2$

$\Rightarrow 3 = 2x^2$ (only the value is taken

$\Rightarrow x^2 = \frac{3}{2}$ as $x > 0$ is the domain of $\log_3^x$)

$\Rightarrow x = \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2} \Rightarrow solution\ set = \left\{\frac{\sqrt{6}}{2}\right\}$

15. A **Given:** $p(t) = a\, e^{kt}$

$a = 100$

$p(2) = 200$

Required: P(10)= ?

$P(2) = 200 \Rightarrow 200 = 100\,(e)^{2k}$

$\Rightarrow 2 = e^{2k}$

$\Rightarrow \ln 2 = 2k$

$\Rightarrow K = \frac{1}{2}\ln 2$

$P(10) = 100\, e^{10k}$

$= 100 \times e^{10\left(\frac{1}{2}\ln 2\right)}$

$= 100 \times e^{5\ln 2}$

$= 100 \times e^{\ln 25}$

$= 100 \times e^{\ln 32}$

$= 100 \times 32$

$= 3{,}200$

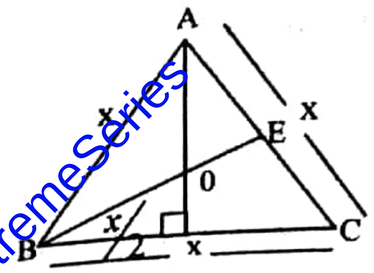
16. C **Given:** ΔABC – equilateral

AD and BE are medians

$OD = \sqrt{3}$

Required: area of ΔABC

AD & BE are medians & ΔABC is equilateral



$\Rightarrow$ AD ⊥ BC (Ad is altitude)

$\Rightarrow AO = \frac{2}{3}$ AD and OD $\frac{1}{3}$ AD

$\Rightarrow \sqrt{3} = \frac{1}{3}$ AD

$\Rightarrow AD = 3\sqrt{3}$

Using, Pythagorean relations on ΔABC, $x^2 - \left(\frac{x}{2}\right)^2 = (AD)^2$

$\Rightarrow \frac{3}{4}x^2 = (\sqrt{3})^2$

$\Rightarrow x^2 = \frac{4\times 27}{3}$

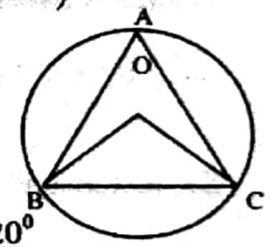
$\Rightarrow x^2 = 36$

$\Rightarrow x = 6$

$\therefore$ area of $\Delta = \frac{1}{2} \times 6 \times 3\sqrt{3} = 9\sqrt{3}$

17. D. **Given:** circle 0 with $m(\hat{A}) + m(B\hat{O}C) = 120^0$ $m(D\hat{B}A) = 30^0$

Required: m (OCA)



$m(\hat{A}) = \frac{1}{2}(\overline{BC})$ and $m(B\hat{O}C) = m(\overline{BC})$

$\Rightarrow 2m(\hat{A}) = m(B\hat{O}C)$

$m(\hat{A}) + m(B\hat{O}C) = 120^0 \Rightarrow m(\hat{A}) + 2m(\hat{A}) = 120^0$

$\Rightarrow 3m(\hat{A}) = 120^0$

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$\Rightarrow m(\hat{A}) = 40^0$

Thus, $m(B\hat{O}C) = 80^0$

Since ΔOBC is isosceles, $m(B\hat{O}C) = m(O\hat{C}B) = 50^0$

Since $m(O\hat{B}A) = 30^0$, $m(A\hat{B}C) = 80^0$

Which means $m(A\hat{C}B) = 60^0$ ($60^0+80^0+40^0=180^0$)

So, $m(O\hat{C}A) = 60^0 - 50^0 = 10^0$

18. **A Given:** A(1,0), B (-3,0), c(1, 2)

Required: Equation of l through A and $\perp$ to $\overline{BC}$

Slope of BC $= \frac{2-0}{1-(-3)} = \frac{2}{4} = 1/2$

$\Rightarrow$ slope of l = -2 (negative reciprocal)

So equation of l is $\frac{y-0}{x-1} = -2$

$\Rightarrow y = -2x + 2$

$\Rightarrow \underline{y + 2x = 2}$

19. **B Given:** - l contains (1, 5)
- inclination of l is 135^0

Required: distance from l to (-1, 1)

Inclination of l is $135^0 \Rightarrow$ slope of l = tan 135^0
$= -1.$

Thus, equation of l is $\frac{y-5}{x-1} = -1$

$\Rightarrow y - 5 = -x + 1$

$\Rightarrow x + y - 6 = 0$

$a = 1, b = 1, c = -6$

distance between a point (x_0, y_0) and line

$ax + by + c = 0$ is given by $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$

So, distance is $\frac{|a(-1) + 1(1) - 6|}{\sqrt{1^2 + 1^2}} = \frac{6}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{6\sqrt{2}}{2} = 3\sqrt{2}$

20. **C Required:** Equation of circle with center at (2, 1) and tangent to the y – axis (x = 0)

r is the distance b/n x = 0 and (2, -1).

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So, $r = \frac{|1(2) + 0(-1) + 0|}{\sqrt{1^2 + 0^2}} = \frac{2}{1} = 2$

The equation of the circle is then,

$(x - 2)^2 + (y + 1)^2 = 2^2$

21. **B Given:** - a parabolic arch of height 3m and base width 3m.
- origin at the vertex

Required: equation of the parabola

$Y = 4px^2$

Since $(3/2, -3)$ is on the parabola,

$-3 = 4p\,(3/2)^2$

$-3 = 4p\,(9/4)^2$

$\Rightarrow \frac{-3}{9} = p \qquad \Rightarrow P = -1/3$

$\therefore y = 4px^2 \Rightarrow y = 4(-1/3)x^2$

$\Rightarrow 3y = -4x^2$

$\Rightarrow 3y + 4x^2 = 0$

22. **A Given:** An ellipse with center (1, 4)
- vertices at (10, 4) & (1, 2)

Required: equation of the ellipse

Center (1, 4) and one vertex (10, 4) $\Rightarrow$ a=10–1= 9

Center (1,4) and one vertex (1,2) $\Rightarrow$ b = 4 – 2 = 2

$\therefore$ Equation is $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

$\Rightarrow \frac{(x-1)^2}{9^2} + \frac{(y-4)^2}{2^2} = 1$

$\Rightarrow \frac{(x-1)^2}{81} + \frac{(y-4)^2}{4} = 1$

Multiplying both sides by 234, we get:

$4(x - 1)^2 + 81\,(y - 4)^2 = 324$

23. B **Required:** Length of the minor axis of $x^2 + 9y^2 - 2x = 0$

$x^2 + 9y^2 - 2x = 0 \Rightarrow x^2 - 2x + 1 + 9y^2 = 0 + 1$

$$\Rightarrow (x-1)^2 + 9y^2 = 1$$
$$\Rightarrow \frac{(x-1)^2}{1^2} + \frac{y^2}{\left(\frac{1}{3}\right)^2} = 1$$

This is an ellipse with center (1, 0) and length of major axis $2\times1 = 2$ and length of minor axis $= 2\left(\frac{1}{3}\right) = \frac{2}{3}$

24. D **Required:** correct statement about $x^2 - 4y^2 = -1$

$$x^2 - 4y^2 = -1 \Rightarrow 4y^2 - x^2 = 1$$
$$\Rightarrow \frac{y^2}{\left(\frac{1}{2}\right)^2} - \frac{x^2}{1^2} = 1$$

$a = \frac{1}{2}$, $b = \frac{1}{2}$

$C^2 = a^2 + b^2$
$= \frac{1}{4} + 1$
$= \frac{5}{4}$
$C = \pm \frac{\sqrt{5}}{2}$

Hyperbola with foci arc at $(0, -\frac{\sqrt{5}}{2})$ and $(0, \frac{\sqrt{5}}{2})$

25. D **Given:** sphere $(x-1)^2 + (y-2)^2 + (z+1)^2 = 6$
Required: point of intersection of the sphere and y – axis
Let the intersection be at (0, a, 0)
$\Rightarrow (0-1)^2 + (a-2)^2 + (0+1)^2 = 6$
$\Rightarrow 1 + a^2 - 4a + 4 + 1 = 6$
$\Rightarrow a^2 - 4a = 0$
$\Rightarrow a(a-4) = 0$
$\Rightarrow a = 0$ or $a = 4$
Thus the intersection point is (0, 0, 0) or (0, 4, 0)

26. C **Required:** Solution set of $x^2 - 4x + 5 = 0$ in the set of complex numbers.

$$x = \frac{(-4) \pm \sqrt{(-4)^2 - 4(1)(5)}}{2(1)}$$

$$x = \frac{4 \pm \sqrt{-4}}{2}$$
$$x_1 = \frac{4+2i}{2} \text{ and } x_2 = \frac{4+2i}{2}$$
$x_1 = 2 + i$ and $x_2 = 2 - i$
Solution set = {2 + i, 2 – i}

27. A **Given:** a complex no Z such that

$$\left|\frac{3Z}{-4+3i}\right| = (2-i)\left(\overline{2-i}\right)$$

Required: Modulus of Z or |Z|

$$\left|\frac{3Z}{-4+3i}\right| = (2-i)\left(\overline{2-i}\right) \Rightarrow \frac{3|Z|}{|-4+3i|} = 2^2 + 1^2$$
$$\Rightarrow \frac{3|Z|}{\sqrt{(-4)^2 + 3^2}} = 5$$
$$\Rightarrow \frac{3|Z|}{5} = 5$$
$$\Rightarrow |Z| = \frac{25}{3}$$

28. B **Required:** Solution set of $Z^3 - 8 = 0$ in the complex n°s.

$$Z^3 - 8 = 0 \Rightarrow (Z-2)(Z^2 + 2Z + 4) = 0$$
$$\Rightarrow Z = 2 \text{ or } Z = \frac{-2 \pm \sqrt{4 - 4(4)(1)}}{2}$$
$$Z = \frac{-2 \pm \sqrt{-12}}{2}$$
$$Z = \frac{-2 \pm 2\sqrt{3}\,i}{2}$$
$\Rightarrow Z_1 = 2, Z_2 = -1 + \sqrt{3}\,i, Z_3 = -1 - \sqrt{3}\,i$
Solution set = $\{2, -1 + \sqrt{3}\,i, -1 - \sqrt{3}\,i\}$
From the choices given, the set given is B is the same as the above solution set because:

$$2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) = 2\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = -1 + i\sqrt{3} \text{ and}$$

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$$2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)=2\left(-\tfrac{1}{2}+i\left(\frac{\sqrt{3}}{2}\right)\right)=-1+i\sqrt{3}$$

29. **A** **Given:** the term of a sequence $a_k=\dfrac{(-1)^k(2-k)}{2k-1}$

Required: a_{k+1} when k is even

If k is even, $k=2n$, $n\in N$

$$a_{k+1}=a_{2n+1}=\frac{(-1)^{2n+1}(2-(2n+1))}{2(2n+1)-1}$$

$$=\frac{(-1)^{2n1}\cdot(-1)(2-2n-1)}{4n+2-1}$$

$$=\frac{(-1)(1-2n)}{4n+1}$$

$$a_{2n+1}=\frac{-1+2n}{4n+1}$$

Replacing n by $k/2$, we get: $a_{k+1}=\dfrac{-1+2(k/2)}{4(k/2)+1}$

$$=\frac{-1+k}{2k+1}=\frac{k-1}{2k+1}$$

30. **B** **Given:** Geometric progression $\{G_n\}_{n=1}^{\infty}$

With $G_{10}=192$, $G_6=12$

Required: G_2

Note: $\boxed{G_n=G_1r^{n-1}}$

$G_{10}=G_1r^9\Rightarrow 1.92=G_1r^9\ldots(1)$

$G_6=G_1r^5\Rightarrow 12=G_1r^5\ldots(2)$

Dividing (1) by (2), we get $=\dfrac{192}{12}=\dfrac{r^9}{r^5}$

$\Rightarrow 16=r^4$

$\Rightarrow r=2$ or $r=-2$

From (2), $12=G_1r^5$

$\Rightarrow 12=G_1(2)^5$ or $12=G_1(-2)^5$

$\Rightarrow 12=G_1(32)$ or $12=-32G_1$

$\Rightarrow G_1=\dfrac{12}{32}$ or $G_1=\dfrac{-12}{32}$

$\Rightarrow G_1=\dfrac{3}{8}$ or $G_1=\dfrac{-3}{8}$

When $r=2$, $G_2=G_1r=\dfrac{3}{8}\times 2=3/4$

When $r=2$, $G_2=G_1r=\dfrac{-3}{8}\times -2=3/4$

31. **C** **Given:** Sequence $\{a_n\}$, $n\geq 1$ with $a_1=2$ and

$a_{n+1}=a_n+4$ for all $n\geq 1$

Required: sum $a_3+a_4+a_5+\ldots+a_{35}$

$a_{n+1}=a_n+4\Rightarrow a_2=a_1+4\Rightarrow a_2=2+4=6$

$a_3=a_2+4\Rightarrow a_3=6+4=10$

The sequence is arithmetic with $a_1=2$ & $d=4$

$$S_{35}=\frac{35}{2}(2(2)+(35-1)\times 4)=\frac{35}{2}[4+34\times 4]=\frac{34}{2}[140]=2450$$

The sum $a_3+a_4+a_5+\ldots+a_{35}=S_{35}-(a_1+a_2)$

Thus, $a_3+a_4+a_5+\ldots+a_{35}=2450-(2+6)$

$=2450-8$

$=2442$

32. **D** **Required:** 50mg the series $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^n}{(3^n)^2}$

When $n=1$, $\dfrac{(-1)^1}{(3^1)^2}=\dfrac{-1}{9}=a_1$

When $n=2$, $\dfrac{(-1)^2}{(3^2)^2}=\dfrac{1}{9^2}=a_2$

When $n=3$, $\dfrac{(-1)^3}{(3^3)^3}=\dfrac{-1}{9^3}=a_3$

Since $a_2/a_1=\dfrac{1/9^2}{-1/9}=-1/9$ and $a_3/a_2=\dfrac{-1/9^3}{1/9^2}=-1/9$

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The series is geometric with $G1 = -1/9$ and $r = -1/9$ which converges to

$$\frac{G_1}{1-r}$$

Thus $\sum_{n=1}^{\infty} \frac{(-1)^n}{(3^n)^2} = \frac{G_1}{1-r} = \frac{-1/9}{1-(-1/9)} = \frac{\frac{-1}{9}}{1+\frac{1}{9}} = \frac{\frac{-1}{9}}{\frac{10}{9}} = -1/10$

33. **D Given:** $p(x) = x^2 + x$ is positive
Required: equivalent statement to $-(\exists x)p(x)$
$\neg(\exists x)p(x) \Leftrightarrow \neg(\exists x)\,(x^2 + x > 0) \Leftrightarrow (\forall x)\,\neg(x^2 + x > 0)$
$\Leftrightarrow \forall x\,(x^2 + x \le 0)$

34. **C Given:** $(P \wedge Q) \Rightarrow (\neg S \vee R)$ is False
Required: True statement
$(P \wedge Q) \Rightarrow (\neg S \vee R)$ is False $\Rightarrow P \wedge Q$ is T and $\neg S \vee R$ is F that is P $\equiv$ T Q $\equiv$ T S $\equiv$ T and R $\equiv$ F
(because $\neg S \vee R \equiv S \Rightarrow R$)
A. $S \Rightarrow R \equiv T \Rightarrow F \equiv F$
B. $\neg P \vee \neg S \equiv F \vee F = F$
C. $(P \wedge \neg Q) \Leftrightarrow R \equiv F \Leftrightarrow F \equiv T$
D. $P \wedge R \equiv T \wedge F \equiv F$
So, only the statement in C) is true

35. **B Required:** Value for $\lim_{n\to\infty} \frac{2n^2 - n + 1}{6n^2 + 1}$

$$\lim_{n\to\infty} \frac{2n^2 - n + 1}{6n^2 + 1} = \lim_{n\to\infty} \frac{\frac{2n^2}{n^2} - \frac{n}{n^2} + \frac{1}{n^2}}{\frac{6n^2}{n^2} + \frac{1}{n^2}}$$

$$= \lim_{n\to\infty} \frac{-\frac{1}{n} + \frac{1}{n^2}}{6 + \frac{1}{n^2}}$$

$$= \frac{2}{6} \quad \left(as\ n \to \infty,\ 1/n\ \&\ 1/n^2 \to 0\right)$$

$$= 1/3$$

36. **C Required:** identifying the divergent sequence
A. $\left\{(\sqrt{2})^{-n}\right\}$ is a convergent geometric sequence with
$G_1 = \frac{1}{\sqrt{2}} = r$
B. $\left\{5^{1/n}\right\}$ is a convergent because sequence

37. **A Required:** true statement about $\left\{2 + \frac{(-1)^n}{n+1}\right\}_{n=1}^{\infty}$

$n = 1,\ 2 + \frac{(-1)^1}{1+1} = 2 - \frac{1}{2} = \frac{3}{2}$

$n = 2,\ 2 + \frac{(-1)^2}{2+1} = 2 + \frac{1}{3} = 7/3$

$n = 3,\ 2 + \frac{(-1)^3}{3+1} = 2 - \frac{1}{4} = 7/4$

$n = 4,\ 2 + \frac{(-1)^4}{4+1} = 2 + \frac{1}{5} = 11/5$

Thus shows that the terms of the sequence are call less than or equal to $7/3$. That is, lub of the sequence is $7/3$.

38. **C Required:** equivalence to $\lim_{x\to 1} \frac{x^3 + x - 2}{x^2 - x}$

$$\lim_{x\to 1} \frac{x^3 + x - 2}{x^2 - x} = \lim_{x\to 1} \frac{(x-1)(x^2 + x + 2)}{x(x-1)}$$

$$= \lim_{x\to 1} \frac{x^2 + x + 2}{x}$$

$$= \frac{1 + 1 + 2}{1} = 4$$

39. **B Required:** equivalence to $\lim_{x\to 0} \frac{|x| - xe^x}{x}$

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$$\lim_{x\to 0-}\frac{|x|-xe^x}{x}=\lim_{x\to 0-}\frac{-x-xe^x}{x}$$

$$=\lim_{x\to 0-}\frac{-\left(-1-e^x\right)}{x}$$

$$=\lim_{x\to 0-}\frac{-1-e^x}{1}$$

$$=-1-e^0$$

$$=-1-1=-2$$

40. **C Given**: $f(x)=\begin{cases}a-\cos x, & x\le 0\\ 3e^x-a, & x>0\end{cases}$

Required: value of a at which f is continuous

$$\lim_{x\to 0+}f(x)=\lim_{x\to 0+}3e^x-a=3e^0-a=3-a$$

$$\lim_{x\to 0+}f(x)=\lim_{x\to 0-}a-\cos x=a-\cos 0=a-1$$

$$\lim_{x\to 0+}f(x)=\lim_{x\to 0-}f(x)\Rightarrow 3-a=a-1$$

$$\Rightarrow 4=2a$$

$$\Rightarrow a=2$$

41. **C Given**: $f(x)=x^3$

Required: value for $\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}$

$$\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}=\lim_{h\to 0}\frac{(2+h)^3-2^3}{h}$$

$$=\lim_{h\to 0}\frac{8+12h+6h^2+h^3-8}{h}$$

$$=\lim_{h\to 0}\frac{h(12+6h+h^2)}{h}$$

$$=\lim_{h\to 0}12+6h+h^2=12+6(0)+0^2$$

$$=12$$

42. **A Given**: $f(x)=\begin{cases}x-\pi & x\le 0\\ 1-2\sin x, & x>0\end{cases}$

Required: tangent line to f(x) at $(\pi, f(\pi))$

The slope of the tangent line is $f'(\pi)$.

Since, $f'(x)=(1-2\sin x)'$ for $x>0=-2\cos x$

$f'(\pi)=-2\cos\pi=-2\times -1=x$

And; $f(\pi)=1-2\sin\pi=1-2\times 0=1$

Equation of tangent line is $y-f(\pi)=f'(\pi)(x-\pi)$

$y-1=2(x-\pi)$

$y=2x-2\pi+1$

43. **C Given**: a differentiable function f(x) graph of if = f'(x)

Required: true statement

- since $f'(x)\ge 0$ for $(-\infty, -1)$ and $(5, \infty)$, f(x) increases on $(-\infty, -1]\cup$
- Since $f'(x)\le 0$ for $(-1, 5)$ and f(x) decreases on $[-1, 5)$
- f'(x) changes its sign from –ve to +ve at x = 5, thus f(x) has min at x = 5.
- f(x) changes its sign from +ve to -5 at x = -1, f(x) has max at m = -1. so only C is true.

44. **B Given**: $h(x)=x+3^x$

$f(x)=(f\circ h)(x)$

$f'(4)=-1$

Required: f'(1)

Since $f(x)=(f\circ h)(x)$, $f'(x)=f'(h(x)).h'(x)$

But $h'(x)=(x+3^x)'=1+3^x\ln^3$

So, $f'(x)=f'(h(x)).(1+3^x\ln^3)$

$f'(1)=f'(h(1)).(1+3^1\ln^3)$

$=f'(4).(1+3\ln^3)$ $\quad (h(1)=1+3^1=4)$

$=-1(1+3\ln^3)$

$=-1-3\ln 3$

45. **D Given**: $f(x)=\ln(2+\sec x)$

Required: f'(x)

$$f(x)=\ln(2+\sec x)\Rightarrow f'(x)=\frac{1}{2+\sec x}\cdot(2+\sec x)'$$

$$\Rightarrow f'(x)=\frac{1}{2+\sec x}\cdot\left(\frac{\sin x}{\cos^2 x}\right)$$

$$=\frac{1}{2+\frac{1}{\cos x}}\cdot\left(\frac{\sin x}{\cos^2 x}\right)$$

$$=\frac{\cos x}{1+2\cos x}\cdot\frac{\sin x}{(\cos x)(\cos x)}$$

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$$= \frac{1}{1+2\cos x} \cdot \tan x$$
$$= \frac{\tan x}{1+2\cos x}$$

46. **D Given:** $f(x) = \begin{cases} x-1, & x<1 \\ x^2-x, & x \geq 1 \end{cases}$

Required: set of all critical numbers

$$\lim_{x\to 1+} \frac{f(x)-f(1)}{x-1} = \lim_{x\to 1+} \frac{x^2-x-0}{x-1} = \lim_{x\to 1+} \frac{x(x-1)}{x-1}$$
$$= \lim_{x\to 1+} x = 1$$

$$\lim_{x\to 1-} \frac{f(x)-f(1)}{x-1} = \lim_{x\to 1} \frac{x-1}{x-1} = \lim_{x\to 1-} 1 = 1$$

$$\lim_{x\to 1-} f(x) = \lim_{x\to 1-} f(x) = 1$$

Thus, f(x) has no critical numbers

47. **A Given:** rate as $r(x)=kx\,(a-x), 0 \leq x \leq a, a>0, k>0$

Required: values of x where the rate increases

Note: when $r'(x) \geq 0$ for each $x \in (z, b)$ then r(x) is increasing on [a, b]

Thus, $r'(x) = [kx\,(x-x)]$
$= k(a-x) + (kx)(-1)$
$= ka - kx - kx$
$= ka - 2kx$

$r'(x) \geq 0 \Rightarrow ka - 2kx \geq 0$
$\Rightarrow ka \geq 2kx$
$\Rightarrow a/2 \geq x$

Therefore, r(x) is increasing on $0 \leq x \leq a/2$

48. **C Given:** $f(x) = x3 - 12x + 1$ on [0, 3]

Required: Range of f(x).

$f'(x) = 0 \Rightarrow 3x^2 - 12 = 0 \Rightarrow 3x^2 = 12 \Rightarrow x = \pm 2$

But, only $2 \in [0, 3]$ is the critical no.

So, $f(0) = 0^3 - 12(0) + 1 = 1$
$f(2) = 2^3 - 12(2) + 1 = -15$
$f(3) = 2^3 - 12(3) + 1 = -18$

1 is the local max while – 15 is the local min.

So, the range of f(x) is [-15, 1]

49. **B Given:** F(x) and G(x) as ant derivatives of f(x) in [a, b].

Required: a necessarily true statement

Note: If F(x) is an antiderivative of f(x) then
F(x)+C is also an antiderivative for some C.

Thus F(x) and G(x) are not necessarily the same.

But, it can be said that G(x) = F(x) + C or F(x) = G(x)+C
$\Rightarrow$ G(x) – F(x) = C or F(x) – G(x) = C

Therefore B) is a true statement.

50. **B Required:** value for $\int \frac{(e^x+1)^2}{e^x} dx$

Let $\frac{1}{e^x} = u \Rightarrow du = -\frac{1}{e^x}$

$\Rightarrow -du = \frac{dx}{e^x}$ and $e^x = \frac{1}{u}$

Replacing this in the integral given, we get

$$\int \frac{(e^x+1)^2 dx}{e^x} = \int \left(\frac{1}{u}+1\right)^2 (-du) = -\int \left(\frac{1}{u}+1\right)^2 du$$
$$= -\int \left(\frac{1}{u^2} + \frac{2}{u} + 1\right) du$$
$$= -\left(\int \frac{1}{u^2} du + 2\int \frac{1}{u} du + \int du\right)$$
$$= -\left(\frac{-1}{u} + 2\ln u + u\right) + C$$
$$= \frac{1}{u} - 2\ln u + u + C$$

Replacing u by $\frac{1}{e^x}$, we get

$$\frac{1}{1/e^x} - 2\ln - \frac{1}{e^x} + C$$
$= e^x - 2(-x) - e^{-x} + C$
$= e^x - e^{-x} + 2x + C$

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51. **D Given**: Equation of line l: $(x - 1\ y + 3).(2, 1) = 0$
Required: Equation of circle whose center is (-2, 1) and tangent to l

$l = (x - 1, y + 3).(2, -1) = 0 \Rightarrow 2x - 2 + -y - 3 = 0$
$\Rightarrow 2x - y - 5 = 0$

Distance between (-2, 1) and l is radius of the circle

$$r = \frac{|2(-2) + (-1)(1) - 5|}{\sqrt{2^2 + (-1)^2}} = \frac{|-10|}{\sqrt{5}} = \frac{10}{\sqrt{5}} = 2\sqrt{5}$$

Thus, equation of circle is $(x + 2)^2 + (y - 1)^2 = (2\sqrt{5})^2$
$\Rightarrow (x + 2)^2 + (y - 1)^2 = 20$

52. **A Given:** An ellipse $\frac{1}{4}(x - 1)^2 + (y + 2)^2 = 1$

Required: Image of the ellipse after $R_{90}{}^0$
Followed by reflection in the x – axis

Center of ellipse is C(1, -2) and a = 2, b = 1
After 90^0 anticlockwise rotation C' (2,1)and a=1,b=2
Reflection through the x – axis results C"(2, -1)and
a = 1, b = 1

Thus, equation of image ellipse is $\frac{(x-2)^2}{1^2} + \frac{(y+1)^2}{2^2} = 1$

$\Rightarrow 4(x - 2)^2 + (y + 1)^2 = 4$

53. **B Given:** $\theta = \arccos\left(\frac{1}{2}\right)$

Required: value of $\tan\theta$

$\theta = \arccos\left(\frac{1}{2}\right) \Rightarrow \cos\theta\ \frac{1}{2}$

Since $\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - \frac{1}{4}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$

So, $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$

54. **C Given:** $x = \arcsin\left(\frac{3}{4}\right)$

Required: value of cos 2x.

$x = \arcsin\left(\frac{3}{4}\right) \Rightarrow \sin x = \frac{3}{4}$

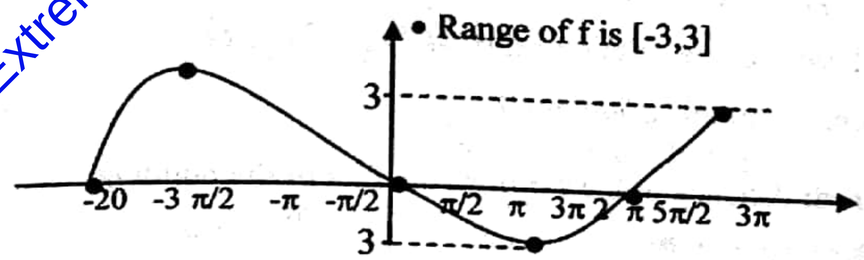
Since $\cos 2x = \cos^2 x - \sin^2 x - \sin^2 x = 1 - 2\sin^2 x$,

$$\cos 2x = 1 - 2\left(\frac{3}{4}\right)^2$$
$$= 1 - 2 \times \frac{9}{16}$$
$$= 1 - \frac{9}{8}$$
$$= \frac{-1}{8}$$

55. **D. Given:** $f(x) = -3\sin\left(\frac{1}{2}x\right)$

Required: true statement about f.

$f(x) = -3\sin\left(\frac{1}{2}x\right) \Rightarrow$ • amplitude $= |-3| = 3$



The sketch of f(x) for $[-2\pi, 3\pi]$ shows f is increasing on $[\pi, 3\pi]$

56. **A Required:** equivalent to $\tan\left(x - \frac{\pi}{4}\right)$, $\cos x \neq 0$

$$\tan\left(x - \frac{\pi}{4}\right) = \frac{\tan x - \tan\frac{\pi}{4}}{1 + \tan x \tan\frac{\pi}{4}}$$
$$= \frac{\tan x - 1}{1 + \tan x}$$
$$= \frac{\frac{\sin x}{\cos x} - 1}{1 + \frac{\sin x}{\cos x}}$$

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$$= \frac{\frac{\sin x - \cos x}{\cos x}}{\frac{\cos x + \sin x}{\cos x}}$$

$$= \frac{\sin x - \cos x}{\sin x + \cos x}$$

57. **A Given:** $\cos^2 x + \sin x \cos x = 1$

Required: Solution set on $[0, 2\pi]$

$\cos^2 x + \sin x \cos x = 1 \Rightarrow \sin x \cos x = 1 - \cos^2 x$

$\Rightarrow \sin x \cos x = \sin^2 x$

For $\sin x \neq 0$, $\cos x = \sin x$

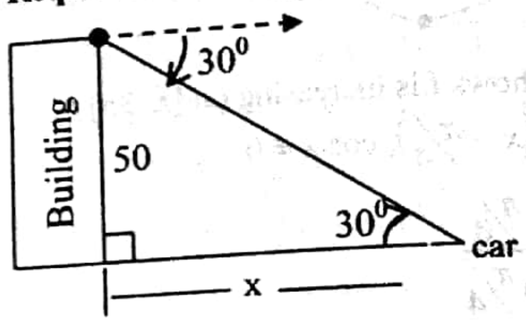
$\Rightarrow x = \pi/4$ *or* $x = 5\pi/4$ on $[0,2n)$

For $\sin x = 0$, $x = 0$ or $x = \pi$ on $[0, 2\pi)$

Therefore, the solution set is $\{0, \pi/4, 5\pi/4, \pi\}$

58. **D Given:** - a building 50m high
- angle of depression 30^0

Required: distance of the car from the foot of the building.



$$\frac{50}{x} = \tan 30^0$$

$$\frac{50}{x} = \frac{1}{\sqrt{3}}$$

$\Rightarrow x = 50\sqrt{3}$ ms

59. **B Required:** simplified from of $\frac{\cos A}{1+\sin A} + \frac{\cos A}{1-\sin A}$

$$\frac{\cos A}{1+\sin A} + \frac{\cos A}{1-\sin A} = \frac{\cos A(1-\sin A) + \cos A(1+\sin A)}{(1+\sin A)(1-\sin A)}$$

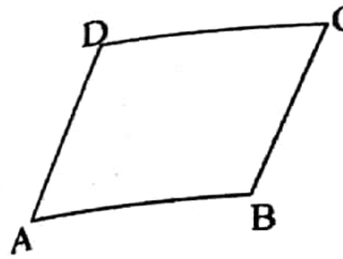
$$= \frac{\cos A - \cos A \sin A + \cos A + \cos A \sin A}{1-\sin A}$$

$$= \frac{2\cos A}{\cos^2 A} = \frac{2\cos A}{(\cos A)(\cos A)}$$

$$= \frac{2}{\cos A} = 2 \sec A$$

60. **D Given:** Parallelogram ABCD

Required: A wrong statement



A. $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} \Rightarrow \overrightarrow{AB} = \overrightarrow{AC} - \overrightarrow{BC}$ true

B. $\overrightarrow{BA} + \overrightarrow{AD} = \overrightarrow{BD}$

$\overrightarrow{BA} + \overrightarrow{AD} = \overrightarrow{BD}$

$-\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{BD}$

$\Rightarrow \overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{BD}$

C. $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$ *and* $\overrightarrow{BD} = \overrightarrow{BA} + \overrightarrow{AD}$

$\Rightarrow \overrightarrow{AC} = \overrightarrow{BD} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{BA} + \overrightarrow{AD}$

$= \overrightarrow{BC} + \overrightarrow{AD}$

$= \overrightarrow{BC} + \overrightarrow{BC} = 2\overrightarrow{BC}$ true

D. Sin $\overrightarrow{BA} + \overrightarrow{AD} = \overrightarrow{BD}$

$\Rightarrow \overrightarrow{AD} = \overrightarrow{BD} - \overrightarrow{BA}$

$= \overrightarrow{BD} + \overrightarrow{AB}$

$= \overrightarrow{BD} - \overrightarrow{CD}$ (not +)

61. **D Given:** A = (0, 1), B = (2, 7), C = (a, b)

$\overrightarrow{AC} = \overrightarrow{AB} + 2\overrightarrow{CA}$

Required: value of (a, b)

$2\overrightarrow{CA} = 2(0 - a, 1 - b) = (-2a, 2-2b)$

$\overrightarrow{AB} = (2 - 0, 7 - 1) = (2, 6)$

$\overrightarrow{AC} = (a - 0, b - 1) = (a, b - 1)$

$\overrightarrow{AC} = \overrightarrow{AB} + 2\overrightarrow{CA} \Rightarrow (a, b - 1) = (2, 6) + (-2a, 2 - 2b)$

$\Rightarrow (a, b - 1) = (2 - 2a, 8 - 2b)$

$\Rightarrow a = 2 - a$ and $b - 1 - 2b$

$\Rightarrow 3a = 2$ and $3b = 9$

$\Rightarrow a = 2/3$ and $b = 3$

Therefore, $(a, b) = (2/3, 3)$

62. **C Given:** $\vec{u} = \vec{2i} - \vec{j}$, $\|\vec{V}\| = 5$, $\vec{u} \perp \vec{v}$

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Required: $\|\vec{u}+3\vec{v}\|$

Let $\vec{v} = a\vec{i} + b\vec{j}$

$\|\vec{v}\| = S \Rightarrow \sqrt{a^2+b^2} = 5 \ldots . (1)$

Sin $\vec{u} \perp \vec{v}$, $\cos 90^0 = \dfrac{a(2)+b(-1)}{\|\vec{u}\|\,\|\vec{v}\|}$

$$\Rightarrow 0 = \frac{2a-b}{\sqrt{5}(5)}$$

$\Rightarrow 2a - b = 0$

$\Rightarrow 2a = b \ldots . (2)$

Replacing (2) in (1), we get $\sqrt{a^2+(2a)^2} = 5$

$\Rightarrow \sqrt{5a^2} = 5$

$\Rightarrow a^2 = 5$

$\Rightarrow a = \pm\sqrt{5}$

$\Rightarrow b = 2a \Rightarrow b = \pm 2\sqrt{5}$

Since $\vec{u}+3\vec{v}$ $(3a+2)\vec{i} + (3b-1)\vec{j}$

$$\|\vec{u}+3\vec{v}\| = \sqrt{(3\sqrt{5})^2+(6\sqrt{5}-1)^2}$$

$$= \sqrt{45+12\sqrt{5}+180-12\sqrt{5}+1}$$

$$= \sqrt{230}$$

63. **A Given:** A translation T that takes
$x^2 + (y+1)^2 = 5$ to $(x-2)^2 + y^2 = 5$

Required: Image of (1, 3) under T

The center for $x^2 + (y+1)^2 = 5$ is (0, -1)

The center for $(x-2)^2 + y^2 = 5$ is (2, 0)

So, (0, -1) $\xrightarrow{T}$ (2, 0)

That is, the translation vector T = (2 – 0, 0 – (-1)) = (2,1)

Therefore, (1, 3) $\xrightarrow[(2,1)]{T}$ (1 + 2, 3 + 1) = (3, 4)

64. **C Given:** equation of l. $x - 2y = 0$

Required: vector equation of a line through (5, 0) and $\perp$ to l.

$l = x - 2y = 0 \Rightarrow y = \frac{1}{2}x$

$\Rightarrow$ a line $\perp$ to l could be $y = -2x$

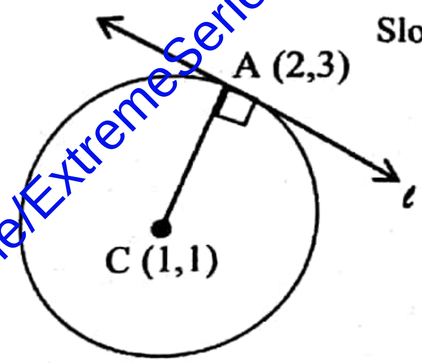
Since (1, -2) lies on $y = -2x$, the parametric vector equation required is

$(x, y) = (5,0) + t(1, -2), t \in R$

That is, $(x, y) = (5+t, -2t, 2t), t \in R$

65. **B Given:** - A circle passing through A (2,3)
- center C (1, 1)

Required: the equation of the tangent line to the circle of A.



Slope of AC $= \dfrac{3-1}{2-1} = \dfrac{2}{1} = 2$

Since AC $\perp$ l, the slope of the line l is $-\frac{1}{2}$

Thus, equation of l is given by

$$\frac{y-3}{x-2} = -\frac{1}{2}$$

$\Rightarrow 2y - 6 = -x + 2$

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1. **D** Given = $g(x) = x - \frac{(x^2+1)}{x^{4+x}}$

Required: Domain of g(x)

Since $x^4 + x = x(x^3 + 1) = x(x+1)(x^2 - x + 1)$

Domain of g(x) is [R\{0 ,-1}

2. **D** Required : Simplified form of $\frac{t-\frac{1}{t}}{1+\frac{t}{1-t}} \div \frac{t^2-2t+1}{t}$,$t \neq 0,1$

$$\frac{t-\frac{1}{t}}{1+\frac{t}{1-t}} \div \frac{t^2-2t+1}{t} = \frac{t^2-1}{\frac{t}{1-t}} \times \frac{t}{(t-1)^2}$$

$$= \frac{(t-1)(t+1)}{t} \times \frac{1-t}{1-t} \times \frac{t}{(t-1)(1-t)}$$

$$= (t+1) = -t-1$$

3. **A** Given $\frac{x-1}{x^{3+x}} = \frac{A}{x} + \frac{Bx+c}{x^2+1}$

Required: values of A ,B , C,

$$\frac{x-1}{x^2+1} = \frac{A}{x} + \frac{BX+C}{x^2+1} \Rightarrow$$

$$\frac{x-1}{x(x^2+1)} = \frac{Ax^2+A+Bx^2+Cx}{x(x^2+1)}$$

$$\Rightarrow \frac{x-1}{x(x^2+1)} = \frac{(A+B)x^2+(x+A}{X(x^2+1)}$$

$$\Rightarrow A + B = 0,\ C = 1 \ \ and\ A = -1$$

$$\Rightarrow B = 1$$

4. **C** Required sclution set of $1-\frac{5}{x^2+4} = \frac{x-1}{x^2-x} - \frac{1}{x}$

$$1-\frac{5}{x^2+4} = \frac{x-1}{x^2-x} - \frac{1}{x} \Rightarrow \frac{x-1}{x(x-1)} - \frac{1}{x} + \frac{5}{x^2+4} - 1 = 0\ ,\ x \neq 0,1$$

$$\frac{(x-1)(x^2+4)-(x-1)(x^2+4)+5x(x-1)-x(x-1)(x^2+4)}{x(x-1)x^2+4)}$$

$$\Rightarrow \frac{5}{x^2+4} - 1 = 0$$

$$\Rightarrow \frac{5}{x^2+4} = 1$$

$$\Rightarrow x^2 + 4 = 5$$

$$\Rightarrow x^2 - 1 = 0$$

$$\Rightarrow x=1, x=-1$$

S.S={-1}

5. **C.** Required: identify a polynomial of degree 3

A. $x^2 - 3x^{-2} + x$ is not a polynomial

B. $x^3 + x^2(1-x) + 1 = x^2 + 1$ a polynomial of degree 2

C. $x^2 + x^2(x-1) + \frac{\pi}{4} = x^2 + x^3 - x^2 + \frac{\pi}{4}$ degree 3

D. $x^3 + 3x^2 + sinx + 1$ is not a polynomial

6. **C** Required identify true statement about

$f(x) = x^5(x-1)^4(x^2+3)$

* 0 has 5 multiply so, A is wrong
* The graph crosses at x= 0 and turns at x=1

Thus B is wrong and C is correct

- f(x) has only two roots (0 and1).so D is wrong.

7. **B.** Given $f(x) = ax^2 + bx^2 - 1$

(x+1) is a factor of f(x)

When f (x) is divided by x-1, rem =4

Required values of a &b

(x+1) is a factor of f(x) => $a(-x)^{10} + b(-1^5 - 1 = 0$

$$\Rightarrow a-b = 1 \ \ldots\ldots\ldots\ldots (1)$$

Rem = 4 , When f is divided by x-1 => f(x) =4

$$\Rightarrow a(1)^{10} + b(1)^5 - 1 = 4$$

$$\Rightarrow a+b=5\ldots\ldots\ldots (2)$$

Solving 1&2 simultaneously, 2a=6 => a=3

=> b=2

8. **C.** Required A function that touches but never crosses the x-axis

A. $f(x) = 1-x^3 = (1-x)(1+x+x^2)$

=> f is crosses the x-axis at x=1

B. $f(x) = x^4 - 1 = (x^4 - 1 = (x^2 - 1)(x^2 + 1)$

=> f crosses at x=1 & at x= -1

C. $f(x) = (x^2 - x)^2$

f touches the x-axis at x=1 (never crosses it)

D. $f(x) = x - x^5 = x(1-x^4) = x(1-x^2)(1+x^2)$

f(x) crosses at x=0,1.&-1

9.D. Required solution set of

$\log(8x^3 - \log(4x) \geq \log x$

For $x > 0$,

$log8x^3 - log^{4x} \geq log^x$

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$\Rightarrow$ Log $\frac{8x^3}{4x} \geq log^x$

$\Rightarrow 2x^2 \geq x$

$\Rightarrow 2x^2 - x \geq 0$

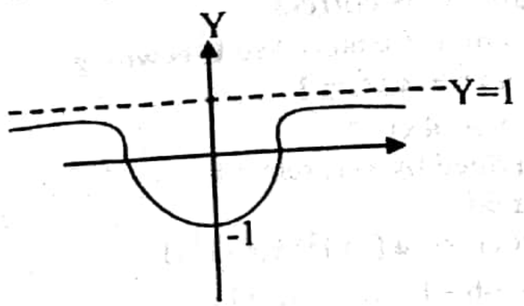
$\Rightarrow x(2x-1) \geq 0$

Using sign chart x=0 Or $x \geq 1/2$

But , x>0 $\Longrightarrow$ Solution set $[1/2, \infty)$

10. A. Given $g(x) = \frac{x^2-1}{x^2+1}$

Required identify wrong statement sketching g(x) as shown below



→ The range of g(x) is [-1,1) so A is wrong

→Since g(x) =g(-x) , g is an event function

→ y=1 is a horizontal asymptote

→ When x gets closer to $-\infty$, g(x) gets closer to 1 (property of H.A)

11. B Required : identify the function that represent the given graph

* The graph of $f(x) = \frac{x}{x-2}$ doesn't have a hole at x=-2 because it is defined at x=-2. Thus, it is not the required graph.

$f(x) = \frac{x^2+2x}{x^2-4} = \frac{x(x+2)}{(x-2)(x+2)} = \frac{x}{x-2}$ for $x \neq -2$

That is, at x=-2 f(x) will have a hole

f(x) =has a vertical asymptote at x =2 and a horizontal asymptote at y=1 (which is similar to the sketched graph)

12. C. Required solution set of $(\sqrt{3})^{-x} = \frac{1}{27}(9^x)$

$(\sqrt{x})^{-x} = (\frac{1}{27})(9^x) \Rightarrow 3^{\frac{-x}{2}} = (3^{-3})(3^{2x})$

$\Longrightarrow 3^{\frac{-x}{2}} = 3^{2x-3}$

$\Rightarrow 2x-3 = \frac{-x}{2}$

$\Rightarrow 4x-6 = -x$

$\Rightarrow 5x = 6$

$\Rightarrow x = \frac{6}{5}$

Solution set {6/5}

13. D Given Circle O, with QP tangent BC $=2\sqrt{3}$

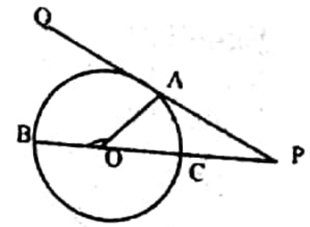
a (sector AOB) $=\pi cm^2$

Required length of AP

BC= $2\sqrt{3}$ => OC= OC=OA=$\sqrt{3}$ cm

A(sector ADB)$=\pi cm^2 \Rightarrow \frac{\theta}{360}\pi \times (\sqrt{3}^{\,2} = \pi$

$\Rightarrow = 120^0 \Rightarrow AoP = 60^0$



As OA $\perp$ AP, Δ OAP is a right triangle

$\Rightarrow \frac{AP}{AO} = \tan(AOP) \Rightarrow \frac{AP}{\sqrt{3}} = \tan 60^0 \Rightarrow \frac{AP}{\sqrt{3}} = \sqrt{3} \Rightarrow AP = 3cm$

14. A Given $P_1(a,b)$, $P_2(b,a)$ With $a \neq b$

Required equation of a line l through P_1 & perpendicular to P_1P_2

Slope of $P_1P_2 = \frac{a-b}{b-a}$

Slope of l=1 is $\frac{y-b}{x-a} = 1$

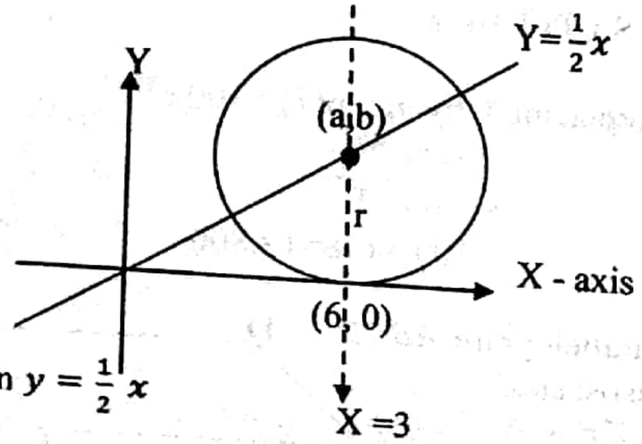
The equation of ℓ as $\ell \perp P_1P_2$

$\Rightarrow$ y-b=x-a

=> =x-a+b

15. D. Required equation of a circle

Given : circle is tangent to x-axis at (6,0)



Center lies on $y = \frac{1}{2}x$

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Let the center has coordinates (a, b)

$$\Rightarrow b = \frac{1}{2}a$$

When $a = 6 \quad b = 3$

$\Rightarrow$ center $(6,3)$

Radius =3 (vertical change)

The equation of a circle is $(x-6)^2 + (y-3)^2 = 3^2$

$$(x-6)^2 + (y-3)^2 = 9$$

16. **B**. Given ellipse with major axis = 10cm and minor axis = 6cm

Required possible equation

Major axis 6cm$\Rightarrow$ 2b =6cm $\Rightarrow$ b=3cm

If the center is at (b,k) the equation could be

$\frac{(x-x)^2}{x^2} + \frac{(y-k)^2}{b^2} = 1$ or $\frac{(x-h)^2}{b^2} + \frac{(1-x)^2}{a^2} = 1$

Therefore, $\frac{(x-h)^2}{25} + \frac{(y-k)}{9} = 1$ or $\frac{(x-h)^2}{9} + \frac{(y-k)^2}{25} = 1$

So if h=1 and k=0, the equation would be

$$\frac{(x-1)^2}{25} + \frac{y^2}{9} = 1 \text{ or } \frac{(x-1)^2}{9} + \frac{y^2}{25} = 1$$

(Vertical and Horizontal axis respectively)

17. A. Given $P(t) = 100\,e^{kt}$, k= constant

t: time is days

Required: P(7) =?

$P(2) = 400 \Rightarrow 100\,e^{2k} = 400$

$\Rightarrow e^{2k} = 4$

$\Rightarrow \ln 4 = 2k$

$\Rightarrow k = \frac{\ln 4}{2} = \ln^{4\,1/2} = \ln 2$

So, $P(7) 100\,e^{7k}$

Replacing k by ln2, $p(7) = 100\,e^{7\ln 2}$

$= 100\,e^{\ln 2^7}$

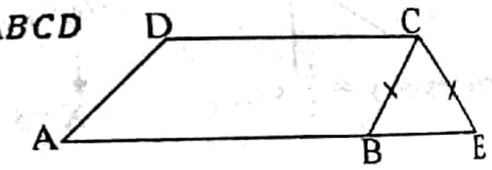
$= 100\,l^{\ln 128}$

$= 100 \times 128 = 12{,}800$

18. Given parallelogram $ABCD$

ΔBCE isosceles

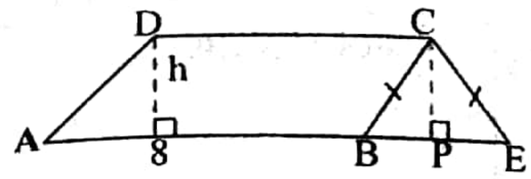
$AB = 8, CE = 6$



$\alpha(ABCD)$ 40

Required $\alpha(\Delta BCE)$

Let the altitude from C to $\overline{BE}$ be h



Area of (ABCD) = 40 square units

$\Rightarrow 40 = 8h$

$\Rightarrow h = 5$ *units*

considering $(BCPE), 6^2 - h^2 = (\overline{PE})^2$

$\Rightarrow 6^2 - 5^2 = (\overline{PE})^2$

$\Rightarrow \sqrt{36-25} = PE$

$= \sqrt{11} = PE = BP$

$\Rightarrow \sqrt{36-25} = PE$

$\Rightarrow \sqrt{11} = PE = BP$

$\Rightarrow BE = 2\sqrt{11} \times 5$

$\therefore a(\Delta BCE) = \frac{1}{2} BE \times h$

$= \frac{1}{2} \times \sqrt{11} \times 5$

$= 5\sqrt{11}$ *Square units*

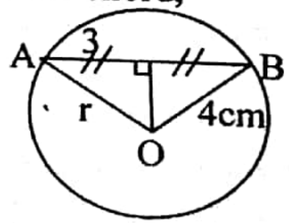
19. B Given Chord at a distance of 4cm from the center

length of chord = 6cm

Required: r = ?

Let AB be the chord

As the segment perpendicular to the chord, from the center, bisects the chord,



$= \sqrt{3^2+4^2}$

$= 5cm$

20. C Given A Parallelogram ABDC M is mid point of AB

Required : necessarily true statement

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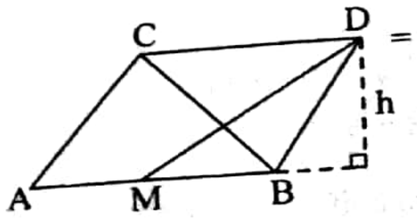
Area of $\Delta ABC = \frac{1}{2} \times AB \times h$

Area of $\Delta BMD = \frac{1}{2} \times AB \times h$

$= \frac{1}{2}(\frac{1}{2}AB)\,h$ as $BM = \frac{1}{2}AB$

$= \frac{1}{2}$ area of ΔABC



21. **B** Given sequence $\{a_n\}_{n=1}^{\infty}$ with
$a_{n+1} = -na_n$, $a_1 = 1$, $n = 1,2,3.....$
Required: to identify a wrong statement
$a_1 = 1$
$a_2 = a_{1+1} = -1_{a1} = -1(1) = -1$
$a_3 = a_{2+1} = -2_{a2} = -2(-1) = 2$
$a_4 = a_{3+1} = -3_{a3} = -3(2) = -6$
$a_5 = a_{4+1} = -4_{a4} = -4(-6) = 24$
$\Rightarrow$ an >0 if n is odd & an<0 if n is even, Thus only B is false

22. **D** Given $Sn = 4n^2$, $n = 1,2,3.....$
Required $= a_3 = ?$
$S_n = 4n^2 \Rightarrow S_1 = 4(1)^2 = 4 = a_1$
$\Rightarrow S_2 = 4(2)^2 = 16 = a_1 + a_2$
$\Rightarrow a_2 = 12$
$S_3 = 4(3)2 = 36 = a_1 + a_2 + a_3$
$36 = 4 + 12 + a_3$
$\Rightarrow$ $a_3 = 20$

23. **A.** Given sequence $\{a_n\}$ $n \geq 1$
$a_1 = 2$, $a_n = 3a_{n-1}$ for all $n \geq 2$
Required
$$\sum_{i=1}^{20} ai$$
$an = 3a_{n-1} \Rightarrow a_2 = 3a_1 = 3(2) = 6$
$a_3 = 3a_2 = 3(6) = 18$
$a4 = 3a_3 = 3(18) = 54$
this is a geometric sequence with $a_1 = 2$ and $r = 3$

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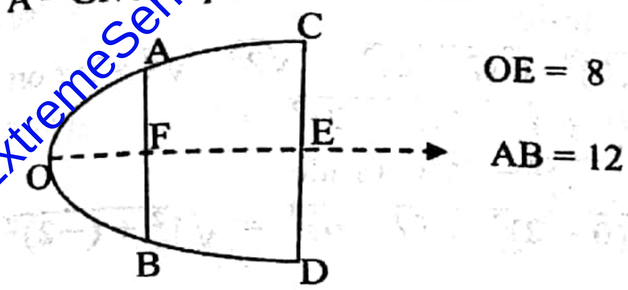
$\Rightarrow S_n = a_1 \frac{(n-1)}{r-1} = \frac{2(3^{20}-1)}{3-1} = \underline{x^{20}-1}$

24. **D** Given $a_0 = 6.4 \times 10^8$
$an = \frac{1}{2} a_n - 1$
Required $a_{20} = ?$
$$an = \frac{1}{2} an - 1 \Rightarrow \frac{a_n}{a_n} = \frac{1}{2} = r$$
$$a_{20} = a_0 r^{20}$$
$$= (6.4 \times 10^8) \times (\tfrac{1}{2})^{20}$$
$$= \frac{64 \times 10^7}{2^{20}} = \frac{64 \times x 10^7}{2^6 \times 2^{14}} = \frac{10^7}{2^{14}} = 2^{-14} \times 10^7$$

25. A – Given a parabolic reflector



Required CD
Let O lies at the origin and OE on the x – axis
Since AF = 6 , 2P, (P = focal length)
P = 3
Thus, the equation of the parabola will be
$$y^2 = 4PX \Rightarrow y^2 = 4(3)X$$
$$y^2 = 12X$$
$$OE = 8cm \Rightarrow X = 8 \text{ on } CD$$
Replacing X we get $y^2 = 4(3)X$
$$\Rightarrow y^2 = 96$$
$$\Rightarrow y = 4\sqrt{6}$$
$$Therefore, CD = 2y$$
$$= 8\sqrt{6}$$

26. **B** Given $3x^2 - 6x = y^2$
Required which conic section the equation represents
$3x^2 - 6x = y^2 \Rightarrow 3x^2 - 6x - y^2 = 0$
$\Rightarrow 3(x^2 - 2x) - y^2 = 0$

$\Rightarrow 3(x-x^2-2x+1)-y^2=3$ adding 3 both sides

$\Rightarrow 3(x-1)^2-y^2=3$

$\Rightarrow \frac{(x-1)^2}{1}-\frac{y^2}{3}=1$

$\Rightarrow \frac{(x-1)^2}{(1)^2}-\frac{y^2}{(\sqrt{3})^2}=1$

a hyperbola centered at (1,0)

27. **D** Given a sphere $x^2+y^2+Z^2+4=4$

Required :- identify a point outside the sphere.

$\Rightarrow x^2+y^2+(Z-2)^2=2^2$

Center (0,0,2,) and r=2

$OA=\sqrt{(0-0)^2+(p+1)^2+(2-1)^2}=\sqrt{1^2+1^2}$

$=\sqrt{2}<$ r inside

$OB=\sqrt{(0-0)^2+(0)^2+(2-2)^2}=\sqrt{2^2}=\sqrt{4}=2=$r on

$OC=\sqrt{(0-1)^2+(0+1)^2+(2-3)^2}=$

$\sqrt{(-1)^2+(1)^2+(-1)^2}=\sqrt{3}<$r inside

$OA=\sqrt{(0+1)^2+(0-2)^2+(2-0)^2}=\sqrt{1^2+(-2)^2+2}$

$=\sqrt{9}=3>$r outside

28 . **C** Given $f(x)=\frac{\sqrt{x}-2}{\frac{-4}{x}+1}$

Required

$$\lim_{x\to 4^+} f(x)$$

$$\frac{\sqrt{x}-2}{\frac{-4}{x}+1}=\frac{\sqrt{x}-2}{\frac{x-4}{x}}=\frac{x(\sqrt{x}-2)}{x-4}\,\frac{(\sqrt{x}+2)}{(\sqrt{x}+2)}=\frac{x(x-4)}{(x-4)(\sqrt{x}+2)}$$

$$=\frac{x}{\sqrt{x}+2}$$

$$\therefore \lim_{x\to 4^+} f(x)=\lim_{x\to 4^+}\frac{x}{\sqrt{x}+2}$$

$$=\frac{4}{2+2}=\frac{4}{4}$$

$$=1$$

29. **B** - Required

$$\lim_{x\to 2}\frac{6x-12}{x^2-x-2}$$

$$\lim_{x\to 2}\frac{6x-12}{x^2-x-2}=\lim_{x\to 2}\frac{6(x-2)}{(x-1)(x-2)}$$

$$=\lim_{x\to 2}\frac{6}{x+1}$$

$$=\frac{6}{3}=2$$

30. Required to evaluate

$$\lim_{x\to\pi}\frac{x\sin 2\pi}{x-\pi}$$

since $\frac{x\sin 2\pi}{x-\pi}$ is an indeterminate form $\frac{0}{0}$

We use la'Hopital's Rule.

That is

That is $\lim_{x\to\pi}\frac{x\sin 2\pi}{x-\pi}=\frac{\lim_{x\to\pi}\frac{[x\sin 2\pi}{x-\pi}]}{\lim_{x\to\pi}\frac{d[x-\pi]}{dx}}$

$$=\lim_{x\to\pi}\frac{(\sin 2x+2x\cos 2x)}{1}$$

$$=\sin 2\pi+2\pi\cos 2\pi$$

$$=0+2\pi(1)$$

$$=2\pi$$

31. **C**

Required: $-\lim_{x\to 0^-}\frac{|x|-2xe^x}{x}$

As $|x|=-x$ for $x<0$

$$\lim_{x\to 0^-}\frac{|x|-2xe^x}{x}=\lim_{x\to 0}\frac{-x-2xe^x}{x}$$

$$=\lim_{x\to 0}\frac{x(-1-2xe^x)}{x}$$

$$\Rightarrow -1-2$$

$$=-3$$

32. **A**

Required $\lim_{n\to\infty}\frac{n+\cos 3n}{2-n}$

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$$\lim_{n\to\infty}\frac{n+\cos 3n}{2-n}=\lim_{n\to\infty}\frac{\frac{n}{n}+\frac{\cos 3n}{n}}{\frac{2}{n}-\frac{n}{n}}\quad \text{dividing all by } n$$

$$\lim_{n\to\infty}\frac{1+\frac{\cos 3n}{n}}{\frac{2}{n}-1}=\frac{1}{-1}=-1$$

33. **B** Given $f(x)=g(\sqrt{4x+1})\ g'(3)=\frac{1}{2}$

Required $=f'(2)$

Let $\sqrt{(4x+1)}=h(x)$

$\Rightarrow f(x)=g(h(x)) \Rightarrow f'(x)=g'(h(x)\ h'(x)$

$h(x)=\sqrt{4x+1} \Rightarrow h(x)=\frac{4}{2\sqrt{4x+1}}=\frac{2}{\sqrt{4x+1}}$

$$h(2)=\frac{2}{3}\ldots\ldots\ldots(1)$$

$h(x)=\sqrt{4x+1} \Rightarrow h(2)=\sqrt{9}=3\ldots\ldots(2)$

$f(2)=g(h(2))\ h(2)$

$f(2)=g(3).h(2)$ (replacing 1 &2)

$$=\frac{1}{2}\cdot\frac{2}{3}=\frac{1}{3}$$

34. **C** Given $f(x)=la(x^2+1)$

Required $f'(x)$

$$f(x)=\frac{1}{x^2+1}.2x=\frac{2x}{x^2+1}$$

$$f(x)=\frac{2(x^2+1)-2x-24}{(x^2+1)^2}$$

$$=\frac{2x^2+2-4x^2}{(x^2+1)^2}$$

$$=-\frac{2x^2+2}{(x^2+1)^2}$$

$$=\frac{2(x^2-1)}{(x^2+1)^2}$$

35. **B** Given $f(x)=x/x/, x\in(-\infty,\infty)$

Required to identify true statement

$$f(x)=x/x/ \Rightarrow f(x)=\begin{cases}x^2, x>0\\0, x=0\\-x^2, x<0\end{cases}$$

- $f(0)=0$ so A is wrong
- $f(x)=\begin{cases}2x,, x>0\\0, x=0\\-2x, x<0\end{cases} \Rightarrow f(x)=2/x/, x\in R$

So B is true

$\Rightarrow f(x)$ is differentiable in $(-\infty,\infty)$

36. **C** Given $f(x)=x^4-6x^2+1$ on $[-2,3]$

Required local max value of f

$f(x)=4x^3-12x$

$f(x)=0 \Rightarrow 4x^3-12x=0 \Rightarrow 4x(x^2-3)=0$

$\Rightarrow x=0\ x=\pm\sqrt{3}$

$f(-2)=16-24+1=-7$ $\quad f(\sqrt{3})=9-18+1=-8$

$f(3)=81-54+1=28$ $\quad f(-\sqrt{3}=9-18+1=-8$

$f(0)=1$ Thus, $f(3)=28$ is the local max value

37. **A** Given $f(x)\begin{cases}3a-x, x<1\\b+1, x=1\\ax, x>1\end{cases}$ f is continuous at x=1

Required value of b

$Lim\ f(x)=lim\ ax$

$x\to 1^+ \quad x\to 1^+=a$

$lim\ f(x)=lim\ 3a-x=3a-1$

$x\to 1^- \quad x\to 1^-$

$lim\ f(x)=lim\ f(x) \Rightarrow a=3a-1$

$x\to 1^- \quad x\to 1^- \quad \Rightarrow=\frac{1}{2}$

Since f is continuous at $x=1, ½=b+1$

$$\Rightarrow b=-\frac{1}{2}$$

38. **D** Given $f(x)=\begin{cases}\frac{x-x^2}{x-1}, x<0\\3-x, x\geq 0\end{cases}$

Required Interval where f is continuous

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$$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} 3 - x = 3$$

$$\lim_{x\to 0^-} f(x) = \lim_{x\to 0^-} \frac{x - x^2}{x-1} = \lim_{x\to 0^-} -x = 0$$

$$\Rightarrow f \text{ is continuous on } [0,\infty]$$

39. **B** Given $f(x)\ x^5 + 2x$

Required $\lim_{h\to 0} \frac{f(1+h) - f(1)}{h}$

$$\lim_{h\to 0} \frac{f(1+h) - f(1)}{h} = f'(1)$$

since $f'(x) = 5x^4 + 2$

$f'(1) = 5(1)^4 + 2 = 7$

40. **B** Given $f(x) = e^{3x-1}$

$g(x)\ \tan x$

$F(x) = x f(x) + f(g(x))$

Required. F(0)

$F(x) = x f(x) + f(g(x))$

$= xe^{3x-1} + e^{(3\tan x)-1}$

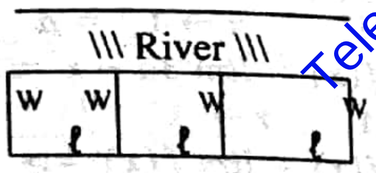
$F'(x) = e^{3x-1} + 3xe^{3x-1} + e^{(3\tan x)-1}(3\sec^2 x)$

$F(0) = e^{0-1} + 0 + e^{0-1}(3\sec^2 0)$

$= e^{-1} + e^{-1}(3)$

$= 4e^{-1}$

41. **D** Given



Perimeter of fence = 1200

Required value of l & w that macimize the total area

$Perimeter = 1200m \Rightarrow 4w + 3l = 1200$

$\Rightarrow 3l = 1200 - 4w$

$\Rightarrow l = \frac{1200-400}{3}$

$Total\ area = 3lw$

$= 3(\frac{1200-4w)^w}{3})$

$= 1200w - 4w^2$

$Total\ area = 3lw$

$= 3(\frac{1200-4w)^w}{3})$

$= 1200w - 4w^2$

Total area is maximum for $w = -\frac{1200}{2(-4)} = \frac{-1200}{-8} = 150$

From $l = \frac{1200-4w}{3}$, $l = \frac{1200-4(150)}{3} = \frac{600}{3} = 200$

Therefore $w = 150\ m$ & $l = 200m$

42. **C** Given Cylinder with

$H = 10cm$

$\frac{\Delta r}{\Delta t} = 3cm/min$

Required $\frac{\Delta r}{\Delta t}$ $when\ r = 8$

Since volume of cylinder $is\ \pi r^2 h$,

$\frac{\Delta r}{\Delta t} = 2\pi h\, r\, \frac{\Delta r}{\Delta t}$ (differentiating with respect to time)

$= 2\pi \times 10cm \times 8cm \times 3cm/min$

$= 480\ \pi cm^3/min$

43. **C** Given $f(x) = \frac{3}{x\sqrt{x}}$

Required Antiderivative of f(x)

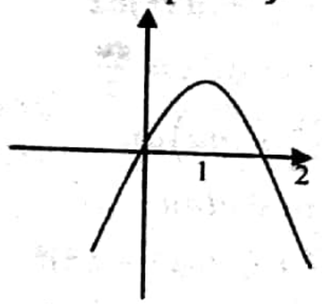
$F(x) \int \frac{3}{x\sqrt{x}}\, dx = \int 3x^{\frac{-3}{2}}\, dx \;.\; \frac{3x^{\frac{-3}{2}+1}}{\frac{-3}{2}+1} + c = \frac{3x^{\frac{-1}{2}}}{\frac{-1}{2}} + c$

$= \frac{-6}{\sqrt{x}} + c$

$If\ c = 1 \quad F(x) = \frac{-6}{\sqrt{x}} + 1$

44. **B** Given $f(x)$ is differentiable at $(-\infty,\infty)$

Graph of $y = f(x)$



Required identify a true statement

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The graph shows that $f(x) = 0$ at $x = 0$ and $x = 2$
$\Rightarrow$ $f(x)$ has local extreme at $x = 0$ or $x = 2$
Thus, $f(x)$ can have max at $x = 2$ which gives B is TRUE
As the graph shows $f(x) < 0$ for all $x \in (-\infty, 0) \cup (2, \infty)$
$\Rightarrow$ $f(x)$ is decreasing in these region which shows A&C are wrong

45. C Given $f(x)\ x^4 - \frac{16}{3}x^3 + 6x^2$

Required Range of f

$$f'(x) = 4x^3 - 16x^2 + 12x$$
$$= 4x(x^2 - 4x + 3)$$
$$= 4x(x-1)(x-3)$$
$$f(x) = 0 \Rightarrow x = 0, x = 1, x = 3$$
$$f(0) = 0, f(1) = \frac{5}{3}, f(3) = -9$$
$$\Rightarrow local\ min = -9$$

As the leading coefficient is + ve, the range of f is $[-9, \infty]$

46. B Required to evaluate $\int \frac{dx}{e^{-x}(1+e^x)}$

$$\int_0^1 \frac{dx}{e^{-x}(1+e^x)} = \int_0^1 \frac{e^x\ dx}{(1+e^x)}$$

Let $e^2 = u \Rightarrow du = e^x\ dx$

$$\Rightarrow dx = \frac{du}{e^x} = \frac{du}{u}$$
$$\Rightarrow \int_0^1 \frac{e^2 dx}{1+e^x} = \int_0^1 \frac{udu}{u(1+u)} = \int_0^1 \frac{du}{1+u}$$
$$= ln(1+u)\ \Big|_0^1 = ln(1+e^x)\ \Big|_0^1 = ln(1+e) - ln_2$$

47. D Required To evaluate $\int_1^2 x(\frac{1}{x^2} + 2lnx)\ dx$

$$\int_1^2 x(\frac{1}{x^2} + 2lnx)\ dx = \int_1^2 x(\frac{1}{x^2} + lnx^2)dx$$

let $x^2 = u \Rightarrow du = 2xdx \Rightarrow xdx = \frac{du}{2}$

Thus $\int_1^2 x(\frac{1}{x^2} + 2ln\,x)dx = \frac{1}{2}\int_1^2 \left(\frac{1}{u} + lnu\right)du$

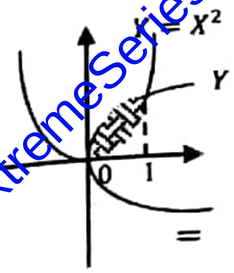
$$= \frac{1}{2}[lnu + ulnu - u]^2$$
$$= \frac{1}{2}[lnx^2 + x^2 lnx^2 - x^2]^2$$
$$= \frac{1}{2}[lnx^2 + (1+x^2) - x^2\ \Big|_1^2$$

$$= \frac{1}{2}[2lnx + (1+x^2) - x^2\ \Big|_1^2$$
$$= \frac{1}{2}[2ln2 + (1+2^2) - 2^2] - \frac{1}{2}\ [2\ln(1+1^2) - 1^2]$$
$$= 5ln2 - 2 - \frac{1}{2}(-1) = 5ln2 - 2 + \frac{1}{2}$$
$$= 5ln2 - \frac{3}{2}$$

48. B Given $4 = x^2$ & $4 = \sqrt{x}$

Required Area bounded between the curves

Area $= \int_0^1 (\sqrt{x} - x^2)dx$



$$= \int_0^1 \sqrt{x}\ dx - \int_0^1 x^2\ dx$$
$$= \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right] - \left[\frac{x^3}{3}\right]$$
$$= \frac{1}{\frac{3}{2}} - \frac{1}{3} = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$$

49. C Required to evaluate $\int \frac{x^2-1}{x^3+x}\ dx$

$$\int \frac{x^2-1}{x^3+x}\ dx = \int \frac{x^2-1}{x(x^2+1)}\ dx$$

Let $u = x^2 + 1 \Rightarrow du = 2 \times dx$ and $x - 1^2 = u - 2$

Replacing these

$$\int \frac{x^2-1}{x(x^2+1)}\ dx = \int \frac{u-2}{(2u-2)u} = \int \frac{1}{u}\,du - \frac{1}{2}\int \frac{1}{u-1}$$
$$= lnu - \frac{1}{2}\ln(u-1) + c$$

Replacing u, we get

$$\int \frac{x^2-1}{x(x^2+1)}\ dx = \ln(x^2+1) - \frac{1}{2}\ lnx^2 + c$$
$$= \ln(x^2+1) - ln|x| + c$$

50. B . Required to evaluate $\int (2 + sin^2x)\ cot\ x\ dx$

$$\int (2 + sin^2x)cotdx = \int (2 + sin^2x)\frac{cosx}{sinx}\ dx$$

Let $u = sinx \Rightarrow du = cosxdx$

$$\Rightarrow \int \frac{(2+u^2)}{u}\,du = \int \frac{2}{u}\,du + \int udu$$

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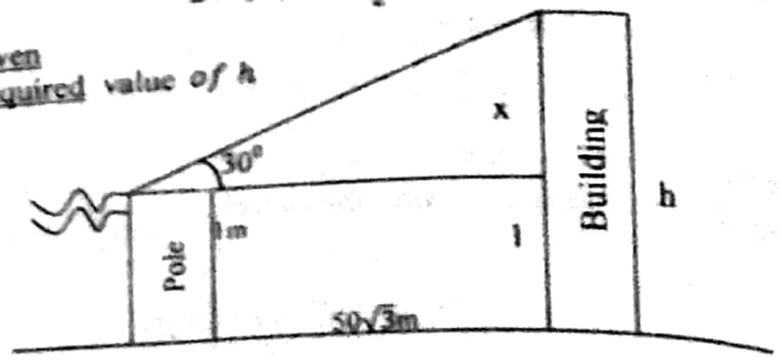
$= lnu + \frac{u^2}{2} + c$

$= 2\ln|\sin| + \frac{1}{2}\sin^2 x + c$

51. **B** Given

Required value of h



$\frac{x}{50\sqrt{3}\,m} = \tan 30^0$

$x = 50\sqrt{3} \times \frac{1}{\sqrt{3}} = 50m$

$h = x + 1 \Rightarrow h = 50 + 1$

$h = 51m$

52. Given $\vec{A} = (3,-3), \vec{B} = (1,-3)$

Required= A unit vector in the direction of

$\vec{C} = 3\vec{A} - \vec{B}$

$\vec{C} = 3A - B = 3(3,-3) - (1,-3) = (8,-6)$

But $\sqrt{8^{2+(-6)^2}} = \sqrt{100} = 10$

unit vector is $(\frac{8}{10}, \frac{-6}{10}) = (\frac{4}{5}, \frac{-3}{5})$

53. No answer Given $\vec{U}$&$\vec{V}$ have direction

$\vec{V} = (-3,4)$

$||\vec{u}|| = 33||\vec{v}||$

Required $||\vec{u} - \frac{3}{2}\vec{v} - 2j||$

$|\vec{u}| = 33|\vec{v}| \Rightarrow \vec{u} = -99i + 132j$

$\therefore \vec{u} - \frac{3}{2}\vec{v} - 2j = -99i + 132j + \frac{9}{2}i - 6j - 2j$

$= -\frac{189}{2}i + 124j$

$|u - \frac{3}{2}v - 2j| = \sqrt{(-\frac{189}{2})^2 + (124)^2}$

$= \frac{1}{2}\sqrt{97225}$

$= \frac{5}{2}\sqrt{3889}$

54. **A** Given $A = 2i - j$ $B = i + j$, θ between A&B

Required $\cos\theta$

$Cos\theta \frac{a.b}{|a||b|} = \frac{x_1x_2 + y_1 + y_2}{\sqrt{x^2+y^2}.\sqrt{x^2+x^2}}$

$\Rightarrow \cos\theta = \frac{2(1) + (-1)(1)}{\sqrt{2^2+(-1)^2}\sqrt{1^2+1^2}} = \frac{2-1}{\sqrt{5}.\sqrt{2}} = \frac{1}{\sqrt{10}}$

55. **C** Given $\theta \sin(\frac{1}{2})$

Required $\tan 2\theta$

$\theta = arc\sin(\frac{1}{2}) \Rightarrow \sin\theta = \frac{1}{2}$

Since $\tan 2\theta = \frac{\sin 2\theta}{\cos 2\theta} = \frac{2\sin\theta\cos\theta}{\cos^2\theta - \sin^2\theta}$

From $\sin^2\theta + \cos^2\theta = 1$, we have

$\cos\theta = \sqrt{1 - \sin^2\theta}$

$\Rightarrow \cos\theta = \sqrt{1 - \frac{1}{4}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$

$\tan 2\theta = \frac{2(\frac{1}{2})(\frac{\sqrt{3}}{2})}{\frac{3}{4} - \frac{1}{4}} = \frac{\frac{\sqrt{3}}{2}}{\frac{2}{4}} = \sqrt{3}$

56. **B** Given $f(x) = 2 - 3\sin(\frac{x+\pi}{4})$

Required amplitude & period

Amplitude $= -3$

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Period $T = \frac{2\pi}{w} = \frac{2\pi}{\frac{1}{4}} = 8\pi$

57. C Required simplified form of $(1 + tan^2x)$

$sin\,(x - \frac{3}{2}\pi)$

Since $sin(x - \frac{3}{2}\pi) = sinx\,cos\frac{3}{2}\pi + cosxsin\frac{3}{2}\pi = -cosx$

$(1 + tan^2x)\,sin(x - \frac{3}{2}\pi) = (1 + \frac{sin^2x}{cos^2x}\,(-cosx)$

$= -cosx - \frac{sin^2x}{cosx}$

$= -\frac{cos^2x - sin^2x}{cosx}$

$= \frac{-1}{cosx}$

$= -sec\,x$

58. No answer

Required solution set of $\sqrt{3}\sin 2x = \cos 2x$ on $[0,\pi]$

Let $2x = y \Rightarrow \sqrt{3}\,siny = \cos y$

But $cosy = \pm\sqrt{1 - sin^2y}$

That is, $\sqrt{3}\sin y = \pm\sqrt{1 - sin^2y}$

$3sin^2y = 1 - sin^2y$

$4\,sin^2y = 1$

$\Rightarrow sisin^2y = \frac{1}{4}$

$\Rightarrow siny = \frac{1}{2}\ or\ siny = \frac{-1}{2}$

$\Rightarrow y\,\pi/6$ as the domain is $[0,\pi]$

$\Rightarrow 2x = \frac{\pi}{6}$

$\Rightarrow x = \frac{\pi}{12}$ $\Rightarrow$ solution set $\{\frac{\pi}{12}\}$

59. C Required $A = (-1,1,0)$

$B = (1,3,2)$ and points of the diameter

Required radius

$r = \frac{d}{2} = 1/2\sqrt{(1+x)^2 + (3-1^2) + (2-0)^2} =$

$1/2\sqrt{2^2 + 2^2 + 2^2} = 1/2\sqrt{12}$

$= \sqrt{\frac{12}{4}} = \sqrt{3}$

60. B Given center of a sphare at $(-1,3,2)$

Radius $= 3$

Required true statement about intersection of sphere with z-axis

A. Distance between center & $(0,0,3)$ is

$\sqrt{1^2 + (-3)^2 + x^2} = \sqrt{1+9+1} = \sqrt{11}$

$\neq r \Rightarrow$ $(0,0,3)$ is not anitersect

B. Asphere will have two intersection points with z-axis

(one +ve z axis & the other on-ve z-axis)

So, B is true

61. D Given $(\Gamma PvQ) \Rightarrow (R\Lambda\Gamma R)$ is TRUE

Required a nessirly true statement

Since $R\Lambda\Gamma R$ is never true for any R

$(\Gamma PvQ) \Rightarrow (R\Lambda\Gamma R)$ *is False*

Thus ΓPvQ must be False

That is Q is False and P is True

Therefore A) R=> Q is false as Q is False

B) (Q=> P) vR can be false if R is true.

C) P$\Lambda\,\urcorner$R is false if R is true

D) (Q Λ R) => P is always true as P is True.

62. B

Given proof of a statement

If x is rational X & Y is irrational, then $x - y$ is irrational.

Required what type of proof followed

The proof is proof by c contradiction and it started with the opposite assumption and ended up in a contradiction.

63. D

Given paraboloa $y = x^2 + 1$

Line $y = -x$

Required reflection image of the parabola through the line

When $y = x^2 + 1$ reflected along $y = -x$, we get

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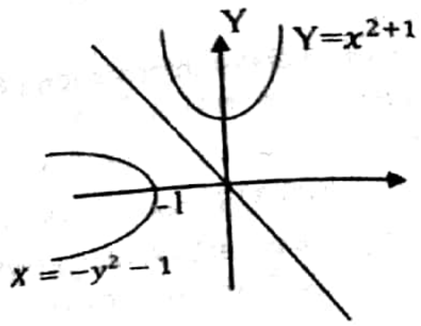


64. D Given $x^2+y^2=2$

$l:(x,y)=(2,0)+t(-1,1)$

Required identify true statement

$l(x,y)=(2,0)+(-1,1) \Rightarrow x=2-t \quad y=t$

$\Rightarrow t=2-x$

$\Rightarrow y=2-x$ rectangular form of l

Replacing is in the circle $x^2+y^2=2$, we get

$x^2+(2-x)^2=2 \Rightarrow x^2+4-4x+x^2=2$

$\Rightarrow x^2-4x+2=0$

$\Rightarrow x^2-2x+1=0$

$\Rightarrow (x-1)^2=0$

$\Rightarrow x=1$

$y=2-x$ *That is* $y=1$

$\Rightarrow$ l is tangent to the circle at (1,1) (only one point)

$x^2+y^2=2 \Rightarrow r=\sqrt{2}$ = distance between center & l

65. A Given mid point of a line segment at M(-3,3,4) one end point is A (0,-1, 3)

Required co-ordinates of the other end

Let the other end is at (P,Q, R)

$\Rightarrow \frac{p+(0)}{2}=-3$, $\frac{Q+(-1)}{2}=3$ *and* $\frac{R+3}{2}=4$

$\Rightarrow p=-6$, $Q=7$ *and* $R=5$

Therefore, the other end is at $(-6,7,5)$

1. $x \Rightarrow |x| = -x$ then $f(x)=\frac{x-|x|}{x}=\frac{x-(-x)}{x}=\frac{2x}{x}=2$

$\therefore f(x)=2$ for $x<0$

Answer B

2. $f(x)=\frac{\sqrt{x+2}}{x+2}$ and $g(x)=\frac{1}{x}-2$

then $f(x(g)=\frac{\sqrt{g(x)+2}}{g(x)+2}$ (g(x) in place of x in f(x))

$=\frac{\sqrt{\frac{1}{2}-2+2}}{g(x)+2}=\frac{\sqrt{\frac{1}{x}}}{\frac{1}{x}}=\left(\frac{1}{x}\right)^{\frac{1}{2}-1}=\left(\frac{1}{x}\right)^{-\frac{1}{2}}=\frac{1}{\sqrt{x}}$

$=\frac{1}{\sqrt{x}}\cdot\frac{\sqrt{x}}{\sqrt{x}}=\frac{x}{\sqrt{x}}$

Answer D

3. $f(x)=\ln\left(\frac{x}{x-1}+2\right)$, for x>1 then we need $f^{-1}(x)$.

$f:x=\ln\left(\frac{x}{x-1}+2\right)$ implies $f^{1}:x=\ln\left(\frac{y}{y-1}2\right)$ (interchange x & y)

Then making y as subject we set $y=f^{-1}(x)$.

$\Rightarrow x=\left(\frac{x}{x-1}+2\right) \Rightarrow \left(\frac{y}{y-1}2\right)e^x \Rightarrow \left(\frac{y}{y-1}\right)e^x-2$

$\Rightarrow y=(y-1)(e^x-2) \Rightarrow y=ye^x-2y-e^x+2$

$\Rightarrow y-ye^x+2y=-e^x+2 \Rightarrow y(2-e^x)=-\frac{e^x+2}{e^x-2}$

Answer A

4. $\frac{4-(a-2)^2}{a^4+2a^2+1}\div\frac{1-\frac{1}{a^2+1}}{a}=\frac{4-(a^2-4a+4)}{a^4+2a^2+1}\div\frac{a^2+1-1}{a(a^2+1)}$

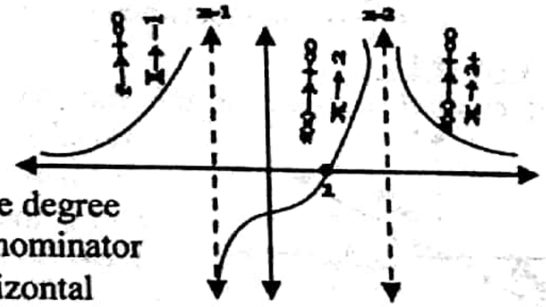
$=\frac{-a^2+4a}{a^4+2a^2+1}\times\frac{a(a^2+1)}{a^2}$

$=\frac{a(-a+4)}{(a^2+1)^x}\times\frac{a(a^2+1)}{a^2}$

$=\frac{4-a}{a^2+1}$

Answer D

5. $f(x)=\frac{x-1}{(x-2)^2(x+1)}$



A. As $x\to\pm\infty$ $f(x)\to 0$ since the degree of numerator is less than denominator in lowest term $\therefore$ it has horizontal asymptote y=0. No oblique asymptote

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B. If f(x) meets its horizontal asymptote it it's at x where $\lim_{x\to\infty} f(x) \approx f(x)$

$\Rightarrow \lim_{x\to\infty} \frac{x-1}{(x-2)^2} = \frac{x-1^{x\to\infty}}{(x-2)^2(x+1)}$

$\Rightarrow 0 = \frac{x-1}{(x-2)^2(x+1)} \Rightarrow x-1=0 \therefore \underline{x=1}$

$\therefore$ f meet its horizontal asymptote at (1,f(1)) = (1,0) to answer C and D let us use sign chart as follows:

	-1 ←	1	2	
x-1	-	-	+	+
x+1	-	+	+	+
$(x-2)^2$	+	+	+	+
f(x)	+	-	+	+

$\Rightarrow f(x) \to +\infty$ as $x \to -1$, $x^{\to 1}$ and $x \to 2$

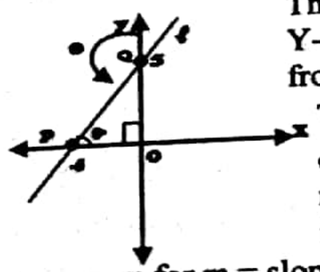
$\Rightarrow f(x) \to -\infty$ as $x \to -1^+$ and $x \to 1^-$

C. As $x \to -1^+$, if $(x) \to -\infty$ (right of x=-1) in the table

D. As $x \to 2^-$,, $f(x) \to +\infty$ (f is positive from both left and right side of x=2)

Answer D

6. ℓ is line through (0,5) and (-5,0),<QPO=α and <YQP=θ



Then we need θ between Y-axis and ℓ means counter –clockwise from y-axis to line ℓ (as in the fig.) i.e. θ

Then in $\triangle PQO$ $\theta = \alpha + \frac{\pi}{2}$ [because θ is external angle at vertex Q, which is sum of remote interior angles.] and is angle of inclination of line ℓ

$\Rightarrow \tan\theta = m$ for m = slope of ℓ

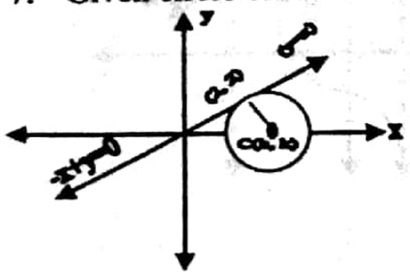
$\Rightarrow \tan\theta = \frac{0-5}{-5-0} = 1 \Rightarrow \theta = \tan^{-1}(1) = \pi/4$

$\therefore \theta = \alpha + \frac{\pi}{2} = \frac{\pi}{4} + \frac{\pi}{2} = \frac{3\pi}{4}$

$\therefore \theta = \frac{3\pi}{4}$ is angle between y-axis and line ℓ through (-5, 0), (0,5)

Answer D

7. Given circle center on x-axis and y=x is tangent



At (2,2) (as in fig)

$\Rightarrow$ Then we need equation of the circle

$\Rightarrow$ Let the center be at (=(h,k)

Then C= (h,k)=(h,0) since k=0

i) $(x-h)^2 + (y-k)^2 = r^2$ will be $(x-h)^2 + y^2 = r^2$ since (2,2) is on the circle : $(2-h)^2 + 2^2 = r^2$

$\Rightarrow 4 - 4h + h^2 + 4 = r^2 \Rightarrow \boxed{h^2 - 4h + 8 = r^2}$ – ①

ii) by distance between point and line $r = \frac{|AX+BY+cl|}{\sqrt{A^2+B^2}}$

then for A = -1, B=1, C=0 since y=x $\Rightarrow$ -x+y =0

and x =h, y=k = 0

$\Rightarrow r = \frac{|.h+1(0)+0|}{\sqrt{(-1)^2+(1)^2}} \Rightarrow r = \frac{|-h|}{\sqrt{2}}$

Squaring both side $\boxed{r^2 = \frac{h^2}{2}}$ ——— ②

$\therefore$ Combining ① and ② $h^2 - 4h + 8 = r^2 = \frac{h^2}{2}$

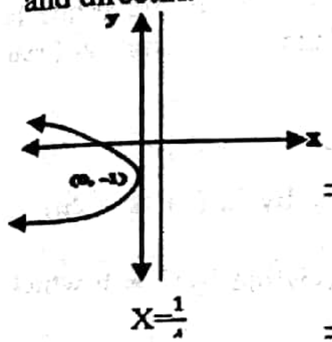
$\Rightarrow h^2 - 4h + 8 = \frac{h^2}{2} \Rightarrow h^2 - 8h + 16 = 0 \Rightarrow (h-4)^2 = 0$

$\therefore$ h = 4 then using $r^2 = \frac{h^2}{2} \Rightarrow r^2 = 8$

$\therefore$ The circle $(x-h)^2 + (y-k)^2 = r^2$ for h = 4, k = 0 and $r^2 = 8$ will be $(x-4)^2 + y^2 = 8$

Answer C

8. Given parabola $x + y^2 + 2y + 1 = 0$, then we need to find vertex = (h, k) and directix.



$\Rightarrow$ Since it is parabola we complete the square of 2nd Degree (i.e. y)

$x + y^2 + 2y + 1 = 0 \Rightarrow x + (y+1)^2 = 0$

$\Rightarrow (y+1)^2 = -x \therefore$ vertex = (h,k) = (0,-1)

$\Rightarrow$ It is horizontally open left parabola since 4p=-1 (negative sign)

Then focal length p will be $4p = 1$, $p = \frac{1}{4}$

$\Rightarrow$ directix is the line $x = h + p = 0 + \frac{1}{4} = \frac{1}{4}$

$\therefore$ Vertex = (0, -1) directrix : $x = \frac{1}{4}$

Answer C

9. Let us consider as horizontal ellipse as in fig below (center at origin)

Given: - $e = 0.206$

$2a = 1.16 \times 10^8$ km

$\Rightarrow a$ = length of half major axis = OV (or ov^1)

$\Rightarrow e$ = eccentricity $= \frac{c}{a}$ (ratio of distance between foci to vertices)

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$\therefore$ The maximum distance between mercury and the sun (in the figure) is Fv^1

$\Rightarrow Fv^1 = a + c$

$= a + ea$ because $e\frac{c}{a}$ gives $c = ea$

$= a(1 + e)$

$= \frac{1.16x10^8}{2}(1+0.206)$ km

$\approx 6.995 \times 10^7$km

Answer D

10. Given $p \Rightarrow 7q$ is false (F)

$p \Rightarrow 7q \equiv F$ gives $p \equiv T$ and $7q \equiv F$

Because implication is false only if $T \Rightarrow F$

$\therefore p \equiv T$ and $q \equiv T$

A. $7q \wedge (p \Rightarrow q) \equiv 7T \wedge (T \Rightarrow T) \equiv F \wedge T \equiv F$

B. $7p \vee (q \Rightarrow 7p)$ $7T \vee (T \Rightarrow 7T) \equiv F \vee (T \Rightarrow F) \equiv F \vee F$ F

C. $(7q \vee p) \Leftrightarrow q \equiv (7T \vee T) \Leftrightarrow T \equiv (F \vee T) \Leftrightarrow T \equiv T \equiv T$

D. $(p \wedge q) \Leftrightarrow 7p \equiv (T \wedge T) \Leftrightarrow 7T \equiv T \Leftrightarrow F \equiv F$

11. Given: $e(x) \equiv$ x is even; $p(x) \equiv$ x is prime ; $d(x) \equiv$ x is divisible by 2 U=IN. Then we need statement with false truth value.

A) $((\exists x)[e(x) \wedge p(x)]$ There is some x so that x is even and prime is satisfied. (Or means some even are prime). This is true since at least we have x = 2

Then $\therefore e(2) \wedge p(2) \equiv T$

B) $\forall x(e(x) \Rightarrow d(x)) \equiv$ All even are divisible by 2

This is true since even numbers are divisibly by 2. (i.e x = 2n) $\therefore e(2n) \Rightarrow d(2n) \equiv T$ for $n \in$

C) $(\exists x)(e(x) \wedge \neg d(x) \equiv$ some evens are not divisible by $2 \equiv F$ which is also negation of $\forall x(e(x) \Rightarrow d(x))$

Because $\neg \forall x(e(x) \Rightarrow d(x)) \equiv \neg(\forall x)\ \neg(e(x) \Rightarrow d(x)) \equiv$

$\equiv \exists x(\neg(e(x) \vee d(x))$ Since $p \Rightarrow q \equiv \neg p \vee q$

$\equiv \exists x(e(x) \vee \neg d(x)$ By demorgans rule

$\equiv F$ (negation of B) $\therefore e(2n) \wedge \neg d(2n) \equiv F, n \in IN$

D) $\forall x(e(x) \vee \neg d(x)) \equiv$ Every x in IN is even or not divisible by 2

This is true since natural numbers are even or odd (not divisible by 2)

Answer C

12. Given: Argument: "production is high if rain continues rain doesn't continue. Therefore either production is low or rain continues."

:p: production is low q; rain continues

Then the argument is symbolized as follows:

Premises: $\frac{\text{production is high}}{\neg p}$ $\frac{\text{rain continues}}{\equiv q} \equiv q \Rightarrow \neg p$ (True)

$\therefore \frac{\text{Rain doesn't continue}}{\equiv \neg q} \equiv \neg q$ (True)

Conclusion: Either $\frac{\text{production is low}}{\equiv p}$ or $\frac{\text{rain continues}}{\equiv q} \equiv p \vee q$ (?)

$\therefore$ Symbolically the argument will be 78, $8 \Rightarrow 7p$ $p \vee q$

i) $\neg q \equiv T$ (premise)

$\neg(\neg q) \equiv \neg T$ (negation)

$q \equiv F$

ii) $q \Rightarrow \neg p \equiv T$ (premise)

$F \Rightarrow \neg p \equiv T$ (since $q \equiv F$)

So $F \Rightarrow \neg p \equiv$ is True it $\neg p \equiv T$ or $\neg p \equiv F$

Therefore p can be true or False. q is always F

If $p \equiv F$ Then $p \vee q \equiv FVF \equiv F$ (invalid)

If $p \equiv T$ Then $pVq \equiv TVF \equiv T$ (valid)

The argument has been invalid due to q = F and when $p \equiv F$

$\therefore$ Invalid due to row 4

Answer C

13. If means of populations of $n_1, n_2, n_3 \ldots n_k$ members /elements are $\overline{x_1}, \overline{x_2}, \overline{x_3}, \ldots, \overline{x_k}$ respectively then the mean of poputation of $n_1 + n_2 + n_3 + \ldots n_k$ will be $\overline{x}$ such that

$\overline{x} = \frac{n_1\overline{x_1} + n_2\overline{x_2} + \ldots + n_k\overline{x_k}}{n_1 + n_2 + \ldots + n_k}$

$\therefore$ The mean $\overline{x}$ of 11A, 11B and 11C together will be $\overline{x} = \frac{n_A\overline{x_A} + n_B\overline{x_B} + n_c\overline{x_B}}{n_A + n_B + n_c}$ $n_A = 28, n_B = 20, n_c = 22$ $x_A = 60, \overline{x_c} = 70$

$= \frac{28(60) + 20(70) + 22(70)}{28 + 20 + 22}$

$= \frac{4620}{70} = 66$

Answer A

14. Given table of data (frequency distribution table) below

Age	8	10	11	12	13
Number of students	5	15	8	10	2

$\Rightarrow \sum_{i=1}^{5} f_i x_i = n\overline{x}$ (sum of age

$\Rightarrow n = \sum_{i=1}^{5} f_i = 40$

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A) Median $= \begin{cases} \left(\frac{n+1}{2}\right)^{th} \text{ them if n is odd} \\ \frac{\left(\frac{n}{2}\right)^{th} item \cdot}{2} \text{ if n is even} \end{cases}$

Now since n = 40 (sum of frequencies) total no. of students

$$\text{Median} = \frac{\left(\frac{40}{2}\right)^{th} them + \left(\frac{40}{2}+1\right)^{th} them}{2}$$

$$= \frac{20^{th}\ item + 21^{th}\ item}{2}$$, 20^{th} item is in 2^{nd} class (=10). and 21^{st} item is in 3^{rd} class (=11)

$$= \frac{10+11}{1} = \underline{10.5} \text{ (True)}$$

B) Mode = 10 (highest frequent value in 2^{nd} column), that is $f_2 = 15$; True

C) $$\text{Mean} = \frac{\sum_{i=1}^{5} f_i x_i}{\sum_{i=1}^{5} f_i} = \frac{5(8)+15(10)+8(11)+10(12)+2(13)}{5+15+8+10+2}$$

$$= \frac{40+150+88+120+26}{40}$$

$$= \frac{384}{40} = 9.6 \neq 10.5 \text{ (False)}$$

D) Range = maximum value – minimum value
= 13-8 = 5 (True)

Answer C

15. Given the data as follows we need S.D (standard deviation)

Xi	1	3	4	5	7
Frequency f_i	2	5	6	5	2

, mean = $\bar{x} = 4$

$$\text{S.D} = \sqrt{\frac{\sum_{i=1}^{5} f_i(x_i - \bar{x})^2}{\sum_{i=1}^{5} f_i}} = \sqrt{\frac{2(1-4)^2 + 5\,(3_{4)}2 + 6(4-4)^2 + 5(5-4)^2 + 2(7-4)^2}{2+5+6+6+2}}$$

$$= \frac{\sqrt{18+5+0+5+18}}{20} = \sqrt{2.3}$$

Answer C

16. Given 0, 1, 2, 3, 4, 5 and we need number of three digits where repetition and 0 can't be first digit

$\Rightarrow$

5- ways	5 – ways	4- ways
except used	except used Except 0	in cell 1 in cell 1 and 2

(1, 2, 3, 4, 5)

Therefore 5 ways followed by 5 ways followed 69 4= 5x5x4 = 100 ways (produce rule)

- OR simply 5 ways followed by arranging / permutation of 2 out of 5=5xp(5,2) = 100

Answer A

17. Given : 3 different mathematics books (3M)
: 3 different physics books (3p)
: 1 History book (1H)

Total combination of 3 books of any type out of 3+3+1= 7 books is = $C(7,3) = \frac{7!}{3!(7-3)!} = \frac{7!}{3!4!} = 35$ —— ①

$\Rightarrow$ Number of choosing 1M and followed by 3(of 3 p and 1H) is C(3M, 1M) followed by C(3p + 1H, 2) . By product rule

$\Rightarrow C(3, 1)\ C(4,2) = \frac{3!}{1!(3-1)!} \cdot \frac{4!}{2!(4-2)!} = 18$ (number of ways to choose 1M and 2 of 3 p and 1H) —— ②

Probability of (1M, 2 P or 1H) = $\frac{18}{35}$

Answer D

18. Probability of D_2 of that already D_1 happened is

$$P(D_2/D_1) = \frac{P(D_1 n\ D_2)}{P(D_1)}$$

Then $P(D_1 n\ D_2) = \frac{2\%x}{x} = 2\%$ if x is total no. of items

$P(D_1) = \frac{5\%x}{x} = 5\%$ if x is total no. of items

Then $P(D_2 \backslash D_1) = \frac{P(D_1 n\ D_2)}{P(D_1)} = \frac{2\%}{5\%} = 0.4$

Answer D

19. Given A $\begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix}$ and B= $\begin{pmatrix} 1 & -y \\ 0 & 3 \\ 1-x & 2 \end{pmatrix}$

Then we need value of y such that $A + 2B^T = 0$

$$\Rightarrow A + 2B^T = 0 \Rightarrow \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix} + 2\begin{pmatrix} 1 & -y \\ 0 & 3 \\ 1-x & 2 \end{pmatrix} = 0$$

$$\Rightarrow \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix} + 2\begin{pmatrix} 1 & 0 & 1-x \\ -y & 3 & 2 \end{pmatrix} = 0$$

(transpose)

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$\Rightarrow \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 2-2x \\ -2y & 6 & 4 \end{pmatrix} = 0$ (constant multiple)

$\Rightarrow \begin{pmatrix} -2+2 & 0+0 & x+2-2x \\ 2y+(-2y) & x+y+6 & -4+4 \end{pmatrix} = 0$ (Matris addition)

$\Rightarrow \begin{pmatrix} 0 & 0 & 2-x \\ 0 & x+y+6 & 0 \end{pmatrix} = 0$

$\Rightarrow 2 - x = 0$ and $x+y+6=0$ (since 0 is of any order)

$\Rightarrow x = 2$ Then $x+y+6=$ gives $x+y+6=0$

$\therefore y=-8$

Answer C

20. Given $A=\begin{pmatrix} 2 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & -1 & \frac{1}{2} \end{pmatrix}$ and $|B| = \frac{1}{10}$

$\Rightarrow$ Then we need to find value of $|2AB0^T|$.

$\Rightarrow$ First let's evaluate $|A|$, determinant of A, by expanding first raw, we evaluate $|A|$ as:

$\Rightarrow |A| = 2\begin{vmatrix} 5 & 0 \\ -1 & \frac{1}{2} \end{vmatrix} - 0\begin{vmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{vmatrix} - 0\begin{vmatrix} 1 & 5 \\ 0 & -1 \end{vmatrix}$

$= 2(s\left(\frac{1}{2}\right) - 0\,(-1) - 0\,(1\left(\frac{1}{2}\right)) - 0(0)) - 0\,(1(-1)-5(0)) = 5$

Then $|2AB^T| \neq 2|AB^T|$ (Take care here)

$|2AB^T| = |2A|\,|B^T|$ (property determinant)

$2^3\,|A|\,|B^T|$ (Because A is matrix of 3- rows so every row times 2 means multiplying the determinant by 2 for 3 times or by 2^3)

$= 8(5)\left(\frac{1}{10}\right) = 4$

Answer B

Note: If A is n by matrix then for $k \neq 0$, $|KA| = K^n|A|$ (n- is number of rows /column).

21. Given systems of linear equations

$$\begin{cases} x+y+2z=1 \\ x+2y+z=2 \\ -2x-2y-4z=-2 \end{cases}$$

Then we need the solution set. Now let's use echelon row operation

i) $\begin{pmatrix} 1 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \\ -2 & -2 & -4 & -2 \end{pmatrix}$ Augmented matrix

ii) Then by pivoting $R_2 \rightarrow -R_1 + R_2$

And $R_3 \rightarrow ZR_1 + R_3$

$\begin{pmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

iii) Such case, if number of variables is greater than number of equations

The system has infinitely many solutions

$\Rightarrow \begin{cases} x+y+2z=1 \\ y-z=1 \end{cases}$

Let $z=k$ for $k \in (-\infty,\infty)$

Then in $y-z=1$ substitute $z=k$

$\Rightarrow y - K = 1 \Rightarrow \underline{y = k+1}$

Again in $x+y+2z =$ putting $z = k$ and $y = k+1$

We get $x + k + k + 2k = 1 \Rightarrow \underline{x = -3k}$

$\therefore$ Solution set = $\{(x, y, z): x=-3k, y= k+1, z= k \text{ for } k \in IR\}$

Or = $\{-3k, k+1, k): k \in (-\infty,\infty)\}$ Answer C

22. Given $\begin{pmatrix} \propto & 2 & f \\ 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ t \\ z \end{pmatrix} = \begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix}$ and determinant of coefficient matrix, that is $\begin{vmatrix} a & 2 & \beta \\ 2 & 1 & 3 \\ -1 & 0 & 1 \end{vmatrix} = -5$

? Then we need x

Using Cramer's rule $x = \frac{\Delta_x}{\Delta}$ where Δ_x= determinant of matrix gaind by replacing the x- column (1st column) of coefficient matrix by constant matrix. Δ= determinant of coefficient matrix (i.e. $\Delta = -5$) given Then

$\Delta_x = \begin{vmatrix} -5 & 2 & \beta \\ 5 & 1 & 3 \\ 0 & 0 & 1 \end{vmatrix}$ now we expand the 3rd raw since it makes

Computation easier due to no. of zeros is highest.

$\Rightarrow \Delta_x = 0\begin{vmatrix} 2 & p \\ 1 & 3 \end{vmatrix} - 0\begin{vmatrix} -5 & p \\ 5 & 3 \end{vmatrix} + 1\begin{vmatrix} -5 & 2 \\ 5 & 1 \end{vmatrix}$

$= 0 - 0 + 1\,(-5(1)-2(5)) = \underline{-15}$

$\therefore x = \frac{\Delta_x}{\Delta} = \frac{-15}{-5} = \underline{3}$ Answer A

23. Given: equation $x^2 - 2x + 5 = 0 \Rightarrow$ using quadratic formula for a = 1, b= -2 , c = 5 $\Rightarrow x_{1,2} = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ gives

$\Rightarrow x_{1,2} = \frac{2 \pm \sqrt{4-20}}{2} = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm \sqrt{-1}\sqrt{16}}{2} = \frac{2 \pm 4\sqrt{-1}}{2}$

$\Rightarrow x_{1,2} = \frac{x(1 \pm 2\sqrt{-1})}{1} = 1 \pm 2i$ because $i = \sqrt{-1}$ or $i^2 = -1$

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$\therefore$ In set of complex numbers solution set of x^2-2x+5=0 is the set of {1-2i, 1+2i}

Answer C

24. Now recall (a+bi) (a-bi) = $a^2 + b^2$ and $\overline{a+bi}$ = a-bi

$\therefore \frac{4-3i}{3+4i} + \overline{1-2i} = \frac{4-3i}{3+4i} \cdot \frac{3-4i}{3-4i} + (1+21)$ (realizing denominator and Conjugate)

$= \frac{(4-3i)(3-4i)}{3^2+4^2} + (1+2i)$

$= \frac{12-16i-9i+12i^2}{25} + (1+2i)$

$= \frac{-25i}{25} + (1+2i)$ since $1^{-2} = -1$

= -I + 1+2i = <u>1+i</u>

Answer A

25. Given : - ZϵC Then we need solution set of $Z^3 - iz = 0$ in C

$\Rightarrow Z^3 - iz = 0$

$\Rightarrow Z(Z^2 - i) = 0$ (factorization)

This is true if z= 0 of $Z^2 - i = 0$

Then for $Z^2 - i = 0 \Rightarrow Z^2 = i \Rightarrow z = \sqrt{i}$

Now we need $\sqrt{i}$ in a+bi form

$\Rightarrow$ Let $\sqrt{i}$ = a + bi

$\Rightarrow$ i = a^2 - a^2 + 2abi

This is true if 0 = a^2 - b^2 and 1 = 2ab since I = 0+1-i

Then 1 = 2ab gives b = $\frac{1}{2a}$ replacing in a^2 - b^2= 0

$a^2 - (\frac{1}{2a})^2 = 0 \Rightarrow 4a^4 - 1 = 0 \therefore a = \pm\frac{1}{\sqrt{2}}$

Then $b^2 = a^2 \Rightarrow |b| = |a| \Rightarrow b = \pm a$

If a = $\frac{1}{\sqrt{2}}$ then b = $\frac{1}{2a} = \frac{1}{2(\frac{1}{\sqrt{2}})} = \frac{1}{\sqrt{2}}$

$\sqrt{i}$ = a + bi will bew

$\sqrt{i} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$ or $\sqrt{i} = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$

$\therefore \sqrt{i} = \pm\frac{1}{\sqrt{2}} \pm \frac{1}{\sqrt{2}}i = \pm\frac{1}{\sqrt{2}}(1+i)$

$\therefore S.S = \left\{0, \pm\frac{1}{\sqrt{2}}(1+i)\right\}$

Answer D

26. Given: sequence a_n= n+cos nπ, nϵIN as n^{th} term formula . Then

A. Fifth term = a_5 = 5+cos 5π = 5+(1) = 4(not 6)

B. 10^{th} term = a_{10} = 10+ cos 10π=10 + cos 5(2 π) = 10 + cos 0

= 10+1 = 1 (not 9)

C. $a_7 + a_8$ = 7+cos 7π + 8 + cos 8π

= 15+ cos (π + 3(2 π) + cos (0+4(2 π)

= 15+ cos π + cos 0= 15+ (-1) +1= 15 (True)

D. $a_7 + a_6$ = 7+cos 7 π – (6 + cos 6 π-(6+cos 6 π)

= 7+ cos (π+3(2 π) – 6 cos (0+3(2 π)) = 7+ cos π-6-cos 0

= - 1 (not 1)

Answer C

27. Arithmetic progression $\{A_n\}$ with $A_1 = 5_1$ $A_5 = 21$

Then we need value of $\sum_{n=1}^{20} A_n$ (the sum of the 1st 20 terms)

$\Rightarrow$ Common difference d = $\frac{A_m - A_n}{m-n}$ if A_m , A_n are given

$\Rightarrow$ d = $\frac{21-5}{5-1}$ for $A_m = A_5 = 21$ and $A_n = A_1 = 5$

= 4 (common difference)

Then using $\sum_{k=1}^{n} A_k = \frac{n}{2}(2A_1 + (n-1)d)$ for n = 20, A_1 = 5, d= 4

$\Rightarrow \sum_{n=1}^{20} A_n = \frac{20}{2}(2(5) + (20-1)4$

= 10(86) = 860

Answer C

28. $\sum_{n=0}^{\infty} 2^n 2^{2-n} = \sum_{n=0}^{\infty} \frac{2^n \cdot 3^2}{3n} = \sum_{n=0}^{\infty} 9(2/3)^n = 9\sum_{n=0}^{\infty} (2/3)^n$ (geometric series). B/C

Now consider $G_n = (2/3)^n$ Then $G_{n+1} = (2/3)^{n+1}$

Then $\frac{G_{n+1}}{G_n} = \frac{(2/3)^{n+1}}{(2/3)^n} = 2/3$ (common ratio)

$\Rightarrow$ r= 2/3 since $|r| < 1$ the series converges

And $G_o = (2/3)^0 = (2/3)^0 = 1$(at n=0, Lower limit)

$\therefore 9\sum_{n=0}^{\infty} (2/3)^n = 9\left(\frac{G_o}{1-r}\right)$ (we use G_o as first element / term)

$= 9\left(\frac{1}{1-2/3}\right) = 27$ Answer C

29. Monthly profit = 3000 Birr $\Rightarrow$ yearly = 12x3000 = 36000 Birr

$\Rightarrow$ it is arithmetic since increment is constant 36,000 yearly

Then if initial investment = A_o, Total amount in 5 year = Birr 230,000 (given)

Then if 1st year profit = A1 = 36,000

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2^{nd} year profit $= A_2 = A_1 + 36{,}000$ ———

In 5 year then total profit, $P_T = 5A_1$ (constant profit)

$(5(36{,}000) = 180,\ 0000$ (total profit)

$\therefore$ Total amount = initial + Total profit (at 5^{th} year)

$230{,}000 = A_o + 180{,}000 \Rightarrow 50{,}000$

Answer B

30. $\lim_{x\to\infty} \frac{3\sqrt{n}-6n+5}{4n+1} = \lim_{x\to\infty} \frac{\frac{3\sqrt{n}}{n} - \frac{6n}{n} + \frac{5}{n}}{\frac{4n}{n} + \frac{1}{n}}$ (convergency theorem)

$= \lim_{x\to\infty} \frac{\frac{3\sqrt{n}}{n} - \frac{6n}{n} + \frac{5}{n}}{4+h} = \frac{\lim_{x\to\infty}\left(\frac{3}{\sqrt{n}} - 6 + \frac{5}{n}\right)}{\lim_{x\to\infty}\left(\frac{3}{\sqrt{n}}\frac{1}{n}\right)}$

$= \frac{\lim_{x\to\infty}\frac{3}{\sqrt{n}} - \lim_{x\to\infty} 6 + \lim_{x\to\infty}\frac{5}{n}}{\lim_{x\to\infty} 4 + \lim_{x\to\infty}\frac{1}{n}} = \frac{0-6_0}{4+0} = \frac{-3}{2}$

Answer A

31. $\lim_{x\to 1^-} \frac{x^4-1}{|x-1|} = ?$, here $x \neq 1$

Let $f(x)\ \frac{x^4-1}{|x-1|}$ since $|x-1| = \begin{cases} x-1\ for\ x>1 \\ 1-x\ for\ x<1 \end{cases}$

Also $f(x) = \begin{cases} \frac{x^4-1}{x-1}, it\ x>1 \\ \frac{x^4-1}{x-1}, it\ x<1 \end{cases} = \begin{cases} (x+1)(x^2+1), x>1 \\ -(1-x)(x^2+1), x<1 \end{cases}$

$[because\ x^4-1 = (x^2-1)(x^2+1)$

$= (x-1)(x+1)(x^2+1)]$

Therefore $f|x1| = -(x+1)(x^2+1)$ is we need that is $x<1$ or $x\to 1^-$

$\therefore \lim_{x\to 1^-} \frac{x^4-1}{|x-1|} = \lim_{x\to 1^-} -(x+1)(x^2+1) = -(1+1)(1^2+1) = -4$

Answer D

32. For $f(x) = \begin{cases} \frac{a}{x^2+3}, if\ x \leq 1 \\ \frac{x-1}{\sqrt{x-1}}, if\ x>1 \end{cases}$ to be continuous at x+1

i) f(1) must be defined. Yes it is defined

Since $f(1) = \frac{a}{1^2+3} = \frac{1}{4}a$(is real value)

ii) $\lim_{x\to 1} f(x)$ must exist . Here since f is junction

at x =1 we assume side limits exist and are equal as follows

$\Rightarrow \lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x)$

$\Rightarrow \lim_{x\to 1^-} \frac{a}{x^2+3} = \lim_{x\to 1^+} \frac{x-1}{\sqrt{x-1}}$; $(x\to 1^- \equiv x<1,\ x\ x\to 1^+ \equiv x>1)$

In left side it takes x = 1 but in the right side

It is $\frac{0}{0}\ form\ \frac{x-1}{\sqrt{x-1}} = \frac{x-1}{\sqrt{x-1}} x \frac{\sqrt{x+1}}{\sqrt{x+1}} = \frac{(x-1)(\sqrt{x+1}}{x-1} = \sqrt{x}+1$

Then $\lim_{x\to 1^-} \frac{a}{x^2+3} = \lim_{x\to 1^+} \frac{x-1}{\sqrt{x-1}}$ will be

$\Rightarrow \lim_{x\to 1^-} \frac{a}{x^2+3} = \lim_{x\to 1^+} \sqrt{x}+1$ (now put x = 1)

$\Rightarrow \frac{a}{x^2+3} = \sqrt{x}+1 \Rightarrow \frac{a}{4} = 2 \Rightarrow \underline{a=8}$

iii) no need to see if $f(x) = \lim_{x\to 1} f(x)$ (3^{rd} condition of continuity)

Answer D

33. $\lim_{x\to\infty} \frac{xCSC3x}{x+1} = ?$

Let $f(x) = \frac{xCSC3x}{x+1}$ Then f(0) is $\frac{\infty}{1}$ form since denominator is non – zero let us search equibalent espression taking x = 0 (but $x\neq 0$)

$f(x) = \frac{xCSC3x}{x+1} = \frac{x.\frac{1}{\sin 3x}}{x+1} = \frac{x}{(x+1)\sin 3x}$ (because $CSC = \frac{1}{sin}$

$= \frac{3x}{(2+1)3\sin 3x}$ (multiplying by $\frac{3}{3}$)

$= \frac{1}{3(x+1)} \cdot \frac{3x}{\sin 3x}\ for\ x \neq 0 (or\ x \neq n\pi))\ for\ n \in$

Now $\lim_{x\to 0} f(x) = \lim_{x\to 0} \frac{xCSC3x}{x+1} = \lim \frac{1}{3(x+1)} \cdot \frac{3x}{\sin 3x}$

$= \lim_{x\to 0} \frac{1}{(x+1)} \cdot \lim_{x\to 0} \frac{3x}{\sin 3x}$ (since both exist)

$= \frac{1}{3(0+1)} .1$ because $\lim_{x\to 0} \frac{ax}{\sin ax} = \lim_{x\to 0} \frac{1}{\frac{\sin ax}{ax}}$

$= \frac{1}{3}$

$= \frac{1}{\lim_{x\to 0} \frac{\sin ax}{ax}} = \frac{1}{\lim_{y\to 0} \frac{\sin y}{y}}$

$= \frac{1}{1} = 1$, for y= ax

Answer A

34. $\lim_{y\to\infty} \left(1+\frac{1}{x}\right)^{2x+4} = \lim_{y\to 0} \left(1+\frac{1}{x}\right)^{2x} \cdot \left(1+\frac{1}{x}\right)^4$ (by $a^{m+n} = a^m . a^n$)

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$= \lim_{y\to 0}\left((1+\frac{1}{x})^x\right)^{2x}\cdot\left(1+\frac{1}{x}\right)^4$ (by $(a^n)^m = a^{nm}$)

$= \lim_{y\to 0}\left((1+\frac{1}{x})^x\right)^{2}\cdot \lim_{y\to \infty}\left(1+\frac{1}{x}\right)^4$ (Both limits exist)

$= \left[\lim_{y\to \infty}(1+\frac{1}{x})^x\right]^2\cdot \lim_{y\to \infty}\left(1+\frac{1}{x}\right)^4$ (Continuity of fog)

$= e^2 . (1+0)^4 = e^2$

Answer C

35. $f(x) = \pi^2$ implies f is constant function whose derivative is zero
$f^1(1) = 0$

Answer D

36. $f(x) = \frac{6x}{x+a}$ Then we need a so that $f^1(a) = 1$,
so find $f^1(x)$ and set $f^1(a) = 1$

$\Rightarrow f^1(x) = \left(\frac{6x}{x+a}\right) = \frac{(6x)^1(x+a)-(x+a)^1(6x)}{(x+a)^2}$

$= \frac{6(x+a)-6x}{(x+a)^2} = \frac{6a}{(x+a)^2}$

Answer C

37. A) False, since continuous functions may fail
Differentiability, for instance $f(x) = |x|$ is continuous at x=0, but $f^1(0)$ doesn't exist
Also functions like $g(x) = |in\ x|$

At x = 1,

38. $g(x) \frac{f(x)}{x+1} \div (f(x))^2$, f(x)=8, $f^1(x)$=2 (given)
Then $g^1(x)$=?. Let's first evaluate $g^1(x)$ Then put x = 1

$\Rightarrow g^1(x) = \left(\frac{f(x)}{x+1} + (f(x))^2\right)^1$ (sum rule),$(f+g)^1 = f^1+g^1$

$\Rightarrow \frac{f^1(x)(x+1)-(x+1)^1 f(x)}{(x+1)^2} + 2f(x).\ f^1(x)$ [Quotient rule $\left(\frac{t}{p}\right)^1 = \frac{f^1g-g^1f}{p^2}$
Plus, and chain rule $(f^2)^1 = 2f.\ f^1$]

$\therefore\ g^1(x) = \frac{(x+1)f^1(x)-f(x)}{(x+1)^2} + 2f(x)\ f^1(x)$

Now putting x = 1, $g^1(1) = \frac{(x+1)f^1(1)-f(1)}{(1+1)^2} + 2f(1)f^1(1)$

$= \frac{2f^1(1)-f(1)}{4} + 2f(1)\ f^1(1)$

$= \frac{2(2)-8}{4} + 2(8)(2)$ (given)

$= \underline{31}$

Answer B

39. $f(x) = xe^{3x} - \cos(2x)$ then to find $f^{11}(0)$ first find $f^1(x)$

$\Rightarrow f^1(x) = (xe^{3x} - cos(2x))^1$

$\Rightarrow f^1(x) = (xe^{3x})^1 - (\cos(2x))^1$ (since $(f-g)^1 = f^1-g^1$)

$= (x)^1e^{3x}+(e^{3x})^1x - (-\sin(2x)^1$

$= [wing(f.\ g)^1 = f^1g + g^1f$ product rule and $(f(g(x)))^1 = f^1(g(x)).\ g^1(x)$, chain rule]

$f^1(x) = e^{3x} + 3e^{3x}x + 2\sin 2x$ (since $(e^{3x})^1 = e^{3x}(3x)^1 = 3e^{3x}$)

Again $f^{11}(x) = (f^1(x))^1 = (e^{3x} + 3xe^{3x} + 2\sin 2x\ )^1$

$= (e^{3x})^1 + (3xe^{3x})^1 + (2\sin 2x)^1$ (since $(f+g)^1 = f^1 + g^1$)

$= (e^{3x}(3x)^1 + (3x)^1 + e^{3x} + (e^{3x})^1 3x + |(2\cos 2x)\ (2x)^1$

[Chain rule + product rule + chain rule]

$= 3e^{3x} + 3e^{3x} + (3x)^1\ e^{3x}3x + 4\cos 2x$ [chain rule on $(e^{3x})^1$]

$\therefore\ f^{11}(x) = 6e^{3x} + 9xe^{3x} + 4\cos 2x$

Then for x = 0 $f^{11}(0) = 6e^{3(0)} + 9(0)e^{3(0)} + 4\cos 2(0)$

$= 6+0+4(1) = 10$

Answer D

40. $f(x) = \frac{1}{3}x^3 - \frac{1}{2}x^2 - 2x + 1$

local extremes are at x where $f^1(x) = 0$ since
$f^1(x)$ Exist for all X∈IR.

$\Rightarrow f^1(x) = x^2 - x - 2$ Then $f^1(x) = 0 \Rightarrow x^2 - x - 2 = 0$

$\Rightarrow (x-2)(x+1) = 0 \Rightarrow x = 2$ or $x = -1$

Then by second derivative test

$f^{11}(x) = = 2x-1$

At $x = 2 \Rightarrow f^{11}(2) = 2(2)-1 = 3>0$

$f^{11}(2) > 0 \Rightarrow f(2)$ is local minimum value

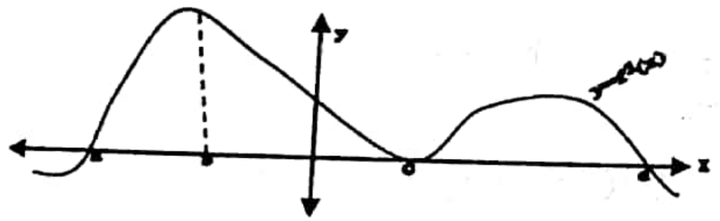
at $x = -1 \Rightarrow f^{11}(-1) = 2(-1) - 1 = -3<0$

$f^{11}(-1)<0 \Rightarrow f(-1)$ is local maximum value

Therefore f has local maximum at x=-1

Answer B

41. given The following graph of



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Let us put how $f^1(x)$ be haves in sign chart from the graph

	- ∞	a	b	c	d	∞
$f^1(x)$	---	+++	++++	++++	---	

$f^1(x) > 0$ for x ∈ (a,c) ∪(c,d) so here f(x) is increasing

A) *True, because at* $x = c$ *and* $x = d$, $f^1(x) = 0$ means increasing, or Decreasing. Because increasing implies $x_1 \leq x_2 \Rightarrow f^1(x_1) \leq f(x_2)$

or

F is increasing whenever $f^1(x) \geq 0$, only.

∴ f is increasing in [a,d] and its subsets like [c,d] (since *is* above x – Axis means $f^1(x) \geq 0$

B) False, since $f^1(x)$ is non – negative between a and d f is non-Decreasing everywhere in [a, d].

C) At x =d $f^1(x)$ changes from +ve to –ve, Therefore f(d) is local Maximum value. (Not minimum).

E) False, because $f^1(x)$ has no sign change. There fore since $f^1(x)$ exist (C, f(c)is stationary point).

Answer A

42. Let A – shows Area

Then given i) A (rectangle) = ℓw (initially)

ii) A (reserve) = w^2 (needed area)

iii) $2w+2\ell = 400$ (perimeter of initial rectangle)

∴ $w + \ell = 200 \Rightarrow \boxed{w = 200 - \ell}$

Then we need to maximize, the remaining area, A= A (rectangle) – A (reserve)

$\Rightarrow A(\ell, w) = \ell w - w^2$ Then since $w = 200\ \ell$

$A^1(\ell) = \ell(200 - \ell) - (200 - \ell)^1$

$= 2(200 - \ell) + (-1)(2\ell - 200)$ (productive rule)

$A^1(\ell) = 600 - 4\ell$

Then $A^1(\ell) = 0$ implies $600 - 4\ell = 0 \Rightarrow \ell = \underline{150}$

∴ Remaining area is maximum if $\ell = \underline{150}$

Answer D

43. $F(x) = \ell n(x^2 + 1)$ concave up ward if $f^{11}(x) > 0$

$\Rightarrow f^1(x) = (\ell n\, x^2 + 1)^1$

$= \frac{1}{x^2+1}(x^2+1)^1$ (chain rule)

$f^1(x) = \frac{2x}{x^2+1}$

Again $f^{11}(x) = (f^1(x))^1 = \left(\frac{2x}{x^2+1}\right)^1$

$= \frac{(2(x))^1(x^2+1) - (x^2+1)^1 2x}{(x^2+1)^2}$ (Quotient rule)

$= \frac{2x^2+2-4x^2}{(x^2+1)^2} = \frac{2-2x^2}{(x^2+1)^2}$

Then fore $f^{11}(x) = \frac{2-2x^2}{(x^2+1)^2}$

Then $f^{11}(x) >= \Rightarrow 2 - 2x^2 > 0$, $(x^2 + 1)^2$ is always =ve

$\Rightarrow 2 > 2x^2 \Rightarrow x^2 < 1$

$\Rightarrow \sqrt{x^2} < \sqrt{1} \Rightarrow |x| < 1;\ -1 < x < 1$

∴ f is concave up in (-1 1, 1)

But in the choice it is given [-1, 1] including inflection points' value of x, which is possible to say but not logical ' Answer D

44. Volume V, of cylinder with radius r (of base) and height h is $\boxed{V=\pi r^2 h}$ we need is $\frac{dh}{dt}$?

⇒ The water poured in to it means only h, increases (r is constant)

Therefore change of V is as a function of change of

$\Rightarrow \frac{dv}{dh} = \frac{d}{dh}(\pi r^2 h) =) \pi r^2 \frac{dh}{dh} = \pi r^2$ ---

$\therefore \frac{dv}{dh} = \pi r^2$ ---- ① ∴ $\boxed{V^1(h) = \pi r^2 = 25\pi, \text{ for all h}}$

Then for the related rate for V → h → t

$\frac{dv}{dt} = \frac{dv}{dh} \cdot \frac{dh}{dt}$ (Chain rule)

$\Rightarrow \frac{dv}{dt} = \pi r^2 \cdot \frac{dh}{dt}$ (Using result of①)

$\Rightarrow \frac{dh}{dt} = \frac{1}{\pi r^2} \cdot \frac{dv}{dt}$ (Dividing by πr^2)

$\Rightarrow \frac{dh}{dt} = \left(\frac{1}{\pi(5m)^2}\right) \cdot \left(10\frac{m^3}{min}\right)$ Given $\frac{dv}{dt} = 10\frac{m^3}{min}$, r =5m)

$= \frac{2}{5\pi}$ m/min Answer B

45. f(x) = tan x, Anti –derivative is element of indefinite integral. so find its indefinite integral first

$\Rightarrow \int f(x)dx = \int \tan x dx \int \frac{\sin x}{\cos x} dx$

Now let cos x = U ⇒ - sin x dx = dU

⇒ Sin x ds = -dU

Then $\int \frac{\sin x}{\cos x} dx = \int \frac{-dU}{U} = -\int \frac{dU}{U} = \ell n|U| + c$

$\therefore \int \tan x\, dx = -\ell n|\cos x| + c$

$= \ell n|\cos x|^{-1} + c$

$= \ell n\left|\frac{1}{\cos x}\right|^{-1} + c$

$= \ell n|\sec x| + c$

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If we take c=0 we get one of its anti-derivative $\ell n|sec|$

$\Rightarrow$ If we take $\ell n|sec|$, odd sec x>0

Though D is not full answer, we say → Answer D

46. $\int \frac{x}{x^2+2x+1}dx = \int \frac{x}{(x+1)^2}\,dx = \int \frac{A}{x+1}\,dx + \int \frac{B}{(x+1)^2}\,dx$

$= A\int \frac{dx}{x+1} + \quad B\int (x+1)^{-2}\,dx$

$+ A|n|\,|x+1| + B\frac{(x+1)^{-1}}{-1} + c$

$= A|n|\,|x+1| - B\,(x+1)^{-1} + c$ - - - ①

Then to get A and B decompose $\frac{x}{(x+1)^2}$ to latrial fraction as

$\frac{x}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2} = \frac{A(x+1)+B}{(x+1)^2} = \frac{Ax+A+B}{(x+1)^2}$

$\Rightarrow x = Ax + A + B \;\therefore A = 1$ and $A + B = 0$

$\Rightarrow B = -A = -1$ (Then replacing in①)

$\therefore \int \frac{x}{(x+1)^2}\,dx = |n|\,|x+1| - \frac{1}{x+1} + c$ Answer B

47. $\int \frac{e^{2x}-4x}{xe^{2x}}\,dx = \int \left(\frac{e^{2x}}{xe^{2x}} - \frac{4x}{xe^{2x}}\right)dx$

$= \int \left(\frac{1}{x} - \frac{4}{e^{2x}}\right)dx$

$= \int \left(\frac{1}{x}\,dx - \right)\frac{4}{e^{2x}}\,dx$

$= |n|x]\,C_1 - 4\int e^{-2x}\,dx$ *for some* C_1

Let $-2x = y \Rightarrow -2dx = dy \Rightarrow dx = \frac{1}{-2}dy$

$= |n|x]\,C_1 - 4\int e^{y}\left(-\frac{1}{2}\,dy\right)$

$= |n|x]\,C_1 + 2\int e^{y}\,dy$

$= |n|x]\,C_1 + C_2$, for some C_2

$= |n|x]\,C_1 + 2e^{-2x} + (2$

$= |n|x] + 2e^{-2x} + C$ for $C = C_1 + C_2$ Answer A

48. $\int_0^1 \frac{(x_1)^2}{x+1}dx = \int_0^1 (x\text{-}3) + \frac{4}{x+1})\,dx$ (by long division)

$= \int_0^1 (x_3)\,dx + 4\int_0^1 \frac{1}{x+1}\,dx$ (Addition on definit integeral

$= \left(\left[\frac{x^2}{2} - 3x\right]\right)_0^1 + 4\,(|n|x+1]$

$= \left(\frac{1^2}{2} - 3(1)\right) + 4\,(\ell n2 - 1n1)$

$= \frac{-5}{2} + 4\,\ell n2$, because $|n| = 0$ Answer C

49. $f(x) = \begin{cases} \sin x, x \le 0 \\ 3x\sqrt{x^2+1}, x > 0 \end{cases}$

First let's just find $\int f(x)$ *for both pieces*

$\int \sin x dx = -\cos x + C_1$, for som C_1 this is for $x \le 0$

$\int 3x\sqrt{x^2+1}dx$ let $x^2 + U \Rightarrow 2xdx = dU$

$\Rightarrow xdx = \frac{1}{2}\,dU$

$= 3\int \sqrt{x+1}\,xdx = 3\int \sqrt{x+1}\,xdx$; $2xdx =$

Then $\int 3x\sqrt{x^2+1}\,dx = 3\int \sqrt{x^2+1}\,xdx = 3\int \sqrt{U}\,\frac{1}{2}\,dU = \frac{3}{2}U^{\frac{1}{2}}dU$

$= \frac{3}{2}\left(U^{\frac{1}{2}}\right)dU$

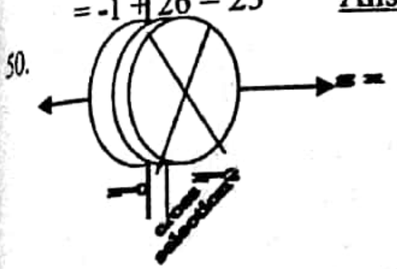
$= (x^{2+1})^{\frac{3}{2}} + C_2$ since $U = x^2 + 2$

$\boxed{\int 3x\sqrt{x^2+1}\,dx = (x^{2+1})^{\frac{3}{2}} + C_2}$ This is for x> 0

Then $\int_{\frac{-\pi}{2}}^{\sqrt{8}} f(x)dx = \underbrace{\int_{\frac{-\pi}{2}}^{0} f(x)dx}_{x \le 0} + \underbrace{\int_{0}^{\sqrt{8}} f(x)d}_{x > 0}$ + since $0 \in \left(\frac{-\pi}{2}, \sqrt{8}\right)$

$= [-\cos x_C_1\,]^0 + \left[(x^2 + 1^{3/2} + (2\right]_0^{\sqrt{8}}$

$= -1 + 26 = 25$ Answer B

50. 

The solid formed that is with disc cross section ofradius

$r(x)\ \sqrt{x+1}$

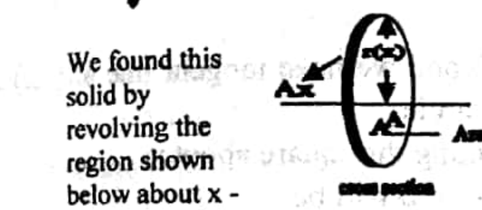
$\therefore A(x) = \pi\,(r(x))^2$

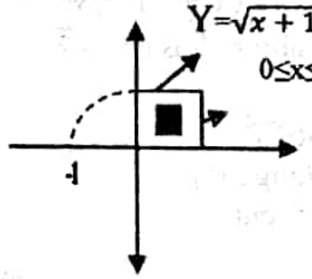
$A(x) = \pi\left(\sqrt{x+1}\right)^2$

$= \pi\,(x+1)$

$\therefore \Delta V = A(x)\,\Delta x$

We found this solid by revolving the region shown below about x - axis





$\int_2^0 A(x)dx = \int_2^0 \pi(x)dx = \left[\pi\left(\frac{x}{2}\right)^2 + x\right]_0^2$

$= 4\pi$

Answer A

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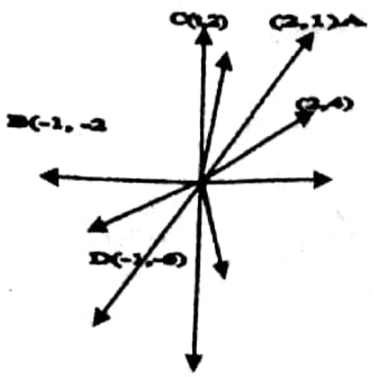
51. The vector $\vec{V}$ (a,b) lies on the line whose slope is $m=\frac{b}{a}$

Then the slope of the line through (0,0) and (2,4) is $m = \frac{\Delta y}{\Delta x} = \frac{4-0}{2-0} = 2$



$\therefore \frac{b}{a} = 2 \Rightarrow b = 2a$ (as in the figure)

Therefore any vector of the form

$\vec{V} = ca, 2a)$ for $a \neq 0$

$= a(1,2)$

So for $a = -1 \Rightarrow \vec{V} = -1\ (1,2) = (-1, -2)$

$\therefore$ Answer B

52. Given: $\vec{i}$ and $\vec{j}$ standard unit vectors in +ve x and y direction vector $\overrightarrow{AB}$ for A= (-1, 0) , B = (2,2)

Then we need unit vector in the direction of V that is $\frac{\vec{V}}{|\vec{V}|}$, such that $\vec{V} = 3\overrightarrow{AB} - 3\vec{i} + 2\vec{j}$

$\Rightarrow$ Then $\overrightarrow{AB} = (x_B - x_A, y_A - y_B)$

$= (2 - (-1), 2-0) = (3,2)$

$= 3\vec{i} + 2\vec{j}$ (in component form)

$\Rightarrow \vec{V} = 3\overrightarrow{AB} - 3\vec{i} + 2\vec{j}$

$= 3(3\vec{i} + 2\vec{j}) - 3\vec{i} + 2\vec{j} = 6\vec{i} + 8\vec{j}$

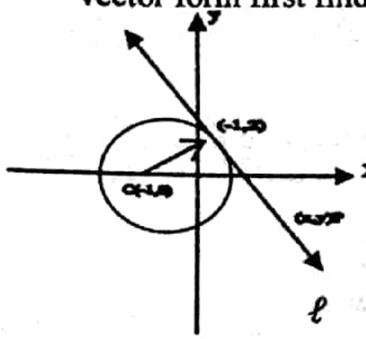
Then $|\vec{V}| = \sqrt{6^2 + 8^2}\ 10$

$\therefore \frac{\vec{V}}{|\vec{V}|} = \frac{6\vec{i} + 8\vec{j}}{10} = \frac{3}{5}\vec{i} + \frac{4}{5}\vec{j}$

$= \left(\frac{3}{5}, \frac{4}{5}\right)$

Answer A

53. Given circle C: $x^2 + y^2 + 2x - 7 = 0$ and we need tangent line at (1,2) in vector form first find center of the circle



$\Rightarrow$ by completing the square about x

$x^2 + y^2 + 2x - 7 = 0$ will be

$x^2 + 2x + 1 - 1 + y^2 - 7 = 0 \Rightarrow (x+1)^2 + y^2 = 8$

Center at (h, k) = (-1, 0) and radius $r = 2\sqrt{2}$

Then as in the figure

The directed line segment $\overrightarrow{CT}$

(From center to point of tangency)

Is perpendicular to the tangent

Line ℓ . Therefore for any

$P = (x, y)$ on ℓ $\boxed{\overrightarrow{CT} \cdot \overrightarrow{TP} = 0}$

$\overrightarrow{CT} = (x_T - x_C, y_T - y_C) = (1-(-1), 2-0) = (2,2)$

$\overrightarrow{TP} = (x - x_C, y - y_C) = (x-1, y-2)$

$\Rightarrow 2(x-1) + 2(y-2) = 0 \Rightarrow 2x + 2y - 6 = 0$

ℓ is $x + y - 3 = 0 \Rightarrow \boxed{y = -x+3}$

ℓ is $(x, y) = (a_o, b_o)$ is on $y = -x + 3$ and (a, b) is any point on $y = -x$ (parallel to $y = -x+3$ through (90))

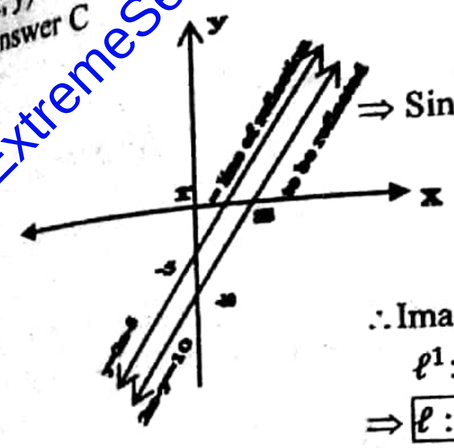
$\Rightarrow$ if $x = \lambda$ then $y = -\lambda \Rightarrow (\lambda, -\lambda)$ is on $y = -x$

$\lambda(1, -1)$ is in the direction of $y = -x + 3$

Then taking (0,3) on $y + -x+3$

$(x, y) = (0, 3) + \lambda(1,-1)$ is vector equation of ℓ

Answer C



$\Rightarrow$ Since ℓ is parallel to ℓ_R also its Image is parallel to ℓ and have Slope = 2 and x-intercept (5,0) Will go to (0,0) shifting 5/2 left Of ℓ_R

$\therefore$ Image in reflection found is

$\ell^1 : \frac{y-0}{x-0} = 2 \Rightarrow y = 2x$

$\Rightarrow \boxed{\ell : -2x + y = 0}$

Again to rotate take two points on $-2x+y = 0$

Let (0, 0) and (1,2)

Rotation of these points about (0,0) by 90^o gives $R_{90^o}(0,0) = (0,0)$

$R_{90^o}(1,2) = (x\cos\theta - y\sin\theta\ ,\ x\sin\theta + y\cos\theta)$

$= (\pi/2 - 2\sin\pi/2\ ,\ 1\sin\pi/2 + 2\cos\pi/2\ )$

$= (-2, 1)$

Then the line through (0, 0) and (-2, 1) is $\frac{y-0}{x-0} = \frac{1-0}{-2-0} \Rightarrow x + 2y = 0$

Answer A

55. $\csc(\pi/2 - x)\cos x - \sin^2 x$

$= \frac{1}{\sin(\pi/2 - x)}\cos x - \sin^2 x$ since $\csc = \frac{1}{\sin}$

$= \frac{1}{\sin\pi/2\ \cos x - \sin x\ \cos\pi/2}(\cos x) - \sin^2 x$ (since $\sin(A-B) =$ $\sin A\cos B - \sin\cos A$)

$= \frac{1}{\cos x - 0}$ $\cos x -$ $\sin^2 x$

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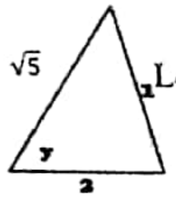
$= 1 - \sin^2 x = \cos^2 x$

Answer B

56. Cos (tan $^{-}$ (0,5)) = ____

Let tan y $= \frac{1}{2}$



Let y is angle as shown in the right – triangle. Then opp = 1, adj= 2

Since tan $= \frac{opp}{adj} = \frac{1}{2}$

$\Rightarrow$ Then hyp $= \sqrt{opp^2 + adj^2} = \sqrt{1^2 + 2^2} = \sqrt{5}$

$\therefore$ cos

57.

Using cosine law

$x^2 = 10^2 + (5\sqrt{6})^2 - 2(10)(5\sqrt{6})^2 \cos 60°$

$= 81$

$\therefore x = 9$

Answer B

58. The radius of the sphere is the distance from the center c(0,1,3) to any point on the surface. So given z-axis intersects at (0, 0, 1) is on the sphere

$\therefore r = \sqrt{(0-0)^2 + (1-0)^2 + (1-3)^2} = \sqrt{5}$

Answer D

59. End point of diameter are A and B means the center is mid- point of $\overline{AB}$

$\Rightarrow \frac{x_A + x_B}{2} = 0 \Rightarrow \frac{1+x_B}{2} = 0 \Rightarrow x_B = -1$

$\frac{y_A + y_B}{2} = 0 \Rightarrow \frac{0+y_B}{2} = 0 \Rightarrow y_B = 0$

$\frac{Z_A + Z_B}{2} = 2 \Rightarrow \frac{-2+Z_B}{2} = -2 \Rightarrow Z_B = -2$

$\therefore B = (x_B, y_B, Z_B) = (-1, 0, -2)$

Answer C

60. We can take the angle between position vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ since $\overrightarrow{OA}$ parallel to ℓ_1 and $\overrightarrow{OB}$ parallel to ℓ_2 for O = (0, 0, 0). For θ is angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$ by dot product

Then $\cos\theta = \frac{\overrightarrow{OA}.\overrightarrow{OB}}{|\overrightarrow{OA}||\overrightarrow{OB}|} = \frac{(1,1,0).(0,1,1)}{|(1,1,0)|\,|(0,1,1)|}$

$= \frac{1(0)+1(1)+0(1)}{\sqrt{1^2+1^2+0^2}.\sqrt{0^2+1^2+1^2}}$

$\therefore \cos\theta = \frac{1}{2} \Rightarrow \theta = \cos^{-1}\left(\frac{1}{2}\right) \Rightarrow 60°$

Answer C

61. If θ is angle between $\vec{a}$ and $\vec{a} - \vec{b}$ Then

$\cos\theta = \frac{\vec{a}.(\vec{a}-\vec{b})}{|\vec{a}|.|\vec{a}-\vec{b}|} = \frac{(-\vec{i}+3\vec{j}).(-\vec{i}+3\vec{j}-(-\vec{i}+\vec{j}))}{|-\vec{i}+3\vec{j}||-\vec{i}+3\vec{j}-(-\vec{i}+\vec{j})|}$

$= \frac{(-1)(01)+0(-1)+3(3)}{\sqrt{(-1)^2+0^2+3^2}} = \frac{9}{10}$ Answer A

62. Given: $\vec{V} = 3\vec{i} - 4\vec{j}$ for $\vec{i}$ and $\vec{k}$ are standard unit vectors in +ve X and Z direction

$\therefore \overrightarrow{AB}$ is vector from A = (0, 1, 2) to B in space

$|\overrightarrow{AB}| = 10$ and $\overrightarrow{AB} \,//\, \vec{V}$

Then we need B.

Let B $= (x_B, y_B, Z_B)$

Then $\overrightarrow{AB} = (x_B - x_A, y_B - y_A, Z_B - Z_A)$

$= (x_B, -0, y_B - 1, Z_B - 2)$

$\Rightarrow$ Since $\overrightarrow{AB} \,//\, \vec{V}$ there exist k ≠0 such that

$\overrightarrow{AB} = k\,\vec{V}$

$\Rightarrow (x_B, y_B - 1, Z_B - 2) = K\,(3, 0, -4)$

$\Rightarrow x_B = 3k,\ y_B = 1,\ Z_B = -4k + 2$

Again since $|\overrightarrow{AB}| = 10$

$\Rightarrow \sqrt{(x_B - x_A)2 + (y_B - y_A)2 + (Z_B - Z_A)2} = 10$

$\Rightarrow \sqrt{(3k-0)^2 + (1-1)^2 + (-4k+2-2)^2} = 10$

$\Rightarrow \sqrt{9k^2 + 16k^2} = 10 \Rightarrow 5\sqrt{k^2} = 10$

$\Rightarrow \sqrt{k^2} = 2 \Rightarrow k = \pm 2$

$\therefore k = 2 \Rightarrow B = (3k, 1, -4k+2)$

$= (6, 1, -6)$

$K = -2 \Rightarrow B = (-6, 1, 10)$

Since (6, 1, -6) is given C $\Rightarrow$ answer C

63. U = {0, 1, 2, 3, 4, ----}

A) $(\forall x)(\exists y)(y > x^2 - 1)$

$y > x^2 - 1 \Rightarrow x^2 < y+1$

So $\forall x \ni y \equiv$ for any $x \in U$ you can get some y in U $\therefore$ True

B) $(\exists x)(\forall y)(y > x^2 - 1)$

$y > x^2 - 1 \Rightarrow x^2 < y+1$

So $(\exists x)(\forall y)$ which is true since Y?

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C) $\exists\, y \forall x \equiv$ There is unique y such that for any x it holds but it is FALSE for instance let x=0

D) $(\exists y)(\forall x)(y \leq x^2 - 1)$ Here enough to obtain at least one example satisfying the statement.

Let x=3, y=2 then $2 \leq 3^2\text{-}1 \Rightarrow 2 \leq 8$ True

Answer C

64. If a statement is true for all n=x+1 whenever it is true for ii) n= 1 and ii) n = x. Then this method is called principle of mathematical induction.

65. Normally $a_n = n^2 - n$ is multiple of 2 for NEIN

Proof (by mathematical induction)

i) Let n= 1 $\Rightarrow a_1 = 1^2 - 1 = 0$ (True)

ii) Assume a_k (for n=k) is divisible by 2

$\Rightarrow a_k = 2m \Rightarrow$ $k^2\text{-}k=2m$ (I.H)

Now we need to check for n= k+1

$\Rightarrow a_{k+1} = (k+1)^2 - (k+1)$

$= k^2 + 2k + 1 - k - 1$

= 2m +2k because from inductive hypothesis

(I.H) $k^2 - k = 2m$

= 2(m+k) divisible by 2

Answer C

But $a_n \neq 3m$ for m (show this using the same step shown above)

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BOOLET CODE:10

1. $a_1 = 2, and\ a_{n+1} = a_n + 4 \quad n \geq 1,$

Then since $a_{n+1} - a_n = 4$ (constant), this is arthimetic with common difference d=4 (by definition).

Hence using $s_n = \frac{n}{2}(a_1 + a_n)$ (n^{th} partial sum)

$$S_{35} = \sum_{n=1}^{35} a_n = \frac{35 - 1 + 1}{2}(a_1 + a_{35}), \text{ then using } a_n = a_1 + (n-1)d$$

$$= \frac{35}{2}(2a_1 + (n-1)d)\ using\ a_1 = 2, n = 35, d = 4$$

$$= \frac{35}{2}(2a_1 + (n-1)d), using\ a_{1=} = 2, n = 35, d = 4$$

$$= \frac{35}{2}(2(2) + (35-1)4)$$

$$= 2450$$

Ans. C

2. $f(x) = x^2 + 2\ell nx$, then

$$\lim_{h \to 0} \frac{f(2+h) - f(2)}{h} = \lim_{h \to 0} \frac{(2+h)^2 + 2\ell n(2+h) - (2^2 + 2\ell n2)}{h}$$

$$= \lim_{h \to 0} \frac{4 + 4h + h^2 + \ell n(2+h)^2 - 4 - \ell n4}{h}$$

$$= \lim_{h \to 0} \left(\frac{4h + h^2}{h} + \frac{\ell n(2+h)^2 - ln4}{h} \right)$$

$$= \lim_{h \to 0} \frac{h(4+h)}{h} + \lim_{h \to 0} \frac{\ell n(2+h)^2 - \ell n4}{h}$$

$$= \lim_{h \to 0} 4 + h + \lim_{h \to 0} \frac{1}{h} \ell n \left(1 + \frac{h}{2}\right)^2, \text{(why?)}$$

$$= \lim_{h \to 0} (4+h) + \lim_{h \to 0} \ell n \left(1 + \frac{h}{2}\right)^{2/h}$$

$$= 4 + 0 + \ell n(\lim_{h \to 0} \left(1 + \frac{h}{2}\right)^{\frac{2}{h}}$$

$$= 4 + 0 + \ell n\left(\lim_{h \to 0}\left(1+\frac{h}{2}\right)^{\frac{2}{h}}\right), \text{(why?)}$$

$$= 4 + \ell n\left(\lim_{x \to 0}(1+x)^{\frac{1}{x}}\right) \text{ lefting } \frac{2}{h} = \frac{1}{x} \text{ or } x = \frac{h}{2}$$

$$\text{and } h \to 0 \Rightarrow x \to 0$$

$$= 4 + \ell n\ e, since \lim_{x \to 0}(1+x)^{\frac{1}{x}}) = e$$

$$= 4 + 1$$

$$= 5$$

Method II

Or shortly :- $\lim_{h \to 0} \frac{f(2+h)-f(2)}{h} = f'(2)$(derivative of f at $x = 2$)

Then $f'(x) = (x^2 + 2\ell nx)' = (x^2)' + (2\ell nx)'$

$$= 2x + 2\frac{1}{x}$$

Hence $f'(2) = 2(2) + 2 \times \frac{1}{2} = 4 + 1 = 5$

Ans. A

3. $f(x) = e^{2x} + x - 3\cos x,$

$f'(x) = [e^{2x} + x - 3cosx]'$ *(differentiating both sides)*

$= (e^{2x})' + x' - (3cosx)'$, *using* $(f+g)' = f' + g'$

$= 2e^{2x} + 1 + 3sinx,$

since $(e^{2x})' = (2x)'e^{2x} = 2e^{2x}$ *(chain rule)*

And $(3cosx)' = 3(cosx)' = (-sinx) = -3sinx$

Again

$f''(x) = (f'(x))' = (2e^{2x} + 1 + 3sinx)'$

$= (2e^{2x})' + 0 + 3(sinx)'$, *since* $(cf)' = cf'$

$= 2(2e^{2x}) + 3cosx$

$= 4e^{2x} + 3cosx$

$\therefore f''(x) = 4e^{2x} + 3cosx$

Ans. D

4. $f(x) = x^4 + 4x$ increases in $x \in I$ *(interval), where* $f'(x) \geq 0$

Then $f'(x) \geq 0 \Rightarrow (x^4 + 4x)' \geq 0$

$\Rightarrow 4x^3 + 4 \geq 0$

$\Rightarrow 4(x^3 + 1) \geq 0$

$\Rightarrow x^3 + 1 \geq 0$

Then since $x^3 + 1 = (x+1)(x^2 - x + 1) \geq 0$

Then $x^3 + 1 \geq 0 \Rightarrow (x+1)(x^2 - x + 1) \geq 0$

$\Rightarrow x + 1 \geq 0$ because $x^2 - x + 1 >$

0, $\forall x \in \mathbb{R}$

$\Rightarrow x \geq -1$

$\therefore f$ *is increasing in* $[-1, \infty)$

Ans. C

5. $|3+4i| - \frac{25i}{3+4i} = \sqrt{3^2+4^2} - \frac{25i}{3+4i} \times \frac{3-4i}{3-4i}$

$= 5 - \frac{75i+100}{3^2+4^2}$

$= 5 - \frac{75i+100}{25}$

$= 5 - (3i + 4)$

$= 1 - 3i$

Explanation

- $|a+bi| = \sqrt{a^2+b^2}$
- $\frac{a+bi}{c+di} = \frac{(a+bi)(c-di)}{c^2+d^2}$

Ans. D

6. $= (\cos\frac{\pi}{10} + i\sin\frac{\pi}{10})^5$, given $z = \cos\frac{\pi}{10} + \sin\frac{\pi}{10}$

$= cos\,5\left(\frac{\pi}{10}\right) + i\sin 5\left(\frac{\pi}{2}\right)$, *by demovries's rule*

$= cos\frac{\pi}{2} + isin\frac{\pi}{2}$

$= 0 + i(1)$

$= i$

Ans. C

7. $|x| + 2x = \begin{cases} -x + 2x \text{ if } x \\ x + 2x \text{ if } x \geq 0 \end{cases} = \begin{cases} x \text{ if } x < 0 \\ 3x \text{ if } x \geq 0 \end{cases}$

Hence for $x < 0, |x| + 2x = x$

Ans. D

8. Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ and $AX\text{-}A^T = 2A$, for 2×2 matrix X.

Then to find x, let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then since A is symmetric matrix, $A = A^T$

Then $AX - A^T = 2A$

$\Rightarrow AX = 2A + AT$

$= 2A + A$, *since* $A = A^T$

$= 3A$

Then $AX = 3A$ *gives*

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = 3\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1.a + 2.c & 1.b + 2.d \\ 2.a + 3.d & 2.b + 3.d \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 9 \end{bmatrix}.$$

Matrix multiplication in left and scalar multiplication in right.

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Then equality holds if corresponding entries are equal.
Hence $a+2c=3, b+2d=6, 2a+3c=b$ and $2b+3d=9$

From $\begin{cases} a+2d=6 \\ 2b+3d=9 \end{cases} \Rightarrow \frac{\begin{cases}2a+4c=6\\2a+3c=9\end{cases}}{c=0, and\ a=3}$

From $\begin{cases} b+2d=6 \\ 2b+3d=9 \end{cases} \Rightarrow \frac{\begin{cases}2b+4d=12\\2b+3d=9\end{cases}}{d=3, and\ b=0}$

Therefore $X=\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$

Another method

Using AX=3A, both side by left multiplying A^{-1}

$A^{-1}Ax=A^{-1}(3A)$

$\Rightarrow (A^{-1}A)x = A^{-1}(3A)$, Associatively of matrix product

$\Rightarrow I_2X = A^{-1}(3A), since\ A^{-1}A = I_2 (identity)$

$\Rightarrow X = A^{-1}(3A), since\ IX = X$

$\Rightarrow X = \frac{1}{|A|}.A\,(3A), using\ A = \frac{1}{|A|}\,A$, Where A is adjoint matrix (transpose of cofactor matrix)

Then $A^{-1} = \frac{1}{1.3-2.2}\begin{bmatrix} 3 & -2 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ 2 & -3 \end{bmatrix}$

and $3A = 3\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 9 \end{bmatrix}$

Hence $X = A^{-1}(3A)$

$= \begin{bmatrix} 3 & -2 \\ 2 & -3 \end{bmatrix}\begin{bmatrix} 3 & 6 \\ 6 & 9 \end{bmatrix}$

$= \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$

Ans. D

9. Given $AB = 2I$ and $|B| = 6\ for\ 3\times 3$ matrix A and B with identity 3×3 matrix I.

To find $|A^T|$, set

$AB = 2I \Rightarrow |AB| = |2I|$

$\Rightarrow |AB| = |2I|$, property of determinant

$\Rightarrow |A^T||B| = 2|I|$ because $|A| = |A^T|$

$\Rightarrow |A^T|.6 = 2(1)$, determinant of identity is 1.

$\Rightarrow |A^T| = \frac{1}{3}$

Ans. A

10. 3,10,17,24,31,− − −goes with common difference d=7 arithmetically then using a_1=3 and $a_n=a_1+(n-1)$ (n^{th} term formula)

$a_{50} = 3 + (50-1)7$

$= 346$

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Ans. B

11. For $x^2+y+1=0$ i.e $(x^2+1)+y=0$

C) If we take any $x\in\mathbb{R}$, we can find $y=-(x^2+1)$ such that $(x^2+1)+y=(x^2+1)+\big(-(x^2+1)\big)=0$

Hence for all x, there is $y=-x^2-1$ such that $x^2+y+1=0$

$\therefore (\forall x)(\exists y)(x^2+y+1=0)$ is True

D) $(\exists x)(\forall y) \equiv$ There is an /unique x for all values of y.

But we can't find a single x, for every y such that

$x^2+y+1=0$

$\therefore (\exists y)(\forall x)(x^2+y+1=0)$ is False

For instance if $y=0, then\ x^2+y+1=x^2+1\neq 0, \forall x.$

E) Also putting as $x^2=-y-1$, if y varies also x varies. Hence $(\exists y)(\forall x)(x^2+y+1=0)$ is False

F) Putting $x^2+y+1=0$ as $x^2=-y-1=0$ and take y = 0, we get $x^2=-1$ (which is False)

$\therefore (\forall y)(\exists x)(x^2+y+1=0)$ is False

Ans. A

12. Simplifying $f(x)=\frac{x^2-9}{x-3}$ for $x\neq 3$, we get

$f(x)=x+3\ for\ x\neq 3$

$\therefore f(x)=\frac{x^2-9}{x-3}$ has hole (not vertical asymptote) at x=3

OR we can examine using limit as follows

A) $\lim\limits_{x\to -1^+} \ell n\,(x+1) = -\infty$

B) $\lim\limits_{x\to -2} \frac{x^2+1}{x^3-8} = \pm\infty$

C) $\lim\limits_{x\to 3} \frac{x^2-9}{x-3} = 6$

D) $\lim\limits_{x\to 0} \frac{x-1}{x^3-x} = \pm\infty$

$\therefore$ Only f in (c) has no vertical asymptotes since the limit at zero of denominator exists.

Ans. C

13. Given $p \Rightarrow (r \vee \urcorner q) \equiv$ False.

Hence, $P \equiv True\ and\ (r \vee \urcorner q) \equiv$ False, because in case of '$\Rightarrow$' only $T \Rightarrow F\ is\ False$

Then $r \vee \urcorner q \equiv False$ gives both $r\ and\ \urcorner q$ are False because $FVF \equiv F(only)$

Then $\urcorner q \equiv F$ gives $\urcorner(\urcorner q) \equiv \urcorner F \Rightarrow q \equiv T$

Therefore $P \equiv T, r \equiv F\ and\ q \equiv T$

So A) $P \Rightarrow r \equiv T \Rightarrow F \equiv F$

B) $\urcorner r \Rightarrow q \equiv \urcorner F \Rightarrow T \equiv T \Rightarrow T \equiv T$

C) $\neg P \Leftrightarrow q \equiv \neg T \Leftrightarrow T \equiv F \Leftrightarrow T \equiv F$
D) $q \Leftrightarrow r \equiv T \Leftrightarrow F \equiv F$

Ans. B

14. $f(x) = \frac{1}{e^x+1}$, to find $f^{-1}(x)$, *set* $f: y = \frac{1}{e^x+1}$, for $y = f(x)$ then f^{-1} is interchanging x and y, as $f^{-1}: x = \frac{1}{e^y+1}$ for $y = f^{-}(x)$, hence solving for y as follows.

$$x = \frac{1}{e^y+1} \Rightarrow x(e^y+1) = 1$$
$$\Rightarrow xe^y += 1$$
$$\Rightarrow e^y = \frac{1-x}{x}$$
$$\Rightarrow y = \log e \frac{1-x}{x} = \ell n\left(\frac{1-x}{x}\right)$$

$\therefore f^{-1}(x) = \ell n\left(\frac{1-x}{x}\right) = \ell n|1-x| - \ell n|x|$, use absolute value or tell $0 < x < 1$.

Ans.

15. $x^2 - 6x + y^2 + k = 0$ a circle with radius r=2. So by completing the square with respect to x gives,

$x^2 - 6x + 9 - 9y^2 + k = 0 \Rightarrow \boxed{(x-3)^2 + y^2 = 9 - k}$

Also since $\boxed{(x-3)^2 + y^2 = r^2}$ in standard form, we get

$r^2 = 9 - k \Rightarrow k = 9 - r^2 = 9 - 2^2 = 5 \ \therefore k = 5$

Ans. B

16. If θ is angle of inclination and m is slop of a line then $m = tan\theta$. Hence $m = tan\frac{3}{4}\pi = \tan\left(\pi - \frac{3}{4}\pi\right) = -tan\pi = -1$. Then if (x, y) is any point on the line, by point slope form of equation of a line, since (0.1)is on the line, $m = \frac{y-y_1}{x-x_1} \Leftrightarrow -1 = \frac{y-1}{x-0} \Rightarrow y = -x + 1$ the line.

Ans. A

17. $g(x) = xf(x) - \sqrt{f(x)}$

$\Rightarrow g'(x) = \left(xf(x) - \sqrt{f(x)}\right)'$, *(dervative both side)*

$= \left(x(x)\right)' - \left(\sqrt{f(x)}\right)'$, *since* $(f+g)' = f' + g'$

$= x'f(x) + \left(f(x)\right)'x - \frac{1}{2\sqrt{f(x)}}.\left(f(x)\right)'$, *by* $(f.g)^2$

$= f'g + g'f$

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In addition, chain rule on $\left(\sqrt{f(x)}\right)' = \frac{(f(x))'}{2\sqrt{f(x)}}$

$= f(x) + xf'(x) - \frac{f'(x)}{2\sqrt{f(x)}}$, *since* $\left(f(x)\right)' = f'(x) \& x' = 1$

then replacing $x = 2$

$$g'(2) = f(2) + 2f'(2) = \frac{f'(2)}{2\sqrt{f(2)}}$$
$$= 4 + 2(4) - \frac{4}{2\sqrt{4}}$$
$$= 11$$

Ans. A

Note: For such case as Q. 17 better to use libenz notation.

i.e. $\frac{dg}{dx} = \frac{d}{dx}(xf(x) - \frac{d}{dx}\sqrt{f(x)}$

$$= \frac{dx}{dx}f(x) + \frac{df}{dx}x - \frac{d\sqrt{f(x)}}{d(fx)}.\frac{df(x)}{dx}$$
$$= 1.f(x) + x\frac{df}{dx} - \frac{1}{2\sqrt{f(x)}}.\frac{df}{dx}$$

Hence for $x = 2$

$$\frac{dg}{dx/(x=2)} = 1.f(2) + 2.\frac{df}{dx(x=2)} - \frac{1}{2\sqrt{f(2)}}.\frac{df}{dx(x=2)} = 11$$

18. $\sum_{n=1}^{\infty}\frac{2^{2n+1}}{5^{n-1}} = \sum_{n=1}^{\infty}\frac{2.4^n}{5^{-1}.5^n} = \sum_{n=1}^{\infty} 10\left(\frac{4}{5}\right)^n = 10\sum_{n=1}^{\infty}\left(\frac{4}{5}\right)^n$ ---(1)

Then setting $G_n = \left(\frac{4}{5}\right)^n$ the common ratio is

$$r = \frac{G_{n+1}}{G_n} = \frac{\left(\frac{4}{5}\right)^{n+1}}{\left(\frac{4}{5}\right)^n} = \frac{4}{5}(common\ ratio)$$

So $|r| < 1$ implies the given series is convergent infinite geometric series, whose sum is $\frac{G_1}{1-r}$ then $G_1 = \left(\frac{4}{5}\right)^1 = \frac{4}{5}$. Substitute in,

$10\sum_{n=1}^{\infty}\left(\frac{4}{5}\right)^n = 10\left(\frac{G_1}{1-r}\right) = 10\left(\frac{\frac{4}{5}}{1-\frac{4}{5}}\right) = 40$

$$\therefore \sum_{n=1}^{\infty}\frac{2^{2n+1}}{5^{n-1}} = 40$$

Ans. A

19. To evaluate $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{-\frac{x}{2}}$, we use $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x} = e$

Then $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{-\frac{x}{2}} = \lim_{x\to\infty}\left(\left(1+\frac{1}{x}\right)^{x}\right)^{-\frac{1}{2}}$

$= \left(\lim_{x\to\infty}\left(\left(1+\frac{1}{x}\right)^{x}\right)^{-\frac{1}{2}}\right)$, continuity of $\left(1+\frac{1}{x}\right)^{-\frac{x}{2}}$ as $x\to\infty$

$= e^{\frac{-1}{2}} = \frac{1}{\sqrt{e}}$

Ans. A

20.

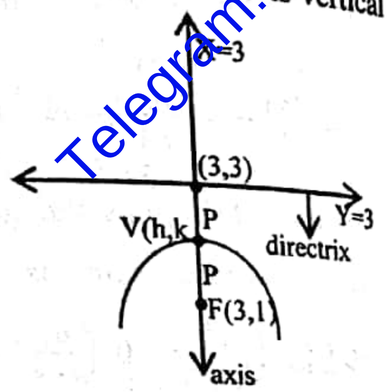
Number of students (f_i)	Score (x_i)
$f_1 = 15$	$x_1 = 6$
$f_2 = 20$	$x_2 = 7$
$f_3 = 10$	$x_3 = 8$
$f_4 = 5$	$x_4 = 10$
Sum of No of student=50	—

Average score means mean score. Hence

$$\bar{x} = \frac{\sum_{i=1}^{4} f_i x_i}{\sum_{i=1}^{4} f_i} = \frac{f_1x_1+f_2x_2+f_3x_3+f_4x_4}{f_1+f_2+f_3+f_4} = \frac{15(6)+20(7)+10(8)+5(10)}{15+20+10+5} = \frac{360}{50} = 7.2$$

Ans. C

21. Given a parabola P with focus, F(3,-1) and directrix y=3. To find the equation of P, the directrix is y=3, ie Horizontal line, hence P is vertical parabola of the form $(x-h)^2 = -4P(y-k)$, Where (h,k) is vertex and P is focal length, (see the figure). It is open down since $y_f < y_{directrix}$. Intersection of axis and directrix is (3,3). Then by mid – point formula,



$$(h,k) = \left(\frac{3+3}{2}, \frac{3+(-1)}{2}\right) = (3,1)$$

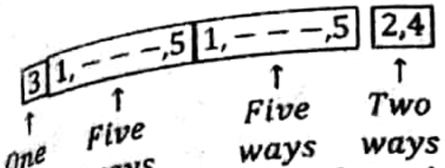
$\therefore h = 3, k = 1 \& P = -|y_v - y_f| = -|k - y_F| = -|1-1) = -2$

$\therefore P = -2$ the –ve sign of P shows, open down.

$\therefore (x-h)^2 = -4P(y-k)$ gives $(x-3)^2 = -8(y-1)$

Ans. B

22. Let the number has digits place as



(The last digit must be 2 or 4 since it is even)

Then by product principle of counting, we have

(one way) × (5 ways) × (5 ways) × (2 ways) = 1 × 5 × 5 × 2 = 50

∴ 50 four digit even numbers starting with 3 can be formed using 1,2,3,4 and 5.

Ans. B

23. Horizontal transverse axis = 24 ⟹ Horizontal hyperbola length of transverse axis $2a = 24 \therefore \boxed{a = 12}$ and the asymptote formula, $y = \pm\frac{b}{a}(x-h)+k$ gives

$$y = \frac{5}{12}(x-h)+k$$

Hence coefficients of x are equal, gives

$$\frac{b}{a} = \frac{5}{12} \Rightarrow \boxed{b = \frac{5}{12}a} \Rightarrow b = \frac{5}{12}(12) = 5 \therefore \boxed{b=5}$$

Then using $c^2 = a^2 + b^2$, we get $c = \sqrt{a^2+b^2}$ and gives

$$c = \sqrt{5^2+12^2} = \sqrt{169} = 13 \therefore \boxed{c=13}$$

Hence eccentricity, $e = \frac{c}{a} = \frac{13}{12}$

Ans. C

24. If $f(x)$ is anti-derivative of $f(x)$, then $\int f(x)dx = f(x)+c$

Hence $\int f(x)dx = \int \frac{2}{2x^2+4x+1}dx = \int \frac{2}{(2x+1)^2}dx$ let $U = 2x+1$ then $dU = 2dx$ gives $\frac{1}{2}dU$=dx then substitution gives

$$\int f(x)dx = \int \frac{2}{(2x+1)^2}dx = \frac{\int \frac{2}{U^2}}{2}dU = \int U^{-2}\,dU = \frac{U^{-2+1}}{-2+1}+c$$

$$\Rightarrow \int f(x)dx = U^{-1}+c, \text{ then since } U = 2x+1$$

$$= -(2x+1)^{-1}+c$$

$$= -\frac{1}{2x+1}+c$$

Then since anti-derivative means for some c, set c=0

$$\therefore f(x) = -\frac{1}{2x+1} \text{ is anti – derivative}$$

Ans. C

25. $f(x) = \frac{1}{4}x^4 - 2x^2$ is differentiable function

$\Rightarrow f'(x) = x^3 - 4x$ then $f'(x) = 0 \Rightarrow x^3 - 4x = 0$

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$\Rightarrow x(x-2)(x+2)=0$

$\Rightarrow x=0\ or\ x=2\ or\ x=-2$

Critical numbers are $-2,0,2$ *then* f has local maximum at critical c *if* $f''(c)<0$.(this is second derivative test).

So $f''(x)3x^2-4, gives\ f''(-2)=3(-2)^2-4=8$

$f''(0)=3(0)^2-4=-4$

$f''(2)=3(2)-4=8$

$\therefore f''(0)<0 \Rightarrow f\ has\ local\ maximum\ at\ x=0$

Note:-You can use sign chart of f' and see f' changes from $+ve\ to-ve\ at\ x=0$.(this first derivative test)

26. $z^3-iz^2+2z=0$ for z is a complex number Ans. B

$\Rightarrow z(z^2-iz+2)=0$

$\Rightarrow z=0\ or\ z^2-iz+2=0$

So for $z^2-iz+2=0, taking\ a=1, b=-i\ and\ c=2$

Then $z_{1,2}=-b\pm\sqrt{b^2-4ac}=\frac{-(-i)\pm\sqrt{(-i)^2-4(1)(2)}}{2}$

$=\frac{i\pm\sqrt{-1-8}}{2}=\frac{i\pm\sqrt{-9}}{2}$

Hence $Z_{1,2}=\frac{i\pm3i}{2}\ since\sqrt{-9}=3i$

$=2i\ or-i$

$\therefore Solution\ set\ of\ z^3-iz^2+2z=0\ is\ \{0,-i,2i\}$

27. P_o will be $P=P_o(1+r)^t$ in t years at a rate of r. Ans. C

So $P(10)=80\ million\ (1+0.02)^{10}$, since r=20%=0..2

$=80(1.02)^{10} million$

$=80(1.22) million, given\ (1.02)^{10}=1.22$

$=97.6\ million$

28. The rate of change of the volume $\frac{dv}{dt}$ Ans. D

Hence $\frac{dv}{dt}=\frac{d}{dt}(2000-40t+0.2t^2)$

$=\frac{d(2000)}{dt}-\frac{d(40t)}{dt}+\frac{d(0.2t^2)}{dt}$

$=0=40+0.4t$

$\therefore \frac{dv}{dt}=-40+0.4t$ ←Rate of change of Vat any time t.Then at $t=40\ seconds$

$\frac{\frac{dv}{dt}}{t}=40=(-40+0.4(40))\frac{cm^3}{sec}$

$=(-40+16)\frac{cm^3}{sec}$

$=-24\frac{cm^3}{sec}$ $(-ve\ sign\ shows\ decreasing)$

Ans. D

29. $\lim_{n\to\infty}\frac{1-n-3n^2}{6n^2+1}=\lim_{n\to\infty}\frac{\frac{1}{n^2}-\frac{n}{n^2}-\frac{3n^2}{n^2}}{\frac{6n^2}{n^2}+\frac{1}{n^2}}=\lim_{n\to\infty}\frac{\frac{1}{n^2}-\frac{n}{n^2}-3}{6+\frac{1}{n^2}}=\lim_{n\to\infty}\frac{\left(\frac{1}{n^2}-\frac{n}{n^2}-3\right)}{\left(6+\frac{1}{n^2}\right)}$

$\frac{\lim_{n\to\infty}\frac{1}{n^2}-\lim_{n\to\infty}\frac{n}{n^2}-\lim_{n\to\infty}3}{\lim_{n\to\infty}6+\lim_{n\to\infty}\frac{1}{n^2}}=\frac{0-0-3}{6+0}=-\frac{1}{2}$

Ans. B

30. $\frac{2}{x}-\frac{x-2}{x^2-2x}=1-\frac{2x-2}{3x-2}, x\neq0,2,\frac{2}{3}$

$\Rightarrow\frac{2}{x}-\frac{x-2}{x(x-2)}=\frac{3x-2}{3x-2}-\frac{2x-2}{3x-2}, x\neq0,2,\frac{2}{3}$

$\Rightarrow\frac{2}{x}-\frac{1}{x}=\frac{3x-2-(2x-2)}{3x-2}, x\neq0,2,\frac{2}{3}$

$\Rightarrow\frac{2-1}{x}=\frac{x}{3x-2}, x\neq0,2,\frac{2}{3}$

$\Rightarrow\frac{1}{x}=\frac{x}{3x-2}, x\neq0,2,\frac{2}{3}$

Then by cross multiplication $1(3x-2)=(x), for\ x\neq0,2,\frac{2}{3}$

$\Rightarrow x^2-3x+2=0, for\ x\neq0,2,\frac{2}{3}$

$\Rightarrow(x-2)(x-1)=0, for\ x\neq0,2,\frac{2}{3}$

$\Rightarrow x=2\ or\ x=1$

But $x\neq0,2\ and\ \frac{2}{3}$ therefore solution set $=\{1\}$

Ans. D

31. f is continuous at x=0 means

i) $f(0) is\ defined$

So $f(0)=3^0+k=1+k\ (defined)$

ii) $\lim_{x\to0^-}f(x)=\lim_{x\to0^+}f(x), then$

$\Rightarrow\lim_{x\to0^-}3^x+k=\lim_{x\to0^+}3\frac{\sin(2x)}{x}$

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$$\Rightarrow 3^0 + k = 3 \lim_{x \to 0^+} \frac{2\sin(2x)}{2x}$$

$$= 6 \lim_{x \to 0^+} \frac{\sin 2x}{2x}$$

$$= 6(1), since \lim_{x \to 0} \frac{\sin ax}{ax} = 1, for\ a \neq 0$$

$$\therefore 3^0 + k = 6 \Rightarrow 1 + k = 6 \ \therefore k = 5$$

32. $\int (1+x)3^x dx = \int (3^x + x3^x)\, dx$

$$= \int 3^x\, dx + \int x3^x\, dx - - - (1)$$

Then $\int 3^x\, dx = \frac{3^x}{\ln 3} + c, for\ constant\ c - - - (2)$

$\int x3^x dx$ needs "by part"

Then, let $U = x$ and $dv = 3^x\, dx$

Then $dU = dx$ and $v = \int 3^x\, dx = \frac{3^x}{\ln 3}$

Then by $\int UdV = UV - \int VdU$,

$$\int x3^x dx = \frac{x3^x}{\ln 3} - \int \frac{3^x}{ln3} dx$$

$$= \frac{x}{ln3} 3^x - \frac{1}{ln3} \int 3^x dx$$

$$= \frac{x}{ln3} 3^x - \frac{1}{ln3} \cdot \frac{3^x}{ln3} + c_2 \ for\ some\ c_2$$

$$= \frac{x}{ln3}\, 3^x - \frac{1.3^x}{(ln3)^2} + c_2 - - - (3)$$

Substituing (2) and (3) in (1), gives

$$\int (1+x)3^x dx = \frac{3^x}{ln3} + \frac{x3^x}{(ln3)^2} + c$$

$$= (1+x)3^x \log_3^e - 3^x \left(\log_3^e\right)^2 + c,$$

$$Since \frac{1}{ln3} = \frac{1}{\log_3^e} = \log_3^e$$

Ans. C

33. B=6, G=4 (no of Boys and Girls) , then in choosing 3, a probability of getting at least one girl is given by:

$$\Rightarrow P(at\ least\ 1\ G\ of\ 3) = P(1G, 2B) + P(2G, 1B) + P(3G, 0B)$$

$$= \frac{c(4,1).c(6.2)}{c(10,3)} + \frac{c(4,2).c(6.1)}{c(10,3)} + \frac{c(4.3).c(6,0)}{c(10,3)}$$

$$= \frac{\frac{4!}{1!3!} \cdot \frac{6!}{4!2!}}{\frac{10!}{7!3!}} + \frac{\frac{4!}{2!2!} \cdot \frac{6!}{5!1!}}{\frac{10!}{7!3!}} + \frac{\frac{4!}{3!1!} \cdot \frac{6!}{6!0!}}{\frac{10!}{7!3!}}$$

$$= \frac{4.5}{120} + \frac{6.6}{120} + \frac{4.1}{120}$$

$$= \frac{100}{120} = \frac{5}{6}$$

Ans. A

34. Let the students be $x_1, x_2, - - - x_6$ and let the two specified students be x_5 and x_6.

- $6\ of\ them\ can\ sit\ in\ round\ in\ (6-1)! = 5!\ ways$
- If two of them must sit together then we have $2(5-1)! = 2(4!)$ ways.

Hence the probability that the two are not together ,

$$P = \frac{5! - 2(4!)}{5!}$$

$$= 1 - \frac{2}{5}\ (by\ complement) = \frac{3}{5}$$

Ans. A

35. $f(x) = \ln(x-1)\ and\ g(x) = \sqrt{1-2x}$,

Then $fog(x) = f(g(x)) = \ln(g(x)-1) = ln\left(\sqrt{1-2x} - 1\right)$

Let $h(x) = ln\left(\sqrt{1-2x} - 1\right)\ for\ h(x) = fog(x)$

Then $\boxed{dom.\ fog = domh(x) \cap domg(x)} - - -*$

Now h(x) is defined if $\sqrt{1-2x} - 1 > 0$

$$\Rightarrow \sqrt{1-2x} > 1$$

$$\Rightarrow 1 - 2x > 1\ (squaring)$$

$$\Rightarrow x < 1 \equiv (-\infty, 1) \leftarrow domh(x)$$

And $g(x) is\ defined\ if\ 1 - 2x \geq 0$

$$\Rightarrow x \leq \frac{1}{2} \equiv \left(-\infty, \frac{1}{2}\right] \leftarrow dom\ g(x)$$

Then using*

$$domfog = dom\ h(x) \cap domg(x)$$

$$= (-\infty, 1) \cap \left(-\infty, \frac{1}{2}\right]$$

$$= \left(-\infty, \frac{1}{2}\right] = \left\{x \in \mathbb{R}: x \leq \frac{1}{2}\right\}$$

Ans. B

36. No of students, $n = 8 + 12 + 20 + 6 + 4 = 50$

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Class (C_i)	Class Interval (Mark)	Class boundary	No of Students (frequency)f_i	Cumulative frequency Cf_i
C_1	55 – 64	54.5-64.5	8	8
C_2	65 – 74	64.5-74.5	12	8+12=20
C_3	75 - 84	74.5-84.5	20	20+20=40
C_4	85-94	84.5-94.5	6	40+6=46
C_5	95 – 100	94.5-100.5	4	46+4=50

$C_3(3rd\ class)$ is the median c lass, since $cf_{c_3} \geq \frac{50}{2} = 25$, because $c_3 = 40 \leftarrow cummulative\ frequency\ of\ C_3$.

Then in $c_3 \rightarrow B_2 = 74.5 \leftarrow lower\ boundary\ of\ median\ class.$

$\rightarrow f_{c_3} = 20 \leftarrow frequency\ of\ median\ class$

$\rightarrow cf_b = 20 \leftarrow cummulative\ frequency\ of\ the\ class\ before\ median\ class.$

Hence $md = Bi + \left(\frac{\frac{n}{2} - cf_b}{cf_{c_3}}\right) i$ where $i = 84.5 - 74.5$

$= 10(median\ class\ size)$

$= 74.5 + \left(\frac{\frac{50}{2} - 20}{20}\right) 10$

$= 74.5 + \frac{5}{2}$

$= 77.0 \quad \therefore md = 77.0$

Ans. C

37. To minimize the material of the box means to minimize its total surface area.

Since it is open in one side, Area of base, $A_b = x^2$

$\therefore \boxed{A_b = x^2} \leftarrow$ Square base of side x – units

And lateral face area, $\boxed{A_L = 4xy} \leftarrow$ Area of four rectangular face of dimension x by y.

Then total surface area,

A_T will be, $A_T = A_b + A_L$

$= x^2 + 4xy \leftarrow A_T$ in terms of x and y

Now to eliminate y, using the volume,

$\Rightarrow V = A_b h$

$\Rightarrow 32 = x^2\, y\ hence \boxed{y = \frac{32}{x^2}}$

$Then\ A_T(x, y) = x^2 + 4xy, will\ be$

$A_T(x) = x^2 + 4x\left(\frac{32}{x^2}\right)$

$= x^2 + \frac{128}{x}$

$Then\ A_T(x)$ is minimum, at x=c where $A_T'(c) = 0$

So $A_T'(x) = \left(x^2 + \frac{128}{x}\right)'$

$= 2x - \frac{128}{x^2}$

Hence $A_T'(x) = 0 \Rightarrow 2x - \frac{128}{x^2} = 0$

$\Rightarrow x^3 = 64 \quad \therefore x = 4$

And using $y = \frac{32}{x^2}, y = \frac{32}{42} = 2$

Ans. A

38. $\int \frac{1}{x^2+x}\, dx = \int \frac{1}{x(x+1)}\, dx$

Then by partial fraction (decomposition),

$\frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}$

$\Rightarrow \frac{1}{x(x+1)} = \frac{A(x+1) + Bx}{x(x+1)} \Rightarrow 1 = (A+B)x + A$

$\Rightarrow A + B = 0\ and\ A = 1$

$\Rightarrow B = -1\ and\ A = 1$

Then $\int \frac{1}{x(x+1)} dx = \int \left(\frac{1}{x} - \frac{1}{x+1}\right) dx$

$= \int \frac{1}{x} dx - \int \frac{1}{x+1} dx$

$= ln|x| - ln|x+1| + C\ for\ some\ C.$

$\therefore \int \frac{1}{x^2+x} dx = ln|x| - ln|x+1| + C$

Ans. C

39. $P \equiv He\ loves\ her,\ q \equiv She\ will\ marry\ him$

Then if he does not love her, she will not marry him$\equiv \neg p \Rightarrow \neg q$

$He\ loves\ her \equiv P$

$\overline{Therefore, she\ will\ marry\ him \equiv q}$

Hence, the representation of the argument will be $\boxed{\neg p \Rightarrow \neg q, P \vdash q}$

Then premises are always true,

So $\neg p \Rightarrow \neg q\ and\ P\ are\ True$

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Then $\rightarrow \neg p \Rightarrow \neg q \equiv T(primise)$

$\rightarrow \neg T \Rightarrow \neg q \equiv T, (P \equiv T)$

$\rightarrow F \Rightarrow \neg p \equiv T$

Hence for $F \Rightarrow \neg q$ to be true means, $\neg q$ can be either True or False since $F \Rightarrow F$ and $F \Rightarrow T$ are both true. Therefore q can also be both True or False. $\therefore$ *Invalid*

Note:- You can also show using $[(\neg p \Rightarrow \neg q) \wedge P] \Rightarrow q$ is not Tautology. Ans. B

40. First using integration by part, for $\int x\cos x dx$

Let $U = x, dV = \cos x dx$

$\Rightarrow dU = dx\ V = \sin x$

$$Hence \int x\cos x dx = UV - \int V dU$$

$$= x\sin x - \int \sin x dx$$

$$= x\sin x + \cos x + c$$

Then $\int_0^{\frac{\pi}{2}} 2x\cos x\, dx = 2\int_0^{\frac{\pi}{2}} x\cos x dx$

$$= 2[x\sin x + \cos x]_0^{\frac{\pi}{2}}$$

$$= 2\left[\left(\frac{\pi}{2}\sin\frac{\pi}{2} + \cos\frac{\pi}{2}\right) - (0.\sin 0 + \cos 0)\right]$$

$$= 2\left[\frac{\pi}{2} + 0 - (0 + 1)\right]$$

$$= 2\left(\frac{\pi}{2} - 1\right)$$

$$= \pi - 2$$

Ans. A

41. i) 5 white , 6 red, 4 black

$$P(w) = \frac{n(w)}{n(w+r+b)} = \frac{5}{15} = \frac{1}{3} - - - (1)$$

ii) 4 white , 6 red , 4 black (already 1 white is drawn)

$$P(r_1) = \frac{n(r)}{n(w+r+b)} = \frac{6}{14} = \frac{3}{7} - - - (2)$$

iii) 4 white , 5 red, 4 black (again 1-red is drawn)

$$P(r_2) = \frac{n(r)}{n(w+r+b)} = \frac{5}{4+5+4} = \frac{5}{13} - - - (3)$$

Therefore since it is drawing three balls one after the other without replacement, we use product of (1), (2) and (3)

$P(w, r, r) = P(w).P(r_1).P(r_2)$

$$= \frac{1}{3} \times \frac{3}{7} \times \frac{5}{13} = \frac{5}{91}$$

Ans. D

42. The equation of a tangent line to $f(x)$ at $(a, f(a))$ is

$$y = f'(a)(x - a) + f(a)$$

Then $f(x) = \frac{1}{x+1} + \cos x$, at $(0, f(0))$ gives $a = 0$

$$\Rightarrow f'(0) = -\frac{1}{(0+1)^2} - \sin 0 = -1$$

$$and\ f(0) = \frac{1}{0+1} + \cos 0 = 2$$

Hence $y = f'(a)(x - a) + f(a)$, gives

$y = f'(0)(x - 0) + 2 = -x + 2$

$x + y = 2$ is the tangent line.

Ans. C

43. $\lim_{x \to 1} \frac{\sqrt{x}-1}{x^2-1} = \lim_{x \to 1} \frac{\sqrt{x}-1}{(x-1)(x+1)}$

$$= \lim_{x \to 1} \frac{\sqrt{x} - 1}{(\sqrt{x} - 1)(\sqrt{x} + 1)(x + 1)}, since\ (\sqrt{x} - 1)(\sqrt{x} + 1) = x - 1$$

$$= \lim_{x \to 1} \frac{1}{(\sqrt{x} - 1)(x + 1)}$$

$$= \frac{1}{(\sqrt{1} - 1)(1 + 1)} = \frac{1}{4}$$

Ans. C

44. $\lim_{x \to 1} \frac{1-x}{1-\frac{1}{x^2}} = \lim_{x \to 1} \frac{1-x}{\frac{x^2-1}{x}}$

$$= \lim_{x \to 1} \frac{x(1-x)}{(x-1)(x+1)}$$

$$= \lim_{x \to 1} -\frac{x}{x+1}$$

$$= -\frac{1}{2}$$

Ans. C

45. $\int \frac{\ell n(xe^x)}{x} dx = \int \frac{\ell nx + \ell n e^x}{x} dx$

$$= \int \frac{\ell nx}{x} dx + \int \frac{\ell n e^x}{x} dx - - - (1)$$

Now for the addend integrals

i) For $\int \frac{\ell nx}{x} dx$

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Let $U = lnx$ and $dV = \frac{dx}{x}$

Then $dU = \frac{dx}{x}$ and $V = lnx$

Hence the $\int UdV = UV - \int VdU$ gives

$$\int \frac{\ell nx}{x}dx = (lnx)^2 - \int \frac{lnx}{x}dx$$

$\Rightarrow 2\int \frac{lnx}{x}dx = (lnx)^2$ (Collecting like integrals)

$$\Rightarrow \boxed{\int \frac{lnx}{x}dx = \frac{1}{2}(lnx)^2 + C_1} \quad ---(2)$$

ii) For $\int \frac{lne^x}{x}dx$, since $lne^x = xlne = x(1) = x$

Then $\int \frac{lne^x}{x}dx = \int \frac{x}{x}dx = \int dx = x + C_2$

$$\Rightarrow \boxed{\int \frac{\ln(ex^x)}{x}dx = x + C_2}$$

Therefore $\int \frac{\ln(ex^x)}{x}dx = \frac{1}{2}(lnx)^2 + C$ for $C = C_1 + C_2$

Ans. B

46. Given $\begin{vmatrix} \alpha & 1 & 1 \\ 1 & 2 & 4 \\ 5 & -1 & 1 \end{vmatrix} = 2 \Rightarrow \alpha\begin{vmatrix} 2 & 4 \\ -1 & 1 \end{vmatrix} - \begin{vmatrix} 1 & 4 \\ 5 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 5 & -1 \end{vmatrix} = 2$

$\Rightarrow \alpha(2+4) - (1-20) + (-1-10) = 2$

$\Rightarrow 8\alpha + 19 - 11 = 2$

$\Rightarrow \alpha = -\frac{3}{4}$

Hence equation will be,
$$\begin{cases} -\frac{3}{4}x + y + z = 1 & ---(1) \\ x + 2y + 4z = 0 & ---(2) \\ 5x - y + z = 0 & ---(3) \end{cases}$$

From (3), $y = 5x + z$, substituting *in* (1)*and* (2)

$$\begin{cases} -\frac{3}{4}x + 5x + z + z = 1 \\ x + 2(5x + z) + 4z = 0 \end{cases} \Rightarrow \begin{cases} \frac{17}{4}x + 2z = 1 \\ 11x + 6z = 0 \end{cases} \Rightarrow \begin{cases} \frac{41}{4}x + 6z = 3 \\ 11x + 6z = 0 \end{cases}$$

$$-\frac{3}{4}x = 3$$

$$\therefore x = -4 \text{ and } z = \frac{22}{3}$$

$$\text{Hence } y = 5x + z = 5(-4) + \frac{22}{3} = -\frac{38}{3}$$

$$\therefore x = -4, y = -\frac{38}{3} \text{ and } z = \frac{22}{3}$$

since $\left(-4, \frac{-38}{3}, \frac{22}{3}\right)$ is not in the choice, let us put x, y and z in terms of $\alpha = -\frac{3}{4}$

$$\text{Since: } \alpha = \frac{-3}{4} = \frac{3}{-4} = \frac{3}{x} \Rightarrow x = \frac{3}{\alpha}$$

$$\text{And } -\frac{38}{3} = \frac{19}{2\alpha} \text{ and } \frac{22}{3} = -\frac{11}{2\alpha} \text{ for } \alpha = -\frac{3}{4}$$

Hence, also $\left(-4, -\frac{38}{3}, \frac{22}{3}\right) = \left(\frac{3}{\alpha}, \frac{19}{2\alpha}, -\frac{11}{2\alpha}\right)$

Ans. ___

47. A) f id decreasing where $f' \leq 0$.

$\Rightarrow f$ is decreasing in $(-2,0] \cup [2,\infty)$

B) f has a local minimum at $x = c$ *if* f' changes from $-ve$ *to* $+ve$.

$\Rightarrow f$ has local minimum at x=0

C) f is concave down where $f'' < 0$, that is also means where f' is decreasing

$\Rightarrow f$ *is concave down in* $(-\infty,-1) \cup (1,\infty)$

D) f is concave up where $f'' > 0$ (*or* f' *increases*)

$\Rightarrow f$ *is concave up in* $(-1,1)$

Ans. D

48. $f(x) = x|x| = \begin{cases} x^2, if\ x \geq 0 \\ -x, if\ x < 0 \end{cases}$ since f is continuous and $f'(0) = 2(0) = 0$ from right &

$f'(0) = -2(0) = 0$ from left, we can use

Then $f'(x) = \begin{cases} 2x, if\ x \geq 0 \\ -2x, if x < 0 \end{cases} = 2|x|$

Ans. B

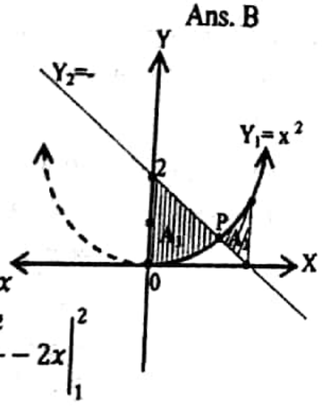
49. Intersection of $y_1 = x^2$ and $y_2 = -x + 2$ is at x, where , $y_1 = y_2$, that is $x^2 = -x + 2$

$\Rightarrow x^2 + x - 2 = 0 \Rightarrow x_1 = -2$ & $x_2 = 1$

Hence $x = 1$ *at point P.*

Then $A_1 = \int_0^1 (y_2 - y_1)dx$ & $A_2 = \int_1^2 (y_2 - y_1)dx$

$$\Rightarrow A_1 = -\frac{x^2}{2} + 2x - \frac{x^3}{3}\Big|_0^1 \text{ and } A_2 = -\frac{x^3}{3} + \frac{x^2}{2} - 2x\Big|_1^2$$



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$\Rightarrow A_1 = \frac{7}{6}$ and $A_2 = \frac{11}{6}$

Therefore $A = A_1 + A_2$

$= \frac{7}{6} + \frac{11}{6} = 3$ square unit

Ans. A

50. Let $p(x) = ax^2 + x + b$

If $(-1,5) \in P(x)$, then $P(-1) = a(-1)^2 + (-1) + b = 5$

$\Rightarrow \boxed{a + b = 6} - - - (1)$

If $(2,-1) \in P(x)$, then $P(2) = a(2)^2 + 2 + b = -1$

$\Rightarrow \boxed{4a + b = -3} - - - (2)$

Combining (1) and (2)

$$\begin{cases} a + b = 6 \\ 4a + b = -3 \end{cases}$$

$-3a = 9 \Rightarrow a = -3$ & $b = 9$

$\therefore b = 9$

Ans. A

51. $f(x) = a\ \sin(kx + b) + c$, has

- Period, $P = \frac{2\pi}{k}$
- Amplitude, $A = |a|$

Hence for $f(x) = 2 - \frac{1}{2}\ sin\left(\frac{\pi}{2}x\right)$, taking $k = \frac{\pi}{2}, a = -\frac{1}{2}$

$$P = \frac{2\pi}{k} = \frac{2\pi}{\frac{\pi}{2}} = 4 \text{ and } A = |a| = \left|-\frac{1}{2}\right| = \frac{1}{2}$$

$\therefore \frac{1}{2}$ and 4 are amplitude and period of $f(x)$ respectively.

Ans. C

52. $\sec\left(\frac{\pi}{2} - x\right) sin^3x + \cos 2x$

Now since the simplification is defined for any $x \neq (1-n)\pi, n \in \mathbb{Z}$ let us consider $x \in \left(0, \frac{\pi}{2}\right)$, then $\sec\left(\frac{\pi}{2} - x\right) = cscx$, (co-function relation) and recall $cos2x = cos^2x - sin^2x$ and $cscx = \frac{1}{sinx}$.

Hence $\sec\left(\frac{\pi}{2} - x\right) sin^3x + cos2x$

$= cscx sin^3x + cos^2 - sin^2x$

$= \frac{1}{sinx}.sin^3x + cos^2x - sin^2x$

$= sin^2x + cos^2x - sin^2x$

$= cos^2x$

Ans. C

53. $\vec{A} = \overrightarrow{PQ} = Q - P = (1,0,2) - (1,2,1) = (0,-2,1)$

$\therefore \boxed{\vec{A} = (0,-2,1)}$

$\vec{B}$ is parallel to $\vec{A} = \overrightarrow{PQ}$ means there is $m \in \mathbb{R}$, such that, $\vec{B} = m\vec{A}$, hence $\vec{B} = m\vec{A}$

$= m(0,-2,1)$

$= (0,-2m,m)$

$\therefore \boxed{\vec{B} = (0,-2m,m)}$

Then $\vec{A}.\vec{B} = -10$, implies

$(0,-2,1).(0,-2m,m) = -10$

$\Rightarrow 0.0 + (-2).(-2m) + 1.m = -10$

$\Rightarrow \boxed{m = -2}$

Therefore A) $\vec{A}$ and $\vec{B}$ have opposite direction since $\vec{A} = m\vec{B}$ for $m < 0$.

B) $\|\vec{B}\| = \|m\vec{A}\| = |m|\|\vec{A}\|$, but $m = -2$ gives $|m| = 2$

$\therefore \|\vec{B}\| = 2\|\vec{A}\|$

C) Shown in (B), that $\|\vec{B}\| \neq \frac{1}{10}\|\vec{A}\|$

D) Shown in (B), that $\|\vec{B}\| = 2\|\vec{A}\|$

Ans. D

54. $cos^2x + \frac{1}{2}\ sin2x = 1$ for $x \in [0,2\pi)$, given

$\Rightarrow cos^2x - 1 + \frac{1}{2}\ sin2x = 0$

$\Rightarrow -(1 - cos^2x) + \frac{1}{2}\ (2\sin x\, cosx) = 0$

$\Rightarrow -sin^2x + sinxcosx = 0$

$\Rightarrow sinx(-sinx + cosx) = 0$

$\Rightarrow sinx = 0$ or $-sinx + cosx = 0$

$\Rightarrow sinx = 0$ or $sinx = cosx$

$\Rightarrow sinx = 0$ or $tanx = 1$

Hence, $sinx = 0$ gives $x = sin^{-1}0 \Rightarrow x = 0$ or π in $[0,2\pi]$ and, $tanx = 1$ gives $x = tan^{-1}(1) \Rightarrow x = \frac{\pi}{4}$ or $\frac{5}{4}\pi$ in $[0,2\pi]$ therefore solution set $= \left\{0, \frac{\pi}{4}, \pi, \frac{5}{4}\pi\right\}$

Ans. A

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55. $\vec{U} = (-3, x)$ and $\vec{V} = (x, y-2)$ then if $\vec{U} + \vec{V} = 3\vec{U} - \frac{1}{2}\vec{V}$, we get

$(-3, x) + (x, y-2) = 3(-3, x) - \frac{1}{2}(x, y-2)$

$\Rightarrow (-3 + x, x + y - 2) = (-9 - \frac{1}{2}x, 3x - \frac{1}{2}y + 1$

Then since $(a, b) = (c, d)$ iff $a = c$ and $b = d$

also $-3x + x = -9 - \frac{1}{2}x$ *and* $x + y - 2 = 3x - \frac{1}{2}y + 1$

$\Rightarrow x = -4$ *and* $\frac{3}{2}y - 2x = 3$

$\Rightarrow \frac{3}{2}y - 2(-4) = 3$

$\therefore y = -\frac{10}{3}$

Ans. B

56. First put $x^2 + y^2 + z^2 - 2x + 6z + 9 = 0$ in standard form by completing square, with respect to variables having 2nd degree terms (x and z).

$\Rightarrow (x-1)^2 + y^2 + (z+3)^2 = 1$ *(in standard form)*

$\Rightarrow$ *center*, $c = (1,0,-3)$ *and radius*, $r = 1$

Then let us evaluate distance between, the center, c and points in A,B, C and D.

B) $d_1 = \sqrt{\Delta^2 x + \Delta^2 y + \Delta^2 z} = \sqrt{(1-1)^2 + (0-0)^2 + (-3-0)^2} = 3$

C) $d_2 = \sqrt{(1-0)^2 + (0-0)^2 + (-3-0)^2} = \sqrt{10}$

D) $d_3 = \sqrt{(1-0)^2 + (0-(-1))^2 + (-3-0)^2} = \sqrt{11}$

E) $d_4 = \sqrt{(1-0)^2 + (0-0)^2 + (-3-(-1))^2} = \sqrt{5}$

Then the shortest distance is between the center $C(1,0,-3)$ and a point $(0,0,-1)$ in (D). i.e $d = \sqrt{5}$. Hence $(0,0,-1)$ is closest.

Ans. D

57. A principle of Mathematical Induction says, for a statement $P(n)$, where $n \in \mathbb{N}$,

i) If P(1) is true (i.e P is true for n=1), and

ii) P(n+1) is true whenever P(n) is true, then, P(n) is true for all $n \in \mathbb{N}$.

Ans. D

58. $cot(arcsinx)$ *for* $0 < x < 1$, means $\cot(\sin_x^{-1})$ *for* $0 < x < 1$.

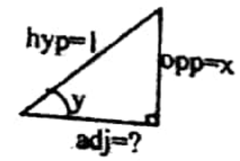
First: let $y = \sin_x^{-1}$, $\left(recall\ for\ 0 < x < 1, 0 < y < \frac{\pi}{2}\right)$

Then $y = sinx^{-1} \Rightarrow x = siny$ so

let $siny = x$ be put as $siny = \frac{opp}{hyp} = \frac{x}{1}$ and y be angle as shown in the figure. Then $adj = \sqrt{hyp^2 - opp^2}$

$= \sqrt{1^2 - x^2}$

$= \sqrt{1 - x^2}$



Therefore, since we set $= sinx^{-1}y_1$

$\Rightarrow \cot(sinx^{-1}) = coty$

$= \frac{adj}{opp}$

$= \frac{\sqrt{1-x^2}}{x}$

Ans. B

59. $\vec{A} = 2\vec{j} - \vec{k}$, $\vec{B} = 5\vec{i} + 15\vec{k}$, then the unit vector in the direction of $\vec{A} + \frac{1}{5}\vec{B}$ is

$$U = \frac{\vec{A} + \frac{1}{5}\vec{B}}{\left\|\vec{A} + \frac{1}{5}\vec{B}\right\|}$$

$$= \frac{2\vec{j} - \vec{k} + \frac{1}{5}(5\vec{i} + 15\vec{k})}{\left\|2\vec{j} - \vec{k} + \frac{1}{5}(5\vec{i} + 15\vec{k})\right\|}$$

$$= \frac{\vec{i} + 2\vec{j} + 2\vec{k}}{\left\|\vec{j} - 2\vec{j} + 2\vec{k}\right\|}$$

$$= \frac{\vec{i} + 2\vec{j} + 2\vec{k}}{\sqrt{1^2 + 2^2 + 2^2}}$$

$$= \frac{1}{3}\vec{i} + \frac{2}{3}\vec{j} + \frac{2}{3}\vec{k}$$

Ans. B

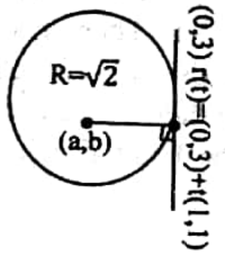
60. Let (a, b) be the center of the circle as in the figure shown

⇒ And the line $r(t) = (0,3) + t(1,1)$ is tangent at $(0,3)$, of a circle of radius , $R = \sqrt{2}$

⇒ By distance formula ,

$(a-0)^2 + (b-3)^2 = \sqrt{2}^2$

⇒ $\boxed{a^2 + (b-3)^2 = 2}$ − − − (1)



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$\Rightarrow$ Let $t = 1$, then $r(1) = (0,3) + 1(1,1) = (1,4)$ is on the line hence since the line is perpendicular to the radius $(a-0, b-3)$.x
$(1-0, 4-3) = 0$
$\Rightarrow (a, b-3)(1,1) = 0$
$\Rightarrow a + b - 3 = 0$
$\Rightarrow \boxed{a+b=3} - - - (2)$
Then combining (1) and (2)
$a^2 + (b-3)^2 = 2$ *and* $a + b = 3$
From $a + b = 3$ *we get* $b = 3 - a$
Then $a^2 + (3 - a - 3)^2 = 2$
$\Rightarrow a^2 = 1 \Rightarrow a = \pm 1$
If $a = -1$, $b = 3 - a = 4$
So possible points are $(1,2)$ and $(-1,4)$

Ans. D

61. Let $P \equiv$ *He was lazy.*
$q \equiv$ *He likes the class room*
$r \equiv$ *He passed the exam*
Then describing the premises as follows,
"He was lazy or he did not like the classroom" $\equiv P \vee \neg q$
"If he was lazy, he could not pass the exam" $\equiv P \Rightarrow \neg q$
"He passed the exam" $\equiv r$
Let Q be the conclusion, then the arguments is valid if $[(P \vee \neg q) \wedge (P \Rightarrow \neg r) \wedge r] \Rightarrow Q$ *is True (Tautology)*
Hence, since premises are true, also Q must be true.
Then identify truth value of p,q and r as follows
→ $r \equiv T$(*premise*)
→ $p \Rightarrow \neg r \equiv T$ (*primise*)
→ $p \Rightarrow \neg T \equiv T$ (*since* $r \equiv T$)
→ $p \Rightarrow F \equiv F$
→ Hence, p must be F.
And → $p \vee \neg q \equiv T$(*premise*)
→ $F \vee \neg q \equiv T$ *since* $p \equiv F$
Hence $\neg q$ must be T so that q is F.
→ Then checking for truth value of choices.

E) He did like the classroom$\equiv q \equiv F$
F) He didn't like the classroom$\equiv \neg q \equiv \neg F \equiv T$
G) If he didn't like the classroom ,he did like the classroom$\equiv \neg p \Rightarrow q$
$\equiv \neg F \Rightarrow F$
$\equiv T \Rightarrow F \equiv F$

H) He was not lazy and he did like the classroom$\equiv \neg p \wedge q$
$\equiv \neg F \wedge F$
$\equiv T \wedge F \equiv F$

Therefore, the only conclusion, (i.e True) is in B.
$\Rightarrow$ *The argument is valied if* (B)*is the conclusion.* i.e $Q = \neg p$

Ans. B

62. We simply translate and rotate the center of the ellipse first.
- $C(-2,1)$ is the center of the ellipse.
Then T: $(2,1) \xrightarrow{U_T} (4,0)$ gives $U_T = (2,-1)$ ← Translation vector.
Hence, $C(-2,1) \xrightarrow{(2,-1)} (-2,1) + (2,-1) = (0,0)$
- Then to rotate $(0,0)$ by any θ means still the image is at (0,0).
i.e $(0,0) \xrightarrow{\theta^{90^0}} (0,0)$
Therefore the center of the image ellipse is (0,0). But note that the given was vertical ellipse, $(x+2)^2 + \frac{(y-1)^2}{2} = 1$, hence rotation by 90^0 makes it horizontal ellipse with center (0,0) and $\frac{x^2}{2} + y^2 = 1$ is the image ellipse.
$\Rightarrow x^2 + 2y^2 = 2$

Ans. A

63. $\vec{a} = 2\vec{i} + (x-1)\vec{j} + \vec{k}$ and $\vec{c} = \vec{i} - \vec{j} + y\vec{k}$
- $\vec{a}.\vec{c} = 0$ gives $(2,(x-1),1).(1,-1,y) = 0$
$\Rightarrow 2 - (x-1) + y = 0$
$\Rightarrow \boxed{-x + y = -3} - - - (1)$
- $\|\vec{a}\| = 3$ gives $\sqrt{2^2 + (x-1)^2 + 1^2} = 3$
$\Rightarrow (x-1)^2 = 4$
$\Rightarrow \quad x - 1 = \pm 2$
$\Rightarrow x = 3$ *or* $x = -1$
Then if $x = 3$, using $-x + y = -3$ from (1)
$-3 + y = -3 \Rightarrow \boxed{y = 0}$
or $x = -1 \Rightarrow -(-1) + y = -3$
$\Rightarrow \boxed{y = -4}$

Ans. A

64. Given $\vec{A} \perp \vec{B}$, we need $cos\theta$ for θ between $\vec{A}$ and $\vec{A} - \vec{B}$.
Then $cos\theta = \frac{\vec{A}.(\vec{A}-\vec{B})}{\|\vec{A}\|\|\vec{A}-\vec{B}\|}$

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$$= \frac{\vec{A}.\vec{A} - \vec{A}.\vec{B}}{\|\vec{A}\|\|\vec{A} - \vec{B}\|}$$

$$= \frac{\|\vec{A}\|^2 - 0}{\|\vec{A}\|\|\vec{A} - \vec{B}\|}, since\ \vec{A}.\vec{B} = 0\ \&\ \|\vec{A}\|^2 = \vec{A}.\vec{A}$$

$$= \frac{\|\vec{A}\|}{\|\vec{A} - \vec{B}\|}$$

65. A) $\|\vec{A}\| = \|\vec{B}\|$ gives only their length is equal, no information about direction. Ans. B

$$\therefore \vec{A} = \vec{B}\ may\ fail$$
$$\therefore (A) is\ False$$

B) $\|k\vec{A}\| = |k|\|\vec{A}\| = \begin{cases} k\|\vec{B}\|\ if\ k \geq 0 \\ -k\|\vec{A}\| if\ k < 0 \end{cases}$ for any real number k.

$$\therefore B\ is\ false$$

C) For $\vec{A} \parallel \vec{B}$, $\vec{A}.\vec{B} = 0$ only if $\vec{A} = \vec{0}$ or $\vec{B} = 0$

$$\therefore (C) is\ false$$

D) If $\vec{u}$ is a unit vector in the direction of $\vec{A}$, then

$$\vec{u} = \frac{\vec{A}}{\|\vec{A}\|}\ (definition)$$

$$= \vec{A}\,\vec{u} = \vec{A}.\frac{\vec{A}}{\|\vec{A}\|} = \frac{\vec{A}.\vec{A}}{\|\vec{A}\|}$$

$$= \frac{\|\vec{A}\|^2}{\|\vec{A}\|} = \|\vec{A}\|$$

$$\therefore \vec{A}.\vec{U} = \|\vec{A}\|$$

$$\therefore D\ is\ True$$

Ans. D

1. A circular cylinder with base radius $r = 2m$ and height $h = 3m$ has a volume V given by,

$$V = \pi r^2 h = \pi(2m)^2 3m = 12\pi m^3 \ldots\ldots\ldots(1)$$

And the water is pumped into it at a rate of $2\,m^3/min$ implies that the rate of change of V is

$$\frac{dV}{dt} = 2\,m^3/min \text{ or simply} \frac{dV}{dt} = 2.$$
$$\Rightarrow dV = 2dt$$

Thus, if $V = U$ is the volume of the water at any time $t = \tau$, then

$$\Rightarrow dU = 2d\tau$$
$$\Rightarrow \int_{V_0}^{V} dU = \int_{t_0}^{t} 2d\tau \ldots\ldots\ldots (2)$$

Where,
$V_0 = 0$, the initial volume (because the tank was empty),
$V = 12\pi m^3$, final volume (from (1)),
$t_0 = 0$, (initial time), and
t = Final time taken for the tank to be full (this is what we need!).

Thus (2) implies $\int_0^{12\pi} dU = \int_0^t 2d\tau \Rightarrow (U)|_0^{12\pi} = (2\tau)|_0^t$

$$\Rightarrow 12\pi - 0 = 2t - 2(0)$$
$$\Rightarrow t = 6\pi\ min.$$

Answer: C

2. First, we extract for B as follows:

$(2A + B)^T = A^T A$, (given)
$\Rightarrow ((2A + B)^T)^T = (A^T A)^T$, (transposing both sides).
$\Rightarrow 2A + B = (A^T A)^T$, (because $(C^T)^T = C$)
$\Rightarrow B = (A^T A)^T - 2A$,
$\Rightarrow B = A^T A - 2A$, (because $A^T A$ is symmetric matrix)
Thus, $B = A^T A - 2A$ ..... (1)

Let us then, find $A^T A$ as:

$$A^T A = \begin{pmatrix} 2 & 0 & -1 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}^T \begin{pmatrix} 2 & 0 & -1 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

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$$= \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 & -1 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 5 & 2 & -2 \\ 2 & 4 & 0 \\ -2 & 0 & 2 \end{pmatrix}.$$

Then using (1),

$$B = A^T A - 2A = \begin{pmatrix} 5 & 2 & -2 \\ 2 & 4 & 0 \\ -2 & 0 & 2 \end{pmatrix} - 2\begin{pmatrix} 2 & 0 & -1 \\ 1 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{pmatrix}.$$

Answer: D

3. **Definition:** A sequence $G_1, G_2, G_3, G_4, \ldots$ of non-zero real numbers is said to be a geometric sequence if there is non-zero real number r such that

$$\frac{G_2}{G_1} = \frac{G_3}{G_2} = \frac{G_5}{G_3} = \cdots = r.$$

Therefore, only choice (A) satisfies this definition, because

$$\Rightarrow \frac{1}{3} = \frac{\frac{1}{3}}{1} = \frac{\frac{1}{9}}{\frac{1}{3}} = \frac{\frac{1}{27}}{\frac{1}{9}} = \cdots = \frac{1}{3}.$$

Answer: A

4. A convergent sequence converges to its limit. So

$$\lim_{n\to\infty} \frac{(n-1)(2n+1)}{1-n^2} = \lim_{n\to\infty} \frac{2n^2 - n - 1}{1-n^2}$$

$$= \lim_{n\to\infty} \frac{\frac{2n^2}{n^2} - \frac{n}{n^2} - \frac{1}{n^2}}{\frac{n^2}{n^2} - \frac{n^2}{n^2}}$$

$$= \lim_{n\to\infty} \frac{2 - \frac{1}{n} - \frac{1}{n^2}}{\frac{1}{n^2} - 1}$$

$$= \frac{\lim_{n\to\infty} 2 - \frac{1}{n} - \frac{1}{n^2}}{\lim_{n\to\infty} \frac{1}{n^2} - 1}$$

$$= \frac{\lim_{n\to\infty} 2 - \lim_{n\to\infty} \frac{1}{n} - \lim_{n\to\infty} \frac{1}{n^2}}{\lim_{n\to\infty} \frac{1}{n^2} - \lim_{n\to\infty} 1} = \frac{2-0-0}{0-1} = -2.$$

Answer: B

5. If $F(x)$ is the anti-derivative of $f(x) = 1 - \frac{2}{x^2}$, then $F(x) = \int f(x)dx$.

Thus, $F(x) = \int f(x)dx$

$$= \int (1 - \frac{2}{x^2})dx$$

$$= \int 1dx - \int \frac{2}{x^2}dx$$

$= x + C_1 - 2\int \frac{1}{x^2}dx$, for a constant of integration C_1.

$= x + C_1 - 2(-\frac{1}{x} + C_2)$, for a constant of integration C_2.

$= x + \frac{2}{x} + C$, for $C = C_1 - 2C_2$.

Hence, $F(x) = x + \frac{2}{x} + C$, for a constant C.

So, $F(1) = 3 + C$.....(1)

To find the value of C we combine (1) and the given condition

$F(1) = 0$, and so

$$3 + C = 0 \Rightarrow C = -3.$$

Thus, $F(x) = x + \frac{2}{x} - 3$ so that $F(2) = 2 + \frac{2}{2} - 3 = 0$.

Answer: A

6. **Definition:** A polynomial expression in x is any expression that can be put in the form

$$a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_nx^n,$$

where $n \in \mathbb{N}$ and $x, a_0, a_1, a_2, a_3 \ldots a_n \in \mathbb{R}$.

Since any real number is a polynomial expression, and $\frac{2+\pi}{1+\pi^2} \in \mathbb{R}$ setting $a_0 = \frac{2+\pi}{1+\pi^2}$ and $a_1 = a_2 = a_3 = \cdots = a_n = 0$,

$\frac{2+\pi}{1+\pi^2}$ is a polynomial expression.

Answer: C

7. Recall that the distance, d between a line $Ax + By + C = 0$ and a point (x_0, y_0) is given by

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}} \ldots\ldots(1)$$

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If (x, y) is any point on a line ℓ that passes through $(1,0)$and $(0,1)$, then its equation is given by $\ell: \frac{y-0}{x-1} = \frac{0-1}{1-0}$.
Hence, the general form (i.e. $Ax + By + C = 0$ form) of equation of the line ℓ through $(1,0)$ and $(0,1)$ is:
$x + y - 1 = 0$, where $A = 1, B = 1$, and $C = -1$.
Then, for $(x_0, y_0) = (0,0)$ using (1)

$$d = \frac{|1(0)+1(0)-1|}{\sqrt{1^2+1^2}} = \frac{1}{\sqrt{2}} = \frac{1}{2}\sqrt{2}.$$

Answer: A

8. First, the frequency distribution table is as in table 1 below

Index	1	2	3	4	5
Value of x_i	10	α	5	20	15
Frequency of $x_i = f_i$	2	2	3	2	1

Table 1

$$\text{Mean, } \bar{x} = \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i} = \frac{f_1x_1+f_2x_2+f_3x_3+f_4x_5+f_5x_5}{f_1+f_2+f_3+f_4+f_5}$$
$$= \frac{2(10) + 2(\alpha) + 3(5) + 2(20) + 1(15)}{2+2+3+2+1}$$
$$= \frac{90 + 2\alpha}{10}$$

$$\text{Thus, } \bar{x} = \frac{90+2\alpha}{10} \Rightarrow 2\alpha = 10\bar{x} - 90 \Rightarrow \alpha = 5\bar{x} - 45$$

Answer: D

9. Marginal cost is the rate of change of production cost, and so it is the derivative of the cost.
Let $M(x)$ =Marginal cost of producing x iron sheets per day,

$$M(x) = \frac{d}{dx}C(x) = \frac{d}{dx}(1{,}000 + 100x - 0.5x^2)$$
$$= \frac{d}{dx}(1{,}000) + \frac{d}{dx}(100x) + \frac{d}{dx}(-0.5x^2)$$
$$= 100 - x.$$

Hence, the Marginal cost, $M(x) = 100 - x$.
Then for a production of $x = 80$ iron sheets, the Marginal cost is

$$M(80) = 100 - 80 = 20.$$

Answer: B

10. $|z|^2 = |x + yi|^2 = (x + yi)^2 = x^2 + y^2$, and
$(z - \bar{z})^2 = \left(x + yi - \overline{(x + yi)}\right)^2$
$= (x + yi - (x - yi))^2 = (2yi)^2 = -4y^2$.
Hence, $|z|^2 = x^2 + y^2$, and $(z - \bar{z})^2 = -4y^2$
Then, substituting in $|z|^2 + \frac{1}{2}(z - \bar{z})^2 = 1$

$$\Rightarrow x^2 + y^2 + \frac{1}{2}(-4y^2) = 1 \Rightarrow x^2 + y^2 - 2y^2 = 1$$
$$\Rightarrow x^2 - y^2 = 1.$$

Answer: A

11. First we evaluate class marks of each interval as in table 2.
Then, the Mean $\bar{x}$ is

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}$$

Class mark: x_i
$x_1 = \frac{3+7}{2} = 5$
$x_2 = \frac{8+12}{2} = 10$
$x_3 = \frac{13+17}{2} = 15$
$x_4 = \frac{18+22}{2} = 20$

$$= \frac{f_1x_1 + f_2x_2 + f_3x_3 + f_4x_4}{f_1 + f_2 + f_3 + f_4}$$
$$= \frac{2(5) + 2(10) + 10(15) + 6(20)}{2+2+10+6}$$
$$= 15.$$

Now, building a table, for values of $f_i(x_i - \bar{x})^2$ in each interval as in table 3.

$f_i(x_i - \bar{x})^2$
$2(5-15)^2 = 200$
$2(10-15)^2 = 50$
$10(15-15)^2 = 0$
$6(20-15)^2 = 150$

Then, the standard deviation S is

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$$S=\sqrt{\frac{f_1(x_1-\bar{x})^2+f_2(x_2-\bar{x})^2+f_3(x_3-\bar{x})^2+f_4(x_4-\bar{x})^2}{f_1+f_2+f_3+f_4}}$$

$$=\sqrt{\frac{200+50+0+150}{2+2+10+6}}=\sqrt{20}=2\sqrt{5}.$$

Answer: A

12. Let F=False and T=True. Then
$(p\vee q)\Rightarrow(\neg r\wedge r)\equiv T$means $(p\vee q)\Rightarrow F\equiv T$, because $\neg r\wedge r\equiv F$,
and$(p\vee q)\Rightarrow F\equiv T$means $p\vee q\equiv F$, because $F\Rightarrow F\equiv T$.
"Disjunction is F if both statements are F."
Hence, $p\equiv F$and $q\equiv F$.
You may see that-Choice (A) fails to be T if $r\equiv T$, and
-Choices (B) and (C) fail to be true if $r\equiv F$.
So $\neg p\equiv T$ implies for whatever $r$is, $\neg p\vee r\equiv T$.

Answer: D

13. $\sum_{n=1}^{\infty}\left(\frac{2^n+5^n}{10^n}\right)=\sum_{n=1}^{\infty}\left(\left(\frac{1}{5}\right)^n+\left(\frac{1}{2}\right)^n\right)$

$$=\sum_{n=1}^{\infty}\left(\frac{1}{5}\right)^n+\sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^n \quad \text{.....(1)}$$

Then (i) $\sum_{n=1}^{\infty}\left(\frac{1}{5}\right)^n$, is geometric series with $G_n=\left(\frac{1}{5}\right)^n$ and

$$G_1=\left(\frac{1}{5}\right)^1=\frac{1}{5}.$$

So $r=\frac{G_{n+1}}{G_n}=\frac{\left(\frac{1}{5}\right)^{n+1}}{\left(\frac{1}{5}\right)^n}=\frac{1}{5}$.

Since, $|r|<1$ the sum $\sum_{n=1}^{\infty}\left(\frac{1}{5}\right)^n$ exists (i.e. converges), and

$$\sum_{n=1}^{\infty}\left(\frac{1}{5}\right)^n=\frac{G_1}{1-r}=\frac{\frac{1}{5}}{1-\frac{1}{5}}=\frac{1}{4} \quad \text{.....(2)}$$

(ii) In the same way for the same reason,

$$\sum_{n=1}^{\infty}\left(\frac{1}{2}\right)^n=1 \quad \text{.....(3)}$$

Then, inserting (2) and (3) in to (1) gives

$$\sum_{n=1}^{\infty}\left(\frac{2^n+5^n}{10^n}\right)=\frac{1}{4}+1=\frac{5}{4}.$$

Answer: C

14. $f(x)=\frac{x+1}{x-1}\Rightarrow f(a)=\frac{a+1}{a-1}$
And using the given $f(a)=5$,
$\frac{a+1}{a-1}=5\Rightarrow a+1=5(a-1)\Rightarrow a=\frac{3}{2}$.
Therefore,$f(2a)=f\left(2\left(\frac{3}{2}\right)\right)=f(3)=\frac{3+1}{3-1}=2$.

Answer: A

15. Sety $=f(x)$. Then $f(x)=\sqrt[3]{1+e^{-x}}$ implies
$y=\sqrt[3]{1+e^{-x}}$.
Then, interchanging $x$and y, the inverse is
$x=\sqrt[3]{1+e^{-y}}$.
Now, we makey which is $y=f^{-1}(x)$ a subject as,

$$x=\sqrt[3]{1+e^{-y}}\Rightarrow x^3=(\sqrt[3]{1+e^{-y}})^3$$
$$\Rightarrow x^3=1+e^{-y}$$
$$\Rightarrow e^{-y}=x^3-1$$
$$\Rightarrow -y=ln(x^3-1)\text{for}\ x^3-1>0.$$
$$\Rightarrow y=-ln(x^3-1)=ln(x^3-1)^{-1}.$$

Therefore,$f^{-1}(x)=ln(\frac{1}{x^3-1})$ for$x^3-1>0$.

Answer: A

16. Given equation of a parabola$y^2+8x+6y+25=0$, We need to find the directrix.
Since y is of second degree thestandard form of equation of this parabola is
$4p(x-h)=(y-k)^2$with directrix$x=h-p$.
Then completing the square with respect to y goes as:

$$y^2+8x+6y+25=0\Rightarrow y^2+6y+9=-8x-25+9$$
$$\Rightarrow (y+3)^2=-8x-16$$
$$\Rightarrow (y+3)^2=4(-2)(x+2)$$

Hence, $h=-2, p=-2$, and $k=-3$.
Therefore, the directrix is the line,$x=h-p=-2-(-2)=0$.

Answer: C

NOTE: Also we may use the general form of equation of a horizontal parabola $Cy^2+Dx+Ey+F=0$ with directrix,

$$x=\frac{E^2-4CF+D^2}{4CD}.$$

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Where as in our case $C = 1, D = 8, E = 6$ and $F = 25$ and

$$x = \frac{6^2-4(1)(25)+8^2}{4(1)(8)} = 0.$$

Answer: C

17. First not thatfactorizing the denominator and simplifying,

$\frac{(f(x)-g(x))(g(x)-2f(x))}{g(x)^2-f(x)^2} = -\frac{g(x)-2f(x)}{g(x)+f(x)}$, for $g(x) \neq \pm f(x)$.

Since $g(3) \neq \pm f(3)$ (i.e. $11 \neq \pm 5$) the required limit exists and so,

$$\text{Lim}_{x\to 3}\left(\frac{(f(x)-g(x))(g(x)-2f(x))}{g(x)^2-f(x)^2}\right) = \lim_{x\to 3}\left(-\frac{g(x)-2f(x)}{g(x)+f(x)}\right),$$

$$= -\frac{\lim_{x\to 3}(g(x)-2f(x))}{\lim_{x\to 3}(g(x)+f(x))}$$

$$= -\frac{\lim_{x\to 3} g(x) - 2\lim_{x\to 3} f(x)}{\lim_{x\to 3} g(x) + \lim_{x\to 3} f(x)}$$

$$= -\frac{11-2(5)}{11+5} = -\frac{1}{16}.$$

Answer: B

18. $f'(x) = \left(\frac{x^2}{1+xg(x)}\right)'$

$= \frac{(x^2)'(1+xg(x))-x^2(1+xg(x))'}{(1+xg(x))^2}$, (By quotient rule)

$= \frac{2x(1+xg(x))-x^2(1+xg(x))'}{(1+xg(x))^2}$ ..... (1)

Now let us evaluate $(1 + xg(x))'$ first, as

$(1 + xg(x))' = 1' + (xg(x))'$, (By sum rule)

$= 0 + x'g(x) + xg'(x)$, (By product rule)

$= g(x) + xg'(x)$..... (2)

Substituting (2) in to (1) gives,

$f'(x) = \frac{2x(1+xg(x))-x^2(g(x)+xg'(x))}{(1+xg(x))^2} = \frac{2x+x^2g(x)-x^3g'(x)}{(1+xg(x))^2}$.

So $f'(2) = \frac{2(2)+(2)^2g(2)-(2)^3g'(2)}{(1+2g(2))^2} = \frac{4+4(1)-8(10)}{(1+2(1))^2} = \frac{-72}{9} = -8.$

Answer: A

19. First simplifying w as,

$$w = \frac{16i}{1+i} + (1-3i)^2 = \frac{16i}{1+i} + (-8-6i)$$

$$= \frac{16i + (1+i)(-8-6i)}{1+i}$$

$$= \frac{-2+2i}{1+i}$$

$$= \frac{-2+2i}{1+i} \times \frac{1-i}{1-i} = 2i.$$

Hence, $|w| = |2i| = 2$ and $\bar{w} = \bar{2i} = -2i$.

Therefore, $z = |w| + \bar{w} = 2 - 2i$.

Answer: D

20. **Properties of determinant**

(i) $A = B \Rightarrow \det(A) = \det(B)$

(ii) $\det(AB) = \det(A)\det(B)$

(iii) If I is the identity (or any diagonal) matrix of order k, then $\det(aI) = a^k \det(I)$, for $a \in \mathbb{R}$.

(iv) $\det(A) = \det(A^T)$

(v) $\det(I) = 1$, if I is the identity matrix of any order.

Now $A^TM = 2I \Rightarrow \det(A^TM) = \det(2I)$ (By property (i))

$\Rightarrow \det(A^T)\det(M) = 2^3\det(I)$, (By property (ii) & (iii))

$\Rightarrow \det(A^T)\det(M) = 8(1)$, (By property (v))

$$\Rightarrow \det(A^T) = \frac{8}{\det(M)}$$

Then, combining with property (iv), we get that,

$$\det(A) = \det(A^T) = \frac{8}{\det(M)} \text{.....(1)}$$

Now we finddet(M), by expanding the first raw as

$$\text{Det}(M) = \det\left(\begin{pmatrix} 0 & 1 & 2 \\ 3 & -1 & 0 \\ 5 & 2 & 4 \end{pmatrix}\right)$$

$$= 0 \cdot (-1)^{1+1}\begin{vmatrix} -1 & 0 \\ 2 & 4 \end{vmatrix} + 1 \cdot (-1)^{1+2}\begin{vmatrix} 3 & 0 \\ 5 & 4 \end{vmatrix} + 2 \cdot (-1)^{1+3}\begin{vmatrix} 3 & -1 \\ 5 & 2 \end{vmatrix}$$

$$= -1 \cdot \big(3(4) - 5(0)\big) + 2 \cdot \big(3(2) - 5(-1)\big) = 10.$$

Then using (1),$\det(A) = \frac{8}{10} = 0.8$.

Answer: C

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21. If the first place is to be occupied by one of the B's then the rest will be occupied by 2A, 2B, 2E, 1Y.

So by product rule one of the B's followed by

$$\frac{7!}{2!\,2!\,2!\,1!} = 1 \times \frac{7!}{2!\,2!\,2!\,1!} = 630$$

And, if the first place is to be occupied by Y then the rest will be occupied by 2A, 3B, 2E.

So by product rule Y followed by $\frac{7!}{3!2!2!} = 1 \times \frac{7!}{3!2!2!} = 210$

Therefore, the codes begin with B or Y=630+210=840.

Answer: A

22. Since the determinant of the coefficient matrix is zero,

$$\text{i.e.}\Delta = \begin{vmatrix} 1 & -1 & 1 \\ -1 & 5 & -4 \\ 2 & 2 & -1 \end{vmatrix} = 0,$$

The system does not have a unique solution.

But the system may have infinitely many solutions if

$$\Delta = 0, \Delta_x = 0,\ \Delta_y = 0, \text{ and } \Delta_z = 0.$$

Then,

$$\Delta_x = \begin{vmatrix} 1 & -1 & 1 \\ 1 & 5 & -4 \\ k & 2 & -1 \end{vmatrix} = 4 - k, \text{ and } \Delta_x = 0 \Rightarrow 4 - k = 0 \Rightarrow k = 4$$

$$\Delta_y = \begin{vmatrix} 1 & 1 & 1 \\ -1 & 1 & -4 \\ 2 & k & -1 \end{vmatrix} = 3k - 12, \text{ \& } \Delta_y = 0 \Rightarrow 3k - 12 = 0 \Rightarrow k = 4$$

$$\Delta_z = \begin{vmatrix} 1 & -1 & 1 \\ -1 & 5 & 1 \\ 2 & 2 & k \end{vmatrix} = 4k - 16, \text{ \& } \Delta_z = 0 \Rightarrow 4k - 16 = 0 \Rightarrow k = 4$$

Hence, the system of equations given above has a solution if $k = 4$.

Answer: D

23. Since f is continuous, but defined in different expressions at $x = 0$, side limits exist and are equal. Hence,

$$\Rightarrow \lim_{x \to 0^-} f(x) = \lim_{x \to 0^+} f(x) \quad \ldots\ldots\ldots(1)$$

Then, $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} a\frac{\sin x}{x-|x|}$, because $x \to 0^-$ implies $x < 0$.

$$= \lim_{x \to 0^-} a\frac{\sin x}{x-(-x)}, \text{ because } x < 0 \text{ implies } |x| = -x.$$

$$= \lim_{x \to 0^-} a\frac{\sin x}{2x} = \frac{a}{2}\left(\lim_{x \to 0^-} \frac{\sin x}{x}\right) = \frac{a}{2}(1) = \frac{a}{2}.$$

And, $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (e^{-x} + \cos x)$

$$= \lim_{x \to 0^+} e^{-x} + \lim_{x \to 0^+} \cos x = e^{-0} + \cos 1 = 2.$$

Then (1) gives $\frac{a}{2} = 2$ so that $a = 4$.

Answer: A

NOTE: We may also use $\lim_{x \to 0^-} f(x) = f(0)$.

The points of intersection are (-1,1), and (1,1).

$$A = A_1 + A_2$$

$$= 2\int_{-1}^{0} (-x^2 + 2 - (-x))dx + \int_{1}^{2} (x - (-x^2 + 2))dx$$

$$= 2\int_{-1}^{0} (2 + x - x^2)dx + \int_{1}^{2} (x^2 + x - 2))dx$$

$$= 2\left[2x + \frac{1}{2}x^2 - \frac{1}{3}x^3\right]_{-1}^{0} + \left[\frac{1}{3}x^3 + \frac{1}{2}x^2 - 2x\right]_{1}^{2}$$

$$= \frac{7}{3} + \frac{11}{6} = \frac{25}{6}.$$

Answer: B

25. $f'(x) = \frac{d}{dx}\left(\int_0^{x^2+\pi} \frac{dt}{1+\sin t}\right)$

$$= \frac{1}{1+\sin(x^2+\pi)} \frac{d}{dx}(x^2 + \pi), \text{ (By chain rule)}$$

$$= \frac{2x}{1+\sin(x^2+\pi)}.$$

Answer: B

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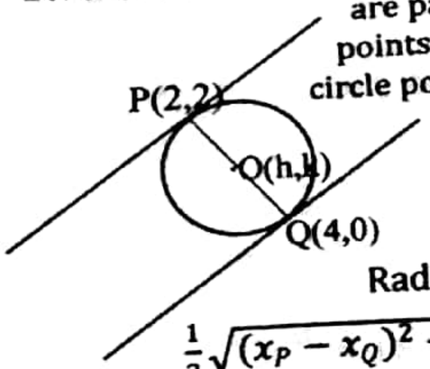
26. Since the two lines $y = x$ and $y = x - 4$ have the same slop they are parallel, so that points of tangencies are end points of a diameter(i.e. $\overline{PQ}$)and the center of the circle point $O(h,k)$is the mid-point of $\overline{PQ}$. Hence,



$$(h,k) = \left(\frac{x_P + x_Q}{2}, \frac{y_P + y_Q}{2}\right)$$

$$= \left(\frac{2+4}{2}, \frac{2+0}{2}\right) = (3,1)\text{is center and}$$

Radius, $r = \frac{1}{2}(diameter) = \frac{1}{2}PQ =$

$$\frac{1}{2}\sqrt{(x_P - x_Q)^2 + (y_P - y_Q)^2}$$

$$= \frac{1}{2}\sqrt{(2-4)^2 + (2-0)^2} =$$

$\sqrt{2}$.

Therefore, equation of the circle is $(x-3)^2 + (y-1)^2 = 2$.

Answer: C

27. A function f continuous at $c \in \mathbb{R}$, has a critical number c if either $f'(c) = 0$ or does not exist.

$$f(x) = \frac{1}{3}x^3 - |4x-1| = \begin{cases} \frac{1}{3}x^3 - 4x + 1 & if\ x \geq \frac{1}{4} \\ \frac{1}{3}x^3 + 4x - 1 & if\ x < \frac{1}{4} \end{cases}$$

$$\Rightarrow f'(x) = \begin{cases} x^2 - 4 & if\ x > \frac{1}{4} \\ x^2 + 4 & if\ x < \frac{1}{4} \end{cases}$$

i) For $x > \frac{1}{4}$, $f'(x) = 0 \Rightarrow x^2 - 4 = 0 \Rightarrow x = \pm 2$, but $x > \frac{1}{4}$.

So the only critical number for $x > \frac{1}{4}$is $x = 2$.

ii) For $x < \frac{1}{4}$, $f'(x) \neq 0$ because$x^2 + 4 \neq 0$ for real x.

So no critical number for $x < \frac{1}{4}$.

iii)At $x = \frac{1}{4}$,$f'\left(\frac{1}{4}\right)$ does not exist because

$$\lim_{x \to \frac{1}{4}^-} \frac{f(x) - f\left(\frac{1}{4}\right)}{x - \frac{1}{4}} = \frac{65}{16} \neq \lim_{x \to \frac{1}{4}^+} \frac{f(x) - f\left(\frac{1}{4}\right)}{x - \frac{1}{4}} = -\frac{63}{16}$$

Therefore the set of all critical numbers=$\{\frac{1}{4}, 2\}$.

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Answer: A

28. $\int_0^{\frac{\pi}{2}} \frac{x - \sin x}{\sec x} dx = \int_0^{\frac{\pi}{2}} (x - \sin x)\cos x\, dx$, because $\frac{1}{\sec x} = \cos x$

$$= \int_0^{\frac{\pi}{2}} x\cos x\, dx - \int_0^{\frac{\pi}{2}} \sin x \cos x\, dx \text{.....(1)}$$

Now evaluating the right side integrals separately as,

i) For $\int_0^{\frac{\pi}{2}} x\cos x\, dx$ using integration by '*parts*'

Let $= x$, and $dv = cosx\, dx$, then

$du = dx$,and $\int dv = \int cosx dx \Leftrightarrow v = sinx$

Thus $\int \overset{u}{\overbrace{x}} \overset{dv}{\overbrace{cosx dx}} = \overset{u}{\overbrace{x}}\overset{v}{\overbrace{sinx}} - \int \overset{v}{\overbrace{sinx}}\overset{du}{\overbrace{dx}}$

$= xsinx + cosx + C$, for a constant C

So $\int_0^{\frac{\pi}{2}} x\cos x\, dx = (xsinx + cosx + C)\Big|_0^{\frac{\pi}{2}}$

$$= \frac{\pi}{2}sin\frac{\pi}{2} + cos\frac{\pi}{2} - (0sin0 + cos0)$$

$$= \frac{\pi}{2} - 1 \text{.....(2)}$$

ii)For $\int_0^{\frac{\pi}{2}} \sin x \cos x\, dx = \int_0^{\frac{\pi}{2}} \frac{1}{2}\sin 2x\, dx$, because $\sin 2x =$ $2\sin x \cos x$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \sin 2x\, dx$$

$= \frac{1}{2}(-\frac{1}{2}cos2x)\Big|_0^{\frac{\pi}{2}}$, because $\int \sin 2x\, dx = -\frac{1}{2}cos2x + C$

$$= -\frac{1}{4}cos2(\tfrac{\pi}{2}) - (-\frac{1}{4}cos2(0))$$

$$= \frac{1}{2} \text{.....(3)}$$

Therefore, by substituting (2) and) in to (1)

$\int_0^{\frac{\pi}{2}} \frac{x - \sin x}{\sec x} dx = \frac{\pi}{2} - 1 - \frac{1}{2} = \frac{\pi - 3}{2}$.

Answer: A

29. $f(x) = \frac{x^3 - x}{x^2 - x} = \frac{x(x-1)(x+1)}{x(x-1)} = x + 1$, for $x \neq 0,1$

Then,$\lim_{x \to a} f(x) = \lim_{x \to a} x + 1 = a + 1$, for any $a \in \mathbb{R}$.

Hence, $\lim_{x \to a} f(x)$ exists for any $a \in \mathbb{R}$ implies that f has no vertical asymptote.

Answer: D

30. $f'(x) = \left(\frac{1+\sin x}{\cos x}\right)' = \frac{(1+\sin x)'cosx - (cosx)'(1+sinx)}{\cos^2 x}$

$$= \frac{\cos^2 x + \sin^2 x + \sin x}{\cos^2 x} = \frac{1+\sin x}{\cos^2 x}.$$

31. Let Q_i, D_i, and P_i be respectively the i^{th}-quartile, decile, and percentile of a data $x_1, x_2, x_3, \ldots, x_n$ in increasing order. Then, $[x_1, P_{25}] \subset [x_1, D_3] \Rightarrow D_3 > P_{25}$. **Answer: B**

32. The slope of the tangent line to the graph of f at $(a, f(a))$ is $m = f'(a)$. **Answer: B**

And $f'(x) = \left(x^2\sqrt{2x+12}\right)'$

$= (x^2)'\sqrt{2x+12} + \left(\sqrt{2x+12}\right)'x^2$ (Product rule)

$= 2x\sqrt{2x+12} + \frac{1}{2\sqrt{2x+12}}(2x+12)'x^2$ (Chain rule)

$= 2x\sqrt{2x+12} + \frac{2x^2}{2\sqrt{2x+12}}$

$= \frac{5x^2+24x}{\sqrt{2x+12}}$ for $x > -6$.

So for $a = 2, f'(2) = \frac{5(2)^2+24(2)}{\sqrt{2(2)+12}} = 17$.

33. Since $f''(x) = (f'(x))'$, first we evaluate f'. **Answer: D**

$f'(x) = (e^{2x}\sin x)' = (e^{2x})'\sin x + e^{2x}(\sin x)'$, (By product rule)

$= e^{2x}(2x)'\sin x + e^{2x}(\sin x)'$, (By chain rule)

$= 2e^{2x}\sin x + e^{2x}\cos x$

$= e^{2x}(2\sin x + \cos x)$.

Then,

$f''(x) = \left(f'(x)\right)'$

$= \left(e^{2x}(2\sin x + \cos x)\right)'$

$= (e^{2x})'(2\sin x + \cos x) + e^{2x}(2\sin x + \cos x)'$, (By product rule)

$= 2e^{2x}(2\sin x + \cos x) + e^{2x}(2\cos x - \sin x)$, (By chain and sum rules)

$= e^{2x}(2(2\sin x + \cos x) + (2\cos x - \sin x))$

$= e^{2x}(3\sin x + 4\cos x)$.

Answer: C

34. A horizontal hyperbola $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ with $a^2 + b^2 = c^2$ has asymptotes given by lines $y = k \pm \frac{b}{a}(x-h)$.

Thus, for $(x-2)^2 - (y-2)^2 = 1$ we have $h = 2 = k$ and $a = 1 = b$.

$y = k \pm \frac{b}{a}(x-h) = 2 \pm \frac{1}{1}(x-2) = x$ or $4 - x$.

Therefore, equations of asymptotes to the hyperbola are, $y = x$, and $y = 4 - x$. **Answer: A**

35. $\sum_{n=1}^{\infty}(-1)^n 3^{-2n}$ implies that $G_n = (-1)^n 3^{-2n}$.

$r = \frac{G_{n+1}}{G_n} = \frac{(-1)^{n+1}3^{-2(n+1)}}{(-1)^n 3^{-2n}} = -\frac{1}{9}$.

So $|r| < 1$ the series converges geometrically to $\frac{G_1}{1-r}$.

Then for $G_1 = (-1)^1 3^{-2(1)} = -\frac{1}{9}$,

$\sum_{n=1}^{\infty}(-1)^n 3^{-2n} = \frac{G_1}{1-r} = \frac{-\frac{1}{9}}{1-\left(-\frac{1}{9}\right)} = -\frac{1}{10} = -0.1$.

Answer: C

36. Vertex at $(h, k) = (1, 0)$, and y-int $= (0, 2)$.

Equation of the form $4p(y-k) = (x-h)2$

$\Rightarrow 4p(y-0) = (x-1)^2$

$\Rightarrow 4py = (x-1)^2$ ..... (1)

Then by y-intercept at $(x, y) = (0, 2)$ in (1),

$4p(2) = (0-1)^2$ i.e. $p = \frac{1}{8}$.

Substitution of $p = \frac{1}{8}$ in (1) gives $y = 2(x-1)^2$. **Answer: C**

37. Number of elements in sample, $n(S) = 500$,
Number of elements with defect 1, $n(D_1) = 40$,
Number of elements with defect 2, $n(D_2) = 60$, and
Number of elements with defect 1 and 2, $n(D_1 \cap D_2) = 25$.
Then, Number of elements with defect is

$n(D_1 \cup D_2) = n(D_1) + n(D_2) - n(D_1 \cap D_2)$

$= 40 + 60 - 25 = 75$.

$p(non\ defective) = 1 - p(n(D_1 \cup D_2))$

$= 1 - p(n(D_1 \cup D_2))$

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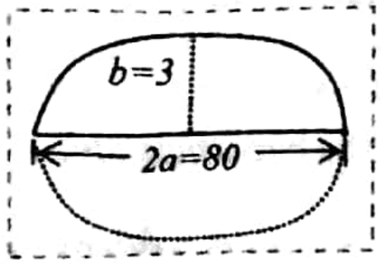
$$= 1 - \frac{75}{500} = \frac{85}{100} = 0.85\,.$$

Answer: C

38. This is a horizontal ellipse as shown below.



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{x^2}{40^2} + \frac{y^2}{30^2} = 1$$

$$\Rightarrow \frac{x^2}{1600} + \frac{y^2}{900} = 1.$$

Answer: B

39. A) False .Becauseif$\Delta = 0$ and $\Delta_x = \Delta_y = \Delta_z = 0$, Infinitely many solutions exist.

B) False. Because row reduction is suitable since the system is reduced to system of two liner equations.

C) True. For instance let

$$A = \begin{pmatrix} a_1 & kb_1 & c_2 \\ a_2 & kb_2 & c_2 \\ a_3 & kb_3 & c_3 \end{pmatrix} \text{ and } \det(A) \neq 0 \text{ then} AX = b \text{ for}$$

$$b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \text{ implies } X = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

(The Explanation is Left to the reader.)

D) False. Because the system is a homogeneous but non zero solution may exist.

Answer: C

40. $y = \sin(3x^2) \Rightarrow \frac{dy}{dx} = \frac{d(\sin(3x^2))}{dx}$

$= \cos(3x^2)\frac{d(3x^2)}{dx}$, (By chain rule)

$= 6x\cos(3x^2)$

Then,

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}(6x\cos(3x^2))$$

$= \cos(3x^2)\frac{d}{dx}(6x) + 6x\frac{d}{dx}(\cos(3x^2))$, (By product rule)

$= 6\cos(3x^2) + 6x\sin(3x^2)\frac{d}{dx}(3x^2)$, (By chain rule)

$= 6\cos(3x^2) + 6x\sin(3x^2)\,6x$

$= 6\cos(3x^2) + 36x^2\sin(3x^2)$.

Answer: C

41. Given $n(S) = 10$and $A \subseteq S$.

$$p(n(A) \geq 3) = 1 - p(n(A) < 3)$$

$$= 1 - \frac{C(10,0)+C(10,1)+C(10,2)}{2^{10}} = 1 - \frac{1+10+45}{2^{10}} = \frac{121}{128}.$$

Answer: C

42. First, we put $\frac{n+3}{n+1}$ as

$$\frac{n+3}{n+1} = \frac{n+1+2}{n+1} = \frac{n+1}{n+1} + \frac{2}{n+1} = 1 + \frac{2}{n+1}.$$

Then, $\lim_{n \to \infty}\left(\frac{n+3}{n+1}\right)^n = \lim_{n \to \infty}\left(1 + \frac{2}{n+1}\right)^n$ .....(1)

Let $\frac{2}{n+1} = \frac{1}{m}$, then $n = 2m - 1$ and as $n \to \infty$ also $m \to \infty$.

So substitution in (1) gives,

$$\lim_{n \to \infty}\left(\frac{n+3}{n+1}\right)^n = \lim_{n \to \infty}\left(1 + \frac{2}{n+1}\right)^n$$

$$= \lim_{m \to \infty}\left(1 + \frac{1}{m}\right)^{2m-1}$$

$$= \lim_{m \to \infty}\left(\left(1 + \frac{1}{m}\right)^{2m}\left(1 + \frac{1}{m}\right)^{-1}\right)$$

$$= \lim_{m \to \infty}\left(\left(\left(1 + \frac{1}{m}\right)^{m}\right)^2\left(1 + \frac{1}{m}\right)^{-1}\right)$$

$$= \left(\lim_{m \to \infty}\left(1 + \frac{1}{m}\right)^{m}\right)^2\left(\lim_{m \to \infty}\left(1 + \frac{1}{m}\right)\right)^{-1}$$

$= e^2(1) = e^2$.

Answer: C

43. First, we put $\frac{x}{x+2}$ as

$$\frac{x}{x+2} = \frac{x+2-2}{x+2} = \frac{x+2}{x+2} + \frac{-2}{x+2} = 1 + \frac{-2}{x+1}.$$

Let $\frac{-2}{x+1} = \frac{1}{m}$, then $x = -2m - 1$ and as $x \to \infty$ also $m \to -\infty$.

Then, $\lim_{x \to \infty}\left(\frac{x}{x+2}\right)^{-3x} = \lim_{x \to \infty}\left(1 + \frac{-2}{x+1}\right)^{-3x}$

$$= \lim_{m \to -\infty}\left(1 + \frac{1}{m}\right)^{-3(-2m-1)}$$

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$$= \lim_{m \to -\infty} \left(\left(1+\frac{1}{m}\right)^{6m}\left(1+\frac{1}{m}\right)^{3}\right)$$
$$= \left(\lim_{m \to -\infty}\left(1+\frac{1}{m}\right)^{m}\right)^{6}\left(\lim_{m \to \infty}\left(1+\frac{1}{m}\right)\right)^{3}$$
$$= e^6(1) = e^6.$$

NOTE: In questions 42 and 43 we can use the formula $\lim_{x \to \pm\infty}\left(\frac{ax+b}{ax+c}\right)^{dx+e} = e^{\frac{d(b-c)}{a}}$.

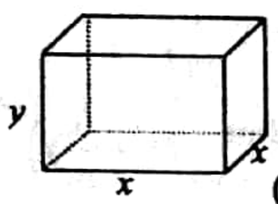
Answer: A

44. Given square base box and open from top with total surface area $1{,}200cm^2$ as in the figure below.
Let x=base side and y=altitude
Then,

Lateral face area + base area = 1200
$$\Rightarrow 4xy + x^2 = 1200$$
$$\Rightarrow y = \frac{1200-x^2}{4x} \quad \ldots\ldots\ldots(1)$$

If V is the volume, then $Volume$ = (base area)(altitude)
$$\Rightarrow V(x,y) = x^2y$$

So V interms of x using (1),



$$V(x) = x^2\left(\frac{1200-x^2}{4x}\right), \text{(By (1))}$$
$$\Rightarrow V(x) = 300x - \frac{1}{4}x^3 \quad \ldots\ldots\ldots(2)$$

Since V is differentiable for all $x \in (0, 20\sqrt{3})$ (why?), it attains its extreme values at x where $V'(x) = 0$.

$$V'(x) = 0 \Rightarrow \left(300x - \frac{1}{4}x^3\right)' = 0 \Rightarrow 300 - \frac{3}{4}x^2 = 0 \Rightarrow x = 20.$$

Hence using (2) the maximum volume is
$$V(20) = 300(20) - \frac{1}{4}(20)^3 = 4000cm^3.$$

Answer: A

45. First we find a function $g(x)$ such that $f(x) = g(x)$ in the domain of f as
$$f(x) = \frac{x^2-1}{x-x^2} = \frac{(x-1)(x+1)}{x(1-x)}, \text{ here } x \neq 0,1.$$
$$= \frac{-(x+1)}{x} = g(x).$$

Hence $f(x) = \frac{-(x+1)}{x} = g(x)$ for $x \neq 0,1$ but note that, $\lim_{x \to a} f(x) = \lim_{x \to a} g(x)$ for $x \neq 0$. Then

A) False. $x = 0$ is the only vertical asymptote because, $\lim_{x \to 0^+} f(x)$ or $\lim_{x \to 0^+} f(x)$ does not exist, since
$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} g(x)$$
$$= \lim_{x \to 0^+} \frac{-(x+1)}{x} = -\infty.$$

Also, show that $\lim_{x \to 0^+} f(x) = \infty$.

But $\lim_{x \to 1} f(x)$ exists, because
$$\lim_{x \to 1} f(x) = \lim_{x \to 1} g(x)$$
$$= \lim_{x \to 1} \frac{-(x+1)}{x} = \frac{-(1+1)}{1} = -2.$$

B) False. The horizontal asymptote (if it exists) is the line
$$y = \lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \frac{x^2-1}{x-x^2}$$
$$= \lim_{x \to \pm\infty} \frac{1-\frac{1}{x^2}}{\frac{x}{x^2}-1} = \frac{\lim_{x \to \pm\infty} 1-\frac{1}{x^2}}{\lim_{x \to \pm\infty} \frac{1}{x}-1} = \frac{-1}{1} = 1.$$

Thus, The horizontal asymptote is the line, $y = -1$.

C) False. f does not have oblique asymptote since the denominator and the numerator are of equal degree, 1.

D) True. This is from the result of (B).

Answer: D

46. Let T≡True and F≡False. Then,

A) Not valid. Because, $p \Longrightarrow q$ and q are true imply $p \Longrightarrow T$ is true. Then, by the rule of implication p can be T or F.

B) Not valid. Because,
$p \Leftrightarrow q$ and $p \Longrightarrow q$ may imply that both p and q are F.

C) Valid. $\neg p \wedge q$ is T implies both $\neg p$ and q are T. Then,
$r \Longrightarrow q$ and q are T imply $T \Longrightarrow r$ is T.
But $T \Longrightarrow r$ is T only if r is T.

D) Not valid. Because, $\neg p$ is T implies both p is F. Then,
$p \vee q$ is T and p is F imply $F \vee q$ is T so that q is T.
$r \Longrightarrow q$ and q are T imply $r \Longrightarrow T$ is true.
Then, by the rule of implication r can be T or F.

Answer: C

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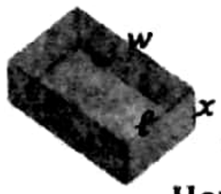
47. We cut out squares with dimension x from each 4-corneras in figure below, then folding sides we get the open top rectangular box with volume

$V = hlw$, where $h = x$, and

$\ell = 72 - 2x = w$.



Note that $x \in (0\ cm, 36\ cm)$, since a box is formed only if $0 < 72 - 2x < 72$.

Then, $V(x) = hlw = x(72 - 2x)^2$.

Hence, $V(x) = 5184x - 288x^2 + 4x^3$.

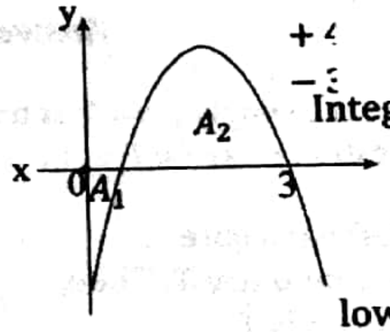
Since $V(x)$ is differentiable on $(0,72)$, the maximum value of V occurs at $x$where$V'(x) = 0$.Then $V'(x) = 0$ gives

$$\Rightarrow 5184 - 576x + 12x^2 = 0 \Rightarrow 432 - 48x + x^2 = 0$$
$$\Rightarrow (x - 12)(x - 36) = 0$$
$$\Rightarrow x = 12 \text{ or } x = 36$$

But from above, since $x \in (0\ cm, 36\ cm)$ we consider only $x = 12\ cm$.

Answer: B

48. We can see from the figure below that:



$\ln A_1$, for $x \in [0,1], f(x) \leq 0$,

lower limit $= 0$

upper limit$=1$ and

Integrand$= 0 - f(x) = x^2 - 4x + 3$.

Hence,

$$A_1 = \int_0^1 [x^2 - 4x + 3]dx = \left[\frac{1}{3}x^3 - 2x^2 + 3x\right]_0^1 = \frac{4}{3}$$

$\ln A_2$, for$x \in [1,3], f(x) \geq 0$,

lower limit $= 1$ and upper limit$=3$ and

Integrand$= f(x) - 0 = -x^2 + 4x - 3$.

Hence,

$$A_2 = \int_1^3 [-x^2 + 4x - 3]dx = \left[-\frac{1}{3}x^3 + 2x^2 - 3x\right]_1^3 = \frac{4}{3}$$

Therefore, $A = A_1 + A_2 = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$.

Answer: D

49. $\int \frac{x+\ln(x+1)}{(x+1)^2}dx = \int \frac{x}{(x+1)^2}dx + \int \frac{\ln(x+1)}{(x+1)^2}dx$.....(1)

Now, we evaluate integrals in the right individually and add.

$$\int \frac{x}{(x+1)^2}dx = \int \left(\frac{A}{x+1} + \frac{B}{(x+1)^2}\right)dx$$
$$= \int \left(\frac{A}{x+1}\right)dx + \int \left(\frac{B}{(x+1)^2}\right)dx$$
$$= A\int \frac{dx}{x+1} + B\int \frac{dx}{(x+1)^2}$$
$$= A\ln(x+1) - \frac{B}{x+1} + C_1, \text{ for constant of integration} C_1.$$

To get A and $B$by decomposition,

$$\frac{x}{(x+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2} = \frac{Ax+A+B}{(x+1)^2} \Rightarrow x = Ax + A + B \Rightarrow A = 1, B = -1.$$

Thus, $\int \frac{x}{(x+1)^2}dx = \ln(x+1) + \frac{1}{x+1} + C_1$.....(2)

For $\int \frac{\ln(x+1)}{(x+1)^2}dx$ using integration by part, $\int UdV = UV - \int VdU$,

Let $U = \ln(x+1)$and $= \frac{dx}{(x+1)^2}$. Then,

$dU = \frac{dx}{x+1}$and$V = \int dV = \int \frac{dx}{(x+1)^2} = \frac{-1}{x+1}$

So $\int \frac{\ln(x+1)}{(x+1)^2}dx = \int \overbrace{\ln(x+1)}^{U}\overbrace{\frac{dx}{(x+1)^2}}^{U}$

$$= \overbrace{\ln(x+1)}^{U}\overbrace{\left(\frac{-1}{x+1}\right)}^{V} - \int \overbrace{\frac{-1}{x+1}}^{V}\overbrace{\left(\frac{dx}{x+1}\right)}^{dU}$$
$$= -\frac{\ln(x+1)}{x+1} + \int \frac{1}{(x+1)^2}dx = -\frac{\ln(x+1)}{x+1} - \frac{1}{x+1} + C_2,$$

for a constant of integrationC_2.....(3)

Then substitutions of (2) and (3) in to (1) gives,

$$\int \frac{x+\ln(x+1)}{(x+1)^2}dx = \int \frac{x}{(x+1)^2}dx + \int \frac{\ln(x+1)}{(x+1)^2}dx$$
$$= \ln(x+1) + \frac{1}{x+1} + C_1 - \frac{\ln(x+1)}{x+1} - \frac{1}{x+1} + C_2$$
$$= \ln(x+1) - \frac{\ln(x+1)}{x+1} + C \text{for } C = C_1 + C_2.$$
$$= \ln(x+1)\left(1 - \frac{1}{x+1}\right) + C = \frac{x\ln(x+1)}{x+1} + C.$$

Answer: D

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50. $F'(x) = (f(2x+2)g(1-x^2))'$

$= (f(2x+2))\,g(1-x^2) + f(2x+2)(g(1-x^2))'$ (Product rule)

$= f'(2x+2)(2x+2)'\,g(1-x^2)$

$+f(2x+2)g'(1-x^2)(1-x^2)'$ (Chain rule)

$= 2f'(2x+2)g(1-x^2) - 2xg'(1-x^2)f(2x+2)$

Then,

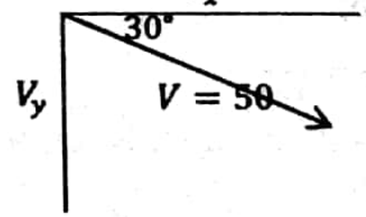
$F'(0) = 2f'(2(0)+2)g(1-(0)^2) - 2(0)g'(1-(0)^2)f(2(0)+2)$

$= 2f'(2)g(1) = 4(-5) = -20.$

Answer: B

51. Consider an air plane is descending at a speed of 50 mile per hour at an angle of below the horizontal as shown in the figure below. Then

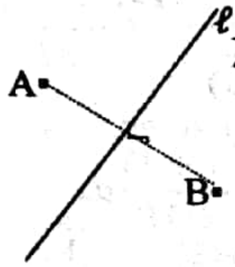
$V_x = 50\cos 30° = 25\sqrt{3}$ and



$V_y = 50\sin 30° = 25$

Answer: A

52. Let the point $A(2,5)$ be reflected under a line ℓ to the point $B(-3,1)$ as shown in the figure below.



Then, the point $C(a,b)$ the intersection of ℓ and $\overline{AB}$ is the md-point of $\overline{AB}$, and so

$$C(a,b) = \left(\frac{2+(-3)}{2}, \frac{5+1}{2}\right)$$

$\Rightarrow a = -\frac{1}{2}$ and $b = 3$.

And, $\overline{AB} \perp \ell$ gives, slop of ℓ,

$$m = -\frac{1}{\text{slop of } \overline{AB}} = -\frac{1}{\frac{5-1}{2-(-1)}} = -\frac{3}{4}.$$

Thus, since ℓ contains point $C(-\frac{1}{2}, 3)$ and has slop $m = -\frac{3}{4}$, if (x,y) is any point on ℓ, by point-slop formula ℓ will be

$y - 3 = m(x - (-\frac{1}{2}))$ for $m = -\frac{3}{4}$.

$\Rightarrow 10x + 8y = 19.$

Answer: C

53. By completing the square with x, $x^2 + 2x + y^2 = 8$ is

$(x+1)^2 + y^2 = 9.$

so that, center$= C(-1,0)$ and radius, $r = 3$units.

Then, the distance d between the center$= C(-1,0)$ and the line $3x + 4y + 14 = 0$ in (B) is

$$d = \frac{|3(-1)+4(0)1+14|}{\sqrt{(3)^2+(4)^2}} = 2.$$

Hence, $r > d$ implies the line $3x + 4y + 14 = 0$ cuts the circle at exactly two points.

Answer: B

54. $\vec{A} = 2\vec{\imath} - \vec{\jmath} + 2\vec{k}$ implies $|\vec{A}| = \sqrt{2^2 + (-1)^2 + 2^2} = 3.$

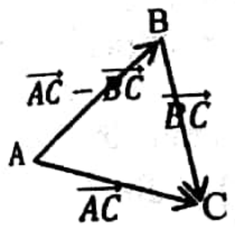
And, since $\vec{U}$ is a unit vector, then $|\vec{U}| = 1.$

Then, $|\vec{A} + \vec{U}|^2 = |\vec{A}|^2 + 2|\vec{A}||\vec{U}| + |\vec{U}|^2$

$= 3^2 + 2(3)(1) + 1^2 = 16.$

Answer: A

55. Let points $A = (-2,3)$, $B = (3,1)$ and C be given in a plane. Then $\overrightarrow{AC} - \overrightarrow{BC}$ is as in figure below. Then



$\overrightarrow{AC} - \overrightarrow{BC} = \overrightarrow{AC} - (-\overrightarrow{CB})$

$= \overrightarrow{AC} + \overrightarrow{CB}$

$= \overrightarrow{AB}$

$= (3,1) - (-2,3) = (5,-2).$

Answer: B

56. Let ℓ be a line through $(-1,2)$ and parallel to vector $\vec{V} = (1,-1)$.

ℓ is parallel to vector $\vec{V} = (1,-1) \Rightarrow$ Slop of ℓ is, $m = \frac{-1}{1} = -1.$

Then if (x,y) is any point on ℓ, using the point $(-1,2)$, equation of ℓ, by point-slop formula will be

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$y-2=m(x-(-1))$ for $m=-1$.
$\Rightarrow x+y-1=0$.

Answer: B

57. Let the line segment $\overline{QP}$ in a space be given with mid-point $R(0,0,k)$ where $Q(a,b,c)$ and $P(2,-1,0)$.
By distance formula

$$RP=\frac{1}{2}PQ=3=\frac{1}{2}\sqrt{(2-0)^2+(-1-0)^2+(0-k)^2}$$
$$\Rightarrow 3=\frac{1}{2}\sqrt{5+k^2}\Rightarrow k=\pm 2$$

But since R is on negative z axis, $k=-2$. And $R=R(0,0,-2)$.
Then, by mid-point formula,

$$R(0,0,-2)=\left(\frac{2+a}{2},\frac{-1+b}{2},\frac{0+c}{2}\right)$$
$$\Rightarrow \frac{2+a}{2}=0,\frac{-1+b}{2}=0, and\ \frac{0+c}{2}=-2$$

$\Rightarrow a=-2, b=1, and\ c=-4$. Therefore, $Q=Q(-2,1,-4)$.

Answer: D

58. $\neg(\forall_x(P(x)\Rightarrow Q(x))=\neg(\forall_x)\neg(P(x)\Rightarrow Q(x))$
$=\exists_x\neg(\neg P(x)\vee Q(x))$
$=\exists_x(\neg(\neg P(x))\vee\neg Q(x))$
$=\exists_x(P(x)\wedge\neg Q(x))$

Answer: D

59. Let the distance,
From ground to bottom of lower window $=y_1$,
From ground to bottom of upper window $=y_2$, and
From bottom of lower window to bottom of upper window $=h$.
Then, $h=y_2-y_1=10\sqrt{3}\tan 60^0-10\sqrt{3}\tan 30^0$
$=10\sqrt{3}(\sqrt{3})-10\sqrt{3}\left(\frac{1}{\sqrt{3}}\right)=20$.

Answer: B

60. **Principle of mathematical induction :** For a given assertion $P(n)$, involving a natural number n, if
(A) $P(1)$ is true, and
(B) $P(k+1)$, whenever $P(k+1)$ is true for a natural number $k>1$

Then the assertion $P(n)$ is true for every natural number n.
So in our case if $P(n)$ ="For any natural number n, $n!<10^n$"
Step 1=(A) and Step 4=(B) are sufficient to conclude the assertion.(i.e. Steps (2) and (3) can be omitted).

Answer: A

61. Step-1 Take two points on the line
$\ell:(x,y)=(-1,0)+t(3,6), t\in\Re$.
For instance, let $t=0,1$. Then
$t=0\Rightarrow(x,y)=(-1,0)+0(3,6)=(-1,0)=A$
$t=1\Rightarrow(x,y)=(-1,0)+1(3,6)=(2,6)=B$
i.e. $A=(-1,0)$, and $B=(2,6)$.

Step-2 Translate points A and B with the translation vector that translates point $(1,0)$ to a point $(0,1)$. If $\vec{U}$ is the translation vector of a translation, T that translates point $(1,0)$ to a point $(0,1)$, then

$$\vec{U}=(0-1,1-0)=(-1,1).$$

But here the line through points $A=(-1,0)$, and $=(2,6)$, $y=2x+2$ is parallel to the line of reflection $y=2x$.
Hence enough to transform one of the points A, or B and use slope, $m=2$.
So $T(A)=T((-1,0))=(-1,0)+\vec{U}=(-1,0)+(-1,1)=(-2,1)$.
i.e. $T(A)=(-2,1)$

Step-3 Reflect $T(A)=(-2,1)$ under the reflection M, through the line $y=2x$.
The perpendicular line to $y=2x$ through $T(A)=(-2,1)$ with slop $=-\frac{1}{2}$ is the line $m: y=-\frac{1}{2}x$.
The intersection of m and $y=2x$ is the solution to the system

$$\begin{cases} y=-\frac{1}{2}x \\ y=2x \end{cases}$$ i.e $C(0,0)$.

Since $C(0,0)$ is mid-point between $T(A)=(-2,1)$ and its image $M(T(A))$ under M, if $M(T(A))=M((-2,1))=(a,b)$ by mid-point formula, then
$(0,0)=\left(\frac{-2+a}{2},\frac{1+b}{2}\right)\Rightarrow(a,b)=(2,-1)$.

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Step-4 Find the line through $(2,-1)$ with slope, $m = 2$.i.e.$y = 2x - 5$

Answer: D

62. Given $\theta = 2\arctan\frac{1}{2}$.

Let $x = \arctan\frac{1}{2}$. Then,$\tan x = \frac{1}{2} = \frac{opposit}{adjacent}$, so that $opposit = 1$ and $adjacent = 2$(as in right triangle below).

$$hypothenous = \sqrt{opposit^2 + adjacent^2}$$

$$= \sqrt{1^2 + 2^2} = \sqrt{5}$$

And, $\cos x = \frac{adjacen\ t}{hypot\ henous} = \frac{2}{\sqrt{5}}$,

$$\sin x = \frac{opposit}{hypot\ henous} = \frac{1}{\sqrt{5}}$$

Thus,$\sec\theta = \sec(2\arctan\frac{1}{2}) = \sec(2x)$, because $x = \arctan\frac{1}{2}$.

$$= \frac{1}{\cos(2x)}, \text{(By reciprocal identity)}$$

$$= \frac{1}{\cos^2 x - \sin^2 x} = \frac{1}{(\frac{2}{\sqrt{5}})^2 - (\frac{1}{\sqrt{5}})^2} = \frac{5}{3}$$

Answer: C

63. Since translation is a rigid motion the translation of the circle can be considered as the translation of the center.

Let T be a translation that takes the circle $x^2 + y^2 - 2x + 6y + 3 = 0$to $(x+2)^2 + (y-4)^2 = 7$.

The circle $x^2 + y^2 - 2x + 6y + 3 = 0$ with $D = -2$ and $E = 6$ has a center $= \left(\frac{-D}{2}\frac{-E}{2}\right) = (1,-3)$.

And the image circle $(x+2)^2 + (y-4)^2 = 7$ has center $(-2,4)$.

Thus, $T(1,-3) = (-2,4)$, (i.e. T takes $(1,-3)$ to $(-2,4)$)

So the translation vector is $\vec{U} = (-2-1, 4-(-3)) = (-3,7)$.

Thus, $T(0,0) = (0,0) + \vec{U} = (0,0) + (-3,7)$

$= (-3,7)$(i.e. T takes $(0,0)$ to $(-3,7)$).

Answer: A

64. Given a line ℓthat passes through a point $(0,2)$ and parallel to a vector $= \vec{i} + 3\vec{j}$, and a circle $(x-2)^2 + (y-1)^2 = 5$.

ℓisParallel to a vector $V = \vec{i} + 3\vec{j} \Rightarrow$Slop of $\ell$is $m = \frac{3}{1} = 3$.

So equation of $\ell$is $3x - y + 2 = 0$.

Then the distance d, between the center $C(2,1)$ and ℓ is

$$d = \frac{|3(2)+(-1)1+2|}{\sqrt{(3)^2+(-1)^2}} = \frac{7}{\sqrt{10}}.$$

Note that the radius of the given circle is $r = \sqrt{5}$.

Then, we can see that $\frac{7}{\sqrt{10}} < \sqrt{5}$ by squaring both sides.

Therefore,$d < r$ implies ℓ intersects the circle at two distinct points.

Answer: C

65. Given $\theta = \frac{\pi}{3}$, the angle between $\vec{A} = 3\vec{i} - 4\vec{j}$ and$\vec{B}$.

Let $\vec{U}$ be the unit vector in the direction of$\vec{B}$.

Then

$$\vec{A}\cdot(\vec{A} - 2\vec{U}) = \vec{A}\cdot\left(\vec{A} - 2\frac{\vec{B}}{\|\vec{B}\|}\right), \text{because}\vec{U} = \frac{\vec{B}}{\|\vec{B}\|}$$

$$= \left(\vec{A}\cdot\vec{A} - 2\frac{\vec{A}\cdot\vec{B}}{\|\vec{B}\|}\right)$$

$$= (\vec{A}\cdot\vec{A} - 2\|\vec{A}\|\cos\theta), \text{because}\vec{A}\cdot\vec{B} = \|\vec{A}\|\|\vec{B}\|\cos\theta$$

$\vec{A}\cdot\vec{A} = (3\vec{i} - 4\vec{j})\cdot(3\vec{i} - 4\vec{j}) = 25$, and $\|\vec{A}\| = \sqrt{3^2 + 4^2} = 5$

Thus, $\vec{A}\cdot(\vec{A} - 2\vec{U}) = (\vec{A}\cdot\vec{A} - 2\|\vec{A}\|\cos\theta) = \left(25 - 2(5)\cos\frac{\pi}{3}\right) = 20$.

Answer: A

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UNIVERSITY ENTRANCE EXAMINATION (UEE)
PHYSICS 2001/2009

Code 04

Constant	Symbol	Value
Acceleration due to gravity	g	$10m/s^2$
Permittivity of vacuum	ε_0	$8.85 \times 10^{-12}F/m$
Faraday's Constant	F	$96500Cmol^{-1}$
Charge of an electron	e	1.6×10^{-19}Coulomb
$\sin 30^\circ = \cos 60^\circ = 0.5$		
$\sin 37^\circ = \cos 53^\circ = 0.6$		
$\sin 53^\circ = \cos 37^\circ = 0.8$		
$\cos 60^\circ = \sin 30^\circ = 0.866$		

1. What is the SI unit of electric field?
   A. Am^{-1} B. Vm^{-1} C.NC^{-1} D. Cm

2. Suppose vectors $\vec{A}=2(\hat{i}+\hat{j})$ and $\vec{B}=2\hat{i}+3\hat{j}+\hat{k}$. Which of the following is unit vector perpendicular to both vectors?
   A. $\frac{1}{\sqrt{3}}(-\hat{i}+\hat{j}+\hat{k})$ C. $\frac{1}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$
   B. $\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}-\hat{k})$ D. $\frac{1}{\sqrt{3}}(\hat{i}-\hat{j}+\hat{k})$

3. Which of the following expression is not correct in a clockwise system?
   A. $\hat{k}\times\hat{i}=\hat{j}$ C. $\hat{i}\times\hat{j}=-\hat{j}\times\hat{i}$
   B. $i\times\hat{i}=1$ D. $\hat{i}\times\hat{k}=-\hat{j}$

4. What is electric potential energy of a system of 4 point charges, each charge q located at the corners of a square of side a?
   A. $\frac{\sqrt{2}kq}{a}(2\sqrt{2}+1)$ C. $\frac{\sqrt{2}kq^2}{a}(2\sqrt{2}+1)$
   B. $\frac{\sqrt{2}kq^2}{a}(2\sqrt{2}-1)$ D. $\frac{\sqrt{2}kq}{a}(2\sqrt{2}-1)$

5. A hoop of mass M and radius R is at rest at the top of an incline as shown. The hoop rolls down the plane without slipping. If the hoop reaches bottom, then what is its angular momentum about its center of mass?

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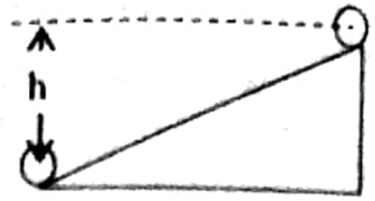
A. $MR\sqrt{gh}$
B. $MR\sqrt{gh}/2$
C. $MR\sqrt{2gh}$
D. Mgh



6. A long straight wire carrying a steady current of 3 A flowing in a direction parallel to unit vector $\frac{1}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$ is placed in a magnetic field $\vec{B}=(0.75\hat{i}+0.4\hat{k})T$. What is the magnetic force per unit length of the wire?
   A. $(0.69\hat{i}-0.61\hat{j}-1.30\hat{k})\frac{N}{M}$ C. $(0.49\hat{i}-0.51\hat{j}-1.37\hat{k})\frac{N}{M}$
   B. $(0.39\hat{i}-0.41\hat{j}-2.30\hat{k})\frac{N}{M}$ D. $(0.59\hat{i}-0.71\hat{j}-0.13\hat{k})\frac{N}{M}$

7. Which one of the followings is correct about a damped oscillation:
   A. runs down and stops.
   B. oscillates with resonance frequency.
   C. vibrate at a frequency higher than its normal frequency.
   D. Oscillates with increasing amplitude.

8. An object has a constant acceleration of $3.2m/s^2$. At a certain instant its velocity is 9.6m/s. What is the ratio of its velocities at 2.5 second later to 2.5 second earlier?
   A. 111 B. 11 C. 11.1 D. 1.1

9. A circle of radius r is drawn with center as charge +q. A charge q_o is taken from one point to another diametrically opposite point along the circumference of the circle. What is the value of the work done?
   A. $\frac{1}{4\pi\varepsilon_o}\frac{q_o}{r^2}\times 2r$ C. $\frac{1}{4\pi\varepsilon_o}\frac{qq_o}{r^2}\times 2r$
   B. $\frac{1}{4\pi\varepsilon_o}\frac{qq_o}{r}\times 2r$ D. zero

10. Given the following information; the body surface area is $1.8m^2$; the clothing is 1.0cm thick; skin surface temperature is 33°C, whereas the outer surface of the clothing is 1.0C; the thermal conductivity of the

clothing is 0.04W/m.K. What is the rate at which body heat flows out through the clothing of a man?

A. 22.04W B. 23.04W C. 2.30W D. 230.4W

11. A force acts on a 3kg particle in such away that the position of the particle as a function of time is given by $x = 3t - 4t^2$, where x is in meters and t is in seconds. What is the work done by the force during the first 4.0 second?

A. 1218J B. 1228J C. 1248J D. 1258J

12. In a purely inductive AC circuit L = 35 mH and the rms voltage is 165V. What is the inductive reactance in Ω and rms current in ampere in the circuit if the frequency is 40Hz, respectively?

A. 8.79, 18.77 C. 7.69, 20.77

B. 7.79, 20.77 D. 9.89, 18.77

13. When the voltage applied to a capacitor increases from 80V to 110V, the charge on the capacitor increases by 9×10^{-5}C. What is the capacitance of the capacitor?

A. 4μF B. 3μF C. 2μF D. 5μF

14. Dire Dawa is approximately 300km east of Addis Ababa. A plane leaves Addis Ababa flying due east at 500km/hr while the wind is blowing to the south at 100km/hr. What is the velocity of the plane relative to the ground?

A. 510km/hr north of east C. 490km/hr north of west

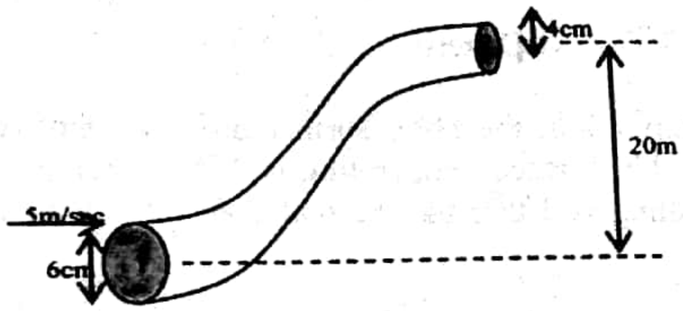
B. 510km/hr south of east D. 490km/hr south of west

15. Consider a uniformly charged insulating spherical shell of inner radius R_1 and outer radius R_2. What are the electric fields at $r > R_2$ and $r < R_1$ respectively?

A. $\frac{q}{4\pi\varepsilon_o r^2}, 0$ C. $\frac{q}{4\pi\varepsilon_o R_1^2}, 0$

B. $\frac{q}{4\pi\varepsilon_o R^2}, 0$ D. $\frac{q}{4\pi\varepsilon_o (R_1^2 - R_1^2)}, 0$

16. Water flows through a pipe as shown below. The water speed through the lower pipe is 5m/s and a pressure gauge reads 75kPa. What is the reading of the pressure gauge on the upper pipe?



A. 75kPa

B. 55kPa

C. 24.6kPa

D. 4.6kPa

17. The cross-sectional area of a water pipe entering the basement is $4 \times 10^{-4}m^2$. The pipe carries water into the basement at a velocity of 6m/s. The pressure at the basement is $3 \times 10^{+5}N/m^2$. What are the speed and pressure when it reaches the 2nd floor, which is 8m above the basement if the cross-sectional area of a water pipe is half of its value at the basement?

A. 12m/s, $1.12 \times 10^{+5}N/m^2$ C. 12m/s, $2.12 \times 10^{+5}N/m^2$

B. 24m/s, $1.12 \times 10^{+5}N/m^2$ D. 18m/s, $2.12 \times 10^{+5}N/m^2$

18. Two masses 2kg and 3kg are attached to the end of a light string passing over a pulley, fixed at the top. What are the tension and acceleration in the string in terms of g?

A. $\left(\frac{12g}{5}, \frac{g}{5}\right)$ C. $\left(\frac{21g}{8}, \frac{g}{8}\right)$

B. $\left(\frac{5g}{8}, \frac{g}{5}\right)$ D. $\left(\frac{7g}{8}, \frac{g}{8}\right)$

19. A copper wire is stretched so that its length increases and its diameter decrease. As a result its:

A. resistance decreases and its resistivity remains the same.

B. resistivity decreases and its resistance remains the same.

C. resistance increases and its resistivity remains the same.

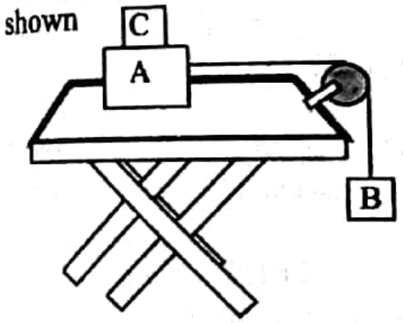
D. resistivity increases and its resistance remains the same.

20. An external resistance R is connected to a cell of internal resistance r, the maximum current flows in the external resistance when one of the following conditions is satisfied.

A. R = r C. R < r

B. R > r D. None of the above

21. Two masses A and B of 10kg and 5kg respectively are connected with a string passing over a frictionless pulley fixed at the corner of a table as shown



The coefficient of friction of A with the table is 0.2. What is the value of the minimum mass of C that may be placed on A to prevent it from moving?

A. 15kg B. 10kg

C. 5kg D. 0kg

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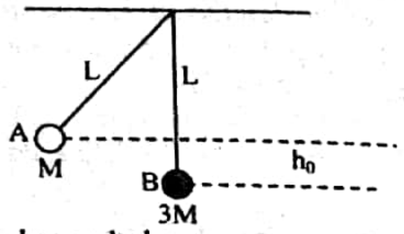
22. A large storage tank is filled to a height h_o. If the tank is punctured at a height h from the bottom of the tank, how far from the tank will the stream land?

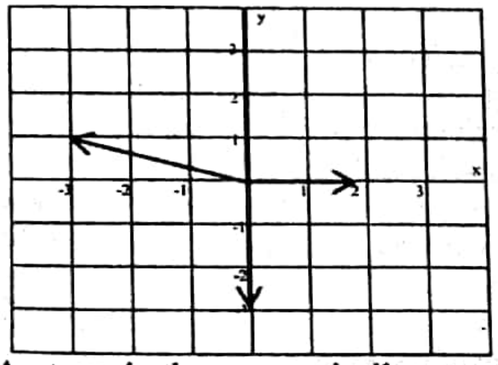
A. $2\sqrt{h(h_o+h)}$ B. $\sqrt{h(h_o-h)}$ C. $\sqrt{h(h_o+h)}$ D. $2\sqrt{h(h_o-h)}$

23. Two spherical masses, A and B, of mass M and 3M respectively, hand from the ceiling on strings of length L. Sphere A is drawn aside so that it is raised to a height shown in the figure below, and then released. Sphere A collides with sphere B; they stick together and swing to a maximum height h. What would be the value of this height?

A. $h_o/16$
B. $h_o/8$
C. $h_o/4$
D. $h_o/2$



24. If the three vectors shown below, represent forces acting on an object, then what is the expression of the vector that represents the net force?



A. $-\hat{i}-2\hat{j}$

B. $-2\hat{i}-\hat{j}$

C. $\hat{i}+2\hat{j}$

D. $2\hat{i}+\hat{j}$

25. A stone is thrown vertically upward. On its way up it passes point A with speed V and point B, 3m higher than A, with speed V/2. What is the speed V in m/s and the maximum height reached by the stone above point B in meters respectively?

A. 7.75, 0.5 C. 7.75, 0.75
B. 7.15, 0.75 D. 7.85, 1.15

26. Three capacitors of capacitances 8μF, 12μF, and 24μF are connected (a) in series and then (b) in parallel. What is the ratio of the equivalent capacitance in case (a) to that in case (b)?

A. 1:11 B. 11:1 C. 1:1 D.1:3

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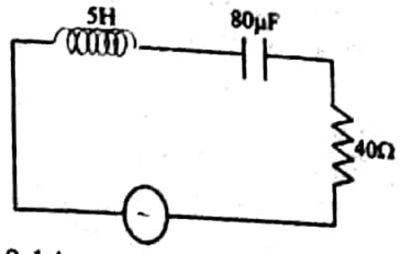
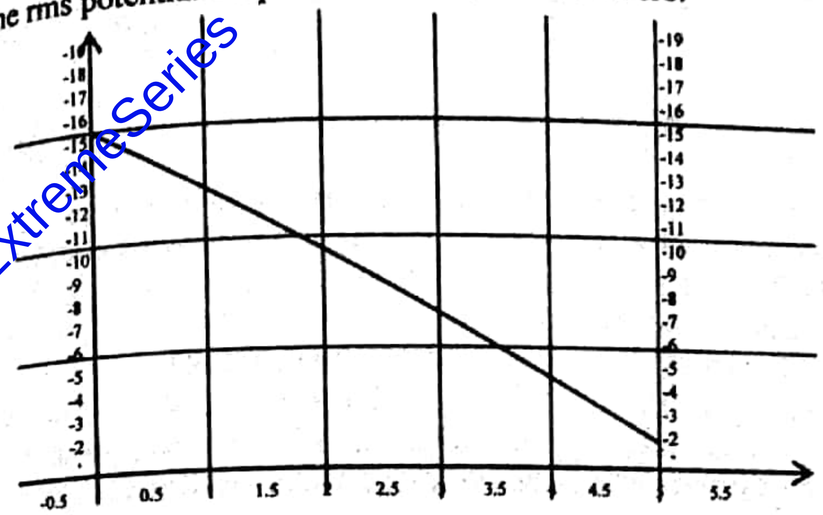
27. The figure below shows a series LCR circuit connected to a variable frequency 230V source. Which statement is **not** correct about the circuit?



A. The source frequency that drives the circuit into resonance is $25/\pi$.
B. The impedance of the circuit is 40Ω.
C. The current amplitude at resonance is 8.1A.
D. The rms potential drop across R at resonance is zero.



28. How far does the object move in the first two seconds?

A. 13m B. 26m C. -26m D. -13m

29. A 5kg block is pulled along a horizontal frictionless floor by a cord that exerts a force F = 12N at angle of 30° above the horizontal. If the force F is slowly increased, then what are the values of the force in Newton and acceleration in m/s^2 of the block just before it is lifted of the floor, respectively?

A. 116, 21 B. 100, 20 C. 116, 20 D. 100, 21

30. A certain capacitor is charged to a potential V. If you wish to increase its stored energy by 10%, then by what percentage should you increase V?

A. 4.88% B. 5.92% C. 6.98% D. 5.13%

31. A circular loop of wire expands due to rise in temperature so that area enclosed by the loop increases slowly with time according to $A = A_o + bt$, where b is a constant given by $b = 1 \times 10^{-6} m^2/s$. What is the induced

emf if the circuit is placed in a uniform magnetic field of magnitude B = 0.4T and is perpendicular to the plane formed by the circuit?

A. 3.0×10^{-5}V
C. 3.0×10^{-6}V
B. 4.0×10^{-6}V
D. 4.0×10^{-7}V

32. Consider a uniformly charged conducting spherical shell of inner radius R_1 and outer radius R_2 carrying a total of negative charge q. A point positive charge of magnitude q is placed at the center of the spherical shell. What are the electric fields at $r < R_1$, $R_1 < r < R_2$, and $r > R_2$, respectively?

A. $\frac{q}{4\pi\varepsilon_0 r^2}, 0, \frac{q}{4\pi\varepsilon_0 (R_2^2 - R_1^2)}$
C. $\frac{q}{4\pi\varepsilon_0 r^2}, 0, 0$
B. $\frac{q}{4\pi\varepsilon_0 R_2^2}, 0, \frac{q}{4\pi\varepsilon_0 r^2}$
D. $\frac{q}{4\pi\varepsilon_0 (R_2^2 - R_1^2)}, 0, 0$

33. A solar cell generates a potential difference of 0.1v when a 500Ω resistor is connected across it and a potential difference of 0.15 when a 1000Ω resistor is substituted. What are the emf and the internal resistance of the solar cell?

A. 0.3V, 1500Ω
C. 0.3V, 1000Ω
B. 0.23V, 1000Ω
D. 0.35V, 1300Ω

34. If a hollow metal sphere placed on an insulating stand is charged positively, then what is the electric potential inside the sphere?
   A. zero.
   B. the same as on the surface.
   C. greater than the potential on the surface.
   D. Less than the potential on the surface.

35. Which one of the following statements is correct?
   A. In a Cartesian system of coordinates, $|A_x\hat{i}| > |\vec{A}|$, for any $\vec{A}$.
   B. For two non – zero vector $\vec{A}$ and $\vec{B}$, $|\vec{A}+\vec{B}|$ is necessarily greater than $|\vec{A}|$.
   C. For two non – zero vectors $\vec{A}$ and $\vec{B}$, $\vec{A}-\vec{B}=\vec{B}-\vec{A}$.
   D. For three non – coplanar vectors $\vec{A}, \vec{B}$ and $\vec{C}$, $\vec{A}+\vec{B}+\vec{C}$ can not be a null vector.

36. A newly discovered planet has twice the mass of an three times the radius of the earth. What is the free – fall acceleration at its surface in terms of the free – fall acceleration g at the surface of the earth?

A. $\frac{4}{3}g$.
B. $\frac{3}{4}g$.
C. $\frac{2}{9}g$.
D. $\frac{2}{3}g$.

37. Which of the following is **not** true?
   A. Physical quantities can be classified as fundamental or derived.
   B. All fundamental physical quantities are scalars.
   C. Vector quantities in the same direction can be added or subtracted just like numbers.
   D. All derived physical quantities are vectors.

38. If the x and y coordinates of a particle at any time t is given by $x = 7t + 4t^2$ and $y = 5t$ where x and y are in m and t is s, then what is the acceleration of the particle at t = 5?

A. zero
B. $8m/s^2$
C. $20m/s^2$
D. $40m/s^2$

39. Why is it logical to say that the potential of a grounded object is zero?
   A. the earth is an insulator.
   B. the net charge on earth is practically zero.
   C. a grounded object can neither give nor take charge from the earth.
   D. a grounded object is at a higher potential than the earth.

40. Particle A has less mass than particle B. Both are pushed forward across a frictionless surface by equal forces for 1s. Which of the following statements is correct if bots start from rest?
   A. The amount of work done on A is greater than the work done on B.
   B. The final speed of A is less than that of B.
   C. The impulses delivered to both particles are the same.
   D. The change in kinetic energy is the same for both particles.

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UNIVERSITY ENTRANCE EXAMINATION (UEE) PHYSICS
SENE 2002/JUNE 2010

Booklet Code: 25

Directions: This exam contains a total of 50 questions. Each of the questions is followed by four possible alternatives. Read each question and carefully **blacken** only the letter of your choice on the separate answer sheet provided.

Constant	Symbol	Value
Acceleration due to gravity	g	$10\ m/s^2$
Permittivity of vacuum	ε_0	8.85×10^{-12} F/m
Faraday's constant	F	96500C mol^{-1}
Charge of an electron	e	1.6×10^{-19} Coulomb
Universal Gravitational Constant	G	$6.67\times10^{-11}\ Nm^2/Kg^2$
$\mu_0 = 4\Pi\times10^{-7}$ Tm/A		
$\pi = 3.14$		
$\sin30^0 = \cos60^0 = 0.5$		
$\sin37^0 = \cos53^0 = 0.6$		
$\sin53^0 = \cos37^0 = 0.8$		
$\mathrm{Sin}60^0 = \cos30^0 = 0.866$		

1. A long, straight wire of radius R carries a steady current I that is uniformly distributed through the cross-section of the wire. What is the magnetic field a distance r from the center of the wire in the regions r > R?

   A. $\frac{\mu_0 I}{2\pi R^2} r$ C. $\frac{\mu_0 I}{\pi r}$

   B. $\frac{\mu_0 I}{2\pi r^2} R$ D. $\frac{\mu_0 I}{2\pi r}$

2. In a LRC series circuit, suppose R = 300Ω, L = 60 mH, C = 0.5 μF, V = 50V, and ω =10,000 rad/s. What is the average power delivered to the entire circuit?

   A. 1.5W B. 2.3W C. 3.5W D. 1.0W

3. The vector product of two non-collinear vectors $\bar{A}$ and $\bar{B}$

   A. Is perpendicular to the plane formed by $\bar{A}$ and $\bar{B}$

   B. Is on the plane formed by $\bar{A}$ and $\bar{B}$
   C. Must be zero
   D. Is a scalar quantity

4. Temperature of a body is a measure of the
   A. Heat content of a body
   B. Average molecular kinetic energy
   C. Average pressure in a gas
   D. Heat of vaporization of a liquid

5. Two negative point charges are 2 meters apart and repel each other with a force of 2 Newton. When the distance between the charges is doubled, the force between them is
   A. One fourth as great C. twice as great
   B. One half as great D. four times as great

6. The terminal voltage of a cell supplying energy to a circuit is usually less than its emf because of the cell's
   A. Size C. mass
   B. Internal resistance D. energy

7. The peak voltage of an alternating emf is 141 volts. The effective value of the voltage in volts is
   A. 70.5 B. 100 C. 141 D. 200

8. The voltage output of a generator is given by V = (300 V) sin (wt). What is the rms current in the circuit when this generator is connected to a 100 – Ω resistor?
   A. 1.50A B. 2.12A C. 1.12A D. 1.41A

9. Which of the following is **NOT** true of vector product?
   A. $\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$ C. $\vec{A}\perp\vec{B} \Rightarrow |\vec{A}\times\vec{B}| = AB$
   B. $\vec{A}\times(\vec{B}+\vec{C}) = \vec{B}\times\vec{A}+\vec{A}\times\vec{C}$ D. $\vec{A}\|\vec{B} \Rightarrow \vec{A}\times\vec{B} = 0$

10. If a single constant force F acts on a particle of mass m and the particle starts from rest, what is the instantaneous power delivered by the force at time t?
    A. $2\frac{F^2}{m}t$ B. $\frac{F^2}{2m}t$ C. $\frac{F^2}{m}t$ D. $3\frac{F^2}{m}t$

11. Which of the following statement is **NOT** true?
    A. If a heat Q is transferred from a hot reservoir at temperature T_h to a cold reservoir at temperature T_C, the entropy of the cold reservoir increases.

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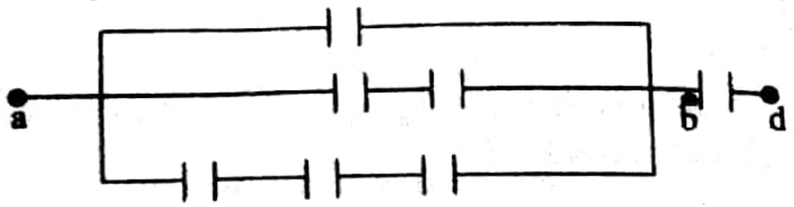
B. The increases in entropy of the cold reservoir is greater than the decrease in entropy of the hot reservoir.
C. The change in entropy of the system between any two equilibrium states does not depend only on the initial and final states.
D. Metals are good heat conductors because of their large number of electrons

12. Consider the network of identical capacitors, each having capacitance C between terminal a and b and a capacitor of capacitance $\frac{11}{5}$C between terminal b and d shown in figure below. What is the effective capacitance between terminal a and d?



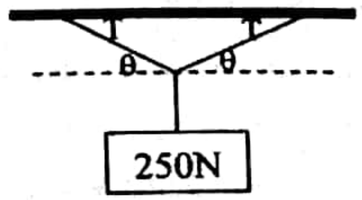
A. $\frac{121}{30}$C
B. 1C
C. $\frac{11}{6}$C
D. 3C

13. An athlete accelerates at $1.9\,m/s^2$ for 2.2 sec at the beginning of a race. What is the speed of the athlete at the end of a 3000m race?
A. $0.86\,m/s$ B. $1.16\,m/s$ C. $4.18\,m/s$ D. $9.20\,m/s$

14. A body is rotated with a uniform speed in a circle of radius 1m. What is the centripetal acceleration in m/s^2 if it completes 12 rotations in 3 seconds?
A. $36\pi^2$ B. $64\pi^2$ C. $72\pi^2$ D. $144\pi^2$

15. If a car moves round a circular road of radius r at a constant speed
A. Its velocity changes and the acceleration is v/r^2
B. There is no force on the car since its speed is constant
C. There is no velocity change since the speed is constant
D. The force per unit mass on the car is towards the center and is v^2/r

16. Traffic light weighting 250N are suspended from a cable as shown. Find the tension T in the string if $\theta_1 = \theta_2 = 30^0$
A. 120N
B. 165N
C. 196N
D. 250N



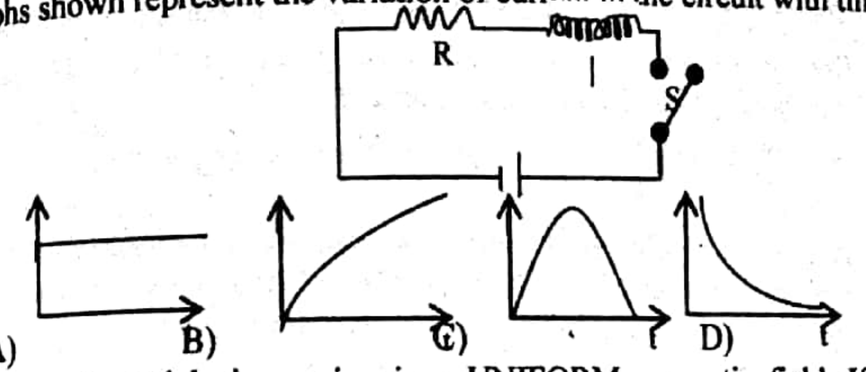
17. The period of oscillation of a particle undergoing simple harmonic motion is:
A. Independent of the amplitude of the motion.
B. Directly proportional to the frequency of oscillation
C. Independent of the frequency of oscillation
D. Directly proportional to mass of oscillating objec

18. Young's modulus is the
A. Ratio of stress to strain
B. Force per unit area
C. ratio of strain to stress
D. force per unit elongation

19. Which of the following combinations of properties would be most desirable for a thermal insulator?
A. High specific heat and low conductivity
B. Low specific heat and high conductivity
C. High specific heat and high conductivity
D. Low specific heat and low conductivity.

20. AN air-filled parallel-plate capacitor has a capacitance of 1pF. The plate separation is doubled and the space between them is completely filled with Wax. If the capacitance increases to 2pF, what is the dielectric constant of wax?
A. 2.0 B. 4.0 C. 6.0 D. 8.0

21. In the circuit shown below the switch is on at t = 0. which one of the graphs shown represent the variation of current in the circuit with time?



22. A changed particle is moving in a UNIFORM magnetic field. If the direction of motion of the charged particle is parallel to the magnetic field, the shape of the charged particle's path will be
A. Circular B. Straight C. spiral D. elliptical

23. An induced e.m.f. is obtained between the ends of a horizontal steel axle X of a train moving due east. This is because
A. X point due east

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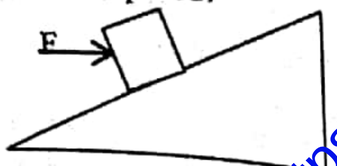
B. The earth's magnetic field has a horizontal component
C. X moves parallel to the earth's field
D. The earth has a vertical magnetic component

24. Which device functions with the principle of mutual induction?
A. Electromagnet C. Electric motor
B. Transformer D. Transistor

25. A capacitance C = 2μF and inductance L = 10H and coil resistance 5Ω are in series in a circuit. When an alternating current of 2A rms flows, the power in watts in the circuit is
A. 100 B. 50 C. 20 D. 2

26. A car with initial speed of $4\,^{m}/_{s}$ starts accelerating in a straight line at $3.2\,^{m}/_{s^2}$ for 10 seconds. If the brakes are then applied and it travels a further 20m before stopping, what is the acceleration in $^{m}/_{s^2}$?
A. -48.6 B. -16.2 C. -8.1 D. -32.4

27. A 100kg box is pushed at a constant speed up the smooth 30^0 inclined plane shown below. What horizontal force F is required?
A. 577.35N
B. 1000.35N
C. C. 1732.35N
D. D. 1154.70



28. A ball rolls off the edge of a horizontal table top 4m high. It strikes the floor at a point 5m horizontally away from the edge of the table. How long was the ball in the air in seconds?
A. $\frac{3}{\sqrt{5}}$ B. $\frac{1}{\sqrt{5}}$ C. $\frac{2}{\sqrt{3}}$ D. $\frac{2}{\sqrt{5}}$

29. A boat is traveling at 15 km/hr with respect to the water of a river in the upstream direction. The water itself is flowing at 6 km/hr with respect to the ground. A child on the boat walks from rear seat to front seat at 5 km/hr with respect to the boat. What are the velocities of the boat and the child in $^{km}/_{hr}$ with respect to the ground respectively?
A. 6,12 B. 5,9 C. 21,12 D. 9,14

30. A body has moved from initial position having a coordinate of (0,1,1.5) meters to a final position with coordinate given by (1.2, 2, 1.5) meters under the influence of a constant force given by $\vec{F} = (-\hat{i} + 2\hat{j} + 3\hat{k})$N.

What is the work done by the force in moving the body from its initial position to its final position?
A. 8J B. 0.8J C. 4J D. 0.4J

31. The angular position of a swinging door is described by θ = 5+ 10t + $2t^2$ rad. What is the angular speed and acceleration of the door at t = 3 s?
A. $22\,^{rad}/_{s}, 4\,^{rad}/_{s}$ C. $22\,^{rad}/_{s}, 12\,^{rad}/_{s^2}$
B. $-17\,^{rad}/_{s}, 4\,^{rad}/_{s^2}$ D. $12\,^{rad}/_{s}, 3\,^{rad}/_{s^2}$

32. An open tank containing a liquid of density ρ has a small hole in its side at a distance y_1 from the bottom. What is the speed at which the fluid leaves the hole when the liquid level is a distance h above the hole?
A. $\sqrt{2gh}$ B. $\sqrt{2gy_1}$ C. $\sqrt{gy}$ D. $\sqrt{gy_1}$

33. A gas is compressed at a constant pressure of $8 \times 10^{+4}\,^{N}/_{m^2}$ from a volume of 9×10^{-3} m^3 to a volume of 3×10^{-3} m^3. In the process 400J of heat energy flows out of the gas. What are the work done by the gas and the change in internal energy of the gas?
A. -567J, -167J C. -589J, -189J
B. -560J, -160J D. -540K, -140J

34. An object having a net charge of 24μC is placed in a uniform electric field of $610\,^{N}/_{C}$ directed vertically. What is the mass of the object if it floats in this electric field?
A. 14.640kg B. 1.464gm C. 14.460gm D. 14.460kg

35. Which of the following is **NOT** property of a conductor in electrostatic equilibrium?
A. The electric field is zero everywhere inside the conductor.
B. Any net charge on the conductor resides entirely on its surface.
C. On an irregularly shaped conductor, the surface charge density is greatest where the radius of curvature of the surface is the smallest.
D. The electric field just outside the conductor is tangent to its surface.

36. Which of the following statement is **NOT** true?
A. Any device such as battery or generator can be considered as source of emf.
B. The terminal voltage can be equal to emf in real system.
C. Emf forces electrons move in a direction opposite to the electrostatic force.
D. Emf can be defined as work done per unit charge.

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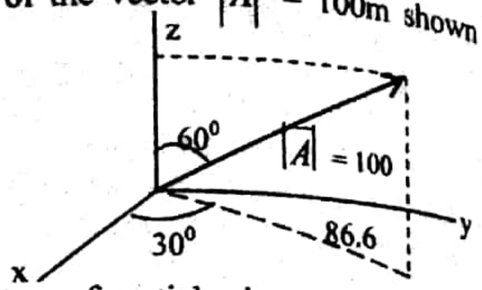
37. Two resistors connected in series have an equivalent resistance of 7Ω. When they are connected in parallel, their equivalent resistance is $\frac{10}{7}$ Ω. What is the resistance of each resistor?

A. 5.5Ω, 1.5Ω B. 6Ω, 1Ω C. 5Ω, 2Ω D. 4Ω, 3Ω

38. The x, y, and z components of the vector $|\vec{A}| = 100$m shown in the figure below, respectively are

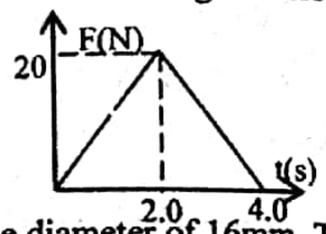


A. 86.6, 21.6, 37.5
B. 50.0, 43.3, 75.0
C. 75.0, 43.3, 50.0
D. 37.5, 21.6, 86.6

39. The net force acting on a system of particles is zero. Which one of the following quantities may vary with time?

A. The momentum of the system
B. The kinetic energy of the system
C. The velocity of the center of mass of the system
D. The position of the center of mass of the system

40. A force F acts on a ball initially at rest on a smooth surface for a time t. the variation of F with t is shown in the figure below. The momentum of the ball, in Ns, after 4 sec is



A. 10
B. 20
C. 30
D. 40

41. A garden hose has an inside diameter of 16mm. The house can fill a 10 liter bucket in 20 seconds. If the diameter of the nozzle is reduced to 8cm,

A. Water exits with a speed twice the speed inside the hose.
B. A 10 litter bucket can be filled in 10 seconds.
C. A 20 litter bucket can be filled in 40 seconds.
D. Water exits with a speed half the speed inside the hose

42. When stretched beyond its elastic limit, a metal rod such as steel

A. Becomes elastic
B. Has no energy
C. Obeys Hook's law
D. Becomes plastic

43. Coulomb's law for the force between electric charges most closely resembles

A. The law of conservation of energy
B. Newton's second law of motion
C. Newton's law of gravitation
D. The law of conservation of mass

44. Household light bulbs are normally connected in parallel to a power supply. Suppose a 40w and a 60W light bulbs are, instead, connected in series. Which bulb is brighter?

A. The 60W bulb.
B. The 40 W bulb.
C. The bulbs are equally bright.
D. The bulbs will not give light in series connection

45. In a 100m race, Usain Bolt and Assafa Powel cross the finish line in a dead heat, both taking 10s. Accelerating uniformly, Usain Bolt took 3s and Assafa Powel took 2s to attain maximum speed, which they maintained for the rest of the race. What was the acceleration of Usain Bolt and Assafa Powel respectively?

A. $4.32\,m/s^2$, $2.25\,m/s^2$ C. $4.92\,m/s^2$, $6.45\,m/s^2$
B. $3.92\,m/s^2$, $5.55\,m/s^2$ D. $4.92\,m/s^2$, $5.35\,m/s^2$

46. A running man has half the kinetic energy that a boy of half his mass has. The man speeds up by $1\,m/s$ and then has the same kinetic energy as the boy. What is the original speed of the man in m/s?

A. $\frac{1}{\sqrt{2}+1}$ B. $\frac{1}{\sqrt{2}-1}$ C. $\frac{1}{\sqrt{2}+2}$ D. $\frac{1}{\sqrt{2}-2}$

47. Consider a long cylindrical charge distribution of radius R with a uniform charge density ρ. What is the magnitude of electric field at distance r from the axis where $r < R$?

A. $\frac{\rho r}{3\varepsilon_0}$ B. $\frac{\rho r}{2\varepsilon_0}$ C. $\frac{\rho r}{7\varepsilon_0}$ D. $\frac{\rho r}{5\varepsilon_0}$

48. What is the ratio of power dissipated by the combined network of resistors, each with resistance R, between terminals a and b to the power dissipated by a resistor of resistance R between terminals b and c, shown in the figure below. The two terminals a and c of the network are connected to a battery such that current I flows from terminal a to terminal c?

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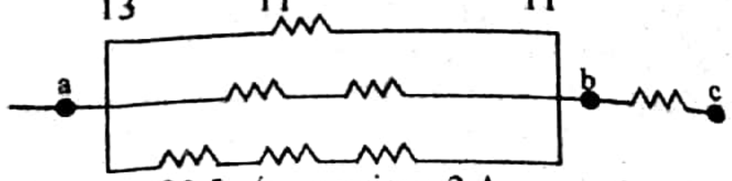
A. $\frac{7}{13}$ B. $\frac{6}{13}$ C. $\frac{7}{11}$ D. $\frac{6}{11}$



49. A wire with mass of 0.5 g/cm carries a 2 A current horizontally towards the south. What are the direction and magnitude of the minimum magnetic field needed to lift this wire vertically upward?
A. 0.25 T towards west
B. 0.25 T towards east
C. 0.50 T towards west
D. 0.50T towards east

50. Two parallel copper wires are 1cm apart. Lightning sends a 10,000A pulse of current along each conductor. What is the force per unit length on one conductor?
A. $1000\,N/m$
B. $200\,N/m$
C. $20\,N/m$
D. $2000\,N/m$

Uniform circular motion is a
A. Motion with constant velocity
B. Motion in one dimension
C. Motion with acceleration of constant magnitude
D. Motion in which net force is constant

2. One of the following statements is NOT characteristics of electric field.
A. An electric field is the region around a charged object where another charged object will experience a force
B. Lines of electric field have a direction which move away from negative and towards positive
C. The direction of electric field is the direction of force acting on a small positive test charge.
D. Lines of electric flux do not cross over each other

3. Which of the following is a vector quantity?
A. Power
B. Momentum
C. Temperature
D. Electric current

4. An object of mass m moving with initial speed of 15 $\frac{m}{s}$ got an acceleration of $10\frac{m}{s^2}$ after a constant force F is applied to it. How long does it take for the object to reach 90 meters from its initial position?
A, 3.22s B. 0.75s C. 3s D. 2.5s

5. The zeroth law of thermodynamics states that
A. Two bodies that are separately in thermal equilibrium with a third body must be in thermal equilibrium with each other.
B. The increase in internal energy of a system is equal to the sum of the energy entering the system through heating, and the work done on the system
C. no process is possible in which there is an overall decrease in the entropy of the universe.
D. no process or cycle is perfectly reversible and no refrigerator or heat engine can be perfectly efficient.

6. Which of the following statement is NOT true?
A. The magnitude of average angular velocity is directly proportional to the angular displacement during a given time.

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B. Average angular velocity and instantaneous angular velocity are the same for uniform angular motion
C. The direction of the average angular velocity is determined by the right hand rule
D. The direction of average angular velocity is parallel to a tangent line along the trajectory of the object.

7. Which one of the following statements is true about action andreaction forces referred to in Newton's third law of motion?
A. They act upon the same body
B. They are equal in magnitude but need not have the same line of action.
C. They act upon two different bodies
D. They are not equal in magnitude but have the same line of action

8. A body has moved from its initial position at origin along x – axis to a final position x= 5 meters under the influence of a force given by F = (4x)iNewtons. What is the work done by th force in moving the body from its initial position to it final postion?
A. 100J B. 50 J C. 25 J D. 20J

9. A wave front is
A. a line that represents the direction of propagation of light
B. the locus of all points of a travelling wave that are in phase
C. the minimum distance between two points that are in phase
D. the number of waves crossing a given plane in a unit time

10. Which of the following statements is NOT true?
A. All electric current can be induced in a circuit with a change in the cross- sectional area of the wire.
B. The induced emf in a circuit is directly proportional to the time rate of change of magnetic flux
C. Emf can be induced in a circuit by change in the area enclosed by the loop.
D. An electric current can be induced in a circuit with a changing of magnetic field.

11. A cart moving at a speed V collides with an identical stationary cart on an air track and the two stick together after collision. What is their velocity after collision?
A. -0.5V B. Zero C. 0.5V D. V

12. A 4kg block is given an initial speed of 8 $\frac{m}{s}$ at the bottom of 30° inclined plane. The frictional force that retards its motion is 15 N. If

the block is directed up the inclined plane, how far will it move before it stops?
A. 3.35 m B. 3.66m C. 4.40m D. 4.67m

13. Which pair of forces CANNOT give a resultant of 5 N?
A. 1N and 4N C. 3N and 4 N
B. 1N and 6N D. 1N and 7 N

14. A sample of gas is expanded to twice is original volume of 1 m^3 in a quasi- static process for which p = αV, with $\alpha = 5\times 10^{+5}\ \frac{N}{m^5}$. How much work was done by the expanding gas?
A. 0.5 MJ B. 1.0 MJ C. 7.5 MJ D. 1.5MJ

15. A ball is thrown down ward(not dropped) from the top of a tower. After being released, its down ward.
A. Velocity will be uniform
B. Acceleration will be equal to g
C. Acceleration will be greater than g
D. Acceleration will be smaller than g

16. A block slides on a rough 45° incline. The coefficient of friction between block and incline is μ_K. What is the ratio of acceleration when the block accelerates down the inline to the acceleration when the block is projected up the incline?
A. $\frac{\mu k-1}{\mu k+1}$ B. $\frac{\mu k}{\frac{\sqrt{2}}{\mu k+1}}$ C. $\frac{1-\mu k}{1+\mu k}$ D. $\frac{1+\mu k}{1-\mu k}$

17. When a small mass m is suspended from one end of a vertical string and then whirled in a horizontal circle at a constant speed V,
A. The string stays vertical
B. The string becomes inclined to the vertical
C. There is no force on m except its weight
D. The angle of inclination to the vertical of the string does not depend on v

18. Suppose $\vec{A}$ = 3i – 2j, $\vec{B}$ = -i -4j and C are three vectors in xy- plane with the property that A + B – C = 0. Which of the following is a unit vector along C?
A. 2i- 6j B. -2i + 6j C. $-\sqrt{\frac{1}{10}}i + \frac{6}{\sqrt{10}}j$. $\sqrt{\frac{1}{10}}i - \frac{3}{\sqrt{10}}$ j

19. Three blocks of mass m_1, m_2, and m_3 are connected by thread and placed on a smooth horizontal surfaces as shown in the figure below. A force F is applied on the block of mass m_3 in forward direction as

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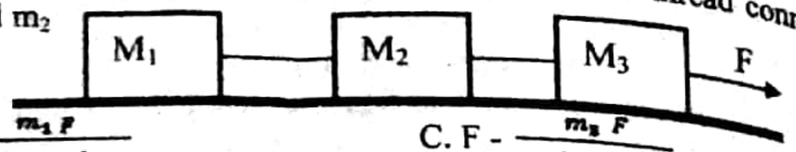
shown. Neglecting friction, find the tension in the thread connecting m_1 and m_2



A. $\frac{m_1 F}{m_1+m_2+m_3}$

C. F - $\frac{m_3 F}{m_1+m_2+m_3}$

B. $\frac{m_2 F}{m_1+m_2+m_3}$

D. $\frac{(m_2+m_3)F}{m_1+m_2+m_3}$

20. Suppose $\vec{A} = 3\hat{i} - 2\hat{j} + \hat{k}$, $\vec{B} = -\hat{i} - 4\hat{j} + 3\hat{k}$ and $\vec{C}$ is a vector lying along x – axis with the property that $\vec{A}.\ (\vec{B} + \vec{C}) = 0$. Then, vector $\vec{C}$ is given by

A. $\frac{3}{\sqrt{10}}\hat{j}$ B. $\frac{-11}{3}\hat{i}$ C. $\frac{6}{\sqrt{10}}\hat{j}$ D. $\frac{-8}{3}\hat{j}$

21. The necessary and sufficient condition for rigid body to be in equilibrium is the sum of

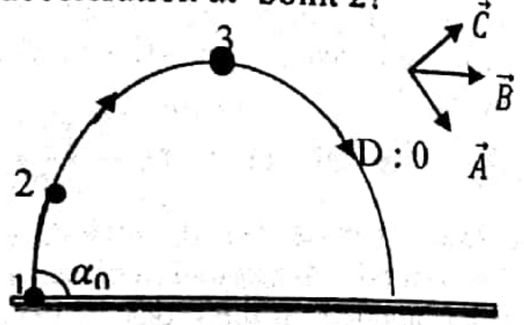
A. All torques acting on it is zero.
B. All forces acting on it should be zero
C. All forces should be equal in magnitude, but opposite in direction to the sum of all torques acting on it
D. All forces and the sum of all torques should be zero.

22. What is the inductance of an L C circuit that oscillates at 120 Hz when the capacitance is $8.0\,\mu F$?

A. 0.22 μ H B. 0.22H C. 2.19 H D. 2.19 μ H

23. A bullet is fired from cannon at point 1 and follows the trajectory shown in the figure below. Air resistance may be neglected. Four possible vectors are also shown in the figure: the letter $\vec{D}$ represents the zero vector. Which vector best represents the bullets acceleration at point 2?

A. $\vec{A}$
B. $\vec{B}$
C. $\vec{C}$
D. $\vec{D}$



24. A car initially at rest and at the origin has moved a distance of5, 10, 15, 20, 25, 30, 40, 50, 60, 70 meters during the 1, 2, 3 .... 10 seconds periods respectively. What is the average speed during the first 5 seconds and the second half period from t= 6sec to $t = 10$ seconds in $\frac{m}{s}$ respectively?

A. 4, 10 B. 4, 9 C. 5, 9 D. 5,10

25. A disk rolls without slipping along a horizontal surface. The center of the disk has a translational speed v. The uppermost point on the disk has a translational speed

A. zero B. V C. 2v D. 2π

26. Light with frequency of 6×10^5GHz passes through two narrow slits of width 0.2 mm apart and forms and interference pattern on a screen 2.0 m away from the slits. What is the distance between the fringes in the interference pattern ? (The speed of light $c = 3 \times 10^8$ m/s)

A. 3mm B. 5mm C. 4mm D. 3.5 mm

27. The angular momentum of a particle is

A. independent of the specific origin of coordinates
B. zero when its position and momentum vectors are paralle
C. zero when its position and momentum vectors are perpendicular
D. not zero for non zero position and momentum vectors

28. A small car and a bus move with velocities given by $\vec{V}_c = (3\hat{i} - 2\hat{j})\frac{m}{s}$ and $v_b = (-5\hat{i} - 4\hat{j})\frac{m}{s}$ respectively according to and observer on the ground. What is the velocity of the bus according to a car driver?

A. $(-2\hat{i} + 6\hat{j})\frac{m}{s}$ C. $(-8\hat{i} + 2\hat{j})\frac{m}{s}$

B. $(-2\hat{i} - 6\hat{j})\frac{m}{s}$ D. $(-8\hat{i} - 2\hat{j})\frac{m}{s}$

29. Water flows through a pipe whose cross- sectional area is 50 cm^2 into another pipe whose cross-sectional area is $10cm^2$. Both pipes are horizontal. If the velocity of the water in the large pipe is $12\frac{m}{s}$, the velocity in the small pipe is

A. 0.24 $\frac{m}{s}$ B. 1.2 $\frac{m}{s}$ C. 60 $\frac{m}{s}$ D. 30 $\frac{m}{s}$

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30. A Parallel- plate capacitor with a plate separation d has a capacitance C in the absence of a dielectric. What is the capacitance when a slab of dielectric material of dielectric constant k and thickness $\frac{1}{3}$ d is inserted between the plates?

A. $\frac{3k}{2K+1}$ C B. $\frac{2k+1}{3K}$ C C. $\frac{3k}{2K-1}$ C D. $\frac{2k-1}{3K}$ C

31. The acceleration of a simple harmonic oscillator
A. Remains constant during its motion
B. Remains non- zero
C. Attains its maximum value when its amplitude is zero.
D. Attains minimum value when its velocity is maximum

32. The resistance of an Ohmic conductors is independent of
A. the size of the material C. the temperature
B. the mature of the material D. the voltage applied

33. As shown below, an ideal massless spring is compressed1 m by a force of 100N. This same spring is placed at the bottom of a frictionless inclined plane which makes an angle of θ 30^0 with the horizontal. A 10kg mass M is released from rest at top of the incline and is brought to rest momentarily after compressing the spring2.0 m Through what distance does the mass slide before coming to rest?
A, 1m B. 2m C. 4m D. 6m

34. When the intensity of light shining on a metal increases,
A. The energy of photoelectrons, emitted, will increase
B. The number of photoelectrons, emitted, will increase
C. The threshold frequency for photoelectron emission will increase
D. The photoelectrons emitted will be deflected

35. Consider the circuit shown below, What are the current in 20Ω resistor and the potential difference between points A and B?

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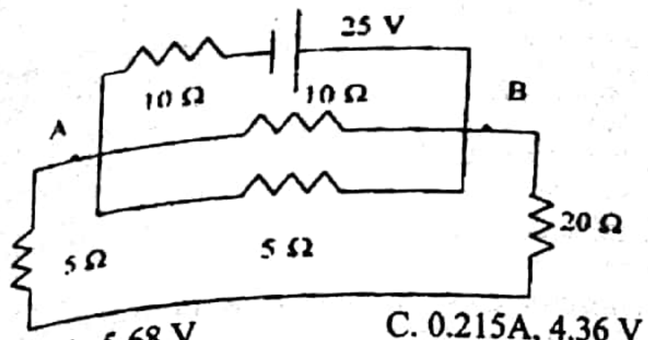
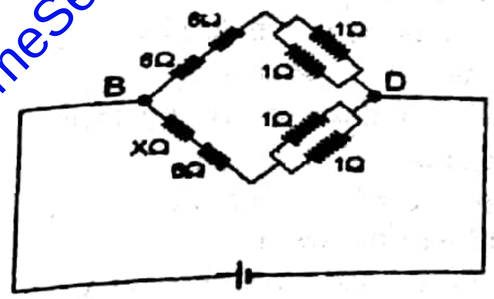


A. 0.227 A, 5.68 V C. 0.215A, 4.36 V
B. 0.251 A, 436 V D. 0.227A, 4.36 V

36. For what value of X will the potential difference between B and D be zero in the arrangement shown below?



A. 2Ω B. 3Ω C. 4Ω D. 6Ω

37. Through a pipe 15 cm in diameter, water is pumped from the river outside of Addis Ababa to one of the sub- cities of Addis Ababa. The river is at an elevation of 564 m and the sub- city is at an elevation of 2096 m. What is the minimum pressure at which the water must be pumped if it is to arrive at the sub- city?
A. 15.32 MPa B. 15.01 Mpa C. 14.99MPa D. 16.03 MPa

38. The phase angle in an alternating currnt ciruit is zero degree when the circuit contains
A. Resistance only
B. Capacitance only
C. Inductance only
D. Series combination of resistance and inductance

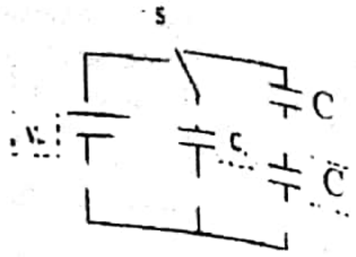
39. When switch S is thrown to the left in the figure shown below the plates of the capacitor C_1 acquire a potential difference V_0 .C_2 and C_3 are initially uncharged. The switch is now thrown to the right. What are the final charges q_1 on the capacitor C_1?

A. $\frac{C_1C_2+C_2C_3}{C_1C_2+C_1C_3+C_2C_3})\,C_1V_0$

B. $\frac{C_1C_2+C_1C_3}{C_1C_2+C_1C_3+C_2C_3})\,C_3V_0$

C. $(\frac{C_1C_2+C_1C_3}{C_1C_2+C_1C_3+C_2C_3})\,C_2V_0$

D. $(\frac{C_1C_2+C_1C_3}{C_1C_2+C_1C_3+C_2C_3})\,C_1V_0$



40 The x and y coordinates of a particle at any time t is given by $x = 9t + 5t^2$ and $y = 5t + 8$ where x and y are in meters and t in seconds. The acceleration of the particle at t = 5 seconds is

A. Zero B. $10m/s^2$ C. $14\ m/s^2$ D. $20m/s^2$

41. The second law of thermodynamic does. Not imply that:

A. there is not any process in which there is a decrease in the entropy of the universe overall

B. the complete conversion of energy from a hot source into work is not possible

C. heat flows from colder body to hotter body as in the refrigerator without work done on the system

D. perfectly reversible processes are impossible

42. A key phenomenon that demonstrates the particle nature of light is

A. photoelectric effect C. dispersion of light

B. Doppler effect D. interference

43. Two discs of moments of inertia I_1 and I_2 about their respective axes (which are normal to the disc and passing through the centre) and rotating with angular speeds W_1 and W_2 are brought into contact face- to-face with their axes of rotation coincident . Take $W_1 \neq W_2$. What is the angular speed of the two- disc system after contact?

A. $\frac{I_1\omega_1+2I_2\omega_2}{I_1+2I_2}$ C. $\frac{I_1\omega_1+2I_2\omega_2}{I_1-2I_2}$

B. $\frac{I_1\omega_1+I_2\omega_2}{I_1+I_2}$ D. $\frac{I_1\omega_1+I_2\omega_2}{I_1-I_2}$

44. A particle that undergoes simple harmonic motion has a period of 0.4 s and an amplitude of 12 mm. The maximum velocity of the particle is

A. $3\frac{cm}{s}$ B. $9.42\frac{cm}{s}$ C. $12\frac{cm}{s}$ D. $18.84\frac{cm}{s}$

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45. When does a light bulb carry more current?

A. Just when it is turned on during which the filament has low temperature and high resistance.

B. Long after it is turned on as the filament warms and has high resistance

C. Just when it is turned on during which the filament has low resistance

D. Long after it is turned on as the filament warms and has low resistance

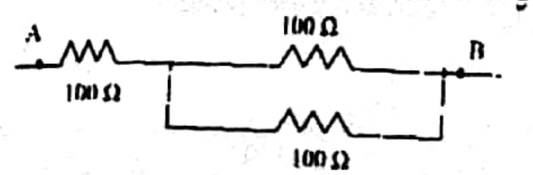
46. The impedance of an 8 microfarad capacitor at 1000Hz is

A. 600 Ohm B. 22.7 Ohm C. 19.9 Ohm D. 0.44 Ohm

47. Three 100-Ω resistors are connected as shown in the figure below. The maximum power that can be delivered to any one resistor is 25.0 W. What is the maximum voltage that can be applied between the terminals A and B

A. 50V B. 75 V

C. 100V D. 125V



48. Which of the following statements follows from kinetic theory of gases?

A. The pressure of the gas is proportional to the average of square of molecular speed of particles

B. The molecules undergo inelastic collisions with each other and with the walls of the container

C. The forces between molecules of gas are strong at all times

D. Newton laws of motion is a consequence of kinetic theory of gases

49. As a satellite moves in an elliptical orbit around the earth, the gravitational attraction of the earth for the satellite:

A. Is greater than the satellite's gravitational attraction for the earth.

B. Is less than the satellite's gravitational attraction for the earth.

C. Varies inversely as the square to the distance between the earth and the satellite.

D. Remains constant

50. A uniform ladder of length l and weight W = 50 N rests against a smooth, vertical wall. If the coefficient of static friction between ladder and ground is $\mu = 0.5$, find the minimum $> \theta_{min}$ between the ladder and ground such that the ladder will not slip.

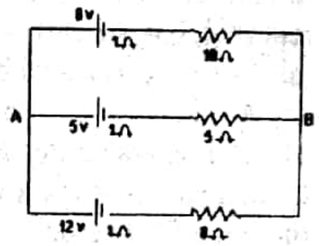
A. 45^0 B. 53^0 C. 60^0. D. 37^0

UNIVERSITY ENTRANCE EXMINATION /UEE/
PHYSICS SENE 2004/JUNE 2012

Directions: Each of the following questions is followed by four possible alternatives. Read each question and carefully **blacken** only the letter of your best choice on the separate answer sheet provided.
You may refer the information given below when you work on some of the questions.

Constant	Symbol	Value
Acceleration due to gravity	g	$10m/s^2$
Permittivity of vacuum	ϵ_o	8.85×10^{-12} F/m
Faraday' constant	F	$96500Cmol^{-1}$
Charge of an electron	e	$1.6x10^{-19}$ Coulomb
Universal Gravitational constant	G	$6.67 \times 10^{-11} \frac{Nm2}{Kg2}$
$\sin 15^0=\cos 75^\circ$ 0.2588		
$\sin 75^0=\cos 15^\circ = 0.9659$		
$\sin 30^0=\cos 60^\circ =0.5$		
$\sin 37^0=\cos 53^\circ = 0.6$		
$\sin 53^0=\cos 37^\circ =0.8$		
$\sin 60^0=\cos 30^0 = 0.866$; $\sin 45^\circ=\cos 45^\circ = \frac{1}{\sqrt{2}} = 0.707$		
$\mu 0 = 4\pi x10^{-7}$ Tm/A, $\pi = 3.14$, tan /53.1/=1.33		

1. An ion accelerates through a potential difference of 115 V and experiences an increase in kinetic energy of 7.37×10^{-17} J. What is the charge on the ion?
   A. $6.41. \times 10^{-19}$C
   B. 2.32×10^{-12} C
   C. 4.51×10^{-15} C
   D. 2×10^{-21} C
2. What is magnitude of the electric force between two protons separated by 2×10^{-15}m?
   A. 67.5N B. 115.0N C.57.5N D. 83.4N
3. If the amplitude of a sine wave is double, the intensity
   A. Remains the same
   B. Increase by a factor of 2
   C. Increases by a factor of 4
   D. Increases by a factor of 16
4. The temperature of a gas is due to
   A. The kinetic energy of the molecules
   B. The potential energy of the molecules
   C. The attractive force between the molecules
   D. The repulsive forces between the molecules
5. Suppose $A = 3i - 2j$ and $B = -i - aj$ are two vectors in xy –plane. What is the value of a, such that A+B=2i?
   A. 3 B. 2 C. -2 D. -3
6. Which statement is correct about a purely inductive circuit?
   A. The current leads the voltage by $\pi/2$
   B. The current is in phase with the voltage
   C. The current lags behind the voltage by $\pi/2$
   D. The current lags by $\pi/2$ at low frequencies and leads by $\pi/2$ a high frequencies.
7. What is the SI unit of Electric field?
   A. Am^{-1} B. Vm^{-1} C. Cm D. NC^{-1}
8. The speed of a mass is increased by two fold. By what factor does the kinetic energy change?
   A. 2 B. 4 C.8 D.16
9. A projectile is fired at 60° with the horizontal. At the highest point of its path, the vertical component of its velocity is momentarily zero. What is the magnitude of its acceleration at this point?
   A. $20m/s^2$ B. $10m/s^2$ C. $5m/s^2$ D. $0m/s^2$
10. The sum and difference of two non-zero vectors A and B are equal in magnitude. What can you conclude about these two vectors?
    A. $\vec{A}$ and $\vec{B}$ have the same direction
    B. $\vec{A}$ and $\vec{B}$ have opposite directions
    C. $\vec{A}$ and $\vec{B}$ have the same magnitude
    D. $\vec{A}$ and are perpendicular to each other
11. Determine $\vec{B}$ the current in the 5Ω resistor in the figures shown below
    A. 7.85A
    B. 1.09A
    C. 0.48A
    D. 0.18A



12. A cell supplies a current of 0.9A through a 2Ω resistor and a current of 0.3A through a 7Ω resistor. The internal resistance of the cell is ?
    A. 0.5Ω B. 1.0Ω C. 1.2Ω D. 2.0Ω
13. Which phenomenon is responsible for the colors seen on a thin oil film on the water surface?

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A. Diffraction C. Superposition
B. interference D. refraction

14. A pipe, 68 cm long, is open at one end and closed at the other. When air is blown across the open end sound is produce at 110Hz. What is the velocity of sound along the pipe?
A. 598.4m/s B. 334.6m/s C. 299.2m/s D. 149.6m/s

15. A 1kg mass is attached to a spring of stiffness constant 100N/m. What is the maximum displacement of the mass from its equilibrium position if its speed is 3m/s at the equilibrium point?
A. 0.1m B. 0.3.m C. 3m D. 10m

16. What does a curve drawn between two points on P-V diagram represent?
A. The state of the system
B. A thermodynamic process
C. Work done in a cyclic process
D. Work done on or by the system

17. What does the value of Reynolds number signify?
A. It describes whether the flow is vertical or horizontal
B. It describes the magnitude of the viscosity of the liquid
C. It describes whether the flow is stream lined or turbulent.
D. It describes the magnitude of the surface tension of the liquid.

18. A car/mass m_1/ is driving at velocity v, when it smashes into an unmoving car /mass m_2/, locking bumpers. Both cars move together at the same velocity. The common velocity will be given by:
A. $\frac{M2V}{M2}$ B. $\frac{m2v}{m1}$ C. $\frac{m1v}{(m1+m2)}$ $\frac{(m1+m2)v}{m1}$

19. A cart on an air track is moving at 0.5 m/s when the air is suddenly turned off. The cart comes to rest after traveling 1m. The experiment is repeated, but the cart is moving at 1m/s when the air is turned off. How far does the cart travel before coming to rest?
A. 1m B. 2m C. 3m D. 4m

20. *A car* accelerates steadily form m/s to 30 m/s in a distance d and a time t. Another car takes a time 3t to accelerate steadily from stationary to the same final velocity. What distance does the second car cover during the new acceleration?
A. 3d B. d C. d/3 D. d/9

21. The total energy of a particle executing simple harmonic Motion is
A. Zero
B. Directly proportional to the amplitude

C. Directly proportional to the square of the amplitude
D. Inversely proportional to the square of the amplitude

22. A table –tennis ball has a diameter of 3.80 cm and average density of 0.084 g/cm^3. What force I required to hold it completely submerged under water?
A. 0.521N B. 0.258N C. 0.300N D. 0.510N

23. The pressure at the bottom of a cylindrical tube filled with water was measured to be 5000 Pa. If the water in the tube were replaced with ethyl alcohol, what would be the new pressure at the bottom of the tube? /The density of ethyl alcohol is 0.8g/cm^3/.
A. 4000Pa B. 4800Pa C. 5000Pa D. 6225Pa

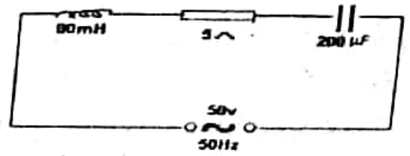
24. A ladder stands on the ground, leaning against a wall. It is safer climbing up the ladder if.
A. The ground is frictionless and the wall is rough
B. The ground is rough and the wall is frictionless
C. The ground is frictionless and the wall is frictionless
D. The ladder leans at 45 degree and the ground is frictionless

25. A wheel starts from rest and rotates with constant angular acceleration to reach an angular speed of $12\frac{rad}{s}$ in 3s. What is the magnitude of the angular acceleration of the wheel in $\frac{rad}{s2}$ and the angle in radians through which it rotates in this time interval respectively?
A. 4,18 B. 3,18 C. 18,5 D.18,4

26. A 4kg block is given an initial speed of 8m/s at the bottom of 30° inclined plane. The frictional force that retards its motions is 15N. If the block is directed up the inclined plane, how far will it move before it stops?
A. 3.66m B. 4.40 C. 3.38.m D. 4.67m

27. An object moving with uniform acceleration has a velocity of 12m/s in the positive x direction when its x coordinate is 3cm.In its x coordinate 2s later is -5cm, what is its acceleration?
A. $-12m/s^2$ B. $-13\ m/s^2$ C. $-16\ m/s^2$ D. $12\ m/s^2$

28. Find the impedance for the RLC circuit shown in the figure below.
A. 1.96 Ω
B. 19.2 Ω
C. 25.5 Ω
D. 648.1Ω



29. What is the main reason why alternation current is used to transmit electricity?

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A. To minimize power gain C. To maximize voltage gain
B. To minimize energy loss D. To maximize current gain

30. Find the magnitude of the force felt by an electron traveling at an angle of 60° to the earth's magnetic field at 1.4×10^7 m/s. /The charge on an electron is 1.6×10^{-19} C and the magnitude of the earth's magnetic field is 5×10^{-5} T)
A. 0N B. 9.7×10^{-17} N C. 1.1×10^{-16} N D 8.6×10^{-17} N

31. The loudness of sound changes from L_1 =30db to L_2 =60db. What is the ratio of intensities I_2/I_1 in the two cases?
A. 2 B.4 C. 600 D. 1000

32. Which statement is **NOT** among the assumptions made in the kinetic theory of gases?=
A. There is no interaction between particles or between particles and the wall except during collision.
B. Collisions between particles, and between particles and walls are perfectly elastic.
C. The volume occupied by the particle is negligible compared to the volume of the gas as a whole
D. The distribution of velocities of particles is uniform.

33. A ladder of length 3m and mass 20kg, leans against a smooth, vertical wall so that the angle between the wall and the ladder is 30°. Determine the magnitude of the normal forces at the contact point on the wall /N_w/ and at the contact point on the ground /N_g/
A. N_w =57.7N, N_g =200N C. N_w=0N, N_g =200N
B. N_w = 200N, N_g =57.7N D. N_w =200N, N_g =0N

34. Which of the following objects has the largest moment of inertia assuming they all have the same mass and radius?
A. A solid disk C. A circular loop
B. A solid sphere D. A solid cylinder.

35. A stationary mass explodes into two parts of masses 0.4kg and 4kg. If the larger mass has a kinetic energy of 100J, what is the kinetic energy of the smaller mass?
A. $4\sqrt{50}J$ B. 100J C. 1000j
D. The change in kinetic energy is the same for both particles.

36. In RLC circuit, given R= 7.6Ω, L =2.2mH and C= 1.8 μ F, what is the frequency of the damped oscillation of the circuit?
A. 2.5×10^3 Hz B. 15.72×10^3Hz C. 12.5×10^3 Hz D. 25.0×10^3 Hz

37. Two coils, held in fixed positions, have a mutual inductance of 100 μH. What is the peak voltage in one when the sinusoidal current given by I(t) =(10.0A) sin(1000t) flows in the other?
A. 1.0 μV B.100.0V C. 10.0V D. 1.0V

38. A conduction rod of length l moves with a constant velocity V perpendicular to along, strait wire carrying current I. What is the magnitude of the emf generated between the ends of the rod?
A. $\frac{\mu_0 Vl}{2\pi r}1$ B. $\frac{\mu_0 Vl}{2\pi l}r$ C. $\frac{\mu_0 Vl}{2l}r$ D. $\frac{\mu_0 Vl}{2r}1$

39. A rod is made of two materials connected to each other. Each conductor has a square cross-section of 3mm on a side. The first material has resistivity of $\rho_1 = 4x\ 10^{-3}\Omega m$ and is 25cm long while the second material has resistivity of ρ_1=6 x10-3 Ωm and is 40cm long. What is the resistance between the ends of the rod?
A. 378 Ω B. 267 Ω C. 436 Ω D. 363 Ω

40. When a potential difference of 150V applied to the plates of a parallel-plate capacitor, the plates carry a surface charge density of $30\frac{nC}{cm^2}$. What is the spacing between plates?
A. 2.24 μm B. 5.32 μm C. 12.24 μm D.4.42 μm

41. The maximum power delivered to the load resistance R occurs when the load resistance R and internal resistance r have the following relation:
A. r =2R B. r= R C. r=3R D. r =7R

42. A small object of mass m carries a charge q and is suspended by a thread between the vertical plates of a parallel-plate capacitor. The plate separationis d. If the thread makes an angle θ whit the vertical, what is the potential difference between the plates?
A. $\frac{mgd\tan\theta}{q}$ B. $\frac{md\tan\theta}{q}$ C. $\frac{mgd\sin\theta}{q}$ D. . $\frac{md\sin\theta}{q}$

43. When a soap bubble is in air, it exhibits different colures under sunlight. The absence of a particular color in a region is due to
A. Double refraction.
B. Straight line propagation of light.
C. Destructive interference of light of that color in that region.
D. Constructive interference of light of that color in that region.

44. A point source broadcasts sound into a uniform medium. If the distance from the source is tripled, how does the intensity change?
A. I t remains unchanged B. It becomes three times larger

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C. It becomes one-third as large D. It becomes one-ninth as large

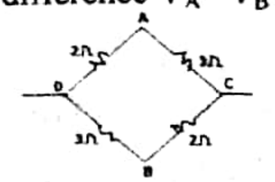
45. Water flows through a fire hose of diameter 6.35 cm at a rate of 0.012 $\frac{m3}{s}$.The fire hose ends in a nozzle of inner diameter 2.20 cm. What is the speed with which the water exits the nozzle?

A.23.4$\frac{m}{s}$ B. 31.6$\frac{m}{s}$ C. 63.2$\frac{m}{s}$ D. 46.8$\frac{m}{s}$

46. A block of mass 0.23 kg is placed on top of a light vertical spring of force constant 5000$\frac{N}{m}$ and pushed downward so that the spring is compressed by 0.10m. After the block is release from rest, it travels upward and then leaves the spring. To what maximum height above the point of release does it rise?

A. 10.2m B. 9.2m C. 8.0m D. 9.0m

47. There is a current of 2A in the system of conductors shown in the figure below. The potential difference $V_A - V_B$ is

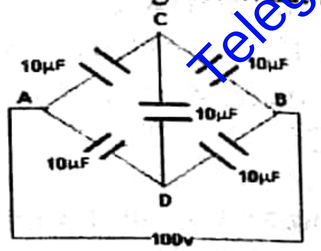


A. -2V
B. -1V
B. +1V
C. +2V

48. A galvanometer many be used as an ammeter by

A. Shunting the galvanometer with a high resistance
B. Connecting a low resistance in parallel with the galvanometer
C. Connecting a low resistance in series.
D. Connecting a high resistance in series

49. Five capacitors $10\mu F$ capacity each is connected to A, D, and C potential of 100 volt as shown in the next figure. What is the equivalent capacitance between A and B?



A. $10\mu F$
B. $20\ \mu F$
C. $30\ \mu F$
D. $60\ \mu F$

50. What happens if the monochromatic light used in Young's double slit experiment is replaced by white light?

A. No fringes are observed
B. All bright fringes become white
C. All bright fringes have colors between violet red.
D. Only the central fringe is white, all other fringes are colored.

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Constant	Symbol	Value
Acceleration due to gravity	G	10 m/s^2
Permitivity of vacuum	Eo	$8.85x10^{-12}F/m$
Charge of an electron	E	$1.6x10^{-19}C$
Sin 30° = cos 60°=0.5		
Sin37° = cos 53°=0.6		
Sin 53° = cos 37°=0.8		
Cos 60° = sin 30°=0.87		

1. Which of the following is correct about the meaning of Photon?
A. quantum of matter C. negatively charged particle
B. positively charged particle D. quantum of light

2. Consider a series RC circuit for which R = 1.0 MΩ, C= 5 μF and ε=30 V. What is the time constant of the circuit and the maximum charge on the capacitor after the switch is closed respectively?
A. 5 seconds, 150 μC C. 10 seconds, 160 μC
B. 7 seconds, 155 μC D. 9 seconds, 135Mc

3. What is the direction t which a fish must push the water with its fins in order to propel eastward?
A. eastward B. upward C. westward D. downward

4. Suppose two molecules of different mass are kept at the same temperature. Which one the following statements is NOT true?
A. Both remain at rest. B. Both move with the same speed.
C. The lighter molecule moves faster. D. The heavier molecule moves faster.

5. One person drops a ball from the top of a building while another person at the bottom observes its motion. The two people disagree on which of the following?
A. the value of the total energy of the ball.
B. the change in potential energy of the ball.
C. the value of the gravitational potential energy of the ball earth system.

D. the kinetic energy of the ball at some point in its motion if one of them is walking.

6. Two non-zero vectors $\vec{A}$ and $\vec{B}$ are related by $\vec{A} = c(\vec{B})$ where c is a scalar. If the two vectors have opposite directions, then one of the following is true about c?
   A. c is a positive number
   B. c is a negative number
   C. c=1
   D. c=0

7. On which of the following does the induced emf in a coil NOT depend when a magnet is inserted into it?
   A. Resisrance of the coil
   B. Number of turns of the coil
   C. Speed of approach of the magnet
   D. Magnetic moment of the magne

8. A wave has a speed of 50 m/s and a wavelength of 25,000 m. What is the frequency of the wave?
   A. 2.0×10^{-3}HZ
   C. 5.0×10^{-2}HZ
   C. 6.0×10^{1}HZ
   D. 1.8×10^{6}HZ

9. An oscillating system is used to create a succession of wave on a water surface. The rate of wave generation is one every 0.33 seconds. What is the period of these wave?
   A. 0.3 seconds B. 1.0 seconds C. 2.0 seconds D. 3.0 seconds

10. A vector that represents the position of an object in relation to another object is called
    A. Unit vectors
    B. Position vectors
    C. Copplanar vectors
    D. Collinear vectors

11. In a reversible thermodynamic process, the system
    A. is always close to equilibrium states
    B. might never be close to any equilibrium states
    C. is clcse to equilibrium states only at the beginning and end
    D. is closeto equilibrium states throughout, except at the beginning and end

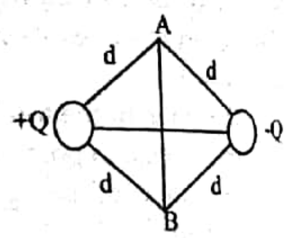
12. Points A and B are each a distance d from each of two equal and opposite charges as shown below. What will be the work required to move a negative charge q from A to B?

A. Zero

B. $\frac{qQ}{4\pi\varepsilon_o d^2}$

C. $\frac{qQ}{4\pi\varepsilon_o d}$

D. $\frac{qQ}{4\pi\varepsilon_o(4d^2)}$



13. An air-filled parallel-plate capacitor has a capacitance of 1Pf. The plate separation is then doubled and a wax dielectric is inserted, completely filling the space between the plates. As a result, the capacitance becomes 2 Pf. What is the dielectric constant of the wax?
    A. 8.0 B. 4.0 C. 2.0 D. 0.5

14. A step-up transformer is used to
    A. increase the power
    B. decrease the power
    C. increase the voltage
    D. derease the voltage

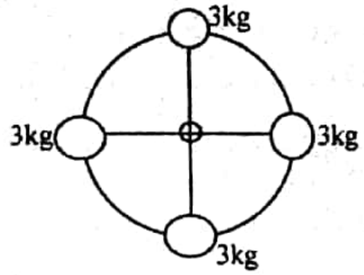
15. A 220-V power line is protected by a 15-A fuse. What is the maximum number of "220 V, 500 W" light bulbs that can be operated at full brightness from this line?
    A. 2 B. 4 C. 6 D. 8

16. If $\vec{A} = 8\vec{i} + \vec{j} - 2\vec{k}$ *and* $\vec{B} = 5\vec{i} - 3\vec{j} + \vec{k}$, then, what is the value of $\vec{B}X\vec{A}$?
    A. $5\vec{i} + 4\vec{j} - 3\vec{k}$
    B. $-3\vec{i} - 4\vec{j} + 3\vec{k}$
    C. $-5\vec{i} - 18\vec{j} - 29\vec{k}$
    D. $5\vec{i} + 18\vec{j} + 29\vec{k}$

17. The coordinate of a particle in meters is given by $X(t) = 25t - 3t^2$, where the time t is in seconds. At what value of t will the partile become momentarily at rest?
    A. 2.78 s B. 1.67 s C. 0.6 s D. 0.36 s

18. A woman driving a 2000 kg car along a level road at 30 m/s takes her foot off the gas to see how far her car will roll before t slows to a stop. She discovers that it takes 150 m. what is the average force of friction acting on the car?
    A. 9000 N B. 6000 N C. 3000 N D. 400 N

19. The radius of the wheel shown below is 0.75 m. what is its moment of inertia?
    A. 2.25 kg m^2
    B. 3.38 kgm^2
    C. 6.75 kg m^2
    D. 9.00 kgm^2



20. An ordinary refrigerator is turned on and its door is open in a thermally insulated kitchen. What is the temperature of the room?
    A. increases according to the first law of thermodynamics
    B. decreases according to the first law of thermodynamics

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C. remains constant according to the first law of thermodynamics
D. remains constant according to the second law of thermodynamics

21. The langth of an aluminum rod at a temperature of 20°C is 3 m. What will be the ratio of change in the length of the rod to its original length if its temperature is raised to 120°c? The coefficient of linear expansion of aluminum is $3.31 \times 10^{-5}/K$

A. 23% B. 3.2% C. 3% D. 0.23%

22. A very large open tank is full of oil whose density is $800\ {}^{kg}/_{m}$. If the absolute pressure at the bottom of the tank is 1.81×10^5 Pa, them what is the depth of the oil? (Note that atmospheric pressure is 1.01×10^5 Pa)

A. 100.0 m B. 75.0 m C. 10.0 m D. 9.5 m

23. A 1.5 kg particle moves in the xy-plane with velocity $\vec{v} = (4.2\vec{i} - 3.6\vec{j})\ {}^{m}/_{s}$ What is its angular momentum when its position vector is $\vec{r} = (1.5\vec{i} + 2.2\vec{j})m$?

A. $-22\vec{k}\ kgm^2/s$
B. $14.6\vec{k}\ kgm^2/s$
C. $-14.6\vec{k}\ kgm^2/s$
D. $22\vec{k}\ kgm^2/s$

24. A 2 kg object experiences two concurrent forces of 100 N at 30° above the positive x-axis and 80 N at 60° above the negative x-axis. What is the magnitude of acceleration of the object?

A. 68 m/s^2 B. 64 m/s^2 C. 48 m/s^2 D. 46 m/s^2

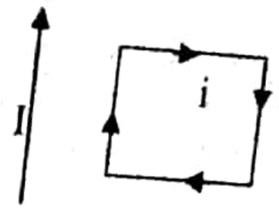
25. A driver moving at a constant peed of 20 m/s sees an accident up ahead and hits the brakes. If the car decelerates at a constant rate of $-5\ m/s^2$, how far does the car go before it comes to a stop?

A. 120 m B. 100 m C. 40 m D. 20 m

26. The energy spectrum emmited by a black body at different temperature is as shown in figure below. Which of the following statements is NOT correct about the black body energy spectrum?

A. A. $T_1 < T_2 < T_3 < T_4$
B. B. $\lambda_1 > \lambda_2 > \lambda_3 > \lambda_4$
C. Black body radiate energy over the entire electromagnetic spectrum
D. The wavelength corresponding to the maximum intensity increases with temperature

27. A rectangular loop carrying a current is situatd near a long straight wire such that the wire is parallel to one of its sides of the loop and it is in a plane of the loop. If a steady current is established in the wire as shown in figure, the loop will



A. remain stationary.
B. move toward the wire.
C. move away from the wire.
D. rotate about an axis parallel to the wire.

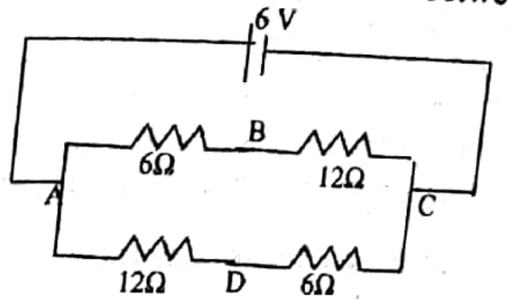
28. Which device uses the principle of electromagnetic induction for its operation?

A. Ac generator C. Cathode ray tube
B. Moving coil meter D. Light emmiting diode

29. An electron enters a region of uniform electric field of $5 \times 10^3 N/C$. What is the force on the electron?

A. 8×10^{-22} N C. 8×10^{-16} N
B. 5×10^{-22} N D. 1.88×10^{22} N

30. The circuit shown below, what is the potential difference between points B and D?



A. 1 V
B. 2V
C. 3 V
D. 4 V

31. The planet Jupiter is about 318 times as massive as Earth, yet on its surface you would weigh only about 2.4 times your weight on earth. Why is this happened?

A. your mass is 132.5 times less on Jupiter.
B. you are 132.5 times more weightless there.
C. Jupiter is significantly farther from the sun.
D. Jupiter's radius is about 11.5 times the Earth's radius.

32. A uniform ball of radius r and mass m starts from rest and rolls down a frictionless inline of height h. When the ball reaches the ground, how fast is it going?

A. $\sqrt{\frac{10}{7}gh}$ B. $\sqrt{\frac{9}{10}gh}$ C. $\sqrt{\frac{10}{9}gh}$ D. $\sqrt{\frac{7}{10}gh}$

33. A ladder of length 3 m and mass 20 kg leans against a smooth, vertical wall so that the angle between the wall and the ladder is 30°. What are

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the magnitudes of the normal forces at the contact point on the wall(N_w) and at the contact point on the ground?

A. N_w =0 N, N_g=200 N
B. N_w=200 N, N_g=0 N
C. N_w=57.7 N, N_g=200 N
D. N_w=200 N, N_g=57.7 N

34. A container resting on the floor of an elevator contains an incompressible fluid of density ρ. When the elevator has an upward acceleration of magnitude 'a' then what will be the pressure difference between two points in the fluid separated by a vertical distance Δh?

A. ρaΔh B. ρ(g+a)Δh C. ρgΔh D. ρgaΔh

35. If two particles have equal momenta, then which one of the following statements is TRUE about their kinetic energies?

A. Their kinetic energies are always equal.
B. Their kinetic energies are never equal.
C. Their kinetic energies are equal when their masses are the same.
D. Their kinetic energies are equal when the speed of one of the particle is half the speed of the other and their masses are the same.

36. In Young's double slit experiment both the separation between the slits and the distance between the slits and the screen are halved then, what is the frige width?

A. halved B. unchanged C. zero D. doubled

37. What is the capacitance of a capacitor that should be connected with 2 other capacitors having capacitance of 4μFand8μFall in series to produce an effective capacitance of $\frac{24}{11}\mu F$?

A. 12 μF B. 10 μF C. 8 μF D. 4 μF

38. A particle of mass m is fired in to a magnetic field of strength B at a speed v. The particle travels in a circular path inside the field with a radius r. Which of the following expressions gives the magnitude of the charge on the particle?

A. $\frac{mr}{vB}$ B. $\frac{mv^2}{Br}$ C. $\frac{vB}{mr}$ D. $\frac{mv}{Br}$

39. An alternating current is represented by $I(t)=10.0A\sin(100\pi)$. What are the amplitude and frequency of the alternating current?

A. 100 A, 75 HZ
B. 10.0 A, 100 HZ
C. 10.0 A, 50 HZ
D. 5.0 A, 25 HZ

40. A series RLC circuit has a resistance of 45 Ω and an impedance of 75 Ω. What average power is delivered to this circuit when ΔV_{rms}=210 V?

A. 573 W B. 353 W C. 255 W D. 247 W

41. A thin film of oil (n=1.25) is located on a smooth wet floor. When viewed perpendicular to the floor, the film refects most strongly red light at 640 nm and reflects no blue light at 512 nm. How thick is the oil film?

A. 640 nm B. 612 nm C. 548 nm D. 512 nm

42. Which one of the following changes to parallel plate capacitor would not increase energy stored in the capacitor at a fixed voltage across the plates?

A. increasing the area of the plates
B. increasing the dielectric constant
C. decreasing the charges on the pates
D. increasing the distance between the plates

43. Suppose that a voltage source produces 240 V for a moment. By what percentage does the power out put of 220 V light bulb increases?

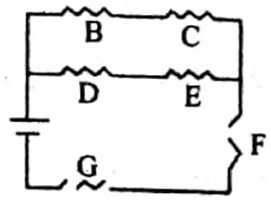
A. 38% B. 36% C. 19% D. 18%

44. A wire carries a steady current of 2.4 A. A straight section of the wire is 0.75 m long and lies along the x-axis within a uniform magnetic field B= 1.6 T along the +z axis. If the current is in the +x direction, what is the magnetic force on the section of the wire?

A. 2.88 j N B. 1.62 j N C. -1.62 j N D. -2.88 j N

45. If all the three rresistors in the circuit shown in the figure have equal resistances, and the current flowing into resistor B is 4 A, then what is the current flowing into resistor G?

A. 16 A
B. 8 A
C. 2 A
D. 1 A



46. Refraction of a wave does NOT involve change in its

A. speed.
B. wavelength.
C. frequency.
D. speed and wavelength.

47. Gas at a pressure P is contained in a vessel. If the masses of all the molecules are doubled and their speeds are halfed, then what will be the resulting pressure?

A. 2P B. $\frac{P}{3}$ C. $\frac{P}{2}$ D. 3P

48. Water flows through a fire hose of diameter 6.35 cm at a rate of 0.012 m^3/s. The fire hose ends in a nozzle of inner diameter 2.20 cm. What is the speed with which the water exits the nozzle?

A. 63.2 m/s B. 46.8 m/s C. 31.6 m/s D. 23.4 m/s

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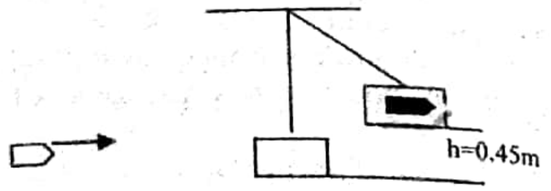
49. A bullet of mass 10 gm, moving horizontally, strikes and embedded itself in a box of mass 1 kg suspended from a light string as shown in the figure below. If the composite mass rises to a height, h=0.45 m, then what is the speed of the bullet before collision?

A. 303 m/s
B. 300 m/s
C. 90.9 m/s
D. 90 m/s



50. A rocket moving in free space has a speed of $3x10^3$ m/s relative to the Earth. Its engines are turned on, and fuel is ejected in a direction opposite the rocket's motion at a speed of $5x10^3$ m/s relative to the rocket. What is the speed of the rocket relative to the Earth once the rocket's mass is reduced to half its mass before ignition?

A. $7x10^3$ m/s
B. $5x10^3$ m/s
C. $4x10^3$ m/s
D. $2x10^3$ m/s

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ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (EUEE)
PHYSICS 2006/2014

DIRECTIONS:- Each of the following questions is following by four possible alternatives. Read each question and carefully.
You may refer to the information given below when you work on some of the question.

Constant	Symbol	Value
Acceleration due to gravity	g	$10m/s^2$
Permittivity of vacuum	ε_0	$8085\times10^{-12}F/m$
Faraday's constant	F	$96500Cml^{-1}$
Charge of an electron	e	1.6×10^{-19} Coulomb
Universal Gravitational constant	G	6.67×10^{-19} Coulomb
Pi	π	3.14159
Density of water	ρ	$1000kg/m^3$
Atmospheric Pressure	P	101.3kPa
Plank's constant	h	6.63×10^{-34} J.s
Electron mass	me	9.11×10^{-31}
Sin 30^0=cos60^0=0.5		
Sin 37^0=cos53^0=0.6		
Sin 45^0=cos45^0=0.707		
Sin 53^0=cos37^0=0.8		
Sin 60^0=cos30^0=0.866		
$\frac{1}{\sqrt{2}} = 0.707$		

1. The distance between two point charges is doubled. What will happen to the force of interaction between the charges?
   A. The force will be doubled.
   B. The force will be reduced by calf.
   C. The force will be reduced by a factor of ¼.
   D. The force will not be affected.
2. What does the slope of a velocity-time graph represent?
   A. Distance
   B. Acceleration
   C. Displacement
   D. Speed
3. Moving 2.5×10^{-6} coulomb of charge from point A to point B in an electric field requires 6.3×10^{-4} joule of work. The potential difference between points A and B is approximately
   A. $4.0 \times 10^{-3}V$
   C. $1.6 \times 10^{-19}V$

B. $2.5 \times 10^{-2} V$ D. $1.0 \times 10^{-14} V$

4. A transverse sinusoidal wave is traveling on a string. Which statement is correct about a point on the string?
   A. The point moves in the same direction as the wave.
   B. The point moves in simple harmonic motion with a different frequency than that of the wave.
   C. The point moves in simple harmonic motion with the same angular frequency as the wave.
   D. The point moves in uniform circular motion with a different angular speed than the wave.
5. Two none zero vectors D and E have precisely equal magnitudes. For the magnitude of D + E to be 3 times larger than the magnitude of D-E, what must be the angle between D and E?
   A. 30^0 B. 37^0 C. 53^0 D. 60^0
6. Which statement is correct about the pressure under a fluid in a container?
   A. It depends on the volume of the fluid.
   B. It depends on the density of the fluid.
   C. It depends on the shape of the container.
   D. It is independent of the depth of the fluid.
7. A capacitor made of parallel plates is charged by connecting it to a battery. The distance between the plates is much less than the surface area of each plate. After fully charged if the plates are disconnected from the battery and then pulled apart a little, what will happen to the total stored energy, the electric field, and the potential difference between the plates?
   A. The electric field between the plates increase.
   B. The potential difference between the plates decrease.
   C. The total energy stored in the capacitor increase.
   D. The electric field, the potential difference and the stored energy remain the same.
8. Which one of the following statements is true about the motion of a particle in a circular path?
   A. The centripetal acceleration is constant if the particle's speed is constant.
   B. The tangential acceleration can be perpendicular to the velocity vector of the particle.
   C. The centripetal acceleration is always in the direction perpendicular to the velocity vector of the particle.

D. The acceleration is always perpendicular to the velocity vector of the particle.

9. Two rectangular loops of wire lie in the plane of this paper as shown in Fig. 1 If the current I in the outer loop is counterclockwise and increases with time, which one of the following statements is correct about the current induced in the inner loop?
   A. A current is induced in the inner loop in the clockwise direction.
   B. A current is induced in the inner loop in the counter wise direction.
   C. The induced current in the inner loop c field inside the area bounded by the inner loop in the same direction of the magnetic field that the outer loop produces in this same area.
   D. It is zero.

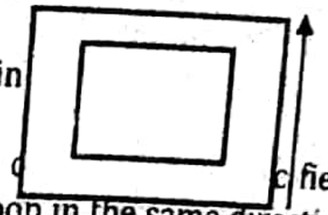

Fig. 1

10. Two men, A and B, are loading identical boxes onto a truck. A lifts his box staring up from the ground to the bed of the truck, whereas B slides his box up a rough ramp to the truck, which statement is correct about the change in the potential energy on the box Earth system?
    A. A and B change the potential energy by the same amount.
    B. A changes more potential energy than B.
    C. B changes more potential energy than A.
    D. None of those statements is necessarily true because the force of friction and the angle of the incline is unknown.
11. Two projectiles are fired from ground level at equal speed but different angles. One is fired at an angle of 30 degrees and the other at 60 degrees. The projectile to hit the ground first will be the one fired at (neglect air resistance)
    A. 60 degrees
    B. 30 degrees
    C. Both hit at the same time.
    D. Cannot be determined from
12. A planet of mass m is circling a star in an elliptical orbital as shown below.
    It its velocity at point A is 60,000m/s and $r_b = 8ra$ what is its velocity at point B?
    A. 7,500m/s C. 60,000m/s
    B. 8,571m/s D. 788,000m/s

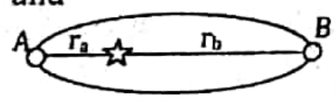

Fig 2

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13. A small radio receiver uses a battery that delivers a constant current of 25mA for 3 hours. Calculate the total charge delivered by the battery.
A. 1080000C B. 270000 C. 540C D. 270C

14. A cup of coffee at a temperature of 80°C is placed on table in a 20°C room. What happens to the thermal energy and molecular motion of the coffee when it is left on the table?
A. The thermal energy decreases and the molecules move more slowly.
B. The thermal energy increases and the molecules move more slowly.
C. The thermal energy decreases and the molecules move faster.
D. The thermal energy increases and the molecules move faster.

15. The vector $2\hat{\imath} + \hat{\jmath} + \hat{k}$ is perpendicular to $\hat{\imath} + 4\hat{\jmath} + \lambda$ if λ is equal to
A. -2 B. -1 C. 2 D. 3

16. A garden hose has and inside diameter of 16cm. The hose can fill a 10 liter bucket in 20 seconds. If the diameter of the nozzle is reduced to 8 cm,
A. Water exists with a speed twice the speed inside the hose.
B. A 10 litter bucked can be filled in 10 seconds.
C. A 20 litter bucket can be filled in 40 seconds.
D. Water exists with a speed half the speed inside the hose.

17. A mass suspended from the end of a spring vibrates up and down 20 times in 36 s. What are the frequency and period of the vibration?
A. 1.5Hz, 0.67S C. 0.0012Hz,864s
B. 0.67Hz, 1.5s D. 864Hz, 0.0012s

18. Determine the total impedance for the circuit shown below

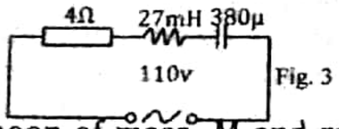

Fig. 3

A. 0 Ω C. 27 Ω
B. 4Ω D. 31 Ω

19. A hoop of mass M and radius R is at rest at the top of an incline plane of height h. The hoop rolls down the plane without slipping. When the hoop reaches bottom, its angular momentum about its center of mass is:
A. Mgh B. $MR\sqrt{2gh}$ C. $MR\sqrt{\frac{gh}{2}}$ D. $MR\sqrt{gh}$

20. What is the resistance of a copper cable if it has a cross-sectional area of $1cm^2$ and a length of 2km? The resistivity of copper is $2 \times 10^{-8}\Omega m$.
A. 0.0004 B. 0.004Ω C. 0.04 Ω D. 0.4 Ω

21. When a load of 500kg is hanging from a steel wire of length 3m and cross sectional area $0.20cm^2$, the wire stretches beyond its no-load length. If the Young's modulus for the wire made of steel is Y=2X 10^{11} Pa, what is the change in length of the wire beyond its no-load length?
A. 3.75mm B. 2.75mm C. 1.75mm D. 0.75

22. As shown in Figure 5, two blocks are connected by a string of negligible mass passing over a pulley of radius r=0.2m and moment of inertia I. The block on the frictionless incline is moving with a constant acceleration $a = 2.0\frac{m}{s^2}$ up along the incline. What is the moment of inertia of the pulley?

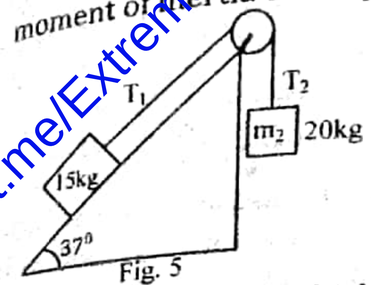

Fig. 5

A. 0.8 kg m²
B. 1.25 kg m²
C. 4 kg m²
D. 40 kg m²

23. A table- tennis ball which is spherical in shape has a diameter of 6 cm and average density of 0.400 g/cm³. What force is required to hold it completely submerged under water?
A. 0.68 N downward C. 1.13 N downward
B. 0.68 N upward D. 1.13 N upward

24. A viewing screen is separated from a double slit by 4.80 m. The distance between the two slits is 0.030 mm. Monochromatic light is directed toward the double slit and forms and interference pattern on the screen. The first dark fringe is 4.50 cm from the center line on the screen. What is the wavelength of the light?
A. 1250nm B. 1875nm C. 2500nm D. 5625nm

25. A horizontal pipe of 10 cm in diameter has a smooth reduction to a pipe of 5cm in diameter. If the pressure of the water in the larger pipe is 80kPa and the pressure in the smaller pipe is 6kPa, at what rate does the water flow through the smaller pipe?
A. 1.63 m/s B. 6.53m/s C. 13.6m/s D. 42.67m/s

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26. An aluminum block of mass m_1=3kg and a copper block of mass m_2=5kg are connected by a light string over a frictionless pulley. They sit on a steel surface as shown in fig 4, where $\theta = 53^0$. The coefficient of kinetic friction between the steel surface and each block in 0.3. If the blocks are released from rest what is the tension T in the string?

A. 29.40 N C. 12.25 N

B. 17.25 N D. 8.60N

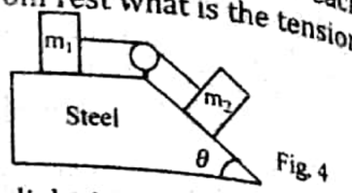

Fig. 4

27. An object of mass 3kg hangs from three light inextensible wires as shown in Figure 6. Two of the wires make angles $\theta_1 = 53^0$ and $\theta_2 = 37^0$ with the horizontal. If the system is in equilibrium, what will be the tensions T_1 and T_2 in the wires?

A. T_1=18N and T_2=24N

B. T_1= T_2=18N

C. T_1= T_2=24N

D. T_1=24N and T_2=18N

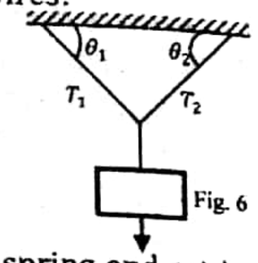

Fig. 6

28. An object of mass m is hung from a spring and set into oscillation. The frequency of the oscillation is measured and recorded as f. If the object of mass m is replaced with an object of mass 2m and set into oscillation, what will be the frequency of the motion?

A. $f/\sqrt{2}$ B. f C. $(\sqrt{2})f$ D. $2f$

29. A placekicker must kick a football from a point which is at a distance of 36.0 m from the goal. When kicked, the ball leaves the ground with a speed of 20.0 m/s at an angle of 37^0 to the horizontal. If the ball hits the crossbar of the goal at a height h and bounces back what will be the height of the crossbar?

A. 2.45m B. 2.85 C. 3.00m D. 3.15m

30. A body moving with constant acceleration covers the distance between two points 60m apart in 5s. Its velocity as it passes the second point is 15m/s. What is the acceleration?

A. 3m/s² B. 2.4m/s² C. 1.8m/s² D. 1.2m/s²

31. Which of the following combinations of properties would be most desirable for a heat sink in a machine?

A. High specific heat and low conductivity.

B. Low specific heat and low conductivity.

C. High specific heat and high conductivity.

D. Low specific heat and low conductivity.

32. A 6V battery is connected across the primary coil of a transformer having 50 turns. If the secondary coil of the transformer has 100 turns, what voltage appears across the secondary?

A. 24.0 V B. 12.0 V C. 6.00V D. 3.00V

33. The change in entropy is zero for:

A. reversible isobaric processes.

B. reversible adiabatic processes.

C. reversible isothermal processes.

D. reversible processes during which no work is done.

34. Two particles with masses 2m and 3m are moving toward each other along the x-axis with the same initial speed v. Particle 2m is traveling to the left, and particle 3m is traveling to the right. They undergo an elation glancing collision such that particle 2m is moving in the negative y direction after the collision. What are the x component of the final velocity of particle 3m and the kinetic energy of particle 2m, respectively?

A. 0.33v and 0.7mV² C. 0.67v and 0.71mV²

B. 0.33v and 1.41mV² D. 0.67v and 1.4mV²

35. Two separate coherent sources produce waves whose wavelengths are 0.10m. Which statement is correct about points that are 3.2m away from one source?

A. Constructive interference occurs if the points are 3.0m from the other source.

B. Constructive interference occurs if the points are 3.05m from the other source.

C. Constructive interference occurs if the points are 2.95m from the other source.

D. Destructive interference occurs if the points are 3.2m from the other source.

36. A coil formed by wrapping 50 turns of wire in the shape of a square is positioned in a magnetic field so that the normal to the plane of the coil makes an angle 37^0 with the direction of the field. When the magnetic field is increased uniformly from 200 mT to 600mT in 4.00s, an emf of magnitude 160.0 mv is induced in the coil. What is the total length of the wire in the coil?

A. 100m B. 200m C. 300m D. 400m

37. The half-life of radium 224 is 3.5 days. What fraction of the sample

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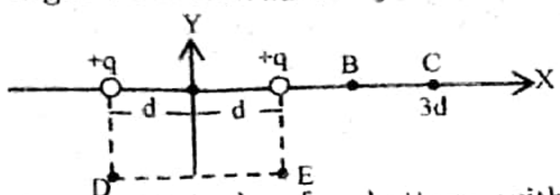
remains undecided after fourteen days?
(If needed use ln2=-ln(1/2)≈0.693)
A. 1/2 B. 1/4 C. 1/8 D. 1/16

38. Two positive charges of magnitude q are each a distance d from the origin A of a coordinate system as shown below



The magnitude of the electric field is minimum at point .
A. A C. C
B. B D. D

39. If the terminals of a battery with zero internal resistance are connected across two identical resistors in series, the total power delivered by the battery is 8 W. If the same is connected across the same resistors in parallel, what is the total power delivered by the battery?
A. 2W B. 4 W C. 16W D. 32W

40. Wire B has twice the length and twice the radius wire A. Bothe wires are made from the same material. If wire A has a resistance R, what is the resistance of wire B?
A. 4R B. 2R C. R/2 D. R/4

41. The sound level at a distance of 4.00 m from a source is 120dB. At what distance is the sound level 100 dB?
A. 0.40m B. 4.80m C. 40m D. 48m

42. An ideal gas is at a temperature of 300k. If we wish to double the root mean square (rm.s) speed of the molecules of the gas, to what value must we raise its temperature?
A. 450 B. 600k C. 800k D. 1200k

43. Suppose that the index of refraction of a soap – bubble film is 1.25, what is the minimum thickness of the soap-bubble film that results in constructive interference in the reflected light if the film is illuminated with light whose wavelength in free space is $\lambda = 600nm$?
120nm B. 150nm C. 300nm D. 450nm

44. A wire carries a steady current of 2.4 A. A straight section of the wire is 0.75m long and lies in xy-plane along the x axis within a uniform magnetic field, B=1.6kT. If the current is in the positive x direction, what is the magnetic force on the section of wire?
A. 2.88N, in the negative y direction.
B. 2.88N, in the positive y direction.
C. 3.84 N, in the negative y direction.

D. 3.84 N, in the positive y direction.

45. Two different capacitors of capacitances $C_1 = 2\mu F$ and $C_2 = 3\mu F$ connected in series across a 120V supply line. The charged capacitors are disconnected from the line and from each other, and reconnected with terminals of like sign together. What will be the final charges on C_1 and C_2 respectively?
A. $57.63\mu C$ and $86.4\mu C$ C. $144\mu C$ across each
B. $115.2\mu C$ and $127.8\mu C$ D. $288\mu C$ across each

46. A glass flask whose volume is exactly 1,000cm³ at 0°C is totally filled with 1000 cm³ of mercury at this temperature. when flask and mercury are heated to 100°C, 15.2cm³of mercury overflow. If the coefficient of volume expansion of mercury is 1.82×10^{-4} per Celsius degree, what should be the coefficient of volume expansion of the glass?
A. $1.52 \times 10^{-4}/C^0$ C. $3.00 \times 10^{-4}/C^0$
B. $1.82 \times 10^{-4}/C^0$ D. $1.00 \times 10^{-5}/C^0$

47. A heat engine is being designed to have a Carnot efficiency of 65.0% when operating between two energy reservoirs. If the temperature of the cold reservoir is 20.0°C, what must be the temperature of the hot reservoir?
A. 57^0C B. 310^0C C. 564^0C D. 807^0C

48. In Figure 8, the current in the 16Ω resistor is :
A. 0.64 A
B. 0.15 A
C. 0.49 A
D. 1.28 A

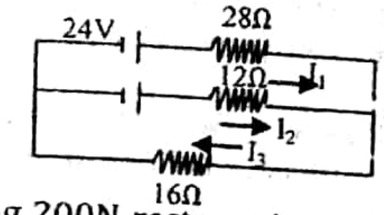


Fig. 8

49. A 10.0 m uniform ladder weighing 200N rests against a frictionless wall. The ladder makes 53⁰ angles with the horizontal. if the ladder is just on the verge of slipping when a 600.0N firefighter is 6.00m along the ladder from the bottom, what is the coefficient of static friction between the ladder and ground?
A. 0.3240 B. 0.4313 C. 0.5750 D. 0.7667

50. A block of mass 1.6 kg is attached to a horizontal spring that has a force constant of 200N/m. The spring is compressed 2.0 cm and is then released from rest. At what speed will the block pass through the equilibrium position if a constant friction force of 10.0N retards its motion from the moment it is released?
A. 70m/s B. 50m/s C. 70m/s D. 50cm/s

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1. Solution
The SI unit of Electric field is N/C.
Answer: C

2. Solution:-
The unit vector perpendicular to both vectors $\vec{A}$ *and* $\vec{B}$ is obtained as:-

$$\vec{n} = \frac{A \times B}{|A \times B|}, \quad \text{Where:- } A \times B = 2i - 2j + 2k \ \&$$

$$\vec{n} = \frac{1(2i - 2j + 2k)}{2\sqrt{3}} \qquad |A \times B| = \sqrt{2^2 + 2^2 + 2^2} = \sqrt{12} = 2\sqrt{3}$$

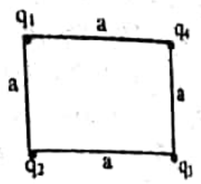
$$\vec{n} = \frac{1(i - j + k)}{\sqrt{3}}$$

Answer: D

3. Solution:
The cross product of two similar vectors is zero. Because the angle between them is zero & $\sin\theta^0 = 0$.
Hence choice 'B' not correct.
Answer: C

4. Solution
Let the charges be $q_1 = q_2 = q_3 = q_4 = q$ as shown below. To calculate the EPE of the system of four charges you need to calculate and add the work done to bring each charge from infinity to the place where each is found.
Hence: EPE = $W_1 + W_2 + W_3 + W_4$
But W_1 = work done to bring 'q_1' from infinity is zero

$$W_2 = v_1 q_2 = \frac{kq_1 \cdot q_2}{a} = \frac{kq^2}{a}; v_1 \equiv \text{potential of '}q_1\text{' where '}q_2\text{' is found.}$$



$$W_3 = (v_1 + v_2)q_3$$
$$W_4 = (v_1 + v_2 + v_3)q_4$$
Finally we get EPE = $W_1 + W_2 + W_3 + W_4 =$

$$\frac{kq^2}{a}(4 + \sqrt{2}) = \frac{\sqrt{2}kq^2}{a}(2\sqrt{2} + 1)$$

Answer: C

5. Solution

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By the law of conservation of energy the gravitational potential energy of the hoop will be into its translational and rotational kinetic energies at the bottom.

$\rightarrow GPE = \frac{1}{2}mv^2 + \frac{1}{2}Iw^2$, For a hoop, $I = mr^2$

$\rightarrow mgh = \frac{1}{2}mv^2 + \frac{1}{2}mr^2w^2$,

$\rightarrow 2gh = 2v^2$ since $v = wr$

$\rightarrow v = \sqrt{gh}$

$\rightarrow$ At the bottom angular momentum $\equiv Iw = mrv = mr\sqrt{gh}$

Answer: A

6. Solution:
The magnetic force on the unit length of the wire is equal to the vector product given by:-

$$F = l \times B$$
$$= 3A\frac{1}{\sqrt{3}}(i + j + k) \times (0.75i + 0.4k)$$
$$= \left(\frac{3}{\sqrt{3}}i + \frac{3}{\sqrt{3}}j + \frac{3}{\sqrt{3}}k\right) \times (0.75i + 0.4k)$$
$$F = (0.69i + 0.61j - 1.3k)\text{N/m}$$

Answer: A

7. Solution:
A damped oscillation is that oscillation which continuously looses power and oscillates with decreasing amplitude until it comes to a stop.
Answer: A

8. Solution:
Let 't' be that instant of time at which the velocity of the object is 9.6m/s.
This means: $9.6\text{m/s} = v_i + 3.2t$
Assuming the initial velocity of the body to be v_i the velocity 2.5sec earlier is given by:-

$V_1 = V_i + 3.2(t + 2.5)$ .....(i)

And the velocity 2.5sec later is given by:-

$V_2 = V_i + 3.2\,(t + 2.5)$.....(ii)

Substitute $V_i = 9.6\text{m/sec} - 3.2t$ in equations (i) and (ii).

Then solving for V_2 & V_1 will give you $V_1 = 1.6m/s$ & $V_2 = 17.6$. The ration $\frac{v_2}{v_1} = 11$

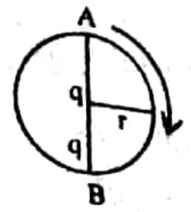
Answer: B

9. Solution:

The work done on the charge 'q_0' in moving it from 'A' to 'B' is given by:-

$W = (V_A - V_B)q_0$. But the potential due to 'q' at 'A' & 'B' are equal. Hence $V_A - V_B = 0$

Therefore the work done is zero



Answer: D

10. Solution:

The rate of heat flow is calculated using the relation:-

$$\frac{\Delta O}{\Delta t} = KA\frac{\Delta T}{\Delta L}$$

$$= \left(0.04\frac{w}{mk}\right)(1.8m^2)\left(\frac{32^0 c}{10^{-2}m^2}\right)$$

$$\frac{\Delta O}{\Delta t} = 23.04w$$

Answer: B

11. Solution:

The work done on the particle is given by:-

$W = F \cdot S$, where $F = m \cdot a$

$W = m \cdot a \cdot s$, where $a = \frac{d^2x}{dt^2} = -8m/\sec^2$

$= (3kg)(-8m/s^2)(3t - 4t^2)$,

$= (3kg)(-8m/s^2)(12 - 64)$meter, $t = 4$sec

$W = 1248$Joule

Answer: C

12. Solution:

For a purely inductive AC circuit, the self inductance and the inductive reactance are related by:-

$X_L = 2\pi fL = (2\pi)(40Hz)(35mH)$

$X_L = 8.79\Omega$

And the rms current:- $I_{rms} = \frac{V_{rms}}{X_L} = \frac{165v}{8.79\Omega} = 18.77A$

Answer: A

13. Solution:

Let the initial and final voltages be V_1=80v and V_2 = 110v. If the charge on the capacitor changes by 9×10^{-5}C means $Q_2 - Q_1 = 9\times10^{-5}$C. Since the capacitance 'C' of the capacitor is constant we can write:-

$$C = \frac{Q_1}{v_1} \& C = \frac{Q_2}{v_2}$$

$$\Rightarrow \frac{Q_1}{v_1} = \frac{Q_2}{v_2}$$

$$\Rightarrow \frac{Q_2 - 9\times10^{-5}C}{80v} = \frac{Q_2}{110v}$$

$\Rightarrow 110Q_2 - (110v)(9 \times 10^{-5}C) = 80Q_2$

$\Rightarrow 30Q_2 = 990 \times 10^{-5}$C

$Q_2 = 330\mu c$

Finally:- $C = \frac{Q_2}{v_2} = \frac{330\mu c}{110v} = 3\mu F$

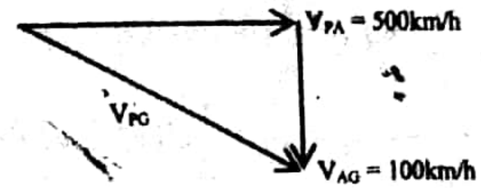
Answer: B

14. Solution:

The motions of the plane and the wind are shown in the figure here.

The velocity of the airplane relative to the ground V_{PG} is equal to the vector some of V_{PA} and V_{AG}

That is: $V_{PG} = V_{PA} + V_{AG}$



$$|V_{PG}| = \sqrt{(V_{PA})^2 + (V_{AG})^2} = \sqrt{(500km/h)^2 + (100km/h)^2}$$

$$= 100\sqrt{26}km/h, \text{ south of east}$$

$$|V_{PG}| = 510km/h, \text{ south of east}$$

Answer: B

15. Solution:

Since there is no charge inside the spherical shell the electric field is zero inside, where $r < R_1$. But at a distance $r > R_2$, the electric field is the same as that due to a point charge 'Q' located at the center of the sphere. Hence

$$Q = \frac{q}{4\pi \in_0 r^2}$$

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Answer: A

16. Solution:-

Use bernoulli's equation:-

$$\rightarrow p_1 + \frac{1}{2}dv_1^2 + dgh_1 = p_2 + \frac{1}{2}dv_2^2 + dgh_2$$

$$\rightarrow p_2 = p_1 + \frac{1}{2}d(v_1^2 - v_2^2) + dg(h_1 - h_2),\ but\ v_2 = \frac{A_1 v_1}{A_2} = \frac{9}{4}v_1$$

$$\rightarrow p_2 = 75kpa + 500kg/m^3(v_1^2 - \frac{81}{16}v_1^2) + (-200{,}000pa)$$

$$\rightarrow p_2 = 25.8kpa$$

No answer is given

17. Solution:

Using the equation of continuity we get the velocity at the 2nd floor as:-

$A_1V_1 = A_2V_2$

$$\Rightarrow V_a = \frac{A_1V_1}{A_2},\ but\ \ A_1 = 2A_2$$

$$\Rightarrow V_2 = 12m/s$$

To get the pressure at the 2nd floor we use bernoulli's equation:-

$$\rightarrow p_1 + \frac{1}{2}dv_1^2 + dgh_1 = p_2 + \frac{1}{2}dv_2^2 + dgh_2,\ d \equiv density$$

$$\rightarrow p_2 = p_1 + \frac{1}{2}d(v_1^2 - v_2^2) + dg(h_1 - h_2)$$

$$\rightarrow p_2 = 1.66 \times 10^5\ N/m^2$$

No answer is given

18. Solution:

The problem is as shown in the figure

The two blocks move with the same acceleration. Hence we can write Newton's 2nd for each block as:-

$F_1 = m_2g - T = m_2a$ .....(1)

$F_2 = T - m_1g = m_1a$ .....(2)

Solving (1) & (2) simultaneously will give us, the value of 'T' & 'a' to be $T = \frac{12g}{5}$ & $a = \frac{g}{5}$

Answer: A

19. Solution:

The resistance of the copper wire is directly proportional to the length and inversely to the area of the wire.

Hence when the wire increases in length resistance increases. When the cross sectional area decreases the resistance again increases. There is no change in the resistively of the wire. So resistance increases and resistively remains the same.

Answer: C

20. Solution:

According to the relationship $I = \frac{\epsilon}{R+r}$, the value of 'I' becomes maximum if the external resistor 'R' has value less than the internal resistor. That is R < r.

Answer: C

21. Solution:

If the blocks do not move the system is all in equilibrium.

This means:- The tensions in the rope is equal to the weight of mass B. That is T = 50N. And also the forces 'F_f' and 'T' on mass 'A' should also be equal.

Therefore:- $F_f = T$

$\rightarrow F_N \cdot \mu = T$

$\rightarrow (m_A + m_C)g\mu = 50$

$\rightarrow m_Ag\mu + m_Cg\mu = 50$

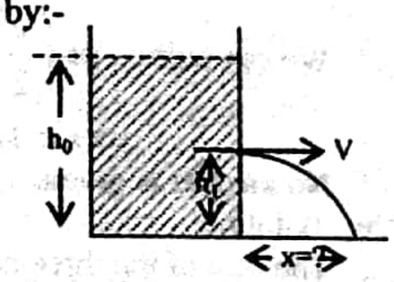
$$m_C = \frac{50N - m_Ag\mu}{g\mu} = \frac{50N - 20N}{2N/kg} = 15kg$$

Answer: A

22. Solution:

The velocity of efflux of the liquid is given by:-

$$v = \sqrt{2g(h_0 - h)}$$



To get 'x' we use the Kinematic equation.

x = v.t, where $t = \sqrt{\frac{2h}{g}}$

Then:- $x = v \cdot t = \sqrt{2g(h_0 - h)} \times \sqrt{\frac{2h}{g}}$

$$x = \sqrt{4h(h_0 - h)} = 2\sqrt{h(h_0 - h)}$$

Answer: D

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23. Solution:
As the sphere 'A' collides with sphere 'B', it transfers all of its GPE to the GPE of the combined mass of spheres A and B. that is:-

GPE(sphere A) = GPE(combined mass)

$\rightarrow mgh_0 = (m + 3m)gh$

$\rightarrow mgh_0 = 4mgh$

$\rightarrow h = \frac{h_0}{4}$

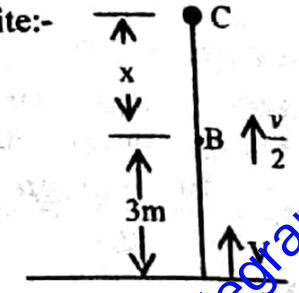
The combined mass will be raised to one-fourth of the original height h_0.

Answer: C

24. Solution:
The Expression that represent the net force is –i -2j. This is obtained by considering the x and y components of the force vector in the second quadrant shown in the figure and the adding them.

Answer: A

25. Solution:
Taking the velocities of 'B' and 'A' on the way up and using the formula $v_f^2 - v_i^2 = 2as$ we can write:-



$\Rightarrow \left(\frac{v}{2}\right)^2 - v^2 = -2gh$

$\Rightarrow \frac{v^2}{4} - v^2 = -6g$

$\Rightarrow v = \sqrt{8g} = 8.94 m/s$

Since at 'C' [maximum point] $v_f = 0$. Hence

We can write:- $\Rightarrow \left(\frac{v}{2}\right)^2 = -2gx$

$\Rightarrow x = 1m$

No answer is given.

26. Solution:
The sum of the three capacitors in series is

$$\frac{1}{C_T} = \frac{1}{8} + \frac{1}{12} + \frac{1}{24} \rightarrow C_T = 4\mu F.$$

The sum of the three capacitors in parallel is
$C_T = C_1 + C_2 + C_3 = 44\mu F$.

Then the ration, $\frac{C_T(series)}{C_T(parallel)} = \frac{4\mu F}{44\mu F} = \frac{1}{11}$ or 1:11

Answer: A

27. Solution:
At resonance it is the rms voltage at the inductor and capacitor combination which is zero. But the rms voltage at the resistor is not zero. Also at resonance the correct amplitude is $I = \frac{V}{R} = 5.75A$ and not 8.1A. Hence both 'C' and 'D' are not correct.

28. Solution:
Take any two points on the graph to find the slope. The slope is equal to the acceleration of the body. Take points (0, 15) and (5, 1).

The slope $= a = \frac{1m/s - 15m/s}{5.sec - 1sec} = -2.8 m/sec^2$

In the given two seconds the moving body has v_i = 15m/s & v_f = 9m/sec. Now we can use:-

$$s = \frac{v_f^2 - v_i^2}{2a} = \frac{(9m/s)^2 - (15m/s)^2}{-5.6m/s^2} = 26m.$$

Answer: B

29. Solution:
Let the new force after the slow increase be F_n. Just before the block is lifted with acceleration, the vertical component of F_n most be equal to the weight of the block.

That is:- $F_n \sin 30 = 50N$

$F_n (0.5) = (50N)$

$F_n = 100N$

The new force is 100N and the acceleration is:-

$$a = \frac{F_n}{m} = \frac{100N}{5kg} = 20N/kg$$

Answer: B

30. Solution:
The initial electric energy stored in the capacitor is $E_1 = ½c v_1^2$. the final electric energy is $E_2 = E_1 + 10\%E_1$.

This means:- $E_2 = E_1 + 0.1E_1 = 1.1E$

$E_2 = (½c v_1^2)(1.1) = ½c(1.1 v_1^2)$

$\rightarrow E_2 = (½c v_2^2)$ means $v_2 = \sqrt{1.1} v_1$

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Then the percent increase in the potential difference is:-

$$\%\ \text{increase} = \left(\frac{v_2 - v_1}{v_1}\right)\times 100\% = \left(\frac{\sqrt{1.1}v_1 - v_1}{v_1}\right)\times 100\%$$

$$= (\sqrt{1.1} - 1)\times 100\% = 4.88$$

Answer: A

31. Solution

The induced emf in the circuit can be calculated using the formula:

$$Emf = \frac{\Delta\Phi}{\Delta t} = \frac{\Delta BA}{\Delta t}$$

$$\rightarrow = \frac{B\Delta A}{\Delta t}, \quad 'B'\ is\ const\ ant$$

$$= B\frac{dA(t)}{dt}$$

$$= B\frac{d}{dt}(A_0 + bt)$$

$$= Bb = (0.4T)\left(10^{-6}m^2/\sec\right)$$

$$Emf = 4\times 10^{-7}v$$

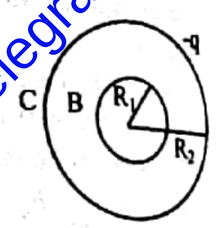
Answer: D

32. Solution:

At point 'A' for $r < R_1$ the field is $\frac{q}{4\pi\epsilon_0 r^2}$

At point 'B' (inside the shell) the field is zero.

At point 'C' for $R_1 < r < R_2$, since the total net change is zero the field is also zero.



Answer: C

33. Solution:

Use the formula $\epsilon = I(r + R)$

For the solar cell, in the first case, $R = 500\Omega$ & $V = IR = 0.1v$. this means:-

$$\epsilon = I(r + R)$$

$$= \frac{v}{R}(r + R), \qquad V = IR = 0.1v$$

$$\epsilon = \frac{0.1A}{500}(r + 500\Omega) \ldots\ldots\ldots\ldots\ldots(1)$$

In the second case, $R = 1000\Omega$ and $V = 0.15v$.

This means:- $\epsilon = I(r + R)$

$$= \frac{v}{R}(r + R)$$

$$\epsilon = \frac{0.15A}{1000}(r + 1000\Omega) \ldots\ldots\ldots\ldots\ldots(2)$$

Now equate (1) and (2) and solve for r:-

$$\rightarrow \frac{0.1A}{500}(r + 500\Omega) = \frac{0.15A}{1000}(r + 1000\Omega)$$

$$\rightarrow 0.0002r + 0.1v = 0.00015r + 0.15v$$

$$\rightarrow 0.0002r - 0.00015r = 0.15v - 0.1v$$

$$\rightarrow 0.00005r = 0.05v$$

$$\rightarrow r = 1000v$$

Substituting 'r' in (1) will give you $\epsilon = 0.3v$.

Answer: C

34. Solution:

The electric potential for a hollow metal sphere inside the sphere is the same as the potential on its surface.

Answer: B

35. Solution:

If the three vectors are not coplanar, their sum can not be zero. This is true. The choices A, B & C are not correct.

Answer:- D

36. Solution:

The acceleration due to gravity on the surface of any planet is given by the relation:-

$$\rightarrow g_p = \frac{Gm_p}{r_p^2}, \qquad m_p = mass\ of\ planet\ \&\ r_p = radius\ of\ planet$$

$$\rightarrow g_p = \frac{G2m_e}{(3r_e)^2}, \qquad since\ \ m_p = 2m_e\ \&\ r_p = 3r_e$$

$$\rightarrow g_p = \left(\frac{Gm_e}{r_e^2}\right)\left(\frac{2}{9}\right), \quad but \quad \frac{Gm_e}{r_e^2} = g$$

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Hence $\Rightarrow g_p = \frac{2}{9}g$. **Answer:- C**

37. Solution:
All derived physical quantities are not vectors. Density is a derived physical quantity but it is not a vector quantity. Hence 'D' is false.
Answer: D

38. Solution:
The acceleration of the particle is equal to the second derivative of its position function.

That is:- $\rightarrow \vec{a}_x = \frac{d^2x(t)}{dt^2} = \frac{d^2(7t+4t^2)}{dt^2} = 8m/s^2$

$\rightarrow$ In the direction of 'y' axis the derivative is zero and so there is no acceleration in the 'y' axis.

Answer: B

39. Solution:
The earth is considered practically as a huge charge reservoir (storage) in which a charge can be either given or removed from it. This means its net charge is practically zero and any object grounded (Earthed) has a zero potential.
Answer: B

40. Solution:
If both masses are pushed with the same force (F) for the same period of time (t = 1sec) then by the formula of impulse I = F.t, the impulse delivered to the masses (particles) are the same.
Answer: C

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1. **Solution:** The magnetic field generated by a current in a long straight wire circles the wire. The magnitude of the magnetic field $\vec{B}$ of the wire at a point 'r' away from the center is given by:- $B = \frac{\mu_0 I}{2\pi r}$

Answer D.

2. **Solution:** In an LRC series circuit, the average power delivered to the entire circuit is given by:-

$P_{av} = V_{rms} I_{rms} \cos\phi$, where $V_{rms} = 50_v$, & $I_{vms} = \frac{V_{rms}}{Z}$

Remember that: $Z = \sqrt{R^2 + (X_L - X_C)^2}$ where $X_L = WL$ & $X_C = \frac{1}{wc}$

Then: $P_{av} = V_{rms} I_{rms} \cos\phi$, where $\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = 53^0$

$= (50V)(0.1A)\ Cos\ (53)$

$P_{av} = (50V)(0.1A)(0.6) = 3w$

No answer is given

3. **Solution:** The vector product of two non-collinear vectors $\vec{A}$ and $\vec{B}$ is perpendicular to the plane formed by $\vec{A}$ and $\vec{B}$.
Answer A.

4. **Solution:** The temperature of a body measures the average KE of the molecules of a body.
Answer B.

5. **Solution:** The initial separation $r_1 = 2m$ & the initial repulsion is $F_1 = 2N$.

Hence by coulomb's in w you can write

$F_1 = \frac{kq_1 q_2}{r_1^2} = \frac{kq_1 q_2}{4m^2} = 2N$

If the separation is doubled, i.e. $r_2 = 2r_1$, then

$F_2 = \frac{kq_1 q_2}{r_2^2} = \frac{kq_1 q_2}{4r_1^2}$ $\qquad F_2 = \frac{kq_1 q_2}{r_2^2}\left(\frac{1}{4}\right) = \frac{F_1}{4}$

Answer A.

6. **Solution:** The terminal voltage of a cell supplying energy to a circuit is usually less than its emf. This is because some amount of energy is wasted inside the cell because of the cell's internal resistance

Answer B.

7. **Solution:** The peak (maximum) value and the effective value are related by: $V_m = V_{ef} \times \sqrt{2}$

$$\Rightarrow V_{ef} = \frac{V_m}{\sqrt{2}} = \frac{141}{\sqrt{2}} v = 100v$$

Answer B.

8. **Solution:** Here $I_{rms} = \frac{V_{rms}}{R}$, but $V_{rms} = \frac{V_m}{\sqrt{2}}$

Since $V_m = 300V$ then $I_{rms} = \frac{V_m}{\sqrt{2}\,R} = \frac{300v}{100\sqrt{2}\Omega}$

$I_{rms} = 2.12A$ **Answer B.**

9. **Solution:** The distributive property of vector multiplication over addition shown in choice 'B' is not correct. The others are correct
Answer B.

10. **Solution:** The power delivered to the particle is equal to the rate of change of the particle's KE.

That is: $P = \frac{\Delta KE}{t}$, $V_i = 0$ & $a = \frac{V}{t}$

By the second law:- $F = ma = \frac{mv}{t}$

Then $P = \frac{F^2}{2m}t$, (after substitution)

Answer B.

11. **Solution:** Metals are good conductors of heat not because they have large number of electrons. It is because there are free outer most valence electrons from each atom of the metal.
Answer D.

12. **Solution:** The total capacitance between terminals 'a' and 'b' is calculated as follows

The sum of the capacitors is the middle branch is $\frac{c}{2}$. The sun of the capacitors in the bottom branch is $\frac{c}{3}$. The sun of the capacitors in the three branches connected in parallel is calculated as:-

$$C_T = C + \frac{C}{2} + \frac{C}{3} = \frac{11}{6}C$$

Finally the total capacitance between 'a' and 'd' is equal to:

$$\frac{1}{11/6\,C} + \frac{1}{11/5\,C} = \frac{1}{C_f} \Rightarrow C_f = 1C$$

Answer B.

13. **Solution:** Assume the athlet starts from rest. Then he accelerates at $1.9m/s^2$ for only 2.2sec.
His velocity after 2.2sec is:-

$$v_f = v_0 + at$$
$$= 0 + (1.9m/s^2)(2.2sec)$$
$$v_f = 4.18m/s$$

since there is no acceleration after 2.2sec, even at the end of the 3000m his velocity is 4.18m/s,
Answer: C

14. **Solution:** The centripetal acceleration of a rotating body is given by the formula:-

$$a_c = \frac{v^2}{r} = w^2 r$$

A body completes 12 rotations in 3 seconds, means it has an angular velocity equal to

$$w = \frac{12rev}{3sec} = \frac{4rev}{sec} = 8\pi rad/sec$$

Hence $a_c = w^2 r = 64\pi^2 m/s^2$
Answer: B

15. **Solution:** If 0 car moves around a circular road with constant speed, because of the change in direction there is a change in velocity. The change in velocity results in the centripetal acceleration. And that is the centripetal force per unit mass of the car given by v^2/r.
Answer: D

16. **Solution:** The system of the traffic lights is in equilibrium. You can apply the first condition of equilibrium at the junction point of the ropes.

This is:- $\Sigma F_x = 0 = T\cos(30) + -(T\cos 30) = 0$ [Horizontal forces cancel each other]

$\rightarrow \Sigma F_y = 0 = T\sin 30 + T\sin 30 + (-250N) = 0$

$2T\sin 30 = 250$

$T = 250N$

Answer: D

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17. **Solution:** The period of oscillation of a particle undergoing SHM is independent of the amplitude of motion (oscillation)
Answer: A

18. **Solution:** Young's modulus is the ratio of tensile stress to strain.
Answer: A

19. **Solution:** Materials that we want to use for thermal insulation should have low thermal conductivity and high specific heat.
Answer: A

20. **Solution:** The capacitance of a capacitor with a dielectric is given by the formula:- $C = \frac{k \in_0 A}{d}$

As the dielectric was air) $k_a = 1$ and

$$C_1 = 1PF = \frac{\in_0 A}{d} \quad \text{.....(1)}$$

As the dielectric is wax we write:-

$$C_2 = k_w \frac{\in_0 A}{2d} = 2PF \quad \text{.....(2)}$$

Combining the two equations we get that:-

$$\Rightarrow k_w = \frac{(4PF)d}{\in_0 A}, \qquad but \quad \in_0 A = (1PF)d$$

$$\Rightarrow k_w = \frac{(4PF)d}{(1PF)d} = 4.0$$

Answer:- B

21. **Solution:** At t = 0 before the switch is closed the current is also zero. After some time the current increases above zero when the switch is closed. Then after it will not be zero as long as the switch is on. It increases like the graph in choice B.
Answer:- B

22. **Solution:-** If the direction of motion of a charged particle is parallel to the magnetic field then there will be no magnetic force exerted on the particle. Hence the path of the charged particle is a straight line.
Answer:- B

23. **Solution:** The emf that is created in the axle is a motional emf. This happens when a metal (axle) crosses a magnetic field of the earth. The axle should cut the magnetic field of the earth. This means the earth most have a vertical magnetic component.
Answer:- D

24. **Solution:** The transformer has two coils in both of which magnetic field changes. The increase or decrease of potential difference by the transformer functions with the principle of mutual induction.
Answer: B

25. **Solution:** The average power dissipated in the circuit is given by:-
$P_{av} = \left(I_{rms}^2\right)(R)$
$P_{av} = (2A)^2(5\Omega) = 20watt$
Answer: C

26. **Solution:** With the given initial velocity 4m/s and acceleration $3.2m/s^2$, the car will attain a velocity of $v_f = v_0 + at = 36m/s$ after 10seconds. If the brakes are applied on the car, it will come to a stop after moving a distance of 20m as stated.

This means:- $a = \frac{v_f^2 - v_i^2}{2s}$

Where: $v_i = 36m/s$, $v_f = 0$ and $s = 20m/s$.

That after substitution you will get the acceleration to be
$a = 32.4m/s^2$
Answer: D

27. **Solution:** If the box is pushed up the inclined plane with constant speed, the forces along the inclined plane on the box should be equal. The component of the force 'F' that pushes the box up is 'F cos 30". It is equal to the weight component down the inclined plane which is equal to mg sin 30.
Hence:- $F\cos(30) = mg\sin(30)$

$$F = \frac{mg\sin(30)}{\cos(30)} = mg\tan(30)$$

$$F = 577.35N$$

Answer: A

28. **Solution:-** According to the properties of the horizontal throw, the time taken to fall vertically 4m down is the same as the time taken to cover the horizontal distance 5m.

Hence we can use:- $h = \frac{1}{2}gt^2$

$$\Rightarrow t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{8m}{10m/s^2}} = \frac{2}{\sqrt{5}} \text{ sec}$$

Answer: D

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29. **Solution:** The boat and the river are moving in the opposite direction. The child and the boat are in the same direction. The given quantities are $V_{br} = 15km/h$, $V_{rg} = 6km/h$ and $V_{cb} = 5km/h$.

Then for the boat and river we have:-

$$\rightarrow V_{br} = V_{bg} = V_{rg}$$
$$\rightarrow V_{bg} = V_{br} - V_{rg}$$
$$= 9km/h$$

For the child and boat we have:-

$$\rightarrow V_{cb} = V_{cg} - V_{bg}$$
$$\rightarrow V_{cg} = V_{cb} + V_{bg}$$
$$= 14km/h$$

Answer: D

30. **Solution:** The work done on the body is equal to the scalar product of the force and the displacement.

That is:- $W = \vec{F} \cdot \vec{S}$

But:- $\vec{F} = -i + 2j + 3k$ and

$$\vec{S} = \vec{S}_f - \vec{S}_i = (1 \cdot 2 - 0)i + (2 - 1)j + (1.5 - 1.5)j$$

$$\vec{S} = 1.2i + j$$

Then: $W = \vec{F} \cdot \vec{S} = F_x S_x + F_y S_y + F_Z S_Z$

$$W = -1.2 + 2 + 0 = 0.8J$$

Answer: B

31. **Solution:** Given angular displacement (θ) to get the angular velocity (ω) we take derivative of 'θ' with respect to 't' as follows.

$$W = \frac{d\theta}{dt} = \frac{d}{dt}(5 + 10t + 2t^2) = 4t + 10$$

At t = 3sec w(t = 3sec) = 4(3) + 10 = 22 rad/sec

To get acceleration (α) we take the 2nd derivative of the displacement as follows.

$$\alpha = \frac{d^2\theta}{dt^2} = \frac{d^2(5 + 10t + 2t^2)}{dt^2} = 4 rad/s^2$$

Answer: A

32. **Solution:** Torricelli's theorem states that the speed of the liquid coming of out of the tank filled to a height of 'h' above the opening is exactly the same as if the liquid has fallen through the same height like a freely falling stone hence the velocity is given by:- $V = \sqrt{2gh}$

Answer: A

33. **Solution:** The gas is compressed and so work is done on it and $W = P\Delta V = -480J$. Heat is taken out of the gas and so $Q = -400J$. By the first law of thermodynamics we have:-

$$\Delta Q = \Delta u + w$$
$$\rightarrow \Delta u = \Delta Q - w$$
$$= -400J - (-480J)$$
$$\Delta u = 80J$$

No answer is given

34. **Solution**: The electric field is directed upward. This means the electric force on a positive change is also upward. For the charge to float in the field, the upward electric force 'F_e' must be the same as the downward gravitational force on the charge.

That is:- $\rightarrow F_e = F_g$

$$\rightarrow E_q = m_g$$

$$\rightarrow m = \frac{E_q}{g} = \frac{610\,N/C \times 24 \times 10^{-6} C}{10\,N/kg}$$

$$\rightarrow m = 1.484gm$$

Answer: B

35. **Solution**: The electric field just outside the conductor is not tangent to surface of the conductor. The field is perpendicular to the surface. If the field is tangent there will be a component of the field in the conductor which can not be true

Answer: D

36. **Solution**: In reality there is an internal resistance of a source of emf. Hence terminal voltage and emf can not be equal.

Answer: D

37. **Solution:** Initially $R_1 + R_2 = 7\Omega$. (In series)

Next $\frac{R_1 R_2}{R_1 + R_2} = \frac{10}{7}\Omega$ (In Parallel)

Solving the above two equations simultaneously you will get:

$R_1 = 5\Omega$ and $R_2 = 2\Omega$

Answer: C

38. **Solution**:- As can be seen from the diagram, the 'z' component of the vector $\vec{A}$ is equal to $A_Z = |A| \cos 60^0 = 50m$. the projection of vector $\vec{A}$ on the xy plane Ap is known to be 85.5m. to get A_x & A_y we decompose Ap using the angle 30^0.

Hence:- $Ax = Ap \cos 30^0 = (88.6)(0.866) \cong 75m$

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$Ay = Ap \sin 30 = (8.6.6)(0.5) = 43.3m$

Answer: C

39. **Solution**:- If net force is zero then acceleration is zero. This means velocity does not change. As a result KE does not change with time. But the position of the c.m. of the particles may very with time.

Answer: D

40. **Solution**:- From the graph we see that the force is variable. Hence the area under the F – t graph is equal to the impulse (I) or the change in momentum.

$$\text{Area} = \frac{1}{2}Ft = \Delta p$$

$$\frac{1}{2}Ft = p_f - p_i, \text{ but } p_i = 0$$

$$\Rightarrow p_f = \frac{1}{2}(20N)(4sec) = 40$$

Answer: D

41. **Solution**:- If the diameter of the hose is reduced it should be 8mm. using equation of continuity you can get the speed of water in the reduced nozzle is 2 litter/sec. this means the speed of water is four times as it exits.

No answer is given

42. **Solution**: When a material like steel is stretched beyond its elastic (yield) point it becomes plastic. This means the steel does not regain its original shape if stretched beyond the limit.

Answer: D

43. **Solution**: Coulomb's law for the force between electric charges is $F_e = \frac{kq_1 q_2}{r^2}$. Newton's law of gravitation for the force between two masses is $F_g = \frac{Gm_1 M_2}{r^2}$. Hence these two laws resemble each other.

Answer: C

44. **Solution**: In a series connection the same current passes through both bulbs. To work with full capacity the bow bulb needs more EE than the 40w bulb. Hence the 40w bulb is more brighter than the 60w bulb.

Answer: B

45. **Solution**: Assume both athlets start from rest.
    - Bolt moves the initial distance 'x' in 3sec and the next distance 100-x in 7seconds.

Hence: i) $x = \frac{1}{2}\bar{a}_b t_1^2$ where, $t_1 = 3sec$ and

ii) $100 - x = v_f \cdot t_2 = \bar{a}_b \cdot t_1 \cdot t_2$ where $t_2 = 7sec$

If you solve the above two equations for $\bar{a}_b$ you will get that

$\bar{a}_b = 3.92m/s$

Similarly

- Powel moves the initial distance 'x' in 2 seconds and the next distance 100-x in 8 seconds.

Hence (i) $x = \frac{1}{2}\bar{a}_p t_1^2$ where $t_1 = 2sec$ and

(ii) $100 - x = v_p \cdot t_2 = a_p \cdot t_1 \cdot t_2$ where $t_2 = 8sec$

Solving (i) and (ii) for $\bar{a}_p$ you will get $\bar{a}_p = 5.55m/s^2$

Answer: B

46. **Solution**:- The mass of the man is twice the mass of the boy

Then initially:- $KE_m = \frac{KE_b}{2}$ **and**

$$\rightarrow \frac{1}{2}m_m v_m^2 = \frac{1}{2}\left(\frac{1}{2}m_b v_b\right), \text{ but } m_m = 2m_b$$

$$\rightarrow V_m^2 = \frac{1}{4}V_b^2 \ldots\ldots (1)$$

Next: $\rightarrow KE_m = KE_b$ when V_m increase by 1.

$$\rightarrow \frac{1}{2}m_m(v_m + 1)^2 = \frac{1}{2}m_b V_b^2$$

$$\rightarrow \frac{1}{2}(2m_b)(V_m + 1)^2 = \frac{1}{2}m_b V_b^2$$

$$\rightarrow (V_m + 1)^2 = \frac{V_b^2}{2} \ldots . (2)$$

Solving equations (1) and (2) simultaneously gives you that original velocity of man (V_b) is equal to $\left(1+\sqrt{2}\right)^m/_s$. This is equal to $\left(\frac{1}{\sqrt{2}-1}\right)^m/_s$.

Answer: B

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